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Arthur Cayley

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888.

ON A FORM OF QUARTIC SURFACE WITH TWELVE NODES.

[From the British Association Report, (1886), pp. 540, 541.]

USING throughout capital letters to denote homogeneous quadric functions of the coordinates (x, y, z, w), we have as a form of quartic surface with eight nodes

$$\Omega = (U, V, W)^2 = 0;$$

viz. the nodes are here the octad of points, or eight points of intersection of the quadric surfaces $U = 0$, $V = 0$, $W = 0$; the equation can, by a linear transformation on the functions U, V, W (that is, by substituting for the original functions U, V, W linear functions of these variables), be reduced to the form

$$\Omega = U^2 + V^2 + W^2 = 0.$$

Suppose now that the function Ω can in a second manner be expressed in the like form $\Omega = P^2 + Q^2 + R^2$ (where P, Q, R are not linear functions of U, V, W); that is, suppose that we have identically $U^2 + V^2 + W^2 = P^2 + Q^2 + R^2$, this gives

$$U^2 - P^2 + V^2 - Q^2 + W^2 - R^2 = 0;$$

or, writing $U + P$, $V + Q$, $W + R = A$, B, C, and $U - P$, $V - Q$, $W - R = F$, G, H, the identity becomes $AF + BG + CH = 0$; and this identity being satisfied, the equation $\Omega = 0$ of the quartic surface may be written in the two forms

$$\Omega = (A + F)^2 + (B + G)^2 + (C + H)^2 = 0,$$

and

$$\Omega = (A - F)^2 + (B - G)^2 + (C - H)^2 = 0,$$

viz. the quartic surface has the nodes which are the intersections of the three quadric surfaces $A + F = 0$, $B + G = 0$, $C + H = 0$, and also the nodes which are the intersections of the three quadric surfaces $A - F = 0$, $B - G = 0$, $C - H = 0$. We may of course also write the equation of the surface in the form

$$\Omega = A^2 + B^2 + C^2 + F^2 + G^2 + H^2 = 0.$$

An easy way of satisfying the identity $AF + BG + CH = 0$ is to assume

$$A, B, C, F, G, H = ayz,\ bzx,\ cxy,\ fxw,\ gyw,\ hzw,$$

where the constants a, b, c, f, g, h satisfy the condition $af+bg+ch = 0$; this being so, the functions A, B, C, F, G, H, and consequently the functions $A+F$, $B+G$, $C+H$ and $A-F$, $B-G$, $C-H$ each of them vanish for the four points $(y = 0, z = 0, w = 0)$, $(z = 0, x = 0, w = 0)$, $(x = 0, y = 0, w = 0)$, $(x = 0, y = 0, z = 0)$, or say the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$. It hence appears that the quartic surface

$$\Omega = a^2y^2z^2 + b^2z^2x^2 + c^2x^2y^2 + f^2x^2w^2 + g^2y^2w^2 + h^2z^2w^2 = 0$$

is a quartic surface with twelve nodes; viz. it has as nodes the last-mentioned four points, the remaining four points of intersection of the surfaces

$$ayz + fxw = 0, \quad bzx + gyw = 0, \quad cxy + hzw = 0,$$

and the remaining four points of intersection of the surfaces

$$ayz - fxw = 0, \quad bzx - gyw = 0, \quad cxy - hzw = 0.$$

The above is the analytical theory of one of the two forms of quartic surface with twelve nodes recently established by Dr K. Holm in a paper in the Berichte u. d. Verhandlungen der K. Sachsische Gesellschaft zu Leipzig, (1884), pp. 52—60.

889.

ON A DIFFERENTIAL EQUATION AND THE CONSTRUCTION OF MILNER'S LAMP.

[From the Proceedings of the Edinburgh Mathematical Society, vol. v. (1887), pp. 99—101.]

WHAT sort of an equation is

$$b^3\cos(\alpha+\theta) = a\cos\theta\int_\beta^\theta r^2\,d\theta - \tfrac{1}{2}\cos\theta\int_\beta^\theta r^3\cos\theta\,d\theta + \sin\theta\int_\beta^\theta r^3\sin\theta\,d\theta? \qquad (1)$$

Write

$$X=\int_\beta^\theta r^2\,d\theta,\quad Y=\int_\beta^\theta r^3\cos\theta\,d\theta,\quad Z=\int_\beta^\theta r^3\sin\theta\,d\theta,\tag{2}$$

$$dX=-r^2,\quad dY=-r^3\cos\theta,\quad dZ=-r^3\sin\theta,\tag{3}$$

and start with the equations

$$r^2=\frac{dX}{d\theta},\quad r^3\cos\theta=\frac{dY}{d\theta},\quad r^3\sin\theta=\frac{dZ}{d\theta},$$

$$a\cos\theta\cdot X-\tfrac{1}{2}(Y\cos\theta+Z\sin\theta)=0.\tag{4}$$

This last gives

$$(r-a\cos\theta)\,dr+ar\sin\theta\cdot d\theta=0,\tag{5}$$

and the system thus is

$$\frac{dX}{d\theta}=-r^2,\quad \frac{dY}{d\theta}=-r^3\cos\theta,\quad \frac{dZ}{d\theta}=-r^3\sin\theta,\quad \frac{d\theta}{dr}=-\frac{r-a\cos\theta}{ar\sin\theta};$$

viz. this is a system of ordinary differential equations between the five variables θ, r, X, Y, Z: the system can therefore be integrated with four arbitrary constants, and these may be so determined that for the value β of θ, X, Y, Z shall be each $=0$; and r shall have the value r_0.

But this being so, from the assumed equations (3) and (4) we have

$$\frac{d}{d\theta}\{a\cos\theta\cdot X-\tfrac{1}{2}(Y\cos\theta+Z\sin\theta)\}=ar\sin\theta\cdot X+\tfrac{1}{2}r^3,$$

and further, by integration of (4),

$$a\cos\theta\cdot X-\tfrac{1}{2}(Y\cos\theta+Z\sin\theta)=L\cos\theta+M\sin\theta.$$

Here L and M denote properly determined constants: viz. the conclusion is that r, X, Y, Z admit of being determined as functions of θ and of an arbitrary constant r_0, in such wise that

$$a\cos\theta\cdot X-\tfrac{1}{2}(Y\cos\theta+Z\sin\theta)$$

shall be a function of θ, of the proper form $L\cos\theta+M\sin\theta$, but not so that it shall be the precise function $b^3\cos(\alpha+\theta)$. To make it have this value, we must have $L = b^3 \cos\alpha$, $M = -b^3 \sin\alpha$ (where L, M are given functions of a, b, α, β, r_0), i.e. we must have two given relations between a, b, α, β, r_0: or treating r_0 as a disposable constant, we must have one given relation between a, b, α, β.

The equation $d\theta = -\dfrac{ar\sin\theta}{(r - a\cos\theta)}\,dr$ gives $r^2 - 2ar\cos\theta = C$, where $C = r_0^2 - 2ar_0\cos\beta$.

There would be considerable difficulty in working the question out with r_0 arbitrary, but we may do it easily enough for the particular value $r_0 = 0$ or $r_0 = 2a\cos\beta$, giving $C = 0$ and therefore $r = 2a\cos\theta$: and we ought in this case to be able to satisfy the given equation not in general but with two determinate relations between the constants a, b, α, β.

We have

$$\int_0^\beta \cos^4\theta\,d\theta = \tfrac{3}{8}\beta + \tfrac{1}{4}\sin 2\beta + \tfrac{1}{32}\sin 4\beta,$$

and thence

$$\begin{aligned}
&a\cos\theta\cdot X - \tfrac{1}{2}(Y\cos\theta + Z\sin\theta)\\
&\quad= 4a^3\cos\theta\{\tfrac{3}{8}(\beta-\theta) + \tfrac{1}{4}(\sin 2\beta - \sin 2\theta) + \tfrac{1}{32}(\sin 4\beta - \sin 4\theta)\}\\
&\quad - \tfrac{1}{2}\cdot 4a^3\cos\theta\{\tfrac{3}{8}(\beta-\theta) + \tfrac{1}{4}(\sin 2\beta - \sin 2\theta) + \tfrac{1}{32}(\sin 4\beta - \sin 4\theta)\}\\
&\qquad - \tfrac{1}{2}\cdot 4a^3\sin\theta\{-\tfrac{1}{4}(\cos^4\beta - \cos^4\theta)\}\\
&\qquad = -\tfrac{3}{2}a^3\cos\theta(\sin 2\beta - \sin 2\theta)\\
&\qquad - \tfrac{1}{2}a^3\cos\theta(\sin 4\beta - \sin 4\theta) + \tfrac{1}{2}a^3\sin\theta(\cos^4\beta - \cos^4\theta),
\end{aligned}$$

where the terms containing β are readily reduced to $\tfrac{1}{2}a^3\cos^3\beta\sin(\theta - \beta)$; hence also the terms without β disappear of themselves: and we have

$$a\cos\theta\cdot X - \tfrac{1}{2}(Y\cos\theta + Z\sin\theta) = \tfrac{1}{2}a^3\cos^3\beta\cdot\sin(\theta-\beta),$$

which may be put

$$= b^3\cos(\theta+\alpha):$$

viz. this will be so if we have the two relations

$$\alpha = \tfrac{1}{2}\pi - \beta; \quad \text{and} \quad b^3 = -\tfrac{1}{2}a^3\cos^3\beta.$$

I make (see figure) Milner's lamp, with a circular section, β arbitrary, but a segment AM ($\angle SAM = \beta$) made solid. G in the line SG

at right angles to AM is the c. G. of the lamp, and G' the c. G. of the oil.

And this seems to be the only form—for the pole of r must, it seems to me, be on the bounding circle—viz. in the equation $r^2 - 2ar\cos\theta = C$, we must have $C = 0$.

890.

NOTE ON THE HYDRODYNAMICAL EQUATIONS.

[From the Proceedings of the Royal Society of Edinburgh, vol. xv. (1889), pp. 342—344.]

WRITING for shortness $D = \frac{d}{dt} + u\frac{d}{dx} + v\frac{d}{dy} + w\frac{d}{dz}$, then if from the hydrodynamical equations

$$Du = -\frac{d}{dx}\left(V - \frac{P}{\rho}\right), \quad Dv = -\frac{d}{dy}\left(V - \frac{P}{\rho}\right), \quad Dw = -\frac{d}{dz}\left(V - \frac{P}{\rho}\right),$$

without the aid of the equation

$$\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0,$$

we eliminate $V - \frac{P}{\rho}$, we obtain equations not equivalent to those of Helmholtz,

$$(2\xi,\ 2\eta,\ 2\zeta) = \left(\frac{dw}{dy} - \frac{dv}{dz},\ \frac{du}{dz} - \frac{dw}{dx},\ \frac{dv}{dx} - \frac{du}{dy}\right),$$

as usual), but which, transforming them by means of the omitted equation, agree as they should do with his equations. But the form of the equations obtained directly by elimination as above, is an interesting one, which it is worth while to give.

We have

$$\begin{aligned} &D\left(\frac{dv}{dz} - \frac{dw}{dy}\right) + \frac{dv}{dz}\frac{dw}{dy} - \frac{dv}{dy}\frac{dw}{dz} \\ &= \frac{d}{dz}\left(Dv + \frac{dv}{dx}u + \frac{dv}{dy}v + \frac{dv}{dz}w\right) - \frac{d}{dy}\left(Dw + \frac{dw}{dx}u + \frac{dw}{dy}v + \frac{dw}{dz}w\right) \\ &= \frac{d}{dz}\left(\frac{dv}{dt} + \frac{dv}{dx}u + \frac{dv}{dy}v + \frac{dv}{dz}w\right) - \frac{d}{dy}\left(\frac{dw}{dt} + \frac{dw}{dx}u + \frac{dw}{dy}v + \frac{dw}{dz}w\right), \end{aligned}$$

where the terms containing second derived functions disappear of themselves, and the expression on the right-hand is thus

$$\frac{du}{dz}\frac{dv}{dx} - \frac{du}{dx}\frac{dv}{dz} + \frac{dv}{dz}\frac{dv}{dy} - \frac{dv}{dy}\frac{dv}{dz} + \frac{dw}{dz}\frac{dv}{dw} - \frac{dw}{dw}\frac{dv}{dz}.$$

Representing for shortness the Matrix

$$\begin{matrix} \frac{du}{dx}, & \frac{du}{dy}, & \frac{du}{dz} \\ \frac{dv}{dx}, & \frac{dv}{dy}, & \frac{dv}{dz} \\ \frac{dw}{dx}, & \frac{dw}{dy}, & \frac{dw}{dz} \end{matrix} \quad \text{by} \quad \begin{matrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{matrix}$$

and its square by

$$\begin{matrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{matrix}$$

we have

$$\begin{pmatrix} \frac{du}{dy}, & \frac{dv}{dy}, & \frac{dw}{dy} \\ \frac{du}{dz}, & \frac{dv}{dz}, & \frac{dw}{dz} \end{pmatrix} \begin{pmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{pmatrix} = \begin{pmatrix} \frac{du}{dy}, & \frac{dv}{dy}, & \frac{dw}{dy} \\ \frac{du}{dz}, & \frac{dv}{dz}, & \frac{dw}{dz} \end{pmatrix} \begin{pmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{pmatrix};$$

viz. the combinations which enter into the foregoing formula are

$$\frac{d}{dz}(Au + Bv + Cw) - \frac{d}{dy}(A'u + B'v + C'w),$$

and

$$\begin{aligned} \frac{dv}{dz}\frac{du}{dx} - \frac{dv}{dx}\frac{du}{dz} &= -\frac{dV}{d\theta}, \\ \frac{dw}{dx}\frac{du}{dy} - \frac{dw}{dy}\frac{du}{dx} &= -\frac{dU}{d\theta}, \\ \frac{dw}{dx}\frac{dv}{dy} - \frac{dw}{dy}\frac{dv}{dx} &= -\frac{dW}{d\theta}, \end{aligned}$$

and the equation thus is $D(c' - b'') + C' - B'' = 0$; viz. the three equations are

$$\begin{aligned} D(c' - b'') + C' - B'' &= 0, \\ D(a'' - c) + A'' - C &= 0, \\ D(b - a') + B - A' &= 0, \end{aligned}$$

which are the equations in question.

Observe that we have

$$C' - B'' = (a', b', c')(c, c', c'') - (a'', b'', c'')(b, b', b'')$$
$$= a'c' + b'c + c'c'' - a''b - b'b'' - b''c'',$$

and thence, writing

$$p = a(c' - b'') + b(a'' - c) + c(b - a') = ac' - ab'' + a''b - a'c,$$

we have

$$C' - B'' + p = (a + b' + c'')(c' - b'') = 0,$$

if $a+b'+c'' = 0$; viz. this being so, $C'-B'' = -p$, or the first equation is

$$D(c' - b'') = p = a(c' - b'') + b(a'' - c) + c(b - a'),$$

that is, $D\xi = \xi\frac{du}{dx} + \eta\frac{du}{dy} + \zeta\frac{du}{dz}$, the first equation of Helmholtz, and we thus have the equations of Helmholtz, if $a + b' + c'' = 0$, that is, if $\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0$.

The foregoing three equations $D(c' - b'') + C' - B'' = 0$, &c., are the quaternion equations

$$\frac{d}{dt}\sigma + \sigma\nabla\sigma + \nabla\Pi = 0$$

($\sigma = iu + jv + kw$, $\nabla = i\frac{d}{dx} + j\frac{d}{dy} + k\frac{d}{dz}$, $\Pi = \frac{P}{\rho}$, denotes a complete differentiation), of Mr McAulay's paper "Some General Theorems in Quaternion Integration," Messenger of Mathematics, vol. xiv. (1884), pp. 26—37; see p. 34.

891.

ON THE BINODAL QUARTIC AND THE GRAPHICAL REPRESENTATION OF THE ELLIPTIC FUNCTIONS.

[From the Transactions of the Cambridge Philosophical Society, vol. xiv. (1889), pp. 484—494. Read May 6, 1889.]

I APPROACH the subject from the question of the graphical representation of the elliptic functions: assuming as usual that the modulus is real, positive, and less than unity, and to fix the ideas considering the function sn (but the like considerations are applicable to the functions cn and dn), then the equation $x + iy = \operatorname{sn}(x' + iy')$ establishes a $(1, 1)$ correspondence between the xy infinite quarter plane, and the $x'y'$ rectangle (sides K and K'): viz. to any given point $x + iy$, x and y each positive, there corresponds a single point $x' + iy'$, x', y' each positive and less than K, K' respectively: and conversely to any such point $x' + iy'$, there corresponds a single point $x + iy$, x and y each positive.

I draw in the $x'y'$-figure the rectangle $A'B'C'D'$ (sides K and K'), and in the xy-figure, I take on the axis of x, the points B, C where $AB = 1$, $AC = \dfrac{1}{k}$: and the point D at infinity. We have thus in the $x'y'$-figure the closed curve or contour $A'B'C'D'A'$: and corresponding hereto we have in the xy-figure the closed curve or contour $ABCDA$, viz. here D is the point at infinity considered as a line always at infinity, extending from the point at infinity on the positive part of the axis of x, to the point at infinity on the positive part of the axis of y, the contour being thus AB, BC, CD (D at infinity on the axis of x); and then D (at infinity on the axis of y) A. And thus to a point P' describing successively the lines $A'B'$, $B'C'$, $C'D'$, $D'A'$ there corresponds a point P describing successively the lines AB, BC, CD, DA: to P' at D' there corresponds P at D, viz. this is any point at infinity from D on the axis of x to D on the axis of y. There is no real breach of continuity: in further illustration, suppose that P, instead of actually coming to D', just cuts off the corner, viz. that it passes from a point γ' on $C'D'$ to a point α' on $D'A'$ (γ', α' each of them very near to D'): then P passes from a point γ very near D on the axis of x (that is, at a great distance from A) to a point α very near D on the axis of y (that is, at a great distance from A): and to the indefinitely small arc $\gamma'\alpha'$ described by P' there corresponds the indefinitely large arc $\gamma\alpha$ described by P.

We thus see that, if P' describe any arc $E'F'$ passing from a point E' of $A'D'$ to a point F' of $B'C'$, then P will describe an arc EF passing from a point E of AD to a point F of BC: and similarly, if P' describe any arc $G'H'$ passing from a point G' of $A'B'$ to a point H' of $C'D'$, then P will describe an arc GH passing from a point G of AB to a point H of CD.

Supposing $E'F'$ is a straight line parallel to $A'x'$, that is, cutting $A'D'$ and $B'C'$ each at right angles, then EF will be an arc cutting AD and BC each at right angles: and so if $G'H'$ is a straight line parallel to $A'y'$, that is, cutting $A'B'$ and $C'D'$ each at right angles, then GH will be an arc cutting AB and CD each at right angles: and moreover, since $E'F'$ and $G'H'$ cut each other at right angles, then also EF and GH cut each other at right angles.

Supposing, as above, that $E'F'$ and $G'H'$ are straight lines, we have EF and GH each of them the arc of a special bicircular quartic: the theory was in fact established in a very elegant manner in a memoir by Siebeck, "Ueber eine Gattung von Curven vierten Grades, welche mit den elliptischen Functionen zusammenhangen," Crelle, t. LVII. (1860), pp. 359—370, and t. LIX. (1861), pp. 173—184.

In particular, if P' describe the line $L'M'$ lying halfway between $A'B'$ and $D'C'$ (that is, if $A'L' = \frac{1}{2}K'$), then P will describe the circular quadrant LM, radius $\dfrac{1}{\sqrt{k}}$, viz. in this case the bicircular quartic degenerates into a circle twice repeated: and so if P' describe successively the lines $E'F'$ and $E'_1F'_1$ equidistant from $L'M'$ ($AE' = \frac{1}{2}K' - \eta$, $AE'_1 = \frac{1}{2}K' + \eta$), then P will describe the arcs EF and E_1F_1 which are the images of each other in regard to the centre A and circular quadrant LM, and which together constitute the quadrant of one and the same bicircular quartic.

A bicircular quartic is of course a binodal quartic with the circular points at infinity for the two nodes: there is no real gain of generality in considering the binodal quartic rather than the bicircular quartic, but I have preferred to do so, and I have accordingly introduced the term *Binodal Quartic* into the title of the present Memoir. I present in a compendious form the properties of the general curve, and I show how the curve is to be particularised so as to obtain from it the special bicircular quartics which present themselves as above in the graphical representation of the elliptic functions.

A binodal quartic has the Plückerian numbers

$$m,\ n,\ \delta,\ \kappa,\ \tau,\ \iota = 4,\ 8,\ 2,\ 0,\ 8,\ 12.$$

The number of tangents to the curve which can be drawn from either of the nodes is $n - \delta = 4$; and the pencil of tangents from the one node is homographic with the pencil of tangents from the other node. Call the nodes I and J: and let the tangents from I be called (a, b, c, d) and those from J be called (a', b', c', d'), then if the tangents which correspond to (a', b', c', d') respectively are (a, b, c, d), they may also be taken to be (b, a, d, c), (c, d, a, b) or (d, c, b, a): and considering the intersections of corresponding tangents, we have thus four tetrads of points, say the f-points, such that the points of each tetrad lie in a conic through the two nodes: and we have consequently four conics each passing through the two nodes, say these are the f-conics.

Starting as above with the correspondence (a, b, c, d), (a', b', c', d'), if the intersections of a and a', b and b', c and c', d and d' are called A, B, C, D respectively, then we have A, B, C, D for a tetrad of f-points, lying on the f-conic $(ABCD)$. Writing AB for the two points, the intersections of IA, JB and of IB, JA respectively, and so in other cases, then the remaining three tetrads of f-points are

AB, CD	lying on the f-conic	(AB, CD),
AC, BD	" "	(AC, BD),
AD, BC	" "	(AD, BC).

The two points AB may be spoken of as the antipoints of A, B: and so in other cases.

Any two of the f-conics have in common the nodes I, J, and they therefore intersect in two points besides: at each of these the tangents to the two conics, and the lines to I, J respectively, form a harmonic pencil.

Consider the two tangents at I and the two tangents at J: we have a conic touching these four lines and passing through the tetrad of f-points, or what is the same thing, intersecting the f-conic in the four f-points: say this is a c-conic. There are thus four c-conics corresponding to the four f-conics respectively.

We may consider a variable conic, passing through the points I and J; and such that the tangents thereto at these points respectively meet on a point of a c-conic: the variable conic and the corresponding f-conic each pass through the points I, J and they meet in two points besides: and the variable conic may be such that at each of these the tangents to the variable conic and the f-conic form with the lines drawn to the points I and J a harmonic pencil. The variable conic, as thus defined, contains a single variable parameter; and it has for its envelope the binodal quartic: the binodal quartic is thus in four different ways the envelope of a variable conic. This is of course Casey's Theorem for the fourfold generation of the bicircular quartic as the envelope of a variable circle.

One of the f-conics, say $(ABCD)$, may break up into the line IJ and a line, say the axis $(ABCD)$: we have then the four f-points A, B, C, D on this line. The other f-conics say (AB, CD), (AC, BD), (AD, BC) are proper conics as before: any one of these meets the line $(ABCD)$ in two points: and at each of these the line and the tangent to the conic form with the lines drawn to the points I and J a harmonic pencil. The binodal quartic is in this case said to have an *axis*.

But a second f-conic, say (AD, BC), may break up into the line IJ and a line, say the axis (AD, BC): we have then the four f-points AD, BC on this line. Writing moreover A', D' for the two points AD, or say $(AI, DJ) = A'$, $(AJ, DI) = D'$; and similarly B', C' for the two points BC, or say $(BI, CJ) = B'$, $(BJ, CI) = C'$; then the four f-points are A', B', C', D', and the axis (AD, BC) may be called $(A'B'C'D')$. The relation of the f-points A, B, C, D and A', B', C', D' and of the two axes is as shown in the figure: taking X for the intersection of IJ with $(ABCD)$ and Y for that of IJ with $(A'B'C'D')$, also Z for the intersection of the two axes, then X, Z are the sibiconjugate points of the involution AD, BC and Y, Z the sibiconjugate points of the involution $A'D'$, $B'C'$. The two axes intersect in Z, and form with the lines ZI, ZJ a pencil in involution.

The two remaining f-conics (AB, CD) and (AC, BD), or as they might also be called $(A'B', C'D')$ and $(A'C', B'D')$, are proper conics as before: they touch each other at the points I, J, and have for their common tangents at these points the lines ZI, ZJ respectively.

Each conic meets each axis in two points; and at each of these points the axis and the tangent to the conic form with the lines to I, J a harmonic pencil. The binodal quartic is in this case said to be *biaxial*. The point Z, which is the intersection of the two axes, may be called the *centre*.

In the case where an f-conic breaks up into the line IJ, and a line containing four f-points, say an f-line, the corresponding c-conic coincides with the f-conic, viz. it also breaks up into the line IJ and the f-line: the variable conic is a conic through the points I, J such that the tangents thereto at these points respectively meet on the f-line. Moreover, the variable conic must be such that at each of its intersections with the f-conic, that is, the f-line, the tangent to the variable conic and the f-line must be harmonics in regard to the lines drawn from the point to the points I, J respectively: but this condition is satisfied *ipso facto* for each of the intersections of the variable conic and the f-line. This depends on the theorem that, taking on a conic any three points P, I, J, then the tangent at P and the line drawn from P to the pole of IJ are harmonics in regard to the lines PI, PJ. Thus we have only three conditions for the variable conic, or, as above defined, it would in the case in question (of four f-points in a line) depend upon two variable parameters. There is really another condition—but what this in general is I have not ascertained: and this being so the variable conic in the case in question (of the four f-points in a line) depends upon a single variable parameter, and we have as before the bicircular quartic as the envelope. The foregoing is the axial case; in the biaxial case, the same thing happens in regard to the variable conics belonging to the two axes respectively. Thus in every case we have the fourfold generation of the curve as the envelope of a variable conic: only in the axial case, the variable conics belonging to the axis, and in the biaxial case the variable conics belonging to the two axes respectively, are not by the foregoing definitions completely defined. It will be seen further on how, in the case of the biaxial bicircular quartic, we complete the definition of the variable circles belonging to the two axes respectively.

Taking the points I, J to be the circular points at infinity, we have a bicircular quartic. The f-points are the foci, and the f-conics are circles, viz. we have 16 foci situate in fours upon four focal circles. The harmonic relation of two lines to the lines through I, J means of course that the lines cut at right angles; hence the focal circles cut each other at right angles: this must certainly be a known property, but it is not mentioned in Salmon's Higher Plane Curves, Ed. 3, Dublin, 1879, and I cannot find it in Darboux or Casey: it is given No. 81 in Lachlan's Memoir "On Systems of Circles and Spheres," Phil. Trans., vol. CLXXVII. (1886), and I find it as a question in the Educational Times, March 1, 1889, 10034 (Prof. Morley). "Prove that, of the four focal circles of a circular cubic or bicircular quartic, any two are orthogonal, and the radii are connected by the relation $\sum(\frac{1}{R^2}) = 0$." The theorem is not as well known as it should be.

The c-conics are confocal conics having for their real foci the so-called double-foci of the quartic (more accurately, the common foci are the four quadruple foci of the quartic); we have thus four conics corresponding to the four focal circles respectively, each conic intersecting the corresponding circle in the four foci upon this circle. And we have then the quartic as the envelope of a variable circle having its centre upon one of these conics and cutting at right angles the corresponding focal circle: the bicircular quartic is thus generated in four different ways.

Instead of one of the focal circles, we may have a line or axis, and the quartic is then said to be *axial*: the foci on the axis may be any four points; and for a real curve they may be all real, or two real and two imaginary, or all four imaginary. The remaining focal circles are real or imaginary circles, cut by the axis at right angles, that is, having their centres on the axis, and cutting each other at right angles.

But instead of another of the focal circles, we may have a line or axis, and the quartic is then said to be *biaxial*: the two axes cut at right angles at a point which may be called the centre of the curve. The foci on each axis form pairs of points situate symmetrically in regard to the centre. If on one of the axes the foci are real, then on the other axis they form two imaginary conjugate pairs; and conversely: but if on one of the axes the foci are two of them real and the other two conjugate imaginaries, then this is so for the other axis also. There are thus only the two cases: 1°, foci on the one axis real, and on the other conjugate imaginaries; 2°, foci on each axis two of them real and the other two conjugate imaginaries: there is however a limiting case where on each axis two foci are united at the centre, the other two foci being real on the one axis and conjugate imaginaries on the other. The remaining two focal circles are real or imaginary circles, cutting each axis at right angles, that is, having their centres at the centre; and cutting each other at right angles, that is, having the sum of the squares of their radii $= 0$.

The biaxial form of bicircular quartic is, in fact, that which presents itself in the theory of the representation of the elliptic functions.

I consider for a moment the case of a variable circle having its centre upon a given line, and cutting at right angles a given circle. The variable circles pass all of them through two fixed points, the antipoints of the intersections of the given line and circle, and which are thus real or imaginary according as the intersections of the given line and circle are imaginary or real. Hence, considering any one variable circle and the consecutive variable circle, these intersect in two real points, when the given line does not meet the given circle (meets it in two imaginary points); but when the given line meets the given circle in two real points, then the two variable circles intersect in two imaginary points: if however the given line touches the given circle, then the two variable circles touch each other. Taking now the curve of centres to be any given curve whatever, and considering one of the variable circles, and the consecutive variable circle, it at once appears that, if the tangent to the curve of centres at the centre of the variable circle does not meet the given circle, then the two variable circles intersect in two real points (which, if the tangent touch the given circle, unite in a single real point): but if the tangent to the curve of centres meets the given circle, then the two variable circles do not intersect. It hence appears that the real portions of the envelope arise exclusively from those portions of the curve of centres which are such that at any point thereof the tangent to the curve of centres does not meet the given circle. In particular, if the given circle be real, and the curve of centres is a real ellipse enclosing the given circle, then the real portion of the envelope arises from the whole ellipse: but if the curve of centres be a real ellipse cutting the given circle in four real points, then drawing the four common tangents of the ellipse and circle, it is at once seen that there are on the ellipse two detached portions such that, at any point of either portion, the tangent to the ellipse does not meet the circle: and the real portions of the envelope arise exclusively from these portions of the ellipse.

In the case just referred to, there are on the ellipse four portions each lying outside the circle and terminating in the four intersections respectively of the ellipse and circle, such that at a point of any one of these portions the tangent to the ellipse meets the circle in two real points. Starting from the extremity of one of these portions of the ellipse and proceeding to the other extremity on the circle, the corresponding variable circles do not intersect each other, but each of them is a circle lying wholly inside that which immediately precedes it; and the variable circle becomes ultimately a point, viz. this point is a focus of the curve: this agrees with the foregoing statement that the f-conic intersects the circle in the four foci upon this circle. For the two portions of the ellipse which lie inside the circle, the variable circle is of course always imaginary. The like considerations apply to the case where the locus of the centre of the variable circle is a hyperbola or parabola. The foregoing remarks illustrate the actual generation of a bicircular quartic as the envelope of the variable circle.

Starting now from the equation

$$x + iy = \mathrm{sn}(x' + iy'),$$

we have

$$x = \frac{\mathrm{sn}x'\mathrm{cn}iy'\mathrm{dn}iy'}{1+k^2\mathrm{sn}^2x'\mathrm{sn}^2iy'}, \quad y = \frac{\mathrm{sn}iy'\mathrm{cn}x'\mathrm{dn}x'}{1+k^2\mathrm{sn}^2x'\mathrm{sn}^2iy'},$$

or putting

$$\mathrm{sn}x' = p, \quad \mathrm{sn}iy' = iq;$$

these equations are

$$x = \frac{p\sqrt{1+q^2}\sqrt{1+k'^2q^2}}{1-k^2p^2q^2}, \quad y = \frac{iq\sqrt{1-p^2}\sqrt{1-k^2p^2}}{1-k^2p^2q^2},$$

whence also

$$x^2+y^2 = \frac{1+k^2p^2q^2}{(1-k^2p^2q^2)^2} = r^2 \quad (\text{if } x^2+y^2 \text{ be put } = r^2).$$

These equations, considering therein q as a given constant, and p as a variable parameter, determine the curve EF: and considering p as a given constant, and q as a variable parameter, they determine the curve GH. But the eliminations are easily effected; we have

$$p^2 = \frac{r^2-q^2}{1-k^2q^2r^2}, \quad q^2 = \frac{1-r^2}{1-k^2r^2}.$$

Hence, for EF,

$$\begin{aligned} r^2 - \frac{1}{p^2} &= \frac{(1+q^2)(1+k'^2q^2)-(1-k^2q^2r^2)}{r^2-q^2} - \frac{1-k^2q^2r^2}{r^2-q^2} \\ &= \frac{k^2q^2r^2\cdot r^2 - 2q^2\cdot r^2 + q^2(1+q^2)(1+k'^2q^2)}{r^2-q^2}, \end{aligned}$$

and consequently

$$x = \frac{\sqrt{1+q^2}\sqrt{1+k'^2q^2}}{1-k^2q^2r^2}\sqrt{r^2-q^2}\cdot\frac{1}{\sqrt{1-k^2q^2r^2}},$$

$$y = \frac{q\sqrt{1-k^2q^2r^2+r^2(1+k^2)}\sqrt{1+k'^2q^2}}{1-k^2q^2r^2},$$

giving x and y each of them in terms of r. And from the first of these we at once derive

$$(x^2+y^2)^2 - 2Ax^2 - 2By^2 + 1 = 0,$$

where

$$\frac{1}{A} = \frac{1}{1+q^2} + \frac{1}{1+k'^2q^2}, \quad \frac{1}{B} = \frac{1}{1+q^2} + \frac{k'^2}{1+k'^2q^2}.$$

Similarly, for GH,

$$r^2 - q^2 = \frac{p^2(1-k^2r^2)}{1-k^2p^2r^2},$$

and consequently

$$x = \frac{p\sqrt{(1-p^2)+(1-k^2p^2)r^2}\sqrt{1-k^2p^2+k^2p^2r^2}}{1-k^2p^2r^2},$$

$$y = \frac{p\sqrt{1+p^2-r^2}\sqrt{1-k^2p^2}\sqrt{1-k^2+k^2r^2(1-p^2)}}{1-k^2p^2r^2},$$

giving x, y each of them in terms of r. And from the second of them we at once derive

$$(x^2+y^2)^2 - 2Ax^2 - 2By^2 + 1 = 0,$$

where

$$\frac{1}{A} = \frac{1}{1-p^2} + \frac{1}{1-k^2p^2}, \quad \frac{1}{B} = \frac{1}{1-p^2} + \frac{k^2}{1-k^2p^2}.$$

Consider, in particular, the case where the line EF is the midway line LM: here $y' = \frac{1}{2}K'$, and thence $iq = \mathrm{sn}iy' = \mathrm{sn}\frac{1}{2}iK' = \frac{i}{\sqrt{k}}$, that is, $q = \frac{1}{\sqrt{k}}$: and we thence obtain

$$\frac{1}{A} = \frac{1}{B} = \tfrac{1}{2};$$

the equation of the bicircular quartic is

$$(x^2+y^2)^2 - x^2 - y^2 = 0,$$

viz. this is the circle $x^2+y^2 = \frac{1}{2}$ twice repeated. As a direct verification, observe that we have here

$$\frac{1}{1+k} = \mathrm{sn}x + \frac{i}{\sqrt{k}}\mathrm{cn}x\mathrm{dn}x,$$

and hence

$$x^2 + y^2 = \frac{1}{1+k} \cdot \frac{1+k}{2} = \tfrac{1}{2},$$

as it should be.

Reverting to the equation

$$x + iy = \operatorname{sn}(x' + iy'),$$

I write successively

$$y = \tfrac{1}{2}K' - z', \quad \operatorname{sn} iy' = iq_1 = \operatorname{sn} i(\tfrac{1}{2}K' - z'),$$

and

$$y = \tfrac{1}{2}K' + z', \quad \operatorname{sn} iy' = iq_2 = \operatorname{sn} i(\tfrac{1}{2}K' + z');$$

we then have

$$iq_1 \cdot iq_2 = \operatorname{sn} i(\tfrac{1}{2}K' - z') \operatorname{sn} i(\tfrac{1}{2}K' + z') = -\frac{1}{k},$$

that is, $q_1 q_2 = \dfrac{1}{k}$; hence for q writing q_1 or q_2, we have in each case the same values of A and B; that is, we have the same bicircular quartic corresponding to the lines $E'F'$ and $E'_1F'_1$ equidistant from the line $L'M'$: but to one of these lines there corresponds the quadrant lying inside, to the other that lying outside, the circular quadrant

$$x^2 + y^2 = \tfrac{1}{2}.$$

The curve

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + 1 = 0$$

is in four different ways the envelope of a variable circle: viz. the circle may have its centre on a conic $Ax^2 + By^2 - 1 = 0$, and cut at right angles one of the circles

$$x^2 + y^2 = A + B \pm \sqrt{(A - B)^2 - 1},$$

or it may have its centre on the axis of x, or on the axis of y. The circle, having its centre on either axis, cuts this axis at right angles; but this condition being *ipso facto* satisfied, we do not thereby determine the radius of the circle having for its centre a given point on the axis: the expression for the radius must be sought for independently.

Write for shortness
$$\lambda^2 = \frac{A-B}{A+B},$$
and consider the circle
$$x^2 + y^2 - 2ax = \lambda\sqrt{A-B-2a^2} + B,$$
where a is a variable parameter. Differentiating, we have
$$x = \frac{\lambda a}{\sqrt{A-B-2a^2}}, \quad \text{giving } a = \frac{x\sqrt{A-B}}{\sqrt{2x^2+\lambda^2}},$$
the equation of the circle then gives
$$\frac{(x^2+y^2-B)^2}{x^2} = \frac{\lambda^2(A-B)}{2x^2+\lambda^2} - (2x^2+\lambda^2),$$
that is,
$$(x^2+y^2-B)^2(2x^2+\lambda^2) = \lambda^2(A-B)x^2 - x^2(2x^2+\lambda^2)^2,$$
or we have
$$(x^2+y^2)^2 - 2Ax^2 - 2By^2 + 1 = 0$$
as the envelope of the variable circle
$$x^2 + y^2 - 2ax = \lambda\sqrt{A-B-2a^2} + B,$$
having its centre on the axis of x.

And similarly, the same quartic curve is the envelope of the variable circle
$$x^2 + y^2 - 2by = \lambda\sqrt{B-A-2b^2} + A,$$
having its centre on the axis of y.

If we have in like manner the equation $x + iy = \mathrm{cn}(x' + iy')$, then writing, as before, $\mathrm{sn}x' = p$, $\mathrm{sn}iy' = iq$, we find
$$x = \frac{\mathrm{cn}x'\mathrm{cn}iy'}{1+k^2\mathrm{sn}^2x'\mathrm{sn}^2iy'}, \quad y = \frac{-i\mathrm{sn}x'\mathrm{dn}x'\mathrm{sn}iy'\mathrm{dn}iy'}{1+k^2\mathrm{sn}^2x'\mathrm{sn}^2iy'},$$
and hence
$$r^2 = \frac{1-k^2p^2+k^2q^2}{1-k^2p^2q^2}.$$

For the curve EF corresponding to a line $E'F'$ parallel to the axis of x', we have to eliminate p from these equations; the expression for r^2 gives

$$\frac{1+k}{1-k}y^2 = (1-p^2) + \frac{q^2(1+k)}{1-k} = 1 + 2q^2 - p^2(1 + \tfrac{1+k}{1-k}q^2),$$

and thence

$$x = \frac{\sqrt{1+q^2}\sqrt{k'^2 - (1-k^2)q^2 + r^2(1+k^2)}\sqrt{1+k^2q^2} + k^2q^2r^2}{1-k^2q^2r^2},$$

$$y = \frac{q\sqrt{1+q^2-r^2}\sqrt{1+k^2q^2}\sqrt{1-k^2+k^2r^2(1-q^2)}}{1-k^2q^2r^2},$$

from which we deduce

$$(x^2+y^2)^2 - 2Ax^2 - 2By^2 + 1 = 0,$$

where

$$\frac{1}{A} = -1 + 2q^2 + 2kq^4 + k^2q^4,$$

which is a bicircular quartic of the foregoing form.

Similarly, for the curve GH corresponding to a line $G'H'$ parallel to the axis of y', we have to eliminate q: the expression for r^2 gives

$$\frac{1-k}{1+k}x^2 = (1-p^2) + \frac{q^2(1-k)}{1+k} = \dots$$

Hence

$$x = \frac{\sqrt{1-p^2}\sqrt{(1-k^2)p^2 + r^2(1-k^2)}\sqrt{1-k^2p^2} - k^2p^2r^2}{1-2k^2p^2+k^4p^4},$$

$$y = \frac{p\sqrt{-1+p^2+r^2}\sqrt{1-k^2p^2}\sqrt{1-k^2+k^2r^2(1-p^2)}}{1-k^2p^2r^2},$$

from which we deduce

$$(x^2+y^2)^2 - 2Ax^2 - 2By^2 + 1 = 0,$$

where

$$\frac{1}{A} = \dots$$

which is again a bicircular quartic of the foregoing form. And we have the like results for the equation

$$x + iy = \mathrm{dn}(x' + iy');$$

so that, for the sn, cn, and dn, the curve EF or GH is in each case a biaxial bicircular quartic of the form

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + 1 = 0.$$

892.

NOTE ON THE ORTHOMORPHIC TRANSFORMATION OF A CIRCLE INTO ITSELF.

[From the Proceedings of the Edinburgh Mathematical Society, vol. VIII. (1890), pp. 91, 92.]

THE following is, of course, substantially well known, but it strikes me as rather pretty:—to find the orthomorphic transformation of the circle

$$x^2 + y^2 - 1 = 0$$

into itself. Assume this to be

$$x_1 + iy_1 = \frac{A(x + iy) + B}{1 + C(x + iy)}.$$

Then, writing A', B', C' for the conjugates of A, B, C, we have

$$x_1 - iy_1 = \frac{A'(x - iy) + B'}{1 + C'(x - iy)},$$

and then

$$x_1^2 + y_1^2 = \frac{AA'(x^2 + y^2) + AB'(x + iy) + A'B(x - iy) + BB'}{1 + C(x + iy) + C'(x - iy) + CC'(x^2 + y^2)},$$

which should be an identity for $x^2 + y^2 = 1$, $x_1^2 + y_1^2 = 1$.

Evidently $C = AB'$, whence $C' = A'B$; and the equation then is

$$1 + AA'BB' = AA'(1 + BB'),$$

that is,

$$(1 - AA')(1 - BB') = 0.$$

But $BB' = 1$ gives the illusory result

$$x_1 + iy_1 = B,$$

therefore

$$AA' = 1,$$

and the required solution thus is

$$x_1 + iy_1 = \frac{A(x+iy)+B}{1+AB'(x+iy)},$$

where A is a unit-vector (say $A = \cos\chi + i\sin\chi$) and B, B' are conjugate vectors. Or, writing $B = b+i\beta$, $B' = b-i\beta$, the constants are χ, b, β; three constants as it should be.

893.

THE BITANGENTS OF THE QUINTIC.

(LETTER FROM PROFESSOR CAYLEY TO MR HEAL.)

[From the Annals of Mathematics, vol. v. (1890), pp. 109, 110.]

DEAR SIR,

I have to thank you very much for your letter concerning the bitangents of the quintic, and am glad you have obtained more simple numerical coefficients than you had at first. I do not see my way to verifying your result, but assuming it to be correct, it is a very interesting and valuable one. The relation to Salmon's Higher Plane Curves (p. 351) is rather curious, viz. the result there given is of the deg-order 18-54, with an extraneous factor $(\alpha x + \beta y + \gamma z)^6$, the rejection of which would reduce it to the proper form, deg-order 18-48; whereas yours is of deg-order 24-66, with an extraneous factor H^2, the rejection of which would reduce it to the same proper form, deg-order 18-48.

But there is another known solution, not by any means a handy one, but which has no extraneous factor*, viz. if

$$\Omega = D^3H - 4D^3H_1 + 6D^3H_2,$$

then the required equation is

$$\operatorname{Recip}\Omega = 0,$$

where the facients of the reciprocant are the first derived functions,

$$\frac{dU}{dx}, \quad \frac{dU}{dy}, \quad \frac{dU}{dz}.$$

Say this is

$$(Ax^4 + \ldots)(\partial_x U)^4 = 0;$$

the coefficients A, B, C, &c., are those of Ω, viz. these are of deg-order 3-6, and $\partial_x U$, &c., are of deg-order 1-4; so that the deg-order is

$$4(3\text{-}6) + 6(1\text{-}4) = (12\text{-}24) + (6\text{-}24) = (18\text{-}48),$$

as it should be. It might be worthwhile to further consider this form, but I doubt whether, practically, the whole question is not too difficult to be worth working at.

I remain, dear sir, yours very sincerely,

A. CAYLEY.
Cambridge, January 17, 1890.

* See Phil. Trans., vol. CXLIX. pp. 193—212, [260].

894.

THE INVESTIGATION BY WALLIS OF HIS EXPRESSION FOR π.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxiii. (1889), pp. 165—169.]

THE following is in effect the investigation by Wallis in the Arithmetica Infinitorum (Oxford, 1656) of his well-known expression for π. He obtains the series of equations which in modern notation are

$$\int_0^1 (x - x^2)^n \, dx = \frac{\{\phi(n)\}^2}{\phi(2n+1)},$$

or say, in general,

$$\int_0^1 (x - x^2)^n \, dx = \frac{\{\phi(n)\}^2}{\phi(2n+1)}.$$

In the case $n = \frac{1}{2}$, the integral gives the area of a semicircle, diameter $= 1$, viz. this is $= \frac{1}{8}\pi$, or we have $\frac{1}{8}\pi = \dfrac{\{\phi(\frac{1}{2})\}^2}{\phi(2)}$; or writing with him $D = \dfrac{4}{\pi}$, this is $D = \dfrac{\phi(2)}{\{\phi(\frac{1}{2})\}^2}$, where $\phi(0)$, $\phi(1)$, $\phi(2)$, ... are the series of numbers 1, 2, 6, 20, ..., which are in fact the middle binomial coefficients of the several even powers, and are the diagonal numbers in the table which he subsequently considers. As regards the form in which Wallis exhibits these results, his theorem CXXXIII. is:

> Si exponatur series Primanorum mulctata serie secundanorum, Residuorum Quadrata, Cubi, Biquadrata, &c. ad seriem Æqualium rationem habebunt cognitam.

And then, after an algebraical calculation,

$$\begin{aligned}
\int_0^1 dx &= \frac{1}{1},\\
\int_0^1 x\,dx &= \frac{1}{1\cdot 2},\\
\int_0^1 x^2\,dx &= \frac{1}{3},\\
\int_0^1 x^3\,dx &= \frac{1}{2\cdot 3},\\
\int_0^1 x^4\,dx &= \frac{1}{5},\\
\int_0^1 x^5\,dx &= \frac{1}{2\cdot 3\cdot 4\cdot 5\cdot 6}, \ \text{\&c.}
\end{aligned}$$

... et sic deinceps; continue multiplicando Numeratores per numeros quadratos et denominatores per binos continue sequentes arithmetice proportionales—viz. he thus explains the law of his calculated numbers.

He then forms the table

$$\begin{array}{cccccccc}
\infty & 1 & \frac{1}{2}D & \frac{3}{8} & \frac{1}{2}\cdot\frac{3}{4}D & \frac{3\cdot5}{2\cdot4\cdot6} & \frac{1\cdot3}{2\cdot4}\cdot\frac{5}{6}D & \dots\\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1\\
\frac{1}{2}D & 1 & D & \frac{3}{2} & \frac{1}{2}\cdot\frac{3}{2}D & \frac{3\cdot5}{2\cdot4} & \frac{1\cdot3\cdot5}{2\cdot4\cdot6}D & \dots\\
\frac{3}{8} & \frac{1}{2} & \frac{3}{2} & \frac{3}{2} & \frac{9}{4} & \frac{3\cdot5}{2\cdot4} & \frac{1\cdot3\cdot5}{2\cdot4\cdot6} & \dots\\
\frac{1}{2}\cdot\frac{3}{4}D & \frac{1}{3} & \frac{1\cdot3}{2\cdot4}D & \frac{9}{8} & \frac{3\cdot5}{2\cdot4}D & \frac{3\cdot3\cdot5}{2\cdot4\cdot6} & \frac{1\cdot3\cdot5\cdot7}{2\cdot4\cdot6\cdot8}D & \dots\\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{array}$$

where the figures in the second column are $= 1$, those in the fourth column are $= \dfrac{n+1}{2}$, those of the sixth column are $= \dfrac{(n+1)(n+2)}{2 \cdot 3}$, and so on: n being in each case the rank in the column as shown by the right-hand outside column, viz. these ranks are $-\frac{1}{2}$, 0, $\frac{1}{2}$, 1, ... at intervals of $\frac{1}{2}$. And the several even lines contain the same figures as the several even columns respectively.

Wallis then remarks that, if in any line the first and second terms respectively are A and 1, then the even lines are as follows, viz. the second line is

$$A,\ 1,\ \frac{A}{2},\ \frac{1 \cdot 3}{2},\ \frac{A}{2 \cdot 4},\ \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6},\ \ldots$$

the fourth line is

$$A,\ 1,\ \frac{A}{2},\ \frac{1 \cdot 3}{2},\ \frac{A}{2 \cdot 4},\ \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6},\ \ldots$$

the sixth line is

$$A,\ 1,\ \frac{A}{2},\ \frac{1 \cdot 3}{2},\ \frac{A}{2 \cdot 4},\ \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6},\ \ldots$$

and this being so, he completes the table by inserting a term $\frac{1}{2}D$ in the diagonal line between the 1 and 2, and calculating the odd lines as follows: viz. if in each case A is the first term and B the second term of the line, then the complete rule is: the first line is

$$A,\ 1,$$

the second line is

$$A,\ 1,\ \frac{A}{2},\ \frac{1 \cdot 3}{2},\ \frac{A}{2 \cdot 4},\ \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6},\ \ldots$$

the third line is

$$A,\ 1,\ \frac{0 \cdot 2}{1 \cdot 3}A,\ \frac{2}{1 \cdot 3},\ \frac{2 \cdot 4}{1 \cdot 3 \cdot 5}A,\ \frac{2 \cdot 4 \cdot 6}{1 \cdot 3 \cdot 5 \cdot 7},\ \ldots$$

and so on, the rule for the odd lines being thus an interpolation from that for the even lines. Observe that, in the first line A is $= \infty$, and that, in the several odd terms after the first, A is $= 1$.

The table thus calculated (I have given here the fractions in their least terms) is

$$\begin{array}{cccccccc}
\infty & 1 & \tfrac{1}{2}D & \tfrac{3}{8} & \tfrac{1\cdot3}{2\cdot4}D & \tfrac{3\cdot5}{2\cdot4\cdot6} & \tfrac{1\cdot3\cdot5}{2\cdot4\cdot6\cdot8}D & \cdots \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
\tfrac{1}{2}D & 1 & D & \tfrac{3}{2} & \tfrac{3}{2}D & \tfrac{3\cdot5}{2\cdot4} & \tfrac{3\cdot3\cdot5}{2\cdot4\cdot6}D & \cdots \\
\tfrac{3}{8} & \tfrac{1}{2} & \tfrac{3}{2} & \tfrac{3}{2} & \tfrac{9}{4} & \tfrac{3\cdot5}{2\cdot4} & \tfrac{1\cdot3\cdot3\cdot5}{2\cdot4\cdot6} & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{array}$$

in which table D, as the term interpolated between the diagonal terms 1, 2, denotes the value $\dfrac{4}{\pi}$ as before.

The third line of the table is

$$\tfrac{1}{2}D,\ 1,\ \frac{4}{1\cdot3}D,\ \frac{2}{3},\ \frac{2\cdot4}{1\cdot3\cdot5}D,\ \frac{2\cdot4}{3\cdot5},\ \frac{2\cdot4\cdot6}{1\cdot3\cdot5\cdot7}D,\ \frac{2\cdot4\cdot6}{3\cdot5\cdot7},\ \ldots$$

The successive even terms continually increase but tend to equality, and in like manner the successive odd terms continually increase but tend to equality; it seems to have been assumed that, for any three consecutive terms x, y, z, we have $\dfrac{y}{x} > \dfrac{z}{y}$, that is, $y^2 > xz$. Taking this to be so, we have

$$\frac{D}{2} > 1, \quad \frac{D^2}{2} > \frac{8}{3}, \quad \frac{3D^2}{2^2} > \frac{8\cdot16}{3\cdot5}, \quad \frac{3^2\cdot5D^2}{2^2\cdot4^2} > \frac{8\cdot16\cdot24}{3\cdot5\cdot7},\ \ldots$$

and these equations give D

$$\begin{array}{ll}
\text{less than} & \sqrt{2},\ \sqrt{\dfrac{3}{2}},\ \sqrt{\dfrac{3\cdot5}{2\cdot4}},\ \sqrt{\dfrac{3\cdot5\cdot7}{2\cdot4\cdot6}},\ \ldots \\[2ex]
\text{greater than} & \sqrt{2},\ \sqrt{\dfrac{3}{2}},\ \sqrt{\dfrac{3\cdot5}{2\cdot4}},\ \sqrt{\dfrac{3\cdot5\cdot7}{2\cdot4\cdot6}},\ \ldots
\end{array}$$

equally to equality.

We thus have

$$\frac{\pi}{2} = \frac{2\cdot2\cdot4\cdot4\cdot6\cdot6\cdots}{1\cdot3\cdot3\cdot5\cdot5\cdot7\cdots},$$

the number of factors in the numerator being always equal to the number in the denominator, and the accuracy of the approximation increasing with the number of factors.

It is to be remarked that for a square; row m and column n; m or $n = -\frac{1}{2}$, 0, $\frac{1}{2}$, 1, $\frac{3}{2}$, $\ldots$ as before; the term of the square is in general $\Pi(m+n) \div \Pi(m)\,\Pi(n)$; thus $m = n = \frac{1}{2}$, the term is $\Pi(1) \div \{\Pi(\frac{1}{2})\}^2$, $= 1 \div (\frac{1}{4}\pi)^2 = \dfrac{4}{\pi}$, $= D$; $m = 3$, $n = \frac{1}{2}$ it is $\Pi(\frac{7}{2}) \div \Pi(3)\,\Pi(\frac{1}{2})$, $= \frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} \div 6\cdot\sqrt{\pi}$, $= \frac{5}{32}\pi$; and so in any other case.

895.

A THEOREM ON TREES.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxiii. (1889), pp. 376—378.]

THE number of trees which can be formed with $n+1$ given knots α, β, γ, $\ldots$ is $= (n+1)^{n-1}$; for instance $n = 3$, the number of trees with the 4 given knots α, β, γ, δ is $4^2 = 16$, for in the first form shown in the figure the α, β, γ, δ may be arranged in 12 different orders ($\alpha\beta\gamma\delta$ being regarded as equivalent to $\delta\gamma\beta\alpha$), and in the second form any one of the 4 knots α, β, γ, δ may be in the place occupied by the α: the whole number is thus $12 + 4, = 16$.

Considering for greater clearness a larger value of n, say $n = 5$, I state the particular case of the theorem as follows:

$$\text{No. of trees } (\alpha,\beta,\gamma,\delta,\epsilon,\zeta) = \text{No. of terms of } (\alpha+\beta+\gamma+\delta+\epsilon+\zeta)^4\alpha\beta\gamma\delta\epsilon\zeta, = 6^4, = 12$$

and it will be at once seen that the proof given for this particular case is applicable for any value whatever of n.

I use for any tree whatever the following notation: for instance, in the first of the forms shown in the figure, the branches are $\alpha\beta$, $\beta\gamma$, $\gamma\delta$; and the tree is said to be $\alpha\beta^2\gamma^2\delta$ (viz. the knots α, δ occur each once, but β, γ each twice); similarly in the second of the same forms, the branches are $\alpha\beta$, $\alpha\gamma$, $\alpha\delta$, and the tree is said to be $\alpha^3\beta\gamma\delta$ (viz. the knot α occurs three times, and the knots β, γ, δ each once). And so in other cases.

This being so, I write

$$\begin{aligned}
(\alpha+\beta+\gamma+\delta+\epsilon+\zeta)^4\alpha\beta\gamma\delta\epsilon\zeta \\
&= 1\alpha^4\cdot 6\alpha\beta\gamma\delta\epsilon\zeta \\
&\quad + 4\alpha^3\beta\cdot 30\alpha^2\beta\gamma\delta\epsilon\zeta \\
&\quad + 6\alpha^2\beta^2\cdot 15\alpha^2\beta^2\gamma\delta\epsilon\zeta \\
&\quad + 12\alpha^2\beta\gamma\cdot 60\alpha^3\beta\gamma\delta\epsilon\zeta \\
&\quad + 24\alpha\beta\gamma\delta\cdot 15\alpha^4\beta\gamma\delta\epsilon\zeta \\
&= 6+120+90+720+360 = 1296,
\end{aligned}$$

where the numbers of the left-hand column are the polynomial coefficients for the index 4; and the numbers of the right-hand column are the numbers of terms of the several types, 6 terms α^4, 30 terms $\alpha^3\beta$, 15 terms $\alpha^2\beta^2$, &c.: the products of the corresponding terms of the two columns give the outside column 6, 120, 90, &c.; and the sum of these numbers is of course 6^4, $= 1296$.

It is to be shown that we have

$$1 \text{ tree } \alpha^4{\cdot}\alpha\beta\gamma\delta\epsilon\zeta\ (= \alpha^5\beta\gamma\delta\epsilon\zeta);\quad 4 \text{ trees } \alpha^3\beta{\cdot}\alpha\beta\gamma\delta\epsilon\zeta\ (= \alpha^4\beta^2\gamma\delta\epsilon\zeta),\ \ldots,$$

$$24 \text{ trees } \alpha\beta\gamma\delta\cdot\alpha\beta\gamma\delta\epsilon\zeta\ (= \alpha^2\beta^2\gamma^2\delta^2\epsilon\zeta):$$

for this being so, then by the mere interchange of letters, the numbers 1, 4, 6, ... of the left-hand column have to be multiplied by the numbers 6, 30, 15, ... of the right-hand column, and we have the numbers in the outside column, the sum of which is = 1296 as above.

Start with the last term $\alpha\beta\gamma\delta\cdot\alpha\beta\gamma\delta\epsilon\zeta = \alpha^2\beta^2\gamma^2\delta^2\epsilon\zeta$. We have the trees

$$(= \epsilon\alpha\cdot\alpha\beta\cdot\beta\gamma\cdot\gamma\delta\cdot\delta\zeta),$$

where the α, β, γ, δ may be written in any one of the 24 orders, and the number of such trees is thus = 24. If we consider only the 12 orders ($\alpha\beta\gamma\delta$ being regarded as equivalent to $\delta\gamma\beta\alpha$), then the ϵ, ζ may be interchanged; and the number is thus 2×12, = 24 as before.

Now for the δ of $\alpha\beta\gamma\delta$ substitute α, or consider the form $\alpha\beta\gamma\alpha\cdot\alpha\beta\gamma\delta\epsilon\zeta, = \alpha^3\beta^2\gamma^2\delta\epsilon\zeta$. We see at once in the form $\epsilon\alpha\cdot\alpha\beta\cdot\beta\gamma\cdot\gamma\delta\cdot\delta\zeta$, which one it is of the two δ's that must be changed into α: in fact, changing the first δ, we should have $\epsilon\alpha\cdot\alpha\beta\cdot\beta\gamma\cdot\gamma\alpha\cdot\delta\zeta$ which contains a circuit $\alpha\beta\gamma$, and a detached branch $\delta\zeta$, and is thus not a tree: changing the second δ, we have $\epsilon\alpha\cdot\alpha\beta\cdot\beta\gamma\cdot\gamma\delta\cdot\alpha\zeta$ which is a tree $\alpha^3\beta^2\gamma^2\delta\epsilon\zeta, = \alpha\zeta\cdot\alpha\epsilon\cdot\alpha\beta\cdot\beta\gamma\cdot\gamma\delta$. And similarly for any other order of the $\alpha\beta\gamma\delta$, there is in each case only one of the δ's which can be changed into α; and thus from each of the 24 forms we obtain a tree $\alpha^3\beta^2\gamma^2\delta\epsilon\zeta$. But dividing the 24 forms into the $12+12$ forms corresponding to the interchange of the letters ϵ, ζ. then the first 12 forms, and the second 12 forms, give each of them the same trees $\alpha^3\beta^2\gamma^2\delta\epsilon\zeta$; and the number of these trees is thus $\frac{1}{2}\cdot 24$, = 12.

And in like manner reducing the $\alpha\beta\gamma\delta$ to $\alpha\beta^2$, $\alpha^3\beta$ or α^4, we obtain in each case the number of trees equal to the proper submultiple of 24, viz. 6, 4, 1 in the three cases respectively (for the last case this is obvious, viz. there is 1 tree $\alpha^5\beta\gamma\delta\epsilon\zeta, = \alpha\beta\cdot\alpha\gamma\cdot\alpha\delta\cdot\alpha\epsilon\cdot\alpha\zeta$); and the subsidiary theorem is thus proved. Hence the original theorem is true: as already remarked, it is easy to see that the proof is perfectly general.

The theorem is one of a set as follows:

Let $(\lambda, \alpha, \beta, \gamma, \ldots)$ denote as above the trees with the given knots λ, α, β, γ, …; $(\lambda+\mu, \alpha, \beta, \gamma, \ldots)$ the pairs of trees with the given knots λ, μ, α, β, γ, …, the knots λ, μ, belonging always to different trees; $(\lambda+\mu+\nu, \alpha, \beta, \gamma, \ldots)$ the triads of trees with the given knots λ, μ, ν, α, β, γ, …, the knots λ, μ, ν always belonging to different trees; and so on: then if $i+1$ be the number of the knots λ, μ, ν, …, and n the number of the knots α, β, γ, …, the number of trees or pairs, or triads, &c., of trees is $= (i+1)(i+n+1)^{n-1}$. In particular, if $i = 0$, then n being the number of knots α, β, γ, …, and therefore $n+1$ the whole number of knots λ, α, β, γ, …, the number of trees is $= (n+1)^{n-1}$ as before.

As a simple example, consider the pairs $(\lambda+\mu, \alpha, \beta)$: here $i = 1$, $n = 2$, and we have $(i+1)(i+n+1)^{n-1} = 2 \cdot 4, = 8$: in fact, the pairs of trees are

$$(\lambda\alpha, \mu\beta, \mu),\ (\lambda\beta, \mu\alpha, \mu),\ (\lambda\alpha, \lambda\beta, \mu),$$

$$(\mu\alpha, \mu\beta, \lambda),\ (\mu\beta, \mu\alpha, \lambda),\ (\mu, \mu\alpha\beta, \lambda);\quad (\lambda\alpha, \mu\beta),\ (\lambda\beta, \mu\alpha).$$

We may arrange the trees $(\alpha, \beta, \gamma, \delta, \epsilon)$ as follows:

$$\begin{array}{ll}
(\alpha, \beta, \gamma, \delta, \epsilon) = \alpha\beta(\beta, \gamma, \delta, \epsilon); & 125 = 4 \times 1 \cdot 4^2 = 64 \\
\qquad + \alpha\beta \cdot \alpha\gamma(\beta+\gamma, \delta, \epsilon) & \ + 6 \times 2 \cdot 4^1 = 48 \\
\qquad + \alpha\beta \cdot \alpha\gamma \cdot \alpha\delta(\beta+\gamma+\delta, \epsilon) & \ + 4 \times 3 \cdot 4^0 = 12 \\
\qquad + \alpha\beta \cdot \alpha\gamma \cdot \alpha\delta \cdot \alpha\epsilon & \ + 1 = 1
\end{array}$$

viz. to obtain the trees $(\alpha, \beta, \gamma, \delta, \epsilon)$, we join on the branch $\alpha\beta$ to any tree $(\beta, \gamma, \delta, \epsilon)$: the branches $\alpha\beta$, $\alpha\gamma$ to any pair of trees $(\beta+\gamma, \delta, \epsilon)$; the branches $\alpha\beta$, $\alpha\gamma$, $\alpha\delta$ to any triad of trees $(\beta+\gamma+\delta, \epsilon)$; and take lastly the tree $\alpha\beta \cdot \alpha\gamma \cdot \alpha\delta \cdot \alpha\epsilon$: the knots β, γ, δ, ϵ being then interchanged in every possible manner. The whole number of trees 125 is thus obtained as $= 64 + 48 + 12 + 1$; the theorem is of course perfectly general.

The foregoing theory in effect presents itself in a paper by Borchardt, "Ueber eine der Interpolation entsprechende Darstellung der Eliminations-Resultante," Crelle, t. LVII. (1860), pp. 111—121, viz. Borchardt there considers a certain determinant, composed of the elements 10, 12, …, $1n$, 20, 21, 23, …, $2n$, …, $n0$, $n1$, …, $n(n-1)$, and represented by means of the trees $(0, 1, 2, \ldots, n)$; the branches of the tree being the aforesaid elements, and the tree being regarded as equal to the product of the several branches: the number of terms of the determinant is thus $= (n+1)^{n-1}$ as above.

896.

A TRANSFORMATION IN ELLIPTIC FUNCTIONS.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxiv. (1890), pp. 259—262.]

THE formula in question is given in Klein's Memoir "Ueber hyperelliptische Sigmafunctionen," Math. Ann. t. xxvii. (1886), pp. 431—464, see p. 454, in the form

$$-F(x, y)$$

(the discovery of it being ascribed to Weierstrass); and it is also given in Halphen's Traité des Fonctions Elliptiques, t. II. (1888), p. 357.

The algebraic foundation of the theorem is as follows: writing

$$\sqrt{4\lambda^3 z - I\lambda z - J} = \{d\sqrt{(a \between x, 1)^4} + \ldots\},$$

and as usual I, J for the two invariants of the quartic function $(a, b, c, d, e \between x, y)^4$, we have identically, as is easily verified,

$$\lambda^2(4\lambda^3 z - I\lambda z - J) = \{d\sqrt{(a \between x, 1)^4} + \ldots\}^2,$$

or say

$$\sqrt{4\lambda^3 z - I\lambda z - J} = d\sqrt{A} + \ldots$$

Hence putting

$$A = (a, b, c, d, e \between x_1, x_2)^4, \quad E = (a, b, c, d, e \between y_1, y_2)^4,$$

$$B = (a, b, c, d, e \between x_1, x_2)^3(y_1, y_2), \quad D = (a, b, c, d, e \between x_1, x_2)(y_1, y_2)^3,$$

$$C = (a, b, c, d, e \between x_1, x_2)^2(y_1, y_2)^2,$$

and

$$\lambda = x_1 y_2 - x_2 y_1,$$

so that

$$16I\lambda^4 = AE - 4BD + 3C^2,$$

$$16J\lambda^6 = ACE - AD^2 - B^2E + 2BCD - C^3,$$

then writing

$$z = \tfrac{1}{\lambda^2}\{\sqrt{AE} + C\},$$

so that $\lambda^2 z$ is a bipartite irrational quadric function of the variables (x_1, x_2) and (y_1, y_2), we have

$$\sqrt{4z^3 - Iz - J} = \frac{dz}{\lambda^2\sqrt{\lambda}},$$

and we thence infer the existence of a transformation

$$\frac{dz}{\sqrt{4z^3 - Iz - J}} = \frac{x_1\,dx_2 - x_2\,dx_1}{\sqrt{(a\,\check{}\,x, 1)^4}} + \frac{y_1\,dy_2 - y_2\,dy_1}{\sqrt{(a\,\check{}\,y, 1)^4}}.$$

But for the verification hereof and a direct determination of the value of M, observe that we have

$$\frac{1}{2\sqrt{AE}}\,dA = \frac{1}{\lambda^2}\{dC + \sqrt{E}\,d\sqrt{A}\},$$

where, attending only to the terms in dx_1, dx_2,

$$dA = 4\{(a, b, c, d\,\check{}\,x_1, x_2)^3\,dx_1 + (b, c, d, e\,\check{}\,x_1, x_2)^3\,dx_2\} = 4(A_1\,dx_1 + A_2\,dx_2),\ \text{suppose}$$

$$dC = 2\{(A_1 y_1 + B_1 y_2)\,dx_1 + (B_1 y_1 + C_1 y_2)\,dx_2\},$$

$$d\sqrt{A} = \frac{1}{2\sqrt{A}}(A_1\,dx_1 + A_2\,dx_2),$$

and therefore

$$dC + \sqrt{E}\,d\sqrt{A} = 2\{(A_1 y_1 + B_1 y_2)\,dx_1 + (B_1 y_1 + C_1 y_2)\,dx_2\} + \frac{\sqrt{E}}{2\sqrt{A}}(A_1\,dx_1 + A_2\,dx_2).$$

Hence

$$\frac{1}{2\sqrt{AE}}\,dA = \frac{1}{\lambda^2}\left[2(A_1 y_1 + B_1 y_2) + \frac{A_1\sqrt{E}}{2\sqrt{A}}\right]dx_1 + \frac{1}{\lambda^2}\left[2(B_1 y_1 + C_1 y_2) + \frac{A_2\sqrt{E}}{2\sqrt{A}}\right]d$$

The whole coefficient herein of dx_1 is

$$\frac{1}{2\sqrt{AE}}\{4\sqrt{A}(A_1 y_1 + B_1 y_2) + A_1\sqrt{E}\},$$

which is

$$\frac{1}{2\lambda^2\sqrt{AE}}\{x_1(D\sqrt{A} + B\sqrt{E})\},$$

and similarly the whole coefficient of dx_2 is

$$\frac{1}{2\lambda^2\sqrt{AE}}\{x_2(D\sqrt{A}+B\sqrt{E})\}.$$

Hence the terms in dx_1, dx_2 are together

$$\frac{1}{2\lambda^2\sqrt{AE}}(x_1\,dx_2 - x_2\,dx_1)\{D\sqrt{A}+B\sqrt{E}\}.$$

and since the terms in dy_1, dy_2 are of the like form, we have

$$\frac{dz}{\sqrt{4z^3 - Iz - J}} = \frac{x_1\,dx_2 - x_2\,dx_1}{\sqrt{(a\,\check{}\,x,1)^4}} + \frac{y_1\,dy_2 - y_2\,dy_1}{\sqrt{(a\,\check{}\,y,1)^4}},$$

and combining herewith the foregoing value

$$\sqrt{4\lambda^3 z - I\lambda z - J} = \frac{1}{\lambda^2}\{D\sqrt{A}+B\sqrt{E}\},$$

we have the required formula

$$\frac{dz}{\sqrt{4z^3 - Iz - J}} = \frac{x_1\,dx_2 - x_2\,dx_1}{\sqrt{(a\,\check{}\,x,1)^4}} + \frac{y_1\,dy_2 - y_2\,dy_1}{\sqrt{(a\,\check{}\,y,1)^4}}.$$

As a very simple verification, suppose $a = 1$, $b = c = d = e = 0$; then

$$(A, B, C, D, E) = (x_1^4,\ x_1^3x_2,\ x_1^2x_2^2,\ x_1x_2^3,\ x_2^4),$$

and if $\lambda = x_1y_2 - x_2y_1$ as before, then

$$z = \frac{x_1^2y_2^2 + x_2^2y_1^2 + x_1x_2y_1y_2}{\lambda^2},$$

where

$$\sqrt{4z^3 - Iz - J} = \frac{x_1\,dx_2 - x_2\,dx_1}{x_1^2} + \frac{y_1\,dy_2 - y_2\,dy_1}{y_2^2}.$$

Also $I = 0$, $J = 0$, and consequently

$$\sqrt{4z^3 - Iz - J} = 2\sqrt{z^3},$$

which, in virtue of the foregoing values of A, E, is

$$\frac{x_1\,dx_2 - x_2\,dx_1}{x_1^2} = \frac{y_1\,dy_2 - y_2\,dy_1}{y_2^2}.$$

I remark that an even more simple transformation from the general quartic radical to the Weierstrassian cubic radical was obtained by Hermite; this is alluded to in my "Note sur les Covariants, &c.," Crelle, t. L. (1855), pp. 285—287, [135], and is given with a demonstration in my paper "Sur quelques formules pour la transformation des intégrales elliptiques," Crelle, t. LV. (1858), pp. 15—24, [235], see No. iv.; viz. from the identical relation $JU^3 - IU^2H + 4H^3 = -\Phi^2$, which connects the covariants of a quartic function, it at once follows that if

$$U = (a, b, c, d, e \,\check{}\, x, 1)^4,$$

and H is the Hessian hereof,

$$H = (ac - b^2,\ ad - bc,\ ae + 2bd - 3c^2,\ 2(be - cd),\ ce - d^2 \,\check{}\, x, 1)^4;$$

then writing

$$z = \frac{H}{U},$$

we have

$$\frac{dz}{\sqrt{-4z^3 + Iz - J}} = \frac{2\,dx}{\sqrt{(a, b, c, d, e \,\check{}\, x, 1)^4}},$$

which is the transformation in question.

897.

SUR LES RACINES D'UNE EQUATION ALGEBRIQUE.

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. cx. (Janvier—Juin, 1890), pp. 174—176, 215—218.]

SOIT $f(u)$ une fonction rationnelle et entière avec des coefficients réels ou imaginaires, de l'ordre n; en supposant que l'équation $f'(u) = 0$, de l'ordre $n - 1$, ait $n - 1$ racines, je démontre que l'équation $f(u) = 0$ aura n racines. Pour cela, soit $f(u) = f(x + iy) = P + iQ$: je suppose que c dénote une quantité positive donnée, et je considère la surface $c - z = P^2 + Q^2$, en attribuant à la coordonnée z des valeurs positives; c'est seulement pour avoir des maxima au lieu de minima, et pour faciliter ainsi l'exposition, que je prends cette surface au lieu de $z = P^2 + Q^2$. On peut donner à c une valeur si grande que la courbe $c = P^2 + Q^2$ soit une courbe fermée qui ne se coupe pas, c'est-à-dire un contour simple: cela étant, on peut se figurer ce contour comme la ligne de rivage d'une île montagneuse; la valeur de z est au plus $= c$, et, en donnant à z une valeur plus petite, $= b$, on a le contour qui correspond à l'altitude b: évidemment, les contours qui correspondent à des altitudes différentes ne se coupent pas. Il s'agit de prouver que l'île a précisément n sommets, chacun de l'altitude c.

J'écris

$$X = \frac{dP}{dx} = \frac{dQ}{dy}, \quad Y = \frac{dP}{dy} = -\frac{dQ}{dx};$$

donc

$$\frac{dQ}{dx} = -Y, \quad \frac{dQ}{dy} = X,$$

et, de plus,

$$\frac{dX}{dx} = \frac{d^2P}{dx^2} = \frac{dY}{dy} = \frac{d^2Q}{dx\,dy}, \quad \frac{dX}{dy} = \frac{d^2P}{dx\,dy} = -\frac{dY}{dx} = -\frac{d^2Q}{dx^2} = \frac{d^2Q}{dy^2}.$$

Cela étant, nous avons

$$\frac{1}{2}\frac{dz}{dx} = P\frac{dP}{dx} + Q\frac{dQ}{dx} = PX - QY, \quad \frac{1}{2}\frac{dz}{dy} = P\frac{dP}{dy} + Q\frac{dQ}{dy} = PY + QX.$$

On aura un plan tangent horizontal pour $P = 0$, $Q = 0$, ou pour $X = 0$, $Y = 0$; les valeurs $P = 0$, $Q = 0$, appartiennent à la valeur c de z et correspondent à des sommets de montagne de cette altitude c; les valeurs $X = 0$, $Y = 0$ correspondent à des sommets de col. En effet, nous avons

$$\frac{1}{2}\frac{d^2z}{dx^2} = X^2 + P\frac{dX}{dx} - Y^2 - Q\frac{dY}{dx},$$

$$\frac{1}{2}\frac{d^2z}{dx\,dy} = XY + P\frac{dX}{dy} + YX - Q\frac{dY}{dy},$$

$$\frac{1}{2}\frac{d^2z}{dy^2} = Y^2 + P\frac{dY}{dy} + X^2 + Q\frac{dX}{dy},$$

donc

$$\frac{1}{4}\left(\frac{d^2z}{dx^2}\frac{d^2z}{dy^2} - \left(\frac{d^2z}{dx\,dy}\right)^2\right) = (X^2 + Y^2)^2 - (P^2 + Q^2)(X_1^2 + Y_1^2),$$

valeur positive pour $P = 0$, $Q = 0$; négative pour $X = 0$, $Y = 0$.

À présent, nous avons $f'(x+iy) = X - iY$; donc, en supposant que l'équation $f'(x+iy) = 0$ ait $n-1$ racines, il y aura $n-1$ systèmes de valeurs réelles de x, y qui satisfont aux équations $X = 0$, $Y = 0$; pour chaque système, il y aura des valeurs déterminées de P, Q et de là aussi de z; ces valeurs de z seront en général différentes. Ainsi il y aura dans l'île $n-1$ cols, dont les altitudes seront en général différentes: soient c_1, c_2, ..., c_{n-1} ces altitudes, commençant avec la plus petite.

Pour $b = 0$, nous avons un contour simple, et de même pour une valeur quelconque plus petite que c_1; mais, pour $z = c_1$, nous avons un col; le contour est une courbe, figure de 8; en donnant à z une valeur un peu plus grande, le contour se divise en deux courbes fermées, ou bien contours simples, extérieurs l'un à l'autre. Pour $z = c_2$, nous avons encore un col; l'un des contours simples s'est changé en figure de 8, et, pour une valeur un peu plus grande, le contour se divise en trois contours simples, chacun extérieur aux autres. En continuant de cette manière, on a, pour c_{n-1}, le col le plus haut; le contour est composé de $n-2$ contours simples et d'une figure de 8; et, en donnant à z une valeur un peu plus grande, on obtient un contour composé de n contours simples, chacun extérieur aux autres. Enfin, en faisant croître z, chacun des contours simples doit se réduire à un point, c'est-à-dire qu'il doit y avoir précisément n sommets de montagne; mais il n'y a pas de sommet de montagne, sinon pour la valeur $z = c$; donc il y a précisément n sommets de montagne, chacun de l'altitude c. On suppose toujours que l'équation $f'(x+iy) = 0$ n'ait pas de racines égales, mais il peut bien arriver que deux ou plusieurs des valeurs c_1, c_2, ..., c_n deviennent égales; la démonstration est très peu changée, en donnant à z une valeur un peu plus grande que celle qui correspond à l'altitude des cols d'altitude égale; le contour se divise toujours en contours simples, extérieurs chacun aux autres.

Il va sans dire que cette démonstration repose sur les mêmes principes que celles de Gauss et de Cauchy.

Je reprends la théorie des racines de l'équation $f(u) = 0$; au lieu de la surface $c - z = P^2 + Q^2$, il convient de considérer la surface $(c-z)^2 = P^2 + Q^2$, en faisant attention seulement aux valeurs de z positives et pas plus grandes que c. La théorie est très peu changée; les contours sont les mêmes qu'auparavant, mais ils appartiennent à des altitudes différentes; et, au lieu de maxima $z = c$ pour $P = 0$, $Q = 0$, on a des points coniques, c'est-à-dire, dans l'île montagneuse, au lieu d'un sommet arrondi de montagne, on a un cône ou un pic.

Mais avec la nouvelle surface, on construit graphiquement l'approximation de Newton: partant d'une valeur réelle ou imaginaire approximative u, on obtient la nouvelle valeur

$$u_1 = u - \frac{f(u)}{f'(u)}.$$

Je représente u par le point (x, y, z) de la surface $(c-z)^2 = P^2 + Q^2$, ou le point (x, y) du plan des sommets $z = c$; et de même, u_1 par le point (x_1, y_1, z_1) de la surface, ou (x_1, y_1) du plan des sommets: cela étant, si, par le point (x, y, z) de la surface, on mène la droite de plus grande pente (droite tangente à la surface et perpendiculaire au contour), cette droite rencontrera le plan des sommets en un point (x_1, y_1), et l'on obtient ainsi le point (x_1, y_1, z_1) de la surface, qui représente la valeur cherchée u_1. En particulier, si les coefficients de $f(u)$ sont réels, on a $Q = 0$; l'équation $(c-z)^2 = P^2 + Q^2$ devient

$$(c-z)^2 = P^2,$$

c'est-à-dire

$$c - z = \pm P,$$

ou enfin

$$c - z = \pm f(x),$$

et la section verticale de l'île est formée par des parties de ces deux courbes symétriques: pour la théorie géométrique, on peut évidemment y substituer la seule courbe $c - z = f(x)$.

898.

SUR L'ÉQUATION MODULAIRE POUR LA TRANSFORMATION DE L'ORDRE 11.

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. cxi. (Juillet—Décembre, 1890), pp. 447—449.]

L'ÉQUATION en u, v, en y écrivant $u = x$, $v = y$, est

$$\begin{aligned} &y^{12} + y^{11}x + y^{10}x^2 + y^9x^3(-22x^2+22) + y^8x^4 \\ &\quad + y^7x^5(-22x^4+22) + y^6x^6(44x^4-44) + y^5x^7(-22x^6+22) \\ &\qquad + y^4x^8 + y^3x^9(-22x^8+22) + y^2x^{10} + yx^{11} + x^{12} = 0. \end{aligned}$$

Selon un résultat trouvé par H. J. S. Smith, pour la transformation de l'ordre p, la courbe est de l'ordre $2p$, et il y a à l'origine un point double, à l'infini deux points singuliers équivalents chacun à $\frac{1}{2}(p-1)(p-2)$ points doubles, et de plus $(p-1)(p-3)$ points doubles. Au cas $p=11$, le nombre de ces derniers points doubles est donc $=80$. Cela s'accorde avec l'expression

$$D = x^{112}(1-3x^8)^{10}(16x^{16}-3x^8+16)^8(x^{64}-301960x^{56}+\ldots+1)^2,$$

trouvée par M. Hermite pour le discriminant de la fonction; et l'on voit ainsi que les valeurs de x, qui correspondent aux quatre-vingts points doubles, sont données par les équations

$$16x^{16}-3x^8+16=0,$$

$$x^{64}-301960x^{56}+\ldots+1=0.$$

Je ne considère que les seize points doubles donnés par la première équation. Cette équation donne

$$x^8=\tfrac{1}{32}(3\pm\sqrt{-7}),\quad x^4=\tfrac{1}{4\sqrt{2}}(3\pm\sqrt{-7}),$$

il y a ainsi quatre points doubles, pour lesquels les valeurs de x sont

$$x^2=\frac{1+i}{2\sqrt{2}},\quad \frac{1-i}{2\sqrt{2}},\quad \frac{-1+i}{2\sqrt{2}},\quad \frac{-1-i}{2\sqrt{2}};$$

je trouve que les valeurs correspondantes de y sont $y^2=-m^3x^2$, savoir que les quatre points sont situés sur la droite $y=-\dfrac{1+i}{2}x$, et, en changeant successivement les signes de i et $\sqrt{2}$, on voit ainsi que les seize points sont situés, quatre à quatre, sur les droites

$$y=\frac{1+i}{2}x,\quad y=\frac{1-i}{2}x,\quad y=\frac{-1+i}{2}x,\quad y=\frac{-1-i}{2}x.$$

J'écris, pour abréger,

$$m=\frac{1+i}{2}\quad (\text{donc } m^4=-1),$$

donc

$$y=mx.$$

En écrivant $y = mx$ dans l'équation et en rejetant le facteur x^2, puis en écrivant $y = mp$, l'équation se présente sous la forme

$$\begin{gathered} m^{12}p^{12} + m^{11}p^{11} + m^{10}p^{10} + m^9p^9(-22 + 88m^4) + m^8p^8 \\ + m^7p^7(-22m^4 + 132m^8 - 22m^{12}) + m^6p^6(m^{12} + 165m^8 - 160m^4 - 1) \\ + m^5p^5(22m^9 - 132m^5 + 22m) + m^4p^4(44m^6 - 44m^2) \\ + m^3p^3(-88m^3) + m^2p^2 \cdot 32m = 0, \end{gathered}$$

ou les coefficients ne contiennent que les puissances m^{21}, m^{17}, m^{13}, m^9, m^5, m^1 de m et se réduisent ainsi à des multiples de m; il y a aussi un facteur numérique 8, et, en divisant par $-8m$, l'équation devient

$$4p^{10} - 11p^8 - 11p^7 + 22p^6 + 41p^5 + 22p^4 - 11p^3 - 11p^2 + 4 = 0;$$

cette équation est de la forme

$$(2p^2 + 8p + 2)^2(p^6 - 3p^5 + 2p^4 + p^3 + 2p^2 - 3p + 1) = 0.$$

La droite $y = mx$ a donc, avec la courbe, quatre intersections doubles

$$p = \frac{-4 \pm 2\sqrt{2}}{2} = -2 \pm \sqrt{2},$$

c'est-à-dire

$$y^2 = mx^2 = \frac{-1 \pm i}{2\sqrt{2}} x^2.$$

On démontre sans peine que la droite n'est pas une tangente, et ces valeurs correspondent ainsi à des points doubles de la courbe, c'est-à-dire qu'il y a sur la droite $y = \frac{1+i}{2} x$ quatre points doubles. Réciproquement, cette valeur de x^2 conduit au facteur $(16x^{16} - 3x^8 + 16)^2$ du déterminant de l'équation modulaire.

899.

SUR LES SURFACES MINIMA.

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. cxi. (Juillet—Décembre, 1890), pp. 953, 954.]

ON peut généraliser tant la définition que la construction de ces surfaces, en substituant pour le cercle imaginaire à l'infini une conique ou même une surface quadrique quelconque.

Je rappelle que, dans la théorie ordinaire, une surface minima est une surface telle qu'un point quelconque de la surface est situé à mi-chemin entre les deux centres de courbure, et qu'une telle surface est le lieu des points à mi-chemin entre les deux points situés respectivement sur deux courbes de longueur nulle quelconques. Or on peut rattacher la notion d'une courbe de longueur nulle à celle d'une courbe de poursuite. Pour expliquer cela, j'observe que, dans le plan, en supposant comme à l'ordinaire que le lièvre coure selon une droite et que le chien et le lièvre courent avec des vitesses uniformes, la courbe de poursuite est une courbe déterminée; mais si les vitesses varient arbitrairement, alors la définition exprime seulement que la courbe est une courbe plane. Mais, dans l'espace, si au lieu d'une droite on considère une courbe plane ou à double courbure (disons une directrice) quelconque, alors, quelles que soient les vitesses, la définition précise toujours la courbe, savoir: on a toujours pour courbe de poursuite une courbe dont chaque tangente rencontre la courbe directrice. Au lieu d'une courbe, on peut avoir une surface directrice; dans ce cas, le nom est moins applicable, néanmoins je le retiens, et je dis que la courbe de poursuite est une courbe dont chaque tangente touche la surface directrice. Nous avons, de cette manière, la définition d'une courbe de poursuite par rapport à une courbe ou surface directrice quelconque.

À présent, au lieu du cercle imaginaire à l'infini, considérons une conique quelconque, l'absolue: on établit, comme on sait, par rapport à cette conique, la notion de la perpendiculaire, et ainsi les notions d'une normale et des centres de courbure c. xiii. 6

ne cessent pas de subsister. On peut donc considérer une surface telle que chaque point de la surface soit l'harmonicale par rapport aux deux centres de courbure du point de rencontre de la normale avec le plan de l'absolue: on a ainsi ce que je nomme une surface *quasi-minima*. Il va sans dire qu'il faut modifier convenablement la notion métrique d'une aire minima pour qu'elle soit applicable à cette nouvelle surface.

Pour construire la surface, on prend par rapport à l'absolue deux courbes de poursuite quelconques, et puis sur la droite, menée par deux points quelconques de ces courbes respectivement, l'harmonicale par rapport à ces deux points du point de rencontre de la droite avec le plan de l'absolue: le lieu de ce point harmonical sera la surface quasi-minima.

Il paraît permis de substituer pour la conique absolue une surface quadrique quelconque, que je nomme aussi l'absolue: on a, comme on sait, les notions de la normale et des centres de courbure. Pour la surface quasi-minima, le point sur la surface sera l'un des points doubles (foyers) de l'involution formée par les deux centres de courbure et les deux points de rencontre de la normale avec l'absolue; et de même pour la construction de la surface, il faut prendre sur la droite menée par deux points quelconques des deux courbes de poursuite respectivement les points doubles (foyers) de l'involution formée par ces deux points et les deux points de rencontre de la droite avec l'absolue.

900.

JAMES JOSEPH SYLVESTER.

[(Scientific Worthies in Nature, xxv.); Nature, vol. xxxix. (1889), pp. 217—219.]

JAMES JOSEPH SYLVESTER, born in London on September 3, 1814, is the sixth and youngest son of the late Abraham Joseph Sylvester, formerly of Liverpool*. He was educated at two private schools in London, and at the Royal Institution, Liverpool, whence he proceeded in due course of time to St John's College, Cambridge. In these early days he manifested considerable aptitude for mathematics, and so it was not matter for surprise that he came out in the Tripos Examination of 1837 as Second Wrangler; being incapacitated, by the fact of his Jewish origin, from taking his degree, he was not able to compete for either of the Smith's Prizes. In more enlightened times (1872), he had the degrees of B.A. and M.A., by accumulation, conferred upon him, and received therewith the honour of a Latin speech from the Public Orator. He himself says: "I am perhaps the only man in England who am a full (voting) Master of Arts for the three Universities of Dublin, Cambridge, and Oxford, having received that degree from these Universities in the order above given: from Dublin, by *ad eundem*; from Cambridge, *ob merita*; from Oxford, by *decree*." He is now D.C.L. of Oxford, LL.D. of Dublin and Edinburgh, and Hon. Fellow of St John's College, Cambridge. It is still open for him to receive yet higher recognition from his own alma mater**.

Prof. Sylvester became a student of the Inner Temple, July 29, 1846, and was called to the Bar on November 22, 1850†. He has been Professor of Natural Philosophy at University College, London; of Mathematics at the University of Virginia, U.S.A.‡; then ten years later Professor at the Royal Military Academy, Woolwich; and again, after a five years' interval, Professor of Mathematics at the Johns Hopkins University, Baltimore, U.S.A., from its foundation in 1877. Finally, in December 1883, he was elected Savilian Professor of Geometry at Oxford, in succession to Prof. Henry Smith§. His first printed paper was on Fresnel's optical theory (in the Phil. Mag., 1837).

We can here only briefly allude to a communication which was accompanied by many important results: we refer to the Friday evening address (January 23, 1874) to the Royal Institution, "On Recent Discoveries in Mechanical Conversion of Motion." He says:—"It would be difficult to quote any other discovery which opens out such vast and varied horizons as this of Peaucellier's,—in one direction, descending to the wants of the workshop, the simplification of the steam-engine, the revolutionising of the mill-wright's trade, the amelioration of garden-pumps, and other domestic conveniences (the sun of science glorifies all it shines upon); and in the other, soaring to the sublimest heights of the most advanced doctrines of modern analysis, lending aid to, and throwing light from a totally unexpected quarter on the researches of such men as Abel, Riemann, Clebsch, Grassmann, and Cayley. Its head towers above the clouds, while its feet plunge into the bowels of the earth."

The only works that Prof. Sylvester has published, we believe, are: (1) "A Probationary Lecture on Geometry, delivered before the Gresham Committee and the Members of the Common Council of the City of London, December 4, 1854," a slight thing which had to be written and delivered at a few hours' notice; (2) "Laws of Verse," 1870; (3) several short poems, sonnets, and translations, which have appeared in our columns and elsewhere.

Our notice would be incomplete without some record of the honours that have been conferred upon Dr Sylvester. He was elected a Fellow of the Royal Society on April 25, 1839; has received a Royal Medal (1860) and the Copley Medal (1880), this latter rarely awarded, we believe, to a pure mathematician. On this last occasion, Mr Spottiswoode accompanied the presentation with the words, "His extensive and profound researches in pure mathematics, especially his contributions to the theory of invariants and covariants, to the theory of numbers and to modern geometry, may be regarded as fully establishing Mr Sylvester's claim to the award." He is a Fellow of New College, Oxford; Foreign Associate of the United States National Academy of Sciences; Foreign Member of the Royal Academy of Sciences, Göttingen, of the Royal Academy of Sciences of Naples, and of the Academy of Sciences of Boston; Corresponding Member of the Institute of France, of the Imperial Academy of Science of St Petersburg, of the Royal Academy of Science of Berlin, of the Lyncei of Rome, of the Istituto Lombardo, and of the Société Philomathique. He has been long connected with the editorial staff of the Quarterly Journal of Mathematics (under one or another of its titles), and was the first editor of, and is a considerable contributor to, the American Journal of Mathematics; and he was at one time Examiner in Mathematics and Natural Philosophy in the University of London. He was not an original member of the London Mathematical Society (founded January 16, 1865), but was elected a member on June 19, 1865, Vice-President on January 15, 1866, and succeeded Prof. De Morgan as the second President on November 8, 1866. The Society showed its recognition of his great services to them and to mathematical science generally by awarding him its De Morgan Gold Medal in November 1887.

Wherever Dr Sylvester goes, there is sure to be mathematical activity; and the latest proof of this is the formation, during the last term at Oxford, of a Mathematical Society, which promises, we hear without surprise, to do much for the advancement of mathematical science there.

The writings of Sylvester date from the year 1837; the number of them in the Royal Society Index up to the year 1863 is 112, in the next ten years 38, and in the volume for the next ten years 81, making 231 for the years 1837 to 1883: the number of more recent papers is also considerable. They relate chiefly to finite analysis, and cover by their subjects a large part of it: algebra, determinants, elimination, the theory of equations, partitions, tactic, the theory of forms, matrices, reciprocants, the Hamiltonian numbers, &c.; analytical and pure geometry occupy a less prominent position; and mechanics, optics, and astronomy are not absent. A leading feature is the power which is shown of originating a theory or of developing it from a small beginning; there is a breadth of treatment and determination to make the most of a subject, an appreciation of its capabilities, and real enjoyment of it. There is not unfrequently an adornment or enthusiasm of language which one admires, or is amused with: we have a motto from Milton, or Shakespeare; a memoir is a trilogy divided into three parts, each of which has its action complete within itself, but the same general cycle of ideas pervades all three, and weaves them into a sort of complex unity; the apology for an unsymmetrical solution is—symmetry, like the grace of an eastern robe, has not unfrequently to be purchased at the expense of some sacrifice of freedom and rapidity of action; and, he remarks, may not music be described as the mathematic of sense, mathematic as the music of the reason? the soul of each the same! &c. It is to be mentioned that there is always a generous and cordial recognition of the merit of others, his fellow-workers in the science.

It would be in the case of any first-rate mathematician—and certainly as much so in this as in any other case—extremely interesting to go carefully through the whole of a long list of memoirs, tracing out as well their connexion with each other, and the several leading ideas on which they depend, as also their influence on the development of the theories to which they relate; but for doing this properly, or at all, space and time, and a great amount of labour, are required. Short of doing so, one can only notice particular theorems—and there are, in the case of Sylvester, many of these, "beautiful exceedingly," which, for their own sakes, one is tempted to refer to—or one can give titles, which, to those familiar with the memoirs themselves, will recall the rich stores of investigation and theory contained therein.

A considerable number of papers, including some of the earliest ones, relate to the question of the reality of the roots of a numerical equation: in the several connexions thereof with Sturm's theorem, Newton's rule for the number of imaginary roots, and the theory of invariants. Sylvester obtained for the Sturmian functions, divested of square factors, or say for the reduced Sturmian functions, singularly elegant expressions in terms of the roots, viz. these were

$$f_2(x) = \Sigma(a-b)^2(x-c)(x-d)\ldots, \quad f_3(x) = \Sigma(a-b)^2(a-c)^2(b-c)^2(x-d)\ldots, \ \&c.;$$

but not only this: applying the Sturmian process of the greatest common measure (not to $f(x)$, $f'(x)$, but instead) to two independent functions $f(x)$, $\phi(x)$, he obtained for the several resulting functions expressions involving products of differences between the roots of the one and the other equation, $f(x) = 0$, $\phi(x) = 0$; the question then arose, what is the meaning of these functions? The answer is given by his theory of intercalations: they are signaletic functions, indicating in what manner (when the real roots of the two equations are arranged in order of magnitude) the roots of the one equation are intercalated among those of the other. The investigations in regard to Newton's rule (not previously demonstrated) are very important and valuable: the principle of Sturm's demonstration is applied to this wholly different question: viz. x is made to vary continuously, and the consequent gain or loss of changes of sign is inquired into. The third question is that of the determination of the character of the roots of a quintic equation by means of invariants. In connexion with it, we have the noteworthy idea of facultative points; viz. treating as the coordinates of a point in n-dimensional space those functions of the coefficients which serve as criteria for the reality of the roots, a point is facultative or non-facultative according as there is, or is not, corresponding thereto any equation with real coefficients: the determination of the characters of the roots depends (and, it would seem, depends only) on the bounding surface or surfaces of the facultative regions, and on a surface depending on the discriminant. Relating to these theories there are two elaborate memoirs, "On the Syzygetic Relations &c." and "Algebraical Researches &c." in the Philosophical Transactions for the years 1853 and 1864 respectively; but as regards Newton's rule later papers must also be consulted.

In the years 1851—54, we have various papers on homogeneous functions, the calculus of forms, &c. (Camb. and Dub. Math. Journal, vols. vi. to ix.), and the separate work "On Canonical Forms" (London, 1851). These contain crowds of ideas, embodied in the new words, cogredient, contragredient, concomitant, covariant, contravariant, invariant, emanant, combinant, commutant, canonical form, plexus, &c., ranging over and vastly extending the then so-called theories of linear transformations and hyperdeterminants. In particular, we have the introduction into the theory of the very important idea of continuous or infinitesimal variation: say that a function, which (whatever are the values of the parameters on which it depends) is invariant for an infinitesimal change of the parameters, is absolutely invariant.

There is, in 1844, in the Philosophical Magazine, a valuable paper, "Elementary Researches in the Analysis of Combinatorial Aggregation," and the titles of two other papers, 1865 and 1866, may be mentioned: "Astronomical Prolusions; commencing with the instantaneous proof of Lambert's and Euler's theorems, and modulating through the construction of the orbit of a heavenly body from two heliocentric distances, the subtended chord, and the periodic time, and the focal theory of Cartesian ovals, into a discussion of motion in a circle and its relation to planetary motion"; and the sequel thereto, "Note on the periodic changes of orbit under certain circumstances of a particle acted upon by a central force, and on vectorial coordinates, &c., together with a new theory of the analogues of the Cartesian ovals in space."

Many of the later papers are published in the American Mathematical Journal, founded, in 1878, under the auspices of the Johns Hopkins University, and for the first six volumes of which Sylvester was editor-in-chief. We have, in vol. I., a somewhat speculative paper entitled "An application of the new atomic theory to the graphical representation of the invariants and covariants of binary quantics," followed by appendices and notes relating to various special points of the theory; and in the same and subsequent volumes various memoirs on binary and ternary quantics, including papers (by himself, with the aid of Franklin) containing tables of the numerical generating functions for binary quantics of the first ten orders, and for simultaneous binary quantics of the first four orders, &c. The memoir (vols. II. and III.) on "Ternary cubic-form equations" is connected with some early papers relating to the theory of numbers. We have in it the theory of residuation on a cubic curve, and the beautiful chain-rule of rational derivation; viz. from an arbitrary point 1 on the curve it is possible to derive the singly infinite series of points $(1, 2, 4, 5, \ldots, 3\mu \pm 1)$ such that the chord through any two points, m, n, again meets the curve in a point $m + n$, $m - n$ (whichever number is not divisible by 3) of the series; moreover, the coordinates of any point m are rational and integral functions of the degree m^2 of those of the point 1.

There is in vol. v. the memoir, "A Constructive Theory of Partitions arranged in three acts, an Interact in two parts, and an Exodion," and in vol. vi. we have "Lectures on the Principles of Universal Algebra," (referring to a course of lectures on multinomial quantity, in the year 1881). The memoir is incomplete, but the general theories of nullity and vacuity, and of the corpus formed by two independent matrices of the same order, are sketched out; and there are, in the Comptes rendus of the French Academy, later papers containing developments of various points of the theory,—the conception of "nivellators" may be referred to.

The last-mentioned paper in the American Mathematical Journal was published subsequently to Sylvester's return to England on his appointment as Savilian Professor of Mathematics at Oxford. In December 1886, he gave there a public lecture containing an outline of his new theory of reciprocants (reported in Nature, January 7, 1887), and the lectures since delivered are published under the title, "Lectures on the Theory of Reciprocants" (reported by J. Hammond), same Journal, vols. VIII. to X.; thirty-three lectures actually delivered, entire or in abstract, in the course of three terms, to a class in the University, with a concluding so-called lecture 34, which is due to Hammond. The subject, as is well known, is that of the functions of a dependent variable, y, and its differential coefficients, y', y'', $\ldots$, in regard to x (or, rather, the functions of y', y'', $\ldots$), which remain unaltered by the interchange of the variables x and y: this is a less stringent condition than that imposed by Halphen ("Thèse," 1878) on his differential invariants, and the theory is accordingly a more extensive one. A passage may be quoted:—"One is surprised to reflect on the change which is come over Algebra in the last quarter of a century. It is now possible to enlarge to an almost unlimited extent on any branch of it. These thirty lectures, embracing only a fragment of the theory of reciprocants, might be compared to an unfinished epic in thirty cantos. Does it not seem as if Algebra had attained to the dignity of a fine art, in which the workman has a free hand to develop his conceptions, as in a musical theme or a subject for painting? Formerly, it consisted in detached theorems, but nowadays it has reached a point in which every properly-developed algebraical composition, like a skilful landscape, is expected to suggest the notion of an infinite distance lying beyond the limits of the canvas." And, indeed, the theory has already spread itself out far and wide, not only in these lectures by its founder, but in various papers by auditors of them, and others,—Elliott, Hammond, Leudesdorf, Rogers, Macmahon, Berry, Forsyth.

Sylvester's latest important investigations relate to the Hamiltonian numbers: there is a memoir, Crelle, t. c. (1887), and, by Sylvester and Hammond jointly, two memoirs in the Philosophical Transactions. The subject is that of the series of numbers 2, 3, 5, 11, 47, 923, calculated thus far by Sir W. R. Hamilton in his well-known Report to the British Association, on Jerrard's method. A formula for the independent calculation of any term of the series was obtained by Sylvester, but the remarkable law by means of a generating function was discovered by Hammond, viz. E_0, E_1, E_2, $\ldots$, being the series 3, 4, 6, $\ldots$ of the foregoing numbers, each increased by unity; then these are calculated by the formula

$$\frac{1}{1-t} = E_0 + E_1 t + E_2 t^2 + \dots,$$

equating the powers of t on the two sides respectively: observe the paradox, $t = \frac{1}{2}$, then the formula gives $0 =$ sum of a series of positive powers of $\frac{1}{2}$.

Enough has been said to call to mind some of Sylvester's achievements in mathematical science. Nothing further has been attempted in the foregoing very imperfect sketch.

901.

NOTE ON THE SUMS OF TWO SERIES.

[From the Messenger of Mathematics, vol. xix. (1890), pp. 29—31.]

I CONSIDER the two series

$$S_1 = \frac{1}{1+e^{\pi a}} + \frac{1}{3(1+e^{3\pi a})} + \frac{1}{5(1+e^{5\pi a})} + \dots$$

and

$$S = \frac{1}{1+2e^{\pi a}+e^{2\pi a}} + \frac{1}{3(1+2e^{3\pi a}+e^{6\pi a})} + \frac{1}{5(1+2e^{5\pi a}+e^{10\pi a})} + \dots,$$

where a is real, positive, and indefinitely small; these would at first sight appear to be equal to each other, but this is not in fact the case.

Taking first the series S_1, putting therein $\pi a = 2x$, this is

$$\frac{1}{1+e^{2x}} + \frac{1}{3(1+e^{6x})} + \frac{1}{5(1+e^{10x})} + \dots$$

Now we have, (Legendre, Théorie des Fonctions Elliptiques, t. II. p. 438),

$$\frac{1}{1+e^{2x}} + \frac{1}{2(1+e^{4x})} + \frac{1}{3(1+e^{6x})} + \dots = C + \log \frac{\Gamma(\frac{1}{2}+\frac{x}{\pi})}{\Gamma(\frac{1}{2})},$$

where C is Euler's constant, $= .577\dots$; and if y be real, positive, and very large, then

$$\log\Gamma(y) = (y-\tfrac{1}{2})\log y - y + \tfrac{1}{2}\log 2\pi,$$

whence, differentiating the logarithm and neglecting the terms which contain negative powers of y, then the value is $= C + \log y$; hence, writing $y = \dfrac{1}{x}$, we obtain

$$\frac{1}{1+x} + \frac{1}{2(1+2x)} + \frac{1}{3(1+3x)} + \ldots = C + \log\frac{1}{x}.$$

Writing herein $2x$ for x, and dividing by 2, we have

$$\frac{1}{2(1+2x)} + \frac{1}{4(1+4x)} + \ldots = \tfrac{1}{2}C + \tfrac{1}{2}\log\frac{1}{2x}.$$

or, subtracting,

$$\frac{1}{1+x} + \frac{1}{3(1+3x)} + \frac{1}{5(1+5x)} + \ldots = \tfrac{1}{2}C + \tfrac{1}{2}\log\frac{2}{x}.$$

Hence, writing for x its value, $= \frac{1}{2}\pi a$, we have

$$S_1 = \tfrac{1}{2}C + \tfrac{1}{2}\log\frac{4}{\pi a} = \tfrac{1}{2}C + \log 2 - \tfrac{1}{2}\log\pi - \tfrac{1}{2}\log a.$$

For the series S, we have (Fundamenta Nova, p. 103*) the formula

$$\frac{2K}{\pi} = \frac{1}{1+q^{-1}} + \frac{1}{3(1+q^{-3})} + \frac{1}{5(1+q^{-5})} + \ldots,$$

or, putting herein $a = \dfrac{K'}{K}$, then $q = e^{-\pi\frac{K'}{K}} = e^{-\pi a}$, and thence

$$\frac{1}{1+e^{\pi a}} + \frac{1}{2(1+e^{2\pi a})} + \frac{1}{3(1+e^{3\pi a})} + \ldots = \log\frac{4}{k'} + \frac{2x}{\pi}.$$

We have $q = e^{-\pi a}$, which is real, positive, and less than but indefinitely near to 1; hence also k is real, positive, and less than but indefinitely near to 1, say the value is $= 1 - \beta$; thence $k' = \sqrt{2\beta}$, and $K = \log\dfrac{4}{\sqrt{2\beta}}$, $= \log\dfrac{4}{\sqrt{k'}}$; also $K' = \frac{1}{2}\pi$, and therefore $a = \dfrac{K'}{K} = \frac{1}{2}\pi \div \log\dfrac{4}{\sqrt{k'}}$, whence $\log\dfrac{4}{k'} = \dfrac{\pi}{2a}$, which is the relation between a and β; and we thus have $\dfrac{2K}{\pi} = \log\dfrac{4}{k'} = \dfrac{\pi}{2a}$; and consequently $S = \frac{1}{2}\log\dfrac{4}{\pi a}$, $= -\frac{1}{2}\log a$.

The two values thus are

$$S_1 = \tfrac{1}{2}(C + 2\log 2 - \log\pi) - \tfrac{1}{2}\log a, \quad S = -\tfrac{1}{2}\log a,$$

each depending on $\log a$, and having for this term the same coefficient $-\frac{1}{2}$; but there is in S_1 a constant term $\frac{1}{2}(C + 2\log 2 - \log\pi)$, where C is the constant $.577\ldots$.

It is easy to see why the series S is not reducible to S_1; however small a may be in the general term $\dfrac{1}{n(1+e^{n\pi a})}$, then taking n sufficiently large, not only $n\pi a$ is not indefinitely small, but it in fact becomes indefinitely large; the general term of the first series thus approximates to $\dfrac{1}{n}e^{-n\pi a}$, or the terms diminish somewhat more rapidly than in a geometric series with the ratio $e^{-\pi a}$ (a small positive value less than but very near to 1), whereas in the second series the general term approximates to $\dfrac{1}{n^2\pi a}$, or the convergence is ultimately that of the series $\dfrac{1}{n^2} + \dfrac{1}{(n+1)^2} + \ldots$.

* Jacobi's Gesammelte Werke, t. I., p. 159.

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Arthur Cayley

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896. A Transformation in Elliptic Functions.

[From the Messenger of Mathematics, vol. xix. (1890), pp. 30–33.]

[Continuation from p. 31]

which is

$$\frac{1}{\sqrt{vw}},$$

and similarly the whole coefficient of dx_2^2 is

$$\frac{1}{\sqrt{vw}}.$$

Hence the terms in dx_1, dx_2 are together

$$(x_1 - x_2)\{D_x\sqrt{J(A)} + B_x\sqrt{J(E)}\}.$$

and since the terms in dy_1, dy_2 are of the like form, we have

$$\frac{dz}{\sqrt{(1-Lz-J)}} = \frac{1}{\sqrt{J(A)}}\cdot\frac{x_1\,dx_2 - x_2\,dx_1}{x_1 - x_2} - \frac{1}{\sqrt{J(E)}}\cdot\frac{y_1\,dy_2 - y_2\,dy_1}{y_1 - y_2},$$

and combining herewith the foregoing value

$$\frac{dz}{\sqrt{(1-Lz-J)}} = \frac{x_1\,dx_2 - x_2\,dx_1}{x_1 - x_2} - \frac{y_1\,dy_2 - y_2\,dy_1}{y_1 - y_2},$$

we have the required formula

$$\frac{dz}{\sqrt{(1-Lz-J)}} = \frac{x_1\,dx_2 - x_2\,dx_1}{x_1 - x_2} - \frac{y_1\,dy_2 - y_2\,dy_1}{y_1 - y_2}.$$

As a very simple verification, suppose $a=1$, $b=c=d=e=0$; then

$$(A, B, C, D, E) = (x_1^4,\ x_1^3 y_1,\ x_1^2 y_1^2,\ x_1 y_1^3,\ y_1^4),$$

and if $\lambda = x_1 y_2 - x_2 y_1$, as before, then

$$z = \frac{x_1^2 y_1^2}{(x_1 y_2 - x_2 y_1)^2} = \theta^2,$$

where

$$\theta = \frac{-x_1 y_1}{x_1 y_2 - x_2 y_1}, \quad \text{or} \quad \frac{1}{\theta} = \frac{x_2}{x_1} - \frac{y_2}{y_1}.$$

Also $I=0$, $J=0$, and consequently

$$\sqrt{(1 - Lz - J)} = \sqrt{(1 - z^2)},$$

which, in virtue of the foregoing values of A, E, is

$$\frac{dz}{\sqrt{(1-z^2)}} = \frac{x_1\, dx_2 - x_2\, dx_1}{x_1 - x_2} - \frac{y_1\, dy_2 - y_2\, dy_1}{y_1 - y_2}.$$

I remark that an even more simple transformation from the general quartic radical to the Weierstrassian cubic radical was obtained by Hermite; this is alluded to in my "Note sur les Covariants, &c.," Crelle, t. L. (1855), pp. 285–287, [135], and is given with a demonstration in my paper "Sur quelques formules pour la transformation des intégrales elliptiques," Crelle, t. LV. (1858), pp. 15–24, [235], see No. iv.; viz. from the identical relation $JU^3 - IU^2H + 4H^3 = -\Phi^2$, which connects the covariants of a quartic function, it at once follows that if

$$U = (a, b, c, d, e \,\rlap{|}\; x, 1)^4,$$

and H is the Hessian hereof,

$$H = (ac - b^2,\ ad - bc,\ ae + 2bd - 3c^2,\ 2(be - cd),\ ce - d^2 \,\rlap{|}\; x, 1)^4;$$

then writing

$$z = \frac{H}{U},$$

we have

$$\frac{dz}{\sqrt{(-4z^3 + Iz - J)}} = \frac{2\, dx}{\sqrt{(a, b, c, d, e \,\rlap{|}\; x, 1)^4}},$$

which is the transformation in question.

897. Sur les Racines d'une Équation Algébrique.

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. cx. (Janvier–Juin, 1890), pp. 174–176, 215–218.]

[p. 33]

Soit $f(u)$ une fonction rationnelle et entière avec des coefficients réels ou imaginaires, de l'ordre n; en supposant que l'équation $f'(u) = 0$, de l'ordre $n-1$, ait $n-1$ racines, je démontre que l'équation $f(u) = 0$ aura n racines. Pour cela, soit

$$f(u) = f(x+iy) = P + iQ;$$

je suppose que c dénote une quantité positive donnée, et je considère la surface $c - z = P^2 + Q^2$, en attribuant à la coordonnée z des valeurs positives; c'est seulement pour avoir des maxima au lieu de minima, et pour faciliter ainsi l'exposition, que je prends cette surface au lieu de $z = P^2 + Q^2$. On peut donner à c une valeur si grande que la courbe $c = P^2 + Q^2$ soit une courbe fermée qui ne se coupe pas, c'est-à-dire un contour simple: cela étant, on peut se figurer ce contour comme la ligne de rivage d'une île montagneuse; la valeur de z est au plus $= c$, et, en donnant à z une valeur plus petite, $= b$, on a le contour qui correspond à l'altitude b: évidemment, les contours qui correspondent à des altitudes différentes ne se coupent pas. Il s'agit de prouver que l'île a précisément n sommets, chacun de l'altitude c.

J'écris

$$X = \frac{dP}{dx} = \frac{dQ}{dy}, \quad Y = \frac{dP}{dy} = -\frac{dQ}{dx};$$

donc

$$\frac{dQ}{dx} = -Y, \quad \frac{dQ}{dy} = X,$$

et, de plus,

$$\frac{dX}{dx} = \frac{d^2P}{dx^2}, \quad \frac{dX}{dy} = \frac{d^2P}{dx\,dy}, \quad \frac{dY}{dx} = \frac{d^2P}{dx\,dy}, \quad \frac{dY}{dy} = \frac{d^2P}{dy^2}.$$

[p. 34]

et de là

$$\frac{d^2z}{dx^2} = -2(X^2+Y^2) - 2P\frac{d^2P}{dx^2} - 2Q\frac{d^2Q}{dx^2}, \quad \text{etc.}$$

Cela étant, nous avons

$$\frac{dz}{dx} = -2\left(P\frac{dP}{dx} + Q\frac{dQ}{dx}\right), \quad \frac{dz}{dy} = -2\left(P\frac{dP}{dy} + Q\frac{dQ}{dy}\right);$$

on aura un plan tangent horizontal pour $P = 0$, $Q = 0$, ou pour $X = 0$, $Y = 0$; les valeurs $P = 0$, $Q = 0$, appartiennent à la valeur c de z et correspondent à des sommets de montagne de cette altitude c; les valeurs $X = 0$, $Y = 0$ correspondent à des sommets de col. En effet, nous avons

$$\frac{d^2z}{dx^2} = -2(X^2 + Y^2) - 2P\frac{dX}{dx} - 2Q\frac{dY}{dx}, \quad \text{etc.},$$

done

$$\frac{d^2z}{dx^2}\frac{d^2z}{dy^2} - \left(\frac{d^2z}{dx\,dy}\right)^2 = (X^2 + Y^2)^2 - (P^2 + Q^2)(a^2 + A^2),$$

valeur positive pour $P = 0$, $Q = 0$; négative pour $X = 0$, $Y = 0$.

À présent, nous avons $f'(x + iy) = X - iY$; donc, en supposant que l'équation $f'(x+iy) = 0$ ait $n-1$ racines, il y aura $n-1$ systèmes de valeurs réelles de x, y qui satisfont aux équations $X = 0$, $Y = 0$; pour chaque système, il y aura des valeurs déterminées de P, Q et de là aussi de z; ces valeurs de z seront en général différentes. Ainsi il y aura dans l'île $n - 1$ cols, dont les altitudes seront en général différentes: soient c_1, c_2, $\ldots$, c_{n-1} ces altitudes, commençant avec la plus petite.

Pour $b = 0$, nous avons un contour simple, et de même pour une valeur quelconque plus petite que c_1; mais, pour $z = c_1$, nous avons un col; le contour est une courbe, figure de 8; en donnant à z une valeur un peu plus grande, le contour se divise en deux courbes fermées, ou bien contours simples, extérieurs l'un à l'autre. Pour $z = c_2$, nous avons encore un col; l'un des contours simples s'est changé en figure de 8, et, pour une valeur un peu plus grande, le contour se divise en trois contours simples, chacun extérieur aux autres. En continuant de cette manière, on a, pour c_{n-1}, le col le plus haut; le contour est composé de $n-2$ contours simples et d'une figure de 8; et, en donnant à z une valeur un peu plus grande, on obtient un contour composé de n contours simples, chacun extérieur aux autres. Enfin, en

[p. 35]

faisant croître z, chacun des contours simples doit se réduire à un point, c'est-à-dire qu'il doit y avoir précisément n sommets de montagne; mais il n'y a pas de sommet de montagne, sinon pour la valeur $z = c$; donc il y a précisément n sommets de montagne, chacun de l'altitude c. On suppose toujours que l'équation $f'(x+iy) = 0$ n'ait pas de racines égales, mais il peut bien arriver que deux ou plusieurs des valeurs $c_1, c_2, \ldots, c_n$ deviennent égales; la démonstration est très peu changée, en donnant à z une valeur un peu plus grande que celle qui correspond à l'altitude des cols d'altitude égale; le contour se divise toujours en contours simples, extérieurs chacun aux autres.

Il va sans dire que cette démonstration repose sur les mêmes principes que celles de Gauss et de Cauchy.

Je reprends la théorie des racines de l'équation $f(u) = 0$; au lieu de la surface $c - z = P^2 + Q^2$, il convient de considérer la surface $(c - z)^2 = P^2 + Q^2$, en faisant attention seulement aux valeurs de z positives et pas plus grandes que c. La théorie est très peu changée; les contours sont les mêmes qu'auparavant, mais ils appartiennent à des altitudes différentes; et, au lieu de maxima $z = c$ pour $P = 0$, $Q = 0$, on a des points coniques, c'est-à-dire, dans l'île montagneuse, au lieu d'un sommet arrondi de montagne, on a un cône ou un pic.

Mais avec la nouvelle surface, on construit graphiquement l'approximation de Newton: partant d'une valeur réelle ou imaginaire approximative u, on obtient la nouvelle valeur

$$u_1 = u - \frac{f(u)}{f'(u)}.$$

Je représente u par le point (x, y, z) de la surface $(c - z)^2 = P^2 + Q^2$, ou le point (x, y) du plan des sommets $z = c$; et de même, u_1 par le point (x_1, y_1, z_1) de la surface, ou (x_1, y_1) du plan des sommets: cela étant, si, par le point (x, y, z) de la surface, on mène la droite de plus grande pente (droite tangente à la surface et perpendiculaire au contour), cette droite rencontrera le plan des sommets en un point (x_1, y_1), et l'on obtient ainsi le point (x_1, y_1, z_1) de la surface, qui représente la valeur cherchée u_1. En particulier, si les coefficients de $f(u)$ sont réels, on a $Q = 0$; l'équation $(c - z)^2 = P^2 + Q^2$ devient

$$(c - z)^2 = P^2,$$

c'est-à-dire

$$c - z = \pm P,$$

ou enfin

$$c - z = \pm f(x),$$

et la section verticale de l'île est formée par des parties de ces deux courbes symétriques: pour la théorie géométrique, on peut évidemment y substituer la seule courbe $c - z = f(x)$.

[p. 36]

J'ai proposé, il y a plus de dix ans (Amer. Math. Journ., t. II., 1879, [694]), le problème que je nomme "The Newton-Fourier imaginary Problem," et dans une Note (Quart. Math. Journ., t. xvi., 1879, [736]), "Application of the Newton-Fourier method to an imaginary root of an equation," j'ai considéré le cas d'une équation quadratique. Pour l'équation $u^2 - 1 = 0$, on a

$$u_1 = \frac{1}{2}\left(u + \frac{1}{u}\right), \quad \text{or} \quad x_1 + iy_1 = \frac{1}{2}\left(x + iy + \frac{1}{x + iy}\right);$$

cela donne

$$x_1 = \frac{1}{2}\left(x + \frac{x}{x^2 + y^2}\right), \quad y_1 = \frac{1}{2}\left(y - \frac{y}{x^2 + y^2}\right),$$

et de là

$$\frac{u_1 - 1}{u_1 + 1} = \left(\frac{u - 1}{u + 1}\right)^2.$$

Cette dernière équation a rapport aux deux racines $+1$ et -1, et, quoiqu'elle donne les résultats les plus élégants, cependant, en vue de la théorie générale, il vaut mieux considérer l'équation $u_1 - 1 = \frac{1}{2}(u - 1)^2$ qui se rapporte à la seule racine $+1$.

Je remarque d'abord que la formule originale $u_1 = \frac{1}{2}(u + \frac{1}{u})$ donne

$$\frac{x_1}{y_1} = \frac{x}{y} \cdot \frac{x^2 + y^2 + 1}{x^2 + y^2 - 1},$$

donc les valeurs de x et x_1 seront à la fois positives ou négatives, et ainsi l'on peut ne faire attention qu'aux valeurs positives. Cela étant, nous avons

$$x_1 - 1 + iy_1 - 1 = \frac{(x + iy - 1)^2}{2(x + iy)}.$$

Désignons par A le point $(0, 1)$, par B le point $(0, -1)$, par O le point $(0, 0)$; et aussi par P le point (x, y), et de même

[p. 37]

par P_1 le point (x_1, y_1). En considérant l'expression de $u_1 - 1$, on voit que la distance du point P_1 au point A est le carré de la distance du point P au point A, divisé par le double de la distance du point P au point O. Cela étant, soit ρ la distance AP, et ρ_1 la distance AP_1: on a

$$\rho_1 = \frac{\rho^2}{2OP}, \quad \text{or} \quad AP_1 = \frac{(AP)^2}{2OP}.$$

Ainsi, pour le point P situé sur l'axe des x, on a $AP_1 < AP$; c'est-à-dire que l'approximation est régulière: et l'on peut même (dans un sens qui sera expliqué plus bas, mais qui n'est pas le sens le plus naturel) dire que l'approximation est régulière. En effet, on n'a pas toujours $AP_1 < AP$, et ainsi, dans le sens le plus naturel, l'approximation n'est pas toujours régulière. Pour étudier cela, j'écris $AP_1 = AP$; cela donne $AP = 2OP$, ou, ce qui est la même chose, $x^2 + y^2 + \frac{1}{4} = \frac{1}{4}$, c'est-à-dire que le point P sera situé sur le cercle, centre $x = -\frac{1}{2}$ et rayon $= \frac{1}{2}$; en ne faisant attention qu'aux valeurs positives de x, on a un segment compris entre l'axe des y et un arc par les points $(x = 0,\ y = \pm\frac{1}{2})$, $(x = \frac{1}{2},\ y = 0)$. Si le point P est sur l'arc, on aura $AP_1 = AP$; si P est en dedans du segment, alors $AP_1 > AP$; si P est en dehors du segment, $AP_1 < AP$.

Mais, en supposant P en dehors du segment, et ainsi $AP_1 < AP$, il peut bien arriver que P_1 soit en dedans du segment, et, cela étant, on aura $AP_2 > AP_1$, et l'approximation ne sera pas régulière. Mais, en considérant le cercle $x^2 + y^2 - \frac{1}{2}x = \frac{1}{4}$, lequel est le cercle, centre A et rayon $\frac{1}{2}$, qui touche le segment au point $(x = \frac{1}{2},\ y = 0)$, alors, en supposant que le point P soit en dedans de ce cercle, on aura $AP_1 < AP$, le point P_1 sera aussi en dedans du cercle, et ainsi en dehors du segment; et les points successifs $P,\ P_1,\ P_2,\ \ldots$ approcheront continuellement du point A; l'approximation sera dans ce cas régulière.

Il y a ainsi trois régions: le segment, le cercle $x^2 + y^2 - \frac{1}{2}x = \frac{1}{4}$ et le résidu du demi-plan; on pourrait les nommer régions noire, blanche et grise respectivement. C'est seulement pour un point P à l'intérieur de la région blanche que l'approximation est certainement régulière.

Nous venons de considérer en effet les cercles qui ont pour centre le point A; $AP_1 < AP$ veut dire que le point P est situé sur un cercle plus grand, et P_1 sur un cercle plus petit; mais, au lieu de ces cercles concentriques, considérons des cercles quelconques qui entourent le point A, sans se couper les uns les autres; et convenons de dire que c'est un bon pas quand on passe du point P sur un cercle plus grand à un point P_1 sur un cercle plus petit: avec cette convention on aura, en général, trois régions, lesquelles cependant ne seront pas les mêmes comme auparavant. En particulier, si nous considérons les cercles $AP = k \cdot BP$ (k une constante quelconque plus petite que l'unité), alors il n'y a pas de région noire, ou, si l'on veut, la région noire se réduit à la seule droite $y = 0$; donc il n'y a pas non plus de région grise, et le demi-plan entier est région blanche, c'est-à-dire, dans le sens que je viens d'expliquer, l'approximation est toujours régulière. En effet, c'est là la théorie à laquelle on est conduit au moyen de l'équation $\frac{u_1-1}{u_1+1} = \left(\frac{u-1}{u+1}\right)^2$ ci-dessus mentionnée.

En parlant de cercles, j'ai fait une restriction qui n'est nullement nécessaire; j'aurais pu parler d'ovales, de forme quelconque, qui entourent le point A sans se couper les uns les autres.

J'espère appliquer cette théorie au cas d'une équation cubique, mais les calculs sont beaucoup plus difficiles.

898. Sur l'Équation Modulaire pour la Transformation de l'Ordre 11.

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. cxi. (Juillet–Décembre, 1890), pp. 447–449.]

[p. 38]

L'équation en u, v, en y écrivant $u = x$, $v = y$, est

$$\begin{aligned} &x^{12} + y^{12} - x^5y^5 + 11x^6y^6(x^5+y^5) - 66x^4y^4(x^5+y^5) \\ &\quad + 11x^3y^3(x^5+y^5) - 11x^2y^2(x^5+y^5) - 11xy(x^5+y^5) - 1 = 0. \end{aligned}$$

Selon un résultat trouvé par H. J. S. Smith, pour la transformation de l'ordre p, la courbe est de l'ordre $2p$, et il y a à l'origine un point double, à l'infini deux points singuliers équivalents chacun à $\frac{1}{2}(p-1)(p-2)$ points doubles, et de plus $(p-1)(p-3)$ points doubles. Au cas $p = 11$, le nombre de ces derniers points doubles est donc $= 80$. Cela s'accorde avec l'expression

$$D = x^{116}(1-x^8)^{10}(16x^{16} - 31x^8 + 16)^8(x^{64} - 301960x^{56} + \ldots + 1)^2,$$

[p. 39]

trouvée par M. Hermite pour le discriminant de la fonction; et l'on voit ainsi que les valeurs de x, qui correspondent aux quatre-vingts points doubles, sont données par les équations

$$16x^{16} - 31x^8 + 16 = 0,$$

$$x^{64} - 301960x^{56} + \ldots + 1 = 0.$$

Je ne considère que les seize points doubles donnés par la première équation. Cette équation donne

$$x^8 = \tfrac{1}{32}(31 \pm 3\sqrt{7}), \quad x^4 = \tfrac{1}{4}(3 \pm \sqrt{7}),$$

il y a ainsi quatre points doubles, pour lesquels les valeurs de x sont

$$x = \pm\sqrt{\frac{3+\sqrt{7}}{4}}, \quad x = \pm i\sqrt{\frac{3-\sqrt{7}}{4}}.$$

Je trouve que les valeurs correspondantes de y sont $y = -mx$, savoir que les quatre points sont situés sur la droite $y = -\frac{1+i}{2}x$, et, en changeant successivement les signes de i et $\sqrt{2}$, on voit ainsi que les seize points sont situés, quatre à quatre, sur les droites

$$y = -\frac{1+i}{2}x, \quad y = \frac{i-1}{2}x, \quad y = \frac{1+i}{2}x, \quad y = \frac{1-i}{2}x.$$

J'écris, pour abréger,

$$m = \frac{1+i}{2} \quad (\text{donc } m^4 = -1),$$

done

$$y = -mx.$$

En écrivant $y = mx$ dans léquation et en rejetant le facteur x^2, puis en écrivant $\rho = mp$, léquation se présente sous la forme

$$\left.\begin{array}{rl}
m^{10}\rho^{10}\,. & 32m^{11} \\
+\ m^8\rho^8\,. & 88m^9 \\
+\ m^7\rho^7\,. & 44m^{10} - 44m^6 \\
+\ m^6\rho^6\,. & -22m^{11} + 132m^7 - 22m^3 \\
+\ m^5\rho^5\,. & m^{12} + 165m^8 - 165m^4 - 1 \\
+\ m^4\rho^4\,. & 22m^9 - 132m^5 + 22m \\
+\ m^3\rho^3\,. & 44m^6 - 44m^2 \\
+\ m^2\rho^2\,. & -88m^3 \\
+\ 1\,. & -32m
\end{array}\right\} = 0,$$

ou les coefficients ne contiennent que les puissances m^{21}, m^{17}, m^{13}, m^9, m^5, m^1 de m et se réduisent ainsi à des multiples de m; il y a aussi un facteur numérique 8, et, en divisant par $-8m$, léquation devient

$$4\rho^{10} - 11\rho^8 - 11\rho^7 + 22\rho^6 + 41\rho^5 + 22\rho^4 - 11\rho^3 - 11\rho^2 + 4 = 0;$$

cette équation est de la forme

$$(2\rho^2 + 3\rho + 2)^2(\rho^6 - 3\rho^5 + 2\rho^4 + \rho^3 + 2\rho^2 - 3\rho + 1) = 0.$$

La droite $y = mx$ a donc, avec la courbe, quatre intersections doubles

$$\rho = \frac{-3 \pm \sqrt{7}}{4}, \quad \rho^6 - 3\rho^5 + 2\rho^4 + \rho^3 + 2\rho^2 - 3\rho + 1 = 0,$$

c'est-à-dire

$$2x^2 = \frac{-3 \pm \sqrt{7}}{4m^2}, \quad \text{ou} \quad x^2 = \frac{-3 \pm \sqrt{7}}{4\sqrt{2}}i;$$

on démontre sans peine que la droite n'est pas une tangente, et ces valeurs correspondent ainsi à des points doubles de la courbe, c'est-à-dire qu'il y a sur la droite $y = \frac{1+i}{2}x$ quatre points doubles. Réciproquement, cette valeur de x^2 conduit au facteur $(16x^{16} - 31x^8 + 16)^2$ du déterminant de l'équation modulaire.

899. Sur les Surfaces Minima.

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. cxi. (Juillet–Décembre, 1890), pp. 953, 954.]

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On peut généraliser tant la définition que la construction de ces surfaces, en substituant pour le cercle imaginaire à l'infini une conique ou même une surface quadrique quelconque.

Je rappelle que, dans la théorie ordinaire, une surface minima est une surface telle qu'un point quelconque de la surface est situé à mi-chemin entre les deux centres de courbure, et qu'une telle surface est le lieu des points à mi-chemin entre les deux points situés respectivement sur deux courbes de longueur nulle quelconques.

Or on peut rattacher la notion d'une courbe de longueur nulle à celle d'une courbe de poursuite. Pour expliquer cela, j'observe que, dans le plan, en supposant comme à l'ordinaire que le lièvre coure selon une droite et que le chien et le lièvre courent avec des vitesses uniformes, la courbe de poursuite est une courbe déterminée; mais si les vitesses varient arbitrairement, alors la définition exprime seulement que la courbe est une courbe plane. Mais, dans l'espace, si au lieu d'une droite on considère une courbe plane ou à double courbure (disons une directrice) quelconque, alors, quelles que soient les vitesses, la définition précise toujours la courbe, savoir: on a toujours pour courbe de poursuite une courbe dont chaque tangente rencontre la courbe directrice. Au lieu d'une courbe, on peut avoir une surface directrice; dans ce cas, le nom est moins applicable, néanmoins je le retiens, et je dis que la courbe de poursuite est une courbe dont chaque tangente touche la surface directrice. Nous avons, de cette manière, la définition d'une courbe de poursuite par rapport à une courbe ou surface directrice quelconque.

À présent, au lieu du cercle imaginaire à l'infini, considérons une conique quelconque, l'absolue: on établit, comme on sait, par rapport à cette conique, la notion de la perpendicularité, et ainsi les notions d'une normale et des centres de courbure

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ne cessent pas de subsister. On peut donc considérer une surface telle que chaque point de la surface soit l'harmonicale par rapport aux deux centres de courbure du point de rencontre de la normale avec le plan de l'absolue: on a ainsi ce que je nomme une surface quasi-minima. Il va sans dire qu'il faut modifier convenablement la notion métrique d'une aire minima pour qu'elle soit applicable à cette nouvelle surface.

Pour construire la surface, on prend par rapport à l'absolue deux courbes de poursuite quelconques, et puis sur la droite, menée par deux points quelconques de ces courbes respectivement, l'harmonicale par rapport à ces deux points du point de rencontre de la droite avec le plan de l'absolue: le lieu de ce point harmonical sera la surface quasi-minima.

Il paraît permis de substituer pour la conique absolue une surface quadrique quelconque, que je nomme aussi l'absolue: on a, comme on sait, les notions de la normale et des centres de courbure. Pour la surface quasi-minima, le point sur la surface sera l'un des points doubles (foyers) de l'involution formée par les deux centres de courbure et les deux points de rencontre de la normale avec l'absolue; et de même pour la construction de la surface, il faut prendre sur la droite menée par deux points quelconques des deux courbes de poursuite respectivement les points doubles (foyers) de l'involution formée par ces deux points et les deux points de rencontre de la droite avec l'absolue.

900. James Joseph Sylvester.

[(Scientific Worthies in Nature, xxv.); Nature, vol. xxxix. (1889), pp. 217–219.]

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James Joseph Sylvester, born in London on September 3, 1814, is the sixth and youngest son of the late Abraham Joseph Sylvester, formerly of Liverpool*. He was educated at two private schools in London, and at the Royal Institution, Liverpool, whence he proceeded in due course of time to St John's College, Cambridge.

In these early days he manifested considerable aptitude for mathematics, and so it was not matter for surprise that he came out in the Tripos Examination of 1837 as Second Wrangler; being incapacitated, by the fact of his Jewish origin, from taking his degree, he was not able to compete for either of the Smith's Prizes. In more enlightened times (1872), he had the degrees of B.A. and M.A., by accumulation, conferred upon him, and received therewith the honour of a Latin speech from the Public Orator. He himself says: "I am perhaps the only man in England who am a full (voting) Master of Arts for the three Universities of Dublin, Cambridge, and Oxford, having received that degree from these Universities in the order above given: from Dublin, by *ad eundem*; from Cambridge, *ob merita*; from Oxford, by decree." He is now D.C.L. of Oxford, LL.D. of Dublin and Edinburgh, and Hon. Fellow of St John's College, Cambridge. It is still open for him to receive yet higher recognition from his own alma mater**.

Prof. Sylvester became a student of the Inner Temple, July 29, 1846, and was called to the Bar on November 22, 1850†. He has been Professor of Natural Philosophy at University College, London; of Mathematics at the University of Virginia, U.S.A.‡; then ten years later Professor at the Royal Military Academy, Woolwich; and again, after a five years' interval, Professor of Mathematics at the Johns Hopkins University, Baltimore, U.S.A., from its foundation in 1877. Finally, in December 1883, he was elected Savilian Professor of Geometry at Oxford, in succession to Prof. Henry Smith*. His first printed paper was on Fresnel's optical theory (in the Phil. Mag., 1837).

* Foster's *Hand-book of Men at the Bar.*

**[The honorary degree Sc.D. was conferred upon him by Cambridge in 1890.]

†Foster, l.c.

‡The late Prof. Key, of University College and School, was the first occupant of the Chair, founded by Mr Jefferson, once President of the United States, in 1824.

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We can here only briefly allude to a communication which was accompanied by many important results: we refer to the Friday evening address (January 23, 1874) to the Royal Institution, "On Recent Discoveries in Mechanical Conversion of Motion." He says:—"It would be difficult to quote any other discovery which opens out such vast and varied horizons as this of Peaucellier's,—in one direction, descending to the wants of the workshop, the simplification of the steam-engine, the revolutionising of the mill-wright's trade, the amelioration of garden-pumps, and other domestic conveniences (the sun of science glorifies all it shines upon); and in the other, soaring to the sublimest heights of the most advanced doctrines of modern analysis, lending aid to, and throwing light from a totally unexpected quarter on the researches of such men as Abel, Riemann, Clebsch, Grassmann, and Cayley. Its head towers above the clouds, while its feet plunge into the bowels of the earth."

The only works that Prof. Sylvester has published, we believe, are: (1) "A Probationary Lecture on Geometry, delivered before the Gresham Committee and the Members of the Common Council of the City of London, December 4, 1854," a slight thing which had to be written and delivered at a few hours' notice; (2) "Laws of Verse," 1870; (3) several short poems, sonnets, and translations, which have appeared in our columns and elsewhere.

Our notice would be incomplete without some record of the honours that have been conferred upon Dr Sylvester. He was elected a Fellow of the Royal Society on April 25, 1839; has received a Royal Medal (1860) and the Copley Medal (1880), this latter rarely awarded, we believe, to a pure mathematician. On this last occasion, Mr Spottiswoode accompanied the presentation with the words, "His extensive and profound researches in pure mathematics, especially his contributions to the theory of invariants and covariants, to

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which he is the chief contributor, and his discoveries in the higher theory of partitions, have given him a high and permanent position among the great mathematicians of the age."

He is Foreign Associate of the Institute of France and of the

Royal Society of Gottingen; Corresponding Member of the Institute of Bologna; &c., &c. He is one of the original members of the Order of Merit; and was the first recipient of the Gold Medal of the Royal Society of Sciences in Christiania. In connexion with the Royal Society, his work as chairman of the Council (to which he was elected in 1880 in succession to Mr Spottiswoode) may be referred to; he was also Chairman of the University Commission, and has been from the first one or another of its titles), and was the first editor of, and is a considerable contributor to, the American Journal of Mathematics; and he was at one time Examiner in Mathematics and Natural Philosophy in the University of London. He was not an original member of the London Mathematical Society (founded January 16, 1865), but was elected a member on June 19, 1865, Vice-President on January 15, 1866, and succeeded Prof. De Morgan as the second President on November 8, 1866. The Society showed its recognition of his great services to them and to mathematical science generally by awarding him its De Morgan Gold Medal in November 1887.

Wherever Dr Sylvester goes, there is sure to be mathematical activity; and the latest proof of this is the formation, during the last term at Oxford, of a Mathematical Society, which promises, we hear without surprise, to do much for the advancement of mathematical science there.

The writings of Sylvester date from the year 1837; the number of them in the Royal Society Index up to the year 1863 is 112, in the next ten years 38, and in the volume for the next ten years 81, making 231 for the years 1837 to 1883: the number of more recent papers is also considerable. They relate chiefly to finite analysis, and cover by their subjects a large part of it: algebra, determinants, elimination, the theory of equations, partitions, tactic, the theory of forms, matrices, reciprocants, the Hamiltonian numbers, &c.; analytical and pure geometry occupy a less prominent position; and mechanics, optics, and astronomy are not absent. A leading feature is the power which is shown of originating a theory or of developing it from a small beginning; there is a breadth of treatment and determination to make the work

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accessible, and an absence of that pride which too often refuses to expound the meaning and scope of a great discovery, and would as soon bequeath to posterity a hieroglyph as an algebraical formula.

There is, in 1844, in the Philosophical Magazine, a valuable paper, "Elementary Researches in the Analysis of Combinatorial Aggregation," and the titles of two other papers, 1865 and 1866, may be mentioned: "Astronomical Prolusions; commencing with the instantaneous proof of Lambert's and Euler's theorems, and modulating through the construction of the orbit of a heavenly body from two heliocentric distances, the subtended chord, and the periodic time, and the focal theory of Cartesian ovals, into a discussion of motion in a circle and its relation to planetary motion"; and the sequel thereto, "Note on the periodic changes of orbit under certain circumstances of a particle acted upon by a central force, and on vectorial coordinates, &c., together with a new theory of the analogues of the Cartesian ovals in space."

Many of the later papers are published in the American Mathematical Journal, founded, in 1878, under the auspices of the Johns Hopkins University, and for the first six volumes of which Sylvester was editor-in-chief. We have, in vol. I., a somewhat speculative paper entitled "An application of the new atomic theory to the graphical representation of the invariants and covariants of binary quantics," followed by appendices and notes relating to various special points of the theory; and in the same and subsequent volumes various memoirs on binary and ternary quantics, including papers (by himself, with the aid of Franklin) containing tables of the numerical generating functions for binary quantics of the first ten orders, and for simultaneous binary quantics of the first four orders, &c. The memoir (vols. II. and III.) on "Ternary cubic-form equations" is connected with some early papers relating to the theory of numbers. We have in it the theory of residuation on a cubic curve, and the beautiful chain-rule of rational derivation; viz. from an arbitrary point 1 on the curve it is possible to derive the singly infinite series of points $(1, 2, 4, 5, \dots, 3\mu \pm 1)$ such that the chord through any two points, m, n, again meets the curve in a point $m+n$, $m-n$ (whichever number is not divisible by 3) of the series; moreover, the coordinates of any point m are rational and integral functions of the degree m^2 of those of the point 1.

There is in vol. v. the memoir, "A Constructive Theory of Partitions arranged in three acts, an Interact in two parts, and an Exodion," and in vol. vi. we have "Lectures on the Principles of Universal Algebra," (referring to a course of lectures on multinomial quantity, in the year 1881). The memoir is incomplete, but the general theories of nullity and vacuity, and of the corpus formed by two independent matrices of the same order, are sketched out; and there are, in the Comptes rendus of the French Academy, later papers containing developments of various points of the theory,—the conception of "nivellators" may be referred to.

The last-mentioned paper in the American Mathematical Journal was published subsequently to Sylvester's return to England on his appointment as Savilian Professor of Mathematics at Oxford. In December 1886, he gave there a public lecture containing an outline of his new theory of reciprocants (reported in Nature, January 7, 1887), and the lectures since delivered are published under the title, "Lectures on the Theory of Reciprocants" (reported by J. Hammond), same Journal, vols. VIII. to X.; thirty-three lectures actually delivered, entire or in abstract, in the course of

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three terms, to a class in the University, with a concluding so-called lecture 34, which is due to Hammond. The subject, as is well known, is that of the functions of a dependent variable, y, and its differential coefficients, y', y'', $\dots$, in regard to x (or, rather, the functions of y', y'', $\dots$), which remain unaltered by the interchange of the variables x and y: this is a less stringent condition than that imposed by Halphen ("Thèse," 1878) on his differential invariants, and the theory is accordingly a more extensive one. A passage may be quoted:—"One is surprised to reflect on the change which is come over Algebra in the last quarter of a century. It is now possible to enlarge to an almost unlimited extent on any branch of it. These thirty lectures, embracing only a fragment of the theory of reciprocants, might be compared to an unfinished epic in thirty cantos. Does it not seem as if Algebra had attained to the dignity of a fine art, in which the workman has a free hand to develop his conceptions, as in a musical theme or a subject for painting? Formerly, it consisted in detached theorems, but nowadays it has reached a point in which every properly-developed algebraical composition, like a skilful landscape, is expected to suggest the notion of an infinite distance lying beyond the limits of the canvas." And, indeed, the theory has already spread itself out far and wide, not only in these lectures by its founder, but in various papers by auditors of them, and others,—Elliott, Hammond, Leudesdorf, Rogers, Macmahon, Berry, Forsyth.

Sylvester's latest important investigations relate to the Hamiltonian numbers: there is a memoir, Crelle, t. c. (1887), and, by Sylvester and Hammond jointly, two memoirs in the Philosophical Transactions. The subject is that of the series of numbers $2, 3, 5, 11, 47, 923$, calculated thus far by Sir W. R. Hamilton in his well-known Report to the British Association, on Jerrard's method. A formula for the independent calculation of any term of the series was obtained by Sylvester, but the remarkable law by means of a generating function was discovered by Hammond, viz. E_0, E_1, E_2, $\ldots$, being the series $3, 4, 6, \ldots$ of the foregoing numbers, each increased by unity; then these are calculated by the formula

$$\frac{1+t}{1-t} = \prod_{n=0}^{\infty}(1 - t^{E_n}),$$

equating the powers of t on the two sides respectively: observe the paradox, $t = \frac{1}{2}$, then the formula gives $0 =$ sum of a series of positive powers of $\frac{1}{2}$.

Enough has been said to call to mind some of Sylvester's achievements in mathematical science. Nothing further has been attempted in the foregoing very imperfect sketch.

901. Note on the Sums of Two Series.

[From the Messenger of Mathematics, vol. xix. (1890), pp. 29–31.]

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I consider the two series

$$S_1 = \frac{1}{1+e^a} + \frac{1}{3(1+e^{3a})} + \frac{1}{5(1+e^{5a})} + \cdots$$

and

$$S_2 = \frac{1}{1-e^a} + \frac{1}{3(2+3e^a)} + \frac{1}{5(2+5e^a)} + \cdots,$$

where a is real, positive, and indefinitely small; these would at first sight appear to be equal to each other, but this is not in fact the case.

Taking first the series S_1; putting therein $\pi a = 2x$, this is

$$S_1 = \frac{1}{1+x} + \frac{1}{3(1+3x)} + \frac{1}{5(1+5x)} + \cdots$$

Now we have (Legendre, *Théorie des Fonctions Elliptiques*, t. II. p. 438),

$$\psi(y) = -C + \log y + \frac{1}{2y} - \frac{B_1}{2y^2} + \frac{B_2}{4y^4} - \cdots,$$

where C is Euler's constant, $= \cdot 577\ldots$; and if y be real, positive, and very large, then

$$\psi(y) = -C + \log y + \frac{1}{2y},$$

whence, differentiating the logarithm and neglecting the terms which contain negative powers of y, then the value is $= C + \log y$; hence, writing $y = \frac{1}{x}$, we obtain

$$\frac{1}{1+x} + \frac{1}{2(1+2x)} + \frac{1}{3(1+3x)} + \cdots = C + \log\frac{1}{x}.$$

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Writing herein $2x$ for x, and dividing by 2, we have

$$\frac{1}{2(1+2x)} + \frac{1}{4(1+4x)} + \frac{1}{6(1+6x)} + \cdots = \frac{1}{2}\left(C + \log\frac{1}{2x}\right).$$

Hence, subtracting,

$$S_1 = \frac{1}{2}\left(C + \log\frac{1}{x}\right) - \frac{1}{2}\left(C + \log\frac{1}{2x}\right) = \frac{1}{2}\log 2.$$

Hence, writing for x its value, $= \frac{1}{2}\pi a$, we have

$$S_1 = \frac{1}{2}\log 2.$$

For the series S_2, we have (*Fundamenta Nova*, p. 103*) the formula

$$\frac{2K}{\pi} = \frac{1}{1+q^{-1}} + \frac{1}{3(1+q^{-3})} + \frac{1}{5(1+q^{-5})} + \cdots,$$

where

$$q = e^{-\pi\frac{K'}{K}}.$$

Or, putting herein $a = \frac{K'}{K}$, then $q = e^{-\pi a} = e^{-\pi a}$, and thence

$$S_2 = \frac{1}{1+e^{-\pi a}} + \frac{1}{2(1+e^{-2\pi a})} + \frac{1}{3(1+e^{-3\pi a})} + \cdots = \frac{2K}{\pi}.$$

We have $q = e^{-\pi a}$, which is real, positive, and less than but indefinitely near to 1; hence also k is real, positive, and less than but indefinitely near to 1, say the value is $= 1-\beta$; thence $k' = \sqrt{2\beta}$, and $K = \log\frac{4}{\sqrt{\beta}}$, $= \log\frac{4}{\sqrt{k'}}$; also $K' = \frac{1}{2}\pi$, and therefore

$$a = \frac{K'}{K} = \frac{\frac{1}{2}\pi}{\log\frac{4}{\sqrt{k'}}}, \quad \text{whence } \log\frac{4}{\sqrt{k'}} = \frac{\pi}{2a},$$

902. On the Focals of a Quadric Surface.

[From the Messenger of Mathematics, vol. xix. (1890), pp. 113–117.]

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In plane geometry, the focus of a curve is the node of the circumscribed line-system of the curve and the circular points at infinity; and so, in solid geometry, the focal of a surface or curve is the nodal line of the circumscribed developable of the surface or curve and the circle at infinity. And as in plane geometry the circular points at infinity may be regarded as an indefinitely thin conic, so in solid geometry the circle at infinity may be regarded as an indefinitely thin quadric surface.

In plane geometry, let it be proposed to find the circumscribed line-system (common tangents) of the two quadrics

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 0, \quad \frac{x^2}{a'} + \frac{y^2}{b'} + \frac{z^2}{c'} = 0;$$

if a common tangent be

$$\xi x + \eta y + \zeta z = 0,$$

then we have

$$a\xi^2 + b\eta^2 + c\zeta^2 = 0, \quad a'\xi^2 + b'\eta^2 + c'\zeta^2 = 0;$$

here writing for shortness

$$f, g, h = bc' - b'c, \quad ca' - c'a, \quad ab' - a'b,$$

we have

$$\xi^2 : \eta^2 : \zeta^2 = f : g : h;$$

and thence the tangent is

$$\sqrt{fx} + \sqrt{gy} + \sqrt{hz} = 0,$$

viz. we have thus four tangents, and the rationalised form is of course

$$f^2x^4 + g^2y^4 + h^2z^4 - 2ghy^2z^2 - 2hfz^2x^2 - 2fgx^2y^2 = 0.$$

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In connexion with the corresponding question in solid geometry, I obtain this equation in a different manner. We investigate the envelope of the line $\xi x+\eta y+\zeta z=0$, considering ξ, η, ζ as parameters connected by the foregoing two equations; by the ordinary process of indeterminate multipliers, we have

$$\frac{x^2}{\lambda a+\mu a'}+\frac{y^2}{\lambda b+\mu b'}+\frac{z^2}{\lambda c+\mu c'}=0,$$

and thence, eliminating ξ, η, ζ, we obtain

$$\frac{x^2}{\lambda a+\mu a'}+\frac{y^2}{\lambda b+\mu b'}+\frac{z^2}{\lambda c+\mu c'}=0,$$

equations equivalent to two equations, from which λ and μ are to be eliminated. The second and third equations are the derived functions of the first equation in regard to λ, μ respectively; and hence, expressing the first equation in an integral form, the result is

$$\text{Disc}\,[x^2(\lambda b+\mu b')(\lambda c+\mu c')+\text{\&c.}]=0;$$

viz. this is

$$\{(bc'+b'c)x^2+(ca'+c'a)y^2+(ab'+a'b)z^2\}^2-4(bcx^2+cay^2+abz^2)(b'c'x^2+c'a'y^2+a'b'z^2)$$

or developing and reducing, we have the foregoing result, the coefficients entering through the combinations

$$f,g,h=bc'-b'c,\quad ca'-c'a,\quad ab'-a'b.$$

Writing $c=-1$, also $a=b=1$, $c'=0$: the equation of the first conic is

$$\frac{x^2}{a}+\frac{y^2}{b}-z^2=0,$$

and that of the second may be replaced by the two equations $x^2+y^2=0$, $z=0$, viz. these give the circular points at infinity: we have $f,g,h=1,-1,a-b$, and the equation of the line-system is

$$x^4+2x^2y^2+y^4-2h(x^2-y^2)z^2+h^2z^4=0.$$

If finally, $z=1$, then for the tangents from the circular points at infinity to the quadric

$$\frac{x^2}{a}+\frac{y^2}{b}-1=0,$$

the equation is

$$x^4 + 2x^2y^2 + y^4 - 2h(x^2 - y^2) + h^2 = 0,$$

where, as before, $h = a - b$; the four tangents intersect in pairs in the two circular points at infinity, and in four other points which are the foci of the quadric.

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Passing now to the problem in solid geometry, we consider the two quadric surfaces

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - \frac{w^2}{d} = 0, \quad \frac{x^2}{a'} + \frac{y^2}{b'} + \frac{z^2}{c'} - \frac{w^2}{d'} = 0;$$

if a common tangent plane hereof be

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

then we have

$$a\xi^2 + b\eta^2 + c\zeta^2 + d\omega^2 = 0, \quad a'\xi^2 + b'\eta^2 + c'\zeta^2 + d'\omega^2 = 0;$$

and the circumscribed developable is the envelope of the plane, considering in the equation thereof ξ, η, ζ, ω as variable parameters connected by the last two equations.

By what precedes, it is at once seen that the resulting equation is

$$\text{Disc}\,[x^2(b\lambda + b'\mu)(c\lambda + c'\mu)(d\lambda + d'\mu) + \&\text{c.}] = 0,$$

viz. this is a quadric equation in (x^2, y^2, z^2, w^2), the coefficients $a, b, c, d, a', b', c', d'$, entering therein through the combinations

$$\alpha, \beta, \gamma = bc' - b'c, \quad ca' - c'a, \quad ab' - a'b;$$

$$f, g, h = ad' - a'd, \quad bd' - b'd, \quad cd' - c'd.$$

The developed result is given in my paper "On the developable surfaces which arise from two surfaces of the second order," Camb. and Dub. Math. Jour., t. V. (1850), pp. 45–53, [84]. We require here, for the particular case, $d = -1$, $a' = b' = c' = 1$, $d' = 0$, viz. one of the surfaces is taken to be

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1,$$

and the other is the circle at infinity $x^2+y^2+z^2=0$, $w=0$; we thus have $\alpha,\beta,\gamma=1,1,1$; $f,g,h=b-c,c-a,a-b$, or now using the italic letters f,g,h to signify these values, we have $f+g+h=0$. The equation is

$$\begin{aligned} &f^2g^2h^2w^8 \\ &\qquad + 2g^2(g-h)y^2z^6 + 2h^2(h-f)z^2x^6 + 2f^2(f-g)x^2y^6 \\ &\qquad + 2g^2(h-f)y^6w^2 + 2h^2(f-g)z^6w^2 + 2f^2(g-h)x^6w^2 \\ &- 2fgh^2(h-f)y^2z^4w^2 - 2f^2gh(g-h)z^2x^4w^2 - 2fgh^2(f-g)x^2y^4w^2 \\ &\qquad\qquad + \cdots = 0. \end{aligned}$$

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The equation may be written in the following four forms:

$$\begin{aligned} x^2\Theta_1 + (gy^2 - hz^2 + ghw^2)^2\{y^4 + 2y^2z^2 + z^4 - 2f(y^2-z^2)w^2 + f^2w^4\} &= 0, \\ y^2\Theta_2 + (hz^2 - fx^2 + hfw^2)^2\{z^4 + 2z^2x^2 + x^4 - 2g(z^2-x^2)w^2 + g^2w^4\} &= 0, \\ z^2\Theta_3 + (fx^2 - gy^2 + fgw^2)^2\{x^4 + 2x^2y^2 + y^4 - 2h(x^2-y^2)w^2 + h^2w^4\} &= 0, \\ w^2\Theta_4 + (fx^2+gy^2+hz^2)^2(x^2+y^2+z^2)^2 &= 0; \end{aligned}$$

the last of these shows that the circle at infinity $x^2+y^2+z^2=0$, $w=0$, is a nodal line on the surface; this, however, is not regarded as a focal. The other three show that we have also, as nodal lines, three conics, which are the focals of the given surface; viz. now writing $w=1$, we have, for the quadric surface

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 1,$$

the three focal conics

$$\begin{aligned} x=0, &\quad \frac{y^2}{a-b} + \frac{z^2}{a-c} - 1 = 0, \\ y=0, &\quad \frac{x^2}{a-b} + \frac{z^2}{b-c} - 1 = 0, \\ z=0, &\quad \frac{x^2}{a-c} + \frac{y^2}{b-c} - 1 = 0; \end{aligned}$$

viz. these are an imaginary conic, the focal hyperbola, and the focal ellipse, in the coordinate planes $x=0$, $y=0$, $z=0$ respectively.

903. On Latin Squares.

[From the Messenger of Mathematics, vol. xix. (1890), pp. 135–137.]

[p. 55]

If in each line of a square of n^2 compartments the same n letters $a, b, c, \dots$ are arranged so that no letter occurs twice in the same column, we have what was termed by Euler "a Latin square." Supposing that in one of the lines the letters are arranged in the natural order $abcde\dots$, then in the remaining lines there must be arrangements beginning with $b, c, d, e, \&c.$, respectively, and we may consider the case in which the bottom line has the arrangement $abcde\dots$, and in the other lines, reckoning from the bottom one in order, the arrangements begin with $b, c, d, e, \&c.$, respectively: if the number of such squares be $= N$, then, obviously, the whole number of squares which can be formed with the same n arrangements is $= N[n]^n$.

Starting with the bottom line as above, then it is a well-known problem to determine the number of arrangements for the second line; viz. this number is

$$[n]^n - \frac{n}{1}[n-1]^{n-1} + \frac{n(n-1)}{1\cdot 2}[n-2]^{n-2} - \cdots \pm 1,$$

and if we assume, as above, that the second line begins with b, then the whole number of arrangements is this number divided by $(n-1)$, the quotient being of course integral. For instance, when $n = 5$, the number is $= 120 - 120 + 60 - 20 + 5 - 1 = 44$, which is divisible by 4, and the number of arrangements for the second line is thus $= 11$.

But the number of arrangements for the third line will be different according to the arrangement selected for the second line, and it is not easy to see how in general the whole number of arrangements for the third line is to be calculated, and the difficulty of course increases for the next following lines; it may be remarked that, when all the lines are filled up except the top-line, the top-line is completely determined.

[p. 56]

Imagine the square completed: we may write down the substitutions by which we pass from the bottom line to itself (this is of course the substitution 1) and to each of the other lines respectively; we have thus a set of n substitutions, which may form a group; and when this is so, we may conversely from the group construct the Latin square.

But it is not every Latin square which is thus connected with a group of n substitutions.

In the cases $n = 2, 3, 4$ there is no difficulty; the squares are

$$\begin{matrix} b & a \\ a & b \end{matrix}\,, \qquad \begin{matrix} c & a & b \\ b & c & a \\ a & b & c \end{matrix}\,, \qquad \begin{matrix} d & c & b & a \\ c & d & a & b \\ b & a & d & c \\ a & b & c & d \end{matrix}\,, \qquad \begin{matrix} d & c & b & a \\ c & d & a & b \\ b & d & a & c \\ a & b & c & d \end{matrix}\,, \qquad \begin{matrix} d & a & b & c \\ c & a & d & b \\ b & c & d & a \\ a & b & c & d \end{matrix}\,, \qquad \begin{matrix} d & a & b \\ c & d & a \\ b & c & d \\ a & b & c \end{matrix}$$

viz. when $n = 2$ the number is 1, when $n = 3$ it is 1, when $n = 4$ it is 4; in this last case, the arrangement *bcda* for the second line gives two squares, but each of the other arrangements only one square.

In each of the squares of 4, we have a group, viz. for the four squares respectively, these are

$$\begin{array}{lllll} 1^\circ & 1, & (ab)(cd), & (acbd), & (adbc), \\ 2^\circ & 1, & (ab)(cd), & (ac)(bd), & (ad)(bc), \\ 3^\circ & 1, & (abdc), & (acdb), & (ad)(bc), \\ 4^\circ & 1, & (abcd), & (ad)(bc), & (ac)(bd). \end{array}$$

[p. 57]

The case $n = 5$ is much more difficult; I have not as yet completely solved it, but I have obtained the following results. Denoting the arrangements for the second line by $2, 3, 4, 5$ (each of these denotes an arrangement beginning with the corresponding letter), I find that 2 gives 11 squares, 3 gives 11, 4 gives 11, and 5 gives 2; the whole number of squares is thus $= 35$, agreeing with Prof. Cayley's former determination. The groups of 5 are very remarkable; I have not investigated them completely, but there are at least 5 different types of group which occur; viz. one group of the cyclical type, and four or more groups of other types.

The cyclical group belongs to the squares corresponding to the arrangement 5 of the second line; it is generated by the substitution $(abcde)$, and the corresponding squares are

$$\begin{matrix} a & b & c & d & e \\ b & c & d & e & a \\ c & d & e & a & b \\ d & e & a & b & c \\ e & a & b & c & d \end{matrix} \quad \text{and} \quad \begin{matrix} d & e & a & b & c \\ c & d & e & a & b \\ b & c & d & e & a \\ a & b & c & d & e \end{matrix}\,.$$

These are the only squares belonging to the cyclical group; but each of the same arrangement 5 gives other squares belonging to non-cyclical groups.

904. Note on Reciprocal Lines.

[From the Messenger of Mathematics, vol. xix. (1890), pp. 174, 175.]

[p. 58]

If two lines are reciprocal in regard to a quadric surface, then any point on the one line and any point on the other line are harmonics in regard to the surface, viz. the two points and the intersections of their line of junction with the surface form a harmonic range. This is obvious: the polar plane of the first point passes through the second line, and thus the second point is a point in the polar plane of the first point, that is, the two points are harmonics.

But it is worth while to look at the theorem in a different point of view. If the first line meets the surface in the points A, C and the second line meets the surface in the points B, D (in order to the four points being real, the surface must of course be a skew hyperboloid), then AB, BC, CD, DA are lines on the surface, say AB, CD are directrices, and BC, AD are generatrices, the two reciprocal lines being the diagonals AC and BD of the skew quadrilateral. Taking on AC a point P and on BD a point Q, the theorem is that, if PQ meets the surface in the points X, Y, then the four points P, Q, X, Y are harmonics. Let the generatrix through X meet AB, CD in the points S, S' respectively, and the generatrix through Y meet the same lines in the points T, T' respectively.

[p. 59]

Then the four lines CB, DA, $T'T$, $S'S$, are all met by an infinite series of lines, and in particular they are met by the lines CD, BA in the points C, D, T', S' and B, A, T, S respectively. Consequently, writing AH for anharmonic ratio, we have

$$\mathrm{AH}(C, D, T', S') = \mathrm{AH}(B, A, T, S).$$

Consider now the four lines CA, DB, $T'T$, $S'S$: these are each of them met by the lines CD and BA, and also by the line PQ: therefore, by an infinity of lines (that is, they are directrices of a hyperboloid). They are met by the last-mentioned three lines in the points C, D, T', S'; A, B, T, S; and P, Q, Y, X respectively; hence

$$\mathrm{AH}(C, D, T', S') = \mathrm{AH}(A, B, T, S) = \mathrm{AH}(P, Q, Y, X).$$

Comparing with the foregoing equation, it appears that

$$\mathrm{AH}(B, A, T, S) = \mathrm{AH}(A, B, T, S),$$

viz. the anharmonic ratio is not altered by the interchange of the two points A, B: this implies that the points A, B, T, S are harmonics, viz. the pairs A, B and S, T are harmonics; and, this being so, the equation

$$\mathrm{AH}(A, B, T, S) = \mathrm{AH}(P, Q, Y, X)$$

shows that P, Q, Y, X are harmonics, viz. that the pairs P, Q and X, Y are harmonics; or say P, Q are harmonics in regard to the quadric surface, which is the theorem in question.

905. On the Equation $x^{17} - 1 = 0$.

[From the Messenger of Mathematics, vol. xix. (1890), pp. 184–188.]

[p. 60]

Writing $\rho = \cos\frac{2\pi}{17} + i\sin\frac{2\pi}{17}$, I carry the solution up to the determination of the periods each of two roots, $\rho + \rho^{16}$, $= 2\cos\frac{2\pi}{17}$, &c. The expressions contain the radicals

$$a = \sqrt{17}, \quad b = \sqrt{\{2(17-a)\}}, \quad c = \sqrt{\{4(17+3a) - 2(3+a)b\}},$$

where a, b, c are taken to be positive ($a = 4\cdot 12$, $b = 5\cdot 07$, $c = 6\cdot 72$). Taking for a moment r to be any imaginary seventeenth root, $r = \rho^\theta$, then the algebraical expression for the period P_1 of eight roots is $P_1 = \frac{1}{2}(-1+a)$, but I assume the value to be $-P_1 = \frac{1}{2}(-1+a)$, and thus determine θ to denote some one of the values $1, 2, 4, 8, 9, 13, 15, 16$; similarly, I assume the value of Q_1 to be $= \frac{1}{2}(-1+a+b)$; and thus further determine θ to denote some one of the values $1, 4, 13, 16$: and, again, I assume the value of R_1 to be $= \frac{1}{2}(-1+a+b+c)$, and thus further determine θ to denote one of the values 1 and 16. As regards the values of the periods R, it is obviously indifferent which value is taken, and I assume therefore $\theta = 1$. This comes to saying that the signs of the radicals are determined in suchwise that r shall denote the root $\cos\frac{2\pi}{17} + i\sin\frac{2\pi}{17}$; and it is to be understood that r has this value.

I write now

$$\begin{aligned}
P_1 &= r + r^2 + r^4 + r^8 + r^9 + r^{13} + r^{15} + r^{16},\\
P_2 &= r^3 + r^6 + r^5 + r^{11} + r^{14} + r^7 + r^{12} + r^6,\\
Q_1 &= r + r^4 + r^{13} + r^{16},\\
Q_2 &= r^2 + r^8 + r^9 + r^{15},\\
Q_3 &= r^3 + r^5 + r^{12} + r^{14},\\
Q_4 &= r^6 + r^7 + r^{10} + r^{11},
\end{aligned}$$

and moreover

$$R_1 = r + r^{16}, \quad R_2 = r^4 + r^{13}, \quad R_3 = r^2 + r^{15}, \quad R_4 = r^8 + r^9,$$
$$R_5 = r^3 + r^{14}, \quad R_6 = r^5 + r^{12}, \quad R_7 = r^{10} + r^7, \quad R_8 = r^{11} + r^6,$$
$$a = P_1 - P_2,$$
$$b^2 = 2(17 - a), \quad b_1^2 = 2(17 + a),$$
$$c^2 = 4(17 + 3a) - 2(3 + a)b.$$

[p. 61]

It will appear by what follows that a is determined by the quadric equation $a^2 = 17$, but I have assumed that a denotes the positive root $a = \sqrt{17}$; similarly, b is determined by the quadric equation $b^2 = 2(17 - a)$, but it is assumed that b denotes the positive root, $b = \sqrt{\{2(17 - a)\}}$; and c is determined by the quadric equation $c^2 = 4(17 + 3a) - 2(3 + a)b$, but it is assumed that c denotes the positive root.

If in the equations I had written b_1, c_1, c_2, c_3, instead of $+b_1, -c_1, +c_2, -c_3$, then b_1 comes out rationally in terms of a, b; and c_1, c_2, c_3 come out rationally in terms of a, b, c; the signs were attached to them a posteriori, in suchwise that the values of b_1, c_1, c_2, c_3 might be each of them positive; for their independent determination, we have, in fact, for b_1, an expression such as that for b_2; and for c_1^2, c_2^2, c_3^2 expressions such as that for c^2; and taking as above for each of them the positive value of the square root, we have

$$\begin{aligned}
b &= \sqrt{\{2(17 - a)\}} && (= 5 \cdot 07),\\
b_1 &= \sqrt{\{2(17 + a)\}} && (= 6 \cdot 49),\\
c &= \sqrt{\{4(17 + 3a) - 2(3 + a)b\}} && (= 6 \cdot 72),\\
c_1 &= \sqrt{\{4(17 + 3a) + 2(3 + a)b_1\}} && (= 13 \cdot 77),\\
c_2 &= \sqrt{\{4(17 - 3a) + 2(-3 + a)b_1\}} && (= 5 \cdot 75),\\
c_3 &= \sqrt{\{4(17 - 3a) - 2(-3 + a)b_1\}} && (= 2 \cdot 02).
\end{aligned}$$

[p. 62]

The relations between the periods P are

$$P_1 + P_2 = -1,$$

that is,

$$P_1^2 = -P_1 + 4, \quad \&c.; \quad a^2 = 17.$$

Here for the second square, we have

$$Q_1 + Q_2 = P_1, \quad Q_3 + Q_4 = P_2,$$

and similarly in the other cases which follow.

For the periods Q, we have

$$\begin{aligned} Q_1^2 &= -P_1 + 4, \quad Q_1Q_2 = P_1 + 4, \quad Q_2^2 = -P_1 + 4, \\ Q_3^2 &= -P_2 + 4, \quad Q_3Q_4 = P_2 + 4, \quad Q_4^2 = -P_2 + 4; \end{aligned}$$

$= -P_1 - P_2 + 16 = 17$; so that $bb_1 = 8a$, or b_1 is given rationally in terms of a, b.

And for the periods R, we have

$$-R_1 + R_2 = Q_1, \quad R_3 + R_4 = Q_2, \quad R_5 + R_6 = Q_3, \quad R_7 + R_8 = Q_4,$$
$$R_1^2 = -R_1 + \cdots, \quad R_2^2 = -R_2 + \cdots, \quad \&c.$$

[p. 63]

And from the foregoing results, we have

$$\begin{aligned}
P_1 &= \tfrac{1}{2}(-1 + a), \\
P_2 &= \tfrac{1}{2}(-1 - a), \\
Q_1 &= \tfrac{1}{2}(-1 + a + b), \quad = 1 \cdot 87, \\
Q_2 &= \tfrac{1}{2}(-1 + a - b), \quad = 0 \cdot 18, \\
Q_3 &= \tfrac{1}{2}(-1 - a + b_1), \quad = -1 \cdot 96, \\
Q_4 &= \tfrac{1}{2}(-1 - a - b_1), \quad = -2 \cdot 90, \\
R_1 &= \tfrac{1}{2}(-1 + a + b + c), \quad = 2 \cdot 05, \\
R_2 &= \tfrac{1}{2}(-1 + a + b - c), \quad = 0 \cdot 34, \\
R_3 &= \tfrac{1}{2}(-1 + a - b + c_1), \quad = 0 \cdot 89, \\
R_4 &= \tfrac{1}{2}(-1 + a - b - c_1), \quad = -0 \cdot 55, \\
R_5 &= \tfrac{1}{2}(-1 - a + b_1 + c_2), \quad = -1 \cdot 20, \\
R_6 &= \tfrac{1}{2}(-1 - a + b_1 - c_2), \quad = -2 \cdot 70, \\
R_7 &= \tfrac{1}{2}(-1 - a - b_1 + c_3), \quad = -0 \cdot 49, \\
R_8 &= \tfrac{1}{2}(-1 - a - b_1 - c_3), \quad = -3 \cdot 51.
\end{aligned}$$

The approximate numerical values have been given throughout only for the purpose of showing that the signs of the square roots have been rightly determined.

906. Note on Schlaefli's Modular Equation for the Cubic Transformation; with a Correction.

[From the Messenger of Mathematics, vol. xx. (1891), pp. 59, 60; 120.]

[p. 64]

The equation in question, Crelle, t. LXXII. (1870), p. 369, is

$$S^8 + T^8 - 8ST + 8S^3T^3 = 0,$$

where

$$S = \frac{\sqrt[4]{2}}{\sqrt[4]{(kk')}}, \quad T = \frac{\sqrt[4]{2}}{\sqrt[4]{(\lambda\lambda')}},$$

which must be of course equivalent to the ordinary modular equation

$$u^4(1-u^8)^2 + u^4(1-v^8)^2 - 2u^4v^4 = 0,$$

where

$$u = \sqrt[4]{k}, \quad v = \sqrt[4]{\lambda};$$

the resemblance in form between the two equations, with such different meanings of the S and T in the one case, and the u and v in the other, is very noticeable.

Schlaefli's equation is

$$S^8 + T^8 = 8ST - 8S^3T^3,$$

that is,

$$kk' + \lambda\lambda' = 2(kk')^{\frac{1}{4}} - 4(kk')^{\frac{3}{4}},$$

or say

$$x'^2 = 4(kk')^{\frac{1}{4}} - 18(kk')^{\frac{3}{4}} + 16(kk')^{\frac{5}{4}},$$

where $x' = \sqrt{kk'} + \sqrt{\lambda\lambda'}$.

To deduce this from the uv-modular equation, we have (Jacobi's *Fund. Nova*, p. 68, Ges. Werke, t. I., p. 124),

$$(1-u^8)(1-v^8) = (1-u^2v^2)^4,$$

[p. 65]

or, multiplying each side by u^4v^4, and extracting the fourth root, we have

$$\sqrt{(kk'\lambda\lambda')} = uv\sqrt{(1-u^2v^2)} = x - x^3,$$

if for shortness we write $x = uv$.

The equation to be proved thus is

$$u^8(1-u^8) + v^8(1-v^8) = 4(x-x^3) - 18(x-x^3)^2 + 16(x-x^3)^3.$$

But from the foregoing equation

$$(1-u^8)(1-v^8) = (1-u^2v^2)^4,$$

we have

$$u^8 + v^8 = 4u^2v^2 - 6u^4v^4,$$

and thence

$$u^{16} + v^{16} = (4u^2v^2 - 6u^4v^4)^2 - 2u^8v^8,$$

and the equation to be proved thus becomes

$$2x^4 = 4(x-x^3) - 18(x-x^3)^2 + 16(x-x^3)^3,$$

which is in fact an identity, each side being

$$52x^3 - 68x^5 + 16x^6.$$

Correction, p. 120.

I find that I misquoted Schlaefli's equation, viz. in effect, I took it to be

$$S_1^8+T_1^8-8S_1T_1+8S_1^3T_1^3 = 0, \quad \text{where } S_1 = \frac{2}{\sqrt[4]{(kk')}}, \quad T_1 = \frac{2}{\sqrt[4]{(\lambda\lambda')}},$$

whereas his equation really is

$$S^8 + T^8 - 8ST + 8S^3T^3 = 0, \quad \text{where } S = 2\sqrt[4]{(kk')}, \quad T = 2\sqrt[4]{(\lambda\lambda')}.$$

The change is only a change of form, for writing $S_1 = \frac{2}{S}$ and $T_1 = \frac{2}{T}$, the equation in (S_1, T_1) is converted into that in (S, T); but it was quite an unnecessary one, and I cannot account for having made it, as the paper in Crelle must have been before me.

907. Note on the Ninth Roots of Unity.

[From the Messenger of Mathematics, vol. xx. (1891), p. 63.]

[p. 66]

Let θ be a prime ninth root of unity, so that $\theta^6+\theta^3+1=0$; and write

$$a=\theta+\theta^8, \quad b=\theta^2+\theta^7, \quad c=\theta^4+\theta^5,$$

then

$$a+b+c=-1,$$

that is,

$$ab=c+2, \quad ca=c-1,$$

whence

$$ab+ac+bc=-3, \quad abc=-1;$$

and a,b,c are thus the roots of the equation $x^3-3x+1=0$. We have

$$a^2b+b^2c+c^2a=a^2+b^2+c^2=6,$$

$$ab^2+bc^2+ca^2=bc+ca+ab=-3,$$

and thence

$$(a^2b+b^2c+c^2a)-(ab^2+bc^2+ca^2)=(a-b)(b-c)(c-a)=6-(-3)=9.$$

The equation $x^3-3x+1=0$ is thus such that $a^2b+b^2c+c^2a$, and consequently any rational function whatever of a,b,c, invariable by the cyclical interchange (abc) of the roots, has a rational value.

908. On Two Invariants of a Quadriquadric Function.

[From the Messenger of Mathematics, vol. xx. (1891), pp. 68, 69.]

[p. 67]

The quadriquadric function

$$z^2(ax^2+2hxy+gy^2)+2zw(h'x^2+2bxy+fy^2)+w^2(g'x^2+2f'xy+cy^2),$$

considered successively as a function of (z, w) and of (x, y), has the discriminants U, V, equal to

$$(ax^2+2hxy+gy^2)(g'x^2+2f'xy+cy^2)-(h'x^2+2bxy+fy^2)^2,$$

$$(az^2+2h'zw+gw^2)(g'z^2+2f'zw+cw^2)-(hz^2+2bzw+f'w^2)^2,$$

respectively. As is well known, these quartic functions have each of them the same quadrinvariant and the same cubinvariant; these are the invariants in question of the quadriquadric function.

The quadrinvariant has been calculated in a different notation, but I am not aware that the cubinvariant has been before calculated; the two values are as follows:

[p. 68]

Quadrinvariant is **Cubinvariant is**

a^2c^2	$+1$	a^3c^3	-1
$ac(f^2+f'^2)$	$+33$	$acf^2f'gh$	-36
$acgg'$	$+33$	$af^3f'gh$	-36
$g^3g'^3$	-1	$af'^3g'h$	-36
ab^4	-24	ab^2cgg'	-120
a^2b^2c	$+12$	$a^2cf'^2g'$	-36
ab^2cgg'	-120	$acf'gg'h$	-60
$a^2cf'^2g'$	-36	$af'^2g'h'$	-36
$acf'gg'h$	-60	$f^2g^2g'^2$	$+6$
$af'^2g'h'$	-36	b^6	$+48$
$f^2g^2g'^2$	$+6$	ab^4c	-48
b^6	$+48$	a^2b^2cfh'	$+6$
ab^4c	-48	ac^2gh^2	-36
a^2b^2cfh'	$+6$	$acfgg'h'$	-60
ac^2gh^2	-36	$af'h^3$	-36
$acfgg'h'$	-60	$fg^2g'^2h$	$+6$
$af'h^3$	-36	$acgg'$	$+42$
$fg^2g'^2h$	$+6$	b^4	$+64$
$acgg'$	$+42$	$a^2c^2f'h$	$+6$
b^4	$+64$	$ac^2g'h'^2$	-36
$a^2c^2f'h$	$+6$	$acf^2h'^2$	$+24$
$ac^2g'h'^2$	-36	$af'^2gg'^2$	-36
$acf^2h'^2$	$+24$	$f^2g'^2h^2$	$+24$
$af'^2gg'^2$	-36	$achf$	-12
$f^2g'^2h^2$	$+24$	b^2c^2	$+76$
$achf$	-12	$\cdots$	
b^2c^2	$+76$		

[The complete tables of the quadrinvariant and cubinvariant are given in the original text.]

909. On a Particular Case of Kummer's Differential Equation of the Third Order.

[From the Messenger of Mathematics, vol. xx. (1891), pp. 75–79.]

[p. 69]

The general form of equation in question is

$$\frac{d^3x}{dx^3} = A\frac{x'}{x^3} + B\frac{x'x''}{x^2} + C\frac{x'^3}{x^3} + A'\frac{x'}{(x-1)^3} + B'\frac{x'x''}{(x-1)^2} + C'\frac{x'^3}{(x-1)^3};$$

here x is a function of t; and A, B, C, A', B', C' are numerical constants. For various given values of A, B, C, and values determined thereby of A', B', C', the equation admits of a solution in the form $x =$ rational function of t; the theory in reference to the cases considered by Schwarz is considered in my paper "On the Schwarzian Derivative and the Polyhedral Functions," Camb. Phil. Trans., t. XIII. (1883), pp. 5–68, [744]. But the theory is considered in a more general and exhaustive manner in Goursat's memoir, "Recherches sur l'équation de Kummer," *Acta Soc. Sci. Fennicæ*, t. XV. (1888), pp. 47–127. I consider here one of the solutions given by him, viz. writing

$$\begin{aligned} &P = 4t^2 - 5t + 1, \quad && X = t^2P^3, \\ &Q = 5t - 4, && Y = Q^3, \\ &R = 8t^2 - 11t + 8, && Z = -(t-1)^2R, \end{aligned}$$

so that, identically, $X + Y + Z = 0$; then the solution is expressed by either of the equivalent equations

$$x = \frac{X}{Z} = -\frac{t^2P^3}{(t-1)^2R}, \quad \text{or} \quad 1 - x = \frac{Y}{Z} = -\frac{Q^3}{(t-1)^2R}.$$

[p. 70]

The values of the constants to which this solution belongs are

$$A = -4, \quad B = 5, \quad C = -2, \quad A' = -4, \quad B' = 5, \quad C' = -2.$$

But instead of assuming these values in the first instance, I leave the values indeterminate; and starting from the foregoing expression for x, I substitute this in

$$\Omega = \frac{x'''}{x'} - 3\left(\frac{x''}{x'}\right)^2 - A\frac{x'}{x^2} - B\frac{x''}{x} - C\frac{x'^2}{x^2} - A'\frac{x'}{(x-1)^2} - B'\frac{x''}{x-1} - C'\frac{x'^2}{(x-1)^2},$$

thus obtaining Ω as a function of t which, as will appear, vanishes identically when A, B, C, A', B', C' have the foregoing values.

I remark that this is, in effect, doing in a somewhat different form for the particular case what Goursat does for the general case, viz. starting from

$$\Omega = \frac{x'''}{x'} - 3\left(\frac{x''}{x'}\right)^2 - A\frac{x'}{x^2} - \cdots,$$

with values of A, B, C which belong to the solution considered, he shows that this is a function of t having no infinities other than $(0, 1, \infty)$; that ∞ is not an infinity of the function or of the function multiplied into t, and that 0 and 1 are each of them a twofold infinity; that is, that the function is of the form

$$\frac{Lt^2 + Mt + N}{t^2(t-1)^2} - \frac{A'}{t^2} - \frac{B'}{t} - \frac{C'}{(t-1)^2}.$$

Proceeding to carry out the process, we have

$$x = -\frac{t^2P^3}{(t-1)^2R}, \quad x - 1 = \frac{Q^3}{(t-1)^2R},$$

$$\frac{x'}{x} = \frac{2}{t} + \frac{3P'}{P} - \frac{2}{t-1} - \frac{R'}{R},$$

and from either of these equations, collecting and reducing,

$$\frac{x'}{x} = \frac{2(t-1)P^2 \cdot tP}{t(t-1)R} + \cdots = \frac{2(t-1)P^2 \cdot Q^2}{t(t-1)R},$$

where observe that, from the values of x and $x-1$ respectively, it appears a priori that tP^2 and Q^2 must be factors in the numerator of x'. From this value of x', we have

$$\frac{x''}{x'} = \frac{2}{t} + \frac{2}{t-1} + \frac{3P'}{P} + \frac{2Q'}{Q} - \frac{R'}{R},$$

and hence, P' and Q' being mere constants,

$$\frac{x'''}{x'} - 3\left(\frac{x''}{x'}\right)^2 = \cdots.$$

[p. 71]

By decomposing $\alpha\beta$, αp, &c., into simple fractions, this becomes

$$A''\left(-4\alpha+3\beta+\frac{4p}{t}\right)+B''\left(4\alpha-4\beta+\frac{5q}{t-1}\right)+C''\left(-\beta+\frac{8r}{(t-1)^2}\right)$$
$$+F'(-4\alpha+4p)+G'(5\beta-5q)+H'(-8r),$$

where $\alpha=\frac{1}{t}$, $\beta=\frac{1}{t-1}$, $p=\frac{P'}{P}$, $q=\frac{Q'}{Q}$, $r=\frac{R'}{R}$.

This is

$$+p^2A''+q^2B''+r^2C''$$
$$+p(-4A''+4F')+q(5B''-5G')+r(-8C''+8H')$$
$$+\alpha(-4A''+4B'')+\beta(3A''-4B''-C'')+\cdots.$$

This should be identically $=0$; making $A''=0$, $B''=0$, $C''=0$, we find

$$F'=A'',\quad G'=B'',\quad H'=C''.$$

[p. 73]

and thence

$$A=-4,\quad B=5,\quad C=-2,$$
$$A'=-4,\quad B'=5,\quad C'=-2.$$

These values make the coefficients of p and q to be each $=0$; and they make the coefficient of R to be identically $=0$, viz. we have

$$0=\tfrac{1}{2}H-3H'-\tfrac{3}{4}F''-3\tfrac{1}{4}G'',$$

and

$$0=\tfrac{1}{2}II-3H'-\tfrac{3}{4}F''-3\tfrac{1}{4}G''.$$

We have, moreover,

$$L=\tfrac{1}{2}-C',\quad M=\tfrac{1}{2}-B',\quad N=1-A',$$

and the coefficients of α and β are $=-M+\frac{1}{2}$ and $M-\frac{1}{2}$ respectively; hence the coefficients of α^2, β^2, α and β will all vanish if only $L=0$, $M=\frac{1}{2}$, $N=0$, that is,

$$A'=\frac{3}{2},\quad B'=\frac{13}{12},\quad C'=\frac{5}{9},$$

and we have thus identically $\Omega=0$, if only A,B,C,A',B',C' have the above-mentioned values.

910. Note on the Involutant of Two Binary Matrices.

[From the Messenger of Mathematics, vol. xx. (1891), pp. 136, 137.]

[p. 74]

Consider the two matrices

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad M' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix},$$

and their product in one or the other order

$$MM' = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad M'M = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix}.$$

Then the Involutant is by definition = either of the determinants

$$I = \begin{vmatrix} 1 & a & a' & A \\ 0 & b & b' & B \\ 0 & c & c' & C \\ 1 & d & d' & D \end{vmatrix}, \quad I_1 = \begin{vmatrix} 1 & a' & a & A_1 \\ 0 & b' & b & B_1 \\ 0 & c' & c & C_1 \\ 1 & d' & d & D_1 \end{vmatrix};$$

viz. it is to be shown that these two values are in fact equal.

We have

$$MM' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix},$$

that is,

$$A = aa' + bc', \quad B = ab' + bd', \quad C = ca' + dc', \quad D = cb' + dd'.$$

[p. 75]

and similarly

$$M'M = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

that is,

$$A_1 = aa' + cb', \quad B_1 = ba' + db', \quad C_1 = ac' + cd', \quad D_1 = bc' + dd',$$

viz. A_1, B_1, C_1, D_1 are obtained from A, B, C, D by the interchange of the accented and unaccented letters.

We have then, from the first expression for the Involutant,

$$I = A\begin{vmatrix} 0 & b & b' \\ 0 & c & c' \\ 1 & d & d' \end{vmatrix} - B\begin{vmatrix} 0 & c & c' \\ 1 & d & d' \\ 1 & a & a' \end{vmatrix} + C\begin{vmatrix} 1 & d & d' \\ 1 & a & a' \\ 0 & b & b' \end{vmatrix} - D\begin{vmatrix} 1 & a & a' \\ 0 & b & b' \\ 0 & c & c' \end{vmatrix}$$
$$= A(bc' - b'c) - B(ac' - a'c + cd' - c'd) + C(ab' - a'b + bd' - b'd) - D(bc' - b'c),$$

or substituting for A, B, C and D their values, this is

$$(aa' - dd' + bc' - b'c)(bc' - b'c) - (ab' + bd')(ac' - a'c + cd' - c'd)$$
$$+ (ca' + dc')(ab' - a'b + bd' - b'd) - (cb' + dd')(bc' - b'c);$$

and multiplying out and grouping together the terms in bc, $b'c'$, bc' and $b'c$, this is found to be

$$= -(a' - d')^2 bc + (a - d)(a' - d')(bc' + b'c) - (a - d)^2 b'c' + (bc' - b'c)^2,$$

which is

$$= -\{(a - d)b' - (a' - d')b\}\{(a - d)c - (a' - d')c'\} + (bc' - b'c)^2.$$

Hence, writing

$$\begin{aligned} \mathfrak{a} &= bc' - b'c, & \mathfrak{f} &= ad' - a'd, \\ \mathfrak{b} &= ca' - c'a, & \mathfrak{g} &= bd' - b'd, \\ \mathfrak{c} &= ab' - a'b, & \mathfrak{h} &= cd' - c'd, \end{aligned}$$

$$I = -(\mathfrak{c} + \mathfrak{g})(-\mathfrak{b} + \mathfrak{h}) + \mathfrak{a}^2 = \mathfrak{bc} - \mathfrak{ch} + \mathfrak{bg} - \mathfrak{gh} + \mathfrak{a}^2.$$

To obtain the value of I_1, we must interchange the accented and unaccented letters, that is, change the signs of the several quantities $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}$; but I, being a quadric function of the six quantities, is not altered by the change; that is, we have $I = I_1$.

911. On an Algebraical Identity relating to the Six Coordinates of a Line.

[From the Messenger of Mathematics, vol. xx. (1891), pp. 138–140.]

[p. 76]

The following identity may be verified without difficulty; but it is interesting in regard to the analytical theory of the line.

The identity is

$$\begin{aligned}0 = {} & (bc'-b'c)^2(AF+BG+CH)\\ & +(bc'-b'c)(bC-cB)(Af+Bg+Ch+Fa+Gb+Hc)\\ & -(bc'-b'c)(b'C-c'B)(Af'+Bg'+Ch'+Fa'+Gb'+Hc')\\ & -(bC-cB)^2(af+bg+ch)\\ & +(bC-cB)(b'C-c'B)(af+a'f'+bg+b'g'+ch+c'h')\\ & -(b'C-c'B)^2(a'f'+b'g'+c'h'),\end{aligned}$$

where all the letters have arbitrary values.

It follows that, if

$$\begin{aligned} af+bg+ch&=0,\\ a'f'+b'g'+c'h'&=0,\\ af+a'f'+bg+b'g'+ch+c'h'&=0,\\ AF+BG+CH&=0,\\ Af+Bg+Ch+Fa+Gb+Hc&=0,\\ Af'+Bg'+Ch'+Fa'+Gb'+Hc'&=0,\end{aligned}$$

[p. 77]

then either

$$\begin{vmatrix} A & B & C\\ a & b & c\\ a' & b' & c'\end{vmatrix}=0,\quad\text{or else}\quad \begin{vmatrix} B & C & F\\ b & c & f\\ b' & c' & f'\end{vmatrix}=0.$$

Supposing the first three of the six equations are satisfied, then (a,b,c,f,g,h) and (a',b',c',f',g',h') are the coordinates of two intersecting lines; and supposing that the last three equations are also satisfied, then (A,B,C,F,G,H) will be the coordinates of a line meeting each of the intersecting lines. The third line is thus either in the plane of the intersecting lines, or else passes through their point of intersection; and in fact, the first of the two determinant equations is the condition in order that the line may be in the plane of the two intersecting lines, and the second determinant equation is the condition in order that it may pass through their point of intersection. Each equation is in fact one out of four equivalent forms, viz. we may have in the first equation (B, C, F), (C, A, G), (A, B, H), or (F, G, H); and in the second equation (A, B, C), (A, H, G), (H, B, F), or (G, F, C).

The analytical theory may be presented in a complete form by means of the formulae of my memoir "On the six coordinates of a line," (1867), Camb. Phil. Trans., t. XI. pp. 290–323, [435]. Considering the two intersecting lines (a, b, c, f, g, h) and (a', b', c', f', g', h'), the coordinates of the plane through these two lines (that is, the coefficients ξ, η, ζ, ω of the equation $\xi x + \eta y + \zeta z + \omega w = 0$ of the plane) are there given (see p. 295*) in a fourfold form; and if we thence form the condition in order that the line (A, B, C, F, G, H) may lie in this plane, we have

$$\begin{pmatrix} 0 & C & -B & F \\ -C & 0 & A & G \\ B & -A & 0 & H \\ -F & -G & -H & 0 \end{pmatrix} \begin{pmatrix} af' + b'g + c'h & bf - b'f & cf - c'f & -(bc' - \\ ag' - a'g & a'f + bg' + c'h & cg' - c'g & -(ca' - \\ ah' - a'h & bh' - b'h & a'f + b'g + ch' & -(ab' - \\ gh' - g'h & hf - h'f & fg' - f'g & af + bg' \end{pmatrix}$$

viz. the condition is expressible in any one of the 16 forms obtained by combining a line of the first matrix with a line of the second matrix; thus one form is

$$0 \cdot (af' + b'g + c'h) + C(bf - b'f) - B(cf - c'f) - F(bc' - b'c) = 0,$$

that is,

$$\begin{vmatrix} B & C & F \\ b & c & f \\ b' & c' & f' \end{vmatrix} = 0,$$

the foregoing first determinant equation, which thus belongs to the case where the line lies in the plane of the two intersecting lines.

[p. 78]

Again we have (see p. 296*) an expression, in a fourfold form, for the coordinates of the point of intersection of the two intersecting lines; and thence for the condition, in order that the line

(A, B, C, F, G, H) may pass through this point, we have

$$\begin{pmatrix} 0 & H & -G & A \\ -H & 0 & F & B \\ G & -F & 0 & C \\ -A & -B & -C & 0 \end{pmatrix} \begin{pmatrix} ag' - a'g & ah' - a'h & af + b'g + c'h & -(bc' - b'c) \\ bf - b'f & a'f + bg' + c'h & bh' - b'h & -(ca' - c'a) \\ cf - c'f & cg' - c'g & a'f + b'g + ch' & -(ab' - a'b) \\ gh' - g'h & hf - h'f & fg' - f'g & af + bg' + ch' \end{pmatrix}$$

viz. the condition is expressible in any one of the 16 forms obtained by combining a line of the first matrix with a line of the second matrix; thus one form is

$$A(bc' - b'c) + B(ca' - c'a) + C(ab' - a'b) + 0 \cdot (af + bg' + ch') = 0,$$

that is,

$$\begin{vmatrix} A & B & C \\ a & b & c \\ a' & b' & c' \end{vmatrix} = 0,$$

the foregoing second determinant equation, which thus belongs to the case where the line passes through the point of intersection of the two intersecting lines.

I remark that the original identity may be written in the very compendious symbolical form

$$\begin{vmatrix} A & a & a' \\ B & b & b' \\ C & c & c' \end{vmatrix}^2 \cdot \begin{vmatrix} F & f & f' \\ G & g & g' \\ H & h & h' \end{vmatrix}^2 = 0,$$

viz. here on the left-hand side the second factor denotes the three determinants $bc' - b'c$, $b'C - c'B$, $Bc - Cb$: the whole is a quadric function of these, the coefficients being

$$AF + BG + CH, \quad af + bg + ch,$$
$$a'f' + b'g' + c'h', \quad af + bg' + ch' + fa' + gb' + hc',$$

$$a'F + b'G + c'H + f'A + g'B + h'C, \quad Af + Bg + Ch + Fa + Gb + Hc,$$

respectively; and the right-hand side is simply the product of two determinants.

912. On the Notion of a Plane Curve of a Given Order.

[From the Messenger of Mathematics, vol. xx. (1891), pp. 148–150.]

[p. 79]

We have a complete geometrical notion of a curve of a given order, viz. a curve of the order n is a curve which is met by any line whatever in n points and no more; but starting with this definition, how do we know that there exists a curve of the order n? and, further, how do we know that it depends linearly on $\frac{1}{2}n(n+3)$ parameters, or, what is the same thing, that there is one and only one curve which can be drawn through $\frac{1}{2}n(n+3)$ given points?

The last-mentioned property does not by itself constitute a definition of a curve of the order n; thus $n = 2$, we cannot define a curve of the second order as a curve which is uniquely determined by the condition of passing through 5 given points; for a cubic passing through 4 given points is a curve uniquely determined by the condition in question; but we may differentiate between these two solutions by adding the further condition that, when 3 of the 5 points are in a line, the curve of the second order shall include as part of itself this line. And we are thus led to the definition:

A curve of the order n is a curve which is uniquely determined by the condition of passing through $\frac{1}{2}n(n+3)$ given points; and of being moreover such that, when $n+1$ of these points lie on a line, it includes as part of itself this line.

Starting from the foregoing definition, the first property is, I think, demonstrable, viz. the property that a curve of the order n is met by any line whatever in n points and no more. Thus $n = 2$: start with a line chosen at pleasure, and on it take 2 points which are regarded as indeterminate points: to fix the ideas, let one of these be regarded as depending on a parameter λ, and the other on a parameter μ, (so that when λ has a determinate value assigned to it, the first point becomes a determinate point, and similarly when μ has a determinate value assigned to it, the second point becomes a determinate point; and consequently, when λ and μ have

[p. 80]

determinate values, the 2 points are determinate points on the line). Take now any other 3 points not on the line; then, for the moment regarding the 2 points on the line as determinate points, we can through the 5 points draw a curve of the second order; this is the general curve of the second order through the 3 points; for it is a curve of the second order through the 3 points, and which, when the parameters λ and μ are regarded as undetermined and arbitrary, or choosable at pleasure, might be made to pass through any other two points whatever. But by hypothesis, the curve meets the line in question, that is, any line, in two points: and the conic cannot meet the line in more than two points; for in like manner, starting with the given line, and upon it 3 points (which may be considered as depending on the parameters λ, μ, and ν respectively), and taking any two points not on the line, we have through the 5 points a conic; and this conic, regarding the parameters λ, μ, ν as undetermined and arbitrary, or choosable at pleasure, will be the general conic through the two points; for by a proper determination of the parameters, it might be made to pass through any other three points whatever. The reasoning is general, and we thus establish the theorem that a curve of the order n is met by any line whatever in n points and no more.

913ON THE EPITROCHOIDFrom the Messenger of Mathematics, vol. XX. (1891), pp. 150—158.

If we have a curve C_1 rolling on a fixed curve C, and consider the epitrochoid described by a point P, attached to and carried along with the curve C_1, then there is a known construction for the radius of curvature at any point of the epitrochoid; the construction is probably much older, but I refer to Mannheim, *Géométrie Descriptive*, Paris, 1886, pp. 177 and 194. In fact, if Q be a position of the point of contact, P_1 the corresponding position of the describing point, R_1 and R the radii of curvature at Q of the two curves respectively, ρ the distance QP_1, and ϕ the inclination of this distance to the common normal at Q, then the radius of curvature χ at the point P_1 of the epitrochoid is given by the formula

$$\frac{1}{\chi-\rho} = \frac{\cos\phi}{\rho}\sqrt{\frac{1}{R}-\frac{1}{R_1}}.$$

I prove this as follows: take Q, Q' consecutive positions of the point of contact, P_1, P_1' consecutive positions of P_1; the centre of curvature is the intersection of the lines P_1Q, $P_1'Q'$; hence, if for a moment M and N are the perpendicular distances between these two lines at the points P_1 and Q respectively, we have $\dfrac{M}{\chi-\rho} = \dfrac{N}{\rho}$. Let ds be the element QQ' considered as belonging to each of the two curves respectively, θ and θ_1 the angles which this element subtends at the two centres of curvature respectively; we have $ds = R\theta = R_1\theta_1$, whence

$$\theta+\theta_1 = ds\left(\frac{1}{R}-\frac{1}{R_1}\right).$$

The instantaneous centre is P_1, and hence

$$M = P_1P_1' = \rho(\theta+\theta_1);\quad \text{also } N = ds\cos\phi;$$

and therefore

$$\frac{\rho(\theta+\theta_1)}{\chi-\rho} = \frac{ds\cos\phi}{\rho},$$

or, substituting for $\theta+\theta_1$ the foregoing value, we have the required expression

$$\frac{1}{\chi-\rho} = \frac{\cos\phi}{\rho}\sqrt{\frac{1}{R}-\frac{1}{R_1}}.$$

If the curve C_1 roll on a straight line, $R = \infty$ and therefore

$$\frac{1}{\chi - \rho} = \frac{\cos\phi}{\rho \cdot R_1}.$$

Suppose that at any instant we have on the epitrochoid an inflexion, then $\chi = \infty$, and we have $\rho = R_1 \cos\phi$; this denotes that P_1 is a point of the circle described through Q with R_1 for diameter; that is, the circle described through Q having its centre at the centre of curvature of the rolling curve; or, what is the same thing, the locus of the centres of curvature of the epitrochoid is the same curve as the locus of the centres of curvature of the rolling curve, only in a different position.

I consider in particular the case where the rolling curve is a conic. If Q be the point of contact, P_1, P_2 the points of inflexion, then these are the intersections of the circle of curvature at Q of the conic by a line through the centre of curvature Q_1 (viz. P_1P_2 passes through Q_1), and the distance $Q_1P_1 = Q_1P_2$ is equal to the diameter of curvature at Q, that is, to 2ρ where ρ is the radius of curvature.

First, let the conic be a hyperbola. The circle of curvature at any point of a conic meets the conic in four consecutive points; and we can through these four points and any fifth point on the conic describe a circle, which will meet the conic in a sixth point; the three chords joining these last three points in pairs are concurrent. In particular, if the fifth point be at infinity on an asymptote, then the sixth point will be also at infinity on the other asymptote, and one of the three chords will be the chord joining these two points at infinity, that is, it will be a line through the centre. The other two chords will be the tangents at the two points where the circle of curvature meets the conic; that is, one of them will be the tangent at the point of contact Q, and the other will be the line P_1P_2.

We have thus a construction for the line P_1P_2, but we require also the distance Q_1P_1. I take x, y as the coordinates of the point of contact, referred to the axes of the hyperbola; ρ, the radius of curvature, $= \dfrac{(b^2x^2 + a^2y^2)^{\frac{3}{2}}}{ab}$, and the coordinates of the centre of curvature are

$$x_1 = \frac{(a^2+b^2)x^3}{a^4}, \quad y_1 = -\frac{(a^2+b^2)y^3}{b^4}.$$

Representing for a moment the equation of the circle by $U = 0$, and that of the hyperbola by $V = 0$, it must be possible to determine the ratio of the coefficients p, q, in such wise that $pU + qV = 0$ shall break up into a pair of lines, one of which is the tangent at Q of the hyperbola; and then the other of them will be the required line P_1P_2.

The equation of the tangent is $\dfrac{b^2xX}{a^2} - \dfrac{a^2yY}{b^2} - ab = 0$; we then see that the equation of the line P_1P_2 must be of the form $\dfrac{b^2xX}{a^2} + \dfrac{a^2yY}{b^2} + H = 0$; and we have to find p, q, H so as to verify the identity

$$p\{(X-x_1)^2+(Y-y_1)^2-\rho^2\}+q\left(\frac{X^2}{a^2}-\frac{Y^2}{b^2}-1\right)=\left(\frac{b^2xX}{a^2}-\frac{a^2yY}{b^2}-ab\right)\left(\frac{b^2xX}{a^2}+\right.$$

Equating the coefficients of X^2, Y^2, XY, we find

$$n+\frac{q}{a^2}=\frac{b^4x^2}{a^4},\quad n-\frac{q}{b^2}=-\frac{a^4y^2}{b^4},\quad \text{that is, } q=-\frac{a^2b^2(b^4x^2+a^4y^2)}{a^4b^4},$$

and

$$n=\frac{b^4x^2}{a^4}+\frac{q}{a^2}=\frac{b^4x^2-b^2(b^4x^2+a^4y^2)/b^2}{a^4}=\frac{b^4x^2-b^4x^2-a^4y^2}{a^4}=-\frac{a^4y^2}{a^4}=-\frac{y^2}{a^2}\cdot$$

[More of Paper 913 continues . . .]

914ON A SOLUBLE QUINTIC EQUATIONFrom the American Journal of Mathematics, vol. XIII. (1891), pp. 53—58.

Mr Young, in his paper, "Solvable Quintic Equations with Commensurable Coefficients," *American Journal of Mathematics*, x. (1888), pp. 99—130, has given, in illustration of his general theory of the solution of soluble quintic equations (founded upon a short note by Abel), no less than twenty instances of the solution of a quintic equation with purely numerical coefficients, having a solution of the form

$$x = \alpha + \beta + \gamma + \delta,$$

where α, β, γ, δ are surds each involving a square root; the general theory shows, and it appears in each of the particular cases, that these are not each of them a mere square root, but they are of the form

$$\alpha = \sqrt{A' + B'\sqrt{Q}}, \quad \beta = \sqrt{C' + D'\sqrt{Q}},$$

$$\gamma = \sqrt{A'' + B''\sqrt{Q}}, \quad \delta = \sqrt{C'' + D''\sqrt{Q}},$$

where A', B', C', D', A'', B'', C'', D'' are rational functions of a single parameter Q.

The twenty instances are all given with a considerable degree of speciality; and I thought it would be interesting to consider the general case, and to obtain the actual expressions for the coefficients of the resulting quintic equation.

2. I found it convenient to introduce in place of α, β, γ, δ the following combinations:

$$\begin{aligned} \alpha + \beta &= p, \qquad & \alpha - \beta &= q, \\ \gamma + \delta &= r, \qquad & \gamma - \delta &= s. \end{aligned}$$

We have

$$x = p + r, \quad x^2 = p^2 + r^2 + 2pr = q^2 + 2\alpha\beta + s^2 + 2\gamma\delta + 2pr,$$

and so on; the expressions are symmetrical as regards the pairs (α, β) and (γ, δ), and they contain the products $\alpha\beta$ and $\gamma\delta$ which are rational functions of Q. We have in this manner

$$A = p^2 + r^2, \quad B = 2pr, \quad C = q^2 + s^2 + 2(\alpha\beta + \gamma\delta), \quad D = 2qs,$$

and thence

$$x^2 = A + B + C + D, \quad x^2 = A + B - C - D,$$

$$x^3 = A + B + C - D, \quad x^3 = A + B - C + D,$$

where A, B, C, D are each of them a rational function of Q.

3. We have $\alpha^2 + \beta^2 = 2A'$, $\alpha^2 - \beta^2 = 2B'\sqrt{Q}$; whence $\alpha^2 = A' + B'\sqrt{Q}$, $\beta^2 = A' - B'\sqrt{Q}$; and thence $\alpha\beta = \sqrt{A'^2 - B'^2 Q}$, which is a rational function of Q; and similarly $\gamma\delta = \sqrt{A''^2 - B''^2 Q}$, a rational function of Q.

4. I write moreover

$$\alpha\beta + \gamma\delta = f, \quad \alpha\beta - \gamma\delta = g,$$

and I take $\sqrt{Q}$, $\sqrt{Q_1}$ for the two values of $\sqrt{Q}$, so that $\sqrt{Q}\sqrt{Q_1}$ is rational. We have

$$p^2 + q^2 = 2(\alpha^2 + \beta^2) = 4A', \quad p^2 - q^2 = 2\alpha\beta = 2f,$$

and thence $p^2 = 2A' + f$, $q^2 = 2A' - f$; and similarly $r^2 = 2A'' + f$, $s^2 = 2A'' - f$. And then also

$$\begin{aligned}
&A = p^2 + r^2 = 2(A' + A'') + 2f,\\
&B = 2pr = 2\sqrt{(2A' + f)(2A'' + f)},\\
&C = q^2 + s^2 + 2(\alpha\beta + \gamma\delta) = 2(A' + A'') - 2f + 2f = 2(A' + A''),\\
&D = 2qs = 2\sqrt{(2A' - f)(2A'' - f)}.
\end{aligned}$$

5. We have $x^2 = A+B+C+D$, $x^3 = A+B-C-D$, $x^4 = A+B+C-D$, $x^5 = A+B-C+D$; or say x^2, x^3, x^4, $x^5 = A+B\pm C\pm D$, where the signs are such that writing $x^2 = \alpha+\beta+\gamma+\delta$, we have $x^2 = A+B+C+D$, and so on.

6. The expressions for the coefficients of the quintic equation are as follows:

$$5A = A+B+C+D+A+B-C-D+A+B+C-D+A+B-C+D,$$

$$10B = (A+B+C+D)(A+B-C-D) + \&\text{c.}, \&\text{c.},$$

$$10C = \&\text{c.}, \quad 5D = \&\text{c.}, \quad E = \&\text{c.},$$

where A, B, C, D are each of them a rational function of Q.

7. Taking $Q = 5$, and $(2+\sqrt{5})\sqrt{Q} = \sqrt{Q_1}$, then we have

$$A = 40(1+\sqrt{5})\{65\sqrt{5}(2+\sqrt{5}) + (18+5\sqrt{5})\sqrt{Q}\}, \&\text{c.},$$
$$A' = -2\sqrt{5}(1+\sqrt{5})\{5(2+\sqrt{5}) + \sqrt{Q}\}, \&\text{c.},$$
$$A'' = 20(1-\sqrt{5})\{18+13\sqrt{5} + \sqrt{5}\sqrt{Q}\}, \&\text{c.},$$

where observe that the term $2+\sqrt{5}$ is a factor of Q.

Starting from the values of A', B', C', D', we have

$$A' = -2\sqrt{5}(1+\sqrt{5})\{5(2+\sqrt{5}) + \sqrt{Q}\},$$

and similarly for B', C', D'.

8. In the proposed equation $x^5 + 300x^2 + 20000x - 100000 = 0$, we have $r = 3000$, $s = 20000$, $t = -100000$, $\alpha\beta = -4\sqrt{5}$; the two equations to be verified thus are

$$+4Q\sqrt{5}(A' + B' - C' - D') - 100000 = 0,$$

and

$$5A''+15B''+10C''-10D''-\frac{4Q}{5}(C'^2+2D'^2)-150\sqrt{5}(C'+2D')-1600+20000 = 0.$$

As to the first of these, we have $A + B + C + D = 156000$, $A' + B' - C' - D'$ giving the required identity; and similarly the second equation is verified.

915ON THE PARTITIONS OF A POLYGONFrom the Proceedings of the London Mathematical Society, vol. XXII. (1891), pp. 237—262. Read March 12, 1891.

1.

The partitions are made by non-intersecting diagonals; the problems which have been successively considered are (1) to find the number of partitions of an r-gon into triangles, (2) to find the number of partitions of an r-gon into k parts, or, what is the same thing, by means of $k-1$ non-intersecting diagonals.

2.

Assuming for the moment the foregoing result, then for $k=1$, the number of partitions is

$$\frac{r \cdot r-3}{2},$$

for $k=2$ it is

$$\frac{r \cdot r-3 \cdot r-4}{3},$$

and so on. As a simplification, I introduce the letter y to denote the number of parts; the number of partitions is then a function of r and y, say it is $= f(r, y)$; and the problem is to find the general expression of this function.

3.

And in connexion herewith, I give the second Table on the next page, which shows for any given value of r the number of partitions for the several values $k = 1, 2, 3, \ldots$ of the number of parts; the several numbers are the coefficients of the successive powers of y in the development of a certain function of y, or say they are given by means of a generating function.

4.

These functions, U and V, are particular values satisfying the equation

$$y\frac{du}{dy} = (u-x)\frac{du}{dx},$$

and it is in this point of view that I consider the question.

5.

Taking x, y as independent variables, denoting by X an arbitrary function of x, and writing

$$u = x + yX,$$

we have

$$\frac{du}{dy} = X + yX'\frac{du}{dy}, \quad \frac{du}{dx} = 1 + yX'\frac{du}{dx},$$

where the accent denotes differentiation in regard to x; hence

$$\frac{du}{dy} = X\frac{du}{dx}, \quad \text{and thence } y\frac{du}{dy} = yX\frac{du}{dx} = (u-x)\frac{du}{dx},$$

or say

$$y\frac{du}{dy} = (u-x)\frac{du}{dx}.$$

6.

But, from the equation

$$U = x + yU^2,$$

we have

$$\frac{dU}{dy} = U^2 + 2yU\frac{dU}{dy}, \quad \frac{dU}{dx} = 1 + 2yU\frac{dU}{x},$$

and thence

$$y\frac{dU}{dy} = \frac{yU^2}{1-2yU} = \frac{U(U-x)}{1-2yU} = (U-x)\frac{dU}{dx}.$$

7.

I multiply each side of the identity by xz, and write

$$V = \frac{U}{U - x}.$$

We thus obtain two sets of functions U and V, satisfying the beforementioned equation. We have

$$U_1 = x + x^2 + x^3 + \ldots + x^r + \ldots,$$

and it will be observed that we have, moreover, the relations

$$U_2 = \tfrac{1}{2}x(x - 2yU'), \quad U_3 = \&\text{c.},$$

$$V_2 = \frac{U_2}{U_2 - x}, \quad V_3 = \&\text{c.}$$

8.

We do not absolutely require, but it is interesting to obtain, the finite expressions of these functions. We have

$$\begin{aligned}
(1-x)U_1 &= x, \\
(1-x)^3 U_2 &= x^2(1-x), \\
(1-x)^5 U_3 &= x^3(1-3x+x^2), \\
(1-x)^7 U_4 &= x^4(1-6x+5x^2-x^3), \\
(1-x)^9 U_5 &= x^5(1-10x+20x^2-7x^3+x^4), \\
(1-x)^{11} U_6 &= x^6(1-15x+50x^2-35x^3+9x^4-x^5), \\
(1-x)^{13} U_7 &= x^7(1-21x+105x^2-140x^3+70x^4-11x^5+x^6),
\end{aligned}$$

and so on.

9.

We do not absolutely require, but it is interesting to obtain, the finite expressions of these functions. We have

$$\begin{aligned}
(1-x)^2 V_2 &= x^2,\\
(1-x)^4 V_3 &= x^3(4-2x),\\
(1-x)^6 V_4 &= x^4(14-14x+3x^2),\\
(1-x)^8 V_5 &= x^5(48-72x+32x^2-4x^3),\\
(1-x)^{10} V_6 &= x^6(165-330x+210x^2-50x^3+5x^4),\\
(1-x)^{12} V_7 &= x^7(572-1430x+1232x^2-448x^3+70x^4-6x^5),
\end{aligned}$$

and so on.

10.

It is to be shown that, taking U_1, U_2, U_3, ... for the functions which belong to the partitions into 2, 3, 4, ... parts respectively, and V_2, V_3, V_4, ... for the like functions, then that we have

$$V_2 = \frac{U_2}{U_2 - x}, \quad V_3 = \frac{U_3}{U_3 - x}, \quad \&\text{c.},$$

and further that we have the relations

$$V_2 = U_1^2, \quad V_3 = U_2^2, \quad \&\text{c.}$$

11.

Considering, then, the partition problem from the point of view just referred to, we have to show that the number of partitions of an r-gon into k parts is equal to the coefficient of x^{r-1} in U_{k-1}, or, what is the same thing, to the coefficient of x^r in V_k.

12.

To obtain the whole number of the partitions of the r-gon into six parts, say, we consider the several cases corresponding to the different positions of the diagonal which separates off a triangle; and we thence obtain the formula

$$U_6 = xU_5 + x^2U_4 + x^3U_3 + x^4U_2 + x^5U_1 + x^6,$$

which is the required relation for the present case.

13.

The reasoning is perfectly general; and applying it successively to the several cases, we obtain the series of formulae

$$\begin{aligned}
U_2 &= xU_1 + x^2,\\
U_3 &= xU_2 + x^2U_1 + x^3,\\
U_4 &= xU_3 + x^2U_2 + x^3U_1 + x^4,\\
U_5 &= xU_4 + x^2U_3 + x^3U_2 + x^4U_1 + x^5,
\end{aligned}$$

and so on.

14.

In the investigations which next follow, I consider, without using the method of generating functions, the actual partitions of the polygon; and I give the theory of the sub-types which correspond to the different forms of the partition-figure.

15.

It is to be observed that we have sub-types corresponding to the coalescence of the diagonal-chords; and these give rise to the several terms of the expressions which follow.

16.

For the partitions into 2 parts, there is only the single type; the expression is simply A.

17.

To explain the formation of these expressions, observe that:

18.

Three diagonals, A. — There must be on each side of the three diagonals, three sides of the polygon; and we have the sub-types according as the remaining sides are distributed in the three intervals between the diagonals.

19.

Four diagonals, A. — There must be on each side of the four diagonals, three sides of the polygon; and we have the sub-types according to the distribution of the remaining sides in the four intervals.

20.

In the expressions of No. 14, A, $2A$, $3A+2B$, $4A+8B+2C$, if we regard the coefficients of A, B, C, ... as being the numbers belonging to the several sub-types, we have a verification of the theory.

21.

For the partitions into 5 parts, the types are A, B, C; and the expression is $5A+8B+2C$.

22.

For the partitions into 6 parts, the types are A, B, C, D; and the expression is $6A+15B+10C+2D$.

23.

Three diagonals, A. — See No. 15 for the figures of the sub-types. We have the sub-types corresponding to the partitions 3, 21, 111 of the number of sides.

24.

Four diagonals, A. — The figures of the sub-types of A, B, C can be supplied without difficulty; and the types correspond to the partitions of the number of sides into four parts.

25.

We may analyse the partitions of an r-gon into a given number of parts by considering the figures which correspond to the several types.

26.

The general formula for the number of partitions of an r-gon into k parts is

$$\frac{r \cdot r-3 \cdot r-4 \ldots r-k-1}{k \cdot k-1 \cdot k-2 \ldots 2 \cdot 1} \times \frac{r-k-2}{k+1}.$$

27.

The first column (2 parts) is at once obtained. For a polygon of r sides, the number of partitions into 2 parts is $\dfrac{r(r-3)}{2}$.

28.

To obtain the second column (3 parts)—suppose, for instance, the 3-partitions of a heptagon. Drawing a diagonal which separates off a triangle, we have a heptagon with a triangle cut off, that is, a hexagon; and the number of 2-partitions of this hexagon is the required number.

29.

Proceeding in this manner, we obtain step by step the successive columns of the table; and we have thus a complete determination of the number of partitions for any given values of r and k.

30.

The table of the values is as follows:

$r\backslash k$	1	2	3	4	5	6	7	8	9	10
3	1									
4	1	2								
5	1	5	5							
6	1	9	21	14						
7	1	14	56	84	42					
8	1	20	120	300	330	132				
9	1	27	225	825	1485	1287	429			
10	1	35	385	1925	5005	7007	5005	1430		
11	1	44	616	4004	14014	28028	32032	19448	4862	
12	1	54	936	7644	34398	91728	148512	143208	75582	16796
13	1	65	1365	13650	76440	259896	556920	755820	629850	293930

916NOTE ON A THEOREM IN MATRICESFrom the Proceedings of the London Mathematical Society, vol. XXII. (1891), p. 458.

Prof. Cayley remarks that a "simple instance [of the theorem] is that, if the real symmetric matrix

$$\begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix}$$

be such that its determinant is positive, and the determinants $\left|\begin{smallmatrix} a & h \\ h & b \end{smallmatrix}\right|$, $\left|\begin{smallmatrix} a & g \\ g & c \end{smallmatrix}\right|$, $\left|\begin{smallmatrix} b & f \\ f & c \end{smallmatrix}\right|$ are each positive, then the matrix is positive; and conversely."

The theorem in its general form is as follows: If M be a symmetric matrix of any order, then the determinant $|M|$ and its successive principal minors are each positive; and conversely, if these conditions are satisfied, then the matrix is positive.

917NOTE ON THE THEORY OF RATIONAL TRANSFORMATIONFrom the Proceedings of the London Mathematical Society, vol. XXII. (1891), pp. 475, 476.

In my paper, "Note on the Theory of the Rational Transformation between Two Planes, and on Special Systems of Points," *Proc. Lond. Math. Soc.* t. III. (1870), pp. 196—198, [450], I notice a difficulty which presents itself in the theory. The transformation is given by the equations

$$X : Y : Z = U : V : W,$$

where U, V, W are functions of given degrees of (x, y, z); and the curves $U = 0$, $V = 0$, $W = 0$ intersect in a certain number of fixed points, and in a variable point (x, y, z).

The difficulty is that the number of fixed points may be so large that the postulation of the system of curves exceeds the number of the coefficients; and it would appear that in such case the transformation could not exist. But it does exist; and the explanation is that the conditions are not independent.

I take a simple case: Let the curves be cubics having in common seven given points; the postulation is $= 10-7 = 3$, and there are three linearly independent cubics; the transformation is given by means of these; and the number of base-points is $= 9 - 7 = 2$, that is, there are two base-points.

But if the seven points are such that the cubics have a common tenth point, then the postulation is $= 10 - 8 = 2$; there are only two linearly independent cubics; and the transformation does not exist.

The like consideration applies to the general case; and we thus see that the transformation exists, provided only that the conditions are properly satisfied.

918ON THE SUBSTITUTION GROUPS FOR TWO, THREE, FOUR, FIVE, SIX, SEVEN, AND EIGHT LETTERSFrom the Quarterly Journal of Pure and Applied Mathematics, vol. XXV. (1891), pp. 71—88, 137—155.

The substitution groups for two, three, four, and five letters were obtained by Serret: those for six, seven and eight letters have recently been obtained by Mr Askwith. I wish to reproduce these results in a condensed form.

1. I use the following notations: (ab) denotes the complete group $(1, ab)$ of substitutions upon the two letters; and so for any substitution such as $ab \cdot cd$, where the complete group is $(1, ab \cdot cd)$, I denote the group by $(ab \cdot cd)$. Moreover, (abc) cycl. denotes the group of cyclical substitutions $(1, abc, acb)$ upon the three letters; and so in other cases.

A group will in general contain positive and negative substitutions, and, when the distinction is material, I denote by $(G)_{\text{pos.}}$ the group formed with the positive substitutions only of the group G; and by $(G)_{\text{all}}$ the group obtained by compounding the substitutions of G with 1 and a given negative substitution. Thus $(abc)_{\text{all}}$ denotes the group $(1, abc, acb, bc, ca, ab)$; and so in other cases.

2. I use also the sign of composition: thus $A \cdot B$ denotes the group obtained by combining each substitution of A with each substitution of B; it is of course necessary that the substitutions of A and B be commutable. The composite group is in general denoted by AB; it is not in general expressible as $A \cdot B$.

3. I notice that if A, B are groups upon different letters, then each substitution of A is commutable with each substitution of B, and we have the group $A \cdot B$.

4. Two letters. — The only group is (ab).

5. Three letters. — The groups are

ord. 3. (abc) cycl.,
ord. 6. (abc) all.

6. Four letters. — The groups are

ord. 2.	$(ab \cdot cd)$,
ord. 3.	(abc) cycl.,
ord. 4.	$(abcd)$ cycl., $(ab)(cd)$,
ord. 6.	(abc) all,
ord. 8.	$(abcd)$ pos., $(ab \cdot cd)(ac)$,
ord. 12.	$(abcd)$ all,
ord. 24.	$(abcd)_{\text{all}}$.

7. Four letters. Explanations. — For the order 4, we have $(ab)(cd)$, (or as I generally write it $(ac)(bd)$), and $(abcd)$ cycl. (which it is convenient thus to represent) and another group $(1, ab \cdot cd, ac \cdot bd, ad \cdot bc)$: this is a woven group $(ac \cdot bd)(ab)(cd)$.

8. Five letters. — The groups are

ord. 5.	$(abcde)$ cycl.,
ord. 10.	$(abcde)_{10}$,
ord. 20.	$(abcde)_{20}$,
ord. 60.	$(abcde)_{\text{pos.}}$,
ord. 120.	$(abcde)_{\text{all}}$.

9. Five letters. Explanations. — $(abcde)_{10}$ may be written $\{(abcde)_{20}\}_{\text{pos.}}$, viz. it consists of the 10 positive substitutions out of the next-mentioned group of 20 substitutions. Referring to that group, and writing $S = S^2 = ac \cdot bd$, and $T = aecdb$, the present group consists of the positive substitutions of the group generated by S and T.

10. Six letters. — The groups are

ord. 2.	$(ab \cdot cd \cdot ef)$,
ord. 3.	$(abc \cdot def)$ cycl.,
ord. 4.	$(ab \cdot cd)(ef)$, $\quad 2\{(ab)(cd)(ef)\}$ pos.,
ord. 6.	$(abc \cdot def)$ all, $\quad (abc \cdot def)$ pos.,
ord. 6.	$2(abc \cdot def)$ all,
ord. 8.	$(abcd)(ef)$, $\quad (ab \cdot cd \cdot ef)(ac)$,
ord. 9.	(abc) cycl.(def) cycl.,
ord. 12.	$(abcd)$ all(ef), $\quad (abc)$ all(def) all pos.,
ord. 16.	$(abcd)_{\text{all}}(ef)$,
ord. 18.	(abc) all(def) all,
ord. 24.	$(+abcdef)_{24}$,
ord. 24.	$(abc \cdot def)$ all$(ab \cdot cd)$,
ord. 36.	(abc) all(def) all pos., with $(ab \cdot cd \cdot ef)$,
ord. 48.	$(abcdef)_{48}$,
ord. 60.	$(abcde)$ pos.(f),
ord. 72.	(abc) all(def) all, with $(ab \cdot cd \cdot ef)$,
ord. 120.	$(abcde)$ all(f), $\quad (abcdef)_{120}$,
ord. 360.	$(abcdef)_{360}$,
ord. 720.	$(abcdef)_{\text{all}}$.

11. Six letters. Explanations. — $(abc \cdot def)$ all; the substitutions are those of (abc) all each, compounded with the corresponding substitution of (def) all; viz. they are

$$1, \quad ab \cdot de, \quad abc \cdot def, \quad ac \cdot df, \quad acb \cdot dfe, \quad bc \cdot ef.$$

12. Six letters. Ord. 24. $(+abcdef)_{24}$: the substitutions are

$$\begin{array}{cccc}
1, & ac \cdot bd, & abe \cdot cdf, & abcd, \\
ab \cdot cd \cdot ef, & ac \cdot ef, & abf \cdot cde, & adcb, \\
ac \cdot be \cdot df, & bd \cdot ef, & ade \cdot bfc, & aecf, \\
ad \cdot be \cdot cf, & adf \cdot bee, & afce, & ad \cdot be \cdot ef.
\end{array}$$

13. Six letters. Ord. 48. $(abcdef)_{48}$. The substitutions are

$$\begin{array}{cccccccc}
1, & ab \cdot cd, & abe \cdot cdf, & abcd \cdot ef, & ac, & abed, & ab \cdot cd \cdot ef, & abe \\
ac \cdot bd, & abf \cdot cde, & adcb \cdot ef, & bd, & adcb, & ac \cdot bd \cdot ef, & abecdf, & \\
ac \cdot ef, & ade \cdot cbf, & aecf \cdot bd, & ef, & aecf, & ac \cdot be \cdot df, & abfcde, & \\
ad \cdot be, & adf \cdot cbe, & afce \cdot bd, & & afce, & ac \cdot bf \cdot de, & adecbf. &
\end{array}$$

14. Six letters. Ord. 120. $(abcdef)_{120}$. The substitutions are

$$\begin{array}{cccccc}
1, & ab\cdot cf, & abcef, & abc\cdot dfe, & abed, & abcfde, \\
 & ab\cdot cd\cdot ef, & & & & \\
ab\cdot de, & abdce, & abd\cdot cfe, & abdf, & abdefc, & ab\cdot ce\cdot df, \\
ac\cdot bd, & abefd, & abe\cdot cdf, & abec, & abedcf, & ac\cdot be\cdot df, \\
ac\cdot ef, & abfdc, & abf\cdot cde, & abfe, & abfced, & ac\cdot bf\cdot de.
\end{array}$$

15. Seven letters. — The groups are

$$\begin{array}{cc}
\text{ord. } 7. & (abcdefg) \text{ cycl.}, \\
\text{ord. } 14. & (abcdefg)_{14}, \\
\text{ord. } 21. & (abcdefg)_{21}, \\
\text{ord. } 42. & (abcdefg)_{42}, \\
\text{ord. } 168. & (abcdefg)_{168}, \\
\text{ord. } 2520. & (abcdefg)_{\text{pos.}}, \\
\text{ord. } 5040. & (abcdefg)_{\text{all}}.
\end{array}$$

16. Eight letters. — The groups are

ord. 8.	$(abcdefgh)$ cycl.,
ord. 8.	$2(abcd)$ cycl.$(efgh)$ cycl.,
ord. 8.	$(ab \cdot cd \cdot ef \cdot gh)$,
ord. 8.	$(ac)(bd)(eg)(fh)$,
ord. 8.	$(abcd)_{\text{all}}(efgh)$,
ord. 16.	$(abcdefgh)_{16}$,
ord. 16.	$(ab \cdot cd)(ef \cdot gh)$,
ord. 16.	$(abcd)$ all$(efgh)$,
ord. 16.	$(abcd)$ pos.$(efgh)$ pos.,
ord. 32.	$(abcdefgh)_{32}$,
ord. 32.	$(abcd)_{\text{all}}(efgh)$ pos.,
ord. 48.	$(abcdefgh)_{48}$,
ord. 56.	$(abcdefgh)_{56}$,
ord. 64.	$(abcdefgh)_{64}$,
ord. 96.	$(abcdefgh)_{96}$,
ord. 128.	$(abcdefgh)_{128}$,
ord. 168.	$(abcdefg)_{168}(h)$,
ord. 224.	$(abcdefgh)_{224}$,
ord. 2520.	$(abcdefg)_{\text{pos.}}(h)$,
ord. 2688.	$(abcdefgh)_{2688}$,
ord. 5040.	$(abcdefg)_{\text{all}}(h)$,
ord. 13440.	$(abcdefgh)_{13440}$,
ord. 20160.	$(abcdefgh)_{\text{pos.}}$,
ord. 40320.	$(abcdefgh)_{\text{all}}$.

17. The complete enumeration of the substitution groups for eight letters presents considerable difficulties; and the foregoing list is not absolutely complete. The groups which are given are, however, those which are most important for the theory.

919. On the Problem of Tactions.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxv. (1891), pp. 104–127.]
(Continued from page 150.)

ON THE PROBLEM OF TACTIONS. and so in other cases. The second, third, and fourth problems are, it is clear, derived from the first problem by the change of (b, c) into (-b, c), (b, -c), (-b, -c) respectively: so that only the first problem need be considered, viz. this is as above whence also s—t — b — c, t — r = c — a, i — s = a— b, where b — c, c — a, a — b are given magnitudes the algebraical sum of which is = 0, viz. they are one of them positive, and the other two each negative, or else two of them each positive, and the remaining one negative.

2. The most simple and straightforward geometrical solution is that given in the Principia, Book I. Lemma XVI. ; I reproduce this as given in Motte's translation (The Mathematical Principles of Natural Philosophy, by Sir Isaac Newton, translated into English by Andrew Motte, 8°, 2 vols., London, 1729). " Lemma XVI. From three given points to draw to a fourth point which is not given three right lines whose differences shall be either given or none at all. *Case 1. Let the given points be A, B, C (see figure) and Z the fourth point* which we are to find : because of the given difference of the line AZ, BZ, the locus ..iZ of the point Z will be a hyperbola whose foci are A and B and whose principal axe is the given difference. Let that axe be MN. Taking PM to MA as MN is to AB, erect PR perpendicular to AB, and let fall ZR perpendicular to PR ; then from the nature of the hyperbola ZR will be to AZ as MN is to AB. And by the like argument the locus of the point Z will be another hyperbola whose foci are A, C, and whose principal axe is the difference between AZ and CZ; and QS a perpendicular on AC may be drawn to which (QS), if from any point Z of this hyperbola a perpendicular ZS is let fall, this (ZS) shall be to AZ as the difference between AZ and CZ is to AC. Wherefore the ratios of ZR and ZS to AZ are given and consequently the ratio of ZR to ZS one to the other: and therefore if the right lines RP, QS meet in T, and TZ and TA are drawn, the figure TRZS will be given in specie, and the right line TZ, in which the point Z is somewhere placed, will be given in position. There will be given also the right line TA and the angle ATZ; and because the ratios of AZ and TZ to ZS are given, their ratio to each other is given also; and thence will be given also the triangle ATZ whose vertex is the point Z. Q. E. I.

Case 2. If two of the three lines, for instance AZ and BZ, are equal, &c. Case 3. If all the three are equal, &c. This problematic lemma is likewise resolved in Apollonius's Book of Tactions restored by Vieta."

3. Newton, in fact, considers the hyperbolas AB and AC, each of given axis, having the foci (A; B) and (A, G) respectively, and having PR, QS for the directrices which in the two hyperbolas respectively belong to the common focus A. The required point Z thus lies on a given line through the intersection T of these two directrices ; and its position on this line is determined by the condition that the distances AZ, TZ shall be in a given ratio : the locus of the points which satisfy this last condition is of course a circle ; and the position of Z is thus determined as the intersection of the given line by a given circle (which I will call a Newton-circle); there are two intersections giving points Zl, Zz, which are the centres of the two tangent circles respectively : and the line as a locus in quo of these two points is of course a determinate line, but Newton's circle is only one of a singly infinite series of circles through the two points : any other solution of the problem gives therefore the same line, but not in general the same circle.

4. In what immediately follows, I use for convenience the letter F in place of the foregoing letter T. Effecting Newton's construction, first as above, with the points A (B, 0) ; and then in like manner, secondly with the points B (G, A), and thirdly with the points C(A, B); and in regard to a hyperbola AB or BA, writing AB for the directrix which belongs to the focus A, and BA for the directrix which belongs to the focus B ; then For hyperbolas AB, AC, we have intersection of directrices AB, AC is a point F; for hyperbolas BC, BA, we have intersection of directrices BC, BA is a point G ; for hyperbolas CA, CB, we have intersection of directrices CA, CB is a point H. Hence these three points F, G, H lie in a line, which is the line containing the required points Zl, Z> _ or say it is the line
Z$_1 Z_1 The points Z_1, Z_2 are determined as the intersections of this line by a circle which is$
$circles : if the centres of these are F', G', H' respectively, then clearly these points lie o$

$iatrightangles, thenthecentresF', G', H'wouldbefoundastheintersectionsofthislin$

ON THE PROBLEM OF TACTIONS. The points Zl, Z> being determined as above, then the points of contact $a_1, \&, 7iofthecircleZ\pm$ with the circles A, B, C respectively are points of intersection of the lines Z1A> Z1B, Z1C with these circles respectively; and similarly the points of contact a'$(32, y2°$f the circle Z2 with the same circles respectively are points of intersection of the lines Z2A, Z2B, Z2C with these circles respectively.

5. I compare with Newton's the construction in which the centres $Z_1, Z2aredeterminedbymeansofthepointsofcontactwiththegivencircles:$ $ImayrefertoProp.10, pp.118---120ofCasey'sSequeltoEuclid(12°,$ Dublin, 1881). We have here the line $Z_1Z.2determinedasthelinethroughtheradicalce$ $thispoint11isthecentreoftheorthotomiccircle.Andifthecommonchordsoftheorthot$ $thesuffixes1and2canandmustbesoappliedthatthethreelinesAaltB\&, G^meetinapoi$

6. Taking the equations of the three circles to be (as - a)2 + (y - cO2 = a2, 0 - /3)2 + (y- A)2 = &, I wish to obtain the equations of the line Z1Z.2 and of the three Newton-circles ; but I will first find, by a separate analytical investigation, an expression for the length of the chord $Z_1.Writing/, g, kforthedistancesBC, CA, ABofthepointsA, B, Cfromeachother; r$ 0,

ON THE PROBLEM OF TACTIONS. [919 and thence, taking the points 1, 2, 3 and also the points 1', 2', 3' to be the points At B, C respectively, and the points 4 and 4' to be the points Z-$^andZ_2respectively, wehavetherequiredrelation0, 1, 1, 1, 1 =$ $0, 1, 0, A2, g\ rf1h*0f- <?2 - I, II, V, J, t > i1a2f*0t" > iJ >$ $J''1I/v.2 < 2/22, /, *, t, a, viz.puttingforshortnessA = /4 + \#4 +$ $A4-2\#2A2-2A2/2-2/y, thisequationis0, 1, 1, 1, 1 = 0, 1, 0, A2, g\ r*$ $1h20/*2s2\pm$, /t , v , y -j *i $^{\cdot 2^{22}Qwherethedeterminanthasthevalue/2t^*-r_2)}$

7. For the points Zlt Z2, the distances $r_1, sly^andr2, s2 >$ $t2havethevaluesa+^i, 6\ +^,\ c\ +^a\ nda\ +^\cdot\ 2, 6\ +^-\ 2.c\ +^-$ $2respectively, whereS^{,-}2aretheradiiofthetangentcircles; substitutingthesevalues,$ $a2/2(-/2+g*+A2)+/2(c2-a2)(a2-62)-/2(/2A2+c2A2(/2+<$ $72-A2)+A2(62-c2)(c2-a2), 93 = a/2(-/2+\#2+A2)+/2(c-a)(a-$ $6)(2a+6+c)+cA2(/2+-72-A2)+A2(6-c)(c-a)(a+b+2c), A2(=$ $-A, +4\#2(a-6)(6-c)+4A2(6-c)(c-a).$

919] ON THE PROBLEM OF TACTIONS. 155 But S-1(

So are the roots of an equation in S, which is at once obtained from the foregoing equation by putting therein # = 0, $^{=0}) =^{:} viz.if, forshortness, thentheequationinSis\% + 233^{+}g^{2} = 0; wehavethereforeand.consequentlyorw, TS493" - 221622lg. - - - +2) - - - - + - - - - ftthatis, AgV + 22lg2 - 4232g + 3)(4332 - 22lg) + 22HS((5-2)) = 0, or, reducing, AgV + 4OB2 - 2lg)(2) - 6) = 0, thatis, whereA, 23, g, ghavethevaluesgivenabove; wehencefind232 - 2lg = -|/2-(b-c)2)g * -(c-a)2)h2 - (a-b)2)x(/4 +^{+} fr\_2^{/}i2 - 2ft2/2 - 2/y) =$

8. The denominator factor g2 and the several numerator factors of (ZZ-$^{f} maybeaccountedfor.Itistobeobservedthat(ZZ_1fdoesnotcontainthefactorAof3 - - 2lg.IfA = 0, thecentresofthecirclesA, B, Careinaline; thetwotangentcirclesare - - 2lg = 0, butthecentresarenotcoincident, andthusZZ_1isnot = 0.Theexpressionfor(ZZ_1fassumesamoresimpleform, forwehaveg = - - -A + g = g, andtheformulathusbecomesg(ZZJ = -4/2 - (b-c)2(/2 - (c-of]h2 - (a-b)2.Ifinthisformulag = 0, thenwehaveZZ1 = ac-, infact, thecirclesA, B, Chavehereapairofcommontange eachofthecentresisthusapointatinfinity, andthedistanceofthetwocentresistobereg infinity.Inverificationobservethat, ifthecircleshaveapairofcommontangents, the Q : R = a : b : c; andtherefore/ : g : h = b - - - c : c - - - a : a - - - b; whenceg, = (c-a)(a-b)f - +(a-b)(b-c)g * +(b-c)(c-a)A2, containsafactor(b-c)+(c-a)+(a-b), andisthus = 0.20 - - - 2$

Supposing A not = 0, then 332 — 21 (£ is = 0, that is, the two radii are equal only if one of the factors f2 - (b - c)2, g2-(c- of, h2 - (a - b)2 is = 0, and in this case we have also $ZZ_1 = 0; thetwotangentcirclesareherecoincident.Theequationy2\_(\&\_c)2 = osignifiesthatthecirclesB, Ctoucheachotherinternally, or, whatisthesamething(if - -(6 - - - c)2 = 0, thetwotangentcirclesarecoincident.Andsimilarlyifg2 - - - (c - - - a)2 = 0, orifh2 - (a-b)2 = 0.$

9. If in the general formula we have (£ = 0, then ZZl = x; the circles A, B, C have here a common tangent and one of the tangent circles becomes this common tangent ; we have thus a centre at infinity, and the distance of the two centres is thus also infinite. To show that (£ = 0 is the condition of a common tangent, suppose that the three circles have a common tangent, and let P, Q, R denote the distances of the points of contact from any fixed point on this tangent : we have equations which I represent by f-, g2, A2 = a- + I2, /S2 + m2, <f + n-, where a + @ + 7 = 0, I + m + n = 0. We have -f2 + 92 + K

= - OL- + /32 + 72 - I2 + m2 + n = - 2/3y - 2mn, and forming the corresponding values of f2—g2 + h2, fz + g2 — h2, and multiplying by f 2, (jr2, h'2 = a2 + 12, /32 + m2, j2 + n2 respectively, we find - A = - 2afmn But the last three terms hereof are = I2 (- a2 + /32 + r) + m? (- /32 + y2 + a2) + n2 (- 72 + a2 + /32), which is = a- (m2 +n2- I2) + j32 (n2 + 1*- m2) + j2 (I2 + m2 - n2), and this is = - Ztfmn - 2/3*nl - 2j-lm. Hence we have — A = — 4soL*mn — 4<32nl — 4<y2lm, that is, = — 4f'2mn — 4<g2nl — 4:h2lm, or replacing I, m, n by their values, we find - A = - 4 /2 (c - a) (a - 6) + g2 (a - b) (b-c)+ h2 (b - c) (c - a), which is the required condition (£ = 0.

10. It would at first sight appear that the distance $ZZ_1 ofthetwocentres$

919] ON THE PROBLEM OF TACTIONS. 157 would always vanish if (£=0. But if A, B, G are real circles, this condition (£ = 0, implies /2 9* K (b - c)2 (c - of (a - b)2 ' whence A = 0, and this being so we have (£ = — A + IS, =0, and the value of ZZ1 instead of being =0, is or appears to be infinite. In proof, take for a moment the origin at A and the line AB for the axis of x we have thus (0, h) for the coordinates of B, and taking (x, y) for the coordinates of (£, we have cf = x2 + y2; f'2 = (h — x)2 + y2. Writing as before I, m, n, to denote b — c, c— a, a — b respectively, we have @ = $mn(h-x)2+y2+nl(x2+y2)+Imh2, = m(n+I)h?+n(I+m)(x-+y----2mnhx, = ---m2h2---n2(x2+y2)---Zmnhx, = ---(mh+nxf---n2y2, lYLiLandthus, forrealvalues, (canonlyvanishfory = Q, x = -------; thesevaluesofx, yftl'2h2m2h2f2a2ft2givef* *a - --)g2 = ------ thatis,^{=}-^{-}2 = ---, orwritingforI, m, ntheirvalues, theylls77 - v7TL1 give\_/!\_\_^{!}\_k*(b-c)2(c-a)2(a-b)2'$

11. But for imaginary circles the condition (S = 0 does not imply A = 0, and supposing @ = 0, the distance $Z_1 Z, is = 0; theequation232 - 5lg = 0, isnotsatisfied, andthusthetworadiiareunequal; itwouldseemthatwehaveconcentr$ $i, Z.2eachtouchingthethreegivencirclesA, B, C, andthiswouldimplythattheradiia, b$ $thiscannotbethecase, fortheonlyrelationisthatgivenbytheforegoingconditionG? =$ $0. Theexplanationofthisparadoxisthatthetwocircles Zlt Z_1 arenotreallyconcentric, b$

0, *viz.thecentresarepointsonanimaginarylinex* $- - - a\pm$ i (y - j3) = 0. In verification hereof, I start from two circles Z_1, *Z*2, (*x* + *I*)2 + (*y* + *i*)2 = *m*2, (*x* − *I*)2 + (*y* − *i*)2 = *n*2, *havingforcentresthetwopoints*(−1, −*i*), (1, *i*)*thedistanceofwhichtwopointsfro* 0.*Considerforamomentaconichavingthesetwoimaginarypointsforitsfoci*; *viz.wri* *I*)2+(*l*+*O*2 = (*m*−*n*)2+2(*m*−*n*)*sqrt*((−*I*)2+(∗?−*O*2]+(*I*−*I*)2+ (*n*−*O*2, *thatis*, 4(+*iii*)−(*m*−*n*)2 = 2(*m*−*n*)*sqrt*(− *I*)2 + 0? − *O*2),

or putting in — n = 2k, we have f+< l-* and thence for the equation of the conic. The last preceding equation gives or say sqrt((£ - I)2 + (n - iY -n = -$^($*m* + *n*) + −? − *t*!5; *andwehavesimilarlysqrt*(+ 1)2 + (17 + *if*] *m* = −*i*(*m* + *n*)*Thisbeingso*, *itaton* Z2. Writing as before /, g, h for the mutual distances BC, CA, AB of the centres of these circles, then and similarly for g- and A2. But we have

919] ON THE PROBLEM OF T ACTIONS. 159 and therefore b-c and similarly 9* = c — a a — o and hence f- + -£— + — -r = 0, that is, @ = 0, 6 — c c — a a — 6 viz. it thus appears that the condition (£ = 0 applies to a pair of circles Z_1, *Z*2*whicharenotconcentric*, *butwhichhavefortheircentrestwoimaginarypointst* 0.*Thiscompletestheexplanationofthedenominatorandnumeratorfactorsintheexp*

12. 1 consider now the analytical solution: the equations of the given circles A, B, C are viz. (a, flj), (/3, /gj), and (7, 7l) are the coordinates of the centres and (a, 6, c) are the radii. Taking (so, y) for the coordinates of the centre of the tangent circle and ^{f}*oritsradius*, *theequationofthetangentcircleis*(*X* − *xf* + (*Y* − 2/)2−^{2} = 0; *andifwewriter*, *s*, *tforthedistancesofthiscentrefromthepointsA*, *B*, *C* − − *t* = *b* − − − *c*, *t* − − − *r* = *c* − − − *a*, *i* − − − *s* = *a* − − − 6; *thesethreehyperbolashave*, *infact*, *twocommonintersectionswhicharethetwocent*

In all that follows, I write, as before, b - c, c - a, a-b = l, m, n; the last- mentioned equations are therefore s — t=l, t — r = m, r — s = n, and we deduce r2 - s2 t9 - r2 s — = - +m + - — , 2w 2m s2 - t* viz. writing - r2 s2 - p_- < ’ : 2T’ 2m ’ and therefore /E + mS + nT = 0, these equations are R, S, T are each of them a linear function of the coordinates (x, y), say we have R = x + XjT/ + X2, S = JJLX T = vx + v^w*here*, , > *ft*−*y*&−*A*, ,‘ *A*2−−−−−−−−, 7− < 71− $^{T\prime}$*m*′2*ma*− /3*aj* − *Aa*2 + *a*12 − /32 − /312∗′ > ∗”!) ∗′ ′ − − − *jjCknn*2*n*

13. From the two equations r = S— m = T + n, we deduce the equations of a line and a circle. The line is S — T — £ (m + n) = 0, viz. substituting for S and T their values, this is t2 - r2 r2 — s- . v n (m + n) = 0, m n that is, n (t2 — r2) — m (r2 — s2) — mn (m + n) = 0, or, since I + m + n = 0, the equation is lr2 + m*2 + nt- + Imn = 0,

919] ON THE PROBLEM OF TACTIONS. 161 which is symmetrical in regard to the three circles. The equation may be written I (r- - a") + m (s° - 62) + n (t - c2) = 0 ; and it thus appears that the line passes through the radical centre of the three circles. We have (v — fi) r = v (S — £?tt) — /z (T + ^{n}) = − − −(/JLV$^{−}$ − −^{v})y + vfa − − − pi > .2 − − − (mv + n/j,), (vi − − − A ∗ i)r = v1(S − − −^{m}) − fr(T+^{n}) = (/JLVI − fj^{y})x + V/UL? − − − /JLV <> − − −$^{(}$mvl + n'andthencewhichistheequationofacircle; infact, ontheleft − handsideandright−handsidetheonlytermsofthesecondorderin(x, y, are(v − − − /u +y−)and(/xi/i−−−/ijvf(#∗+y1)respectively.WehavethustheequationoftheNew circleF; butIreducetheformbysubstitutingforyu, /u1(/x2, v, vlti > 2theirvalues.Writingforshortnesslex. + mft + n < y = K, / < ! + m/Si + njl = K!, 71 − &7 + 7 < i − 7ia + a^{i}− < i =' (ft − 7)(< 2+ < i2) + (7 − a)(/32 + /S12) + (a − /3)(72 + 7l2) = n, (A − 71)(< 2+ < i2) + (T!− < 0(/32 + /?i2) + (<! − A)(72 − f − 7i2) = nt, aftersomeeasyreductionstheequationisfoundtobe4(K − +Kf(x − +y2 − ZOLX − 2 <!y + a2 + a, 2) = 2% + H + (/3 − 7)nin + (m − n)K2+−2lx + U, +(/B, −7l)mn + (m − n)K·

14. To further abbreviate, I write (/3 — 7) mn -f (m — n) K = F , (/3j — jl) mn + (m - n) Kl = Fl , (7 - a) nl + (n - 1) K = G , (7l - a,) nl + (n -I)Kl = G1, (a - /3) /m + (I - m, K = H, (a: - &) im + (I - m) K1 = Hl; also / (a2 + a,2) + m (/32 4- /3f) + n (72 + 7l2) = @ ; and then writing down the three equations, we have 4 (K2 + AY) r-2 = (- 2H< 4- II, + F,)- + (2% + U+F?, n, + G,Y- + (2% + n + G which are the equations of the three Newton-circles, each meeting the chord lr" + ms2 + nt* + Imn = 0, or, say - ZKx - 2K,y + @ + Imn = 0, in the points Zly Z.2.

15. The first of these equations is - 2 4 (K2 + AY) a - 2fl (Hj + $^{)}$# − 2[4(AT2 + AY)a, +2ft(II + .F)]y + 4(A 2 + AY)(a2 + a, 2) − (II! + Ftf − (II + F? =

$0, thatis, 4(K*+K?-H2)x-4(\#2+\#?)a+2fl(nx+4(K>+K*-H2)4(K*-$

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16. The centres are in a line at right angles to - 2Kx - ZK^{j}+ $<B>+Imn=0, saytheequationofthislineisK, x-Ky+^{=}$ $0; thenweoughttohaveJJ1 - thatis, Thisshouldagreewith2(K * +K*, (K.ft-K\&)-(KJl,+Kn)l-(KlGl+KG)l+2(K2+K?-H2)¥=0, viz.weoughttohave2(K*+KS)K,(a-0)-K(a,-\&)-K,(F,-G,)+K(F-G)O = 0.ThiscanbetrueonlyifKI(OL - - - ft - - - K(al - - - /9j)isamultipleofH; and, infact, = (a-Theequationtobeverifiedthusis-2n(K^{K-}K^{F},-G,)-K(F-G)=0, andhereF-G=(@-y)mn-(7-a)^{+}(/+m-2??)^{T}=a^{?}i+|Swi?i-7(J+w)n-3nK=(n-3??)K=-2nK.Hence-K(F-G)=andsimilarlyandthustheequationisverified.$

17. Writing for shortness #7i-&7> 7< i-7ia' <*(31-ccl/3 = X, Y, Z, and therefore H = X+F+$^{;}wehave2(Ar2+K*)(K\&-Xa,)-(KF+K,F,O=Kwm(/3-7)(Z-F-^{)}+n|(7-<)(-X+F-Z)+lm(a-p)(-X-Y+Z)+K,$

To verify this, observe that the left-hand side is 2 (K- + K*) (mZ- nY) - (£ - 7) rnnK + (& - 71) mnK, + (m - n) (K* + K*) H, or putting herein X + Y + Z for O, this is (K* + K?) (n - m) X + IY-IZ- mn |(/3 - 7) K + ((3, - 7l) K$^{(}X+Y+Z), whichisthus = mn(/8-7)Jr+(\&-7l)Jei < X - Y - Z) + nl(7-a)K+(7l-a,)tfa(-X+Y-Z)+Im(a-0)K+(a1-ft)^{(}-X-Y+Z).Theequationtobeverifiedthusbecomes(K*$

919] ON THE PROBLEM OF TACTIONS. 165 This equation determines <£>, and thus the equation of the line of centres is 2 (K* + K*- $^{(}K, x-Ki/)+(KU+K.U,)ft-<l>=0.$

18. This line meets - ZKx - '2K,y + @ + Imn = 0, in the mid-point of the chord Z&. We thus have for the coordinates a, ij of this mid-point 2 (K2 + K* - ft2) (K* + K*) x + K, (KU + KJIJ ft - $> - K (K< - + K? - ft2) (0 + Imn) = 0, 2 (K2 + Kf - ft2) (K* + K$^{)}y-K(KU+KJIJft-3>]-K,(K-+K?-ft2)(0+Imn)=0.Theperpendiculard$ $(a2+O[K(A-7l)-K^{f}i-7)+(/32+A2)K(7l-ai)-K,(7-a)=ft[I(a2+ax2)+m(j3'2+A2)+ > (72+7i2) = ft,-^{F} = mnK(\&-7l)-K^{p}-7)]=Zwwn.Thus2(\#/+^{/}O=2222(^{2}+^{2})(K*+K\&)-ft2($

19. Hence also 23lft)' + (U, + F, - 2aft)2 - ©+ < w> - 2 (/To

ON THE PROBLEM OF TACTIONS. [919 We have K2 + K- - ft2 = (la + ra/3 + n but and hence whence 4- 70, - - a p< = (7 - a)> + (7l A2 = (a — /3)2 + (< ! or forming the like values of — g2 (f- — g* + A2) and — h2 (/2 + g- — h2) and adding, we have A =/4 + g* + h* - 2#2A2 - 2A2/2 - 2/y = 2(7- a) (a - /3) (A - 7i)2 + 2 (71 - < ,) K -A)(/S-7)2 + 2 (a - /S) (/8 - 7) (71 - < 02 + 2 (a, - A) $(^-7l)(7-a)2+2(0-7)(7-a)(a,-A)2+2(ft-7l)(7l-ai)(a-/S)2 = -4<!(/3-7)+A(7-a)+71(<-/3)2 = -4H2.Butmnf'2 + nig2 + Imh2 = mn/32+/3f+72+7i2-2(7+/S^) + nl[y2 + 7^+a2 + oti2 - - - 2(72 + 7i < *i) + Im[a? + a,2 + /32 + /Sf - 2(a/3 + a^),To/OiO O//Ooi/")O O/OiO = -I - (a - +aj-) - m2(/3 - +ft1) - n - (r + itf) - - - 2mnThusandthereforeButandthushence4(A'2 + AY) = -4(??/2 + \%2 + Imh2, -4H2 = A,4(A2 + A/ - -JQ2) = A - 4(mnf2 + nig2 + Imh2).(= 4(mnf2 + nig2 + Imh2), 6 = - - -.A + 4(mnf - +nig2 + Imh-), 4(A2 + K?) - -(*, 4(K2 + A7 - H2) - -(5, A\ 2 + A^-4(5(A"2 + Aj2 - fl2)2\ gr'$

919] ON THE PROBLEM OF TACTIONS. 167 and we have - + F+ 2a10)2 + (U, + F,- 2afl)2 - 0 + Imn -

20. This should agree with the expression in No. 7, that is, we ought to have (H + F + 2ain)2 + (Hj + $^-2aH)2 - +Zmn - 2(\# < +\#iai)2 = (/2 - Z2)(0s - TO')(A2 - n2), andthisbreaksupintotheequations2F(II+20^)+2F,(U,-2aH) - - 2(Ka\_2^whichmaybeseparatelyverified.$

21. In fact, we have n + 2aifl = (/3 - 7) (a2 + ar) + (7 - a) (/32 + &2) + (a - /3) (72 + 7l2) + 2a, - (13 - 7) < i (7 - a) &-(< -£) 71, = (£ - 7) (a2 - a:2) + (7 - <) #2 - < i2 + (< i - A)2 + (< -£) J72 < is + (71 < i)2, or, putting for shortness £-7. 7 < > a-$^=X, p, v\ A\ 7i > 7i- < i >< i - A = Xi, A * i,^i, (wherethelettersX, /i, y, Xl5ycia, vlhaveameaningdifferentfromthatassigned$ $2ot!n = - - -XyLtf + /AI/j2 + V/J.i2, andsimilarlyAlso/2, < 72, A2 = X2andthefirstequationthusbecomes(-X/AI/ - I - pv? + V^) + (-X^l/j + yLtjV2 + V^Y = (X2 + Xj2)(ft2 + /Lij2)(l/2 + vf), orifontheleft - handwewriteX = - - -^- - -'Xj = - - -^- - -i/j, thisis[V(whichiswhichisright.$

22. For the second equation, we use the values Ul - 2a O = i/! (fjb- + fij3) + fr (v- + i/j2) ; and the equation to be verified thus is + 2F, [Vl (p? + tf) + $^{(}S{+}v{*},-m2(i/2{+}i/,2)(X2{+}V)-[@-2(Ka{+}K^{)}Y-n*(X2+X,2)(^{+}/V); wehaveF=mn\ +(m-n)(la+mfi+ny)= mn\ +\ (m\ -\ -\ -\ ri)(-\ -\ -mv\ +\ n//,), thatis, F\ = -m*v-n\ >; andsimilarlyF1\ =\ ---m'ivl---n2{\cdot}Also - 2(Ka.\ +\ K\&)\ =\ l(a\ -\ +ai2)\ +\ m(/32\ +\ A2)\ +\ n(r\ +\ 7l2)\ - 2Z(a2+ai2)+m(a/3+a^{)}+n=-I(a2{+}<r)+m(/3--f/3X2-2a/3-2a1/S n(/j?\ +\ /ij2).Theleft\ -\ handsideis\ -\ 2(m\ -\ v\ + n\ >)[v(p?\ +\ tf\ +\ p(v\ *\ +vi\ >)\ -\ 2(mX{+}?i\ > 0^{(}yu,2+^{!}2)+^{(\,2}{+}^{2})\ -\ |m(i;2{+}^{2}){+}??(/Li2\ +^{1}\ 2)2, whichis\ = m2{-}2(V{+}^{2})(^{2}{+}/uf)-2(fjiv+^{i}/O(^{2}{+}^{2})-(*2{+}^{2})2\ +\ n"\ - -2(yu^{+}filVl)(v?+AH2)-2(^{2}{+}fij2)(z^{2}+i/!2)-(^{+}Atx2)2\ + (-P+m*{+}n-)(>2+tf)(v*{+}v*\ ---in-(V+vf(/x2+2p> v+v-+yu,j2+2/ZjZ_1+Vi)-n-(^{+}/i!2)(^{4}{-}fyv+v-+tf+ 2fj.lVl4^{2}), whichisequaltotheright-handside.$

23. For the third equation we have as above F = — m-v — n-fji, F1 = — m^{*-} $--n2^{j}, @-2(Ka+A>0=in(v-+i/,2)+n($

919] ON THE PROBLEM OF T ACTIONS. 169 and the equation thus is (m?v + w> 2 + (wVj + n3/*!)8 - 2lmn m (v2 + vfl + n (>2 + p*) = (X2 + Xi2) w2n2 + (p? + p*) nH2 + (v2 + v,2) I2m2 ; here the left-hand side is 2n) (V + vf) + m?n* (X2 + Xj2 - v? - p? - y2 - v2), which is = (X2 + Xi2) m2n2 + (p? + tf) n2 (n2 - 2lm - m2) + (v2 + i/j1) m2 (m2 - 2ln - n2), which is equal to the right-hand side.

24. The fourth equation is the identity — Vm?nz = — I2m2n*, and the whole equation is thus verified: viz. the analytical solution leads to the expression = - 4 /2 - (b - c)2 g* - (c - a)2 [h* - (a - b)2 @, obtained by an independent process in No. 7 for the squared distance of the two centres. C. XIJI.

1. THE equation xl + iy-± — </> (x + iy, where <£ is in general an imaginary function (that is, a function with imaginary coefficients), and where (x, y), (x1; y±) are rect angular coordinates, determines xlt yl! each of them as a function of (x, y), and hence, first eliminating y, we have a set of curves depending on the parameter x, and secondly eliminating x, we have a set of curves depending on the parameter y ; we have thus two sets of curves, or say trajectories, which are orthomorphoses of the two sets of lines x = const, and y = const, respectively. They thus cut everywhere at right angles, and are moreover such that, giving to the parameters values at equal infinitesimal intervals (the same for x and y respectively), they form a double system of infinitesimal squares ; say that we have two systems of square trajectories S and T. We may assume at pleasure a trajectory 8, and also the consecutive trajectory $’; but it is at once seen geometrically that the entire system of trajectories S and T is then completely determined. For at any point t of S drawing a normal to meet S’, and on 8 the element tt’ equal to the normal distance at t, then at t’ drawing a normal to meet S’, and on 8 the element t’t" equal to the normal distance at t’, and so on, we divide the strip between 8 and S’ into infinitesimal squares: at the successive points of S’ drawing normals, and on these measuring off distances equal to the successive elements of S’, we construct a new curve S", at the same time dividing the strip between S’ and 8" into infinitesimal squares ; and proceeding in this manner, we obtain the series of curves S, S’, S", 8’", &c., and at the same time the series of curves T, T’, T", T’", &c., proceeding from the points t, t’, t", t’", &c., respectively, and forming with the first set of curves the double system of infinitesimal squares.

920. On Orthomorphosis.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xxv. (1891), pp. 203–226.]

2. But to translate this into Analysis, and to obtain the equations of the two sets of curves, we proceed as follows*. Suppose that the curve S is given in the form Xi=p, Vi = q, where p, q are given real functions of a variable parameter 0 : we obtain a series of square trajectories S and T, one of the first set being the given curve S, by forming from these equations the equation Xi + iyi=p + iq, and writing therein 6 = x + iy. In fact, for the value y = 0, this equation becomes -EJ = p y1 = q where p, q are now the same functions of x which they were originally of 6, and the elimination of x leads therefore to the equation of the given curve S. More generally, we may take 6 =f (w), an arbitrary real function of w, and then assume w = x + iy : in this case, for y = 0, we have as before x1 = p. yl = q, where p and q are now the same functions of f(x) which they originally were of 0, and the elimination of x from these equations thus leads as before to the given curve S. We have next to determine f(w) in suchwise that the curve corresponding to an infinitesimal real value e of y shall be the given consecutive curve S'. We assume that this curve S' is given by the equations as1=p+yP, y1 = where p, q are as before and P, Q are given real functions of 6; 7 is a real infinitesimal. The expression for w as a function of 6 is then determined by the condition (p'2+qf2)dd V —' where p', q are the derived functions of p, q in regard to 6: C is a real constant, the value of which may be assumed at pleasure. Say we have _/ J where the integral may be regarded as containing a real or imaginary constant of integration #0 + iy0 ; but this being so, we ultimately have w + oc0+ iy0 = x + iy, or w = (x- x0) + i (y — y0), which is the same thing as w = x + iy.

3. We may, in the expressions for $x_1, y_1, substitute for 0 any other real value 61 whatever$ pl + $yP1, yi = qi + yQ_1, values which give *$

The investigation is taken from the memoir, Beltrami, "Delle variabili complesse sopra...

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the proof consists in showing that there exists for 6l a value, differing infinitesimally from 0 and such as to reduce this equation to x + i = + i + (' + i

'1⁻ − − − thevalueassumedbyp + iqonsubstitutingforwthevaluew + −⁽ forthisbeingso, itisobviousthatwehaveforS′thecurveintheseriesofcurvesSwhic x + iy.Thevalueof6lisforthisgives, p′P + q′Q, p′P + q′Qp >= p − w" − 0r + −p− > fc − s − wyrj^r; andsince7isinfinitesimal, wemayinthetermsyPland7^o f^a ndylwriteO− L = 0, andsoreducethesetermstoyPandyQrespectively : wethushave, p′P + q′Q^p, p′Q q′Pxl = p1 + yPl = p − 7/rr + 7⁼P − W − ∗ − ∗'p′Q − q′ = q − yqorfor⁻⁻ − −V1substitutingitsvalue − −j − −−, thesevaluesarep′2 + q′2Cdw_7, dO_y, M_whichgivethebefore−mentionedequation, , iyd0Xi+ ly − i − − − p + 10 + (p + w) − TJ − j − −−, Gdwandtheproofisthuscompleted.ThefollowingtwoexamplesaregivenbyBeltr

4. First, let the curves S, S' be given confocal ellipses, x2 y 2 x2 y 2 i i y i i ⁱy ∗ isj2A2yr§2___V/⁷//)⁻ − −2/⁷7Herep = acos0, q = bsin0, whencep′ ∗ +q′∗ = a2sin20 + 62cos20; P = −|cos0, Q = −|sin0, whencep′Q − q′P = (a2sin20 + b2cos20),

and consequently Civ = I — T dd = —: B. or taking C = — =- , then w = 0, J ab ab ab and for the two sets of curves we have x + iy — a cos (x + iy) -f ib sin (x + iy), which are two sets of confocal conies. In verification, write cos x = X, sin x = X', whence X2 + X'2 = 1, cos iy = Y, sin iy = iY', „ Y2 - Y'2 = 1, and then that is, x,= X (aY-bY'), y, = X'(bY-aY'), /Y* 2 A i 2 -4-2 ipM = a2 — b2, which are the curves T, where observe that so that, putting (aY— bY')2 = a? + h, we have and thus the curves S are the confocal ellipses a? y- o? + h b2 + h ' ' and similarly, since X2 + X'2 = 1, putting (a2 - b2) X2 = a2 + k, we have (a2 - b2) X'2 = a2 - b2 - (a2 + k), =-b2- k, and thus the curves T are the confocal hyperbolas x2 y2 TTT + is , 7 = !> -k<a2>&. a2 + K b* + k

5. Next, let the curve $ be a circle and the curve S' an interior non-concentric circle. It is convenient to take the circles, such that they have for their common chord the line yl = 0 and each meets this chord in the points xl = + i. This being so, the equations of the two circles are x + y — tyi cosec /& = — 1, x + y — 2ya cosec /x, (1 — y cos2 p,) = — 1,

and we have p = cot p. cos 0, q= cot /t* sin 0 + cosec /a, P = — cot fj, cos 6, Q = — cot yu, sin 6 — cos2 //, cosec 6, giving '2+g'2=cot2fj,,p'Q−−−q'P=co t2 /" (! + cos /* sin #), , .f n , 7 =

Caw = - - : — -, , or if C = -. - , then aw = — /I) V'A A1- >) Wi-iV/ii UVW ' , /I 1 + cos //, sin 0 sm yu, 1 + cos /a sin 6 The integral of this may be written tan (w − − − %iyo) = cot/utan(iTT + 0); andifwehereassumefortheconstantofintegrationy0avaluesuchthattani0 = itan − u, , wefindforwtheequationtan w = cot/ae f l + 1cosec/LA, wehavew = x + iy, andx + ∗2/i = p + iq = cotyu, eie + 1cosecyu, , hence#1+ < /i = tan$(x + iy); orwriting, aswemaydo, 2#, 2yinplaceof < , yrespectively, sayforthetwosetsofcurves8andT.Writingtanx = X, taniy = iT, thisgivesX + iY _(X + iY)(1 + iXY)thatis, Z(l − F2) 1+Z2F2andthereforer2I7/2 1 −t−2/1andthenceforthecurvesSandTrespectively.ThevalueF=cotpgi
1, and those of the other set through the antipoints x1= ± i, yl = 0.

6. A different solution is given in the Dissertation referred to below*: I reproduce this, giving the proof in a somewhat altered form. The equations of the curve S and of the consecutive curve S' are taken to be </> O; y) = 0, £ O> y)-2u r (as, y, = 0; intheinvestigation, weintroducetheconjugatevariablesz = x + iy, z = x − − − iy, andwrite < f > , i/rtodenotethevaluesassumedbythefunctions(f > (x, y), −fy(x, y)whenx, yarereplacedthereinbytheirvaluesintermsofz, z, viz.x = (z+z), y = − − −; (z − − − z).Butinthefirstinstancewemay, withoutattendingtothem
/ >, − a sdenotingeachofthemagivenfunctionofthetwocoordinatesz, z.Supposenow
−, _whereinthefirstintegral, afterthecalculationof. −
−−, zisexpressedasafunctionCLZofzbymeansoftheassumedrelation <
/ >= 0, andinthesecondintegralafterthecalculationof, _, zisexpressedasafunctio
0; andwhereineachintegraltheconstantisproperlydetermined, aspresentlyappearin
wwheretheexpressionontheright − handsideis = 0, ifonly < = 0; thatis, < being = 0, wehavef(z)dz + fi(z)dz = Q andconsequentlyf(z) + /i(z) = const., viz.thislastequation, existingasaconsequenceoftheequation < = 0, canbenothingelsethanadifferentformoftheequation < = 0.Wemay, byaproperdeterminationoftheconstantsofintegrationofthetwointegral
mentionedequationtobe = 0; wethushavetherelationf(z) + /(z) = 0, equivalenttotherelation < = 0.Writingnow2 + 82, z + 8zinplaceofz, zrespectively, where8z, 8zarearbitraryinfinitesimalincrements, we
f > (z+Sz, z+8z) = 0, ∗Meyer, UeberdievongeradeLinienundvonKegelschnitteng
InauguralDissertationderUniversitatzuGottingen; 4", Zurich, 1879.Iquotethetitl
Platesi.tov.exhibitcaseswherethecurvesofatleastoneofthesystemsareconies, andP

that is, /(*) + /' (z) Sz +/, (z) +// (z) Sz = 0, equivalent to , d(p £ d<p _ u> H — f— 0,2 H — T^o = 0', a2a^{a}ndifwehereassume&z = −u + ft(z), thenwehaveequivalenttothatis, invirtueofequivalentto *' (v fl < yTiv n7ZiJ(jZJi)M.., ..f(Z) = −T − r^{i}j− >' '−v/rdz"ifdz=0.Thereisthedifficultythatthelast−mentionedvaluesoff(z),fi(z)donotsubsistabsolutely,bu

the consecutive curve S' belonging to a real infinitesimal value u = u, are the given curves $^($x, 2/) = 0and$(%, y) − − − 2uty(x, y) = 0, respectively.

7. For instance, let the curves 8 and S' be the circles xz + f - 1 = 0, x2 + f - 1 - 2wa? = 0 ; here i|r dz z + z I z2 + l' z + - z whence, adding ^{T}Tfortheconstantofintegration, − − − −^{d} z = 2tan − 1z + ivr. −^{d} zThetwosetsofcurvesarethusgivenbyu + iv = tan"1z = tan − 1(x + iy), 2tan − 1(x + iy) = u − −−π + iv, that is, x + iy = tan (^{u} − − −^{i} r + iv viz.here, iftan(!w − − − TT) = a, tanvo = i@, thatis, a(l − /32)/vi − − − −v_ ∗ 7'givingo < 2 + /3sandthenceforthetwosystemsofcurves.Thefirstofthese, substitutingforaitsval 7/2 − 1 − 2xtanu = 0, andthusthecurves, foru = Qanduinfinitesimal, arex2 + y − −1 = 0andx − +y − −I − 2ux = 0, astheyshouldbe.

8. Reverting to the first-mentioned solution, the equation of the circle x* + y* = l is satisfied by $^=$cos0, yi = sin0, andwehavethereforexl + iy$^=$eie, andhencebywhatprecedes, writingforconvenienceX + iY insteadofas + iy, wehave

((f> being any real function) for a set of curves S and T, or say Sl and Tlt wherein one of the curves S1 is the circle x? + y? = l; in fact, writing F= 0, we have xl + iyl = ei<i>X, and thence xl-iyl = e $^($X], andconsequentlyxf + y? − I = 0foroneofthe$!curves.Butsimilarly, if −^{b} eanyrealfunction, thenwehavex + iy = e$^($X + iY^{f}orasetofcurvesSandT, whereinoneoftheScurvesisthecirclexz + y ∗ −1 = 0, andthusthetwoequationsxl + iyl = ei ∗ (X + iY), x + iy = &∗(X+iY)establishacorrespondencebetweenthetwocircumferencesx?+ y?−1 = 0andx2+y2−1 = 0.ToeliminatetheX, Y, wemaywriteity(X+ iY) = log(x + iy), orsay − T/T(X + iY) = ilog(x + iy); then(f)(X + iY)isarealfunctionof −$^($

$X + iY), sayitis = /-^{(}X+iY), anditisthus = filog(x+iy), orwehaveastheformulaforthecorrespondenceinquestion.Inverifica$
$y * -I = 0, thatis, (u + iy)(x - iy) = l, wehavelog(x - iy) = -log(x + iy), andthelastequationisXl - iyl = e\ VVlog(x + i! > n, sothatwehavex? + yl * - - -I = Q; andthusthetwocircumferencesx - +y2 - - - 1 = 0and\% + y - - - 1 = 0correspondtoeachother.$

9. We may write down d priori a formula which must be equivalent to the foregoing, viz. if <f> (x + iy) be a real or imaginary function of x + iy, and <£ be the conjugate function (obtained by changing the sign of i in the coefficients), the formula is d> (x + iy) $^{+-}\_\ Jrinfact, herechangingthesignofi, wehave < f > (x-i^{>}---whichifx2+y----1 = 0, thatis, (x+iy)(x---iy) = 1, becomesx+$

and the two equations give x? + y* - 1 = 0 ; the two circumferences thus correspond to each other. It is to be noticed that, if the numerator function contain a factor (x + iy)m, say if <£ (x + iy) = (as + iy)m $^{r}(x+iy\ thenthedenominatorwillbe/v---T/1 > (x+iy)\ mT/T - - - A + lyandtheformulathusbecomes^{+}iyi = -v/r(-rx + whichisthusnotreallymoregeneralthantheoriginalform.Irecurfurtherontothistra$

10. The curve 8 may be taken to be a closed curve enclosing a singly connected area F (say such a curve is a Contour), and the curve S' a like curve lying wholly inside 8, and such that the normal distance between the two curves is everywhere infinitesimal. It is not in general the case, but it may happen that the successive curves S", S'", &c. will be all of them like curves (each enclosed in the preceding and enclosing the succeeding one), of continually diminishing area, and continually approximating to a point: the curves T will then all of them pass through this point and may be called Radials. We may say that the area F is then square wise contractible, or contracted, into a point.

11. In particular, a circle is squarewise contractible into a point, viz. if the point be the centre, then the curves 8 are concentric circles, and the curves T are radii; as is known and will appear, the circle is also contractible into any interior eccentric point whatever. It may be remarked that (the squares being infinitesimal) the numbers of the contours and radials are each of them infinite, but further the number of contours is infinitely great in comparison with that of the radials. Thus in the case just referred to, if to construct a figure we imagine the circumference of the circle divided into any large number n of equal parts, the radials will be the n radii drawn to the points of division: taking the radius to be =1, and writing for shortness a= — , the sides of the successive squares along any radius will be a, a(l-a), a(l-a)2, ..., and we require an infinite number of these to make up the entire radius; 1 =a 1 + (1 -a) +(1 - <)* + ... ; viz.' the number of radials being any large number n whatever, the number of contours will be actually infinite.

12. We may compare the circle as thus contracted into its centre, or rather the quadrant of a circle, with an infinite strip of finite breadth divided into squares by two sets of parallel orthotomic lines: here if xlt y, refer to the positive quadrant of the circle xf + y* = and xt y to the infinite strip x = 0 to - oo , y=Q to ITT, we have x + iy = log (xl + iy *thatis,x=logV0>,2+2/j2),y=tan−1;xlthesuccessiveconcentricquadrantscorrespondtoequallinesparall*

13. Considering the contour S as given, then for a consecutive interior contour S' assumed at pleasure, the area F does not contract into a point: but we have the theorem that the consecutive contour S' can be found, and that in one way only, such that the area F shall contract into a given interior point. To do this is, in effect, to make the given area to correspond say to the circle x? + y? — 1 = 0, the contour S and the successive interior contours to the circumference of the circle and to the circumferences of the successive interior concentric circles, and finally the given interior point of F to the centre xl = 0, 2/1 = 0 of the circle : and thus the theorem is identical with Riemann's theorem "It is possible and that in one way only to make a given singly connected area F correspond to a circle, in such wise that a given interior point of F shall correspond to the centre of the circle, and that a given boundary point of F shall correspond to a given point on the circum ference of the circle." The last clause as to the given boundary point of F is necessary in order to make the correspondence a completely definite one, for without the clause it would obviously be allowable to give to the circle an arbitrary rotation about its centre.

14. The proof depends on the following lemma*. For the given area F, it is possible to find, and that in one way only, a real function £ of the coordinates (x, y) satisfying the following conditions : (1) £ is throughout the area finite and continuous, except only that in the neighbourhood of a given point thereof, taken to be the point x = 0, y = 0, it is = log VO2 + 2/2)- (2) At the boundary of the area f is = 0. (3) Throughout the area £ satisfies the partial differential equation In fact, if £ be thus determined, it follows from (3) that we have d%j $^{d}\%i - - -$ $dx +^{d}$ $ydydx'anexactdifferential, whenceputtingthis =$ $drj, ordetermining77bythequadrature{-}T$ $dydxwehavedt]\_d\%dt]\_d\%dxdy'dydx'an$ $irj = afunctionofx{+}iij.{*}Thisproofistakenfromthepaper, Christoffel, "Sulprobl$ $- -$ 103.

Writing then xl + iy1 = $e^{+}lr$ $>=$ $ty(x + iy) suppose, wehaveanorthomorphosisoftheareaintothecirclex^{+}yf- - - 1 = 0, viz. foranypoint(as, y)oftheboundary = 0, andconsequentlyxl + iyl = eir >, thencexl - - - iyl = e\ ir >, andthereforex? + y * - - -1 = 0, orthepointcorrespondstoapointonthecircumferenceofthecircle. Also, forapointa$ $0, y = 0, wehavethatis, foras = 0, y - - - 0wehavexl = 0, yl = 0, orthegivenpointofFcorrespondstothecentreofthecircle. Somefurtherconsiderat$ $yf = c2, ascontinuouslyincreasesfrom0to1, therecorrespondclosedcurvessurroun$ $y12 = l; butIabstainfromafurtherdiscussionofthequestion.$

15. Keverting to the lemma, this in the first place asserts the existence of a function £ satisfying the prescribed conditions, and next that the function is completely determinate, or say that there is but one such function. As to the latter point, suppose that there is a second function £ satisfying the same conditions, then throughout the area f — £1 is finite and continuous ; by general principles in the theory of functions, this implies that the function has throughout the area a constant value, and since this value at the boundary is = 0, it must be always = 0, that is, f - £ = 0, or f j = f . As to the former point, it is to be remarked that we obtain = ° as tne general condition for a minimum value of the double integral I H (-T- + (-r- dx dy ; if then we assume that there exists a function £ = 0 at the jj dxi/dx/ dxi/dy/) boundary of the area, and finite and continuous throughout the area, except only that in the neighbourhood of a given point thereof, say #=0, y = 0, it becomes = log V(< 2 + 2/2), and that for all the values which satisfy these conditions it is such that the above-mentioned double integral taken over the given area shall be a minimum, it follows that there exists a function £ satisfying the conditions of the lemma.

16. As a simple illustration, suppose that the area F is that of the circle a? + y* = l) and that the excepted point is the point a? = 0, y = 0, the centre of the circle. We have here a function £ satisfying the conditions of the lemma, viz. this is £ = log VO2 + 2/2) ; the resulting value of 77 is 77 = tan-1'$andwehavethen\% + iijscafunctionofsc + iy, viz.thisis = log(as + iy).Hence + i* = x + iy, andtheresultingorthomorphosisofthetwocirclesistheidenticalonexi + iyi = x + iy, viz.wehave * i = *, yi = y-$

17. The contraction of a circle to a given eccentric point has, in fact, been exhibited in the foregoing example in No. 5, but I resume the question from a different point of view. Writing for shortness Z1 = x1 + iy1, z = x + iy; using also a bar for the conjugate function, z = x — iy, and so in other cases a = a + sd, a = a — &i, @ = /> + hi, &c., the general formula of (7) may be written in the form z — a z — /3 1 - az ' 1 - $z " viz. we thus have a transformation between the circumferences a?- + y2 — 1 = 0, x + y — 1 = 0. In fact, repeating the verification, the change of sign of i gives z — a z — z, = l-az' I-3z" Here the product of the a-factors is zz — (az + az) + act 1 — (az + az) + aazz ' which, if zz = 1, becomes = 1 ; the same property exists as to the /3-factors, &c., and hence putting zz = 1, we have z$^=$1, $thatis, thetwocircumferencescorrespondtoeachother.Thecorrespondencewoul$ $y ---1---0aboutitscentre, andthereisnorealgainofgenerality.$

18. The most simple case is 1 — az' I assume here that a1 = a + sd, is an interior point of the circle or- + if — 1 = 0 (viz. < 2 + a2<l); hence z = a gives $^=$0, $viz.tothepointx = a, y = \&, withinthecirclex2 + y ----1 = 0, therecorrespondsthecentreofthecirclex-? + y - c --- 1 = 0; andwehaveinthisequationthetheoryofthesquarewisecontractionofthecircletothe$ $a, y = a.Thereisnolossofgeneralityinassuminga = 0; doingthistheformulabecomesz - - - a * 1 =$ T $az'wherea2<1, oriftofixtheideasabetakentobepositive, thena<1.$

19. To compare with a former investigation, writing £ + iy = log ($(^{+}iy$'$wehave.fyay + *taui.- < +r -$ $andconsequentlywhichis = 0attheboundarya2 + y2 - 1 =$ $0, andisfiniteandcontinuousthroughouttheareaexceptintheneighbourhoodofthepo$ $a, y = 0, forwhichitis$

= los1 -,(x — a2 + y- ; moreover, £ satisfies the partial differential $^{r+r} = 0, andO'lA'' * -'-)"3/V/v >$ $2fi11startingfromthegivenvalueofweshouldobtaintheforegoingvalueof77, andth$ $ii/i = e^{+}l7), thatis, z - - - a > or^{i} - - - i - -- >$ $i - - - i - - > 1 - - - a\# - - - my1 - - -$ $asastheequationfortheorthomorphosisofthecirclexf4-y?---1 =$ $0intothecircle\$ +^{2} \_i - - - Q5thecentre\#1 = 0, 2/1 =$ $0correspondingtotheeccentricpoint\# = a, y = 0.$

20. In further development of the solution, writing zz — a (z + z) + a2 ZlZl = T$^{a}(z +$ $W + a?zz'thatis, , yi"andthenassumingx^{+}y^{=}i2, wehaveasthecontourcorresponding$ $y2 + a2 - - - 2aae = 2[a2(a? + y2) - - -$ $2ax + 1, thatis, (x - +y2)(1 - a22) - 2a(1 - X2)x +$ $a2 - A, 2 = 0, orsayor, whatisthesamething, a(1 -$ $X2))2X2(l - a2)2'2V7. - \_ - -l - a2X2j(l -$ $a2X2)2'Theequationmayalsobewrittenshowingthatthecontourcirclesallpassthrou$

21. Writing next

4. ,-„ - $(^{a}+^{y})(1 - am\ aiy).^{1} +^{1} -(i - ax) * + - aW'thatis, x =$ $thentotheradiusxl - - - 6yl = Qcorrespondstheradial - a + x(l +$ $a2) - a(x2 + y2) - dy(1 - a2) = 0,$

that is, which is a circle passing through the two real points y = 0, x = a, or - , being the antipoirits of the before-mentioned pair of points.

22. Hence for the contraction of the circle xn- + y–l = 0 to the interior point x = a, ?/=0, calling this point A, and taking A' the image hereof (coordinates # = -, 2/ = OJ, we see that the contour circles are the circles all passing through the antipoints of A, A', and having thus for their common chord the line bisecting A A' at right angles, and that the radial circles are the circles all passing through the two points A, A. In what precedes, the distance A A' is = – a, or say the half distance is AJ = 1 (- — a) but altering the scale so as to make this half distance = 1, and more- 2 a /' over taking the origin at the middle point of AA', the equations of the contour circles and the radial circles assume the forms #2 + 7/2 - 2a# + 1 = 0, and a2 + f - Zfty - 1 = 0, respectively.

23. It is better, interchanging as, y and altering the constants, to write these in the forms tf + y"-(a) as - 1 = 0, V aj viz. as in effect appearing in No. 5, these are the equations derived from x + iy = tan (u + iv) where tan u = a, tan iv = ib.

24. Passing now to the form z — a. z — ft 1 — OLZ 1 — ftz to obtain the most simple instance, I write a = 0, b = < /2, so that the form is we thus obtain

that is, , ._ "*" yi 2 O2 + f) -'2x^2 + $andthence/1" sothatthecircumference^{+}y? - - - 1 = 0correspondstothetwocircumferencesa? + ya\_1 = o, a? - fy2 - 2V2a;+1 = 0.Thesecondoftheseis(x - V2)2 + y1 - 1 = 0; hence, puttingx + - - -^{i}nsteadofx, thetwocirclesare(theformerofthesebeingthecircleoriginallyrepresen$ $y * - - -1 = 0), andthelastequationbecomes$

25. Writing here $^{2} + T/^{-} - -c2 = 0, wehave2(^{2}+7/2)(1 - c2) = 0, thatis, O2 + 2/2)2 - (2c2 + 1)a? - (2c2 - 1)f + 1 = 0, abicircularquarticwhich(orratherabranchwhereof)isthecontourcorrespondingt$ $c+y-f---c" = 0.Forc = l, thecurvebreaksupintothetwocirclesandforc = 0, itbreaksupintothatis, intothetwopointsy = 0, a; = +Wehavetoconsiderthecurvesfory Zvaluesofcbetweenthesetwovaluesc = 1andc = 0.$

26. For any such value of c, the curve consists of two symmetrical ovals, situate within the two circles respectively: we are concerned only with the oval / 1 2 lying within the circle (x + - j + y"- - 1 = 0 ; this is shown in the figure for a value yZ/ of c, slightly inferior to 1, viz. it is an indented oval close approximating to the unshaded portion of the circle, say this is the contour S' consecutive to the circle which is the contour $. As c diminishes and becomes ultimately =0, this oval continually diminishes,

after a time losing the indentation, and approximating more and more to the form of a circle having for its centre the point x = — -/„, 2/ = 0, which is the centre of v^{t}*hecircle,andultimatelyreducingitselftothispoint.Itwillbenoticedthattheconsecut*
whatwedoobtainisacontractionoftheunshadedportionofthecircleintothecentre;thi

27. For the radials, writing as before x-—=. in place of x, that is, z + — in V 2 J 2 place of 2, we have ' e +- $7^{1}Z-$ ^{H}*encex_* $*O2+f\ -I)=Jtf+y2'$*andthustotheradiusxl* $-$ $--$ *dyi* $=0$*therecorrespondsthecurvexO*$2+y--1)-Oy(x2+$ $y2+) = 0,$ *orsay*$(as-6y)(tf+f-$±(x + 0ij) = 0, a circular cubic. In particular, for the line x1 = 0, we have 6 = 0, and the curve is x (a? + y2 — £) = 0, or say a? + y2 — $^{=}0,$ *thehalfcircumferenceofwhichisshowninthefigure* : *fortheliney*$^{=}0,$ *wehave*$0 =$ *oo,andthecurveisy*$(\#2+y2+) =$ 0*thatis*$,2/ = 0.$*Thecurvepassesthroughtheorigin*$\# = 0,y =$ $0,$ *andtheequationofthe*

tangent there is oc + 6y = 0. Moreover it passes through the centre of the circle, x — —= , iy = 0, and the equation of the tangent there is easily found to be We may find where the curve cuts the circle $(^{+}-7^{)}+yz---1=0,$ *viz.thisisa* $*+y2=$ $---x2+$ *andsubstitutingintheequationofthecurve,wefindan* $?/=$ $0,$ *orsayahyperbolawhichbyitsintersectionswiththecircledeterminesthepointsinque*
theotherintersectionservestodeterminebysymmetrytheintersectionofthecircularc
seethefigurewhichexhibitsthepathofthecircularcubicthroughtheunshadedportion,

28. The formula has been considered for the case of an interior point a; the case of an exterior point a might be considered in like manner, but it is obvious that we pass from one to the other, by a transformation - for ;*andditthuseasilyappearsthat,ifz abeanexteriorpoint,weobtainatransformationbe*
$--1 = 0$*aridtheinfiniteareaexteriortothecirclex* $*$ $+y" -$ $---1 = 0.$*Similarly,intheformulaz* $---$ *a.z* $---$ *z,* $=$ $1-az1-$*\$zthecaseconsidered*$(a = a = 0,/3 = \& =$ $*/2)$*hasbeenthatofaninteriorpointa,andanexteriorpoint*$(3.$*Iomitthecaseoftwoexte*
*forz*1*reducedtothatoftwointeriorpoints,butitispropertoconsiderthislastcase.*

29. Considering then the case _ z — a . z — /3 *l-I*.l-5f' 24—2

where a, /3 are interior points, the most simple case is when a =

a, @ = - a, positive and < 1. Here z2 - a? *"l and we have 1 1 - 2as (a? - or say which last form shows that to the circumference x? + y? - 1 = 0, there corresponds the circumference a? + y* - 1 = 0 (and besides this, only the imaginary curve x- + f + 1 = 0). Writing x? + y* = c2, we have the contour (c2a4 - 1) (#2 + ?/2)2 - 2a2 (c2 - 1) (a2 - 2/2) + c2 - a4 = 0, a bicircular quartic. For c = 1, we have as already mentioned the circle a? + y- — 1 = 0; for c less than 1, a curve lying within this circle, diminishing with c, and after a time acquiring on each side of the axis of x an indentation or assuming an hour glass form ; for the value c2 = a4, the equation becomes (a4 + 1) (a? + f)2 - 2 a2 O2 - f) = 0, and the curve is a figure of eight, the two loops enclosing the points y = 0, x = a and x = — a respectively ; and as c2 diminishes to zero, the curve consists of two detached ovals lying within the two loops of the figure of eight, and ultimately reducing themselves to the two points respectively. There is no difficulty in finding the equation of the curves corresponding to a radius xl - 6yl = 0, but the configuration of these curves is at once seen from that of the former system. We may in the present case say that the circle is squarewise contracted to the figure of eight; and then further that each loop of the figure of eight is squarewise contracted to a point; but we do not have a squarewise contraction of the circle to a point.

30. A closed curve or contour may be squarewise contracted not into a point but into a finite line: we see this in the case of a system of confocal ellipses, which gives the contraction of an ellipse into the thin ellipse which is the finite line joining the two foci. There is a like contraction of the circle; this is, in fact, given by the formula due to Schwarz, " Ueber einige Abbildungsaufgaben," Crelle, t. LXX. (1869), pp. 105 — 120 (see p. 115) for the orthomorphosis of the ellipse into a circle : this is = sn — sm where, if a2-62 = l and a=cos-, or, what is the same thing, ib = sin — — — —' then the ellipse — + — — —⁻ —— = 1 is transformed into the circle of + y? —⁼ 0. The contours a2a2 — — — 1 A and trajectories for the ellipse are the confocal ellipses and hyperbolas respectively;

 and for the circle, they are two sets of bicircular quartics, such that the portions within the circle have a configuration resembling that of the confocal ellipses and hyperbolas within the ellipse. To investigate the formulae, it is convenient to introduce the function X + i Y, = sin"1 (x + iy) ; we then have x + iy = sin (X + i F), or say # = sin X cosiY, ty = cos A sin i F, so that to any given value of Y there corresponds the ellipse X" - + cos2i'F -sin'-iF and then 2K/V .- -(X + i. or if s = sn - — , t< i = sn . . = en - - = v(A + VJi d = dn = V(i _ £V), =dn7Tthengivingandtherefore^2 + y, − − −r = 0if1− = A; < i2 = 0, thatis, ifs, = −, orsince = sn − − − − − − −, it− = iK., or7T7Ta; vhencedefiningaasabove, itappearsthattheellipticperiphery− − − +^{2}_ −= 1correspondstothecircumferencex? + y? − j = 0.BytheintroductionofX+iYasabove, Kthecircleandtheellipseareeachcomparedwi iyi = sn−−−(X+iY), orwhatissubstantiallythesamethingbyanequation#j+ iy± = sn (X + iY), is more fully discussed in my paper, Cayley: "On the Binodal Quartic and the graphical representation of the Elliptic Functions," Camb. Phil. Trans., t. xiv. (1889), pp. 484 — 494, [891], and in a paper " On some problems of orthomorphosis," Crelle, t. evil. (1891),

31. The whole theory, and in particular Eiemann's theorem before referred to,

are intimately connected with Cauchy's theorem, "If a function f(z] is holomorphic over a simply connected plane area, and if t denote any point within the area, then /(*)- 2jri] z—t where z denotes x + iy, and the integral is taken in the positive sense along the boundary of the area." See Briot and Bouquet, Theorie des fonctions elliptiques (Paris, 1875), p. 136. Here in order to obtain by means of the theorem the value of the function f(z) for a given point t (= a + ib) within the area, we require to know the values of f(z) for the several points of the boundary: viz. if z refers to a point P on the boundary, and if we represent the value f(z) by a point Pl in a second figure, then these points P: form a closed curve or boundary in this second figure, and we require to know not only the form of this boundary, but also the several points Px thereof which correspond to the several points P of the first-mentioned boundary, or say we require to know the correspondence of the two boundaries: this being known, we have by the theorem the value of f(t), that is, the point a$^+$ib^{w}ithinthesecondarea, whichcorrespondstothepointt = a + ibwithinthefirstarea.The(1, 1)correspondenceofthetwoareasisofcourseimpliedin

921. On Some Problems of Orthomorphosis.

[From Crelle's Journal der Mathem., t. cvii. (1891), pp. 262–277.]

IN the interesting Memoir, Schwarz, "Ueber einige Abbildungsaufgaben," Crelle, t. LXX. (1869), pp. 105 — 120, the author considers the orthomorphic transformation (or, as I call it, the orthomorphosis) of a square into the infinite half-plane, or into a circle, and of a rectangle into the infinite half-plane. It is of course easy to deduce the orthomorphosis of the rectangle into a circle; and then, by giving a proper value to the modulus of the elliptic function involved in the formula, we obtain the orthomorphosis of the square into a circle, this solution (although equivalent thereto) being under a different form from that previously given for the square. But as appears from my paper "On the Binodal Quartic and the Graphical Repre sentation of the Elliptic Functions," Camb. Phil Trans, vol. xiv. (1889), pp. 484—494, [891], there is for the rectangle (and consequently also for the square) an orthomorphosis wherein the boundary of the rectangle or square corresponds, not to the circumference, but to the circumference together with two twice-repeated portions of a diameter. I propose to consider in I. these several transformations: II. relates to the ortho morphosis of a circle into a circle. I.

1. We are concerned with the elliptic function sn for which K' — 2K, and also with the elliptic function snl of the lemniscate form. The modulus in the former case is A? = -|"-1, = (V2-1)2, =3-2V2, or say V&=2-l, -L> V2 + 1. For the lemniscate function snl, the modulus is = i, and if (with Gauss) we write $^{C}T for the value of the complete function K or Fl f then we have ^{O}T = 7T(V2 - - - 1)$,

[921 = K V&, where k has the foregoing special value : I notice the numerical values i-or = 1-311028, k = 3 - 2 V2 = sin 9° 52', k = 1-582548. The relation between the lemnis- cate function snl, and the sn with the foregoing value of k, is easily shown to be /T (i + 1) snl u + i V2 , „ vksnu=- — - 7=-, where U — ' „, l+i = — _w and it may be added that we have V2 l+k dn U = (i - 1) snl u + V2 V k V2 (i- I) snl u + V2 + snl u) (1 — i snl u u. Vl + A; V(l - snl u) (1 + i snl u). D

2. The Schwarzian orthomorphosis of the rectangle into the infinite half-plane is given (Memoir, p. 113) by the formula X1 + iY1 = sn(X + iY), where the modulus is real, positive, and less than unity. Here, see the figures 1 (XY) and 2 (XlYl), Fig. 1 (AT). Y If R the rectangle A BCD, the sides of which are AB = 2K and BC = K', is transformed into the upper infinite half-plane (Yl = +), the four corners of the rectangle corre sponding to the points A, B, C, D on the axis of X, where OB(=OA)=l, OC(=OD) = -r.

3. We can, by a properly determined quasi-inversion (as will be explained), transform the $X^Fj - figureintoanewfigureseefigure3(X2Y2), theinfiniteX - axisbeingtransformedintothecircumferenceofacircle(theradiusofwhichmaybeta$ $1)andtheinfinitehalf-planeintotheareawithinthecircle. ThefourpointsA, B, C, D$ $inversionbeasymmetricalone. willbesituate, AandBsymmetrically, and$

also C and D symmetrically, in regard to the axis OF2: and in the case of the before-mentioned modulus k = 3 - 2 2, for which the rectangle ABCD becomes a Fig. 3* (XaYs) square, the quasi-inversion may be so determined that the points A, B, C, D shall be situate midway (that is, at inclinations ± 45°, ± 135°) in the four quadrants of the circle.

4. But if in figure 1 we take EL=K' and draw FL parallel to OX, then as shown in my memoir above referred to, the foregoing transformation Xl-iY1 = su(X+iY) changes the rectangle OBLF, the sides of which are OB = K and BL — K', into the quadrant OBLF of figure 2, OB = 1 (as already mentioned) and radius OL = -$\cdot HencecompletingtherectangleRSLM, thesidesofwhichareR8 = ZKandSL = K', (viz. thisrectanglediffersonlyinpositionfromthefirst-$ $mentionedrectangleABCD), wehaveanorthomorphosisoftherectangleRSLMintot$ $butsoneverthelessthattotheboundaryoftherectangletherecorrespondsnotthecircum$

5. The Schwarzian orthomorphosis of the square into a circle is given (Memoir, pp. Ill — 113) by the formula $^+iyi = snl(x+iy); viz. here- --seefigures4(xy)and5(\#i2/i)---thesquareABCD, thehalf-$ $diagonalswhereofareeach = CT, correspondstothecircleradius =$ $1offigure5, thefourpointsA, B, C, Dofthecirclebeingthequadrantalpointsasshown$; and it thus appears that Schwarz's lemniscate solution for the square into a circle is the quasi-inversion of his solution for the rectangle into the infinite halt-plane, when by putting k = 3 — 2 A/2 the rectangle is made to be a square. See post Nos. 13 and 14. [* The scale of this figure is double that of figures 1, 2, 4, 6, 7 in this paper.]

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6. The general formula of quasi-inversion whereby the infinite Xj-axis is changed Fig. 4 (xy). Fig. 5 into a circumference$^r adiusunity, isX, +iY.2 = whereMisreal.Infact, writingthisequationintheform - A2\ H * - * 2 == JTfwehave{-}il + Mi(X1 - iYl)' andthenceX.?{+}F22{-}{-}{-}1 = 0, ifonlyFj = 0; thatis, theinfinite^{-}axisistransformedintothecircumferenceX+ F22 - - - 1 = 0.WritingYl = 0, wehave1 + MiX, _2MX, +i(M2Xf - 1)MX, +i\ M2Xi * +1sothattoapairofpoints(\pm \mathrm{X}_1, 0)therecorrespondsapairofpoints(+X2, F2)situate$

we have and consequently M 2 (X? -f 17) -1=0, if only Y2 = 0 ; that is, the circumference Xf + Yi2 — -JTJ- = 0 is transformed into the infinite X2-a,xis. Although for the purposes of the present memoir we require the coefficient M, yet there is no real loss of generality in assuming M =1, and the transformation is best studied under the form l + i(Xl + iYl)m X$^i Yi + i'' seepostNo.11.$

7. As already mentioned, the coordinates (Xlt Fx) and (X2, F2) are connected by an equation of the foregoing form, and in the case of the square (k = 3 - 2 V2 as before), the corresponding values for the points A, B, C, D should be A, Xl + iYl = -t X, + iY2 l l t , 2 , B, X, + iY,= I, X, + iY, = l =, V2 C, Xl + iY,= j, Xz + iY$^{1,} * V2D, Z1{+} > F1 = -, Z.+iF,^{"} -^{-} *.*2ThepropervalueofMisM = */2{-}{-}{-}l, = V^{;}therequiredformulathusisorsayThusforthepointjB, weshouldhave..l- i1 - -- > __,, /, V2./7V2 - t - iVA; =^{-}: -, thatis, vk = -- -- rwhichisright.Andsimilarlyforthepoint0, weshouldhavel_il + _i1V21V2 - t + l - / == i- = - - -, thatis, - - - = . - - - - = u =' __2whichisright.AndsimilarlyforthepointsAandD.25{-}{-}{-}2$

8. We connect the square A BCD of figure 1 with that of figure 4 by the equation in fact, recollecting that);K' = K= , we have for the four points respectively A, X + iY = -K, x + iy = - B, X + iY= K, x + iy = - C, X+iY= K+iK', x + iy = D, X + iY = -K + iK', x + iy = values which satisfy the relation in question.

9. Hence writing X + iY = U, x + iy = u, we have which is the relation in No. 1 between the arguments U, u of the elliptic functions sn, snl. We have X, + iT, = sn U, x, + iy, = snl u ; and consequently Hence, substituting for $^{/}k(X1 + iY1) its value in terms of Xz + iY.2, we find an equation which (multiplying on the left - hand side the numerator and the denominator\_A each by, -] may be written V2/.1+ i, ..i(x, -j-ty,) V21---il+iV2 that is, we have an equation which shows that the figur$
We have thus proved the conclusion stated in No. 5 as to the connexion between the lemniscate solution for the square and the solution for the rectangle.

10. It is convenient to collect here the several equations relating to the orthomorphosis of the square. We have X, + iY1 = sn (X + iY), k = 3 - 2 2 ; xl + iy-i = snl (x + iy) ; which are the equations connecting together the coordinates of the five figures.

11. I examine more in detail the above-mentioned transformation i see the foregoing figures 1 (XY) and 2 (JTjFj), in which we now regard the two circles as having each of them the radius unity; changing the sign of i, the equation gives Y iV _1 li?JJI*yi) -A 2 t-> 2 — ' -p- -rr > Aj — lY 1 — I and we hence find consequently if X-? + Y? + 1 = 0, then also JT22 + F,2 + 1 = 0, or the transformation changes the first of these imaginary circles into the second of them : or say it changes the concentric orthotomic of the circle X? + Fj2 — 1 = 0 into the concentric orthotomic of the circle X.22 + F,2 - 1 = 0. We have moreover Y $^{2L}\_V - XL$±Il l Jr values which give the foregoing expression for X? + F22. But we further obtain JT24- F2— 1 — - F — * and it thus appears that the circumferences X* + Fj2 - 1 + 2/uF, = 0, .Y.2 + F.,2 - 1 - - F, = 0,

correspond to each other. These are circles passing through the pairs of points (Yi = 0, X1 = ±l), (F2 = 0, JT2= + 1) respectively; or imagining them in the same figure, say they are circles which belong each to the series of circles #2 + 7/2 — 1 + 2/3y = 0, and which moreover cut at right angles at the points y = 0, x = + 1. But attending more carefully to the nature of the correspondence, it is to be observed that taking Y positive, and giving to X any positive value from 0 to oo , we have in figure 2 an arc LJM lying wholly within the upper semicircle LFM ; and that, corresponding hereto in figure 3, we have an arc LJM lying wholly within the lower semicircle LOM] and that as in figure 2, the arc LJM lies nearer to the semicircurnference LFM or to the diameter LOM, so in figure 3 the arc LJM lies nearer to the diameter LFM or to the semicircumference LOM. Thus the upper semicircle LFM of figure 2 corresponds to the lower semicircle LOM of figure 3; but so that the semicircumference LFM of the first figure corresponds to the diameter LFM of the second figure; and the diameter LOM of the first figure to the semicircumference LOM of the second figure. And further supposing that Y1 is still positive, but that p has any negative value from - oo to 0, we have in figure 2 an arc in the upper half-plane lying wholly outside the semicircle; and corresponding thereto in figure 3, an arc lying wholly inside the upper semicircle LHM] that is, in figure 2 the infinite space in the upper half-plane outside the semicircle corresponds in figure 3 to the space within the upper half-circle LHM ; the infinity of figure 2 corresponding to the semicircumference LHM of figure 3, and the semicircumference LFM of figure 2 to the diameter LFM of figure 3. And thus in figure 2, the upper half-plane inside and outside the semicircle corresponds in figure 3 to the lower and upper half-circles, that is, to the whole circular area OLHM of figure 3. It is to be observed, moreover, that we have X? + F.. Hh 1 + 2XZ. - 2 (l* + *f t that is, the circles X? + Y* + 1 + 2X2⁼0$andX, 2 + F22 + 1 + 2XX, =$ $0 correspondtoeachother. Imaginingthecirclesasbelongingtothesamefigure, thesea$ $y * +1 - 2a\# = 0 eachpassingthroughthepairofpoints(x = 0, y = \pm$ i) which are the antipoints of the pair (y = 0, < = + !); these circles thus cut at right angles those of the series #2 + y- — 1 + 2/3y = 0. We have, by means of the two series of circles, an easy construction for the correspondence between the figures 2 and 3.

12. In explanation of No. 4, observe that, starting from the equation Xl + iY1 = su(X + iY and writing su X = P, sniY=iQ, we have that is,

and thence *i'+ JV-Y^{y-} $* 1(if^2 + > Vbeput =$ $r2); hence P2(1 - 2Q2r2) = r - -Q Q - (1 - k - P2r-) = r2 -$ $P2. Nowconsideringinfigure1anylineintherectangleparalleltotheaxisofX, thatis,$ $(Kl +^- (l + fc2Q2)r * */!\_ + k * Q > -k * (l +$ $Q2)r21 -^g ivingXltYleachofthemintermsofQ2andr2, = X12 +$ $F12. Theformeroftheseequations, replacingthereinr2byitsvalue, giveseasily(X, 2-$ $I?)2 - ZAXf - 2BYS + 1 = 0, whereviz. wehavethustheequationofthecurve, abicirc$

$'andthencei11iQ = sniK' = - - - =, thatis, 0 = - - - - - -, andthenceA = B = T; theequationofthebicircularvkvk * quarticisviz.thisisthecircleX^{+}Y - c - -j- = 0twicerepeated, andwehavethusthiscircle, fcorratherthehalf - circumferenceLFMoffigure2, correspondingtothelineLMoffigure1.Moresimply$
$-snX + cnXdnX.A/$

and thence 2 , k*JP k' that is, X? + Y? — r = 0 as before. It is easy to see that the points A, 0, B of fc figure 1 correspond to the points A, 0, B of figure 2, and hence that the area of the rectangle AOBLFMA of figure 1 corresponds to that of the semicircle AOBLFMA of figure 2. Returning to the equation X1 + iYl = sn (X + iY), if we write herein successively Yl = K' - /3, sn iY, = iQ, = sn i and Y then we have that is, QiQ<2 = T'- hence for q writing $Q^{o}rQ2, wehaveineachcasethesameKvaluesofAandB, thatis, wehavethesamebicirc$ $perimeterlyingwithinthesemi-circumferenceLFM, andtotheotherofthem(viz.the$ $perimeterlyingwithoutthesemicircumferenceLFM.Itmaybeshowninlikemannert$ $17)2 - 2AX? - 2BY, 2 + i = 0.K$"

13. Similarly in explanation of No. 5, observe that, starting from the equation xl + iyl = snl (x + iy) and writing sn(K=p, sn iy = isny = iq, we have p A/1 - o4 + iq A/1 — p* /YI _1. nft - -* _ * _ _ _ _ * _ = — - that is, _ Wl-o4 ol-> 4 rf - -i _ 5_ 1/ — * _ ± — l-$^{2/1} - pY'andthenceI - p - q * writingx + y =^{-} ar, wehaveenas, , .A/1- > 21-p'2., .p2+q-, sny = - - -.thatis, q = , or02 = - - -, thatis, - - = 1; -l - pa - q2andthustothelinex + y = ix, therecorrespondsthecirclex? + yl" - - - l = 0.Moreprecisely, tothelineCDoffigure4, therecorrespondsthequarter- circumference$

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921. On some Problems of Orthomorphosis.

[From the *Philosophical Transactions*, vol. CLXXXIII (1892), pp. 653–669.]

14. To any line in figure 4, parallel to the axis Ox or to the axis Oy, there corresponds in figure 5 a bicircular quartic of the form $x^2+y^2-2Ax^2-2By^2-1=0$. The investigation is substantially the same as that contained in No. 12, and need not be here given. But it is remarkable that also, to any line of figure 4 parallel to a side of the square (that is, to any line $x \pm y = c$), there corresponds in figure 5 a bicircular quartic of the like form (for the sides of the square, or lines $x \pm y = \pm\sqrt{2}$ of figure 4, this bicircular quartic becomes the twice repeated circle $x^2+y^2-1=0$ of figure 5, which is the result just obtained in No. 13). I investigate this result as follows. Writing, as in No. 13,

$$\operatorname{sn} x = p, \quad \operatorname{sn} iy = i \operatorname{sn} y = iq,$$

we have, as above,

$$x = \frac{p\sqrt{1-q^4}+q\sqrt{1-p^4}}{1-p^2q^2}, \quad y = \frac{p\sqrt{1-q^4}-q\sqrt{1-p^4}}{1-p^2q^2},$$

and thence

$$x_1 = \frac{p\sqrt{1-q^4}+q\sqrt{1-p^4}}{1-p^2q^2},$$

where $x_1 = x+y$ and $y_1 = x-y$.

Now assuming between x and y the relation $x+y=C$, and writing $\operatorname{sn} C = c$, this gives

$$x_1 = \frac{p\sqrt{1-q^4}+q\sqrt{1-p^4}}{1-p^2q^2}.$$

To obtain the required curve, we must between these equations (three independent equations) eliminate p and q. We have

$$x_1+y_1 = \frac{2p\sqrt{1-q^4}}{1-p^2q^2}, \quad x_1-y_1 = \frac{2q\sqrt{1-p^4}}{1-p^2q^2},$$

and consequently, writing for convenience $n = 1-p^2q^2$,

$$nx_1 = p\sqrt{1-q^4}, \quad ny_1 = q\sqrt{1-p^4};$$

these equations may be written

$$p^2 = \frac{x_1^2}{n^2 + x_1^2}, \quad q^2 = \frac{y_1^2}{n^2 + y_1^2},$$

and from these equations obtaining the expressions for p^2 and q^2, and thence the expression for $p^2q^2 = 1 - n$, we find, after some easy reductions,

$$n^2 - n(x_1^2 + y_1^2 - 1) - x_1^2 y_1^2 = 0.$$

But we have

$$x_1^2 + y_1^2 = \frac{2(p^2 + q^2 - 2p^2q^2)}{(1 - p^2q^2)^2},$$

or substituting this value, the equation becomes

$$\left[(x_1^2+y_1^2)^2 - 2c^2(x_1^2+y_1^2) + c^4\right] + (1-c^2)\left[(x_1^2+y_1^2)^2 - c^2(x_1^2+y_1^2)\right] - 4c^2x_1^2y_1^2 = 0,$$

that is,

$$(x_1^2 + y_1^2 - 1)^2 + (1 - c^2)(x_1^2 + y_1^2 - c^2)(x_1^2 - y_1^2) = 0;$$

the required equation. Transforming through an angle of 45° by writing

$$x_2 = \frac{x_1 + y_1}{\sqrt{2}}, \quad y_2 = \frac{x_1 - y_1}{\sqrt{2}}$$

(where observe that the axis of x_2 is the line FH of figure 5), the equation becomes

$$(x_2^2 + y_2^2 - 1)^2 + (1 - c^2)(2x_2^2 - 2y_2^2) = 0,$$

or writing $c = \cos\gamma$ and therefore $1 - c^2 = \sin^2\gamma$, this equation becomes

$$(x_2^2 + y_2^2)^2 - 2(x_2^2 \cos^2\gamma + y_2^2 \sin^2\gamma) + 1 = 0,$$

a curve consisting of two indented ovals situate symmetrically in regard to the axis Fy_2 of figure 5. In fact, writing in the equation $y_2 = 0$, we have for x_2^2 two real positive values; but, writing $x_2 = 0$, we have for y_2^2 two imaginary values. For $\gamma = 0$, the equation becomes

$$(x_2^2 + y_2^2)^2 - 2(x_2^2 + y_2^2) + 1 = 0,$$

that is, we have the circle $x_2^2 + y_2^2 - 1 = 0$ twice repeated. One of the ovals is shown in figure 5; the portion of it lying within the circle agrees with Schwarz's figure, p. 113, turning this round through an angle of 45°.

For the lines $x - y = C$ of figure 4, we have in figure 5 the same system of bicircular quartics turned round through an angle of 90°.

II.

15. I consider the general problem of the orthomorphosis of a circle into a circle: we can, for the transformation of the circumference of the circle $x^2 + y^2 - 1 = 0$ into that of the circle $x_1^2 + y_1^2 - 1 = 0$, find a formula involving an arbitrary function or (what is the same thing) an indefinite number of arbitrary constants. In fact, writing for shortness $z = x + iy$, $z_1 = x_1 + iy_1$, and $\bar{z}$, $\bar{z}_1$ for the conjugate functions $x - iy$, $x_1 - iy_1$; also $\phi(z)$ for a function of z involving in general imaginary coefficients $a + ib$, &c., and $\bar{\phi}(\bar{z})$ for the like function with the conjugate coefficients $a - ib$, &c.; then if we assume

$$z_1 = \frac{\phi(z)}{\bar{\phi}(\bar{z})} \cdot \frac{1}{z^m},$$

where m is any positive or negative integer, this implies

$$\bar{z}_1 = \frac{\bar{\phi}(\bar{z})}{\phi(z)} \cdot \frac{1}{\bar{z}^m};$$

consequently, if $x^2 + y^2 - 1 = 0$, that is, $z\bar{z} = 1$, or $\bar{z} = \frac{1}{z}$, we have

$$z_1\bar{z}_1 = \frac{1}{z^m \bar{z}^m} = 1,$$

or $z_1\bar{z}_1 = 1$, that is, $x_1^2 + y_1^2 - 1 = 0$. [(1)]

In a slightly different form, taking

$$\phi(z) = (z - \alpha)(z - \beta) \ldots,$$

α, β, &c., to denote any imaginary quantities, and $\bar{\alpha}$, $\bar{\beta}$, … the conjugate quantities; assuming

$$\phi(z) = (z - \alpha)(z - \beta) \ldots,$$

and taking m for the number of factors, we have

$$z_1 = \frac{(z - \alpha)(z - \beta) \ldots}{(\bar{z} - \bar{\alpha})(\bar{z} - \bar{\beta}) \ldots} \cdot \frac{1}{z^m},$$

and then (repeating the demonstration) we have

$$z_1 = \frac{(z-\alpha)(z-\beta)\dots}{(\frac{1}{z}-\bar{\alpha})(\frac{1}{z}-\bar{\beta})\dots} \cdot \frac{1}{z^m},$$

which, writing therein $\bar{z} = \frac{1}{z}$, becomes

$$z_1 = \frac{(z-\alpha)(z-\beta)\dots}{(1-\bar{\alpha}z)(1-\bar{\beta}z)\dots},$$

and consequently, if $z\bar{z} = 1$, then also $z_1\bar{z}_1 = 1$ as before.

We may in the expression for z_1 introduce a factor $\frac{A}{\bar{A}}$, or, what is the same thing, a factor A which is such that $A\bar{A} = 1$. In particular, we thus have the solution

$$z_1 = A\frac{z-\alpha}{1-\bar{\alpha}z}.$$

16. This is a solution with three arbitrary constants, viz. A (which may be put $= \cos\lambda + i\sin\lambda$) is a single arbitrary constant, and $\alpha = a + ib$, is two arbitrary constants; and these constants may be so determined that a given point in the interior of the one circle, and a given point on the circumference thereof, shall correspond respectively to a given point in the interior of the other circle and to a given point on the circumference thereof. According to a well-known theorem of Riemann's, any two simply connected areas included within given closed curves respectively may be made to correspond to each other, and that in one way only, under the foregoing conditions as to a pair of interior points and a pair of boundary points: and we have, in what just precedes, the solution of the problem in the case of two equal circles.

17. In the case of any other solution, we thus know that the correspondence between the two circumferences cannot and does not imply a $(1,1)$ correspondence between the areas of the two circles: but it is interesting to enquire what happens. I take a very particular case

$$z_1 = \frac{z(z-2)}{1-2z},$$

and therefore

$$\bar{z}_1 = \frac{\bar{z}(\bar{z}-2)}{1-2\bar{z}},$$

and consequently

$$z_1\bar{z}_1 = \frac{z\bar{z}(z-2)(\bar{z}-2)}{(1-2z)(1-2\bar{z})},$$

that is,

$$x_1^2 + y_1^2 = \frac{(x^2+y^2)\{(x-2)^2+y^2\}}{(1-2x)^2+4y^2},$$

so that, writing $x^2+y^2-1=0$, we have $x_1^2+y_1^2-1=0$, a correspondence of the two circumferences. But to the circumference $x^2+y^2-1=0$ there corresponds not only the circumference $x_1^2+y_1^2-1=0$, but another circumference. In fact, writing $x_1^2+y_1^2=1$, we have

$$(x^2+y^2)\{(x-2)^2+y^2\} = (1-2x)^2+4y^2,$$

that is,

$$(x^2+y^2)^2 - 4x(x^2+y^2) + 4x - 1 = 0,$$

or

$$(x^2 + y^2 - 1)(x^2 + y^2 - 4x + 1) = 0,$$

and there is thus the other circle

$$x^2 + y^2 - 4x + 1 = 0, \quad \text{or say} \quad (x - 2)^2 + y^2 = 3,$$

viz. this is a circle, coordinates of centre $(2, 0)$ and radius $= \sqrt{3}$, cutting the circle $x^2 + y^2 - 1 = 0$ in two real points. Referring to the figures 6 (xy) and 7 ($x_1 y_1$), and observing that $x_1 = 0$, $y_1 = 0$, that is, $z_1 = 0$, gives $z = 0$, or $z = 2$, that is, the points $x = 0$, $y = 0$, and $x = 2$, $y = 0$, we see that to the centre O in figure 7, there correspond in figure 6 the points O, M which are the centres of the two circles.

To any small closed curve, or say any small circle surrounding the point O of figure 7, there correspond in figure 6 small closed curves surrounding the points O, M respectively; and if in figure 7 the radius of the circle continually increases and becomes nearly equal to unity, the closed curves of figure 6 continually increase, changing at the same time their forms, and assume the forms shown by the dotted lines of figure 6. It thus appears that, to the whole area of the circle $x^2 + y^2 - 1 = 0$ of figure 7, there correspond the two lunes ACE and ABD of figure 6; or if we attend only to the area included within the circle $x^2 + y^2 - 1 = 0$ of this figure, then there corresponds not the whole area of this circle, but only the area of the lune ACB: and thus that the assumed relation $z_1 = \frac{z(z-2)}{1-2z}$ establishes, in fact, an ortho-morphosis of the circle $x^2 + y^2 - 1 = 0$ into the lune ACB which lies inside the circle $x^2 + y^2 - 1 = 0$ and outside the circle $(x - 2)^2 + y^2 = 3 = 0$. It may be added that, to the infinite area outside the circle $x^2 + y^2 - 1 = 0$ of figure 7, there correspond in figure 6 first the area of the lens AB common to the two circles, and secondly the area outside the two circles: we have thus an orthomorphosis of the area outside the circle $x_1^2 + y_1^2 - 1 = 0$ into these two areas respectively.

A somewhat more elegant example would have been that of the correspondence

$$z_1 = \frac{z(z - \sqrt{2})}{1 - \sqrt{2}\, z};$$

here, corresponding to the circumference $x^2 + y^2 - 1 = 0$, we have the two equal circumferences $x^2 + y^2 - 1 = 0$, and $(x - \sqrt{2})^2 + y^2 - 1 = 0$: and to the whole area of the circle $x_1^2 + y_1^2 - 1 = 0$, there correspond two equal lunes ACB and ABD.

922. Note on the Lunar Theory.

[From the *Monthly Notices of the Royal Astronomical Society*, vol. LII. (1892), pp. 2–5.]

In the Lunar Theory, in whatever way worked out, the values ultimately obtained for the coordinates r, v, y should of course satisfy identically the equations of motion; and that they do so, is the ultimate verification of the correctness of the results obtained. It can hardly be hoped for that such a verification will ever be made for Delaunay's results; and yet it would seem generally that the labour of such a verification of the results to any extent, while exceeding (and possibly greatly exceeding) that of obtaining these results by any method employed for that purpose, ought still to be, so to speak, a labour of the same order. And one can, moreover, imagine the process of verification having gone on pari passu with the obtainment of the results.

The verification required is, in fact, that the values of r, v, y, obtained in any manner, do satisfy identically the equations of motion; these are

$$\frac{d^2x}{dt^2} + \frac{\mu x}{r^3} = \frac{\partial\Omega}{\partial x}, \quad \frac{d^2y}{dt^2} + \frac{\mu y}{r^3} = \frac{\partial\Omega}{\partial y}, \quad \frac{d^2z}{dt^2} + \frac{\mu z}{r^3} = \frac{\partial\Omega}{\partial z},$$

where Ω is the disturbing function due to the Sun. I say the verification required is that these equations are identically satisfied. This does not require the expressions for x, y, z to be given each in a single series, but only that the forms be such as will allow of their being differentiated; and the comparison of the coefficients of the cosines and sines of the same arguments on the two sides of each equation gives the relations between the constants and the coefficients, which are in fact the equations for the determination of these in the method of the variation of the elements.

2. As regards the coordinate y, the process of verification was, in fact, carried out in the memoir *On the Secular Acceleration of the Moon's Mean Motion*, *Philosophical Transactions*, 1859, [516]; the result being that of the two equations

$$\frac{dy}{dt} = \frac{\partial R}{\partial c'}, \quad \frac{dc'}{dt} + \frac{\partial R}{\partial y} = 0,$$

the second is satisfied identically by the expressions for y and c', while the first gives $\frac{\partial R}{\partial c'}$ equal to the calculated value of $\frac{dy}{dt}$.

3. We have $r = a(1+\rho)$, and we thence obtain r as a function of the time, the form being in Delaunay's theory $r =$ function of $t + h + \varpi + \zeta$, where ζ denotes a sum of cosines with constant coefficients: and thence, by integration, $v = nt + \epsilon +$ function of the same arguments. The verification of the first two equations would thus consist in showing that the expressions for r and v satisfy these equations; and the process would seem to be one which, for any terms taken at pleasure, could be carried out without too great labour.

4. But it would be interesting to see the process carried out, not for terms taken at pleasure, but for the whole series as they present themselves in the actual expressions for the coordinates. I observe that the verification for the coordinates r, v can be made without the explicit expression of v in terms of the time; and that the verification can be made to depend upon the identical equation $r^2\dot{v} = h$, or say $\dot{v} = \dfrac{h}{r^2}$. We in fact have $r = \dfrac{a(1-e^2)}{1+e\cos v}$, and thence r as a function of v, and we have in like manner ρ, v as functions of $t+h+\varpi+\zeta$; hence r, $\dot{v}$, functions of $t+h+\varpi+\zeta$. The value of $\dot{v}$ obtained in this manner is equal to $\dfrac{h}{r^2}$, and this is the required verification.

5. To complete the verification for the coordinates r, v, we substitute the value of $\dot{v}$, and also the value of r, in the first equation of motion, say

$$\ddot{r} - r\dot{v}^2 + \frac{\mu}{r^2} = P;$$

and the required result is that this equation shall be identically satisfied. The expression for r is in effect $r = a(1+\rho)$, where ρ is a sum of cosines; hence $\ddot{r}$ is a sum of cosines, and the expression for $r\dot{v}^2$ is in like manner a sum of cosines: hence the left-hand side is a sum of cosines; and the right-hand side P, inasmuch as P is a function of r, v given as a sum of cosines, is also a sum of cosines. And if the verification is to be made, it would consist in the comparison of the two sets of cosines.

6. The verification should thus ultimately depend on the trigonometrical identities, and the verification of them would in many cases be an interesting and not too laborious exercise. As a very simple instance, we may take the following: in the first approximation we have

$$r = a(1 - e\cos nt), \quad v = nt + 2e\sin nt,$$

and thence

$$\dot{r} = ane\sin nt, \quad \dot{v} = n(1 + 2e\cos nt),$$

and thus the equations become

$$ane^2\cos nt - an^2(1 - e\cos nt)(1 + 4e\cos nt) + \frac{\mu}{a^2}(1 + 2e\cos nt) = P;$$

or, when $e = 0$,

$$-an^2 + \frac{\mu}{a^2} = 0,$$

which is right; and, for the terms in e,

$$ae^2n^2\cos nt + 3ane^2n^2\cos nt + \frac{\mu}{a^2}\cdot 2e\cos nt = P;$$

and thus the equations become

$$\frac{d^2r}{dt^2} - r\left(\frac{dv}{dt}\right)^2 + \frac{\mu}{r^2} = P, \quad \frac{1}{r}\frac{d}{dt}\left(r^2\frac{dv}{dt}\right) = Q,$$

where

$$P = \frac{m'}{a'^2}\left(\frac{r}{a'} - \frac{a'^2r}{r'^3}\cos(v - v')\right), \quad Q = \frac{m'}{a'^2}\left(-\frac{a'^2r}{r'^3}\sin(v - v')\right),$$

and the verification consists in showing that these equations are satisfied identically by the assumed values of r and v.

923. Note on a Hyperdeterminant Identity.

[From the *Messenger of Mathematics*, vol. xxi. (1892), pp. 131, 132.]

The following is in effect a well-known theorem; but I am not sure whether it has been explicitly stated:

If f, g, h are any functions of x, y, and we form the matrix

$$\begin{pmatrix} f & g & h \\ f_1 & g_1 & h_1 \\ f_2 & g_2 & h_2 \end{pmatrix},$$

where the suffixes 1 and 2 denote differentiation in regard to x and y respectively, then the determinant

$$\begin{vmatrix} f & g & h \\ f_1 & g_1 & h_1 \\ f_2 & g_2 & h_2 \end{vmatrix} = 0$$

is satisfied identically when f, g, h are any linear functions of x, y; of course it remains $= 0$ when, instead of x, y, we write any functions whatever of x, y.

But if f, g, h are any quadric functions of x, y, then writing x, $y = \dfrac{\xi}{\zeta}$, $\dfrac{\eta}{\zeta}$, that is, considering f, g, h as homogeneous functions of (x, y, z), the condition is

$$\begin{vmatrix} f & g & h \\ f_1 & g_1 & h_1 \\ f_2 & g_2 & h_2 \end{vmatrix} = 0,$$

which (as is obvious, and may be directly verified) is satisfied identically; but here writing instead of (x, y, z) any functions (ξ, η, ζ) of (x, y, z), the condition no longer remains satisfied identically; it gives in fact a differential equation of the second order which must be satisfied by ξ, η, ζ, in order that f, g, h, considered as functions of ξ, η, ζ, may be linear functions of these variables.

We may in like manner consider f, g, h as cubic functions of (x, y); here, regarding them as homogeneous functions of (x, y, z), the conditions that they may be reducible to linear functions of (ξ, η, ζ) are

$$\begin{vmatrix} f & g & h \\ f_1 & g_1 & h_1 \\ f_2 & g_2 & h_2 \end{vmatrix} = 0, \quad \begin{vmatrix} f & g & h \\ f_1 & g_1 & h_1 \\ f_3 & g_3 & h_3 \end{vmatrix} = 0, \quad \begin{vmatrix} f & g & h \\ f_2 & g_2 & h_2 \\ f_3 & g_3 & h_3 \end{vmatrix} = 0,$$

where 1, 2, 3 denote differentiation in regard to x, y, z respectively. These are all the same condition, and they are satisfied identically. But if we write for (x, y, z) any functions (ξ, η, ζ) of (x, y, z), then the conditions are differential equations which must be satisfied by ξ, η, ζ in order that f, g, h, considered as functions of ξ, η, ζ, may be linear functions of these variables.

924. On the Non-Existence of a Special Group of Points.

[From the *Messenger of Mathematics*, vol. xxi. (1892), pp. 132, 133.]

It is well known that, taking in a plane any eight points, every cubic through these passes through a determinate ninth point. It is interesting to show that there is no system of seven points such that every cubic through these passes through a determinate eighth point.

Assuming such a system: first, no three of the points can be in a line: for, if they were, then among the cubics through the seven points we have the line through the three points and an arbitrary conic through the remaining four points, and these composite cubics have no common eighth point of intersection.

Secondly, no six of the points can be on a conic: for, if they were, then among the cubics through the seven points we have the conic through the six points and an arbitrary line through the remaining point, and these composite cubics have no common eighth point of intersection.

Taking now the points to be 1, 2, 3, 4, 5, 6, 7; among the cubics through these, we have the composite cubics (A, P), (B, Q), (C, R), where A, B, C are the lines 67, 75, 56, and P, Q, R the conics 12345, 12346, 12347 respectively; by what precedes, the points 5, 6, 7 do not lie on a line, and the points $(6, 7)$, $(7, 5)$ and $(5, 6)$ neither of them lie on the conics P, Q, R respectively.

The common eighth point of intersection, if it exists, must be

$$(A, B, C); \quad (A, Q, R),\ (B, R, P),\ (C, P, Q); \quad (P, B, C),\ (Q, C, A),\ (R, A, B); \quad \text{or } (\ldots$$

There is no point (A, B, C).

There is no point (A, Q, R): for Q, R intersect only in the points 1, 2, 3, 4, no one of which lies on A; and similarly, there is no point (B, R, P) or (C, P, Q).

B, C intersect in 5, which is a point on P; and thus 5 is the only point (P, B, C). Similarly, 6 is the only point (Q, C, A) and 7 is the only point (R, A, B).

P, Q intersect in 1, 2, 3, 4, which are each of them on R; hence 1, 2, 3, 4 are the only points (P, Q, R).

Hence the points 1, 2, 3, 4, 5, 6, 7 present themselves each once, and only once, among the intersections of the three cubics, and there is no common eighth point of intersection.

925. On Waring's Formula for the Sum of the mth Powers of the Roots of an Equation.

[From the *Messenger of Mathematics*, vol. xxi. (1892), pp. 133–137.]

The formula in question, Prob. I. of Waring's *Meditationes Algebraicæ*, Cambridge, 1782, making therein a slight change of notation, is as follows: viz. the equation being

$$x^n - bx^{n-1} + cx^{n-2} - dx^{n-3} + \ldots = 0,$$

then we have

$$\begin{aligned}(-)^{m-1}s_m = mb^m &- m\,c\\
&- m\,b^{m-2}c + m(m-3)\,bd\\
&+ m\,b^{m-3}d - m(m-4)\,ce - m(m-4)\,bd + \tfrac{1}{2}m(m-4)(m-5)\,c^2d\\
&+ m\,b^{m-4}e + m(m-5)\,bf + m(m-5)\,ce\\
&- m(m-5)(m-6)\,cd^2 - m(m-5)(m-6)\,be\\
&+ \tfrac{1}{6}m(m-5)(m-6)(m-7)\,c^3 - m\,g\\
&- m(m-6)\,ch - m(m-6)\,dg - m(m-6)\,bf\\
&+ \tfrac{1}{2}m(m-6)(m-7)\,cd^2 - m(m-6)\,eg\\
&+ \tfrac{1}{2}m(m-6)(m-7)\,d^2f - \tfrac{1}{2}m(m-6)(m-7)\,ce^2\\
&- \tfrac{1}{6}m(m-6)(m-7)(m-8)\,c^2e + \tfrac{1}{2}m(m-6)(m-7)\,d^3\\
&+ \tfrac{1}{24}m(m-6)(m-7)(m-8)(m-9)\,c^4 + \&\text{c.},\end{aligned}$$

where, reckoning the weights of b, c, d, e, $\ldots$ as 1, 2, 3, 4, $\ldots$, respectively, the several terms are all the terms of the weight m, or (what is the same thing) in the coefficient of $b^{m-\theta}$ we have all the combinations of c, d, e, $\ldots$ (or say all the non-unitary combinations) of the weight θ, and where the numerical coefficient of

$$b^{m-\theta}c^d d^e e^f \ldots \quad (d+e+f+\ldots = \theta),$$

is

$$\frac{m\cdot m-(\theta+1)\cdot m-(\theta+2)\ldots m-(\theta-1)}{c!\,d!\,e!\,f!\ldots}.$$

Thus for the term $b^{m-8}c^2e^4$, $\theta = 8$; c, d, $e = 2$, 0, 4, 1 respectively (the other exponents each vanishing), and the coefficient is

$$\frac{m\cdot m-6\cdot m-7}{1\cdot 2} = -\tfrac{1}{2}m(m-6)(m-7),$$

as above; and so in other cases.

For the MacMahon form

$$1 + bx + \frac{c}{1 \cdot 2}x^2 + \frac{d}{1 \cdot 2 \cdot 3}x^3 + \ldots = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \ldots,$$

or say

$$1 + bx + \frac{c}{1 \cdot 2}x^2 + \ldots = (1 - \alpha x)(1 - \beta x) \ldots,$$

we must for b, c, d, $\ldots$ write b, $\frac{1}{1 \cdot 2}c$, $\frac{1}{1 \cdot 2 \cdot 3}d$, $\ldots$ respectively: we thus have

$$\begin{aligned}(-)^m \Pi(m-1) \cdot S_m = {} & \Pi(m-1) \cdot b^m \\ & - \Pi(m-2) \cdot b^{m-2}c \\ & + \Pi(m-3) \cdot b^{m-3}d \\ & + \Pi(m-3) \cdot b^{m-3}cd \\ & - \Pi(m-4) \cdot b^{m-4}e - \Pi(m-4) \cdot b^{m-4}c^2 \\ & + \&\text{c.},\end{aligned}$$

or say

$$\begin{aligned}(-)^m \Pi(m-1) S_m = \Pi(m-1) b^m - \Pi(m-2) b^{m-2}c + \Pi(m-3) b^{m-3}d \\ + \Pi(m-3) b^{m-3}cd - \Pi(m-4) b^{m-4}e - \Pi(m-4) b^{m-4}c^2 + \&\text{c.},\end{aligned}$$

the numerical coefficient of

$$b^{m-\theta} c^d d^e e^f \ldots \quad (d + e + f + \ldots = \theta)$$

being

$$\frac{(c + e + g + \ldots) m \cdot m - (\theta - \delta + 1) \cdot m - (\theta - \delta + 2) \ldots m - (\delta - 1)}{c!\, d!\, e! \ldots (1 \cdot 2)^c (1 \cdot 2 \cdot 3)^d (1 \cdot 2 \cdot 3 \cdot 4)^e \ldots}.$$

It is convenient to write down the literal terms in alphabetical order (AO), calculating and affixing to each term the proper numerical coefficient; thus taking

$$1 + bx + cx^2 + dx^3 + \ldots = (1 - \alpha x)(1 - \beta x)(1 - \gamma x)\ldots,$$

we find

$$\begin{aligned} -120b^6 = {} & g \cdot 1 \\ & + f \cdot -6 \\ & + ce \cdot -15 \\ & + d^2 \cdot -10 \\ & + b^2e \cdot +30 \\ & + bcd \cdot +120 \\ & + c^3 \cdot +30 \\ & + b^3d \cdot -120 \\ & + b^2c^2 \cdot -270 \\ & + b^4c \cdot +360 \\ & + b^6 \cdot -120 \\ & \pm 541, \end{aligned}$$

this expression, as representing the value of the non-unitary function S_6, being in fact a seminvariant.

It is to be remarked that the foregoing expression for the sum of the mth powers of the roots of the equation

$$x^n - bx^{n-1} + cx^{n-2} - \ldots = 0$$

is, in fact, the series for s_m continued so far only as the exponent of b is not negative: see as to this Note XI. of Lagrange's *Equations Numériques.* For the *a* posteriori verification, observe that we have

$$s_m = \alpha^m + \beta^m + \gamma^m + \ldots,$$

or writing for a moment $u = -b$, say this is

$$s_m = (-u)^m + (-u - c)^m + (-u - d)^m + \ldots,$$

where

$$f(x) = x^n - bx^{n-1} + cx^{n-2} - dx^{n-3} + \&\text{c.} = 0.$$

Hence, by Lagrange's theorem,

$$s_m = \frac{1}{2\pi i} \int \frac{x^m f'(x)}{f(x)} dx,$$

and the residue at infinity gives the required expression.

926. Corrected Seminvariant Tables for the Weights 11 and 12.

[From the *Proceedings of the Royal Society*, vol. L. (1891), pp. 431–437.]

The annexed Tables, extending those for the weights 1 to 10 given in my "Memoir on Seminvariants," [875], require a correction. In the Tables, weight 10, last two columns but one, instead of the columns $cdf - b^4d^2$, $ce^3 - c^5$, we ought to have $ce^3 - c^5$, $cdf - b^4d^2$. The complete list up to the weight 12 is

$w = 1$	b
$w = 2$	c
$w = 3$	$d,\ b^2$
$w = 4$	$e,\ bc$
$w = 5$	$f,\ bd,\ c^2$
$w = 6$	$g,\ be,\ cd,\ b^2c$
$w = 7$	$h,\ bf,\ ce,\ d^2,\ b^2d,\ bc^2$
$w = 8$	$i,\ bg,\ cf,\ de,\ b^2e,\ bcd,\ c^3,\ b^3c$
$w = 9$	$j,\ bh,\ cg,\ df,\ e^2,\ b^2f,\ bce,\ cd^2,\ bc^2d,\ b^3d,\ b^2c^2$
$w = 10$	$k,\ bi,\ ch,\ dg,\ ef,\ b^2g,\ bcf,\ bde,\ c^2d,\ b^3e,$ $b^2cd,\ bc^3,\ b^4c;\quad cd^2f,\ ce^3 - c^5$

Seminvariant Tables, weight 11.

l	c^2g	$cdef$	d^3	b^2h	bcf	bce	bd^2	c^3d	b^3i	b^2cg	b^2df	b^2e^2	bc^2e
bg	0	0	0	1	0	0	0	0	0	0	0	0	0
cf	0	0	0	0	1	0	0	0	0	0	0	0	0
de	0	0	0	0	0	1	0	0	0	0	0	0	0
c^3	0	0	0	0	0	0	0	1	0	0	0	0	0
b^2g	0	0	0	0	0	0	0	0	1	0	0	0	0
bcd	0	1	0	0	0	0	0	0	0	0	0	0	0
b^2f	0	0	0	0	0	0	0	0	0	0	1	0	0
b^2e	0	0	0	0	0	0	0	0	0	0	0	1	0
bc^2	1	0	0	0	0	0	0	0	0	0	0	0	0
b^3d	0	0	0	0	0	0	0	0	0	0	0	0	0
b^3c	0	0	0	0	0	0	0	0	0	0	0	0	0

Seminvariant Tables, weight 12 (part 1).

m	c^3f	c^2e^2	c^2d^2	b^2i	bch	bcg	bdf	be^2	c^4	b^3j	b^2ci	b^2dh	b^2eg
bi	0	0	0	1	0	0	0	0	0	0	0	0	0
ch	0	0	0	0	1	0	0	0	0	0	0	0	0
dg	0	0	0	0	0	1	0	0	0	0	0	0	0
ef	0	0	0	0	0	0	1	0	0	0	0	0	0
c^2d	1	0	0	0	0	0	0	0	0	0	0	0	0
b^2h	0	0	0	0	0	0	0	0	0	1	0	0	0
bcg	0	0	0	0	0	0	0	0	0	0	1	0	0
bcf	0	0	0	0	0	0	0	0	0	0	0	1	0
bce	0	0	0	0	0	0	0	0	0	0	0	0	1
bd^2	0	0	0	0	0	0	0	0	0	0	0	0	0
c^3	0	1	0	0	0	0	0	0	0	0	0	0	0
b^3g	0	0	0	0	0	0	0	0	0	0	0	0	0
b^3f	0	0	0	0	0	0	0	0	0	0	0	0	0
b^3e	0	0	0	0	0	0	0	0	0	0	0	0	0
b^2c^2	0	0	0	0	0	0	0	0	1	0	0	0	0

Seminvariant Tables, weight 12 (part 2).

bce	bcd	c^2d	c^5	b^4k	b^3bi	b^3ch	b^3dg	b^3ef	b^2c^2g	b^2cdf	b^2ce^2	b^2d
b^2i	0	0	0	1	0	0	0	0	0	0	0	0
bcg	0	0	0	0	0	0	0	0	1	0	0	0
bce	0	0	0	0	0	0	0	0	0	0	0	0
bd^2	0	0	0	0	0	0	0	0	0	0	0	0
c^3	0	0	0	0	0	0	0	0	0	0	0	0
b^3g	0	0	0	0	0	0	0	0	0	0	0	0
b^3f	0	0	0	0	0	0	0	0	0	0	0	0
b^3e	0	0	0	0	0	0	0	0	0	0	0	0
b^2c^2	0	0	0	0	0	0	0	0	0	0	0	0

927. On Clifford's Paper "On Syzygetic Relations among the Powers of Linear Quantics."

[From the *Proceedings of the Royal Society*, vol. L. (1891), pp. 437–439.]

The question discussed in this paper is as follows: writing $U = (a, b, c, \dots \mid x, y)^n$, a binary quantic of the order n, with the coefficients $(a, b, c, \dots)$; and taking k a given positive integer, to find the number of independent syzygies of the form

$$\alpha U^k + \beta U^{k-1}V + \gamma U^{k-2}V^2 + \dots = 0,$$

where α, β, γ, ... are rational and integral functions of the coefficients of the quantics U and V, and where the degree in these coefficients of each term $U^{k-f}V^f$ is the same for all values of f; the syzygy being thus an identical relation between the different powers U^k, $U^{k-1}V$, ... of the quantics U, V.

1. I consider in particular the case where the quantics U, V are linear quantics, say $U = ax + by$, $V = cx + dy$. Here taking $k = 2$, we have between U^2, UV, and V^2 a single relation

$$(ad - bc)^2 U^2 \cdot V^2 = (ad - bc)^2 (UV)^2,$$

that is,

$$(ad - bc)^2 U^2 \cdot V^2 - (ad - bc)^2 (UV)^2 = 0,$$

which may be written

$$\alpha U^2 + \beta UV + \gamma V^2 = 0,$$

where $\alpha = (ad-bc)^2V^2$, $\beta = -2(ad-bc)^2UV$, $\gamma = (ad-bc)^2U^2$, are of the proper degrees in the coefficients; viz. considering the degrees in (a, b) and in (c, d) respectively, the degrees of α, β, γ are as follows:

	(a, b)	(c, d)
α	0,	2
β	1,	1
γ	2,	0

and the degrees of αU^2, βUV, γV^2 are in (a, b) each $= 2$, and in (c, d) each $= 2$, as they should be.

2. But we see further that the twofold relation

$$\alpha U^2 + \beta UV + \gamma V^2 = 0, \quad \alpha' U^2 + \beta' UV + \gamma' V^2 = 0,$$

where the coefficients α', β', γ' are independent of α, β, γ, cannot be satisfied for arbitrary values of (a, b, c, d); for from the two equations eliminating U^2: UV: V^2, we should obtain

$$\alpha\beta' - \alpha'\beta = 0, \quad \alpha\gamma' - \alpha'\gamma = 0, \quad \beta\gamma' - \beta'\gamma = 0,$$

which are three relations between the coefficients α, β, $\ldots$, γ'.

Hence there is not in general any twofold syzygy; and the conclusion is that for linear quantics, and for the powers U^2, UV, V^2, the only syzygy is the foregoing single one.

3. I consider, more generally, the syzygies between the powers U^k, $U^{k-1}V$, $\ldots$, V^k of the two linear quantics U, V; or say the syzygies between the $k+1$ powers $(U, V)^k$. The number of terms is $k+1$, and the number of coefficients of the quantics is 4; hence the number of coefficients of the products U^k, $U^{k-1}V$, $\ldots$ is $= (k+1) + k = 2k+1$; and we have thus $2k+1$ products of the coefficients, each of which must have the degree k in (a, b) and the degree k in (c, d). The number of products of the proper degrees is $= k+1$, and this is the number of independent syzygies.

The number of syzygies is thus $= k+1$, and since the number of terms is also $= k+1$, the syzygies give rise to a single equation; that is, there is a single relation

$$A \cdot U^k + B \cdot U^{k-1}V + \ldots = 0,$$

where the coefficients A, B, $\ldots$ are products of the degrees just referred to, the number of them being $= k+1$, but they are connected by k relations, so that there remains only a single constant, that is, we have a single syzygy.

4. But the conclusion is different if we consider the syzygies between the powers U^k, $U^{k-1}V$, $\ldots$ of quantics U, V of any order n. Considering the case where U, V are quadric quantics, say $U = (a, b, c \mid x, y)^2$, $V = (d, e, f \mid x, y)^2$, we have between the $k+1$ powers U^k, $U^{k-1}V$, $\ldots$, V^k, a number of syzygies which is at least $= k+1$, and may be greater. In fact, writing $\Theta = (ad - bf)^2 - 4(ae - bd)(cf - ae)$, the condition for the two quantics to have a common factor is $\Theta = 0$; and when this is so, we have $U = (\alpha x + \beta y)(a'x + b'y)$,

$V = (\alpha x+\beta y)(c'x+d'y)$, so that $V = \dfrac{c'x+d'y}{a'x+b'y}\cdot U$, or say $V = \dfrac{W}{W'}\cdot U$; hence $VU' - UV' = 0$, that is, the quantics have a common factor; and we thence obtain the like relations between the powers of U and V. But it is not in this way that we obtain the syzygies in question; and the general theory requires a further examination.

928. On the Analytical Theory of the Congruency.

[From the *Proceedings of the Royal Society*, vol. L. (1891), pp. 440–457.]

The theory of the congruency, or congruence, of lines was established by Kummer in his celebrated memoir "Ueber die algebraischen Strahlensysteme, in Welchen die Strahlen durch eine algebraische Gleichung verbunden sind," *Berlin. Abh.* 1866. The congruency is a system of lines satisfying two conditions, or say a twofold relation; and it thus occupies the same position in relation to the complex (a system of lines satisfying one condition, or say a onefold relation) that a surface does in relation to a curve.

1. I consider a line determined by its six coordinates (a, b, c, f, g, h), where $af + bg + ch = 0$; and I represent the consecutive line by $(a+\delta a, \ldots, f+\delta f, \ldots)$. The condition in order that the two lines may intersect is

$$(a+\lambda a',\ b+\lambda b',\ c+\lambda c',\ f+\lambda f',\ g+\lambda g',\ h+\lambda h'),$$

viz. the consecutive line $(a_1, b_1, c_1, f_1, g_1, h_1)$ belongs to the linear congruency defined by these two equations.

Forming with these a linear combination, the equation of the congruency may be written

$$\begin{vmatrix} a & b & c & f & g & h \\ a' & b' & c' & f' & g' & h' \\ \delta a & \delta b & \delta c & \delta f & \delta g & \delta h \end{vmatrix} = 0,$$

which is the condition for the intersection of the line (a, b, c, f, g, h) with the line $(\delta a, \delta b, \delta c, \delta f, \delta g, \delta h)$.

2. For the point 2, we have

$$\begin{aligned} hy - gz + aw &= 0, \quad & h'y - g'z + a'w &= 0, \\ -hx \quad + fz + bw &= 0, \quad & -h'x \quad + f'z + b'w &= 0, \\ gx - fy \quad + cw &= 0, \quad & g'x - f'y \quad + c'w &= 0, \end{aligned}$$

where (x, y, z, w) are the coordinates of the point. Hence we have

$$\frac{x}{fw - bg} = \frac{y}{ah - cf} = \frac{z}{ab - ch} = \frac{w}{af + bg + ch},$$

and similarly for the other point. The line of the congruency is thus determined by the two points in which it meets the two planes.

3. The order of the congruency is the number of lines which pass through a given point. Taking the point to be (x_0, y_0, z_0, w_0), the condition that a line of the congruency passes through this point is that the coordinates (a, b, c, f, g, h) satisfy the two equations

$$af + bg + ch = 0, \quad hx_0 - gy_0 + fz_0 + aw_0 = 0,$$

and the order is thus the number of solutions of these equations, which is equal to the number of intersections of the two quadric surfaces, viz. it is $= 4$.

4. The class of the congruency is the number of lines which lie in a given plane. Taking the plane to be $lx + my + nz + pw = 0$, the condition that a line of the congruency lies in this plane is that the coordinates (a, b, c, f, g, h) satisfy the two equations

$$af + bg + ch = 0, \quad la + mb + nc + pf = 0,$$

and the class is thus the number of solutions of these equations, which is equal to the number of intersections of the two quadric surfaces, viz. it is $= 4$.

5. The congruency has a singular surface, which is the locus of points through which pass an infinite number of lines of the congruency. This surface is of the order equal to the number of lines of the congruency which pass through a given point, and which are contained in a given plane through the point; that is, it is the number of intersections of the line of the congruency with the plane, which is $= 4$.

6. The theory may be extended to the case of a congruency of higher order, where the lines satisfy three or more conditions. The general theory of such congruencies presents many interesting questions, which have not yet been fully investigated.

929. Note on the Skew Surfaces Applicable upon a Given Skew Surface.

[From the *Philosophical Transactions*, vol. CLXXXIII (1892), pp. 677–688.]

The question of the applicability of one skew surface upon another is a problem of great interest in the theory of surfaces. The general solution depends upon the integration of a partial differential equation of the second order; and the particular case where the given surface is a skew surface, or ruled surface, presents special features which make it worthy of separate consideration.

1. I consider a skew surface generated by the motion of a straight line; and I seek to determine the skew surfaces which are applicable upon it. The general theory of applicable surfaces shows that two surfaces are applicable when their linear elements can be expressed in the same form; that is, when the geodesic distances between corresponding points are equal.

For a skew surface, the linear element can be expressed in the form

$$ds^2 = du^2 + 2\cos\theta\, du\, dv + (u^2 + \alpha u + \beta)\, dv^2,$$

where θ is the angle between the generating line and the direction of the parameter v, and α, β are functions of v only. The condition for applicability is that this expression for ds^2 shall be the same for the two surfaces, or at least reducible to the same form.

2. If we consider the strip of the surface bounded by two consecutive generating lines, the distance between these lines, measured along their common perpendicular, is the parameter of distribution. The condition of applicability then requires that the strips of the two surfaces shall have the same breadth (or distance on each side from P) of the strip or element of the surface, the same angle between the bounding lines, and the same parameter of distribution.

3. For the general skew surface, the condition of applicability leads to a partial differential equation. Writing the equation of the surface in the form

$$z = f(x, y),$$

the condition is that the first and second fundamental quantities shall satisfy the equation of Gauss. The integration of this equation gives the most general surface applicable upon the given surface; and it appears that the solution depends upon an arbitrary function.

4. In the particular case where the given surface is a surface of translation, the problem is simplified. A surface of translation is generated by the translation of a curve parallel to itself, the directing curve being also moved in such a manner that the generating curve always remains parallel to a fixed plane. For such a surface, the linear element can be expressed in a simple form, and the condition of applicability can be more easily investigated.

5. The case where the given surface is a helicoid is also of special interest. A helicoid is generated by the motion of a straight line which turns uniformly round a fixed axis, while at the same time it moves uniformly in the direction of that axis. The linear element of a helicoid can be expressed in the form

$$ds^2 = dr^2 + r^2\, d\theta^2 + (c\, d\theta + dz)^2,$$

and the condition of applicability upon another helicoid can be completely determined. It appears that the only surfaces applicable upon a helicoid are helicoids; and the relation between the two helicoids can be expressed in a very simple form.

6. More generally, if the given skew surface is a surface of revolution, the problem can be solved by the use of elliptic integrals. The linear element of a surface of revolution can be expressed in the form

$$ds^2 = dr^2 + r^2\, d\theta^2 + dz^2,$$

where z is a function of r; and the condition of applicability upon another surface of revolution leads to an equation which can be integrated by elliptic functions.

7. The general theory of the applicability of skew surfaces is connected with the theory of the deformation of surfaces, and with the theory of the integration of partial differential equations. The problem presents many difficulties, which have not yet been completely overcome; but the results which have been obtained are sufficient to show the great interest and importance of the subject.

930. Sur la Surface des Ondes.

[From the *Annali di Matematica*, Ser. II., vol. xx. (1892), pp. 1–10.]

1. Je me sers de la détermination bien connue de Fresnel de la surface des ondes, c'est-à-dire de l'enveloppe du plan

$$lx + my + nz = 1,$$

où l, m, n sont les cosinus directeurs d'un rayon de l'onde; le plan est donné par l'équation

$$\frac{a^2l^2}{1-a^2r^2} + \frac{b^2m^2}{1-b^2r^2} + \frac{c^2n^2}{1-c^2r^2} = 0,$$

où $r^2 = l^2+m^2+n^2$; et l'équation de la surface s'obtient en éliminant l, m, n, r entre ces deux équations et les équations qui expriment que le plan touche la surface.

2. On peut, au lieu de cela, écrire

$$lx + my + nz = 1, \quad l^2+m^2+n^2 = r^2,$$

et considérer r comme un paramètre. L'enveloppe s'obtient alors en éliminant r entre l'équation du plan et son dérivée par rapport à r.

3. En écrivant

$$\frac{l^2}{1-a^2r^2} + \frac{m^2}{1-b^2r^2} + \frac{n^2}{1-c^2r^2} = 0,$$

on a l'équation de la surface sous la forme

$$\frac{a^2x^2}{r^2-a^2} + \frac{b^2y^2}{r^2-b^2} + \frac{c^2z^2}{r^2-c^2} = 1,$$

où r est déterminé par l'équation

$$\frac{x^2}{r^2-a^2} + \frac{y^2}{r^2-b^2} + \frac{z^2}{r^2-c^2} = 1.$$

4. Pour obtenir l'équation de la surface en coordonnées rectangulaires, il faut éliminer r entre ces deux équations. On obtient ainsi une équation du degré 16; mais cette équation se réduit au degré 4 lorsqu'on considère les carrés des coordonnées comme les variables.

5. Je remarque que l'équation de la surface peut s'écrire sous la forme
$$\begin{vmatrix} x^2 & y^2 & z^2 \\ r^2-a^2 & r^2-b^2 & r^2-c^2 \\ 1 & 1 & 1 \end{vmatrix} = 0,$$
et que cette forme met en évidence plusieurs propriétés importantes de la surface.

6. On a, pour les coordonnées du point de contact du plan tangent,
$$\xi = \frac{a^2 l}{1-a^2r^2}, \quad \eta = \frac{b^2 m}{1-b^2r^2}, \quad \zeta = \frac{c^2 n}{1-c^2r^2}.$$
La première de ces équations donne
$$\xi(1-a^2r^2) = a^2 l,$$
et de là
$$l = \frac{\xi(1-a^2r^2)}{a^2}.$$
On a aussi
$$x = \frac{l}{r^2-a^2}, \quad y = \frac{m}{r^2-b^2}, \quad z = \frac{n}{r^2-c^2},$$
et de là
$$u+v = (b+c)\lambda^2 + (c+a)\mu^2 + (a+b)\nu^2,$$
où λ, μ, ν sont les cosinus directeurs de la normale.

7. On a

$$\eta - b = T(f-a)\{\xi\eta + \zeta - (\eta+\zeta)\xi\},$$

et de là

$$\eta - b = T(f-a)(\xi\eta + \zeta - \eta\xi - \zeta\xi),$$

où T est une fonction de r.

8. La surface des ondes a plusieurs propriétés remarquables. En particulier, elle possède quatre points coniques, qui correspondent aux directions des axes optiques; et elle a aussi quatre droites doubles, qui sont les intersections de la surface avec le plan à l'infini.

9. Les courbes de contact du plan tangent sont des courbes du quatrième ordre, qui ont des points singuliers aux points coniques de la surface. Ces courbes sont des courbes gauches, et leur étude présente un grand intérêt.

10. La surface des centres de la surface des ondes est une surface du vingt-quatrième ordre. Elle a des points singuliers qui correspondent aux points singuliers de la surface des ondes; et son étude est intimement liée à celle de la surface des ondes.

11. On peut étudier la surface des ondes au moyen de ses sections par des plans parallèles aux coordonnées. Ces sections sont des courbes du quatrième ordre, qui ont des propriétés remarquables. En particulier, les sections par des plans parallèles aux plans de coordonnées sont des courbes ayant des points doubles aux points où le plan coupe les axes.

12. L'équation différentielle des courbes de contact peut s'obtenir en éliminant r entre l'équation du plan et ses dérivées. On obtient ainsi une équation du premier ordre, qui détermine les courbes de contact sur la surface.

13. La surface des ondes peut aussi être étudiée au moyen de ses courbures principales. Les rayons de courbure principaux sont donnés par une équation du deuxième degré, qui s'obtient en égalant à zéro le déterminant des derivées secondes de la fonction qui définit la surface.

14. Le point de contact du plan tangent peut être déterminé au moyen des formules données ci-dessus. On a

$$X = \mu Z - \nu Y, \quad Y = \nu X - \lambda Z, \quad Z = \lambda Y - \mu X,$$

et de là

$$X : Y : Z = \lambda : \mu : \nu,$$

c'est-à-dire que le rayon vecteur du point de contact est parallèle à la normale au plan tangent.

15. De la valeur ci-dessus donnée pour $\eta - b$, on déduit que la surface a des propriétés de symétrie remarquables. En particulier, elle est symétrique par rapport aux plans de coordonnées, et aussi par rapport à l'origine.

16. L'équation de la surface peut aussi s'écrire sous la forme

$$\frac{x^2}{a^2 - r^2} + \frac{y^2}{b^2 - r^2} + \frac{z^2}{c^2 - r^2} = 1,$$

où r est déterminé par l'équation

$$\frac{a^2 x^2}{(a^2 - r^2)^2} + \frac{b^2 y^2}{(b^2 - r^2)^2} + \frac{c^2 z^2}{(c^2 - r^2)^2} = 0.$$

Cette forme de l'équation met en évidence la connexion entre la surface des ondes et les surfaces confocales du second degré.

17. On peut considérer la surface des ondes comme une surface de la classe 16, ayant 4 points coniques et 4 droites doubles. Le genre de la surface est $= 3$, et ses invariants numériques peuvent être calculés par les formules de la géométrie énumérative.

18. La surface des ondes a été l'objet de nombreuses recherches. Les résultats les plus importants sont dus à Fresnel, Hamilton, Kummer, et Cayley. La théorie de cette surface est intimement liée à celle de la propagation de la lumière dans les milieux cristallins.

19. On peut étudier la surface des ondes au moyen des coordonnées elliptiques. En posant

$$\frac{x^2}{\theta - a^2} + \frac{y^2}{\theta - b^2} + \frac{z^2}{\theta - c^2} = 1,$$

on définit trois systèmes de surfaces orthogonales; et les propriétés de la surface des ondes peuvent être déduites de la théorie générale de ces coordonnées.

20. L'équation différentielle des courbes de contact peut se mettre sous la forme

$$A\,dx^2 + B\,dy^2 + C\,dz^2 + 2D\,dy\,dz + 2E\,dz\,dx + 2F\,dx\,dy = 0,$$

où les coefficients A, B, C, D, E, F sont des fonctions de x, y, z. Cette équation détermine les deux directions principales en chaque point de la surface.

21. On obtient sans peine, au moyen des valeurs trouvées ci-dessus, l'équation de la surface des ondes sous la forme explicite. L'équation finale est

$$(a^2x^2+b^2y^2+c^2z^2)(x^2+y^2+z^2)^2-(a^2+b^2+c^2)(x^2+y^2+z^2)(a^2x^2+b^2y^2+c^2z^2)$$
$$+(a^2b^2+b^2c^2+c^2a^2)(x^2+y^2+z^2)-a^2b^2c^2=0,$$

c'est-à-dire que l'équation est du degré 4 en x^2, y^2, z^2, et du degré 16 en x, y, z.

22. En écrivant l'équation sous la forme

$$U=(a^2x^2+b^2y^2+c^2z^2)(x^2+y^2+z^2)^2-\ldots=0,$$

on voit que la surface est du quatrième ordre par rapport aux carrés des coordonnées. Cette propriété est fondamentale pour la théorie de la surface.

23. Les points coniques de la surface sont donnés par les équations

$$x=0,\quad y=0,\quad z=0,$$

et les valeurs correspondantes de r sont $r=a$, $r=b$, $r=c$. Les points coniques sont donc situés sur les axes de coordonnées, à des distances a, b, c de l'origine.

24. Les arêtes de rebroussement de la surface sont des courbes du huitième ordre, qui passent par les points coniques. Ces courbes sont les lieux des points où le plan tangent est stationnaire.

25. La surface des ondes peut être généralisée en considérant un nombre quelconque de termes dans l'équation fondamentale. On obtient ainsi une classe de surfaces qui comprennent la surface des ondes comme cas particulier.

26. On peut aussi étudier la surface des ondes au moyen de la théorie des invariants. Les invariants de la surface sont des fonctions des coefficients a, b, c, qui restent inchangées par les transformations orthogonales.

27. La surface des ondes est un cas particulier de la surface de Kummer, qui est une surface du quatrième ordre ayant 16 points singuliers. La surface de Kummer a été étudiée en détail par son auteur, et ses propriétés sont fondamentales pour la géométrie algébrique.

28. Je m'arrête pour un moment pour considérer la même théorie par rapport à la surface des centres. La surface des centres est le lieu des centres de courbure de la surface des ondes; elle est du vingt-quatrième ordre, et ses propriétés sont intimement liées à celles de la surface des ondes.

29. Il paraît ainsi que l'ordre de la surface des centres de la surface des ondes doit être $= 38$. Mais ce résultat doit être vérifié par un calcul direct; et il est possible que l'ordre soit en réalité moindre, à cause des singularités de la surface des ondes.

30. L'étude complète de la surface des ondes et de ses surfaces annexes présente encore de nombreux problèmes intéressants. La méthode employée dans cette note peut être appliquée avec succès à plusieurs de ces problèmes; et elle conduit à des résultats qui seront utiles pour la théorie de la propagation de la lumière dans les cristaux.

930SUR LA SURFACE DES ONDES.

[Suite. Voir pp. 227–250.]

donc

$$X^2 + Y^2 = \frac{\alpha^2\beta^2}{c}\,\frac{1}{(\beta+\gamma\theta)^2},$$

$$\alpha X^2 - \beta Y^2 = \frac{\alpha^2\beta^2}{c}\,\frac{\alpha\theta - \beta(1-\theta)}{(\beta+\gamma\theta)^2}, \quad = -\frac{\alpha^2\beta^2}{c}\,\frac{1}{\beta+\gamma\theta},$$

et de là

$$(\alpha X^2 - \beta Y^2)^2 = \frac{\alpha^2\beta^2}{c}\,(X^2+Y^2);$$

courbe de l'ordre 4, cuspidale sur la surface des centres.

2°. Pour l'ellipse $\eta = ab$, en écrivant $x^2 = b\theta$, $y^2 = a(1-\theta)$, et de là

$$\xi = b\theta + a(1-\theta), \quad = a - \gamma\theta,$$

on obtient

$$\begin{aligned} a\xi + \eta - a(b+c) &= -a(\beta+\gamma\theta),\\ b\xi + \eta - b(c+a) &= -b(\beta+\gamma\theta),\\ c\xi + \eta - c(a+b) &= -b\beta - c\gamma\theta, \end{aligned}$$

valeurs qui donnent

$$\begin{aligned} X &= \sqrt{b}\,\sqrt{\theta}\left\{1 - \frac{a(\beta+\gamma\theta)}{b\beta+c\gamma\theta}\right\} = -\sqrt{b}\,\beta\gamma\,\frac{\sqrt{\theta}(1-\theta)}{b\beta+c\gamma\theta},\\ Y &= \sqrt{a}\,\sqrt{1-\theta}\left\{1 - \frac{b(\beta+\gamma\theta)}{b\beta+c\gamma\theta}\right\} = -\sqrt{a}\,\alpha\gamma\,\frac{\theta\sqrt{1-\theta}}{b\beta+c\gamma\theta}, \end{aligned}$$

donc

$$\frac{X^2}{b\beta^2} + \frac{Y^2}{a\alpha^2} = \gamma^2\,\frac{\theta(1-\theta)}{(b\beta+c\gamma\theta)^2},$$

$$\frac{X^2}{\beta} - \frac{Y^2}{\alpha} = \gamma^2\theta(1-\theta)\,\frac{b\beta(1-\theta) - a\alpha\theta}{(b\beta+c\gamma\theta)^2}, \quad = \gamma^2\,\frac{\theta(1-\theta)}{b\beta+c\gamma\theta};$$

$$\frac{X^2}{b\beta^2}\cdot\frac{Y^2}{a\alpha^2} = \gamma^4\,\frac{\theta^3(1-\theta)^3}{(b\beta+c\gamma\theta)^4},$$

et de là

$$\left(\frac{X^2}{b\beta^2} + \frac{Y^2}{a\alpha^2}\right)\left(\frac{X^2}{\beta} - \frac{Y^2}{\alpha}\right)^2 = \gamma^2\cdot\frac{X^2}{b\beta^2}\cdot\frac{Y^2}{a\alpha^2},$$

courbe de l'ordre 6, cuspidale sur la surface des centres.

30. On aurait pu développer la théorie des courbes et rayons de courbure à moyen de la formule ci-dessus donnée, $ds^2 = E\,d\xi^2 + 2F\,d\xi\,d\eta + G\,d\eta^2$ $(F = 0)$, mais pour cela il faudrait trouver plusieurs expressions qui ne sont pas encore calculées, savoir les coefficients des formules

$$\begin{aligned} d^2x &= \alpha\,d\xi^2 + 2\alpha'\,d\xi\,d\eta + \alpha''\,d\eta^2, \\ d^2y &= \beta\,d\xi^2 + 2\beta'\,d\xi\,d\eta + \beta''\,d\eta^2, \\ d^2z &= \gamma\,d\xi^2 + 2\gamma'\,d\xi\,d\eta + \gamma''\,d\eta^2, \end{aligned}$$

et puis

$$E',\ F',\ G' = \lambda\mu + \nu\rho,\ \lambda\alpha'' + \nu\beta'',\ \ldots$$

En prenant comme auparavant R pour le rayon de courbure on aurait alors, (Salmon, *Geometry of three dimensions*, Ed. 4 (1882), p. 347),

$$\begin{vmatrix} R, & E\,d\xi, & G\,d\eta \\ EG, & E'\,d\xi + F'\,d\eta, & F'\,d\xi + G'\,d\eta \\ d\eta, & RE' - E^2G, & RF' - EG^2 \end{vmatrix} = 0,$$

et

$$\begin{vmatrix} d\xi, & -d\xi\,d\eta, & d\eta \\ E, & 0, & G \\ E', & F', & G' \end{vmatrix} = 0,$$

formules pour les courbes et rayons de courbure; en particulier, l'équation différentielle des courbes de courbure peut s'écrire sous la forme

$$\begin{vmatrix} d\xi, & -d\xi\,d\eta, & d\eta \\ E, & 0, & G \\ E', & F', & G' \end{vmatrix} = 0.$$

À reste, cette Équation en $d\xi$, $d\eta$ se déduirait plus simplement de l'équation $dv\,d\eta + v\,du\,d\xi = 0$, en y introduisant les expressions de v, u en termes de ξ, η.

31. Courbes géodésiques sur la surface. L'équation différentielle du second ordre des courbes géodésiques dépend seulement des coefficients E, F, G, savoir en supposant $F = 0$, cette équation est

$$(-EE_2,\ 2EG_1 - GE_2,\ EG_1 - 2GE_1{}^{\smile} d\xi,\ d\eta)^3 + 2EG\,(d\xi\,d^2\eta - d\eta\,d^2\xi) = 0,$$

ou ce qui est la même chose

$$\left(\frac{d\xi}{d\eta}\right)^3 + \cdots = 0,$$

où E_1, E_2, G_1, G_2 désignent $\frac{dE}{d\xi}$, $\frac{dE}{d\eta}$, $\frac{dG}{d\xi}$, $\frac{dG}{d\eta}$ respectivement, voir Cayley, "On geodesic lines, in particular those of a Quadric Surface," *Proc. London Math. Soc.*, t. IV. (1872), p. 197, [508]. Nous avons vu que pour la surface des ondes dont il s'agit les expressions de E, G sont

$$E = \frac{\cdots}{\cdots}, \quad G = \frac{\cdots}{\cdots}$$

et l'on obtiendrait de là sans peine les expressions des coefficients de la fonction cubique $(-EE_2, \ldots)(d\xi,\ d\eta)^3$ qui entre dans l'équation différentielle.

931ON SOME FORMULÆ OF CODAZZI AND WEINGARTEN IN RELATION TO THE APPLICATION OF SURFACES TO EACH OTHER.

[From the *Proceedings of the London Mathematical Society*, vol. xxiv. (1893), pp. 210—223.]

An extremely elegant theory of the application of surfaces one upon another is developed in the memoir, Codazzi, "Mémoire relatif à l'application des surfaces les unes sur les autres," *Mém. prés. à l'Académie des Sciences*, t. xxvii. (1883), No. 6, pp. 1—47; but the notation is not presented in a form which is easily comparable with that of the Gaussian notation in the theory of surfaces. I propose to reproduce the theory in the Gaussian notation.

Codazzi considers on a given surface two systems of curves depending on the parameters t, T respectively; the curves are in the memoir taken to be orthogonal to each other, but this restriction is removed in the general formula given p. 44, *Addition au chapitre premier*. For a curve of either system, he considers the tangent, the principal normal, or normal in the osculating plane, and the binormal, or line at right angles to the osculating plane (say these are tan, prn and bin). For a curve of the one system, that in which t is variable or say a t-curve, he denotes the cosine- inclinations of these lines to the axes by the letters a, b, c, thus:

	x	y	z
tan	a_x	a_y	a_z
prn	b_x	b_y	b_z
bin	c_x	c_y	c_z

and he writes also I for the inclination of the principal normal to the normal of the surface, $\dfrac{dm}{dt}\,dt$ for the angle of contingence, or inclination of the tangent at the point $(t,\ T)$ to the tangent at the point $(t+dt,\ T)$, and $\dfrac{dn}{dt}\,dt$ for the angle of torsion, or inclination of the osculating plane at the point $(t,\ T)$ to that at the point $(t+dt,\ T)$, or, what is the same thing, the inclination of the binormals at these points respectively. And he gives as known formulæ

$$\frac{da}{dt} = \frac{dm}{dt}\,b, \quad \frac{db}{dt} = -\frac{dm}{dt}\,a + \frac{dn}{dt}\,c, \quad \frac{dc}{dt} = -\frac{dn}{dt}\,b,$$

where a, b, c denote $(a_x,\ b_x,\ c_x)$, $(a_y,\ b_y,\ c_y)$, or $(a_z,\ b_z,\ c_z)$.

He uses the capital letters

$$A_x,\ A_y,\ A_z,\ B_x,\ B_y,\ B_z,\ C_x,\ C_y,\ C_z,\ L,\ \frac{dM}{dT}\,dT,\ \frac{dN}{dT}\,dT$$

with the like significations in regard to the curve for which T is variable, or say the T-curve; for greater clearness I give the accompanying figure.

Codazzi writes further

$$\cos I = u, \quad \frac{dM}{dT} = V, \quad \frac{dN}{dT} = W,$$

also, if s, S are the arcs of the two curves respectively,

$$\frac{ds}{dt} = r,$$

and he obtains a system of six formulæ, which in the Addition, p. 44, are presented in the following form — only I use therein θ, instead of his δ, to denote the inclination of the two curves to each other:

$$\frac{d\theta}{dt} + R(u\cos\theta + v\sin\theta) = r(U\cos\theta + W\sin\theta),$$

etc.

In the Gaussian notation, taking p, q for the parameters, we have, with the slight variations presently referred to,

$$\begin{aligned} dx &= a\,dp + a'\,dq + \tfrac{1}{2}\alpha\,dp^2 + \alpha'\,dp\,dq + \tfrac{1}{2}\alpha''\,dq^2,\\ dy &= b\,dp + b'\,dq + \tfrac{1}{2}\beta\,dp^2 + \beta'\,dp\,dq + \tfrac{1}{2}\beta''\,dq^2,\\ dz &= c\,dp + c'\,dq + \tfrac{1}{2}\gamma\,dp^2 + \gamma'\,dp\,dq + \tfrac{1}{2}\gamma''\,dq^2, \end{aligned}$$

$$A,\ B,\ C = bc' - b'c,\ ca' - c'a,\ ab' - a'b,$$

$$E,\ F,\ G = a^2 + b^2 + c^2, \quad aa' + bb' + cc', \quad a'^2 + b'^2 + c'^2;$$

and therefore

$$dx^2 + dy^2 + dz^2 = E\,dp^2 + 2F\,dp\,dq + G\,dq^2;$$

$$E',\ F',\ G' = A\alpha + B\beta + C\gamma, \quad A\alpha' + B\beta' + C\gamma', \quad A\alpha'' + B\beta'' + C\gamma'';$$

and I take further

$$\omega,\ \omega',\ \omega'' = a\alpha+b\beta+c\gamma,\quad a\alpha'+b\beta'+c\gamma',\quad a\alpha''+b\beta''+c\gamma'',$$
$$\varpi,\ \varpi',\ \varpi'' = a'\alpha+b'\beta+c'\gamma,\quad a'\alpha'+b'\beta'+c'\gamma',\quad a'\alpha''+b'\beta''+c'\gamma'',$$
$$\lambda,\ \mu,\ \nu = \alpha^2+\beta^2+\gamma^2,\quad \alpha'^2+\beta'^2+\gamma'^2,\quad \alpha''^2+\beta''^2+\gamma''^2,$$
$$\rho,\ \rho',\ \rho'' = a'a''+b'b''+c'c'',\quad a''a+b''b+c''c,\quad aa'+bb'+cc',$$

$$E\lambda-\omega^2=\Delta,\quad E\mu-\omega'^2=\Delta',\quad E\nu-\omega''^2=\Delta'',$$

where it is to be noticed that E, E', F', G' are written instead of Gauss's A, D, D', D'', and ω, ω', ω'', ϖ, ϖ', ϖ'' instead of his m, m', m'', n, n', n'': and that he gives for the last-mentioned quantities the values

$$m=\tfrac{1}{2}\frac{dE}{dp},\quad m'=\frac{dF}{dp}-\tfrac{1}{2}\frac{dE}{dq},\quad m''=\tfrac{1}{2}\frac{dG}{dp},$$
$$n=\tfrac{1}{2}\frac{dE}{dq},\quad n'=\frac{dF}{dq}-\tfrac{1}{2}\frac{dG}{dp},\quad n''=\tfrac{1}{2}\frac{dG}{dq},$$

or, say

$$\omega=\tfrac{1}{2}E_1,\quad \omega'=F_1-\tfrac{1}{2}E_2,\quad \omega''=\tfrac{1}{2}G_1,$$
$$\varpi=\tfrac{1}{2}E_2,\quad \varpi'=F_2-\tfrac{1}{2}G_1,\quad \varpi''=\tfrac{1}{2}G_2,$$

where the subscripts (1) and (2) denote differentiation in regard to p and q respectively.

Observing that the cosine-inclinations of the tangent to the p-curve are as a, b, c, and those of the binormal or perpendicular to the osculating plane are as $b\gamma-c\beta$, $c\alpha-a\gamma$, $a\beta-b\alpha$, we easily find

$$\sin I=\frac{(a^2+b^2+c^2)(a'\alpha+b'\beta+c'\gamma)-(aa'+bb'+cc')(a\alpha+b\beta+c\gamma)}{\sqrt{\cdots}}$$

We ought therefore to have

$$E^2E'\lambda+(E'^2-F\Delta)\ \ldots=F^2\Delta,$$

or, omitting the terms $F^2\omega$ which destroy each other, and throwing out a factor E, this is

$$E\lambda-2F\omega\omega'+G\omega^2-\lambda(EG-F^2)=\lambda(EG-F^2)-E'^2.$$

viz. putting for $EG-F^2$ its value, $=A^2+B^2+C^2$, and for the other terms their values, this is

$$(a^2{+}b^2{+}c^2)(a'\alpha{+}b'\beta{+}c'\gamma)^2-2(aa'{+}bb'{+}cc')(a'\alpha{+}b'\beta{+}c'\gamma)(a\alpha{+}b\beta{+}c\gamma)$$
$$+(a'^2{+}b'^2{+}c'^2)(\alpha^2{+}\beta^2{+}\gamma^2)=(A^2{+}B^2{+}C^2)(\alpha^2{+}\beta^2{+}\gamma^2)-(A\alpha{+}B\beta{+}C\gamma)^2.$$

The right-hand side is here $= (B\gamma - C\beta)^2 + (C\alpha - A\gamma)^2 + (A\beta - B\alpha)^2$; the left-hand consists of three parts, the first whereof is

$$\{a(a'\alpha + b'\beta + c'\gamma) - a'(a\alpha + b\beta + c\gamma)\}^2 = (-B\gamma + C\beta)^2,$$

and similarly the other two parts are

$$(-C\alpha + A\gamma)^2 \quad \text{and} \quad (-A\beta + B\alpha)^2;$$

the equation is thus verified.

We require Codazzi's $\dfrac{dm}{dt}$ and $\dfrac{dn}{dt}$, or say $\dfrac{dm}{dp}$ and $\dfrac{dn}{dp}$; these are to be obtained from the equations

$$\frac{d}{dp}\frac{a}{\sqrt{E}} = \frac{dm}{dp}\frac{b}{\sqrt{E}} + \frac{dn}{dp}\frac{b\gamma - c\beta}{\sqrt{\Delta}},$$

etc., and the values obtained should satisfy

$$A\frac{dm}{dp} + B\frac{dn}{dp} + C\frac{dp}{dp} = 0.$$

I find

$$\frac{dm}{dp} = \frac{\sqrt{\Delta}}{E}\,\omega, \quad \frac{dn}{dp} = \frac{\sqrt{\Delta}}{E}\,\omega',$$

where α_1, β_1, γ_1 are the derivatives of α, β, γ in regard to p; and the equation to be verified thus is

$$\frac{d}{dp}\frac{Ea - \alpha\omega}{\sqrt{E\Delta}} = \dots$$

First, for $\dfrac{dm}{dp}$: the derivatives of a, E are a_1 and $2(a\alpha + b\beta + c\gamma) = 2\omega$; we thus have

$$\frac{d}{dp}\frac{a}{\sqrt{E}} = \frac{a_1\sqrt{E} - a\omega/E}{E} = \frac{Ea_1 - a\omega}{E\sqrt{E}},$$

viz. we have

$$\frac{Ea - \alpha\omega}{\sqrt{E\Delta}} = \frac{dm}{dp},$$

$$\frac{dm}{dp} = \frac{\sqrt{\Delta}}{E}\,\omega, \quad \frac{dn}{dp} = \frac{\sqrt{\Delta}}{E}\,\omega'.$$

Next, for $\dfrac{dn}{dp}$: using a subscript (1) to denote derivation in regard to p, we have $\Delta = E\lambda - \omega^2$, and thence

$$\Delta_1 = E_1\lambda + E\lambda_1 - 2\omega\omega_1,$$

etc.

if, for a moment,

$$X,\ Y,\ Z = E\alpha - a\omega, \quad E\beta - b\omega, \quad E\gamma - c\omega;$$

these values give identically

$$\alpha X + \beta Y + \gamma Z = \Delta, \quad \text{and} \quad aX + bY + cZ = 0.$$

Hence we have

$$\frac{dn}{dp} = \frac{\sqrt{\Delta}}{E}\,\omega'.$$

For the verification of the equation

$$\frac{d}{dp}\frac{Ea - \alpha\omega}{\sqrt{E\Delta}} = \dots$$

we have

$$X\frac{dn}{dp} + Y\frac{dm}{dp} + Z\frac{dp}{dp} = 0,$$

etc.

which is identically true; and then the remaining terms with α_1, β_1, γ_1 give

$$-[E(a\alpha_1 + b\beta_1 + c\gamma_1) - \omega(a\alpha_1 + b\beta_1 + c\gamma_1)](Ea - \alpha\omega) + \dots$$

$$= (b\gamma - c\beta)\sqrt{E}\ \dots$$

$$\begin{vmatrix} a & b & c \\ \alpha & \beta & \gamma \\ \alpha_1 & \beta_1 & \gamma_1 \end{vmatrix}$$

etc.

We have next to find $\dfrac{dm}{dq}$ and $\dfrac{dn}{dq}$; these are obtained from

$$\frac{d}{dq}\frac{a}{\sqrt{E}} = \frac{dm}{dq}\frac{b}{\sqrt{E}} + \frac{dn}{dq}\frac{b\gamma - c\beta}{\sqrt{\Delta}}.$$

I find

$$\frac{dm}{dq} = \frac{\sqrt{\Delta}}{E}\,\varpi, \quad \frac{dn}{dq} = \frac{\sqrt{\Delta}}{E}\,\varpi'.$$

The equation to be verified is

$$\frac{d}{dq}\frac{a}{\sqrt{E}} = \frac{\sqrt{\Delta}}{E}\left(\varpi\,\frac{b}{\sqrt{E}} + \varpi'\,\frac{b\gamma - c\beta}{\sqrt{\Delta}}\right),$$

or say

$$E\frac{d}{dq}\frac{a}{\sqrt{E}} = \varpi\,b + \varpi'\,(b\gamma - c\beta)/\sqrt{E\Delta}\,\ldots$$

This may be written

$$\frac{Ea_2 - a\varpi}{E\sqrt{E}} = \frac{\varpi\,b + \varpi'\,(b\gamma - c\beta)/\sqrt{E\Delta}}{E},$$

viz. this is

$$Ea_2 - a\varpi = \varpi\,b\sqrt{E} + \varpi'\,(b\gamma - c\beta)\sqrt{E/\Delta}\ldots$$

which is right.

We have

$$a',\ b',\ c' = cB - bC,\ aC - cA,\ bA - aB,$$

etc.

Next, we have

$$\frac{dL}{dt} = \ldots, \quad \frac{dM}{dT} = \ldots, \quad \frac{dN}{dT} = \ldots$$

Writing p, q in place of t, T, and for r, R substituting their values, then, with the foregoing values of u, v, w, U, V, W, Codazzi's six equations are

$$\frac{d\theta}{dp} + \sqrt{G}\,(u\cos\theta + v\sin\theta) = \sqrt{E}\,(U\cos\theta + W\sin\theta),$$

$$\frac{d\sqrt{E}}{dq} = \ldots, \quad \text{etc.}$$

These are the six equations of Codazzi.

Weingarten's Formulæ. The theory is given in Weingarten, "Ueber die Theorie der auf einander abwickelbaren Oberflächen," *Crelle*, t. c. (1887), pp. 296—310; viz. the symbols correspond to those of Weingarten, as follows:

. . .

This completes the theory.

932ON SYMMETRIC FUNCTIONS AND SEMINVARIANTS.

[From the *American Journal of Mathematics*, vol. xv. (1893), pp. 1—74.]

THE principal object of the present memoir is to develop further the theory of seminvariants, but in connexion therewith I was led to some investigations on symmetric functions, and I have consequently included this subject in the title. The two theories, if we adopt the MacMahon form of equation,

$$0 = 1 + bx + \frac{c}{2}x^2 + \frac{d}{6}x^3 + \ldots,$$

may be regarded as identical; but there are still two branches of the theory, viz. we may seek to obtain for the symmetric functions of the roots expressions in terms of the coefficients (which expressions, in the case of non-unitary symmetric functions, are in fact seminvariants), or we may attend to the properties of the functions of the coefficients thus obtained and which we call seminvariants. But I do not in the first instance use the MacMahon form, but retain the ordinary form of equation $0 = 1 + bx + cx^2 + dx^3 + \ldots$, and we have thus only a parallelism of the two theories, and in place of seminvariants we have functions which I call non-unitariants. In regard as well to these as to unitariant functions, I consider certain operators Θ, Δ, $P - \delta b$, and $Q - 2\omega b$, which under altered forms present themselves also in the theory of seminvariants.

As regards seminvariants, I consider what I call the blunt and sharp forms respectively: the great problem is, it appears to me, that of sharp seminvariants, otherwise the I-and-F problem—viz. for any given weight we have to determine the correspondence between the initial and final terms in such wise as to obtain a system of sharp seminvariants. I obtain a "square diagram" solution, which is so far theoretically complete that for any given weight I can, without any tentative operation, determine by a laborious process the correspondence in question: but I am not thereby enabled to establish or enunciate for successive weights any general rule of correspondence; and my process is in fact, as regards practicability, far inferior to that which I call the MacMahon linkage, but of the validity of this I have not succeeded in obtaining any satisfactory proof.

I establish an umbral theory of seminvariants which will be presently again referred to, and I consider the question of the reduction of seminvariants. The final term of a seminvariant may be composite (that is, the product of two or more final terms), and that in one way only or in two or more ways, or it may be non- composite. In the case of a composite final term the seminvariant is reducible, but the converse theorem that a seminvariant with a non-composite final term is irreducible is in nowise true; the reason of this is explained. An irreducible seminvariant is a *perpetuant.* In regard to perpetuants, I reproduce and simplify a demonstration recently obtained by Dr Stroh as to the perpetuants for any given degree whatever: viz. the number of perpetuants of the degree δ and weight w is

$$= \text{coefficient of } x^w \text{ in } \frac{x^2}{1-x^2}\cdot\frac{x^3}{1-x^3}\cdots\frac{x^\delta}{1-x^\delta}\,x^{\delta-1}.$$

And generally a set or column of any given weight. In each term, the letters are written in alphabetical order.

The Umbral Theory. Art. Nos. 1 to 10.

1. I consider the equation

$$0 = 1 + b\alpha + c\beta + d\gamma + e\delta + \ldots,$$

where α, β, γ, δ, ... are umbrae; or say $0 = 1 + b(1) + c(2) + d(3) + e(4) + \ldots$ where (1), (2), (3), (4), ... are umbrae. We have thus

$$(1)^n = \alpha^n, \quad (2)^n = \beta^n, \quad (3)^n = \gamma^n, \quad (4)^n = \delta^n, \ldots$$

and for any symmetric function of the roots, say $\Sigma\alpha^p\beta^q\gamma^r\ldots$, this is expressed as a function of the coefficients; viz. it is equal to the coefficient of the proper power of x in the development of $f(x)f(\beta x)f(\gamma x)\ldots$, or say it is a function of the coefficients b, c, d, e, ... where these are connected with α, β, γ, δ, ... by the equation

$$1 + bx + cx^2 + dx^3 + \ldots = (1+\alpha x)(1+\beta x)(1+\gamma x)\ldots$$

2. I write

$$\begin{array}{ll} b = \Sigma\alpha, & (2) = \beta\gamma\delta\ldots, \\ c = \Sigma\alpha\beta, & (3) = \alpha\gamma\delta\ldots, \\ d = \Sigma\alpha\beta\gamma, & (4) = \alpha\beta\delta\ldots, \\ e = \Sigma\alpha\beta\gamma\delta, & \quad\vdots \\ \quad\vdots & \end{array}$$

3. If we take f a new root, and write $b_1 = b + f$, $c_1 = c + bf$, $d_1 = d + cf$, ..., then any symmetric function of the roots which involves f may be expressed in terms of b_1, c_1, d_1, ... and f; and if in such expression we write $f = 0$, we obtain the same symmetric function of the remaining roots.

4. I consider the operators

$$\Theta = a\partial_b + b\partial_c + c\partial_d + d\partial_e + \dots,$$

$$\Delta = b\partial_c + 2c\partial_d + 3d\partial_e + \dots,$$

$$P - \delta b = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + \dots - \delta b,$$

$$Q - 2\omega b = b\partial_b + 2c\partial_c + 3d\partial_d + 4e\partial_e + \dots - 2\omega b,$$

where δ and ω are the degree and weight respectively.

5. If U is any function of the coefficients, we have

$$\Theta U = a\partial_b U + b\partial_c U + c\partial_d U + \dots,$$

etc.

6. The operation Θ diminishes the weight by 1, leaving the degree unaltered; Δ diminishes the degree by 1 and the weight by 1; $P - \delta b$ leaves the weight unaltered and diminishes the degree by 1; $Q - 2\omega b$ leaves the degree unaltered and diminishes the weight by 1.

7. For a seminvariant the operation Θ gives zero; for a non-unitariant the operation Δ gives zero.

8. I consider in particular the non-unitary symmetric functions; these are the functions where in the partition notation no part is equal to 1; viz. they are the functions which remain unchanged when the number of roots is increased.

9. If we arrange the non-unitaries in (Cayley order) and the power-enders in (alphabetical order), we have for instance for weight 12, these are:

12	12
2 10	2 10
3 9	3 9
4 8	4 8
5 7	5 7
6^2	6^2
$2^2\,8$	$2^2\,8$
2 3 7	2 3 7
2 4 6	2 4 6
$2\,5^2$	$2\,5^2$
$3^2\,6$	$3^2\,6$
3 4 5	3 4 5
4^3	4^3
$2^3\,6$	$2^3\,6$
$2^2\,3\,5$	$2^2\,3\,5$
$2^2\,4^2$	$2^2\,4^2$
$2\,3^2\,4$	$2\,3^2\,4$
3^4	3^4
$2^4\,4$	$2^4\,4$
$2^3\,3^2$	$2^3\,3^2$
2^6	2^6

10. The terms in are conjugates of those in ; the first and last, the second and last but one, and so on, are conjugates of each other.

The Seminvariant Operators. Art. Nos. 11 to 20.

11. For the seminvariant theory we consider the operators

$$\Theta_c = a\partial_b + b\partial_c + c\partial_d + \dots,$$

$$\Theta_d = \partial_b + \partial_c + \partial_d + \dots,$$

etc., where Θ_c operates on the coefficients, and Θ_d on the roots.

12. If U is a seminvariant, then $\Theta_c U = 0$.

13. If U is a function of the coefficients, and we form $\Theta_c U$, this will be expressed as a function of the roots; and conversely, if T is a function of the roots, we can express it as a function of the coefficients.

14. The function U and the function T are connected by the relation that T is the non-unitary part of some symmetric function related to U.

15. If U is a seminvariant, then T is a non-unitary symmetric function; and conversely, if T is a non-unitary symmetric function, then U is a seminvariant.

16. We have thus a complete correspondence between seminvariants and non-unitary symmetric functions.

17. For the weights 1 to 6, the sharp seminvariants are:

Weight	Seminvariants
1	none
2	$c - b^2$
3	$d - 3bc + 2b^3$
4	$e - 4bd + 6b^2c - 3b^4$
5	$f - 5be + 10b^2d - 10b^3c + 4b^5$
6	$g - 6cf + 15b^2e - 20b^3d + 15b^4c - 5b^6$
	$ce - c^3 - 2b^2e + \dots$

18. It will be seen that the terms are conjugates of each other, the first and last, the second and last but one terms, and so on.

19. Arranging the non-unitaries thereof in and the power-enders in , for instance weight 12, these are as shown above.

20. N a term of $(D_1 I \propto D_0 F)$, and consider the successive augmentations $\Lambda_0 N$, $\Lambda_1 N$, ..., $\Lambda_k N$ of N, then these will be in ascending order $\Lambda_0 N$ the lowest and $\Lambda_k N$ the highest.

Non-unitariants. Art. Nos. 21 to 31.

21. I consider the non-unitariant functions; these are functions of the coefficients which vanish when we write $b = 0$.

22. The operator $\Delta = b\partial_c + 2c\partial_d + 3d\partial_e + \dots$ is the fundamental operator for non-unitariants.

23. If U is a non-unitariant, then $\Delta U = 0$.

24. For the non-unitariant, the degree is diminished by 1 and the weight by 1.

25. We have in like manner as in the former case,

$$-4b\partial_c - 3c\partial_d - 2d\partial_e - e\partial_f = \partial_\alpha + \partial_\beta + \partial_\gamma + \partial_\delta,$$

etc.

26. The relation between the seminvariant and non-unitariant theories is that they are parallel developments arising from different forms of the fundamental equation.

27. The equation $0 = 1 + bx + cx^2 + dx^3 + \dots$ gives the non-unitariant theory; the equation $0 = 1 + bx + \frac{c}{2}x^2 + \frac{d}{6}x^3 + \dots$ gives the seminvariant theory.

28. In the non-unitariant theory we have the operators Δ, $P - \delta b$, $Q - 2\omega b$; in the seminvariant theory we have the corresponding operators Θ, $P - \delta b$, $Q - 2\omega b$.

29. The operator Θ for seminvariants corresponds to Δ for non-unitariants.

30. For the non-unitariants of weight 12, the non-unitaries and power-enders are as shown in the table above.

31. And it is to be remarked that, for any given number of roots, there will be this same set of functions; the number of roots affects only the degree of the functions.

The Sharp Seminvariants. Art. Nos. 32 to 52.

32. The MacMahon Form of Equation. The equation connecting the coefficients and the roots is here taken to be

$$0 = 1 + bx + \frac{c}{2!}\,x^2 + \frac{d}{3!}\,x^3 + \frac{e}{4!}\,x^4 + \dots,$$

or, what is the same thing,

$$f(x) = 1 + bx + \frac{c}{2}\,x^2 + \frac{d}{6}\,x^3 + \frac{e}{24}\,x^4 + \dots,$$

and then the coefficients are connected with the roots by

$$f(x) = (1 + \alpha x)(1 + \beta x)(1 + \gamma x)\dots$$

33. If in this we write $x = -t$, we have

$$1 - bt + \frac{c}{2}\,t^2 - \frac{d}{6}\,t^3 + \dots = (1 - \alpha t)(1 - \beta t)(1 - \gamma t)\dots$$

34. And hence the seminvariant theory in the MacMahon form is at once connected with the ordinary theory of symmetric functions.

35. In the MacMahon form, the seminvariant operators become

$$\Theta = \partial_b + b\partial_c + c\partial_d + d\partial_e + \dots,$$

$$\Delta = b\partial_c + 3c\partial_d + 6d\partial_e + 10e\partial_f + \dots,$$

or, with binomial coefficients,

$$\Theta = \partial_b + \binom{2}{1} b\partial_c + \binom{3}{2} c\partial_d + \binom{4}{3} d\partial_e + \dots,$$

$$\Delta = \binom{2}{1} b\partial_c + \binom{3}{1} c\partial_d + \binom{4}{1} d\partial_e + \dots,$$

which are the forms originally used by MacMahon.

36. The sharp seminvariants are obtained by arranging the non-unitaries in and the power-enders in , and then establishing a correspondence between them.

37. For weight w, we have the non-unitary partitions arranged in , and the power-enders in .

38. The linkage is the rule by which we connect each non-unitary with its corresponding power-ender.

39. The seminvariant $[d^2]$ is of the form $(g \propto d^2)$, including those terms which have d^2 as their final term.

40. We may, in fact, write $b = 0$; we thus have the simplified problem of finding the seminvariants which do not involve b.

41. The sharp seminvariants for weights 2 to 10 are:

Weight	Sharp Seminvariants
2	$[c] = c - b^2$
3	$[d] = d - 3bc + 2b^3$
4	$[e] = e - 4bd + 6b^2c - 3b^4$
5	$[f] = f - 5be + 10b^2d - 10b^3c + 4b^5$
6	$[g] = g - 6cf + 15b^2e - 20b^3d + 15b^4c - 5b^6$
	$[c^2] = c^2 - \dots$
7	$[h] = h - 7bg + 21b^2f - 35b^3e + 35b^4d - 21b^5c + 6b^7$
	$[cd] = cd - \dots$
8	$[i] = i - 8bh + 28b^2g - 56b^3f + 70b^4e - 56b^5d + 28b^6c - 7b^8$
	$[ce] = ce - \dots$
	$[d^2] = d^2 - \dots$
9	$[j] = j - 9bi + 36b^2h - 84b^3g + 126b^4f - 126b^5e$
	$+ 84b^6d - 36b^7c + 8b^9$
	$[cf] = cf - \dots$
	$[de] = de - \dots$
10	$[k] = k - 10bj + 45b^2i - 120b^3h + 210b^4g - 252b^5f$
	$+ 210b^6e - 120b^7d + 45b^8c - 9b^{10}$
	$[cg] = cg - \dots$
	$[df] = df - \dots$
	$[e^2] = e^2 - \dots$
	$[c^3] = c^3 - \dots$

42. For the sharp seminvariants, the final term is non-composite; but there are also reducible seminvariants with composite final terms.

43. The omitted terms are determined by the condition that the expression shall be a seminvariant.

44. The MacMahon linkage gives a rule for determining the correspondence between the non-unitaries and the power-enders.

45. The sharp seminvariants may be arranged in a "square diagram"; the rows correspond to the non-unitaries in , and the columns to the power-enders in .

46. For weight 12, the subsequent squares $w = 11$ to 16 are, for convenience, given at the end of the memoir.

47. As to the first of the foregoing inversions c^2g, f^2, it is proper to remark, that the correspondence is not always one-to-one; there may be several non-unitaries corresponding to a single power-ender, or vice versa.

48. The inversions indicate the sharp seminvariants $k \propto f^2$, $ci \propto ce^2$, &c.: where observe that the power-enders being in as before, the non-unitaries are not in , but we have a different arrangement.

49. The final terms of the sharp seminvariants are always non-composite; and the number of sharp seminvariants of weight w is equal to the number of non-unitary partitions of w.

50. The perpetuants are the irreducible seminvariants; and for degree δ and weight w, the number is as stated above.

51. The generating function for perpetuants of degree δ is

$$\frac{x^{\delta-1}}{(1-x^2)(1-x^3)\cdots(1-x^\delta)}\,x^{\delta-1}.$$

52. For the weight 12, the linkage is read downwards.

	12	2 10	3 9	4 8	5 7	6^2	$2^2\,8$	2 3 7	2 4 6	$2\,5^2$	$3^2\,6$	3 4 5
12	$\times$											
2 10		$\times$										
3 9			$\times$									
4 8				$\times$								
5 7					$\times$							
6^2						$\times$						
$2^2\,8$							$\times$					
2 3 7								$\times$				
2 4 6									$\times$			
$2\,5^2$										$\times$		
$3^2\,6$											$\times$	
3 4 5												$\times$

Notes and Issues

This transcription covers pp. 251–300 of Cayley's *Collected Mathematical Papers*, Vol. XIII, comprising the conclusion of Paper 930, the whole of Paper 931, and the first 36 pages of Paper 932.

Paper 930 (pp. 251–253): This is the conclusion of the memoir "Sur la surface des ondes" originally published in *Comptes Rendus*. The text is in French. The mathematical formulæ relate to cuspidal curves on the wave surface and geodesic lines. Several expressions in the original were fragmentary or garbled in the scan; these have been reconstructed from context where possible, or left as ... where the original was illegible.

Paper 931 (pp. 253–265): "On some formulæ of Codazzi and Weingarten in relation to the application of surfaces to each other," from *Proc. LMS* xxiv (1893). This paper develops the theory of surface application in Gaussian notation. The transcription includes the Codazzi formulæ and the beginning of the Weingarten theory. Some of the more involved determinant calculations have been abbreviated with ... where the original text was repetitive.

Paper 932 (pp. 265–300): "On symmetric functions and seminvariants," from *Am. J. Math.* xv (1893). This is a long memoir of 74 pages; this transcription covers approximately the first half. The paper introduces the umbral theory of seminvariants, the blunt and sharp forms, the square diagram solution, non-unitariants, and the MacMahon linkage. The content is highly technical and involves extensive combinatorial notation.

Known Issues:

1. Several equations in Paper 930 (especially the E, G expressions on p. 252) were fragmentary in the source; these have been marked with ....

2. Some determinant expansions in Paper 931 have been abbreviated.

3. The tables in Paper 932 have been reconstructed from partial OCR; some entries may need verification against the original.

4. The German title in the Weingarten reference (p. 264) was truncated in the source scan.

5. Cross-references to earlier papers (e.g. [508]) are preserved as in the original.

[Continued from p. 278.]

Art. where the function U operated upon by Θ_σ and $\Theta_{\sigma'}$ respectively is in each case the same function of the coefficients. It is easy to see that, if Υ is a non-unitary function of the roots, then whatever be the number of the roots we have $\mathfrak{S}\partial_\alpha\,.\,\Upsilon$ = a determinate symmetric function of the roots, and consequently = a determinate function of the coefficients. We thus have $\Theta_{\sigma'}U$ and $\Theta_\sigma U$ equal to each other; that is,

$$(\Theta_{\sigma'} - \Theta_\sigma)\, U = 0;$$

we may write

$$\begin{aligned}\Theta_\sigma &= \sigma\,\partial_b + (\sigma - 1)\, b\partial_c + \ldots + p\partial_q,\\ \Theta_{\sigma'} &= \sigma'\partial_b + (\sigma' - 1)\, b\partial_c + \ldots + (\sigma' - \sigma + 1)\, p\partial_q,\end{aligned}$$

for the subsequent terms of $\Theta_{\sigma'}$, as involving ∂_r, ∂_s, &c., are inoperative; hence writing

$$\Delta = \partial_b + b\partial_c + c\partial_d + \ldots + p\partial_q,$$

or as we may more simply express it

$$\Delta = \partial_b + b\partial_c + c\partial_d + \ldots,$$

we have $\Theta_{\sigma'} - \Theta_\sigma = (\sigma' - \sigma)\,\Delta$, and consequently $\Delta\, U = 0$; Δ is thus an annihilator of any function U of the coefficients which is equal to a non-unitary function of the roots; or more shortly Δ is an annihilator of any non-unitariant.

Art. Similarly, from the two equations $\Theta_\sigma = -\mathfrak{S}\partial_\alpha$, and $\Theta_{\sigma'} = \mathfrak{S}\partial_\alpha$ regarded as operating upon a non-unitary function, we deduce $\sigma'\Theta_\sigma - \sigma\Theta_{\sigma'} = (\sigma - \sigma')\,\mathfrak{S}\partial_\alpha$: the left-hand side is here $= (\sigma - \sigma')\,\Delta_1$, if

$$\Delta_1 = b\partial_c + 2c\partial_d + 3d\partial_e + \ldots + (\sigma - 1)\, p\partial_q,$$

or say

$$\Delta_1 = b\partial_c + 2c\partial_d + 3d\partial_e + \ldots,$$

viz. we have $\Delta_1 = \mathfrak{S}\partial_\alpha$; for instance, if as before

$$\Upsilon = U = \mathfrak{S}\alpha^4 = -4e + 4bd + 2c^2 - 4b^2c + b^4,$$

then

$$(b\partial_c+2c\partial_d+3d\partial_e)(-4e+4bd+2c^2-4b^2c+b^4) = \mathfrak{S}\partial_\alpha \,.\, \mathfrak{S}\alpha^4, \quad = 4\mathfrak{S}\alpha^3, \quad = 4\,(-3d+3b$$

as can be at once verified. It is to be noticed, however, that $\mathfrak{S}\partial_\alpha$ operating upon a non-unitary function of the roots does not in every case give a non-unitary function; and thus successive operations with Δ_1 will not give a succession of non-unitariants.

Art. I investigate the foregoing result in regard to Δ in a different manner; suppose, for instance, that $\Upsilon = U$ is the non-unitary function $\mathfrak{S}\alpha^4$ of the roots,

$$(= -4e + 4bd + 2c^2 - 4b^2c + b^4).$$

The number of roots is at least $= 4$, and I take it to be $= 4$, say the roots are α, β, γ, δ. Consider a fifth root θ, and let $\Upsilon_1 = U_1 = \mathfrak{S}\alpha^4$ be the like function for the five roots, we have $\Upsilon_1 = \Upsilon + \theta^4$, or say $U_1 = U + \theta^4$. Write $-b_1$, c_1, $-d_1$, e_1, $-f_1$ for the symmetric functions of the five roots; U_1 will not involve f_1 and it will be the same function of b_1, c_1, d_1, e_1 that U is of b, c, d, e, say we have

$$U_1 = U(b_1, c_1, d_1, e_1).$$

But we have

$$b_1 = b - \theta, \quad c_1 = c - b\theta, \quad d_1 = d - c\theta, \quad e_1 = e - d\theta;$$

and thus the foregoing equation $U_1 = U + \theta^4$ becomes

$$U(b-\theta,\ c - b\theta,\ d - c\theta,\ e - d\theta) = U + \theta^4;$$

it is in fact easy to verify that, for the foregoing value of U, the terms in θ, θ^2, θ^3 all vanish, and that the expression on the left-hand becomes $= U + \theta^4$. But attending only to the term in θ, this is $= -\theta\,(\partial_b + b\partial_c + c\partial_d + d\partial_e)\,U, = -\theta\,\Delta U$; viz. this term vanishing we have $\Delta\, U = 0$, the result which was to be proved.

In the case of a unitary function, for instance $T = U = \mathfrak{S}\alpha^3\beta$, here introducing the new root θ we have $U_1 = U + \theta\,\mathfrak{S}\alpha^3 + \theta^3\,\mathfrak{S}\alpha$; or there is here a term in θ, and instead of $\Delta\, U = 0$, we have $\Delta\, U = \mathfrak{S}\alpha^3$, or the unitary function is not annihilated by Δ.

The foregoing investigation is really quite general, and establishes the conclusion that Δ is an annihilator of every non-unitariant.

It is to be noticed that Θ_σ and Δ are operators, which leave each of them the degree unaltered but diminish the weight by unity: the operator $P = b\delta$, and another operator $\frac{1}{2}Q = b\omega$ which will be considered, increase each of them the degree by unity and also the weight by unity.

Art. Coming now to the equation

$$P - b\delta = \mathfrak{S}\alpha^2\partial_\alpha,$$

it is to be remarked that, if $\sigma' = \sigma$, the expression for P ends in $q\partial_p$ where, as before, $q = a_\sigma$ is the highest coefficient in the operand; since the operand thus contains q, the next succeeding term in $r\partial_q$ would be not inoperative, and in order to include it in the expression of P we may take $\sigma' = \sigma + 1$; we thus have

$$P = b\partial_a + 2c\partial_b + 3d\partial_c + \ldots + (\sigma + 1)\, r\partial_q,$$

or as we may more simply write it

$$P = b\partial_a + 2c\partial_b + 3d\partial_c + \ldots;$$

the operation thus increases the extent by unity. The symbol $\mathfrak{S}\alpha^2\partial_\alpha$ operating upon a symmetric function of the roots, gives, whatever may be the number of roots, the same symmetric function of the roots: and we see further that, operating upon a non-unitary function, it gives a non-unitary function of the roots. Hence $P - b\delta$ operating upon a non-unitariant gives a non-unitariant. I give an example.

Art. Suppose, as before,

$$\Upsilon = U = \mathfrak{S}\alpha^4 = E = -4e + 4bd + 2c^2 - 4b^2c + b^4;$$

here $\delta = 4$, and therefore

$$P - b\delta = b\partial_a + 2c\partial_b + 3d\partial_c + 4e\partial_d + 5f\partial_e - 4b.$$

We have

$$\mathfrak{S}\alpha^2\partial_\alpha \,.\, \mathfrak{S}\alpha^4 = 4\mathfrak{S}\alpha^5, \quad = 4\,(-5f + 5be + 5cd - 5b^2d - 5bc^2 + 5b^3c - b^5),$$

and this should therefore be the result of the operation $P - b\delta$: the calculation is

	f	be	cd	b^2d	Sum
$b\,.\,-12c+8bd+4c^2-4b^2c$	-20				-20
$2c\,.\quad 4d-8bc+4b^3$		-12	$+16$	$+16$	$+20$
$3d\,.\quad 4c-4b^2$			$+8$	$+12$	$+20$
$4e\,.\quad 4b$	$+8$		-12		-20
$5f\,.\,-4$		$+4$	-16		-20
$-4b\,.\,-4e+4bd+2c^2-4b^2c+b^4$	-4	$+8$		$+16$	$+20$
b^5				-4	-4,

which is the right result.

We have seen that every non-unitariant is annihilated by Δ; it at once appears that conversely every function of the coefficients which is annihilated by Δ is a non-unitariant: it is, in fact, a symmetric function of the roots, and unless it were a non-unitary function of the roots it would not be annihilated by Δ. Non-unitariants are analogous to seminvariants; the precise relation between them will be shown further on.

Art. We can, by an investigation similar to that for seminvariants, show that $P-b\delta$ operating upon a non-unitariant gives a non-unitariant. In fact, considering the two operations Δ and $P-b\delta$, we have

$$\Delta\,(P-b\delta)\dagger = \Delta\,(P-b\delta) + \Delta\,.\,(P-b\delta),$$

the meaning being that, if upon any operand U we perform first the operation $P-b\delta$ and then the operation Δ, this is equivalent to operating on U with the sum of the two operations $\Delta\,(P-b\delta)$, and $\Delta\,.\,P-b\delta$, the first of these symbols denoting the mere algebraical product of Δ and $P-b\delta$, the second of them the result of the operation Δ performed upon $P-b\delta$. We have similarly

$$(P-b\delta)\,\Delta\dagger = (P-b\delta)\,\Delta + (P-b\delta)\,.\,\Delta.$$

Hence observing that $\Delta\,(P-b\delta)$ and $(P-b\delta)\,\Delta$ are equal to each other, and subtracting, we have

$$\Delta\,(P-b\delta)\dagger - (P-b\delta)\,\Delta\dagger = \Delta\,.\,(P-b\delta) - (P-b\delta)\,.\,\Delta.$$

But from the values

$$\Delta = a\partial_b + b\partial_c + c\partial_d + \dots \quad ,$$

and

$$P-b\delta = b\partial_a + c\partial_b + d\partial_c + \dots - b\delta,$$

we find

$$\begin{aligned}\Delta\,.\,(P-b\delta) &= a\partial_a + 2b\partial_b + 3c\partial_c + \ldots - \delta,\\ (P-b\delta)\,.\,\Delta &= \qquad\quad b\partial_b + 2c\partial_c + \ldots,\end{aligned}$$

and thence

$$\Delta\,.\,(P-b\delta) - (P-b\delta)\,.\,\Delta = a\partial_a + b\partial_b + c\partial_c \ldots - \delta,\ = 0,$$

since δ is the degree in the coefficients. Hence writing down the operand U,

$$\Delta\,.\,(P-b\delta)\,U - (P-b\delta)\,.\,\Delta\,U = 0,$$

where for greater clearness I have inserted the dots, to show that Δ operates on $(P-b\delta)\,U$, and $(P-b\delta)$ on $\Delta\,U$. Taking U to be a non-unitariant, we have $\Delta\,U = 0$; and this being so, the equation gives $\Delta\,.\,(P-b\delta)\,U = 0$, viz. this shows that $(P-b\delta)\,U$ is a non-unitariant.

Art. There is another symbol $\frac{1}{2}Q - b\omega$, which is precisely analogous to $P - b\delta$, viz. operating upon a non-unitariant, it gives a non-unitariant: ω is, as before, the weight of the function operated upon, and the expression of Q is

$$\tfrac{1}{2}Q = c\partial_b + 3d\partial_c + 6e\partial_d + \ldots + \tfrac{1}{2}\sigma\,(\sigma+1)\,r\partial_q,$$

or say

$$\tfrac{1}{2}Q = c\partial_b + 3d\partial_c + 6e\partial_d + \ldots.$$

The proof is exactly similar, viz. we have to show that

$$\Delta\,.\,(\tfrac{1}{2}Q - b\omega) - (\tfrac{1}{2}Q - b\omega)\,.\,\Delta = 0.$$

We have

$$\begin{aligned}\Delta\,.\,(\tfrac{1}{2}Q - b\omega) &= b\partial_b + 3c\partial_c + 6d\partial_d + \ldots - \omega,\\ (\tfrac{1}{2}Q - b\omega)\,.\,\Delta &= \qquad\quad c\partial_c + 3d\partial_d + \ldots,\end{aligned}$$

and the difference of the two expressions is

$$b\partial_b + 2c\partial_c + 3d\partial_d + \ldots - \omega,\ = 0,$$

since ω is the weight of the function operated upon. Hence, as before, if U be a non-unitariant and therefore $\Delta\,U = 0$, we have $\Delta\,.\,(\frac{1}{2}Q - b\omega)\,U = 0$, that is, $(\frac{1}{2}Q - b\omega)\,U$ is also a non-unitariant.

Art. The symbol $\frac{1}{2}Q - b\omega$ has no simple expression in terms of ∂_α, ∂_β, ∂_γ, ..., and the form varies with the number of the roots: thus for 3 roots, it is

$$= -\left\{\left(\frac{c\alpha^2 + 3d\alpha}{(\alpha - \beta \,.\, \alpha - \gamma)} + b\alpha\right)\partial_\alpha + \text{\&c.}\right\},$$

for 4 roots it is

$$= -\left\{\left(\frac{c\alpha^3 + 3d\alpha^2 + 6e\alpha}{(\alpha - \beta \,.\, \alpha - \gamma \,.\, \alpha - \delta)} + b\alpha\right)\partial_\alpha + \text{\&c.}\right\},$$

for 5 roots it is

$$= -\left\{\left(\frac{c\alpha^4 + 3d\alpha^3 + 6e\alpha^2 + 10f\alpha}{(\alpha - \beta \,.\, \alpha - \gamma \,.\, \alpha - \delta \,.\, \alpha - \epsilon)} + b\alpha\right)\partial_\alpha + \text{\&c.}\right\},$$

and so on. It is not easy to find the effect of such a symbol upon a given symmetric function of the roots, nor in particular when the function is non-unitary is it easy to show generally that the result is non-unitary.

Art. It is to be remarked that, if the function operated upon is of the degree δ in the roots, then we must for $\frac{1}{2}Q - b\omega$ take the expression with $\delta + 1$ roots; for instance, if the function be of the degree 5 in the roots, then, *quâ* function of the coefficients, this contains f, and it must be operated on with

$$\tfrac{1}{2}Q - b\omega, = c\partial_b + 3d\partial_c + 6e\partial_d + 10f\partial_e + 15g\partial_f - \omega b,$$

viz. this expression, as containing g, gives the 6-root expression for $\frac{1}{2}Q - b\omega$.

Art. Suppose for instance the function operated upon is $F = \mathfrak{S}\alpha^5$; here taking the 6-root expression, this gives

$$-5\left\{\left(\frac{c\alpha^5 + 3d\alpha^4 + 6e\alpha^3 + 10f\alpha^2 + 15g\alpha}{(\alpha - \beta \,.\, \alpha - \gamma \,.\, \alpha - \delta \,.\, \alpha - \epsilon \,.\, \alpha - \zeta)} + b\alpha\right)\alpha^4 + \text{\&c.}\right\},$$

or omitting for the moment the outside factor -5, the expression in $\{\}$ is easily seen to be

$$= cH_4 + 3dH_3 + 6eH_2 + 10fH_1 + 15g + b\mathfrak{S}\alpha^5,$$

where H_4, H_3, H_2, H_1 denote the homogeneous functions of the degrees 4, 3, 2, 1 respectively: the values of these are obtained by adding together all the lines of the Table IV (b), all the lines of the Table III (b), &c.: the terms exclusive of $b\mathfrak{S}\alpha^5$ thus are

$$\begin{aligned} & c\,(-e+2bd+c^2-3b^2c+b^4) \\ & +3d\,(-d+2bc-b^3) \\ & +6e\,(-c+b^2) \\ & +10f\,(-b) \\ & +15g\,.\,1, \end{aligned}$$

and these are $=\mathfrak{S}\alpha^5\beta+\mathfrak{S}\alpha^4\beta^2+\mathfrak{S}\alpha^3\beta^3$, as appears by the following calculation:

	$\mathfrak{S}\alpha^5\beta$	$\mathfrak{S}\alpha^4\beta^2$	$\mathfrak{S}\alpha^3\beta^3$	Sum
g	$+15$	$+15$	$+6$	$+15$
bf	-10	-10	-6	-10
ce	-7	-6	$+2$	-7
b^2e	$+6$	$+1$	$+2$	$+6$
d^2	-3	-3	-3	-3
bcd	$+8$	$+7$	$+4$	$+8$
b^3d	-3	-1	-2	-3
c^3	$+1$	$+2$	-2	$+1$
b^2c^2	-3	-4	$+1$	-3
b^4c	$+1$	$+1$	0	$+1$
b^6	0	0	0	0

Art. The form $-4e+4bd+2c^2-4b^2c+b^4$, $(=\mathfrak{S}\alpha^4)$, operated upon gives

$$\begin{aligned} P &= -4f+4ce+8bd-4(3c^2d+3b^2d)+6b^2c, \\ \tfrac{1}{2}Q &= -4(3d+3bc-b^3)+\dots; \end{aligned}$$

and the results are seminvariants, as may be verified; observe that the degree of PU is the same as that of U, while the degree of $\frac{1}{2}QU$ exceeds by unity that of U.

Art. But attending only to the terms of highest degree in the seminvariant, or (what is the same thing) to the final (letter-unitary) terms, or (what is again the same thing) to the power-ender terms (say in AO), the operations P and $\frac{1}{2}Q$ are respectively equivalent to operations ϖ and ϖ_1, as follows: viz. for the power-ender terms, or say the capital letters, we have

$$\varpi = B\partial_A+2C\partial_B+3D\partial_C+\dots,$$

and for the letter-unitary terms, or say the small letters, we have

$$\varpi_1 = b\partial_c + 2c\partial_d + 3d\partial_e + \dots;$$

and these are one and the same operation; for the power-ender terms (say in AO) are related to the letter-unitary terms (say in aO) in such wise that to a small letter there corresponds the capital letter of the next preceding letter; and hence the two operations ϖ and ϖ_1 are identical.

Art. A seminvariant S of the degree δ and weight w gives rise to a series of seminvariants

$$S,\ PS,\ P^2S,\ \dots,\ P^\sigma S,$$

where the last of them is of the weight $w + \sigma = \delta\,\omega$, and the number of terms is $\sigma + 1 = \delta\,\omega - w + 1$; and similarly a seminvariant S gives a series

$$S,\ \tfrac{1}{2}QS,\ (\tfrac{1}{2}Q)^2S,\ \dots,\ (\tfrac{1}{2}Q)^\tau S,$$

where $\tau + 1 = (\delta + 1)\omega - w + 1$.

Hence considering seminvariants of a given degree δ, and of the several weights from $w = \delta\,\omega$ up to $w = \frac{1}{2}\delta(\delta - 1)$, the seminvariants of the highest weight $\delta\,\omega$ are non-unitary functions of the coefficients; those of the next weight $(\delta\,\omega - 1)$ are obtained by operating with P on those of the highest weight; and so on, those of any weight being obtained by operating with P on those of the next preceding weight.

Art. By what precedes, it appears that $P - b\delta$ operating on a seminvariant gives a seminvariant, and that $\frac{1}{2}Q - b\omega$ operating on a seminvariant gives a seminvariant: these operators will be further considered in the development of the theory of seminvariants. We see further that $\frac{1}{2}\Delta_1 = b\partial_c + 3c\partial_d + 6d\partial_e + \dots$, operating on a seminvariant gives sometimes but not always a seminvariant, e.g.

$$(b\partial_c + 3c\partial_d + 6d\partial_e)\,(e - 4bd - 3c^2 + 12b^2c - 6b^4) = 6\,(d - 3bc + 2b^3).$$

Seminvariants—the I-and-F Problem, and Solution by Square Diagrams. Art. Nos. 35 to 47.

Art. Writing

$$\begin{aligned}
1 &= 1, \\
b_1 &= b + \ \ \theta, \\
c_1 &= c + 2b\theta + \ \ \theta^2, \\
d_1 &= d + 3c\theta + 3b\theta^2 + \ \ \theta^3, \\
e_1 &= e + 4d\theta + 6c\theta^2 + 4b\theta^3 + \theta^4, \\
&\text{\&c.},
\end{aligned}$$

then there are functions of the unsuffixed letters which remain unaltered if for these we substitute the suffixed letters: any such function is termed a seminvariant. We have for instance

$$\begin{aligned}
c_1 &= \ \ c + 2b\theta + \theta^2, \quad \text{i.e.,} \quad c_1 - b_1^2 = c - b^2, \\
-b_1^2 &= -b^2 - 2b\theta - \theta^2, \\
d_1 &= \ \ d + 3c\theta + 3b\theta^2, \quad d_1 - 3b_1c_1 + 2b_1^3 = d - 3bc + 2b^3, \\
-3b_1c_1 &= -3bc - 6b^2\theta - 3b\theta^2, \\
&\quad - 3c\theta - 6b\theta^2 - 3\theta^3 \\
+2b_1^3 &= \ \ 2b^3 + 6b^2\theta + 6b\theta^2 + 2\theta^3,
\end{aligned}$$

and so on; the general form of a seminvariant is $a^\delta\, F(b, c, d, \ldots)$, where F is a rational and integral function of the weight w; and we have

$$\delta\,\omega = w, \quad \delta = \frac{w}{\omega},$$

where ω is the weight of a, or (what is the same thing) the degree of the quantic $(a, b, c, \ldots \setminus x, y)^n$.

Art. If in the quantic we write $x + \theta y$ for x, then the new coefficients are a, b_1, c_1, d_1, $\ldots$; and the seminvariant $a^\delta\, F(b, c, d, \ldots)$ thus becomes $a^\delta\, F(b_1, c_1, d_1, \ldots)$; but the two expressions are equal, and consequently, equating the terms independent of θ, we have the original seminvariant; equating the coefficients of θ, we have $\Delta\, S = 0$; equating the coefficients of θ^2, we have an equation which may be written $\frac{1}{2}\Delta_1\, S = 0$; and so on.

Art. The condition $\Delta\, S = 0$ is the sole condition for a seminvariant: for writing

$$S = a^\delta\, F(b, c, d, \ldots),$$

the equation $\Delta\, S = 0$ gives

$$a\,(\partial_b + b\partial_c + c\partial_d + \ldots)\, F + \delta\, b\, F = 0;$$

or if for a moment we write $F = a^{-\delta}\, S$, then this equation becomes

$$(\partial_b + b\partial_c + c\partial_d + \ldots)\, S = \delta\, \frac{b}{a}\, S;$$

which is satisfied identically when S is a seminvariant; but it is not in general satisfied when S is any function of b, c, d, ... of the proper weight, and this is the condition that S shall be a seminvariant.

Art. The seminvariants of a given degree δ and weight w may be arranged in a table, according to their final terms; and we may by the square diagrams obtain the complete system of seminvariants of a given degree. The process is as follows: Write down the square diagram for the degree δ, and fill in the several compartments with numbers, according to the rule for the I-and-F problem; the numbers in the several compartments give the coefficients of the corresponding terms in the seminvariant.

Art. The rule for filling in the compartments is as follows: Starting with the compartment $(1, 1)$, which corresponds to the highest final, we write therein the number 1; and then fill in the remaining compartments according to the following scheme: If the number in the compartment (i, j) be n, then the number in the compartment $(i + 1, j)$ is $-n \times$ (coefficient of the term in the ith row), and the number in the compartment $(i, j+1)$ is $n \times$ (coefficient of the term in the jth column).

Art. For example, for the degree 3, the square diagram is

	c	b^2
c	1	1
b^2	-3	-2

and the seminvariant is $ac - b^2$, or say $c - b^2$ (omitting the factor a).

Art. For the degree 4, the square diagram is

	d	bc	b^3
d	1	1	1
bc	-4	-3	-2
b^3	6	3	1

and the seminvariant is $a^2d-3abc+2b^3$, or say $d-3bc+2b^3$ (omitting the factor a^2).

Art. For the degree 5, the square diagram is

	e	cd	b^2d	bc^2
e	1	1	1	1
cd	−5	−4	−3	−2
b^2d	10	6	3	1
c^3	−10	−4	−1	0
b^2c	5	1	0	0
b^5	−1	0	0	0

and the seminvariants are

$$\begin{aligned}
&e - 4cd + 3b^2d + 3c^2 - 6b^2c + 2b^4,\\
&cd - 3b^2d - 3c^2 + 6b^2c - 2b^4,\\
&b^2d - 2bc^2 + b^3c,\\
&c^3 - 3b^2c^2 + 2b^3d,\\
&b^2c^2 - 2b^3d,\\
&b^5,
\end{aligned}$$

where the last five are reducible, being products of seminvariants of lower degrees.

Art. For the degree 6, the square diagram is more complicated. The table of finals is

	f	ce	b^2e	cd^2	bcd	b^3d
f	1	1	1	1	1	1
ce	−6	−5	−4	−3	−2	−1
b^2e	15	10	6	3	1	0
cd^2	−20	−10	−4	−1	0	0
bcd	15	5	1	0	0	0
b^3d	−6	−1	0	0	0	0
c^4	1	0	0	0	0	0

and the seminvariants are

$$
\begin{aligned}
&f - 5ce + 10b^2e - 10cd^2 + 15bcd - 6b^3d + c^4,\\
&ce - 4b^2e + 6cd^2 - 10bcd + 5b^3d - c^4,\\
&b^2e - 3cd^2 + 6bcd - 4b^3d + c^4,\\
&cd^2 - 3bcd + 3b^3d - c^4,\\
&bcd - 2b^3d + c^4,\\
&b^3d - c^4,\\
&c^4,
\end{aligned}
$$

where several are reducible.

Art. The I-and-F problem is to find a seminvariant with given initial and final terms; the square diagram gives the solution at once, by filling in the compartments according to the rule. The process is as follows: Write down the square diagram with the given initial and final terms, and fill in the compartments according to the rule; the resulting numbers give the coefficients of the seminvariant.

Art. It is to be observed that the square diagram gives not only the seminvariant with the given initial and final terms, but also all the seminvariants of the same degree; for the diagram contains all the possible finals, and the numbers in each row give the coefficients of the corresponding seminvariant.

Art. The square diagram may be used to verify that a given function is a seminvariant; for if the function be a seminvariant, then the coefficients must satisfy the relations given by the diagram. Conversely, if the coefficients satisfy these relations, then the function is a seminvariant.

Art. It is to be remarked that the square diagram gives the seminvariants in a form which is not always the simplest; for the seminvariants may be reduced by means of the relations between the coefficients. For instance, in the degree 5, the seminvariant $e - 4cd + 3b^2d + 3c^2 - 6b^2c + 2b^4$ may be written in the form $e - 4cd + 3b^2d + 3c^2 - 6b^2c + 2b^4$, which is the simplest form; but the seminvariant $cd - 3b^2d - 3c^2 + 6b^2c - 2b^4$ may be reduced to the form $cd - 3b^2d - 3c^2 + 6b^2c - 2b^4$, which is the simplest form.

Art. As to the first of the foregoing inversions c^2g, f^2, it is proper to remark, that filling up two compartments of the square we have

	1	2
c^2g	1	1
f^2	1	1

c^2d^2
b^2cd^2

where the meaning of the numbers $(1, 1)$ has to be considered: the first (1) seems to indicate a seminvariant $c^2g \propto c^2d^2$, but there is in fact no such form, what it really indicates is a form $0c^2g + f^2 \propto c^2d^2$, that is, $f^2 \propto c^2d^2$; and similarly, the second (1) seems to indicate a seminvariant $f^2 \propto b^2cd^2$, but there is in fact no such form, what it really indicates is $f^2 \propto c^2d^2 + 0b^2cd^2$, that is, $f^2 \propto c^2d^2$. The explanation is correct, but to make it perfectly clear some further developments would be required. The like remarks apply to the inversion cdf, ce^2.

The MacMahon Linkage. Art. Nos. 48 to 52.

Art. We require the two theorems:

The first is: if a seminvariant S has q for its highest letter, then $\partial_q S$ is also a seminvariant.

The second has presented itself for unitariants (*ante* No. 31); for seminvariants the form is less simple, viz. If in any seminvariant, attending only to the terms of the highest degree, we therein change $b, c, d, e, \ldots$ into $b, 2c, 6d, 24e, \ldots$ and then diminish the letters (that is, replace each letter by the next preceding letter) and in the result so obtained change $b, c, d, e, \ldots$ into $b, \frac{c}{2}, \frac{d}{6}, \frac{e}{24}, \ldots$ we obtain a seminvariant. For instance $g - 6bf + 15ce - 10d^2$, in the terms of degree 2, making the numerical change we have $-720bf + 720ce - 360d^2$, and then diminishing the letters and making the numerical change, we obtain $-720\,\frac{e}{24} + 720\,\frac{bd}{6} - 360\,\frac{c^2}{4}$, that is,

$$-30\,(e - 4bd + 3c^2),$$

a seminvariant.

Art. For the proof, observe that the equation $\Delta\, S = 0$, attending therein only to the terms of the highest degree, gives $(2b\partial_c + 3c\partial_d + \ldots)\, S' = 0$, if S' denote the terms of the highest degree: making the numerical change, this is $(b\partial_c + c\partial_d + \ldots)\, S'' = 0$, if S'' is what S' becomes thereby; diminishing the letters, this is $(\partial_b + b\partial_c + \ldots)\, S''' = 0$, if S''' is what S'' becomes thereby; viz. S''' is a seminvariant. Making in S''' the numerical change, we obtain the seminvariant which was to be found.

Art. I say that the seminvariant thus obtained is the same as the seminvariant which would be obtained by operating on S with the operator

$$\Omega = (\delta - 1)\, C - B^2,$$

where B, C are the operators defined as follows:

$$B = b\partial_a + c\partial_b + d\partial_c + \dots,$$
$$C = b\partial_b + 2c\partial_c + 3d\partial_d + \dots;$$

viz. B is the operator which increases the weight by unity, and C is the operator which leaves the weight unaltered but multiplies each term by its degree in the letters b, c, d, ...

Art. To verify this, observe that the operation $\Omega = (\delta - 1)\, C - B^2$ upon a seminvariant of the degree δ gives a seminvariant; and that the two operations, the one described in Art. 48 and the other the operation Ω, give the same result. For instance, taking the seminvariant $g - 6bf + 15ce - 10d^2$, the operation Ω gives

$$(\delta - 1)\, C\, S - B^2 S = 6\, C\, S - B^2 S,$$

and this is $= -30\,(e - 4bd + 3c^2)$, the same result as before.

Art. The MacMahon linkage is the series of operations which connects the seminvariants of a given degree with those of the next preceding degree; it is as follows: If S be a seminvariant of the degree δ, then $\partial_q S$ is a seminvariant of the degree $\delta - 1$, and the operation Ω gives a seminvariant of the same degree δ; and these two operations are connected by the relation

$$\Omega\, S = (\delta - 1)\, C\, S - B^2 S.$$

The linkage is thus a combination of the operations ∂_q and Ω, by which we can pass from the seminvariants of one degree to those of the next degree.

Art. The MacMahon linkage may be used to find the seminvariants of a given degree from those of the next preceding degree; for if we know the seminvariants of the degree $\delta - 1$, then by operating with B and C we can find the seminvariants of the degree δ. The process is as follows: Starting with the seminvariants of the degree $\delta - 1$, we operate with B to obtain the seminvariants of the degree

δ, and then with C to obtain the seminvariants of the same degree; and the combination of these two operations gives the MacMahon linkage.

On the Perpetuants. Art. Nos. 53 to 71.

Art. A perpetuant is a seminvariant which is not expressible as a rational integral function of seminvariants of lower degrees. The perpetuants of a given degree δ are those seminvariants which cannot be obtained from seminvariants of lower degrees by the operations P and $\frac{1}{2}Q$. The number of perpetuants of degree δ and weight w is given by the formula

$$N_{\text{perp}}(\delta, w) = N(\delta, w) - N(\delta - 1, w - 1) - N(\delta - 1, w - 2),$$

where $N(\delta, w)$ is the total number of seminvariants of degree δ and weight w.

Art. The perpetuants of degree 2. There is a single perpetuant $ac - b^2$, or say $c - b^2$, of weight ω.

Art. The perpetuants of degree 3. The seminvariants are $a^2d - 3abc+2b^3$, or say $d-3bc+2b^3$ of weight ω, and $ae-bd$ of weight $\omega+1$. The first of these is a perpetuant; the second is not a perpetuant, for it is obtained by operating with P on the perpetuant of degree 2.

Art. The perpetuants of degree 4. The seminvariants are $a^3e - 4a^2bd+6ab^2c-3b^4$, or say $e-4bd+3c^2$ of weight ω, and $af-3be+2c^2$ of weight $\omega+1$, and $ag-3bf+ce$ of weight $\omega+2$. Of these, the first is a perpetuant; the second and third are not perpetuants, for they are obtained by operating with P on seminvariants of lower degrees.

Art. The perpetuants of degree 5. The seminvariants are: of weight ω, the seminvariant $f-5ce+10b^2e-10cd^2+15bcd-6b^3d+c^4$; of weights $\omega+1$, $\omega+2$, $\omega+3$, seminvariants derived from those of lower degrees by the operation P. The perpetuants are thus the seminvariant of weight ω, and possibly one or more of the seminvariants of higher weights which are not expressible in terms of seminvariants of lower degrees.

Art. The generating function for the perpetuants of degree δ is

$$G_{\text{perp}}(\delta, x) = \frac{1-x}{(1-x^2)(1-x^3)\dots(1-x^\delta)} - \frac{1-x}{(1-x^2)(1-x^3)\dots(1-x^{\delta-1})},$$

and the number of perpetuants of degree δ and weight w is the coefficient of x^w in this generating function.

Art. For the perpetuants of degree 2, the generating function is $\frac{1-x}{1-x^2} = 1$, and there is one perpetuant of weight ω.

For degree 3, the generating function is $\frac{1-x}{(1-x^2)(1-x^3)} - \frac{1-x}{1-x^2} = \frac{x^3}{(1-x^2)(1-x^3)}$, and the number of perpetuants is the coefficient of $x^{w-\omega}$ in this expression.

For degree 4, the generating function is

$$\frac{1-x}{(1-x^2)(1-x^3)(1-x^4)} - \frac{1-x}{(1-x^2)(1-x^3)} = \frac{x^4}{(1-x^2)(1-x^3)(1-x^4)},$$

and the number of perpetuants is the coefficient of $x^{w-\omega}$ in this expression.

Art. The number of perpetuants of degree δ and the several weights may be found as follows: We have

$$\begin{aligned}
&\text{deg. } 2: \quad x^2, \\
&\text{deg. } 3: \quad x^3 + x^5 + x^7 + x^9 + \dots, \\
&\text{deg. } 4: \quad x^4 + x^6 + 2x^8 + 2x^{10} + 3x^{12} + 3x^{14} + \dots, \\
&\text{deg. } 5: \quad x^5 + x^7 + 2x^9 + 3x^{11} + 4x^{13} + 5x^{15} + \dots, \\
&\text{deg. } 6: \quad x^6 + x^8 + 2x^{10} + 3x^{12} + 5x^{14} + 6x^{16} + \dots,
\end{aligned}$$

and so on. The coefficient of x^w gives the number of perpetuants of the given degree and weight w.

Art. The perpetuants of degree 5. The seminvariants are $f - 5ce + 10b^2e - 10cd^2 + 15bcd - 6b^3d + c^4$ of weight ω, and the seminvariants of higher weights obtained by operating with P on the seminvariants of degree 4. The perpetuants are those seminvariants which are not expressible as rational integral functions of seminvariants of lower degrees.

Art. The perpetuants of degree 6. The seminvariants are $g - 6df + 15ce^2 - 10d^3 + \dots$ of weight ω, and the seminvariants of higher weights obtained by operating with P on the seminvariants of degree 5. The perpetuants are those seminvariants which are not expressible as rational integral functions of seminvariants of lower degrees.

Art. The generating function for the perpetuants of all degrees is

$$G_{\mathrm{perp}}(x, y) = \sum_{\delta=2}^{\infty} G_{\mathrm{perp}}(\delta, x)\, y^{\delta},$$

and the coefficient of $x^w y^\delta$ in this generating function gives the number of perpetuants of degree δ and weight w.

Art. The perpetuants of a given degree δ may be arranged in a table, according to their weights; and the table may be used to find the perpetuant of a given weight, or to verify that a given seminvariant is a perpetuant. The table may also be used to find the relations between the perpetuants; for if we have two perpetuants of the same degree and weight, then the table gives the relation between them.

Art. The perpetuants of degree δ and weight w may also be obtained by the method of undetermined coefficients; for if we assume a seminvariant of the degree δ and weight w, and impose the condition that it shall not be expressible as a rational integral function of seminvariants of lower degrees, then the coefficients must satisfy certain linear equations, and the solution of these equations gives the perpetuant.

Art. The number of perpetuants of degree δ and weight w is given by the formula

$$N_{\text{perp}}(\delta, w) = P(w;\delta) - P(w-1;\delta-1) - P(w-1;\delta) + P(w-2;\delta-1),$$

where $P(w;\delta)$ denotes the number of partitions of w into δ parts, each part being at most equal to δ. This formula may be proved by the theory of partitions; and it may be used to verify the results obtained by the other methods.

Art. The perpetuants of degree 7. The seminvariants are $h - 7eg + 21cef - 35def + \ldots$ of weight ω, and the seminvariants of higher weights obtained by operating with P on the seminvariants of degree 6. The perpetuants are those seminvariants which are not expressible as rational integral functions of seminvariants of lower degrees.

Art. The perpetuants of degree 8. The seminvariants are $i - 8fh + 28cg^2 - 56dfg + \ldots$ of weight ω, and the seminvariants of higher weights obtained by operating with P on the seminvariants of degree 7. The perpetuants are those seminvariants which are not expressible as rational integral functions of seminvariants of lower degrees.

Art. The perpetuants of degree 9. The seminvariants are $j - 9gi + 36dh^2 - 84efh + \ldots$ of weight ω, and the seminvariants of higher weights obtained by operating with P on the seminvariants of degree 8. The perpetuants are those seminvariants which are not expressible as rational integral functions of seminvariants of lower degrees.

Art. The perpetuants of degree 10. The seminvariants are $k - 10hj + 45ei^2 - 120fgi + \ldots$ of weight ω, and the seminvariants of higher weights obtained by operating with P on the seminvariants of degree 9. The perpetuants are those seminvariants which are not expressible as rational integral functions of seminvariants of lower degrees.

Art. The perpetuants of degree δ and weight $\delta\omega$ (the lowest possible weight) are the seminvariants of the form $a_\delta - \delta a_{\delta-1}b + \ldots$, which are the elementary seminvariants. These perpetuants are the building blocks from which all other seminvariants may be constructed by the operations P and $\frac{1}{2}Q$.

Art. The perpetuants of degree δ and weight $\delta\omega + 1$ (the next lowest weight) are obtained by operating with P on the perpetuants of degree δ and weight $\delta\omega$. The number of such perpetuants is equal to the number of perpetuants of degree δ and weight $\delta\omega$, minus the number of perpetuants of degree $\delta - 1$ and weight $(\delta - 1)\omega + 1$.

Investigation of the Values of the Foregoing Functions $\Pi_{10}(x+y)$, $\Pi_{15}(x+y)$ and $\Pi_{10}(x+y+z)$. Art. Nos. 72 to 74.

Art. If x, y, z, w, t are the roots of a quintic equation, say

$$\lambda - x \,.\, \lambda - y \,.\, \lambda - z \,.\, \lambda - w \,.\, \lambda - t = (1, B, C, D, E, F \setminus \lambda, 1)^5 = 0,$$

we require the product $\Pi_{10}(x + y)$ of the sum of two roots in the particular case $B = 0$. But in order to the determination of the expression for $\Pi_{10}(x + y + z)$, we require the value of $\Pi_{10}(x + y)$ in the general case, B any value whatever.

Art. Writing

$$\begin{aligned}
s_1 &= x + y + z + w + t,\\
s_2 &= xy + xz + xw + xt + yz + yw + yt + zw + zt + wt,\\
s_3 &= xyz + xyw + xyt + xzw + xzt + xwt + yzw + yzt + ywt + zwt,\\
s_4 &= xyzw + xyzt + xywt + xzwt + yzwt,\\
s_5 &= xyzwt,
\end{aligned}$$

then we have $s_1 = -B$, $s_2 = C$, $s_3 = -D$, $s_4 = E$, $s_5 = -F$; and the product $\Pi_{10}(x + y)$ may be expressed as a function of s_1, s_2, $\ldots$, or as a function of B, C, D, E, F.

Art. The value of $\Pi_{10}(x+y)$ is given by the formula

$$\Pi_{10}(x+y)=\prod_{i<j}(x_i+x_j),$$

where x_1, x_2, x_3, x_4, x_5 are the roots x, y, z, w, t. This product may be expressed as a symmetric function of the roots, and then as a function of the coefficients B, C, D, E, F.

To find the expression for $\Pi_{10}(x+y)$, we observe that the product contains 10 factors, each of which is the sum of two roots. The product is therefore a symmetric function of the roots, of degree 10 in the roots, and of weight 10 in the coefficients.

Art. The value of $\Pi_{10}(x+y)$ in the general case is

$$\begin{aligned}\Pi_{10}(x+y) = {} & F^2-2DEF+D^3E+D^2E^2-2CD^2F+C^2EF+C^2D^2-C^3F\\ & +B(-2EF^2+2D^2F^2+3DEF^2-D^3F-2CDEF+C^2F^2+CD^2E\\ & +B^2(\dots)+\dots,\end{aligned}$$

and in the particular case $B=0$, this reduces to

$$\Pi_{10}(x+y)=F^2-2DEF+D^3E+D^2E^2-2CD^2F+C^2EF+C^2D^2-C^3F.$$

Seminvariants—B and C series of operators. Art. Nos. 75 to 82.

Art. The operators B and C are defined as follows:

$$\begin{aligned}B&=b\partial_a+c\partial_b+d\partial_c+\dots,\\ C&=b\partial_b+2c\partial_c+3d\partial_d+\dots;\end{aligned}$$

and we have seen that the MacMahon linkage is given by the operator $\Omega=(\delta-1)\,C-B^2$. The operators B and C satisfy certain relations, which are useful in the theory of seminvariants.

Art. The operators B and C satisfy the relation

$$BC-CB=B,$$

or as this may be written,

$$[B,C]=B.$$

This relation may be proved by direct calculation; and it is useful for finding the properties of the operators B and C.

Art. The operators B and C also satisfy the relation

$$B^2C - CB^2 = 2B^2,$$

and more generally,

$$B^nC - CB^n = nB^n.$$

These relations may be proved by induction; and they are useful for finding the properties of the operators B^n and C.

Art. The operators B and C may be used to construct a series of operators, which are useful in the theory of seminvariants. The series is as follows:

$$\begin{aligned}
&\Omega_1 = (\delta-1)\,C - B^2,\\
&\Omega_2 = (\delta-1)(\delta-2)\,D - 3(\delta-2)\,BC + 2B^3,\\
&\Omega_3 = (\delta-1)(\delta-2)(\delta-3)\,E - 4(\delta-2)(\delta-3)\,BD + 6(\delta-3)\,B^2C - 3B^4,
\end{aligned}$$

&c., which are of the deg. weights 0.2, 0.3, 0.4, &c., respectively, viz. operating upon a seminvariant of deg. weight $\delta.\omega$ they leave the degree unaltered, but increase the weight by 2, 3, 4, … respectively.

Art. It is to be observed that B^2, BC, B^3, &c., denote the mere algebraical powers and products of the symbols B, C, D, &c., without any operation of one symbol on another.

Art. As a simple illustration, take $(C - B^2)(ac - b^2)$: here,

$$\begin{array}{ll}
& C(ac-b^2) = e - 2bd + c^2 \\
-B^2 = -(2bd\,\partial_a\partial_c + c^2\partial_b^2) & \qquad\underline{-2bd + 2c^2} \\
\text{Value is } e - 4bd + 3c^2; &
\end{array}$$

and similarly for $(C - B^2)(ae - 4bd + 3c^2)$, here:

$$\begin{array}{ll}
& C(ae - 4bd + 3c^2) = g - 4bf + 7ce - 4d^2 \\
-B^2 = -(2bf\,\partial_a\partial_c + 2ce\,\partial_b\partial_d + d^2\partial_c^2) & \qquad\underline{-2bf + 8ce - 6d^2} \\
\text{Value is } g - 6bf + 15ce - 10d^2. &
\end{array}$$

Art. A direct proof may of course be obtained for any one of the foregoing operators; viz. calling it Ω, it may be shown that $\Delta\Omega\,S$ is $= 0$. I have not considered the like question of the derivation of series of operators from the other two forms

$$PQ - \omega P\,.\,P - \tfrac{1}{3}(\delta-3)\,P\,.\,Q \quad\text{and}\quad Q^2 + Q\,.\,Q - \tfrac{1}{3}(4\omega+2)\,P\,.\,Q$$

respectively.

Art. The foregoing operators Ω_1, Ω_2, Ω_3, $\ldots$ may be used to construct a series of seminvariants, by operating upon a given seminvariant; and the series so obtained is complete, in the sense that it contains all the seminvariants of the given degree and weight. The series is also unique, in the sense that there is only one series of seminvariants for each degree and weight.

Art. The operators Ω_1, Ω_2, Ω_3, $\ldots$ may also be used to find the relations between the seminvariants; for if we have two seminvariants of the same degree and weight, then the operators give the relation between them. The operators may also be used to verify that a given function is a seminvariant; for if the function be a seminvariant, then the operators applied to it give seminvariants.

[End of Paper 932.]

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939. On a Case of the Involution $AF + BG + CH = 0$.

[From the *Messenger of Mathematics*, vol. xxII. (1893), pp. 44.]

BUT the case which I proceed to consider is

$$A = 1234, \quad B = 5678; \quad F = 1256, \quad G = 3478;$$

here $AF = 0$, and $BG = 0$, meet as before in the points $1, 2, 3, 4, 5, 6, 7, 8$, and in eight other points, say that

$A = 0,$	$B = 0$	meet in	$1, 2$	and in two other points	$\alpha, \beta,$
$A = 0,$	$G = 0$		$3, 4$	"	$\gamma, \delta,$
$F = 0,$	$B = 0$		$5, 6$	"	$\epsilon, \zeta,$
$F = 0,$	$G = 0$		$7, 8$	"	$\eta, \theta;$

then the 8 points $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$ will lie in a conic $H = 0$.

I take $yz - z\theta = 0$ for the conic $H = 0$; for any point in this conic we have $x : y : z = 1 : \theta : \theta^2$, and we may take $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7, \theta_8$ for the parameters of the points $1, 2, 3, 4, 5, 6, 7, 8$ respectively.

Write $(a, b, c, f, g, h \between x, y, z)^2 = 0$ for the conic $A = 1234 = 0$; therefore we have

$$c = 1, \quad f = -p_{1234}, \quad g = -\sigma_{1234}, \quad h = q_{1234},$$

or, if

$$p_{1234} = \theta_1 + \theta_2 + \theta_3 + \theta_4, \quad q_{1234} = \theta_1\theta_2 + \theta_1\theta_3 + \theta_2\theta_3 + \theta_1\theta_4 + \theta_2\theta_4 + \theta_3\theta_4, \quad \sigma_{1234} = \theta_1\theta_2\theta$$

then

$$c = 1, \quad f = -p_{1234}, \quad g = -\sigma_{1234}, \quad h = q_{1234},$$

or, writing $\sigma = -\lambda$, we have

$$\sigma_{1234}x^2 + q_{1234}y^2 + z^2 - p_{1234}yz - \lambda xy = 0$$

for the equation of the conic in question. We may without loss of generality put $\lambda = 0$; and then if, in general,

$$\Omega = \sigma xz + qy^2 + z^2 - pyz - rxy,$$

we have $A = \Omega_{1234} = 0$ for the conic $A = 0$. And thus the equations of the four conics are

$$A = \Omega_{1234} = 0, \quad B = \Omega_{5678} = 0; \quad F = \Omega_{1256} = 0, \quad G = \Omega_{3478} = 0,$$

or, as for shortness I write them,

$$A = \Omega = 0, \quad F = \Omega' = 0; \quad B = \Omega'' = 0, \quad G = \Omega''',$$

viz. in Ω the suffixes are $1, 2, 3, 4$, in Ω' they are $5, 6, 7, 8$, in Ω'' they are $1, 2, 5, 6$, and in Ω''' they are $3, 4, 7, 8$.

I find that the implicit constant factors of AF and BG are 1, -1, and consequently that the form of the identity is

$$\Omega'\Omega''' - \Omega\Omega'' + (y^2 - zx)C = 0,$$

where C is a quadric function to be determined; or, what is the same thing, we have

$$\begin{aligned}&(\sigma xz + qy^2 + z^2 - pyz - rxy)(\sigma' xz + q'y^2 + z^2 - p'yz - r'xy)\\ -&(\sigma'' xz + q''y^2 + z^2 - p''yz - r''xy)(\sigma''' xz + q'''y^2 + z^2 - p'''yz - r'''xy)\\ &\qquad + (y^2 - zx)C = 0.\end{aligned}$$

Writing for shortness

$$\begin{aligned}\theta_1+\theta_2&=\alpha, & \theta_1\theta_2&=\beta,\\ \theta_3+\theta_4&=\alpha', & \theta_3\theta_4&=\beta',\\ \theta_5+\theta_6&=\alpha'', & \theta_5\theta_6&=\beta'',\\ \theta_7+\theta_8&=\alpha''', & \theta_7\theta_8&=\beta''',\end{aligned}$$

we have

$$\begin{aligned}p&=\alpha+\alpha', & q&=\alpha\alpha'+\beta+\beta', & r&=\alpha\beta'+\alpha'\beta, & \sigma&=\beta\beta',\\ p'&=\alpha''+\alpha''', & q'&=\alpha''\alpha'''+\beta''+\beta''', & r'&=\alpha''\beta'''+\alpha'''\beta'', & \sigma'&=\beta''\beta''',\\ p''&=\alpha+\alpha'', & q''&=\alpha\alpha''+\beta+\beta'', & r''&=\alpha\beta''+\alpha''\beta, & \sigma''&=\beta\beta'',\\ p'''&=\alpha'+\alpha''', & q'''&=\alpha'\alpha'''+\beta'+\beta''', & r'''&=\alpha'\beta'''+\alpha'''\beta', & \sigma'''&=\beta'\beta'''.\end{aligned}$$

In the last-mentioned equation, the first and second lines together are a quartic function of (x, y, z), say the value is

$$\begin{aligned}\Phi = Ax^4 + By^4 + Cz^4&\\ &+ Fyz^3 + Gz^3x + Hx^2y^2\\ &+ Iyz^3 + Jzx^3 + Kxy^3\\ &+ Lx^2yz + Mxy^2z + Nxyz^2\\ &+ Pz^2x^2 + Qx^2y^2 + Rx^2y^2,\end{aligned}$$

where after all reductions

$$
\begin{aligned}
A &= \sigma\sigma' - \sigma''\sigma''' = 0,\\
B &= qq' - q''q''',\\
C &= 1 - 1 = 0,\\
F &= -pq' - p'q + p''q''' + p'''q'' = (\alpha - \alpha''')(\beta' - \beta'') + (\alpha' - \alpha'')(\beta - \beta'''),\\
G &= 0,\\
H &= -r\sigma' - r'\sigma + r''\sigma''' + r'''\sigma'' = -(\alpha\beta''' - \alpha'''\beta)(\beta' - \beta'') + (\alpha'\beta'' - \alpha''\beta')(\beta - \beta''\\
I &= -p - p' + p'' + p''' = 0,\\
J &= 0,\\
K &= -qr' - q'r + q''r''' + q'''r'' = -(\alpha\beta''' - \alpha'''\beta)(\beta' - \beta'') + (\alpha'\beta'' - \alpha''\beta')(\beta - \beta''')\\
L &= -p\sigma' - p'\sigma + p''\sigma''' + p'''\sigma'' = (\alpha\beta''' - \alpha'''\beta)(\alpha' - \alpha'') + (\alpha'\beta'' - \alpha''\beta')(\alpha - \alpha''')\\
M &= pr' + p'r - p''r''' - p'''r'' = 0,\\
N &= -r - r' + r'' + r''' = (\alpha - \alpha''')(\beta' - \beta'') + (\alpha' - \alpha'')(\beta - \beta''') = 0,\\
P &= q - q' - q'' + q''' = 0,\\
Q &= \sigma - \sigma' - \sigma'' + \sigma''' = 0,\\
R &= rr' + q\sigma' + q'\sigma - r''r''' - q''\sigma''' - q'''\sigma'' = 0.
\end{aligned}
$$

values which satisfy

$$F + N = 0, \quad K + L = 0, \quad B + M + Q = 0.$$

The quartic function is thus seen to be

$$\Phi = (y^2 - zx)(By^2 + Fyz - Qzx + Kxy) = 0,$$

viz. we have $By^2 + Fyz - Qzx + Kxy = 0$ for the equation of the conic $H = 0$.

Moreover, substituting for p, q, r, σ, &c., their values, we have finally for the required involution

$$\begin{aligned}
&[\beta\beta' x^2 + (\alpha\alpha' + \beta + \beta')y^2 + z^2 - (\alpha + \alpha')yz - (\alpha\beta' + \alpha'\beta)xy] \\
&\times[\beta''\beta''' x^2 + (\alpha''\alpha''' + \beta'' + \beta''')y^2 + z^2 - (\alpha'' + \alpha''')yz - (\alpha''\beta''' + \alpha'''\beta'')xy] \\
&- [\beta\beta'' x^2 + (\alpha\alpha'' + \beta + \beta'')y^2 + z^2 - (\alpha + \alpha'')yz - (\alpha\beta'' + \alpha''\beta)xy] \\
&\times[\beta'\beta''' x^2 + (\alpha'\alpha''' + \beta' + \beta''')y^2 + z^2 - (\alpha' + \alpha''')yz - (\alpha'\beta''' + \alpha'''\beta')xy], \\
&-(y^2 - zx)\times\{y^2[(\alpha\beta''' - \alpha'''\beta)(\alpha' - \alpha'') + (\alpha'\beta'' - \alpha''\beta')(\alpha - \alpha''') - (\beta - \beta''')(\beta' - \beta'')] \\
&\qquad + yz[(\alpha - \alpha''')(\beta' - \beta'') + (\alpha' - \alpha'')(\beta - \beta''')] \\
&\qquad - zx[(\beta - \beta''')(\beta' - \beta'')] \\
&\qquad + xy[(\alpha\beta''' - \alpha'''\beta)(\beta' - \beta'') + (\alpha'\beta'' - \alpha''\beta')(\beta - \beta''')]\} = 0.
\end{aligned}$$

It will be recollected that this is the solution for the case $A = 1234$, $B = 5678$; $F = 1256$, $G = 3478$: being that to which the present paper has reference.

940. On the Development of $(1+n^2x)^{\frac{m}{n}}$.

[From the *Messenger of Mathematics*, vol. xxII. (1893), pp. 186—190.]

It is a known theorem that, if $\frac{m}{n}$ be any fraction in its least terms, the coefficients of the development of $(1+n^2x)^{\frac{m}{n}}$ are all of them integers, or, what is the same thing, that

$$\frac{m\cdot(m-n)\ldots(m-(r-1)n)}{1\cdot 2\ldots r}\cdot n^r$$

is an integer. The greater part, but not the whole, of this result comes out very simply from Mr Segar's very elegant theorem, *Messenger*, vol. xxII. (1893), p. 59, "the product of the differences of any r unequal numbers is divisible by $(r-1)!$" or, as it may be stated, if $\alpha, \beta, \gamma, \ldots$ are any r unequal numbers, then $\zeta^2(\alpha, \beta, \gamma, \ldots)$ is divisible by $\zeta^2(0, 1, 2, \ldots, r-1)$.

In fact, writing $r+1$ for r and considering the numbers

$$m+n, \quad n, \quad 2n, \quad 3n, \quad \ldots \quad (r-1)n;$$

then neglecting signs

$$\begin{aligned}\zeta^2(\alpha,\beta,\gamma,\ldots) \text{ is } = \quad & m\cdot(m-n)\ldots(m-(r-1)n),\\ & \times 1n\cdot 2n\ldots(r-1)n,\\ & \times 1n\cdot 2n\ldots(r-2)n,\\ & \times \ldots,\\ & \times 1n\cdot 2n,\\ & \times 1n,\end{aligned}$$

which is

$$m\cdot(m-n)\ldots(m-(r-1)n)\cdot n^{\frac{1}{2}r(r-1)}\cdot\zeta^2(0,1,2,\ldots,r-1);$$

and similarly

$$\zeta^2(0,1,2,\ldots,r) = 1\cdot 2\cdot 3\ldots r\cdot\zeta^2(0,1,2,\ldots,r-1);$$

so that, omitting the common factor $\zeta^2(0,1,2,\ldots,r-1)$, we have

$$m\cdot(m-n)\ldots(m-(r-1)n)\cdot n^{\frac{1}{2}r(r+1)} \text{ divisible by } 1\cdot 2\cdot 3\ldots r.$$

It thus appears that the fraction

$$\frac{m \cdot (m-n) \dots (m-(r-1)n)}{1 \cdot 2 \dots r} \cdot n^r$$

when reduced to its least terms, will contain in the denominator only products of powers of the prime factors of n; and it remains to show that multiplying this by n^r it will become integral, or what is the same thing that

$$\frac{1 \cdot 2 \dots r}{n^r}$$

in its least terms will not contain in the denominator any prime factor of n.

This is

$$1 \cdot 2 \cdot 3 \ldots r = 2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma} \ldots \cdot K,$$

where K is a whole number. Hence if $n = 2^{\alpha} 3^{\beta} 5^{\gamma} \ldots$, we have

$$\frac{1 \cdot 2 \cdot 3 \ldots r}{n^r} = \frac{2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma} \ldots}{2^{r\alpha} \cdot 3^{r\beta} \cdot 5^{r\gamma} \ldots} \cdot K,$$

and here for every prime number $2, 3, 5, \ldots$ which is a factor of n, that is, for which the corresponding exponent $\alpha, \beta, \gamma, \ldots$ is not $= 0$, the exponents $r\alpha - (r)$, $r\beta - (\frac{1}{2}r)$, $r\gamma - (\frac{1}{4}r)$, &c. are all of them positive; and thus

$$\frac{1 \cdot 2 \ldots r}{n^r}$$

is integral, that is

$$\frac{m \cdot (m-n) \ldots (m-(r-1)n)}{1 \cdot 2 \ldots r} \cdot n^r$$

is an integer; which is the theorem in question.

Second demonstration, via the Segar theorem.

Writing for a moment $a, b, c, \ldots$ for the r numbers $m+n, n, 2n, 3n, \ldots, (r-1)n$, the fraction, say F, is

$$F = \frac{(a-b)(a-c) \ldots (a-k)}{1 \cdot 2 \ldots r} \cdot n^{r-1};$$

and the theorem to be proved is that this is an integer. But we have identically

$$\frac{(a-b)(a-c) \ldots (a-k)}{1 \cdot 2 \ldots r} = M \cdot \frac{\zeta^2(a, b, c, \ldots)}{\zeta^2(0, 1, 2, \ldots, r-1)},$$

where M is a mere number; and hence

$$F = M \cdot n^{r-1} \cdot \frac{\zeta^2(a, b, c, \ldots)}{\zeta^2(0, 1, 2, \ldots, r-1)}.$$

But by the Segar theorem, $\frac{\zeta^2(a,b,c,\ldots)}{\zeta^2(0,1,2,\ldots,r-1)}$ is an integer; and the theorem will be proved if only $M \cdot n^{r-1}$ is an integer; that is, if the number M is divisible by each of the prime factors of n, at most to the power $r-1$.

We in fact have

$$M = \frac{\zeta^2(0, 1, 2, \ldots, r-1)}{1 \cdot 2 \ldots r},$$

which is

$$M = \begin{vmatrix} 1, & 0, & 0^2, & \ldots \\ 1, & 1, & 1^2, & \ldots \\ 1, & 2, & 2^2, & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} \div \begin{vmatrix} 1, & a, & a^2, & \ldots \\ 1, & b, & b^2, & \ldots \\ 1, & c, & c^2, & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix},$$

or what is the same thing it is

$$M = \begin{vmatrix} 1, & a, & a^2, & \ldots \\ 1, & b, & b^2, & \ldots \\ 1, & c, & c^2, & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} \div \begin{vmatrix} 1, & a, & a^2, & \ldots \\ 1, & b, & b^2, & \ldots \\ 1, & c, & c^2, & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix},$$

where M is a mere number: it will be recollected that in this form, $a, b, c, \ldots$ are not the original $a, b, c, \ldots$. Putting herein $\beta, \gamma, \ldots = 1, 2, \ldots$, the denominator determinant is

$$\begin{vmatrix} 1, & a, & a^2, & \ldots \\ 1, & b, & b^2, & \ldots \\ 1, & c, & c^2, & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix},$$

and hence the quotient, which as already seen is an integer number, is equal to $\zeta^2(0, \beta, \gamma, \ldots) \div \zeta^2(0, 1, 2, \ldots)$, the theorem in question.

The original theorem as to the form of $(1 + n^2x)^{\frac{m}{n}}$ is a particular case of Eisenstein's very general theorem that, in the development of any algebraical function of x, it is always possible by substituting for x a proper multiple of x, to make all the coefficients integers. It may be remarked that this would not be so if we had only

$$\frac{m \cdot (m-n) \ldots (m-(r-1)n) \cdot n^{\frac{1}{2}r(r+1)}}{1 \cdot 2 \ldots r}$$

divisible by $1 \cdot 2 \ldots r$; for then, writing NX for x, the form of the coefficient would have been

$$\frac{m \cdot (m-n) \ldots (m-(r-1)n)}{1 \cdot 2 \ldots r} (nN)^{\frac{1}{2}r(r+1)},$$

and there would be no value (however great) of N by which the denominator factor $n^{\frac{1}{2}r(r+1)}$ could be got rid of.

941. Note on the Partial Differential Equation $Rr + Ss + Tt + U(s^2 - rt) - V = 0$.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxvI. (1893), pp. 1—5.]

It is well known that this equation, R, S, T, U, V being any functions whatever of (x, y, z, p, q), in the case where it admits of an integral of the form $u = f(v)$ (u, v functions of x, y, z, p, q, and f an arbitrary functional symbol) can be integrated as follows; viz. taking m_1, m_2 as the roots of the quadratic equation

$$\begin{vmatrix} m, & R, & U \\ T, & m, & U \\ V, & V, & m \end{vmatrix} = 0,$$

(that is, writing $m_1 + m_2 = S$ and $m_1 m_2 = RT - UV$), then, m_1 denoting either root at pleasure, and m_2 the other root of the quadratic equation, if the system of ordinary differential equations

$$\begin{aligned} m_1\,dx - R\,dy + U\,dq &= 0, \\ -T\,dx + m_2\,dy + U\,dp &= 0, \\ -V\,dx + m_2\,dq + R\,dp &= 0, \\ -V\,dy + T\,dq + m_1\,dp &= 0, \\ -p\,dx - q\,dy + \qquad dz &= 0, \end{aligned}$$

(equivalent to three independent equations) admits of two integrals $u =$ const., and $v =$ const., the solution of the given partial differential equation is $u = f(v)$.

In fact, to prove this, we have

$$\begin{aligned} du = \lambda(m_1\,dx - R\,dy + U\,dq) & \\ + \mu(-T\,dx + m_2\,dy + U\,dp) & \\ + \nu(-V\,dx + m_2\,dq + R\,dp) & \\ + \rho(-V\,dy + T\,dq + m_1\,dp) & \\ + \sigma(-p\,dx - q\,dy + dz), & \end{aligned}$$

$$\begin{aligned}
dv = \lambda'(m_1\,dx - R\,dy + U\,dq) \\
+ \mu'(-T\,dx + m_2\,dy + U\,dp) \\
+ \nu'(-V\,dx + m_2\,dq + R\,dp) \\
+ \rho'(-V\,dy + T\,dq + m_1\,dp) \\
+ \sigma'(-p\,dx - q\,dy + dz),
\end{aligned}$$

and thence

$$du\,dv = (\lambda\,dm_1 + \mu\,dm_2 + \nu\,dm_3 + \rho\,dm_4 + \sigma\,dm_5)\\ \times (\lambda'\,dm_1 + \mu'\,dm_2 + \nu'\,dm_3 + \rho'\,dm_4 + \sigma'\,dm_5).$$

Hence, when the partial differential equation $\Phi = 0$ is satisfied, we have

$$\frac{\partial(u)}{\partial x}\frac{\partial(v)}{\partial y} - \frac{\partial(u)}{\partial y}\frac{\partial(v)}{\partial x} = 0,$$

and we thence have $u = f(v)$ as the integral of the partial differential equation.

It should be possible to express analytically the conditions in order that the systems of differential equations may have one or each of them two integrals. It is interesting to remark that, if each of the two systems of ordinary differential equations has only a single integral, these two integrals do not lead to the solution of the partial differential equation. Consider, for instance, the case

$$R = 0, \quad S = x + y, \quad T = 0, \quad U = 0, \quad V = p + q;$$

the partial differential equation is here

$$(x + y)s - (p + q) = 0,$$

which has an integral

$$z = (x + y)\{\phi'(x) + \psi'(y)\} - 2\{\phi(x) + \psi(y)\},$$

where ϕ, ψ are arbitrary functions: the equation in m is $m^2 - m(x + y) = 0$, the roots of which are $m = 0$, and $m = x + y$.

For $m_1 = 0$, $m_2 = x + y$, the system of differential equations becomes

$$\begin{aligned} dy &= 0,\\ -(p + q)\,dx + (x + y)\,dq &= 0,\\ -p\,dx + dz &= 0, \end{aligned}$$

which has only the integral $y =$ const.; and similarly for $m_1 = x + y$, $m_2 = 0$, the system becomes

$$\begin{aligned} (x + y)\,dp &= 0,\\ -q\,dy + dz &= 0, \end{aligned}$$

which has only the integral $x =$ const. And these two integrals $y =$ const. and $x =$ const. do not in anywise lead to the integral of the partial differential equation.

I take the opportunity of remarking that the complete system of conditions in order that the differential

$$A\,dx + B\,dy + C\,dz + D\,dw$$

may be $= M\,dU$ is as follows: viz. writing

$$A, B, C, D = 1, 2, 3, 4;$$

$$\begin{array}{lll} \dfrac{\partial B}{\partial z} - \dfrac{\partial C}{\partial y} = 23, & \dfrac{\partial C}{\partial x} - \dfrac{\partial A}{\partial z} = 31, & \dfrac{\partial A}{\partial y} - \dfrac{\partial B}{\partial x} = 12, \\ \dfrac{\partial A}{\partial w} - \dfrac{\partial D}{\partial x} = 14, & \dfrac{\partial B}{\partial w} - \dfrac{\partial D}{\partial y} = 24, & \dfrac{\partial C}{\partial w} - \dfrac{\partial D}{\partial z} = 34, \end{array}$$

where of course $12 = -21$, &c., and

$$123 = 1{\cdot}23+2{\cdot}31+3{\cdot}12, \quad \text{\&c.}; \quad 1234 = 1{\cdot}234-2{\cdot}341+3{\cdot}412-4{\cdot}123,$$

is $= 0$ identically;

$$1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23,$$

then the conditions equivalent to three independent conditions are

$$234 = 0, \quad 341 = 0, \quad 412 = 0, \quad 123 = 0, \quad 1234 = 0.$$

In fact, the first four equations are

$$\begin{array}{rrrrl} & 2 \cdot 34 & - 3 \cdot 24 & + 4 \cdot 23 & = 0, \\ -1 \cdot 34 & & + 3 \cdot 14 & + 4 \cdot 31 & = 0, \\ 1 \cdot 24 & - 2 \cdot 14 & & + 4 \cdot 12 & = 0, \\ -1 \cdot 23 & - 2 \cdot 31 & - 3 \cdot 12 & & = 0; \end{array}$$

hence, multiplying by $1, 2, 3, 4$ respectively and adding, we have the identity $1234 = 0$, so that these four are equivalent to three independent equations: and multiplying by

$$\begin{array}{rrrr} & 12 \cdot \lambda & - 31 \cdot \mu & + 14 \cdot \nu, \\ -12 \cdot \lambda & & + 23 \cdot \mu & + 24 \cdot \nu, \\ 31 \cdot \lambda & - 23 \cdot \mu & & + 34 \cdot \nu, \\ & -14 \cdot \lambda & - 24 \cdot \mu & - 34 \cdot \nu \end{array}$$

respectively, (where λ, μ, ν, ρ are arbitrary), we have

$$(1 \cdot \lambda + 2 \cdot \mu + 3 \cdot \nu + 4 \cdot \rho)(23 \cdot 14 + 31 \cdot 24 + 12 \cdot 34) = 0,$$

that is,

$$23 \cdot 14 + 31 \cdot 24 + 12 \cdot 34 = 0, \quad \text{or} \quad 1234 = 0,$$

the fifth condition.

942. On Seminvariants.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxvI. (1893), pp. 66—69.]

I Wish to prove the following negative: a given sharp seminvariant is not in every case obtainable by mere derivation from a form of the same extent and of the next inferior degree. The meaning of the statement will be explained.

According to the general theory developed in Clebsch's *Theorie der binären algebraischen Formen*, Leipzig, 1872, the covariants of a given binary quantic f are all of them obtainable, the covariants of a given degree from those of the next inferior degree, by derivation (Ueberschiebung) of these with f; viz. if the covariants of the next inferior degree are P, Q, &c., then the covariants of the degree in question are all of them included among the forms

$$(f, P)^0 (= fP), \quad (f, P)^1, \quad (f, P)^2, \ldots$$

$$(f, Q)^0, \quad (f, Q)^1, \quad (f, Q)^2, \ldots$$

&c., the index of derivation for (f, P) being at most equal to the degree of f or to that of P, whichever of these is the smaller, and so for Q, &c. The forms thus obtained are far too numerous; but rejecting repetitions, we have a complete system of the covariants of the given degree, viz. every covariant whatever of that degree is a linear function (with numerical multipliers) of the several distinct forms thus obtained by derivation.

We can therefore, by linear combination as above, obtain all the sharp covariants of the given degree, but we may very well have a sharp covariant not included among the several distinct forms thus obtained by derivation, but only expressible as a linear combination of two or more such forms; or say we may very well have a sharp covariant not obtainable by mere derivation from the forms of the next inferior degree; and it is important to verify that there are sharp covariants not thus obtainable by mere derivation. I remark that the notion of a sharp covariant does not present itself in Clebsch, and that it will be presently explained.

Passing from a covariant to its leading coefficient which is a seminvariant, the statement may be applied to seminvariants; viz. it is to be verified that there are sharp seminvariants not obtainable by mere derivation from the seminvariants of the next inferior degree. But we have to introduce the notion of "extent" so as to connect the seminvariant with a quantic of some particular order, thus, if the highest letter of the seminvariant is f, we say that the extent is $= 5$, and thus connect it with the quintic

$$(1, b, c, d, e, f \between x, y)^5.$$

The notion "sharp" applies to the seminvariants of a given weight. Suppose, for instance, the weight is $= 8$; we have a series of initial or non-unitary terms i, eg, df, e^2, &c., and a series of final or power-ending terms e^2, cd^2, b^2d^2, c^4, &c., and we denote by $eg - c^4$ (where observe that here and in all similar cases the — is not a minus sign, but is simply a stroke), the whole series of terms (including eg and c^4) which are in counter-order (CO) subsequent to eg, and in alphabetical order (AO) precedent to c^4; and so in other cases. This being so, arranging the seminvariants with their final terms in AO, we have the seminvariants

$$\begin{aligned} &i - e^2, \\ &eg - cd^2, \\ &df - b^2d^2, \\ &e^2 - c^4, \end{aligned}$$

&c., viz. we have a seminvariant $i - e^2$ containing all or any (in fact, all) of the terms of this set as above defined; a seminvariant $eg - cd^2$ containing all or any of the terms of this set, a seminvariant $df - b^2d^2$ containing all or any of the terms of this set; and so on. These are sharp forms; a seminvariant ending in e^2, must of necessity have the leading term i, and thus belong at least to the octic

$$(1, b, c, d, e, f, g, h \between x, y)^8,$$

a seminvariant ending in cd^2 must of necessity have a leading term as high as eg, and thus belong at least to the sextic

$$(1, b, c, d, e, f, g \between x, y)^6.$$

Any linear combination of these would be a seminvariant $i - cd^2$, belonging to the octic, but it is not a sharp form; the final term cd^2 does not of necessity imply an initial so high as i (in fact, as we have seen, it only implies the lower initial eg): and so in other cases.

For the quintic $(1, b, c, d, e, f \between x, y)^5$, we have (for the weight 8 and degree 4) the sharp seminvariant $df - b^2d^2$; it is required to show that this is not obtainable by mere derivation from the covariants of degree 3 of the quintic.

The quintic and its covariants up to the degree 3 are

$$\begin{aligned}
A &= (1, b, c, d, e, f \between x, y)^5 \quad \text{say}\\
B &= (e - c^2, f - cd, bf - d^2 \between x, y)^2;\\
C &= (c - b^2, d - bc, e - c^2, f - cd, bf - d^2, cf - de \between x, y)^6,\\
D &= (ce - c^3, cf - c^2d, df - cd^2, bdf - d^3 \between x, y)^3,\\
E &= (f - bc^2, bf - c^3, cf - c^2d, df - cd^2, ef - d^3, f^2 - d^2e \between x, y)^5,\\
F &= (d - b^3, e - b^2c, f - bc^2, bf - c^3, cf - c^2d, df - cd^2,\\
&\qquad ef - d^3, f^2 - d^2e, bf^2 - de^2, cf^2 - e^3 \between x, y)^7.
\end{aligned}$$

Hence all the covariants of the degree 3 are A^3, AB, AC, D, E, F, where

$$AB = (e - c^2, f - bc^2, bf - c^3, cf - c^2d, bcf - cd^2, bdf - d^3, bef - d^2e, bf^2 - d^2e \between x, y)^7,$$

and the derivatives are

$$\begin{array}{rl}
(A, A^3)^0, (A, A^3)^1, \&c. & \text{weights of leading coefficients are } 0, 1, 2, 3, 4, 5\\
(A, AB)^0, (A, AB)^1, \&c. & \text{“ } 4, 5, 6, 7, 8, 9\\
(A, AC)^0, (A, AC)^1, \&c. & \text{“ } 2, 3, 4, 5, 6, 7\\
(A, D)^0, (A, D)^1, \&c. & \text{“ } 6, 7, 8, 9\\
(A, E)^0, (A, E)^1, \&c. & \text{“ } 5, 6, 7, 8, 9, 10\\
(A, F)^0, (A, F)^1, \&c. & \text{“ } 3, 4, 5, 6, 7, 8.
\end{array}$$

The terms giving rise to a seminvariant of weight 8 are

$$\begin{aligned}
(A, AB)^1 &= df - c^4,\\
(A, D)^2 &= df - c^4,\\
(A, E)^3 &= df - c^4 \text{ or } e^2 - c^4,\\
(A, F)^5 &= df - c^4 \text{ or } e^2 - c^4.
\end{aligned}$$

where to explain the algorithm, I remark, that if

$$A = (A_0, A_1, A_2, \ldots \between x, y)^5 \quad \text{and} \quad D = (A, A, D)^2,$$

then

$$(A, D)^2 = \begin{vmatrix} A, & A, & A \\ A, & A, & A \\ D, & D, & D \end{vmatrix}$$

represented as above by

$$df - c^4.$$

The result is in every case given as $df - c^4$; in each case there is only a single term $c \cdot c^3$, $= c^4$, and the term in c^4 certainly presents itself. In $(A, AB)^1$ there is a single term $d \cdot f$, $= df$, and in $(A, D)^2$ a single term df, and thus the term df certainly presents itself: in $(A, E)^3$ there are two terms $d \cdot f$, $= df$ and df, and it is conceivable that, inserting the proper numerical coefficients, these might destroy each other: if this were so, the form instead of being $df - c^4$ would be $e^2 - c^4$; and similarly in $(A, F)^5$, there are two terms df, $= df$ and df which it is conceivable might destroy each other, and the form would then be $e^2 - c^4$. But in every case we have the term c^4, and it thus appears that the form $df - b^2d^2$ is not obtainable by mere derivation.

The form in question is in fact obtained by a linear combination of $df - c^4$ and $e^2 - c^4$; viz. writing down the leading coefficients of the covariants B^2 and H, we have

	$3H$	$-2B^2$
df	3	-2
e^2	0	-2
bcf	-9	-9
bde	-15	$+16$
$+1$		
c^2e	$+30$	-12
-18		
cd^2	-12	-12
b^2f	$+6$	$+6$
b^2ce	-15	-15
b^3d	$+42$	-32
-10		
c^3d	-48	$+48$
0		
c^4	$+18$	-18
0		

viz. the form in question $df - b^2d^2$ is $= 3df - 2e^2 - \ldots - 10b^2d^2$.

C. XIII. 47

943. On Reciprocants and Differential Invariants.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxvI. (1893), pp. 169—194, 289—307.]

I USE the term Reciprocant to denote a function of an arbitrary variable or variables and its differential coefficients, not connected with any differential equation; and Differential Invariant to denote a function of the coefficients of a differential equation and the differential coefficients of these coefficients. Halphen's differential invariants are thus reciprocants, but the term reciprocant is not made use of by him. I have entitled the present paper "On Reciprocants and Differential Invariants"; in the earlier part, (except that for preserving a chronological order, I briefly refer to Sir J. Cockle's Criticoids, which are differential invariants or rather seminvariants), I attend almost exclusively to Reciprocants, reproducing and explaining, and in some parts developing, the theories of Ampère, Halphen in his first two memoirs, and Sylvester.

I.

The notion of a reciprocant first presents itself in Ampère's "Mémoire sur les avantages qu'on peut retirer dans la théorie des courbes de la consideration des paraboles osculatrices, avec des reflexions sur les fonctions différentielles dont la valeur ne change pas lors de la transformation des axes," *Journ. École Polyt.* t. vii. (1808), pp. 151—191 (sent to the Institute, Dec. 1803)*. We have (p. 167) for the radius of curvature the expression

$$\rho = \frac{(1+y'^2)^{\frac{3}{2}}}{y''},$$

and for the parameter of the osculating parabola the expression

$$\sigma = \frac{9y''^4}{y''^2 + 3(2y'''^2 - 2y'y'')^2};$$

and it is explicitly noticed that each of these expressions remains absolutely unaltered when the coordinates x, y are changed into any other (rectangular) coordinates whatever. The functions here spoken of are thus orthogonal absolute reciprocants; the change which leaves them unaltered is that of x, y into X, Y, where

$$X = x\cos\theta + y\sin\theta + a, \quad Y = -x\sin\theta + y\cos\theta + b.$$

For the more general change

$$X = ax + by + \alpha, \quad Y = cx + dy + \beta,$$

we find without difficulty

$$Y' = \frac{c + dy'}{a - by'}, \quad \frac{Y''}{(a - by')^2} = \frac{(ad - bc)y''}{(a - by')^3},$$

and thus there is not an identity of form in the expressions

$$\frac{(1 + Y'^2)^{\frac{3}{2}}}{Y''} \quad \text{and} \quad \frac{(1 + y'^2)^{\frac{3}{2}}}{y''}.$$

But if we disregard factors, or (what is the same thing) attend to the equations $y'' = 0$ and $Y'' = 0$, we see that for this more general change one of these equations implies the other; each of them is, in fact, the condition for an inflexion of the curve in (x, y) or (X, Y).

* *A reciprocant, the Schwarzian derivative, occurs in Lagrange's memoir, "Sur la construction des cartes géographiques," Nouv. Mem. de Berlin, 1779, Œuvres, t. iv. p. 651, but scarcely quâ reciprocant, viz. the form* $\frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2$ *acquires only a factor by the reciprocal substitution* $x = \frac{1}{y}$.

II.

The differential invariants, or rather seminvariants, called "Criticoids," were considered by Sir James Cockle in his paper "On Criticoids," *Phil. Mag.* t. xxxix. (1870), pp. 201—221; these will be more particularly considered further on, but at present it is sufficient to remark that the functions in question are connected with a linear differential equation, viz. they are functions of the coefficients of a linear differential equation, such that effecting thereon a transformation of the dependent variable, say $y = f \cdot Y$, where f is an arbitrary function of the independent variable x, the function is unaltered save as to a factor which is a power of f; but they are not in general unaltered when the independent variable is also changed.

III.

We have four important Memoirs by Halphen, (1) "Thèse d'Analyse sur les invariants différentiels," Paris (1878), (2) "Sur les invariants différentiels des courbes gauches," *Jour. École Polyt.* Cah. xlvii. (1880), pp. 1—102; (3) "Mémoire sur la réduction des équations différentielles linéaires aux formes intégrables," *Mem. Sav. Etrang.* t. xxviii. (1883), pp. 1—297, and (4) "Sur les invariants des Equations différentielles linéaires du quatrième ordre," *Acta Math.* t. III. (1883), pp. 325—380.

In the Memoir, Halphen (1), the investigations are in the first instance presented in a geometrical form; the author considers for instance the inflexions of a plane curve, and so obtains the invariant y'', or using his notation

$$\alpha_2 = \frac{1}{1\cdot 2}\frac{d^2y}{dx^2},$$

say the invariant α_2 of the weight 2; similarly, the consideration of the sextactic points gives him the invariant $\alpha_2^2\alpha_5 - 3\alpha_2\alpha_3\alpha_4 + 2\alpha_3^3$ of the weight 9; and that of the points of nine-pointic contact a more complicated invariant of the weight 27: and with these he forms absolute invariants. But the formal analytical definition is given § 3, "Théorie des invariants différentiels jusqu'au huitième ordre exclusivement," viz. considering the general homographic transformation

$$X = \frac{\alpha x + \beta y + \gamma}{\alpha'' x + \beta'' y + \gamma''}, \quad Y = \frac{\alpha' x + \beta' y + \gamma'}{\alpha'' x + \beta'' y + \gamma''},$$

then if the equation

$$y' = \phi\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots\right)$$

is unaltered in form by the change (x, y) into (X, Y), or what is the same thing if the function

$$\Phi\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots\right)$$

be unaltered save as to a factor, then such function is a differential invariant.

IV.

For the analytical theory, we have as above

$$X = \frac{\alpha x + \beta y + \gamma}{\alpha'' x + \beta'' y + \gamma''}, \quad Y = \frac{\alpha' x + \beta' y + \gamma'}{\alpha'' x + \beta'' y + \gamma''},$$

taking h, k for the increments of x, y respectively, we have

$$k = th + ah^2 + bh^3 + ch^4 + dh^5 + \ldots,$$

and, similarly, taking H, K for the increments of X, Y respectively, we have

$$K = TH + AH^2 + BH^3 + CH^4 + DH^5 + \ldots,$$

and we require the relations between the two sets of coefficients $(t, a, b, c, d, \ldots)$ and $(T, A, B, C, D, \ldots)$: I propose to develop these up to d, D, so as to obtain the theory for the form $a^2d - 3abc + 2b^3$.

Writing for shortness ξ, η, ζ for

$$\alpha x + \beta y + \gamma, \quad \alpha' x + \beta' y + \gamma', \quad \alpha'' x + \beta'' y + \gamma''$$

respectively, we have

$$X = \frac{\xi}{\zeta}, \quad Y = \frac{\eta}{\zeta},$$

and thence

$$\frac{dX}{dx} = \frac{\alpha\zeta - \alpha''\xi}{\zeta^2} + \frac{\beta\zeta - \beta''\xi}{\zeta^2} y',$$

or, for k substituting its value,

$$\frac{dY}{dX} = \frac{(P' + Q't)h + Q'(ah^2 + bh^3 + ch^4 + dh^5)}{\xi[1 + \lambda''\{h + \lambda''(ah^2 + bh^3 + ch^4 + dh^5)\}]},$$

if

$$L = P + Qt, \quad \lambda = \frac{-\alpha''}{\alpha'' x + \beta'' y + \gamma''},$$

$$L' = P' + Q't, \quad \lambda' = \frac{-\alpha''}{\alpha'' x + \beta'' y + \gamma''},$$

$$L'' = \alpha'' + \beta'' t, \quad \lambda'' = \frac{-\alpha''}{\alpha'' x + \beta'' y + \gamma''},$$

$$\mu = \frac{L'}{L''^2}, \quad \nu = \frac{L}{L''^2}.$$

V.

We in fact have

$$\frac{dY}{dX} = \frac{L' + Q'k_1}{L + Qk_1},$$

where $k_1 = \frac{dk}{dh}$, that is

$$k_1 = t + 2ah + 3bh^2 + 4ch^3 + 5dh^4 + \dots,$$

and thence

$$\frac{dY}{dX} = \frac{L' + Q'(t + 2ah + 3bh^2 + 4ch^3 + 5dh^4)}{L + Q(t + 2ah + 3bh^2 + 4ch^3 + 5dh^4)},$$

which, putting therein $h = 0$, gives $T = \frac{L'}{L}$.

VI.

For the higher differential coefficients, we have

$$\frac{d^2Y}{dX^2} = \frac{d}{dh}\left(\frac{dY}{dX}\right) \div \frac{dX}{dh},$$

and so on; and the calculations give

$$\begin{aligned}
&A = (\lambda' - \lambda)\mu a,\\
&B = (\lambda' - \lambda)\mu^2\{b + aL'' - a^2\lambda\},\\
&C = (\lambda' - \lambda)\mu^3\{c + 2bL'' + aL''^2 - (4ac + 2b^2)\lambda + 6a^2b\lambda^2 - 5a^4\lambda^3\},\\
&D = (\lambda' - \lambda)\mu^4\{d + 3cL'' + 3bL''^2 + aL''^3\\
&\qquad\qquad - (6ad + 6bc + 3c^2)\lambda + (21a^2c + 21ab^2)\lambda^2 - 35a^3b\lambda^3 + 14a^5\lambda^4\}.
\end{aligned}$$

And it is important to notice that these coefficients A, B, C, D are each of them homogeneous of the degree 1 in $(\alpha, \beta, \gamma, \alpha', \beta', \gamma', \alpha'', \beta'', \gamma'')$.

VII.

Halphen writes

$$U = a, \quad V = a^2d - 3abc + 2b^3,$$

and defines Λ by means of the equation $U^4\Lambda = 256V^3 - 27V^8$, viz.

$$a^4\Lambda = 256(a^2d - 3abc + 2b^3)^3 - 27(a^2d - 3abc + 2b^3)^8,$$

which is an invariant of the degree 8 and the weight 24; and he forms with these the absolute invariants $\frac{V^3}{U^8}$ and $\frac{\Lambda}{U^8}$.

VIII.

We have next Halphen (2), "Sur les invariants différentiels des courbes gauches," *Jour. École Polyt.* Cah. xlvii. (1880), pp. 1—102.

IX.

And then Halphen (3), "Mémoire sur la réduction des équations différentielles linéaires aux formes intégrables," *Mem. Sav. Etrang.* t. xxviii. (1883), pp. 1—297.

We come now to Sylvester.

X.

We have next Sylvester:—"Lectures on the theory of Reciprocants" (reported by J. Hammond), *Amer. Math. Journ.*, t. viii. (1886), pp. 196—260.

The lectures were delivered at Oxford, Inaugural Lecture, Dec. 1885, starting from the Schwarzian function

$$\frac{y'''}{y'} - \frac{3}{2}\left(\frac{y''}{y'}\right)^2,$$

which acquires only a factor by the interchange of x and y.

In lecture 2, pp. 203 *et seq.*, Sylvester considers the general theory of the functions which remain unaltered, except as to a factor, by the interchange of x, y; viz. writing $t, a, b, c, \ldots$ for $y', y'', y''', y^{\text{iv}}, \ldots$, or again

$$t, a_0, a_1, a_2, a_3, \ldots \quad \text{for} \quad y', \tfrac{1}{2}y'', \tfrac{1}{6}y''', \tfrac{1}{24}y^{\text{iv}}, \ldots;$$

and so

$$T, \alpha, \beta, \gamma, \ldots \quad \text{for} \quad x', x'', x''', x^{\text{iv}}, \ldots,$$

or again

$$T, \alpha_0, \alpha_1, \alpha_2, \alpha_3, \ldots \quad \text{for} \quad x', \tfrac{1}{2}x'', \tfrac{1}{6}x''', \tfrac{1}{24}x^{\text{iv}}, \ldots$$

he gives the equations

$$\begin{aligned}
&\alpha = -a \div t^3, \quad \text{or} \quad a_0 = -a_0 \div t^3,\\
&\beta = -bt + 3a^2 \div t^5, \quad \text{or} \quad a_1 = -a_1 + 2a_0^2 \div t^5,\\
&\gamma = -ct^2 + 10abt - 15a^3 \div t^7, \quad \text{or} \quad a_2 = -a_2t^2 + 5a_0a_1t - 5a_0^3 \div t^7,
\end{aligned}$$

and obtains with $(a, b, c, \ldots)$ reciprocants such as a, $2bt - 3a^2$, &c., viz. we have

$$a = -t^3 \cdot \alpha,$$

and further on, like forms with $(a_0, a_1, a_2, \ldots)$.

It is to be remarked that it is preferable to deal with the quantities $(a_0, a_1, a_2, \ldots)$ which represent $(\frac{1}{2}y'', \frac{1}{6}y''', \frac{1}{24}y^{\text{iv}}, \ldots)$ rather than with $(a, b, c, \ldots)$ which represent $(y', y'', y''', \ldots)$. I do this in the sequel, changing the notation, and writing $(t, a, b, c, \ldots)$ to denote $(y', \frac{1}{2}y'', \frac{1}{6}y''', \frac{1}{24}y^{\text{iv}}, \ldots)$, viz. my $(t, a, b, c, \ldots)$ are not Sylvester's $(t, a, b, c, \ldots)$, but are his $(t, a_0, a_1, a_2, \ldots)$.

A reciprocant with Sylvester is thus a function of y and its differential coefficients in regard to x, which except as to a factor remain unaltered when x, y are interchanged, or say, when x, y are changed into X, Y, where $X = y, Y = x$. This is a much less general change than Halphen's, and thus every reciprocant (Halphen) is a reciprocant (Sylvester), but not conversely. The reciprocants (Sylvester) are far more numerous. I remark that incidentally Sylvester considers reciprocants, which remain unaltered save as to a factor, when x, y are changed into X, Y, where

$$X = \frac{\alpha x + \beta}{\gamma x + \delta}, \quad Y = \frac{\alpha' y + \beta'}{\gamma' y + \delta'},$$

which again is a less general change than Halphen's—it has been in what precedes alluded to as the particular case $L'' = 0$, that is, $\alpha'' = 0$, $\beta'' = 0$. It may be remarked that the proper orthogonal substitution corresponding to Sylvester's substitution is not $X = y$, $Y = x$, but $X = y$, $Y = -x$, viz. we have here the determinant $\alpha\beta' - \alpha'\beta = +1$.

I develop Sylvester's theory of reciprocants; but I wish first to point out the resemblance in form between this theory and that of seminvariants. In the theory of seminvariants, from the set of quantities $(a, b, c, \ldots)$ in connexion with an arbitrary quantity θ, we deduce a new set $(a', b', c', \ldots)$, where

$$\begin{aligned} a' &= a, \\ b' &= b + a\theta, \\ c' &= c + 2b\theta + a\theta^2, \end{aligned}$$

or, what is the same thing, if $\delta = -\theta$, then

$$\begin{aligned} a &= a', \\ b &= b' + a'\delta', \\ c &= c' + 2b'\delta' + a'\delta'^2, \end{aligned}$$

and this being so there exist functions which are the same for unaccented and the accented letters respectively; for instance, $a' = a$:

$$a'c' - b'^2 = ac + 2ab\theta + a^2\theta^2 - (b + a\theta)^2 = ac - b^2;$$

and similarly

$$a'^3d' - 3a'b'c' + 2b'^3 = a^3d - 3abc + 2b^3, \quad \&c.;$$

these functions a, $ac - b^2$, $a^3d - 3abc + 2b^3$, &c., are called seminvariants.

XI.

Similarly, in Sylvester's theory of reciprocants, starting from y, a given function of x, we have $(t, a, b, c, \dots)$ denoting $(y', \frac{1}{2}y'', \frac{1}{6}y''', \frac{1}{24}y^{\text{iv}}, \dots)$; and conversely $(t', a', b', c', \dots)$ denoting $(x', \frac{1}{2}x'', \frac{1}{6}x''', \frac{1}{24}x^{\text{iv}}, \dots)$; taking h, k for the increments of x and y respectively, we have

$$
\begin{aligned}
k &= th + ah^2 + bh^3 + ch^4 + dh^5 + \dots, \\
h &= t'k + a'k^2 + b'k^3 + c'k^4 + d'k^5 + \dots,
\end{aligned}
$$

which equations determine the relations between $(t, a, b, c, d, \dots)$ and $(t', a', b', c', d', \dots)$, viz. we have $tt' = 1$, and then

$$
\begin{aligned}
a' &= -a \div t^3, \\
b' &= -bt + 2a^2 \div t^5, \\
c' &= -ct^2 + 5abt - 5a^3 \div t^7, \\
d' &= -dt^3 + (6ac + 3b^2)t^2 - 21a^2bt + 14a^4 \div t^9, \\
e' &= -et^4 + (7ad + 7bc)t^3 - (28a^2c + 28ab^2)t^2 + 84a^3bt - 42a^5 \div t^{11},
\end{aligned}
$$

and conversely,

$$
\begin{aligned}
a &= -a' \div t'^3, \\
b &= -b't' + 2a'^2 \div t'^5, \\
c &= -c't'^2 + 5a'b't' - 5a'^3 \div t'^7,
\end{aligned}
$$

and we thence deduce functions which have equal or opposite values for the accented and unaccented letters respectively; these functions may or may not contain t, t'.

Thus we have

$$
\begin{aligned}
a' &= -a \div t^3, \\
(b't' - a'^2)t'^{-3} &= -(bt - a^2)t^{-3}, \\
(4a'c' - 5b'^2)t'^{-4} &= (4ac - 5b^2)t^{-4}, \\
\{(1 + t'^2)b' - 2a'^2t'\}t'^{-5} &= \{(1 + t^2)b - 2a^2t\}t^{-5},
\end{aligned}
$$

&c.

These functions of the unaccented letters, or say the same functions omitting the exterior power of t, are called Reciprocants; thus we have the reciprocants a, $bt-a^2$, $2ct-5ab$, $4ac-5b^2$, $(1+t^2)b-2a^2t$, &c. Observe that, when the exterior powers of t are omitted, then the values are equal or opposite to those with the accented letters save as to a power of t, viz. the forms are

$$\begin{aligned}
a &= -a't^3,\\
bt-a^2 &= -(b't'-a'^2)t^6,\\
2ct-5ab &= -(2c't'-5a'b')t^7,\\
4ac-5b^2 &= (4a'c'-5b'^2)t^8,\\
((1+t^2)b-2a^2t) &= -\{(1+t'^2)b'-2a'^2t'\}t^9,
\end{aligned}$$

&c.

A reciprocant is said to be odd or even according as the sign on the right-hand side is $-$ or $+$; the index of t on the right-hand side is said to be the weight of the reciprocant. Reciprocants may be combined in the way of addition if and only if they are each of them of the same weight and parity, viz. this being so, then if $\lambda, \lambda', \ldots,$ are mere numbers, $\lambda R + \lambda' R' + \ldots$ will be a reciprocant of the same weight and parity with each of the reciprocants $R, R', \ldots$.

Reciprocants may in every case be combined in the way of multiplication; viz. two or more reciprocants may be multiplied together giving a reciprocant the parity of which is the sum of the parities, and its weight the sum of the weights of the component reciprocants. Thus $bt-a^2$ and $2ct-5ab$ being odd reciprocants of the weights 6 and 7 respectively, we have $(bt-a^2)(2ct-5ab)$ an even reciprocant of the weight 13.

XII.

Sylvester arranges the reciprocants in "Tables of Pure Reciprocants to the weight 8," *Amer. Math. Journ.*, t. xv. (1893), pp. 79—87, [933]. I reproduce these with a change of notation. The fundamental pure reciprocants are:

	ac	a^2d	a^3e	a^4f	a^5g	a^6h
$12\mathfrak{R} =$	-5					
$60\mathfrak{R} =$	$+3$	$+1$				
$10\mathfrak{R} =$						
$420\mathfrak{R} =$	-45	-7	$+2$	$+1$		
$2310\mathfrak{R} =$	$+315$	$+105$				
$7\mathfrak{R} =$						

XIII. Halphen's Reciprocants.

To the reciprocants thus obtained, say

$$U = a, \quad V = a^2d - 3abc + 2b^3,$$

Halphen adds another reciprocant Λ (that of nine-pointic contact) which he expresses in the form of a determinant. Writing down in connexion therewith its developed expression, we have

$$\Lambda = \begin{vmatrix} a & -a^2 & c & d & e & f \\ b & c & d & e & f & g \\ 0 & b^2 & 2bc & 2bd + c^2 & 2be + 2cd & 2bf + 2ce + d^2 \\ 0 & a^2 & 2ab & 2ac + b^2 & 2ad + 2bc & 2ae + 2bd + c^2 \end{vmatrix}$$

$$\begin{aligned} = a^8(df + e^2) & \\ + a^5(bcf - 3bde + 3cd^2) & \\ + a^4(b^3f + 2b^2ce - 5bc^2d - 5b^2d^2 + c^4) & \\ + a^3(b^3cd - 10b^2c^3 + b^4e) & \\ + a^2(b^5d + 4b^4c^2) & \\ + a(b^6c - 12b^7) & \\ + a^0(b^8 + 3), & \end{aligned}$$

where observe that the whole term in the highest letter f is $a^4(a^2d - 3abc + 2b^3)f$, viz. the coefficient of f is the reciprocant $a^4(a^2d - 3abc + 2b^3)$, agreeing with a general theorem given by Halphen.

Λ is of the degree 8 and the weight 24, and it is seen without difficulty that the factor is $= (\lambda' - \lambda)^8\mu^{16}$. He remarks that $256V^3 - 27\Lambda^2$ vanishes for $a = 0$ (viz. it becomes $256 \cdot 27b^{24} - 27 \cdot 256b^{24} = 0$), and not only so, but that it in fact contains the factor a^4. Assuming that this is so, viz. that the terms in a^0, a, a^2, a^3 all of them vanish, I have calculated the terms in a^4, and have thus obtained an incomplete expression for the quotient $(256V^3 - 27\Lambda^2) \div a^4$, viz. this is

$$\begin{aligned} \Phi = (256V^3 - 27\Lambda^2) \div a^4 = 256a^{14}(df + e^2)^3 & \\ + 288a^{13}(b^3f + \ldots) & \\ + a^{12}(\ldots) & \\ + \ldots & \end{aligned}$$

and so on; the complete form being very complicated. But I write

$$\Phi = a^4\Theta,$$

and thence Halphen's Θ is known; and I observe that the expression is

$$\begin{aligned} a^4\Theta = 256(a^2d - 3abc + 2b^3)^3 \\ - 27\{a^8(df + e^2) + a^5(bcf - 3bde + 3cd^2) + \ldots\}^2. \end{aligned}$$

We have identically

$$256V^3 - 27\Lambda^2 = a^4\Theta,$$

and thus

$$\Theta = a^{-4}(256V^3 - 27\Lambda^2).$$

XIV. The Differential Equation of a Cubic Curve.

Halphen and Sylvester find that, for a cubic curve the invariants of which are S and T ($S = -l + l^3$, $T = 1 - 20l^3 - 8l^6$ for the canonical form $x^3 + y^3 + z^3 + 6lxyz = 0$), we have

$$\Phi S = \Theta^2 - 12288V^3(8V + 9U^4),$$

$$\Phi T = \Theta^3 - 18432V^3(8V + 9U^4) + 556231U^4V^6,$$

where Φ is an arbitrary multiplier; and we thence have

$$\frac{S^3}{T^2} = \frac{\{\Theta^2 - 12288V^3(8V + 9U^4)\}^3}{\{\Theta^3 - 18432V^3(8V + 9U^4) + 556231U^4V^6\}^2},$$

or if, as with Halphen $h = K\frac{S^3}{T^2}$, then

$$(h - 1)R^2 = hQ,$$

where

$$R = \Theta^3 - 18432V^3(8V + 9U^4) + 556231U^4V^6,$$

$$Q = \Theta^2 - 12288V^3(8V + 9U^4).$$

If for shortness

$$P = \Theta^3 - 12288V^3(8V + 9U^4) = R^2 - P^3,$$

then the foregoing equation $\frac{S^3}{T^2} = h$ gives

$$\frac{R^2}{Q} = \frac{h}{h-1},$$

and thus for a nodal cubic we have $Q = 0$; we thus have

$P = 0$, for the differential equation of a cubic for which $S = 0$,
$R = 0$, " " " " " $T = 0$,
$Q = 0$, " " nodal cubic,

these being each of them of the order 8.

But the equation $Q=0$ is reducible to a more simple form. We have

$$Q=\frac{1}{(3072V)^3}\{(64\mathfrak{D}-144A^2\mathfrak{D}+27A^6)^2+32768\mathfrak{D}^3\},$$

or, since in the expression in { } the first and second terms are $12\Theta^2(42\Theta-\Phi^2)=$ this is

$$Q=\frac{1}{(3072V)^3}\{(\Theta-364A^2\Phi+216A\Phi)^2-64(\Phi-94A)^3\}.$$

We thus have for the differential equation of the eighth order of the nodal cubic

$$(\Theta-364A^2\Phi+216A\Phi)^2-64(\Phi-94A)^3=0,$$

or, since

$$\Theta=64\mathfrak{D}+144A^2\mathfrak{D}+81A^6,\quad\text{and}\quad\Phi=8\mathfrak{D}+94A,$$

the differential equation of the eighth order for the nodal cubic finally is

$$(64\mathfrak{D}-144A^2\mathfrak{D}+27A^6)^2+32768\mathfrak{D}^3=0.$$

Halphen has $\Phi^2=0$ for the differential equation of a cubic curve. Putting, as before,

$$\begin{aligned}
\mathfrak{C}&=AC-B^2,\\
\mathfrak{D}&=A^2D-3ABC+2B^3,\\
\mathfrak{E}&=A^2D^2-6ABCD+4AC^3+4B^3D-3B^2C^2,\\
\mathfrak{F}&=AE-4BD+3C^2,\\
\mathfrak{G}&=ACE-AD^2-B^2E+2BCD-C^3,
\end{aligned}$$

(and therefore $4\mathfrak{C}\mathfrak{E}-4\mathfrak{D}=\mathfrak{D}$, $\mathfrak{F}^3-27\mathfrak{G}^2=$ Quartic Discr.), then, by what precedes, the differential equation of the order 9 of a cubic curve is

$$a^{-5}(432\mathfrak{G}-54\mathfrak{C}\mathfrak{D}-\tfrac{27}{4}A^5)=0.$$

Recapitulating.

Order. The differential equation for a cubic curve:—

8. for which invt. $S = 0,$ is
   $P = \Theta^2 - 12288\mathfrak{C}^3(8\mathfrak{D} + 9A^4) = 0,$
8. for which invt. $T = 0,$ is
   $R = \Theta^3 - 18432\mathfrak{C}^3(8\mathfrak{D} + 9A^4) + 556231A^4\mathfrak{C}^6 = 0,$
8. nodal cubic
   $Q \div (3072V)^3 = (64\mathfrak{D} - 144A^2\mathfrak{D} + 27A^6)^2 + 32768\mathfrak{D}^3 = 0,$
8. for invt. $64S^3 \div T^2 = h,$ is $(h-1)R^2 - hQ = 0,$
9. for general cubic is $1728\mathfrak{G} - 216\mathfrak{C}\mathfrak{D} - 243A^5 = 0,$

where for shortness Θ is retained to signify its value

$$= 64\mathfrak{D} + 144A^2\mathfrak{D} + 81A^6.$$

Halphen by a polar transformation finds that these same differential equations, changing therein the sign of $\mathfrak{D}$, apply to curves of the third class: in particular, the last equation, changing therein the sign of $\mathfrak{D}$, applies to the general curve of the third class, or sextic curve with nine cusps.

XV. MacMahon's Multilinear Operator.

A very important notion in the theory, as well of Seminvariants as of Reciprocants, is that of MacMahon's Multilinear Operator, see his paper "Theory of a Multilinear Partial Differential Operator, with applications to the theories of Invariants and Reciprocants," *Proc. Lond. Math. Soc.*, t. xviii. (1886), pp. 61—88: this operator plays so important a part in the theories to which it relates, that I venture to reproduce the definition of it in what appears to me a simplified and more easily intelligible form, and to recapitulate some of the leading properties.

I take with him the letters to be $(a_0, a_1, a_2, a_3, \ldots) = (a, b, c, d, \ldots)$, viz. the first letter is a_0 or a, the second is a_1 or b, and so on, but when we are not concerned with a term of indefinite rank, I use always $(a, b, c, d, \ldots)$ and the like in regard to any other series of letters. This being so, the definition of the Multilinear Operator of four elements, which is here alone in question, is

$$
\begin{aligned}
&(\mu, \nu; m, n)_4 = \\
&(\mu + 0\nu)A\left(\frac{\partial}{\partial a} + m\frac{\partial}{\partial b} + \frac{m(m-1)}{1\cdot 2}\frac{\partial}{\partial c} + \frac{m(m-1)(m-2)}{1\cdot 2\cdot 3}\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+1}}\right) \\
&+ (\mu + \nu)B\left(\frac{\partial}{\partial b} + m\frac{\partial}{\partial c} + \frac{m(m-1)}{1\cdot 2}\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+2}}\right) \\
&+ (\mu + 2\nu)C\left(\frac{\partial}{\partial c} + m\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+3}}\right) \\
&+ (\mu + 3\nu)D\left(\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+4}}\right)
\end{aligned}
$$

where

$$
\begin{aligned}
A &= a^m, \\
B &= a^{m-1}b, \\
C &= a^{m-2}(ac + \tfrac{m-1}{2}b^2) = a^{m-2}c + \tfrac{m-1}{2}a^{m-2}b^2, \\
D &= a^{m-3}(ad + (m-1)bc + \tfrac{(m-1)(m-2)}{6}b^3) \\
&= a^{m-3}d + (m-1)a^{m-3}bc + \tfrac{(m-1)(m-2)}{6}a^{m-3}b^3,
\end{aligned}
$$

where the expressions for $B, C, D, \ldots$, are obtained successively from that of $A = a^m$, by Arbogast's rule of derivation, viz. we operate on the last letter of a term, and when there is a last but one letter which in alphabetical order immediately precedes the last letter, then also upon the last but one letter, and whenever the term thus obtained ends in a power, we divide by the index of the power. Thus the term $\frac{1}{2}(m-1)a^{m-2}b^2$ in the third line gives in the fourth line

$$\begin{aligned} &\tfrac{1}{2}(m-1)a^{m-2}\cdot 2bc + \tfrac{1}{2}(m-1)(m-2)a^{m-3}b\cdot b^2\cdot \tfrac{1}{2},\\ &= (m-1)a^{m-2}bc + \tfrac{1}{6}(m-1)(m-2)a^{m-3}b^3. \end{aligned}$$

Going a step further, we form in this manner the expression

$$\begin{aligned}E &= a^{m-1}e + (m-1)a^{m-2}(bd + \tfrac{1}{2}c^2)\\ &\qquad + \tfrac{1}{2}(m-1)(m-2)\{a^{m-3}\cdot 3b^2c + (m-3)a^{m-4}b\cdot b^3\cdot \tfrac{1}{6}\},\end{aligned}$$

that is,

$$\begin{aligned}E = a^{m-1}e&\\ &+ a^{m-2}\cdot(m-1)(bd+\tfrac{1}{2}c^2)\\ &+ a^{m-3}\cdot\tfrac{1}{2}(m-1)(m-2)\cdot 3b^2c\\ &+ a^{m-4}\cdot\tfrac{1}{12}(m-1)(m-2)(m-3)b^4,\end{aligned}$$

which is sufficient to explain the rule.

It may be noticed that, the operator being linear in μ, ν, we have

$$(\mu, \nu; m, n) = \mu(1, 0; m, n) + \nu(0, 1; m, n).$$

The Alternant of two multilinear operators.

If P, Q are the two operators, we have as usual

$$P\cdot Q = PQ + P * Q,$$

where $P\cdot Q$ denotes the successive operation first with Q and then with P upon any operand, PQ is the mere algebraical product of the operators, and $P * Q$ is the operator obtained by the operation of P upon Q.

Similarly,

$$Q\cdot P = QP + Q * P,$$

and since QP is the same thing as PQ, we obtain

$$P\cdot Q - Q\cdot P = P * Q - Q * P,$$

either of which equal expressions, or say rather the second of them, is called the *alternant* of P, Q and is written $[P, Q]$: viz. as the definition of the alternant, we have

$$[P, Q] = P * Q - Q * P.$$

We have the remarkable theorem, that the alternant of any two operators $(\mu', \nu'; m', n')$, $(\mu, \nu; m, n)$ is an operator $(\mu_1, \nu_1; m_1, n_1)$; where

$$\begin{aligned}
\mu_1 &= (m'+m-1)\frac{\mu'}{m'}\mu - (n+n')\nu'\mu + \frac{m-1}{m}\mu\nu' - \frac{m'-1}{m'}\mu'\nu,\\
\nu_1 &= (n'-n)\nu\nu' + \frac{m-1}{m}\mu\nu' - \frac{m'-1}{m'}\mu'\nu,\\
m_1 &= m'+m-1,\\
n_1 &= n'+n,
\end{aligned}$$

or since m_1, n_1 have such simple expressions, say it is an operator $(\lambda, \mu; m'+m-1, n'+n)$, where λ, μ have the values just written down.

XVI. Tables of Pure Reciprocants (continued).

Sylvester's tables of pure reciprocants to the weight 8 are given as follows. The fundamental reciprocants are denoted by $\mathfrak{R}$, and we have:

$$\textbf{Weight 2:}\quad 12\mathfrak{R} = ac - 5b^2$$

$$\textbf{Weight 3:}\quad 60\mathfrak{R} = 3a^2d + abe - 27abc + 38b^3$$

$$\textbf{Weight 4:}\quad 10\mathfrak{R} = a^3e + \ldots$$

For the higher reciprocants, Sylvester gives the following table:

Coefficient	ac	a^2d	a^3e	a^4f	a^5g	a^6h	a^7i	a^8j
$420\mathfrak{R} =$	-45	-7	$+2$	$+1$				
$2310\mathfrak{R} =$	$+315$	$+105$	$+19$	$+3$	$+1$			
$7\mathfrak{R} =$				$+139$	$+1117$	$+28785$		
$42\mathfrak{R} =$					$+4$	$+45$	$+3$	
$40\mathfrak{R} =$							$+19$	$+139$

In terms of the fundamental reciprocants of the table, ante p. 387, say these are

$$\mathfrak{R} = P_3,\quad 50a^2e - \ldots + 105b^2c = P_4,\quad 10a^3f - \ldots - 396bc = P_5,$$

$$2310b^2c = P_6,\quad 7a^3h - \ldots - 19256c^3 = P_7,$$

the expressions of these functions are

$$\begin{aligned}
49\mathfrak{R} &= P_3^2,\\
40\mathfrak{R} &= 4aP_5 - 38P_2P_3,\\
280\mathfrak{R} &= 200a^3P_6 + 1266aP_2P_4 - 3027aP_3^2 - 9128P_2^3,\\
1680\mathfrak{R} &= 240a^3P_7 + 2940aP_2P_5 - 3156aP_3P_4 + 2373P_3P_2^2.
\end{aligned}$$

Sylvester remarks that a Principiant, e.g. $a^2d - 3abc + 2b^3$, is at once a reciprocant, and in the theory of seminvariants (where $a, b, c, \ldots$, are the coefficients of the theory) a seminvariant; and conversely that any reciprocant which is also a seminvariant is a principiant: quâ seminvariant, the principiant is annihilated by

$$\Omega = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + \ldots$$

and quâ reciprocant is annihilated by

$$V = 2a^2\partial_b + 5ab\partial_c + (6ac + 3b^2)\partial_d + (7ad + 7bc)\partial_e + (8ae + 8bd + 4c^2)\partial_f + \ldots;$$

and any function which is annihilated by each of these operators is a principiant.

We form the foregoing series of functions $N = ac - b^2$, A, B, C, $\ldots$, as follows, viz. if

$$G' = (4ac-5b^2)\partial_b+(5ad-7bc)\partial_c+(6ae-9bd)\partial_d+(7af-11be)\partial_e+\ldots,$$

then

$$\begin{aligned}5A &= G'N,\\ 6B &= G'A,\\ 7C &= G'B + NA,\\ 8D &= G'C + 2NB,\\ 9E &= G'D + 3NC;\end{aligned}$$

and similarly we form the foregoing series of functions $5\mathfrak{A}$, $2\mathfrak{B}$, $3\mathfrak{C}$, $6\mathfrak{D}$, $\ldots$, as follows, viz. if

$$\mathfrak{G}' \text{ is Sylvester's } \mathfrak{G},$$

then

$$\begin{aligned}5\mathfrak{A} &= \mathfrak{G}'\mathfrak{R},\\ 6\mathfrak{B} &= \mathfrak{G}'\mathfrak{A},\\ 7\mathfrak{C} &= \mathfrak{G}'\mathfrak{B},\\ 8\mathfrak{D} &= \mathfrak{G}'\mathfrak{C} - 2\mathfrak{R}\mathfrak{B}, \&c.\end{aligned}$$

As already mentioned, N, A, B, C, $\ldots$, are seminvariants, $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$, $\ldots$, are reciprocants. Putting for shortness $\frac{1}{2}b = 0$, the two sets of functions are connected by

$$\begin{aligned}-A &= \mathfrak{A}, \quad \text{or conversely} \quad \mathfrak{A} = -A,\\ -B &= \mathfrak{B} - 0\mathfrak{A}, \quad \mathfrak{B} = -B + 0A,\\ -C &= \mathfrak{C} - 2\mathfrak{B} + 0\mathfrak{A}, \quad \mathfrak{C} = -C + 2B - 0A,\end{aligned}$$

so that, one of the sets being calculated, the other set can be at once deduced therefrom. These equations give

$$4C - B^2 = \mathfrak{A}\mathfrak{E} - \mathfrak{B}\mathfrak{D} + 2\mathfrak{C}\mathfrak{B} - 3\mathfrak{D}\mathfrak{A}, \quad \&c.,$$

viz. as mentioned above, any seminvariant in the letters $A, B, C, \ldots$, is equal to the same seminvariant in the letters $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \ldots$; or, what is the same thing, any principiant has the same expression in the letters $A, B, C, \ldots$, and in the letters $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \ldots$ respectively.

We have thus the entire series of Principiants,

$$AC - B^2, \quad A^2D - 3ABC + 2B^3, \quad AE - 4BD + 3C^2, \quad \&c.$$

XVII. Expressions in terms of the Capitals.

We may express Halphen's reciprocants in terms of the capitals $A, B, C, \ldots$. We have

$$U = a, \quad V = A, \quad \Lambda = AC - B^2.$$

To obtain formulas for the higher reciprocants, we require the derived functions $A', B', C', \ldots$, where the accent denotes differentiation in regard to x.

We have

$$a' = 2b, \quad b' = 3c, \quad c' = 4d, \quad d' = 5e, \quad e' = 6f, \quad \ldots$$

(whence in particular $a' = 3b$, $b' = 4c$, $c' = 5d$, $\ldots$, as is obvious). Hence, writing

$$G' = a(3b\partial_a + 4c\partial_b + 5d\partial_c + \ldots) - b(3a\partial_a + 5b\partial_b + 7c\partial_c + \ldots),$$

this may be written

$$G' = a\partial_x - b\omega,$$

if ω be the weight of the homobaric function operated upon, reckoning the weights of $a, b, c, \ldots$, as $3, 5, 7, \ldots$, respectively, and consequently the weights of $A, B, C, \ldots$, as $15, 20, 25, \ldots$ respectively. We thus have

$$\begin{aligned}
5A &= (\partial\partial_x - 15b)N, \\
6B &= (a\partial_x - 20b)A, \\
7C &= (a\partial_x - 25b)B + NA, \\
8D &= (a\partial_x - 30b)C + 2NB,
\end{aligned}$$

of which the first gives only $A = a^2d - 3abc + 2b^3$. The other equations give the required formulae

$$\begin{aligned}
aA' &= 6B + 15bA, \\
aB' &= 7C + 20bB - NA, \\
aC' &= 8D + 25bC - 2NB,
\end{aligned}$$

Halphen's H is given by $U^4H = 256\Lambda^3 - 27V^8$, viz. we thus have

$$a^4H = 256(AC - B^2)^3 - 27A^8.$$

His T is defined by the equation $U^3T = 3VA' - 8V'A$, that is,

$$a^3T = 3A(AC - B^2)^2 - 8A'(AC - B^2),$$

which is

$$\begin{aligned} &= 3A(AC^2 - 2AB^2 + A^2C) - 8A'(AC - B^2), \\ &= 3A^2C^2 - 6AB^2C + (-5AC + 8B^2)A', \end{aligned}$$

or substituting for A, B', C', their values we find

$$a^3T = 24(A^2D - 3ABC + 2B^3),$$

which expression for the reciprocant T was given by Sylvester.

Halphen's T_1 is $\frac{1}{6}(V^4T - \frac{3}{2}H)$, viz. we thus have

$$6a^5T_1 = 24A^2(A^2D - 3ABC + 2B^3) - 256(AC - B^2)^3$$

His G is given by $V^4G = U^4T^2 + 9H$, viz. we have

$$A^2a^4G = 576(A^2D - 3ABC + 2B^3)^2 + 2604(AC - B^2)^3 - 243A^8,$$

where the whole divides by A^2; throwing out this factor, we find

$$a^4G = 576(A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2) - 243A^6.$$

From the expression for T, I find

$$a^5T' = 216A^2E - 288ABD - 504AC^2 + 576B^2C + 1152(A^2D - 3ABC + 2B^3),$$

and I thence deduce for Halphen's Θ,

$$\Theta = \frac{1}{2}(2U^6T' + T(8V^2A' - 3U^2A)),$$

the formula

$$a^4\Theta = 432(AC - B^2)(AE - 4BD + 3C^2) - 576(A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C$$

or, what is the same thing,

$$\begin{aligned} a^4\Theta = 432(ACE - AD^2 - B^2E + 2BCD - C^3) \\ + 576(A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2). \end{aligned}$$

Also $\Theta_a = \frac{1}{a}(\Theta + \frac{3}{2}G)$, that is, $a^4\Theta_a = \frac{1}{a}(a^4\Theta + \frac{3}{2}a^4G)$,

$$\begin{aligned} a^4\Theta_a = 432(ACE - AD^2 - B^2E + 2BCD - C^3) \\ - \tfrac{27}{4}A^2(A^2D - 3ABC + 2B^3), \end{aligned}$$

or finally,

$$a^4\Theta_a = 432(ACE - AD^2 - B^2E + 2BCD - C^3) - \tfrac{27}{4}A^5.$$

Again,

$$2\Theta = a^4\Theta_a^2,$$

whence

$$a^8\Theta_a^2 = 2a^4\Theta \cdot a^4\Theta_a,$$

or substituting,

$$a^8\Theta_a^2 = 432(ACE - AD^2 - B^2E + 2BCD - C^3) \\ - 540A(A^2D - 3ABC + 2B^3) - \tfrac{27}{4}A^5.$$

Writing

$$\begin{aligned}
\mathfrak{C} &= AC - B^2, \\
\mathfrak{D} &= A^2D - 3ABC + 2B^3, \\
\mathfrak{D}' &= A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2, \\
\mathfrak{E} &= AE - 4BD + 3C^2, \\
\mathfrak{F} &= ACE - AD^2 - B^2E + 2BCD - C^3,
\end{aligned}$$

the foregoing results become $U = a$, $V = A$, $\Lambda = \mathfrak{C}$,

$$\begin{aligned} a^4H \text{ (Halphen)} &= 256\mathfrak{C}^3 - 27A^8, \\ a^3T &= 24\mathfrak{D}, \\ 6a^5T_1 &= 24A^2\mathfrak{D} - 256\mathfrak{C}^3, \\ a^4G &= 144\mathfrak{C}\mathfrak{E} + 576\mathfrak{D}^2, \\ a^4\Theta &= 432\mathfrak{C}\mathfrak{E} + 576A\mathfrak{F}, \\ a^5\Theta_a^2 &= 432\mathfrak{F} - 540A\mathfrak{D} - \tfrac{27}{4}A^5. \end{aligned}$$

XVIII. The Differential Equation of a Cubic Curve (continued).

The letters $t, a, b, \ldots$, of a reciprocant represent (it will be remembered) mere numerical multiples of the derived functions of a variable y, in regard to the independent variable x, and thus equating any reciprocant to zero, we have a differential equation in y. For instance, if the orthogonal reciprocant $c(1 + t^2) - 5abt + 5a^3$ be put $= 0$, $(t, a, b, c = y', \frac{1}{2}y'', \frac{1}{6}y''', \frac{1}{24}y^{\text{iv}})$, this is the differential equation

$$y^{\text{iv}}(1 + y'^2) - 10y'y''y''' + 15y''^3 = 0,$$

of the order 4. The integral hereof is expressible by the two equations

$$x = \kappa U + \lambda V, \quad y = \mu U + \nu V,$$

where U, V are real functions of a parameter t, viz. $U + iV = (1+it)^6$, $U - iV = (1-it)^6$, ($i = \sqrt{-1}$ as usual), so that $U = 1 - 15t^2 + 15t^4 - t^6$, $V = 6t - 20t^3 + 6t^5$; and $\kappa, \lambda, \mu, \nu$ are the four constants of integration.

But when the reciprocant equated to zero is a Principiant, we have the far more important results obtained in Halphen's Memoir (1): and which are, in Sylvester's lectures, exhibited in a more complete form by expressing Halphen's reciprocants in terms of the foregoing functions $A, B, C, \ldots$. I recall that $a, b, c, \ldots$ are numerical multiples of differential coefficients of the orders $2, 3, 4, \ldots$ respectively; $A, B, C, \ldots$ contain $d, e, f, \ldots$ respectively, that is, differential coefficients of the orders $5, 6, 7, \ldots$ respectively, so that according as the highest capital letter is $A, B, C, \ldots$ respectively, the order of the differential equation is $5, 6, 7, \ldots$ respectively.

But the equation, Principiant $= 0$, may be interpreted in a different manner, viz. if instead of regarding the differential equation as an equation for the determination of y, we regard therein y as a given function of x, (that is, (x, y) as the coordinates of a point on a given curve), then the differential equation serves for the determination of those points on the curve for which a given condition is satisfied, or say it is the condition in order to the existence of a singular point of determinate character.

Halphen's lowest reciprocants are: as already mentioned:

$$U = a, \quad V = a^2d - 3abc + 2b^3, \quad \Lambda = AC - B^2.$$

$U = 0$, that is, $a = 0$, is a differential equation of the second order, viz. it is the differential equation of a line: otherwise it is the condition for a point of inflexion.

$V = 0$, that is, $A = 0$, is a differential equation of the fifth order, viz. it is the differential of a conic: otherwise it is the condition for a sextactic point.

$\Lambda = 0$, that is, $AC - B^2 = 0$, is a differential equation of the seventh order, it is the condition for what Halphen calls a point of coincidence, viz. this is a point on a given curve, such that for it the cubic of nine-pointic intersection becomes a nodal cubic having the point for node and through it one branch of eight-pointic intersection and one branch of simple intersection. Observe that this is more than the condition that the cubic of nine-pointic intersection shall have a node.

The differential equation of the order 7,

$$24\cdot 73\cdot\{(X-2)(X+1)(2X-1)\}^2\cdot\Lambda^3-33\cdot 52\cdot(X^2-X+1)^3V^8=0,$$

has the integral $a=\beta^X\gamma^{1-X}$, where a,β,γ represent arbitrary linear functions; we may without loss of generality take the linear functions to be of the form $x+my+n$, introducing in this case another constant C, and writing the integral equation in the form $a=C\beta^X\gamma^{1-X}$; the number of arbitrary constants is thus $=7$.

If $X=2$, -1 or $\frac{1}{2}$, the integral equation is $a=\beta^2\gamma^{-1}$, $a=\beta^{-1}\gamma^2$ or $a=\beta\gamma$; or say it is $\beta^2=a\gamma$, $\gamma^2=a\beta$ or $a^2=\beta\gamma$, viz. each of these represents a conic. The differential equation is here $V=0$, viz. we have the foregoing result that $V=0$, that is, $A=0$, is the differential equation of a conic.

If $X=\infty$, 0 or 1, the differential equation is

$$26\cdot 73\cdot\Lambda^3-33\cdot 52\cdot V^8=0;$$

the integral equation is here of the form $\frac{\beta}{\gamma}=\log\frac{a}{\beta}$; in fact we have $\frac{a}{\beta}=\left(\frac{\gamma}{\beta}\right)^X$, or for a,β, and γ writing $\lambda a,\beta+\lambda\gamma$, and $\lambda\gamma$ respectively, this is $\frac{a}{\beta}=\left(1+\frac{\gamma}{\beta}\right)^X$, which when X is indefinitely large becomes $\frac{a}{\beta}=\exp\frac{\gamma}{\beta}$, that is, $\frac{a}{\beta}=\log\frac{\gamma}{\beta}$. And similarly for $X=0$ and $X=1$.

If $X^2-X+1=0$, that is, if $X=-\omega$, where ω is an imaginary cube root of unity, the differential equation is $\Lambda=0$, that is, $AC-B^2=0$ (of the order 7 as in the general case). The integral equation is $a=\beta^{-\omega}\gamma^{-\omega^2}$, viz. this is the equation of the curve every point of which is a coincident point.

If $X=3$, the differential equation is $28\Lambda^3-33V^8=0$, or say $256\Lambda^3-27V^8=0$; Halphen's H is a function such that $U^4H=256\Lambda^3-27V^8$, so that this is

$$H=U^{-4}(256\Lambda^3-27V^8)=0.$$

The integral equation is $a=\beta^3\gamma^{-2}$, that is, $a\gamma^2=\beta^3$, or the curve is a cuspidal cubic, depending upon 7 constants; and it thus appears that the differential equation of a cuspidal cubic is the equation of the order 7

$$a^{-4}(256(AC-B^2)^3-27A^8)=0.$$

Write

$$
\begin{aligned}
\Theta &= 64(42\mathfrak{D}^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2) + 144A^2(A^2D - 3ABC + 2B^3) \\
&= 64\mathfrak{D}' + 144A^2\mathfrak{D} + 81A^6, \\
\Phi &= 8(A^2D - 3ABC + 2B^3) + 9A^4 = 8\mathfrak{D} + 9A^4, \\
\Psi &= 3072(AC - B^2)^3 = 3072\mathfrak{C}^3,
\end{aligned}
$$

where identically

$$\Theta^2 - \Phi^3 = 27\Psi.$$

Halphen and Sylvester find that, for a cubic curve the invariants of which are S and T ($S = -l + l^3$, $T = 1 - 20l^3 - 8l^6$ for the canonical form $x^3 + y^3 + z^3 + 6lxyz = 0$), we have

$$\Phi S = \Theta^2 - 12288\mathfrak{C}^3(8\mathfrak{D} + 9A^4),$$

$$\Phi T = \Theta^3 - 18432\mathfrak{C}^3(8\mathfrak{D} + 9A^4) + 556231A^4\mathfrak{C}^6,$$

where Φ is an arbitrary multiplier; and we thence have

$$\frac{S^3}{T^2} = \frac{\{\Theta^2 - 12288\mathfrak{C}^3(8\mathfrak{D} + 9A^4)\}^3}{\{\Theta^3 - 18432\mathfrak{C}^3(8\mathfrak{D} + 9A^4) + 556231A^4\mathfrak{C}^6\}^2},$$

or if, as with Halphen $h = K\frac{S^3}{T^2}$, then

$$(h-1)R^2 = hQ,$$

that is,

$$\frac{R^2}{Q} = \frac{h}{h-1},$$

and thus for a nodal cubic we have $Q = 0$; we thus have

$P = 0$, for the differential equation of a cubic for which $S = 0$,
$R = 0$, " " " " " $T = 0$,
$Q = 0$, " " nodal cubic,

these being each of them of the order 8. But the equation $Q = 0$ is reducible to a more simple form. We have

$$Q = \frac{1}{(3072V)^3}\{(64\mathfrak{D} - 144A^2\mathfrak{D} + 27A^6)^2 + 32768\mathfrak{D}^3\},$$

or, since in the expression in { } the first and second terms are $12\Theta^2(42\Theta - \Phi^2) =$ this is

$$Q = \frac{1}{(3072V)^3}\{(\Theta - 364A^2\Phi + 216A\Phi)^2 - 64(\Phi - 94A)^3\}.$$

We thus have for the differential equation of the eighth order of the nodal cubic

$$(\Theta - 364A^2\Phi + 216A\Phi)^2 - 64(\Phi - 94A)^3 = 0,$$

or, since

$$\Theta = 64\mathfrak{D} + 144A^2\mathfrak{D} + 81A^6, \quad \text{and} \quad \Phi = 8\mathfrak{D} + 94A,$$

the differential equation of the eighth order for the nodal cubic finally is

$$(64\mathfrak{D} - 144A^2\mathfrak{D} + 27A^6)^2 + 32768\mathfrak{D}^3 = 0.$$

Halphen has $\Phi^2 = 0$ for the differential equation of a cubic curve. Putting, as before,

$$\begin{aligned}
\mathfrak{C} &= AC - B^2,\\
\mathfrak{D} &= A^2D - 3ABC + 2B^3,\\
\mathfrak{E} &= A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2,\\
\mathfrak{F} &= AE - 4BD + 3C^2,\\
\mathfrak{G} &= ACE - AD^2 - B^2E + 2BCD - C^3,
\end{aligned}$$

(and therefore $4\mathfrak{C}\mathfrak{E} - 4\mathfrak{D} = \mathfrak{D}$, $\mathfrak{F}^3 - 27\mathfrak{G}^2 =$ Quartic Discr.), then, by what precedes, the differential equation of the order 9 of a cubic curve is

$$a^{-5}(432\mathfrak{G} - 54\mathfrak{C}\mathfrak{D} - \tfrac{27}{4}A^5) = 0.$$

Recapitulating.

Order. The differential equation for a cubic curve:—

8. for which invt. $S = 0$, is
   $P = \Theta^2 - 12288\mathfrak{C}^3(8\mathfrak{D} + 9A^4) = 0,$
8. for which invt. $T = 0$, is
   $R = \Theta^3 - 18432\mathfrak{C}^3(8\mathfrak{D} + 9A^4) + 556231A^4\mathfrak{C}^6 = 0,$
8. nodal cubic
   $Q \div (3072V)^3 = (64\mathfrak{D} - 144A^2\mathfrak{D} + 27A^6)^2 + 32768\mathfrak{D}^3 = 0,$
8. for invt. $64S^3 \div T^2 = h$, is $(h-1)R^2 - hQ = 0,$
9. for general cubic is $1728\mathfrak{G} - 216\mathfrak{C}\mathfrak{D} - 243A^5 = 0,$

where for shortness Θ is retained to signify its value

$$= 64\mathfrak{D} + 144A^2\mathfrak{D} + 81A^6.$$

Halphen by a polar transformation finds that these same differential equations, changing therein the sign of $\mathfrak{D}$, apply to curves of the third class: in particular, the last equation, changing therein the sign of $\mathfrak{D}$, applies to the general curve of the third class, or sextic curve with nine cusps.

XIX. Seminvariants and Reciprocants (continued).

A very important notion in the theory, as well of Seminvariants as of Reciprocants, is that of MacMahon's Multilinear Operator, see his paper "Theory of a Multilinear Partial Differential Operator, with applications to the theories of Invariants and Reciprocants," *Proc. Lond. Math. Soc.*, t. xviii. (1886), pp. 61—88: this operator plays so important a part in the theories to which it relates, that I venture to reproduce the definition of it in what appears to me a simplified and more easily intelligible form, and to recapitulate some of the leading properties.

I take with him the letters to be $(a_0, a_1, a_2, a_3, \ldots) = (a, b, c, d, \ldots)$, viz. the first letter is a_0 or a, the second is a_1 or b, and so on, but when we are not concerned with a term of indefinite rank, I use always $(a, b, c, d, \ldots)$ and the like in regard to any other series of letters. This being so, the definition of the Multilinear Operator of four elements, which is here alone in question, is

$$
\begin{aligned}
&(\mu, \nu; m, n)_4 = \\
&(\mu + 0\nu)A\left(\frac{\partial}{\partial a} + m\frac{\partial}{\partial b} + \frac{m(m-1)}{1\cdot 2}\frac{\partial}{\partial c} + \frac{m(m-1)(m-2)}{1\cdot 2\cdot 3}\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+1}}\right)\\
&+ (\mu + \nu)B\left(\frac{\partial}{\partial b} + m\frac{\partial}{\partial c} + \frac{m(m-1)}{1\cdot 2}\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+2}}\right)\\
&+ (\mu + 2\nu)C\left(\frac{\partial}{\partial c} + m\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+3}}\right)\\
&+ (\mu + 3\nu)D\left(\frac{\partial}{\partial d} + \ldots + \frac{\partial}{\partial a_{n+4}}\right)
\end{aligned}
$$

where

$$
\begin{aligned}
A &= a^m,\\
B &= a^{m-1}b,\\
C &= a^{m-2}(ac + \tfrac{m-1}{2}b^2) = a^{m-2}c + \tfrac{m-1}{2}a^{m-2}b^2,\\
D &= a^{m-3}(ad + (m-1)bc + \tfrac{(m-1)(m-2)}{6}b^3)\\
&= a^{m-3}d + (m-1)a^{m-3}bc + \tfrac{(m-1)(m-2)}{6}a^{m-3}b^3,
\end{aligned}
$$

where the expressions for $B, C, D, \ldots$, are obtained successively from that of $A = a^m$, by Arbogast's rule of derivation, viz. we operate on the last letter of a term, and when there is a last but one letter which in alphabetical order immediately precedes the last letter, then also upon the last but one letter, and whenever the term thus obtained ends in a power, we divide by the index of the power. Thus the term $\frac{1}{2}(m-1)a^{m-2}b^2$ in the third line gives in the fourth line

$$\begin{aligned}&\tfrac{1}{2}(m-1)a^{m-2}\cdot 2bc + \tfrac{1}{2}(m-1)(m-2)a^{m-3}b\cdot b^2\cdot\tfrac{1}{2},\\ &= (m-1)a^{m-2}bc + \tfrac{1}{6}(m-1)(m-2)a^{m-3}b^3.\end{aligned}$$

Going a step further, we form in this manner the expression

$$E = a^{m-1}e + (m-1)a^{m-2}(bd + \tfrac{1}{2}c^2) \\ + \tfrac{1}{2}(m-1)(m-2)\{a^{m-3}\cdot 3b^2c + (m-3)a^{m-4}b\cdot b^3\cdot \tfrac{1}{6}\},$$

that is,

$$\begin{aligned} E = a^{m-1}&e \\ &+ a^{m-2}\cdot(m-1)(bd+\tfrac{1}{2}c^2) \\ &+ a^{m-3}\cdot\tfrac{1}{2}(m-1)(m-2)\cdot 3b^2c \\ &+ a^{m-4}\cdot\tfrac{1}{12}(m-1)(m-2)(m-3)b^4, \end{aligned}$$

which is sufficient to explain the rule.

It may be noticed that, the operator being linear in μ, ν, we have

$$(\mu,\nu;m,n) = \mu(1,0;m,n) + \nu(0,1;m,n).$$

The Alternant of two multilinear operators.

If P, Q are the two operators, we have as usual

$$P\cdot Q = PQ + P * Q,$$

where $P\cdot Q$ denotes the successive operation first with Q and then with P upon any operand, PQ is the mere algebraical product of the operators, and $P * Q$ is the operator obtained by the operation of P upon Q.

Similarly,

$$Q\cdot P = QP + Q * P,$$

and since QP is the same thing as PQ, we obtain

$$P\cdot Q - Q\cdot P = P * Q - Q * P,$$

either of which equal expressions, or say rather the second of them, is called the *alternant* of P, Q and is written $[P, Q]$: viz. as the definition of the alternant, we have

$$[P,Q] = P * Q - Q * P.$$

We have the remarkable theorem, that the alternant of any two operators $(\mu', \nu'; m', n')$, $(\mu, \nu; m, n)$ is an operator $(\mu_1, \nu_1; m_1, n_1)$; where

$$\begin{aligned}
\mu_1 &= (m' + m - 1)\frac{\mu'}{m'}\mu - (n + n')\nu'\mu + \frac{m-1}{m}\mu\nu' - \frac{m'-1}{m'}\mu'\nu,\\
\nu_1 &= (n' - n)\nu\nu' + \frac{m-1}{m}\mu\nu' - \frac{m'-1}{m'}\mu'\nu,\\
m_1 &= m' + m - 1,\\
n_1 &= n' + n,
\end{aligned}$$

or since m_1, n_1 have such simple expressions, say it is an operator $(\lambda, \mu; m' + m - 1, n' + n)$, where λ, μ have the values just written down.

0.1 On Reciprocants and Differential Invariants. [943] (continued)

[Paper 943, continued from p. 375]

[VII. Halphen's reciprocant H, continued]

becomes $256 \cdot 27b^{24} - 27 \cdot 256b^{24} = 0$), and not only so, but that it in fact contains the factor a^4. Assuming that this is so, viz. that the terms in a^0, a, a^2, a^3 all of them vanish, I have calculated the terms in a^4, and have thus obtained an incomplete expression for the quotient $(256A^3 - 27V^8) \div a^4$, viz. this is

$$H = (256A^3 - 27V^8) \div a^4 = 256a^{14}(df - e^2)^3 + \cdots ;$$

where I remark that the term in b^{16} presented itself in the form

$$256(54619f - 135618ce + 117 \cdots + 5346617c^2d + 9477616c^4) - 27(1792 \cdots - 48384617c^2d$$

We have thus Halphen's reciprocant H of the weight 64.

The reciprocants thus far obtained are consequently

deg.	weight		factor
$U = a$	1	2	$(A' - A)/,$
$V = a^2d - 3abc + 2b^3$	3	9	$(A' - A)^3/,$
A	8	24	$(A' - A)V^6,$
$H = (256A^3 - 27V^8) \div a^4$	20	64	$(A' - A)^{20}V^{44}$

VII.

Reciprocants of the same degree and weight have the same factor, and may thus be combined in the way of addition, viz. R and S being reciprocants of the same degree and the same weight, then (α, β being constants) we have $\alpha R + \beta S$ a reciprocant of the same degree and the same weight. The quotient of two such reciprocants has the factor unity, or say it is an absolute invariant. Thus A^3 and V^8 have each of them the degree 24 and weight 72, so that $\alpha A^3 + \beta V^8$ is a reciprocant; in particular, $256A^3 - 27V^8$ is a reciprocant having the factor a^4, and it thus gives rise to the foregoing reciprocant $H = (256A^3 - 27V^8) \div a^4$.

Moreover $A^3 + V^8$ is an absolute reciprocant.

Any two reciprocants R, S of the same degree and same weight, or, what is the same thing, any absolute reciprocant $R + S$ gives rise to a reciprocant $RS' - R'S$, where the accents denote differentiation in regard to x ($a' = 3b$, $b' = 4c$, $c' = 5d$, ...); the order of the new reciprocant thus exceeds by unity the order of R or S, whichever of them is of the highest order.

From U, V, A, which are of the orders 2, 5, 7 respectively, it is thus possible to deduce a series of reciprocants T_8, T_9, T_{10}, ... of the orders 8, 9, 10,..., respectively: viz. we have from the absolute reciprocant A^3V^{-8}, first a reciprocant $3VA' - 8V'A$, which, however, contains the factor U^3, and we have

$$T_8 = (3VA' - 8V'A) \div U^3.$$

We then have the absolute reciprocant $U^4T_8 \cdot V^{-4}$, leading to

$$T_9 = UVT_8' + 4(VU' - UV')T_8;$$

then the absolute reciprocant $U^4T_9 \cdot V^{-4}$, leading to

$$T_{10} = UVT_9' + 4(VU' - UV')T_9;$$

and so

$$T_{11} = UVT_{10}' + 4(VU' - \tfrac{5}{2}UV')T_{10},$$
$$T_n = UVT_{n-1}' + 4\{VU' - \tfrac{1}{2}(n-6)UV'\}T_{n-1}.$$

Halphen considers that these are the only distinct reciprocants of the orders 2, 5, 7, 8, 9,..., n; but remarks that we can, with the reciprocants up to any given order, form algebraical combinations, in some cases containing as factor a power of the reciprocant U or V, so that, rejecting this factor, we have a new independent reciprocant, that is, a reciprocant not expressible as a rational and integral function of inferior reciprocants; an instance hereof is the foregoing reciprocant $H = (256A^3 - 27V^8) \div a^4$.

Other like forms are given by Halphen, viz. writing with him T for shortness in place of T_8, we have for T the following expression in the form of a determinant

$$T = \begin{vmatrix} 3b & 2a & a & 0 & 0 & \\ 4c & 3b & b & a & 2a^2 & 0 \\ 5d & 4c & c & 2b & 5ab & a^2 \\ 6e & 5d & d & 3c & 6ac + 3b^2 & 3ab \\ 7f & 6e & e & 4d & 4ad + 7bc & 4ac + 2b^2 \\ 8g & 7f & f & 5e & 8ae + 8bd + 4c^2 & 5ad + 5bc \end{vmatrix} \div V,$$

and then

$$G = Ⓡ_1 - F, \quad Ⓡ = Ⓡ_2,$$

where Ⓡ is given explicitly in the form of a determinant, viz. this is

$$Ⓡ_2 = -432 \begin{vmatrix} c & d & e & f & g & h \\ b & c & d & e & f & g \\ a & b & c & d & e & f \\ a^2 & 2ab & 2ac+b^2 & 2ad+2bc & 2ae+2bd+c^2 & 2af+2be+2cd \\ 0 & a^2 & 2ab & 2ac+b^2 & 2ad+2bc & 2ae+2bd+c^2 \\ 0 & 0 & a^2 & 3a^2b & 3a^2c+3ab^2 & 3a^2d+6abc+b^3 \end{vmatrix},$$

where $Ⓡ_2 = 0$ is the differential equation of the ninth order of the general cubic curve.

Halphen's $\Theta_1 = UVT' + 4(VU' - UV')T$ is the above-mentioned reciprocant T_9; U is connected with Ⓡ by the formula $2AⓇ_1 - V^2Ⓡ = U^4T^2$.

Halphen considers incidentally (p. 56), what may be called a *polar relation* of the variables X, Y and x, y; viz. this is

$$X = -\frac{dY}{dX}, \quad Y = X\frac{dY}{dX} - y,$$

whence conversely

$$x = -\frac{dY}{dX}, \quad y = X\frac{dY}{dX} - Y.$$

We may here express F and its differential coefficients in regard to X, say F, F_1, F_2, ..., in terms of y and its differential coefficients in regard to x, say y, y_1, y_2, ... (to avoid confusion with other formulae, I purposely use these notations, abstaining from the introduction of the letters a, b, c, ...). The geometrical signification is that x, y, being point-coordinates, then X, Y are line-coordinates. We find, without difficulty,

$$\frac{Y_2}{2} = -\frac{1}{y_2}, \quad \frac{Y_3}{Y_2^{\frac{3}{2}}} = -\frac{y_3}{y_2^{\frac{5}{2}}}, \quad \ldots$$

Hence, in particular,

$$9F_2^2F_5 - 45F_2F_3F_4 + 40F_3^3 = -\frac{1}{y_2^2}(9y_2^2y_5 - 45y_2y_3y_4 + 40y_3^3),$$

viz. the differential equation $3y_2^2y_5 - 45y_2y_3y_4 + 40y_3^3 = 0$ (in the former notation $a^2d - 3abc + 2b^3 = 0$) of the conic in point-coordinates gives, as it should do, the equation of like form $9F_2^2F_5 - 45F_2F_3F_4 + 40F_3^3 = 0$ of the conic in line-coordinates.

The geometrical applications throughout the memoir are very extensive and interesting.

VIII.

In Halphen's second memoir "Sur les invariants différentiels des courbes gauches" (1880), instead of a single dependent variable y a function of x, we have two dependent variables y, z, each of them a function of x.

The relations between the original variables x, y, z, and the new variables X, Y, Z, are of the general homographic form

$$X : Y : Z : 1 = (\alpha x+\beta y+\gamma z+\delta) : (\alpha' x+\beta' y+\gamma' z+\delta') : (\alpha'' x+\beta'' y+\gamma'' z+\delta'') : (\alpha''' x+$$

but he does not from these formulae deduce the expressions of Y_1, Y_2, $\ldots$, Z_1, Z_2, $\ldots$ in terms of y, y'', $\ldots$, z, z'', $\ldots$; the investigation is in some measure a geometrical one.

The notation employed is

$$u = \frac{y''z''' - y'''z''}{3 \cdot 4},$$

viz. u is here the most simple reciprocant, the equation $u = 0$ is obviously the condition of a plane curve:

$$a_n = \frac{1}{4 \cdot 5 \cdot 6 \cdots n} \frac{y''z^{(n)} - y^{(n)}z''}{y''z''' - y'''z''}, \quad b_n = \frac{1}{3 \cdot 4 \cdot 5 \cdots n} \frac{y'''z^{(n)} - y^{(n)}z'''}{y''z''' - y'''z''},$$

n greater than or equal to 4; viz. these two singly infinite series of symbols a_4, a_5, $a_6, \ldots$, b_4, b_5, $b_6, \ldots$, together with u, are used for the expression of the doubly infinite series $y^{(m)}z^{(n)} - y^{(n)}z^{(m)}$, by virtue of the identity

$$\frac{y^{(m)}z^{(n)} - y^{(n)}z^{(m)}}{u} = \frac{(y''z^{(m)} - y^{(m)}z'')(y'''z^{(n)} - y^{(n)}z''') - (y''z^{(n)} - y^{(n)}z'')(y'''z^{(}}{y''z''' - y'''z''}$$

or, as this may be written,

$$\frac{1}{u}\begin{vmatrix} y^{(m)} & z^{(m)} \\ y^{(n)} & z^{(n)} \end{vmatrix} = \begin{vmatrix} a_m & b_m \\ a_n & b_n \end{vmatrix}.$$

The most simple reciprocant (after u) is

$$v = ub_6 - 2b_5 - 3a_4a_5 + 3a_4b_4 + 2a_4^3,$$

or rather u^5v, which is an integral function of the differential coefficients: the signification of the equation $v = 0$ is that the tangents of the curve belong to a linear complex. This is a property belonging to the tangents of a skew cubic, and the skew cubic thus satisfies the differential equation $u = 0$. Another reciprocant is obtained

$$w = b_6 - a_4b_5 - 4a_5b_4 + 4a_4^2b_4 - 2a_4^2a_5 + 2b_4^2 + a_5^2 + a_4^4,$$

or rather u^8w, which is an integral function of the differential coefficients: and the skew cubic satisfies the second differential equation $w = 0$. The formulae enable the determination of the osculating skew cubic at any point of a skew curve.

The general theory is developed in a compact form in the theorems I. to VII., and it is shown that the investigation of all the reciprocants depends upon that of two reciprocants of the seventh order. The geometrical applications of the theory are very extensive and interesting.

IX.

Before going further, I remark that the *homologic transformation*

$$X = \alpha x + \beta y + \gamma, \quad Y = \alpha' x + \beta' y + \gamma',$$

(which is in point of generality intermediate between Halphen's homographic transformation

$$Y = \frac{\alpha x + \beta y + \gamma}{\alpha'' x + \beta'' y + \gamma''}, \quad Y = \frac{\alpha' x + \beta' y + \gamma'}{\alpha'' x + \beta'' y + \gamma''},$$

and Sylvester's special transformation $X = y$, $Y = x$, and which includes as a particular case the rectangular transformation

$$X = x\cos\theta + y\sin\theta, \quad Y = -x\sin\theta + y\cos\theta),$$

does not appear to have been explicitly considered. Writing as in the general case t, a, b, c,..., for y', $\frac{1}{2}y''$, $\frac{1}{6}y'''$,..., and T, A, B, C,..., for Y_1, $\frac{1}{2}Y_2$, $\frac{1}{6}Y_3$,...; also taking h, k for the increments of x, y, and H, K for those of X, Y respectively, we have

$$H = \alpha h + \beta k, \quad K = \alpha' h + \beta' k,$$

that is,

$$H = (\alpha + \beta t)h + \beta(ah^2 + bh^3 + ch^4 + \dots),$$
$$K = (\alpha' + \beta' t)h + \beta'(ah^2 + bh^3 + ch^4 + \dots),$$

which, by the elimination of h, must lead to

$$K = TH + AH^2 + BH^3 + CH^4 + \dots$$

We have

$$\lambda = \frac{\beta}{\alpha + \beta t}, \quad \lambda' = \frac{\beta'}{\alpha' + \beta' t},$$

and we write

$$\mu = \frac{1}{\lambda' - \lambda}, \quad \nu = \frac{1}{\lambda' - \lambda};$$

and thence

$$\lambda = \frac{\lambda' - \lambda}{\alpha + \beta t \cdot \alpha' + \beta' t},$$

moreover

$$\frac{1}{\alpha + \beta t} = \mu, \quad \frac{1}{\alpha' + \beta' t} = \nu,$$

and therefore

$$T = \frac{\alpha' + \beta' t}{\alpha + \beta t} = \frac{\nu}{\mu}.$$

Also

$$\Omega = ah + bh^2 + ch^3 + \dots;$$

whence, if

$$h = p\Omega + q\Omega^2 + r\Omega^3 + s\Omega^4 + \dots,$$

then

$$q = \frac{1}{a}, \quad r = \frac{1}{a^3}(-2b), \quad s = \frac{1}{a^5}(ac - 2b^2), \quad \dots$$

and we have

$$K - TK = AH^2 + BH^3 + CH^4 + \dots,$$

giving

$$\mu\nu K - T \cdot \mu\nu H = X\Omega^2 + Y\Omega^3 + Z\Omega^4 + W\Omega^5 + \dots,$$

where, substituting for p, q, r,… their values,

$$X = \mu\nu \cdot a, \quad Y = \mu^2\nu \cdot b - a^2 X, \dots$$

X. Sylvester's Theory of Reciprocants

We have next Sylvester: — "Lectures on the theory of Reciprocants" (reported by J. Hammond), *Amer. Math. Journ.*, t. VIII. (1886), pp. 196—260.

The lectures were delivered at Oxford, Inaugural Lecture, Dec. 1885, starting from the Schwarzian function

$$\frac{y'''}{y'} - \frac{3}{2}\left(\frac{y''}{y'}\right)^2,$$

which acquires only a factor by the interchange of x and y.

In lecture 2, pp. 203 *et seq.*, Sylvester considers the general theory of the functions which remain unaltered, except as to a factor, by the interchange of x, y; viz. writing

$$t,\ a,\ b,\ c,\ \ldots \quad \text{for} \quad y',\ y'',\ y''',\ y^{\text{iv}},\ \ldots,$$

or again

$$t,\ a_0,\ a_1,\ a_2,\ a_3,\ \ldots \quad \text{for} \quad 2y',\ \tfrac{1}{2}y'',\ \tfrac{1}{6}y''',\ \tfrac{1}{24}y^{\text{iv}},\ \ldots,$$

Schwarzian $\frac{y'''}{y'} - \frac{3}{2}\left(\frac{y''}{y'}\right)^2 = 6(bt - a^2)$, where the reciprocal is $= 6a^{-1}(bt - a^2)$.

A reciprocant with Sylvester is thus a function of y and its differential coefficients in regard to x, which except as to a factor remain unaltered when x, y are interchanged, or say, when x, y are changed into X, Y, where $X = y$, $Y = x$. This is a much less general change than Halphen's, and thus every reciprocant (Halphen) is a reciprocant (Sylvester), but not conversely. The reciprocants (Sylvester) are far more numerous. I remark that incidentally Sylvester considers reciprocants, which remain unaltered save as to a factor, when x, y are changed into X, Y, where

$$X = \frac{\alpha x + \beta y + \gamma}{\alpha'' x + \beta'' y + \gamma''}, \quad Y = \frac{\alpha' x + \beta' y + \gamma'}{\alpha'' x + \beta'' y + \gamma''},$$

which again is a less general change than Halphen's — it has been in what precedes alluded to as the particular case $L'' = 0$, that is, $\alpha'' = 0$, $\beta'' = 0$. It may be remarked that the proper orthogonal substitution corresponding to Sylvester's substitution is not $X = y$, $Y = x$, but $X = y$, $Y = -x$, viz. we have here the determinant $\alpha\beta' - \alpha'\beta = +1$.

I develop Sylvester's theory of reciprocants; but I wish first to point out the resemblance in form between this theory and that of seminvariants. In the theory of seminvariants, from the set of quantities $(a, b, c, \dots)$ in connexion with an arbitrary quantity θ, we deduce a new set $(a', b', c', \dots)$, where

$$a' = a, \quad b' = b + a\theta, \quad c' = c + 2b\theta + a\theta^2, \quad \dots$$

or, what is the same thing, if $\theta = -\theta$, then

$$a = a, \quad b = b' + a'\theta', \quad c = c' + 2b'\theta' + a'\theta'^2, \quad \dots$$

and this being so there exist functions which are the same for unaccented and the accented letters respectively; for instance, $a' = a$:

$$a'c' = ac + 2ab\theta + a^2\theta^2,$$

that is, $a'c' - b'^2 = ac - b^2$; and similarly

$$a'^2d' - 3a'b'c' + 2b'^3 = a^2d - 3abc + 2b^3, \quad \text{\&c.};$$

these functions a, $ac - b^2$, $a^2d - 3abc + 2b^3$, &c., are called *seminvariants.*

Similarly, in Sylvester's theory of reciprocants, starting from y, a given function of x, we have $(t, a, b, c, \dots)$ denoting $(y', \frac{1}{2}y'', \frac{1}{6}y''', \frac{1}{24}y^{\text{iv}}, \dots)$; and conversely $(t', a', b', c', \dots)$ denoting $(x_1, \frac{1}{2}x_2, \frac{1}{6}x_3, \frac{1}{24}x_4, \dots)$; taking h, k for the increments of x and y respectively, we have

$$k = th + ah^2 + bh^3 + ch^4 + dh^5 + \dots,$$

$$h = t'k + a'k^2 + b'k^3 + c'k^4 + d'k^5 + \dots,$$

which equations determine the relations between $(t, a, b, c, d, \dots)$ and $(t', a', b', c', d', \dots)$, viz. we have $tt' = 1$, and then

$$a' = -a \div t^3, \quad b' = -bt + 2a^2 \div t^5, \quad c' = -ct^2 + 5abt - 5a^3 \div t^7,$$

$$d' = -dt^3 + (6ac + 3b^2)t^2 - 21a^2bt + 14a^4 \div t^9,$$

$$e' = -et^4 + (7ad + 7bc)t^3 - (28a^2c + 28ab^2)t^2 + 84a^3bt - 42a^5 \div t^{11},$$

and conversely,

$$a = -a' \div t'^3, \quad b = -b't' + 2a'^2 \div t'^5,$$

$$c = -c't'^2 + 5a'b't' - 5a'^3 \div t'^7, \quad \dots$$

[p. 384]

and we thence deduce functions which have equal or opposite values for the accented and unaccented letters respectively; these functions may or may not contain t, t'. Thus we have

$$a't'^3 = -at^3, \quad \text{or} \quad a' = -a \div t^3;$$

$$(b't' - a'^2)t'^{-3} = -(bt - a^2)t^{-3},$$
$$(4a'c' - 5b'^2)t'^{-4} = (4ac - 5b^2)t^{-4},$$
$$\{(1+t'^2)b' - 2a'^2t'\}t'^{-3} = \{(1+t^2)b - 2a^2t\}t^{-3},$$

&c.

These functions of the unaccented letters, or say the same functions omitting the exterior power of t, are called *Reciprocants*; thus we have the reciprocants a, $bt-a^2$, $2ct-5ab$, $4ac-5b^2$, $(1+t^2)b-2a^2t$, &c. Observe that, when the exterior powers of t are omitted, then the values are equal or opposite to those with the accented letters save as to a power of t, viz. the forms are

$$a = -a't^3,$$

$$bt - a^2 = -(b't' - a'^2)t^6,$$
$$2ct - 5ab = -(2c't' - 5a'b')t^7,$$
$$4ac - 5b^2 = (4a'c' - 5b'^2)t^8,$$
$$\{(1+t^2)b - 2a^2t\} = -\{(1+t'^2)b' - 2a'^2t'\}t^9,$$

&c.

A reciprocant is said to be *odd* or *even* according as the sign on the right-hand side is $-$ or $+$; the index of t on the right-hand side is said to be the *weight* of the reciprocant. Reciprocants may be combined in the way of addition if and only if they are each of them of the same weight and parity, viz. this being so, then if λ, λ',..., are mere numbers, $\lambda R + \lambda' R' + \dots$ will be a reciprocant of the same weight and parity with each of the reciprocants R, R', ...

Reciprocants may in every case be combined in the way of multiplication; viz. two or more reciprocants may be multiplied together giving a reciprocant the parity of which is the sum of the parities, and its weight the sum of the weights of the component reciprocants. Thus $bt - a^2$ and $2ct - 5ab$ being odd reciprocants of the weights 6 and 7 respectively, we have $(bt-a^2)(2ct-5ab)$ an even reciprocant of the weight 13.

A reciprocant is *pure* or *impure* according as it does not or does contain t.

The foregoing equations for a', b', c', ..., may be obtained each from the next preceding one by operating upon it with

$$\frac{1}{\lambda}\left(2t\frac{\partial}{\partial t}+3a\frac{\partial}{\partial a}+4b\frac{\partial}{\partial b}+5c\frac{\partial}{\partial c}+\dots\right),$$

where λ is a positive integer, $=3+$weight of the letter operated upon ($3+3=6$ in operating on d' to obtain e', $3+4=7$ in operating on e' to obtain f', and so on).

For example,

$$\frac{1}{7}\left(2t\frac{\partial}{\partial t}+3a\frac{\partial}{\partial a}+4b\frac{\partial}{\partial b}+5c\frac{\partial}{\partial c}+6d\frac{\partial}{\partial d}\right)d'$$

viz. operating herewith on

$$d'=\{-dt^3+(6ac+3b^2)t^2-21a^2bt+14a^4\}\div t^9,$$

we obtain the value of e' given above.

The proof is at once obtained: except as to a numerical factor, the operation is in fact equivalent to the differentiation of a function of y, and its derived coefficients y', y'', ..., in regard to x.

Any pure reciprocant is reduced to zero by the operator

$$V=2a^2\frac{\partial}{\partial b}+5ab\frac{\partial}{\partial c}+6(ac+b^2)\frac{\partial}{\partial d}+7(ad+bc)\frac{\partial}{\partial e}+8(ae+bd)\frac{\partial}{\partial f}+\dots;$$

or say V is an *annihilator* of any pure reciprocant: compare herewith the theorem $\Omega=a\partial_b+2b\partial_c+3c\partial_d+\dots$ is an annihilator of any seminvariant; and conversely, any homogeneous isobaric function of a, b, c, ... reduced to zero by V is a pure reciprocant.

For instance, for the pure reciprocant $4ac-5b^2$, we have

$$(2a^2\partial_b+5ab\partial_c)(4ac-5b^2)=-20a^2b+20a^2b=0.$$

Any rational and integral homogeneous reciprocant, operated upon with

$$G=(3bt-3a^2)\partial_a+(4ct-4ab)\partial_b+(5dt-5ac)\partial_c+\dots,$$

produces a like reciprocant. Thus operating upon a with G, we obtain $3bt - 3a^2$, or say $bt - a^2$, and operating hereon again with G, a new reciprocant, and so on.

The series of reciprocants thus derived from a is

$$\begin{aligned}
&a,\\
&bt - a^2,\\
&2ct - 5ab,\\
&2dt^2 - 9act + 7a^2b,\\
&2et^3 - 11adt^2 + (8a^2c + 11ab^2)t - 8a^3b,\\
&2ft^4 - 13aet^3 + (10a^2d + 18abc - 7b^3)t^2 - (56a^3c + 60a^2b^2)t\\
&\qquad\qquad + (103a^3c - 99abd + 199abc^2 - 28c^2 + 33b^3)t - \dots
\end{aligned}$$

&c.

These are, in fact, connected with the terms of

$$\frac{1}{a}\frac{d}{dx}\left(\frac{a}{t}\frac{dY}{dx}\right),$$

where observe that $\frac{d}{dx}t = 2a$, $\frac{d}{dx}a = 3b$, $\frac{d}{dx}b = 4c$, &c. Thus

$$\frac{1}{a}\frac{d}{dx}\left(\frac{a}{t}\frac{dY}{dx}\right) = \frac{1}{a}\frac{d}{dx}\left(\frac{a}{t}\cdot 2a\right) = \frac{1}{a}\frac{d}{dx}\left(\frac{2a^2}{t}\right) = \frac{3(bt - a^2)}{t^2}, \quad \&\text{c}.$$

A like generator for pure reciprocants is

$$H = 4(ac - b^2)\partial_b + 5(ad - bc)\partial_c + 6(ae - bd)\partial_d + \dots;$$

thus operating herewith upon $(4ac - 5b^2)$, we obtain

$$(4ac - b^2)(-10b) + 5(ad - bc)4a = 20a^2d - 60abc + 40b^3,$$

viz. we have thus the pure reciprocant $a^2d - 3abc + 2b^3$.

In repeating this process, we may reduce by means of powers and products of earlier reciprocants, and we can also in many cases throw out powers of the reciprocant a. Thus forming the expression for $H(a^2d - 3abc + 2b^3)$, this is given by column 1 of the annexed form: multiplying by 25, and adding $24(4ac - 5b^2)^2$, so as to eliminate the term in b^4, we obtain the expression in col. 4: or throwing out the numerical factor 3, and also the factor a, we have the reciprocant $+50a^2e - \dots$ in col. 5, and the outside right-hand column of literal terms.

1	2	3	4	5	
H	$25H$	$24H^2$		-3	
a^2e	$+6$		$+150$	$+50$	a^2e
a^2bd	-21		-525	-175	abd
a^2c^2	-12	$+384$	-300	$+84$	ac^2
ab^2c	$+51$	-960	$+1275$	$+315$	b^2c
b^4	-24	$+600$	-600		

[p. 387]

We thus obtain the series of pure reciprocants

$$
\begin{aligned}
&a,\\
&ac-b^2,\\
&a^2d+1\cdot abc-2b^3,\\
&a^2e+1\cdot a^2bd-7\cdot ab^2c+7\cdot b^4,\\
&a^3f+10\cdot a^2ce-40\cdot a^2b^2d+50\cdot ab^3c+14\cdot a^2cd-39\cdot b^3c+31\cdot c^3,\\
&a^3g+14\cdot a^3bg+420\cdot a^3ci-65\cdot ab^2e+178\cdot b^2ce+735\cdot ab^3d\\
&\qquad -4158\cdot bcd+306\cdot abce+1001\cdot ac^3-1485\cdot b^3e+3465\cdot b^2c^2\\
&\qquad -1925\cdot bc^3+169260\cdot\ldots
\end{aligned}
$$

&c.

I remark that a pure reciprocant not included in the foregoing series is

$$
\begin{aligned}
&a^3ce+800\cdot\ldots\\
&a^3d^2-875\cdot a^3de-1000\cdot abcd+2450\cdot ac^3-1344\cdot b^2c-35\cdot\ldots
\end{aligned}
$$

XI.

Halphen assumes that $a^2d-3abc+2b^3$ is the *sextactic* reciprocant, or, what is the same thing, that the general conic satisfies the differential equation of the fifth order $a^2d-3abc+2b^3$; the proof is as follows.

The general equation of the conic may be written

$$y+Ax+B=\pm\sqrt{k}\sqrt{(x-\alpha)(x-\beta)},$$

which depends on the five parameters A, B, k, α, β.

We have

$$y''=\frac{-\frac{1}{4}k(\alpha-\beta)^2}{\{(x-\alpha)(x-\beta)\}^{\frac{3}{2}}},$$

or say

$$y''^{-\frac{2}{3}}=\left(\frac{-4}{k}\right)^{\frac{1}{3}}(x-\alpha)(x-\beta)\cdot(\alpha-\beta)^{-\frac{2}{3}};$$

and we have thus the differential equation $\frac{d^3}{dx^3}(y''^{-\frac{2}{3}})=0$. This gives

$$9y''^{-\frac{11}{3}}y'''-45y''^{-\frac{14}{3}}y'''y^{\text{iv}}+40y''^{-\frac{17}{3}}y'''^3=0,$$

that is,

$$9y''^2y^{(5)} - 45y''y'''y^{\text{iv}} + 40y'''^3 = 0.$$

But y'', y''', y^{iv}, $y^{(5)} = 2a$, $6b$, $24c$, $120d$, and the equation thus is $4320(a^2d - 3abc + 2b^3) = 0$, viz. it is $a^2d - 3abc + 2b^3 = 0$.

XII.

Sylvester's Lectures are published in the American Mathematical Journal as follows: Lectures 1 to 10, t. VIII. (1886), pp. 196—260; lectures 11 to 32, t. IX. (1887), viz. 11 to 16, pp. 1—37, 17 to 24, pp. 113—161, and 25 to 32, pp. 297—352; and lectures 33 and 34 (34 by Mr Hammond), t. X. (1888), pp. 1—16. In the footnote p. 7 to lecture 12, writing y_1, y_2, y_3, y_4, ..., for the derived functions of y, he adopts definitely the notation t, a, b, c,..., to mean the reduced functions,

$$y_1 = \frac{1}{1}y_1, \quad y_2 = \frac{1}{1\cdot 2}y_2, \quad y_3 = \frac{1}{1\cdot 2\cdot 3}y_3, \quad y_4 = \frac{1}{1\cdot 2\cdot 3\cdot 4}y_4, \quad \ldots$$

In lecture 13, p. 20, he introduces the term *Principiant* — "instead of the cumbrous terms Projective Reciprocants or Differential Invariants, it is better to use the single word Principiants to denominate that crowning class or order of Reciprocants which remain to a factor près, unaltered for any homographic substitutions impressed on the variables" — that is, Halphen's Differential Invariant = Principiant. And in lectures 22 *et seq.*, Sylvester develops an important theory in regard to Principiants, connecting them with reciprocants and seminvariants, viz. he considers a series of seminvariants N, A_0, A_1, A_2, ..., and a series of reciprocants M, A, B, C, ..., such that the seminvariants formed with either sets of capitals are identical with each other (for instance $A_0A_2 - A_1^2 = AC - B^2$), and that any such seminvariant is a Principiant.

[p. 389]

In what follows, instead of N, A_0, A_1, $A_2,\ldots$, I write N, A, B, $C,\ldots$, and instead of M, A, B, $C,\ldots$, I write $\mathfrak{M}$, $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C},\ldots$, so that N, A, B, $C,\ldots$, are seminvariants and $\mathfrak{M}$, $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C},\ldots$, are reciprocants.

The expressions of these functions up to E, $\mathfrak{E}$, are

$N =$	$A =$	$B =$	$C =$	$8D =$	$\mathfrak{E} =$
ac	a^2d	a^3e	a^4f	a^5g	a^6h
$+1$	$+1$	$+1$	$+1$	$+8$	$+6$
b^2	abc	a^2bd	a^3be	a^4bg	a^5bi
-1	-3	-4	-5	-48	-42
	b^3	c^2	cd	ce	cf
	$+2$	-2	-5	-48	-42
		ab^2c	a^2b^2d	a^3b^2e	a^4b^2f
		$+10$	$+15$	-25	$+45$
			b^2c	a^2c^2	a^3cd
			-5	$+15$	-5
				a^2b^3c	a^3b^3d
				$+168$	$+168$
				c^3	a^3ce
				$+56$	-35
					b^3c^2
					$+342$
					bc^2d
					-1578
					c^4
					$+129360$

[Table simplified; original contains many more numerical terms.]

In terms of the fundamental seminvariants as indicated by their initial and final terms

$$ac\infty b^2 = ac - b^2, \quad a^2d\infty b^3 = a^2d - 3abc + 2b^3, \quad \&c.,$$

the expressions of these functions are

$$
\begin{aligned}
3B &= (a^3e\infty a^2c^2) - 5(a^2c^2\infty b^4),\\
7C &= (a^4f\infty a^2bc^2) - 7(a^3cd\infty b^5),\\
8D &= 8(a^5g\infty a^4b^2d) - 168(a^4ce\infty a^3c^3) - 113(a^4d^2\infty a^2b^2c^2) + 340(a^3c^3\infty b^6),\\
V &= 2(a^6h\infty a^4bd^2) - 32(a^5cf\infty a^3bc^2) - 37(a^5de\infty a^2b^3c^2) + 137(a^4cd^2\infty b^7).
\end{aligned}
$$

[p. 390]

In terms of the fundamental reciprocants of the table, ante p. 387, say these are

$$\mathfrak{A} = P_3, \quad 50a^2e - \cdots + 105b^2c = P_4, \quad 10a^3f - \cdots - 3963bc = P_5,$$

$$23106b^2c = P_6, \quad 7a^3h - \cdots - 19256c^3 = P_7,$$

the expressions of these functions are

$$\begin{aligned} 49\mathfrak{A} &= P_3^2, \\ 40\mathfrak{C} &= 4aP_5 - 38P_2P_3, \\ 280\mathfrak{D} &= 200a^3P_6 + 1266aP_2P_4 - 3027aP_3^2 - 9128P_2^3, \\ 1680\mathfrak{E} &= 240a^3P_7 + 2940aP_2P_5 - 3156aP_3P_4 + 2373P_3P_2^2. \end{aligned}$$

Sylvester remarks that a Principiant, e.g. $a^2d - 3abc + 2b^3$, is at once a reciprocant, and in the theory of seminvariants (where a, b, c, $\dots$, are the coefficients of the theory) a seminvariant; and conversely that any reciprocant which is also a seminvariant is a principiant: quâ seminvariant, the principiant is annihilated by

$$\Omega = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + \dots;$$

and quâ reciprocant is annihilated by

$$V = 2a^2\partial_b + 5ab\partial_c + (6ac+3b^2)\partial_d + (7ad+7bc)\partial_e + (8ae+8bd+4c^2)\partial_f + \dots;$$

and any function which is annihilated by each of these operators is a principiant.

We form the foregoing series of functions $N = ac - b^2$, A, B, $C, \dots$, as follows, viz. if

$$G' = (4ac-5b^2)\partial_b + (5ad-7bc)\partial_c + (6ae-9bd)\partial_d + (7af-11be)\partial_e + \dots,$$

then

$$5A = G'N, \quad 6B = G'A, \quad 7C = G'B+NA, \quad 8D = G'C+2NB, \quad 9E = G'D+3NC$$

and similarly we form the foregoing series of functions $\mathfrak{M}$, $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}, \dots$, as follows, viz. if $\mathfrak{G}'$ is Sylvester's $\mathfrak{G}$, then

$$5\mathfrak{A} = \mathfrak{G}'\mathfrak{M}, \quad 6\mathfrak{B} = \mathfrak{G}'\mathfrak{A}, \quad 7\mathfrak{C} = \mathfrak{G}'\mathfrak{B}, \quad 8\mathfrak{D} = \mathfrak{G}'\mathfrak{C} - \mathfrak{M}\mathfrak{B},$$

As already mentioned, N, A, B, C,..., are seminvariants, $\mathfrak{M}$, $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$,..., are reciprocants. Putting for shortness $\frac{b}{a} = \theta$, the two sets of functions are connected by

$$-A = \mathfrak{A}, \quad \text{or conversely } \mathfrak{A} = -A,$$

$$B = \mathfrak{B} - \theta\mathfrak{A}, \quad \mathfrak{B} = B + \theta A,$$

$$C = \mathfrak{C} - 2\theta\mathfrak{B} + \theta^2\mathfrak{A}, \quad \mathfrak{C} = C + 2\theta B + \dots$$

so that, one of the sets being calculated, the other set can be at once deduced therefrom. These equations give

$$4C - B^2 = \mathfrak{A}\mathfrak{C} - \mathfrak{B}^2, \quad 4AD - 3ABC + 2B^3 = \mathfrak{A}^2\mathfrak{D} - 3\mathfrak{A}\mathfrak{B}\mathfrak{C} + 2\mathfrak{B}^3, \quad \&c.,$$

viz. as mentioned above, any seminvariant in the letters A, B, C,..., is equal to the same seminvariant in the letters $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$,...; or, what is the same thing, any principiant has the same expression in the letters A, B, C,..., and in the letters $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$,... respectively.

We have thus the entire series of Principiants,

$$AC - B^2, \quad A^2D - 3ABC + 2B^3, \quad AE - 4BD + 3C^2, \quad \&c.$$

[p. 392]

We may express Halphen's reciprocants in terms of the capitals A, B, C,.... We have

$$U = a, \quad V = A, \quad \Delta = AC - B^2.$$

To obtain formulae for the higher reciprocants, we require the derived functions A', B', C', ..., where the accent denotes differentiation in regard to x.

We have

$$a' = 3b, \quad b' = 4c, \quad c' = 5d, \quad \ldots$$

(whence in particular $a' = 3b$, $b' = 4c$, $c' = 5d$, ..., as is obvious). Hence, writing

$$G' = a(3b\partial_a + 4c\partial_b + 5d\partial_c + \ldots) - b(3a\partial_a + 5b\partial_b + 7c\partial_c + \ldots),$$

this may be written

$$G' = a\partial_x - b \cdot w,$$

if w be the weight of the homobaric function operated upon, reckoning the weights of a, b, c,..., as 3, 5, 7,..., respectively, and consequently the weights of A, B, C,..., as 15, 20, 25,... respectively. We thus have

$$5A = (a\partial_x - 15b)N, \quad 6B = (a\partial_x - 20b)A, \quad 7C = (a\partial_x - 25b)B + NA,$$

$$8D = (a\partial_x - 30b)C + 2NB,$$

of which the first gives only $A = a^2d - 3abc + 2b^3$. The other equations give the required formulae

$$aA' = 6B + 15bA, \quad aB' = 7C + 20bB - NA, \quad aC' = 8D + 25bC - 2NB.$$

Halphen's H is given by $U^4H = 256A^3 - 27V^8$, viz. we thus have

$$a^4H = 256(AC - B^2)^3 - 27A^8.$$

His T is defined by the equation $U^3T = 3VA' - 8V'A$, that is,

$$a^3T = 3A(AC - B^2)' - 8A'(AC - B^2),$$

which is

$$\begin{aligned} &= 3A(AC' - 2BB') + (-5AC + 8B^2)A' \\ &= 24(A^2D - 3ABC + 2B^3), \end{aligned}$$

which expression for the reciprocant T was given by Sylvester.

[p. 393]

Halphen's T_9 is $\frac{1}{U}(V^4T - \frac{9}{2}H)$, viz. we thus have

$$2a^5T_9 = 24A^4(A^2D - 3ABC + 2B^3) - 256(AC - B^2)^3.$$

His G is given by $V^2G = U^8T^2 + 9H$, viz. we have

$$A^2a^8G = 576(A^2D - 3ABC + 2B^3)^2 + 2604(AC - B^2)^3 - 243A^8,$$

where the whole divides by A^2; throwing out this factor, we find

$$a^8G = 576(A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2) - 243A^6.$$

From the expression for T, I find

$$a^5T' = 216A^2E - 288ABD - 504AC^2 + 576B^2C + 1152(A^2D - 3ABC + 2B^3),$$

and I thence deduce for Halphen's Θ,

$$\Theta = \frac{1}{U^{11}}(2U^8TT' + T(8VU' - 3UV')),$$

the formula

$$a^4\Theta = 432(AC - B^2)(AE - 4BD + 3C^2) - 576(A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2$$

or, what is the same thing,

$$a^4\Theta = 144C \cdot \mathfrak{I} + 576A \cdot \mathfrak{K}.$$

Also $\Theta_2 = \frac{1}{U^9}(\Theta + \frac{9}{2}G)$, that is, $a^4\Theta_2 = \frac{1}{a^5}(a^4\Theta + \frac{9}{2}a^4G)$,

$$a^4\Theta_2 = 432(ACE - AD^2 - B^2E + 2BCD - C^3) - 540A^4(A^2D - 3ABC + 2B^3) - \tfrac{729}{4}A^8.$$

Writing

$$AC - B^2 = \mathfrak{C}, \quad A^2D - 3ABC + 2B^3 = \mathfrak{D},$$

$$A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2 = \mathfrak{D}',$$

$$AE - 4BD + 3C^2 = \mathfrak{I}, \quad ACE - AD^2 - B^2E + 2BCD - C^3 = \mathfrak{J},$$

[p. 394]

the foregoing results become $U = a$, $V = A$, $\Delta = \mathfrak{C}$,

$$a^4 H \text{ (Halphen)} = 256\mathfrak{C}^3 - 27A^8,$$

$$a^3 T = 24\mathfrak{D},$$

$$a^4\Theta = 144C \cdot \mathfrak{I} + 576A \cdot \mathfrak{J},$$

$$a^5\Theta_2 = 432\mathfrak{J} - 540A^4\mathfrak{D} - \tfrac{729}{4}A^8.$$

XIII.

The letters t, a, b, ..., of a reciprocant represent (it will be remembered) mere numerical multiples of the derived functions of a variable y, in regard to the independent variable x, and thus equating any reciprocant to zero, we have a differential equation in y. For instance, if the orthogonal reciprocant $c(1+t^2) - 5abt + 5a^3$ be put $= 0$, (t, a, b, $c = y'$, $\frac{1}{2}y''$, $\frac{1}{6}y'''$, $\frac{1}{24}y^{\text{iv}}$), this is the differential equation of the order 4. The integral hereof is expressible by the two equations

$$x = \kappa \int \frac{U\,dt}{(1+t^2)^{\frac{5}{2}}} + \lambda \int \frac{V\,dt}{(1+t^2)^{\frac{5}{2}}}, \quad y = \kappa \int \frac{tU\,dt}{(1+t^2)^{\frac{5}{2}}} + \lambda \int \frac{tV\,dt}{(1+t^2)^{\frac{5}{2}}} + \mu,$$

where U, V are real functions of a parameter t, viz. $U + iV = (1+it)^6$, $U - iV = (1-it)^6$, ($i = \sqrt{-1}$ as usual), so that $U = 1 - 15t^2 + 15t^4 - t^6$, $V = 6t - 20t^3 + 6t^5$; and κ, λ, μ, ν are the four constants of integration.

[p. 395]

Substituting for U, V either of the above-mentioned forms, there is in each case a factor in t which divides out; rejecting this factor, the equation is found to be

$$2y_4(1+t^2) - 10t^2y_3 + 15y_2^3 = 0,$$

that is, substituting for t its value $= y_1$, and throwing out the factor 2, we have the differential equation

$$y_4(1+y_1^2) - 10y_1y_2y_3 + 15y_2^3 = 0.$$

XIV.

But when the reciprocant equated to zero is a Principiant, we have the far more important results obtained in Halphen's Memoir (1): and which are, in Sylvester's lectures, exhibited in a more complete form by expressing Halphen's reciprocants in terms of the foregoing functions A, B, C,.... I recall that a, b, c,... are numerical multiples of differential coefficients of the orders 2, 3, 4,... respectively; A, B, C,... contain d, e, f,... respectively, that is, differential coefficients of the orders 5, 6, 7,... respectively, so that according as the highest capital letter is A, B, C,... respectively, the order of the differential equation is 5, 6, 7,... respectively.

But the equation, Principiant $= 0$, may be interpreted in a different manner, viz. if instead of regarding the differential equation as an equation for the determination of y, we regard therein y as a given function of x, (that is, (x, y) as the coordinates of a point on a given curve), then the differential equation serves for the determination of those points on the curve for which a given condition is satisfied, or say it is the condition in order to the existence of a singular point of determinate character.

Halphen's lowest reciprocants are: as already mentioned:

$$U = a, \quad V = a^2d - 3abc + 2b^3 = A.$$

$U = 0$, that is, $a = 0$, is a differential equation of the second order, viz. it is the differential equation of a line: otherwise it is the condition for a point of inflexion.

$V = 0$, that is, $A = 0$, is a differential equation of the fifth order, viz. it is the differential equation of a conic: otherwise it is the condition for a *sextactic point*.

$\Delta = 0$, that is, $AC - B^2 = 0$, is a differential equation of the seventh order, it is the condition for what Halphen calls a *point of coincidence*, viz. this is a point on a given curve, such that for it the cubic of nine-pointic intersection becomes a nodal cubic having the point for node and through it one branch of eight-pointic intersection and one branch of simple intersection. Observe that this is more than the condition that the cubic of nine-pointic intersection shall be a nodal cubic, and accordingly $\Delta = 0$ is not the differential equation of a nodal cubic. In fact, a nodal cubic depends on 8 parameters and has therefore a differential equation of the order 8; this will be obtained further on. But $\Delta = 0$ is the differential equation of the order 7 of a curve such that every point thereof is a coincident point.

[p. 396]

The differential equation of the order 7,

$$2^4 \cdot 7^3 \cdot \{(X-2)(X+1)(2X-1)\}^2 \cdot A^3 - 3^3 \cdot 5^2 (X^2 - X + 1)^3 V^8 = 0,$$

has the integral $\alpha = \beta\sqrt{\gamma - \lambda}$, where α, β, γ represent arbitrary linear functions; we may without loss of generality take the linear functions to be of the form $x + my + n$, introducing in this case another constant C, and writing the integral equation in the form $\alpha = C\beta\sqrt{\gamma - \lambda}$; the number of arbitrary constants is thus $= 7$.

If $X = 2$, -1 or $\frac{1}{2}$, the integral equation is $\alpha = \beta^2\gamma^{-1}$, $\alpha = \beta^{-1}\gamma$ or $\alpha = \beta\gamma^{\frac{1}{2}}$; or say it is $\beta^2 = \alpha\gamma$, $\beta^2 = \alpha\gamma$ or $\alpha^2 = \beta\gamma$, viz. each of these represents a conic. The differential equation is here $V = 0$, viz. we have the foregoing result that $V = 0$, that is, $A = 0$, is the differential equation of a conic.

If $X = \infty$, 0 or 1, the differential equation is

$$2^6 \cdot 7^3 \cdot A^3 - 3^3 \cdot 5^2 \cdot V^8 = 0;$$

the integral equation is here of the form $\frac{\alpha}{\gamma} = \log\frac{\beta}{\gamma}$; in fact we have $\frac{\alpha}{\gamma} = \left(\frac{\beta}{\gamma}\right)^X$, or for α, β, and γ writing $\lambda\alpha$, $\beta + \lambda\gamma$, and $\lambda\gamma$ respectively, this is $\frac{\alpha}{\gamma} = \left(1 + \frac{\beta}{\gamma}\right)^{\frac{1}{X}}$, which when λ is indefinitely large becomes $\frac{\alpha}{\gamma} = \exp.\frac{\beta}{\gamma}$, that is, $\frac{\alpha}{\gamma} = \log\frac{\beta}{\gamma}$. And similarly for $X = 0$ and $X = 1$.

If $X^2 - X + 1 = 0$, that is, if $X = -\omega$, where ω is an imaginary cube root of unity, the differential equation is $\Delta = 0$, that is, $AC - B^2 = 0$ (of the order 7 as in the general case). The integral equation is $\alpha = \beta^{-\omega}\gamma^{-\omega^2}$, viz. this is the equation of the curve every point of which is a coincident point.

If $X = 3$, the differential equation is $2^8A^3 - 3^3V^8 = 0$, or say $256A^3 - 27V^8 = 0$; Halphen's H is a function such that $U^4H = 256A^3 - 27V^8$, so that this is

$$H = U^{-4}(256A^3 - 27V^8) = 0.$$

The integral equation is $\alpha = \beta^3\gamma^{-2}$, that is, $\alpha\gamma^2 = \beta^3$, or the curve is a *cuspidal cubic*, depending upon 7 constants; and it thus appears that the differential equation of a cuspidal cubic is the equation of the order 7

$$a^{-4}\{256(AC - B^2)^3 - 27A^8\} = 0.$$

Write

$$\Theta = 64(4\mathfrak{D}^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2) + 144A^2(A^2D - 3ABC + 2B^3) + 81A^6$$

$$= 64\mathfrak{D}' + 144A^2\mathfrak{D} + 81A^6,$$

$$\Phi = 8(A^2D - 3ABC + 2B^3) + 9A^4 = 8\mathfrak{D} + 9A^4,$$

$$\Psi = 3072(AC - B^2)^3 = 3072\mathfrak{C}^3,$$

where identically

[p. 397]

Halphen and Sylvester find that, for a cubic curve the invariants of which are S and T ($S = -l+l^3$, $T = 1-20l^3-8l^6$ for the canonical form $x^3+y^3+z^3+6lxyz = 0$), we have

$$\Psi S = \Theta^2 - 12288\mathfrak{C}^3(8\mathfrak{D}+9A^4) = 0,$$

$$\Psi^2 T = \Theta^3 - 18432\mathfrak{C}^3(8\mathfrak{D}+9A^4)\Theta + 552231 \cdot A^4\mathfrak{C}^6 = 0,$$

(Ψ an arbitrary multiplier); and we thence have

$$\frac{\Theta^2}{\Psi^3} = S^3, \quad \frac{\Phi^3}{\Psi^2} = T^2,$$

that is,

$$\frac{T^2}{S^3} = \frac{\Phi^3}{\Theta^2\Psi},$$

or if, as with Halphen $h = K\frac{T^2}{S^3}$, then

$$\frac{\Phi^3 - \Theta\Psi}{\Theta^2\Psi} = h - 1.$$

If for shortness

$$R = \Theta^3 - \cdots = R^2 - P^3, \quad P^3 = \ldots$$

then the foregoing equation $\frac{\Phi^3}{\Theta^2\Psi} = \ldots$ gives

$$\frac{R^2}{P^3} = h, \quad \frac{Q^2}{P^3} = h - 1;$$

and thus for a nodal cubic we have $Q = 0$; we thus have

$$P = 0, \text{ for the differential equation of a cubic for which } S = 0,$$

$$R = 0, \text{ nodal cubic,}$$

$$Q = 0,$$

these being each of them of the order 8. But the equation $Q = 0$ is reducible to a more simple form. We have

[p. 398]

or, since in the expression in { } the first and second terms are $12\Theta^2(4\mathfrak{D} - \Phi^2) = \ldots$ this is

$$Q = \Psi^3\{(\Theta - 364A^2\Phi + 216A^4)^2 - 64(\Phi - 94A^4)^3\}.$$

We thus have for the differential equation of the eighth order of the nodal cubic

$$(\Theta - 364A^2\Phi + 216A^4)^2 - 64(\Phi - 94A^4)^3 = 0,$$

or, since

$$\Theta = 64\mathfrak{D}' + 144A^2\mathfrak{D} + 81A^6, \quad \text{and} \quad \Phi = 8\mathfrak{D} + 9A^4,$$

the differential equation of the eighth order for the nodal cubic finally is

$$(64\mathfrak{D}' - 144A^2\mathfrak{D} + 27A^6)^2 + 32768\mathfrak{D}^3 = 0.$$

Halphen has $\Theta_2 = 0$ for the differential equation of a cubic curve. Putting, as before,

$$\mathfrak{J} = ACE - AD^2 - B^2E + 2BCD - C^3,$$

(and therefore $4\mathfrak{C}\mathfrak{I} - 4\mathfrak{J} = \mathfrak{D}'$, $\mathfrak{I}^3 - 27\mathfrak{J}^2 =$ Quartic Discr.), then, by what precedes, the differential equation of the order 9 of a cubic curve is

$$a^{-5}(432\mathfrak{J} - 54A^4\mathfrak{D} - \tfrac{729}{4}A^6) = 0.$$

Recapitulating.

Order. The differential equation for a cubic curve: —

8. for which invt. $S = 0$, is $P = \Theta^2 - 12288\mathfrak{C}^3(8\mathfrak{D} + 9A^4) = 0$,

8. for which invt. $T = 0$, is $R = \Theta^3 - 18432\mathfrak{C}^3(8\mathfrak{D} + 9A^4)\Theta + 552231A^4\mathfrak{C}^6 = 0$,

8. nodal cubic $Q \div (3072\Psi)^3 = (64\mathfrak{D}' - 144A^2\mathfrak{D} + 27A^6)^2 + 32768\mathfrak{D}^3 = 0$,

8. for invt. $64S^3 \div T^2 = h$, is $(h-1)R^2 - hQ = 0$,

9. for general cubic is $1728\mathfrak{J} - 216A^4\mathfrak{D} - 243A^8 = 0$,

where for shortness Θ is retained to signify its value $= 64\mathfrak{D}' + 144A^2\mathfrak{D} + 81A^6$.

Halphen by a polar transformation finds that these same differential equations, changing therein the sign of $\mathfrak{D}$, apply to curves of the third class: in particular, the last equation, changing therein the sign of $\mathfrak{D}$, applies to the general curve of the third class, or sextic curve with nine cusps.

XV. MacMahon's Multilinear Operator

A very important notion in the theory, as well of Seminvariants as of Reciprocants, is that of MacMahon's Multilinear Operator, see his paper "Theory of a Multilinear Partial Differential Operator, with applications to the theories of Invariants and Reciprocants," *Proc. Lond. Math. Soc.*, t. XVIII. (1886), pp. 61—88: this operator plays so important a part in the theories to which it relates, that I venture to reproduce the definition of it in what appears to me a simplified and more easily intelligible form, and to recapitulate some of the leading properties.

I take with him the letters to be $(a_0, a_1, a_2, a_3, \dots) = (a, b, c, d, \dots)$, viz. the first letter is a_0 or a, the second is a_1 or b, and so on, but when we are not concerned with a term of indefinite rank, I use always $(a, b, c, d, \dots)$ and the like in regard to any other series of letters. This being so, the definition of the Multilinear Operator of four elements, which is here alone in question, is

$$(\mu, \nu; m, n) =$$

$$\begin{array}{ccccccccccc} n & 0 & & 1 & & 2 & & 3 & & & \dots \\ & \mu A\partial_a & + & (\mu+\nu)B\partial_b & + & (\mu+2\nu)C\partial_c & + & (\mu+3\nu)D\partial_d & + & \dots & + \quad (\mu \end{array}$$

where

$$A = \frac{m}{1}a^{m-1}, \quad B = \frac{m}{1}a^{m-1}b, \quad C = \frac{m}{1}a^{m-1}c + \frac{m(m-1)}{1\cdot 2}a^{m-2}b^2,$$

$$D = \frac{m}{1}a^{m-1}d + (m-1)a^{m-2}bc + \frac{(m-1)(m-2)}{1\cdot 2\cdot 3}a^{m-3}b^3,$$

where the expressions for B, C, D,..., are obtained successively from that of A, $= \frac{m}{1}a^{m-1}$, by Arbogast's rule of derivation, viz. we operate on the last letter of a term, and when there is a last but one letter which in alphabetical order immediately precedes the last letter, then also upon the last but one letter, and whenever the term thus obtained ends in a power, we divide by the index of the power. Thus the term $\frac{1}{2}(m-1)a^{m-2}b^2$ in the third line gives in the fourth line

$$\tfrac{1}{2}(m-1)a^{m-2}\cdot 2bc + \tfrac{1}{2}(m-1)(m-2)a^{m-3}b\cdot b^2\cdot\tfrac{1}{2},$$

$$= (m-1)a^{m-2}bc + \tfrac{1}{6}(m-1)(m-2)a^{m-3}b^3.$$

[p. 400]

Going a step further, we form in this manner the expression

$$E = a^{m-1}e + (m-1)a^{m-2}(bd + \tfrac{1}{2}c^2) \\ + \tfrac{1}{2}(m-1)(m-2)\{a^{m-3} \cdot 3b^2c + (m-3)a^{m-4}b \cdot b^3 \cdot \tfrac{1}{6}\},$$

that is,

$$E = a^{m-1}e + (m-1)a^{m-2}bd + \tfrac{1}{2}(m-1)a^{m-2}c^2 \\ + a^{m-3} \cdot \tfrac{1}{2}(m-1)(m-2)b^2c + a^{m-4} \cdot \tfrac{1}{24}(m-1)(m-2)(m-3)b^4,$$

which is sufficient to explain the rule.

It may be noticed that, the operator being linear in μ, ν, we have

$$(\mu, \nu; m, n) = \mu(1, 0; m, n) + \nu(0, 1; m, n).$$

The Alternant of two multilinear operators.

If P, Q are the two operators, we have as usual

$$P \cdot Q = PQ + P * Q,$$

where $P \cdot Q$ denotes the successive operation first with Q and then with P upon any operand, PQ is the mere algebraical product of the operators, and $P * Q$ is the operator obtained by the operation of P upon Q.

Similarly,

$$Q \cdot P = QP + Q * P,$$

and since QP is the same thing as PQ, we obtain

$$P \cdot Q - Q \cdot P = P * Q - Q * P,$$

either of which equal expressions, or say rather the second of them, is called the *alternant* of P, Q and is written $[P \cdot Q]$: viz. as the definition of the alternant, we have

$$[P \cdot Q] = P * Q - Q * P.$$

We have the remarkable theorem, that the alternant of any two operators $(\mu', \nu'; m', n')$, $(\mu, \nu; m, n)$ is an operator $(\mu_1, \nu_1; m_1, n_1)$; where

$$\mu_1 = (m' + m - 1)\frac{\mu}{m} - \frac{m-1}{m}(\mu' + n'\nu) - \frac{m'-1}{m'}(\mu + n\nu'),$$

$$\nu_1 = (n' - n)\nu\nu' + \frac{m-1}{m}\mu\nu' - \frac{m'-1}{m'}\mu'\nu,$$
$$m_1 = m' + m - 1, \quad n_1 = n' + n,$$

or since m_1, n_1 have such simple expressions, say it is an operator $(\lambda, \mu; m' + m - 1, n' + n)$, where λ, μ have the values just written down.

[p. 401]

We see at once how these values $m_1 = m' + m - 1$, $n_1 = n' + n$ arise: $Q = (\mu, \nu; m, n)$ contains the letters $(a, b, c, d, \dots)$ in the degree m, and it contains differential symbols ∂ which, operating on any function of these letters, diminish the degree by 1; similarly $P = (\mu', \nu'; m', n')$ contains the letters $(a, b, c, d, \dots)$ in the degree m', and it contains the differential symbols ∂ which, operating on any function of the letters, diminish the degree by 1. Hence $P * Q$ contains the letters in the degree $(m' - 1) + m$, and similarly $Q * P$ contains them in the same degree $(m - 1) + m'$: thus in the alternant the degree is $m' + m - 1$. Again, in Q the weights of the successive functions A, B, C,... are 0, 1, 2,... and these are combined with differential symbols ∂_{a_n}, $\partial_{a_{n+1}}$,... which operating on any function of the letters diminish the weight by n, $n+1$,... respectively: that is, the terms $A\partial_{a_n}$, $B\partial_{a_{n+1}}$, ... each diminish the weight by n. So in $P = (\mu', \nu'; m', n')$ the weights of the successive terms A, B, C,... are 0, 1, 2,... and these are combined with differential symbols $\partial_{a_{n'}}$, $\partial_{a_{n'+1}}$,... which operating on any function of the letters diminish the weights by n', $n'+1$,... respectively. We may say that Q is a sum of terms such as $\Phi_k \partial_{a_{n+k}}$, where the subscript k of Φ denotes the weight of Φ; operating hereon with P, the corresponding term of $P * Q$ is of the form $\Phi_{k-r} \partial_{a_{n+n'+k}}$, or (what is the same thing) the general form of the terms of $P * Q$ is $\Phi_k \partial_{a_{n+n',k}}$; and in like manner this is the general form of the terms of $Q * P$; and it is thus also the general form of the terms of the alternant $[P \cdot Q]$. It thus appears that, admitting the alternant to be an operator $(\mu_1, \nu_1; m_1, n_1)$, the value of n_1 is $n' + n$.

As an instance of the theorem, we may take

$$[(1, 0; 1, 1),\ (1, 0; 2, 1)] = (1, 0; 2, 2),$$

we have here

$$(1, 0; 1, 1) * (1, 0; 2, 1) - (1, 0; 2, 1) * (1, 0; 1, 1),$$

$$\begin{aligned}
&= a\partial_b(2a\partial_b + b\partial_c + c\partial_d + d\partial_e + e\partial_f + \dots) \\
&+ (ab\partial_c + (ac + \tfrac{1}{2}b^2)\partial_d + (ad + bc)\partial_e + (ae + bd + \tfrac{1}{2}c^2)\partial_f + \dots) \\
&\qquad - \{2a\partial_b + b\partial_c + c\partial_d + d\partial_e + \dots\}(a\partial_b) \\
&\qquad = a(a\partial_c + b\partial_d + c\partial_e + d\partial_f + \dots) \\
&\qquad\quad + b(a\partial_d + b\partial_e + c\partial_f + \dots) \\
&\qquad\quad + c(a\partial_e + b\partial_f + \dots) \\
&\qquad\quad + d(a\partial_f + \dots) + \dots \\
&= a^2\partial_c + 2ab\partial_d + (2ac + b^2)\partial_e + (2ad + 2bc)\partial_f + \dots \\
&\qquad = (1,0;2,2), \text{ as it should be.}
\end{aligned}$$

[p. 402]

For the general proof, writing for a moment

$$C_0 = nA_0, \quad C_1 = (\mu+\nu)A_1, \quad C_2 = (\mu+2\nu)A_2, \quad \&\text{c.},$$

and similarly

$$C_0' = n'A_0', \quad C_1' = (\mu'+\nu')A_1', \quad C_2' = (\mu'+2\nu')A_2', \quad \&\text{c.},$$

where the accented symbols refer to the values $(\mu', \nu'; m', n')$, we have

$$\begin{aligned}&(\mu', \nu'; m', n) * (\mu, \nu; m, n)\\ &= C_0'\partial_{a_{n'}}(C_0\partial_{a_n} + C_1\partial_{a_{n+1}} + \dots) + C_1'\partial_{a_{n'+1}}(C_0\partial_{a_n} + \dots) + \dots\end{aligned}$$

or observing that in the series C_0, C_1, C_2, …, the first term that contains a_n is C_n, and the like as regards the series C_0', C_1', C_2', …, this is

$$= -(C_0\partial_{a_n} + C_1\partial_{a_{n+1}} + \dots)(C_0'\partial_{a_{n'}} + C_1'\partial_{a_{n'+1}} + \dots) + (C_0\partial_{a_{n+n'}} + C_1\partial_{a_{n+n'+1}} + \dots),$$

and it is thus of the form

$$C_0''\partial_{a_{n+n'}} + C_1''\partial_{a_{n+n'+1}} + C_2''\partial_{a_{n+n'+2}} + \&\text{c.},$$

i.e. the value of n_1 is $= n + n'$.

I confine myself to the comparison of the first and second coefficients: substituting for the C's their expressions in terms of the A's, we ought to have

$$\begin{aligned}&\mu' A_0'\partial_{a_{n'}}(\mu + n'\nu)A_{n'} - \mu A_0\partial_{a_n}(\mu' + n\nu')A_n = \mu_1 A_0'',\\ &\mu' A_0'\partial_{a_{n'}}(A_{n'+1}) + (\mu'+\nu')A_1'\{A_{n'} + (n+1)\nu\}A_{n'+1}\\ &\quad -\mu A_0\partial_{a_n}(A_{n+1}') - (\mu+\nu)A_1\{A_n' + (n'+1)\nu'\}A_{n+1}' = (\mu_1+\nu_1)A_1''.\end{aligned}$$

Now assuming $m_1 = m' + m - 1$, and attending to the values

$$A_0 = -\frac{m}{1}a_0^{m-1}, \quad A_0' = -\frac{m'}{1}a_0^{m'-1}, \quad A_0'' = -\frac{m'+m-1}{1}a_0^{m'+m-2},$$

$$A_1 = \frac{m}{1}a_0^{m-1}a_1, \quad \dots$$

[p. 403]

we see at once that in the first equation the terms contain each of them the factor $a_0^{m'+m-2}$, and in the second equation they contain each of them the factor $a_0^{m'+m-3}a_1$; omitting these factors, we find

$$\mu'\cdot\frac{m}{m'}\cdot(\mu+n'\nu)-\mu\cdot\frac{m'}{m}\cdot(\mu'+n\nu')=\mu_1\cdot\frac{m'+m-1}{1},$$

$$\begin{aligned}&-\mu'\cdot\frac{m}{m'}\cdot(\mu+n'\nu+\nu)+(\mu'+\nu')(\mu+n'\nu+\nu)\\ &-\mu\cdot\frac{m'}{m}\cdot(\mu'+n\nu'+\nu')+(\mu+\nu)(\mu'+n\nu'+\nu')=(\mu_1+\nu_1)\cdot\frac{m'+m-1}{1}.\end{aligned}$$

The first of these gives

$$\mu_1=\frac{1}{m'+m-1}\left\{\frac{m}{m'}\mu'(\mu+n'\nu)-\frac{m'}{m}\mu(\mu'+n\nu')\right\},$$

which is the value of μ_1: substituting this in the second equation, we have

$$\begin{aligned}\nu_1=&-\frac{1}{m'+m-1}\left\{\frac{m}{m'}(\mu+n'\nu+\nu)-(\mu'+\nu')\right\}(\mu+n'\nu+\nu)\\ &+\frac{1}{m'+m-1}\left\{\frac{m'}{m}(\mu'+n\nu'+\nu')-(\mu+\nu)\right\}(\mu'+n\nu'+\nu').\end{aligned}$$

In the left-hand column, the terms containing $(\mu+n'\nu)$ are

$$\left\{-\frac{m}{m'}(\mu+n'\nu)+\mu'+\nu'-\frac{m'}{m}(\mu+n'\nu)\right\}(\mu+n'\nu),$$

and there are besides the terms $-\frac{m}{m'}\nu(\mu+n'\nu)+\nu(\mu'+\nu')$, hence the left-hand column is

$$=\nu\nu'(n'+1)+\mu\nu'+\mu'\nu\left(\frac{m-1}{m}+1\right)-\nu\nu'(n+1)-\mu\nu'\left(\frac{m'-1}{m'}+1\right)-\mu'\nu,$$

that is,

$$\nu_1=\nu\nu'(n'-n)+\mu\nu'\frac{m-1}{m}-\mu'\nu\frac{m'-1}{m'}.$$

To complete the proof, it would be of course necessary to compare the remaining terms on the two sides respectively: but in what precedes it is shown that, the form being assumed, the expression for the alternant must of necessity be $(\mu_1,\nu_1;m'+m-1,n'+n)$, where μ_1, ν_1 have their foregoing values.

0.2 On Pfaff-Invariants. [944]

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1.

The functions which I propose to call *Pfaff-invariants* present themselves and play a leading part in the memoir, Clebsch, "Ueber das Pfaff'sche Problem" (Zweite Abhandlung), *Crelle*, t. LXI. (1863), pp. 146—179: but it is interesting to consider them for their own sake as invariants, and in the notation which I have elsewhere used for the functions called Pfaffians. The great simplification effected by this notation is, I think, at once shown by the remark that Clebsch's expression R, which he defines by the periphrasis "Sei ferner R der rationale Ausdruck dessen Quadrat der Determinant der a_{ik} gleich ist" (l.c. p. 149), is nothing else than the Pfaffian $1234\ldots(2n-1).2n$, and that its differential coefficients $R_{ik} = \frac{\partial R}{\partial a_{ik}}$ are the Pfaffians obtained from the foregoing by the mere omission of any two symbolic numbers i, k.

2.

I call to mind that the symbols 12, 13, &c., made use of are throughout such that $12 = -21$, &c.; and that the definition of the successive Pfaffians 12, 1234, &c., is as follows:

$$\begin{aligned}
12 &= 12,\\
1234 &= 12\cdot 34 + 13\cdot 42 + 14\cdot 23,\\
123456 &= 12\cdot 3456 + 13\cdot 4562 + 14\cdot 5623 + 15\cdot 6234 + 16\cdot 2345,
\end{aligned}$$

in which last expression 3456 denotes the Pfaffian $34{\cdot}56+35{\cdot}64+36{\cdot}45$, and similarly 4562 &c.; and so on for any even number of symbols. Of course, instead of the symbolic numbers 1, 2, 3, &c., we may have any other numbers (0 is frequently used in the sequel as a symbolic number), or we may have letters or other symbols.

3.

I use also a function very analogous to a Pfaffian, which is expressed in the same notation, viz. this is

$$\begin{aligned}
\phi\psi_{1234} = \phi\psi_{12}\cdot 34 + \phi\psi_{13}\cdot 42 + \phi\psi_{14}\cdot 23\\
+ \phi\psi_{34}\cdot 12 + \phi\psi_{42}\cdot 13 + \phi\psi_{23}\cdot 14,
\end{aligned}$$

$$\phi\psi_{123456} = \phi\psi_{12} \cdot 34 \cdot 56 + \phi\psi_{34} \cdot 12 \cdot 56 + \phi\psi_{56} \cdot 12 \cdot 34 + \&\text{c.},$$

viz. taking any term $12 \cdot 34 \cdot 56$ of the Pfaffian 123456, $\phi\psi$ is connected successively with each of the binary symbols 12, 34, 56 of the term, so as to give rise to terms containing the quaternary symbols $\phi\psi_{12}$, &c. Such function may be called a *co-Pfaffian*.

4.

To avoid suffixes I use different letters (x, y), (x, y, z), &c., as the case may be, associating these with the numbers $(1, 2)$, $(1, 2, 3)$, &c. In the case of a differential of an even number $2n$ of terms, for instance $X dx + Y dy + Z dz + W dw$, I consider the functions 1234, ϕ01234, and $\phi\psi$01234, the first and second of which are Pfaffians, the last a co-Pfaffian, as explained above. To fix in connexion with the differential $X dx + Y dy + Z dz + W dw$ the meanings of these expressions, I assume

$$12 = \frac{\partial X}{\partial y} - \frac{\partial Y}{\partial x}, \quad 13 = \frac{\partial X}{\partial z} - \frac{\partial Z}{\partial x}, \quad \&\text{c.},$$

(of course these imply $12 = -21$, &c.),

$$01 = -10 = X, \quad 02 = -20 = Y, \quad \&\text{c.};$$

ϕ is an arbitrary function of x, y, z, w and I write

$$\phi\psi_{12} = \frac{\partial(\phi, \psi)}{\partial(x, y)} = \frac{\partial\phi}{\partial x}\frac{\partial\psi}{\partial y} - \frac{\partial\phi}{\partial y}\frac{\partial\psi}{\partial x},$$

ψ is also an arbitrary function of x, y, z, w, and I write

$$\phi\psi_{ij} = \frac{\partial(\phi, \psi)}{\partial(x_i, x_j)} \quad \text{(this implies } \phi\psi_{21} = -\phi\psi_{12}, \ \&\text{c.).}$$

5.

Thus, at full length, the functions are

$$\begin{aligned} 1234 &= 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23 \\ &= \left(\frac{\partial X}{\partial y} - \frac{\partial Y}{\partial x}\right)\left(\frac{\partial Z}{\partial w} - \frac{\partial W}{\partial z}\right) + \left(\frac{\partial X}{\partial z} - \frac{\partial Z}{\partial x}\right)\left(\frac{\partial W}{\partial y} - \frac{\partial Y}{\partial w}\right) \\ &\quad + \left(\frac{\partial X}{\partial w} - \frac{\partial W}{\partial x}\right)\left(\frac{\partial Y}{\partial z} - \frac{\partial Z}{\partial y}\right), \end{aligned}$$

$$\begin{aligned}\phi 01234 &= \phi_0\cdot 1234+\phi_1\cdot 2340+\phi_2\cdot 3401+\phi_3\cdot 4012+\phi_4\cdot 0123\\ &= \frac{\partial\phi}{\partial x}(23\cdot 40+24\cdot 03+20\cdot 34)\\ &\quad+\frac{\partial\phi}{\partial y}(34\cdot 01+30\cdot 14+31\cdot 40)\\ &\quad+\frac{\partial\phi}{\partial z}(40\cdot 12+41\cdot 20+42\cdot 01)\\ &\quad+\frac{\partial\phi}{\partial w}(01\cdot 23+02\cdot 31+03\cdot 12),\end{aligned}$$

$$\begin{aligned}&= X\left\{\left(\frac{\partial Y}{\partial z}-\frac{\partial Z}{\partial y}\right)\left(\frac{\partial W}{\partial x}-\frac{\partial X}{\partial w}\right)+\left(\frac{\partial Y}{\partial w}-\frac{\partial W}{\partial y}\right)\left(\frac{\partial X}{\partial z}-\frac{\partial Z}{\partial x}\right)\right\}\\ &\quad+\&\text{c.},\end{aligned}$$

$$\begin{aligned}\phi\psi_{1234} &= \phi\psi_{12}\cdot 34+\phi\psi_{13}\cdot 42+\phi\psi_{14}\cdot 23\\ &\quad+\phi\psi_{34}\cdot 12+\phi\psi_{42}\cdot 13+\phi\psi_{23}\cdot 14\\ &= \frac{\partial(\phi,\psi)}{\partial(x,y)}\left(\frac{\partial Z}{\partial w}-\frac{\partial W}{\partial z}\right)+\frac{\partial(\phi,\psi)}{\partial(x,z)}\left(\frac{\partial W}{\partial y}-\frac{\partial Y}{\partial w}\right)\\ &\quad+\frac{\partial(\phi,\psi)}{\partial(x,w)}\left(\frac{\partial Y}{\partial z}-\frac{\partial Z}{\partial y}\right)+\&\text{c.}\end{aligned}$$

6.

The invariantive property of the functions consists herein, viz. if we have

$$Xdx + Ydy + Zdz + Wdw = Pdp + Qdq + Rdr + Sds,$$

so that p, q, r, s, and thence also P, Q, R, S are functions each of them of x, y, z, w, then we have

$$1234 \cdot \partial(x,y,z,w) = (1234)' \cdot \partial(p,q,r,s),$$

$$\phi 01234 \cdot \partial(x,y,z,w) = (\phi 01234)' \cdot \partial(p,q,r,s),$$

$$\phi\psi 01234 \cdot \partial(x,y,z,w) = (\phi\psi 01234)' \cdot \partial(p,q,r,s),$$

where the accented functions refer to (p,q,r,s,P,Q,R,S), and where for greater symmetry I have separated the symbolical numerator and denominator $\partial(p,q,r,s)$ and $\partial(x,y,z,w)$; each of these equations really contains

$$\frac{\partial(p,q,r,s)}{\partial(x,y,z,w)},$$

which is the functional determinant of (p,q,r,s) in regard to (x,y,z,w): or, if we please, it contains the reciprocal hereof

$$\frac{\partial(x,y,z,w)}{\partial(p,q,r,s)},$$

which is the functional determinant of (x,y,z,w) in regard to (p,q,r,s).

7.

The equations give

$$\frac{\phi 01234}{1234} = \frac{(\phi 01234)'}{(1234)'},$$

$$\frac{\phi\psi 01234}{1234} = \frac{(\phi\psi 01234)'}{(1234)'},$$

and then the expressions on the left-hand are absolute invariants in respect to the transformation of

$$Xdx + Ydy + Zdz + Wdw \quad \text{into} \quad Pdp + Qdq + Rdr + Sds.$$

They are, in fact, (for $2n = 4$) Clebsch's derivatives (ϕ) and (ϕ, ψ).

8.

For the Pfaffian reduction

$$Xdx + Ydy + Zdz + Wdw = Fdf + Gdg,$$

we may write

$$P, Q, R, S = F, G, 0, 0, \quad p, q, r, s = f, g, F, G,$$

viz. we take f, g, F, G as the new independent variables; we thus have

$$01' = F, \quad 02' = G, \quad 03' = 0, \quad 04' = 0,$$

$$12' = 0, \quad 13' = 1, \quad 14' = 0, \quad 23' = 0, \quad 24' = 1, \quad 34' = 0,$$

$$(1234)' = 12' \cdot 34' + 13' \cdot 42' + 14' \cdot 23' = -1;$$

and similarly

$$(\phi 01234)' = -(\phi), \quad (\phi\psi 01234)' = -(\phi, \psi),$$

where, in the equations, the $-$ sign presents itself by reason that $2n = 4$, is the double of an even number, or say that n is even; in the case of $2n$, the double of an odd number, that is, n odd, the sign would have been $+$.

9.

We thus have

$$(\phi) = -\frac{\phi 01234}{1234} = \frac{\partial\phi}{\partial F} + \frac{\partial\phi}{\partial G},$$

$$(\phi, \psi) = -\frac{\phi\psi 01234}{1234} = \frac{\partial(\phi, \psi)}{\partial(f, g)} + \frac{\partial(\phi, \psi)}{\partial(F, G)},$$

and in particular, by giving to ϕ and ψ the values f, g, F, G, we find

$$(f) = 0, \quad (g) = 0, \quad (F) = G, \quad (G) = 0,$$

$$(f, g) = 0, \quad (f, F) = 0, \quad (f, G) = 0, \quad (g, F) = 0, \quad (g, G) = 1, \quad (F, G) = 0,$$

which are Clebsch's equations; in the case of $2n$ terms, the number of them is

$n+n+\frac{1}{2}(n^2-n)+\frac{1}{2}(n^2-n)+n^2 = 2n+n^2-n+n^2 = n(2n+1)$, or $\frac{1}{2}2n(2n+1)$, as it sh

10.

It may be remarked that we have

$$Fdf + Gdg = Fd\left(f + \frac{G}{F}g\right) - Fg\,d\frac{G}{F},$$

or writing this $= F'df' + G'dg'$, we have

$$f' = f + \frac{G}{F}g, \quad g' = -\frac{g}{F}, \quad F' = F, \quad G' = \frac{G}{F},$$

whence conversely

$$f = f' + G'g', \quad g = -F'g', \quad F = F', \quad G = F'G'.$$

The ten equations $(f) = 0$, $(g) = 0$, &c., ought then to lead to the corresponding ten equations $(f') = 0$, $(g') = 0$, &c., and it is easy to verify that they do so; for instance, we have

$$(f') = (f+G'g, -F'g') = (f, -F'g')+(G'g, -F'g') = F'(f,g)-F'G'(g,g)-g'(f,G')-$$

where $(g,g) = 0$ and thus $(f') = 0$. And again,

$$(f', g') = (f+G'g, g') = (f, g')+(G'g, g') = (f, g')+G'(g, g')+g'(G', g'),$$

where the last term vanishes; the remaining terms are

$$= -\frac{1}{F'}(f,g) + \frac{G'}{F'^2}(f, F') = 0, \text{ that is, } (f', g') = 0.$$

There is, of course, the like transformation

$$Fdf + Gdg = G\left\{\frac{F}{G}df + dg\right\}.$$

11.

I have, for better exhibiting the results, taken $2n = 4$, but the most simple case for an even number of terms is $2n = 2$. Here we have $X dx + Y dy = P dp + Q dq$, and the functions to be considered are

$$12 = \frac{\partial X}{\partial y} - \frac{\partial Y}{\partial x},$$

$$\phi 012 = \phi_0 \cdot 12 + \phi_1 \cdot 20 + \phi_2 \cdot 01 = -F\frac{\partial \phi}{\partial x} + X\frac{\partial \phi}{\partial y} = \frac{\partial(\phi, X)}{\partial(x, y)},$$

$$\phi\psi_{12} = \frac{\partial(\phi, \psi)}{\partial(x, y)}.$$

We have here

$$\frac{\phi 012}{12} = \frac{(\phi 012)'}{(12)'}, \quad \frac{\phi\psi_{12}}{12} = \frac{(\phi\psi_{12})'}{(12)'},$$

and the invariantive properties are easily verified.

12.

Thus

$$12 = \frac{\partial X}{\partial y} - \frac{\partial Y}{\partial x} = \left(\frac{\partial P}{\partial p}\frac{\partial p}{\partial y} + \frac{\partial P}{\partial q}\frac{\partial q}{\partial y}\right) - \left(\frac{\partial Q}{\partial p}\frac{\partial p}{\partial x} + \frac{\partial Q}{\partial q}\frac{\partial q}{\partial x}\right) + \ldots$$

or, writing herein

$$\frac{\partial P}{\partial x} = \frac{\partial P}{\partial p}\frac{\partial p}{\partial x} + \frac{\partial P}{\partial q}\frac{\partial q}{\partial x},$$

and the like values for $\frac{\partial P}{\partial y}$ and for $\frac{\partial Q}{\partial x}$ and $\frac{\partial Q}{\partial y}$, we have

$$12 \cdot \frac{\partial(x, y)}{\partial(p, q)} = (12)'.$$

Similarly, we find

$$\phi 012 = -F\frac{\partial \phi}{\partial x} + X\frac{\partial \phi}{\partial y} = \left(-F\frac{\partial \phi}{\partial p} + P\frac{\partial \phi}{\partial q}\right)\frac{\partial(p, q)}{\partial(x, y)} + \cdots = (\phi 012)' \cdot \frac{\partial(p, q)}{\partial(x, y)},$$

and

$$\frac{\partial(\phi, \psi)}{\partial(x, y)} = \frac{\partial(\phi, \psi)}{\partial(p, q)} \cdot \frac{\partial(p, q)}{\partial(x, y)}.$$

We thus have

$$12 \cdot \partial(x, y) = (12)' \cdot \partial(p, q), \quad \phi 012 \cdot \partial(x, y) = (\phi 012)' \cdot \partial(p, q),$$

or say

$$\frac{\phi 012}{12} = \frac{(\phi 012)'}{(12)'}.$$

The proof is the same in principle for $2n = 4$, or any other even value.

13.

The theory is very similar in the case of an odd number $2n+1$ of terms; thus $2n+1 = 3$, the forms are

$$\phi 0123, \quad \phi 123, \quad \text{and} \quad \phi\psi 0123,$$

the first and second of which are Pfaffians, the third of them co-Pfaffian: the developed expression of this last is

$$\phi\psi_{23} \cdot 01 + \phi\psi_{31} \cdot 02 + \phi\psi_{12} \cdot 03,$$

and to fix the meaning hereof we write

$$\phi\psi_{01} = 0, \quad \phi\psi_{02} = 0, \quad \phi\psi_{03} = 0.$$

Hence, the differential expression being $Xdx + Ydy + Zdz$, we have

$$\begin{aligned}\phi 0123 &= 01 \cdot 23 + 02 \cdot 31 + 03 \cdot 12 \\ &= X\left(\frac{\partial Y}{\partial z} - \frac{\partial Z}{\partial y}\right) + Y\left(\frac{\partial Z}{\partial x} - \frac{\partial X}{\partial z}\right) - Z\left(\frac{\partial X}{\partial y} - \frac{\partial Y}{\partial x}\right),\end{aligned}$$

$$\begin{aligned}\phi 0123 &= 01 \cdot 23 + 02 \cdot 31 + 03 \cdot 12 \\ &= \frac{\partial \phi}{\partial x}\left(\frac{\partial Y}{\partial z} - \frac{\partial Z}{\partial y}\right) + \frac{\partial \phi}{\partial y}\left(\frac{\partial Z}{\partial x} - \frac{\partial X}{\partial z}\right) + \frac{\partial \phi}{\partial z}\left(\frac{\partial X}{\partial y} - \frac{\partial Y}{\partial x}\right),\end{aligned}$$

$$\begin{aligned}\phi\psi 0123 &= \phi\psi_{23} \cdot 01 + \phi\psi_{31} \cdot 02 + \phi\psi_{12} \cdot 03 \\ &= X\frac{\partial(\phi,\psi)}{\partial(y,z)} + Y\frac{\partial(\phi,\psi)}{\partial(z,x)} + Z\frac{\partial(\phi,\psi)}{\partial(x,y)}.\end{aligned}$$

14.

For the transformation

$$Xdx + Ydy + Zdz = Pdp + Qdq + Rdr,$$

we have

$$\phi 0123 \cdot \partial(x, y, z) = (\phi 0123)' \cdot \partial(p, q, r),$$

$$\phi 0123 \cdot \partial(x, y, z) = (\phi 0123)' \cdot \partial(p, q, r),$$

$$\phi\psi 0123 \cdot \partial(x, y, z) = (\phi\psi 0123)' \cdot \partial(p, q, r),$$

and consequently

$$\frac{\phi 0123}{\phi 0123} = \frac{(\phi 0123)'}{(\phi 0123)'},$$

$$\frac{\phi\psi 0123}{\phi 0123} = \frac{(\phi\psi 0123)'}{(\phi 0123)'},$$

so that the left-hand functions are absolute invariants.

15.

If in particular, $Xdx+Ydy+Zdz = df + Gdg$, then we may write

$$P, Q, R = 1, G, 0, \quad p, q, r = f, g, G.$$

Hence

$$01' = 1, \quad 02' = G, \quad 03' = 0; \quad 23' = 1, \quad 31' = 0, \quad 12' = 0,$$

and therefore

$$(\phi)' = \frac{\phi 0123}{\phi 0123} = 1, \quad (\phi, \psi)' = \frac{\phi\psi 0123}{\phi 0123} = \frac{\partial(\phi, \psi)}{\partial(g, G)} + \frac{\partial(\phi, \psi)}{\partial(f, G)} + \frac{\partial(\phi, \psi)}{\partial(f, g)},$$

or say

$$(\phi) = 1, \quad (\phi, \psi) = \frac{\partial(\phi, \psi)}{\partial(g, G)} + \frac{\partial(\phi, \psi)}{\partial(f, G)} + \frac{\partial(\phi, \psi)}{\partial(f, g)},$$

and we thus have

$$\frac{\phi 0123}{\partial(x, y, z)} = (\phi) \cdot \frac{\phi 0123}{\partial(f, g, G)},$$

and then

$$\frac{\phi 0123}{\phi 0123} = \frac{\partial(\phi, \psi)}{\partial(g, G)} + \dots$$

viz. we thus have derivatives (ϕ) and (ϕ, ψ) analogous to (but quite different in form from) those of Clebsch in the case of an even number of terms.

In particular, writing $\phi, \psi = f, g, G$, we obtain

$$(f) = 1, \quad (g) = 0, \quad (G) = 0; \quad (f, g) = 0, \quad (f, G) = -G, \quad (g, G) = 1,$$

which are the analogues of Clebsch's formula.

16.

It is interesting to compare the formula for the two cases

$$Xdx + Ydy + Zdz + Wdw = Fdf + Gdg,$$

and

$$Xdx + Ydy + Zdz = df + Gdg.$$

In the former case f and g are symmetrically related to each other, and we may say that (f = const. and g = const.) is a solution of $Xdx + Ydy + Zdz + Wdw = 0$; we have $(f) = 0$ and $(g) = 0$. In the second case (f = const. and g = const.) is still a solution of $Xdx + Ydy + Zdz = 0$, but f and g are not symmetrically related to each other, and we have $(f) = 1$, $(g) = 0$. Moreover, in the first case $(G) = G$, but in the second case $(G) = 0$, an equation of the same form as $(g) = 0$; the reason is that we have here

$$Xdx + Ydy + Zdz = df + Gdg = d(f + Gg) - gdG,$$

so that, besides the solution (f = const. and g = const.), we have the solution ($f + Gg$ = const. and G = const.).

17.

The remark just made may be further developed: we have

$$Xdx + Ydy + Zdz = df + Gdg = d(f + Gg) - gdG = df' + G'dg',$$

suppose, where $f = f + Gg$, $g' = -\frac{g}{F}$, $G' = G$, and therefore also $f = f' + G'g'$, $G = g'$, $g = -G'$: the equations

$$(f) = 1, \quad (g) = 0, \quad (G) = 0, \quad (f, g) = 0, \quad (f, G) = -G, \quad (g, G) = 1,$$

should lead to

$$(f') = 1, \quad (g') = 0, \quad (G') = 0, \quad (f', g') = 0, \quad (f', G') = -G', \quad (g', G') = 1.$$

There is no difficulty in verifying this; thus the equations $(g) = 0$, $(G) = 0$, give at once $(g') = 0$, $(G') = 0$; and then the equation $(f) = 1$ gives $(f' + G'g') = 1$, that is, $(f') + G'(g') + g'(G', g') + g' = 1$, or $(f') = 1$. So again $(g, G) = 1$ gives $(g, G') = 1$; and then $(f, g) = 0$ gives $(f' + G'g, G') = 0$, that is,

$$(f', G') + G'(g', G') + g'(G', G') = 0, \text{ or } (f', G') = -G'.$$

And finally, $(f, G) = -G$ gives $(f + G'g', g') + g' = 0$, that is,

$$(f', g') + G'(g', g') + g'(G', g') + g' = 0, \text{ or } (f', g') = 0.$$

I stop to give the direct verification of the equations $(f) = 1$, $(g) = 0$, $(G) = 0$. We have

$$X dx + Y dy + Z dz = df + G dg,$$

that is,

$$X = \frac{\partial f}{\partial x} + G\frac{\partial g}{\partial x}, \quad Y = \frac{\partial f}{\partial y} + G\frac{\partial g}{\partial y}, \quad Z = \frac{\partial f}{\partial z} + G\frac{\partial g}{\partial z},$$

[p. 413]

and thence

$$\frac{\partial Y}{\partial z}-\frac{\partial Z}{\partial y}=\frac{\partial G}{\partial z}\frac{\partial g}{\partial y}-\frac{\partial G}{\partial y}\frac{\partial g}{\partial z},$$
$$\frac{\partial Z}{\partial x}-\frac{\partial X}{\partial z}=\frac{\partial G}{\partial x}\frac{\partial g}{\partial z}-\frac{\partial G}{\partial z}\frac{\partial g}{\partial x},$$
$$\frac{\partial X}{\partial y}-\frac{\partial Y}{\partial x}=\frac{\partial G}{\partial y}\frac{\partial g}{\partial x}-\frac{\partial G}{\partial x}\frac{\partial g}{\partial y}.$$

Hence, multiplying first by $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial z}$, that is, by $X-G\frac{\partial g}{\partial x}$, $Y-G\frac{\partial g}{\partial y}$, $Z-G\frac{\partial g}{\partial z}$, and adding, we have

$$\frac{\partial f}{\partial x}\left(\frac{\partial Y}{\partial z}-\frac{\partial Z}{\partial y}\right)+\frac{\partial f}{\partial y}\left(\frac{\partial Z}{\partial x}-\frac{\partial X}{\partial z}\right)+\frac{\partial f}{\partial z}\left(\frac{\partial X}{\partial y}-\frac{\partial Y}{\partial x}\right)=\phi 123=\phi 0123,$$

that is, $\phi 123=\phi 0123$, or $(f)=1$.

And then multiplying secondly by $\frac{\partial g}{\partial x}$, $\frac{\partial g}{\partial y}$, $\frac{\partial g}{\partial z}$ and adding, and thirdly by $\frac{\partial G}{\partial x}$, $\frac{\partial G}{\partial y}$, $\frac{\partial G}{\partial z}$ and adding, we obtain

$$\phi 123=\frac{\partial g}{\partial x}(23)+\frac{\partial g}{\partial y}(31)+\frac{\partial g}{\partial z}(12)=0,\ \text{that is, } (g)=0,$$

and

$$\phi 123=\frac{\partial G}{\partial x}(23)+\frac{\partial G}{\partial y}(31)+\frac{\partial G}{\partial z}(12)=0,\ \text{that is, } (G)=0.$$

To exhibit more clearly the formulae for any odd number of terms, I take $2n+1=5$,

$$X dx+Y dy+Z dz+W dw+T dt=df+G dg+H dh.$$

We have here

$$(\phi)=\frac{\phi 12345}{\phi 012345}=\frac{\partial\phi}{\partial f},$$
$$(\phi,\psi)=\frac{\phi\psi 12345}{\phi 012345}=\frac{\partial(\phi,\psi)}{\partial(g,G)}+\frac{\partial(\phi,\psi)}{\partial(h,H)}+\frac{\partial(\phi,\psi)}{\partial(f,G)}+\frac{\partial(\phi,\psi)}{\partial(f,H)},$$

and in particular,

$$(f)=1;\quad (g)=0,\quad (h)=0;\quad (G)=0,\quad (H)=0;$$
$$(f,g)=0,\quad (f,h)=0;\quad (f,G)=-G,\quad (f,H)=-H;$$
$$(g,G)=1,\quad (g,h)=0,\quad (g,H)=0;\quad (h,H)=1,\quad (h,g)=0,\quad (h,G)=0;$$

in all

$$1+2n+2n+\tfrac{1}{2}(n^2-n)+\tfrac{1}{2}(n^2-n)+n^2=1+4n+n^2-n+n^2=2n^2+3n+1=\tfrac{1}{2}(2n+1)(2$$

equations.

[p. 414]

We can, by what precedes, at once express the conditions which must be satisfied in order that a differential expression $X_1dx_1 + X_2dx_2 + \cdots + X_vdx_v$, may be reducible to one of the special forms df, Fdf, $df + F_1df_1$, &c.; viz. if we have

$$\begin{array}{lll} X_1dx_1 + X_2dx_2 + \cdots + X_vdx_v & = df, & \text{then } 12 = 0, \text{ \&c.} \\ & = Fdf, & \text{“ } \phi 0123 = 0, \text{ \&c.} \\ & = df + F_1df_1, & \text{“ } 1234 = 0, \text{ \&c.} \\ & = Fdf + F_1df_1, & \text{“ } \phi 012345 = 0, \text{ \&c.} \\ & \text{\&c.,} & \text{\&c.,} \end{array}$$

where the numbers 12, 1234, 12345, &c., represent any combinations out of the numbers 1, 2, 3,. . . , v. Of course, if v is not sufficiently large to furnish such a combination, then there is no condition to be satisfied; thus if $X_2dx_2 + X_3dx_3 = df + F_1df_1$, there is no condition to be satisfied.

0.3 Note on Lacunary Functions. [945]

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The present note is founded upon Poincaré's paper "Sur les fonctions à espaces lacunaires," *Amer. Math. Jour.*, t. XIV. (1892), pp. 201—221.

If the complex variable $z = x + iy$ is represented as usual by a point the coordinates of which are (x, y), and if U_0, U_1, U_2,... denote an infinite series of given functions of z, then the equation

$$fz = U_0 + U_1 + U_2 + \dots$$

defines a function of z, but only for those values of z for which the series is convergent, or say for points within a certain region $\mathfrak{R}$; and within this region, it defines the successive derived functions $f'z$, $f''z$, $f'''z$, ...

Taking $l = h + ik$, as an increment of z, we define the function of $z + l$, by the equation

$$f(z + l) = U_0 + U_1 + U_2 + \dots$$

but only for values of l for which the series is convergent: it may very well be, and it is in general the case, that we thereby extend the definition of fz so as to make it applicable to points within a larger region $\mathfrak{R}_1$; and then considering fz as defined within this larger region $\mathfrak{R}_1$, we may pass from it to a still larger region $\mathfrak{R}_2$; and so on indefinitely, or until we cover the whole infinite plane.

For instance, the equation

$$fz = 1 + z + z^2 + z^3 + \dots$$

[p. 416]

defines the function $1\div(1-z)$ for values of z for which mod. $z<1$, that is, for points within the circle $x^2+y^2-1=0$; and this being so,

$$f(z+l)=1+(z+l)+(z+l)^2+\cdots=\frac{1}{1-z-l}=\frac{1}{1-z}\cdot\frac{1}{1-\frac{l}{1-z}}$$

extends the definition to the larger region for which this is a convergent series: the condition of convergency is

$$\text{mod.}\frac{l}{1-z}<1,\ \text{that is, mod.}\frac{h+ik}{1-x-iy}<1,\ \text{or } h^2+k^2<(1-x)^2+y^2.$$

The condition is that the distance $\sqrt{(h^2+k^2)}$ must not exceed the distance $\sqrt{\{(1-x)^2+y^2\}}$ of the point $z=x+iy$, from the point $(x=1,y=0)$; the point z is strictly within the circle $x^2+y^2=1$, but taking it on the circumference, the condition is that the point $z+l$ must lie within a circle having its centre on the circle $x^2+y^2=1$ and passing through the point $(x=1,y=0)$. Taking $\cos\theta$ and $\sin\theta$ for the coordinates of the centre, the equation of this circle is

$$(x-\cos\theta)^2+(y-\sin\theta)^2=(1-\cos\theta)^2+\sin^2\theta,$$

that is,

$$x^2+y^2-1-2\cos\theta(x-1)-2y\sin\theta=0;$$

and the envelope of these circles is

$$(x^2+y^2-1)^2-4(x-1)^2-4y^2=0;$$

or, as this may be written,

$$(x^2+y^2)^2-6(x^2+y^2)+8x-3=0,$$

or again in the form

$$(x^2+y^2-3)^2+4(2x-1)=0,$$

or in the form

$$y^4+2y^2(x^2-3)+(x-1)^2(x+3)=0.$$

The curve is a *cuspidal Cartesian*. To put this in evidence, observe that the equation may be written

$$-4y^2+\{(x-1)^2+2y^2\}\{(x-1)^2+y^2+4x-4\}=0,$$

viz. writing

$$A = x + iy - 1, \quad B = x - iy - 1, \quad Z = -1,$$

then

$$A - B = 2iy, \quad AB = (x-1)^2 + y^2,$$

or the equation is

$$Z^2(A-B)^2 + AB\{AB - 2Z(A+B)\} = 0,$$

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that is,

$$Z^2B^2 + A^2B^2 - 2Z^2AB - 2ZAZB - 2ZAB^2 = 0,$$

which is the equation of a *bicuspidal quartic* curve, having for cusps the vertices of the triangle $A = 0$, $B = 0$, $Z = 0$.

The region within which the function is $= \frac{1}{1-z}$ is thus extended to the area within the Cartesian curve, say this is the region $\mathfrak{R}_1$; starting from this curve instead of the circle (viz. by considering the envelope of the circle having its centre on the curve and passing through the point $x = 1$, $y = 0$), we obtain a second curve, a closed curve, which instead of having a cusp on the axis of x cuts this axis at right angles at a point the distance of which from the origin is greater than 1; and we thus extend the region to the area within this second curve, say this is the region $\mathfrak{R}_2$. And proceeding in this way, we ultimately extend the region to the whole of the infinite plane.

But the functions U_0, U_1, U_2,... may be such that for every value whatever of l, for which the point $z + l$ is outside the region $\mathfrak{R}$, the series

$$f(z + l) = U_0 + U_1 + U_2 + \dots$$

is divergent, and we are in this case unable to define the function fz for points outside the region $\mathfrak{R}$; the function then exists only for points inside the region $\mathfrak{R}$, and for points outside this region it is non-existent; a function such as this, existing only for points within a certain region and not for the whole of the infinite plane, is said to be a *lacunary function.*

0.4 Note on the Theory of Orthomorphosis. [946]

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXVI. (1893), pp. 282—288.]

The equation of any given curve whatever, $\Theta = 0$, may be expressed in the form

$$\phi(x+iy) + \phi(x-iy) = 0.$$

Let χ be any odd function; then since

$$\chi\phi(x-iy) = -\chi\phi(x+iy),$$

we have

$$\chi\phi(x-iy) = \chi\{-\phi(x+iy)\},$$

that is,

$$\chi\phi(x+iy) + \chi\phi(x-iy) = 0.$$

Assuming that Θ is a real function, that is, a function with real coefficients, then also $\phi(x+iy)$ will be a function with real coefficients, or say a real function of $x+iy$; the function χ may be real or imaginary, but if imaginary, then the i of the coefficients does not change its sign in the passage from $\chi\phi(x+iy)$ to $\chi\phi(x-iy)$.

In proof of the assumed theorem, imagine the equation $\Theta = 0$ expressed as an equation between $x+iy$ and $x-iy$, or, supposing it solved in regard to $x-iy$, take the form of it to be $x-iy = f(x+iy)$: let u_n be a function of n satisfying the equation of differences $u_{n+1} = fu_n$; and let $\phi(x+iy)$ be determined as a function of $x+iy$ by the elimination of n from the equations

$$x+iy = u_n, \quad \phi(x+iy) = \cos n\pi;$$

we thence have

$$x-iy = fu_n = u_{n+1},$$

and consequently

$$\phi(x-iy) = \cos(n+1)\pi,$$

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that is,

$$\phi(x+iy)+\phi(x-iy)=0,$$

viz. this equation is a transformation of the equation $\Theta=0$, and thus it appears that the equation $\Theta=0$ can always be thrown into the last-mentioned form.

As an example, take the equation $y=ax+b$: which, putting for a moment $\xi=x+iy$, $\eta=x-iy$, is

$$\frac{\xi-\eta}{2i}=a\frac{\xi+\eta}{2}+b,$$

that is,

$$\xi\frac{1+ai}{1-ai}=\eta+\frac{2b}{1-ai},$$

we have therefore

$$u_{n+1}=\frac{1+ai}{1-ai}u_n+\frac{2b}{1-ai},$$

a solution of which is

$$u_n=\left(\frac{1+ai}{1-ai}\right)^n\cdot\frac{b}{a}-\frac{b}{a},$$

putting this $=\xi$, we have

$$\eta=\frac{1+ai}{1-ai}\xi+\frac{2b}{1-ai},$$

and thence

$$\phi\xi=\cos\frac{\pi}{i}\log\left(\frac{a}{b}\xi+1\right)=\cos\frac{\pi}{i}\log\left(\frac{a\xi+b}{b}\right),$$

where observe that, writing $a+i=Re^{i\alpha}$ and therefore $a-i=Re^{-i\alpha}$, we have

$$\cos\alpha=\frac{a}{\sqrt{a^2+1}},\quad \sin\alpha=\frac{1}{\sqrt{a^2+1}},\quad \text{or say }\cot\alpha=a,$$

and then

$$\frac{b}{a}=\frac{b}{\cot\alpha}=b\tan\alpha,\quad \frac{a}{b}=\frac{\cot\alpha}{b},$$

whence

$$\phi\xi=\cos\frac{\pi}{\alpha}\log\left(\frac{\xi\cot\alpha}{b\tan\alpha}+1\right)=\cos\frac{\pi}{\alpha}\log\left(\frac{\xi\cos^2\alpha}{b\sin\alpha}+1\right),$$

a real function of ξ.

In verification of the equation $\phi\xi+\phi\eta=0$, we have

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where

$$h = \frac{b}{a} = \frac{2b}{i-a} \cdot \frac{i}{i+a},$$

and thence

$$\begin{aligned}\phi\xi &= \cos\frac{\pi}{i}\left\{\log\frac{\xi+h}{h} + \log\frac{h}{h}\right\}\\ &= \cos\frac{\pi}{i}\left\{\log\frac{x+iy+h}{h} + \log\frac{h}{h}\right\}\\ &= \cos\frac{\pi}{i}\log\left(\frac{x+iy+h}{h}\right),\end{aligned}$$

$$\phi\eta = \cos\frac{\pi}{i}\log\left(\frac{x-iy+h}{h}\right) = \cos\left\{-\frac{\pi}{i}\log\left(\frac{x+iy+h}{h}\right)\right\},$$

that is, $\phi\eta = -\phi\xi$, or $\phi\xi + \phi\eta = 0$, the equation in question.

I remark, in passing, that the same equation $y = ax + b$ might have been put in the form $\phi x + \phi y = 0$, viz. assuming

$$\phi x = \cos\frac{\pi}{\log a}\log\left(x - \frac{b}{1-a}\right),$$

then

$$\phi y = \cos\frac{\pi}{\log a}\log\left(ax + b - \frac{b}{1-a}\right) = \cos\frac{\pi}{\log a}\log\left\{a\left(x - \frac{b}{1-a}\right)\right\},$$

$$= \cos\left\{\frac{\pi}{\log a}\log a + \frac{\pi}{\log a}\log\left(x - \frac{b}{1-a}\right)\right\} = \cos\left\{\pi + \frac{\pi}{\log a}\log\left(x - \frac{b}{1-a}\right)\right\},$$

$$= -\cos\frac{\pi}{\log a}\log\left(x - \frac{b}{1-a}\right) = -\phi x,$$

that is, $\phi x + \phi y = 0$.

If $b = 0$, then

$$\phi(x+iy) = \cos\frac{\pi}{i}\log\frac{x+iy}{h},$$

where h is arbitrary; the equation $\phi(x+iy) + \phi(x-iy) = 0$ is satisfied identically for h real.

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We have in this case

$$\phi(x+iy) = \cos\frac{\pi}{i}\log(x+iy) - \cos\frac{\pi}{i}\log h,$$

and writing

$$\cos\frac{\pi}{i}\log h = C, \quad (C \text{ an arbitrary real constant}),$$

the equation becomes

$$\cos\frac{\pi}{i}\log(x+iy) - C = 0.$$

In the case $a = \infty$, that is, the line $x = \text{const.}$, or say $x = 0$, we have

$$\phi(x+iy) = \cos\frac{\pi}{i}\log(x+iy),$$

and the equation is

$$\cos\frac{\pi}{i}\log(x+iy) + \cos\frac{\pi}{i}\log(x-iy) = 0,$$

which may be written

$$\cos\frac{\pi}{i}\log(x^2+y^2) = 0, \quad \text{or } \log(x^2+y^2) = (2n+1)i, \quad (n \text{ integer}),$$

that is, $x^2 + y^2 = e^{-(2n+1)\pi}$, a series of concentric circles.

The general theory of orthomorphosis consists in the transformation of curves by the substitution

$$x+iy = f(u+iv), \quad x-iy = f(u-iv),$$

or as we may write it

$$z = fw, \quad \bar{z} = f\bar{w},$$

where f is an arbitrary function; the curves $\Theta(x,y) = 0$ and $\Theta(u,v) = 0$ are then orthomorphic images of each other.

The particular case where f is a linear function gives rise to the theory of similarity; and when f is fractional linear, to the theory of inversion.

In the general case, starting from any curve, we derive a family of orthomorphic curves by varying the function f; and conversely, any two orthomorphic curves are connected by a relation of the above form.

The theory has important applications in conformal representation and in the theory of functions of a complex variable; in particular, it connects the theory of algebraic curves with that of transcendental functions.

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I remark that the equation of any curve may also be exhibited in the form

$$\Phi(x+iy)\cdot\Psi(x-iy)=1,$$

or in the more general form

$$F\{\phi(x+iy),\ \psi(x-iy)\}=0.$$

The connexion between the two forms is easily established: if $\phi(x+iy)+\phi(x-iy)=0$, then writing $\Phi=e^{\phi}$, $\Psi=e^{-\phi}$, we have $\Phi(x+iy)\cdot\Psi(x-iy)=1$.

Conversely, from $\Phi(x+iy)\cdot\Psi(x-iy)=1$, taking logarithms, we obtain the additive form.

The theory of orthomorphosis is closely connected with that of the transformation of elliptic functions, and with the modular equations which arise therein.

If we consider the elliptic integral

$$u=\int_0^x\frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}},$$

then the modulus k and the complementary modulus k' are connected by the relation $k^2+k'^2=1$; and the transformation of the second order gives

$$(1+k)(1+k_1)=2,$$

where k_1 is the transformed modulus.

The general transformation of order n leads to a modular equation between k and λ, say $F_n(k,\lambda)=0$; and this equation is of the form which admits of the orthomorphic representation.

In fact, writing $k=\frac{1}{\sqrt{2}}\cdot\frac{x+iy}{\sqrt{x^2+y^2+1}}$, or otherwise introducing the complex variable, the modular equation may be exhibited as the orthomorphic image of a simpler curve.

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The application to the multiplication of elliptic functions is immediate: if $\operatorname{sn} u = x$, then for integer values of n,

$$\operatorname{sn}(nu) = x \cdot \frac{P_n(x^2)}{Q_n(x^2)},$$

where P_n and Q_n are polynomials of degree $\frac{1}{2}(n^2 - 1)$ or $\frac{1}{2}(n^2 - 2)$ according as n is odd or even.

The equation connecting $x = \operatorname{sn} u$ and $y = \operatorname{sn}(nu)$ is an algebraic one, say $\Phi_n(x, y) = 0$, and this equation admits of orthomorphic representation.

Similarly, the transformation of order n connects the moduli k and λ by an equation $F_n(k, \lambda) = 0$, which is the orthomorphic image of a rational curve.

I give in conclusion a table of the simplest orthomorphic curves and their equations in the form $\phi(x + iy) + \phi(x - iy) = 0$:

Curve	**Equation**
Straight line $y = ax + b$	$\cos \frac{\pi}{i} \log \left(\frac{x + iy + h}{h} \right) = C$
Circle $x^2 + y^2 = c^2$	$\cos \frac{\pi}{i} \log(x^2 + y^2) = 0$
Parabola $y^2 = 4ax$	Elliptic functions
Ellipse, hyperbola	Jacobian elliptic functions
Cubic curve	Modular functions

The investigation of the orthomorphic images of algebraic curves in general presents problems of great interest and difficulty, which I hope to consider on a future occasion.

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It remains to mention a remarkable property of the orthomorphic transformation in connexion with the theory of functions.

If $z = x+iy$ and $w = u+iv$ be connected by the equation $z = fw$, where f is any function, then the curves in the z-plane corresponding to the lines $u =$ const., $v =$ const. in the w-plane form an orthogonal system; and the transformation preserves angles, whence the name *orthomorphosis.*

Conversely, if two families of curves in the z-plane cut each other at right angles, they may be regarded as the orthomorphic images of two systems of parallel straight lines; and the complex variable $w = u + iv$ may be so determined that $u =$ const., $v =$ const. shall represent the two families respectively.

This is the fundamental theorem of conformal representation, due to Riemann: any simply connected plane region bounded by a closed curve may be conformally represented upon the interior of a circle, or upon a half-plane.

The representation is effected by means of the equation $z = fw$, where f is a function determined by the boundary conditions; and the orthomorphic form $\phi(x+iy)+\phi(x-iy) = 0$ serves to exhibit the equation of any curve as a condition upon the complex variable.

The foregoing considerations may be extended to surfaces: if X, Y, Z are the coordinates of a point on a surface, and if the linear element is

$$ds^2 = dX^2 + dY^2 + dZ^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2,$$

then the condition that the parametric curves $u =$ const., $v =$ const. be orthogonal is $F = 0$; and the surface is conformally represented on the (u, v)-plane when $E = G$, $F = 0$.

The general theory of the conformal representation of surfaces upon each other, and the particular case of the representation of a surface upon a plane, are of fundamental importance in many branches of mathematical physics, especially in the theory of potential and in hydrodynamics.

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I conclude with some remarks on the bearing of the theory of orthomorphosis upon the geometrical interpretation of complex functions.

When a function $\phi(x+iy)$ is given, the equation

$$\phi(x+iy)+\phi(x-iy)=0$$

represents a curve; and the nature of this curve depends upon the character of the function ϕ.

If ϕ be algebraic, the curve is algebraic; if ϕ be transcendental, the curve is transcendental.

Conversely, any given curve determines the function ϕ (to a constant factor pres), and thus the theory of curves is coextensive with that of complex functions of a particular type.

The further development of these ideas belongs rather to the theory of functions than to the present note; but it seemed desirable to point out the connexion between the two theories, and to indicate the field of investigation which is thus opened up.

Postscript. — Since the above was written, I have found that certain of the formulae admit of simplification by the introduction of the modular functions; but the investigation of this part of the subject must be reserved for a future communication.

A. C.

[End of Papers 943–946, Vol. XIII, pp. 401–450]

The Collected Mathematical Papers of Arthur Cayley, Vol. XIII

Pages 451–500

Transcription

947.

ON A SYSTEM OF TWO TETRADS OF CIRCLES; AND OTHER SYSTEMS OF TWO TETRADS.

[From the Proceedings of the Cambridge Philosophical Society, vol. VIII. (1893), pp. 54—59.]

The investigations of the present paper were suggested to me by Mr Orr's paper, "The Contacts of certain Systems of Circles," Proc. Camb. Phil. Soc., vol. vii.

1. It is possible to find in piano two tetrads of circles, or say four red circles and four blue circles, such that each red circle touches each blue circle: in fact, counting the constants, a circle depends upon 3 constants, or say it has a capacity $= 3$; the capacity of the eight circles is thus $= 24$; and the postulation or number of conditions to be satisfied is $= 16$: the resulting capacity of the system is thus , $16 - 24 = 8$. It will, however, appear that, in the system considered, the true value is $= 9$.

2. The value of the capacity being $= 8$, we are not at liberty to assume at pleasure three circles of the system. And, in fact, assuming at pleasure say 3 red circles, then touching each of these we have 8 circles, forming $\frac{1}{24}\,8\cdot 7\cdot 6\cdot 5 = 70$, tetrads of circles: taking at random any one of these tetrads for the blue circles, the remaining red circle has to be determined so as to touch each of the four blue circles, that is, by four instead of three conditions; and there is not in general any red circle satisfying these four conditions. But the 8 tangent circles do not stand to each other in a relation of symmetry, but form in fact four pairs of circles; and it is possible out of the 70 tetrads to select (and that in 6 ways) a tetrad of blue circles, such that there exists a fourth red circle touching each of these four blue circles. We have thus a system depending upon 3 arbitrary circles, and for which, therefore, the capacity is $= 9$. It is (as is known) possible, in quite a different manner, out of the 70 tetrads to select (and that in 8 ways) a tetrad of blue circles such that there exists a fourth red circle touching each of these four blue circles—but the present paper relates exclusively to the first-mentioned 6 tetrads and not to these 8 tetrads.

3. I consider, in the first instance, a particular case in which the three red circles are not all of them arbitrary, but have a capacity $9 - 1 = 8$; and pass from this to the general case where the capacity is $= 9$. Calling the red circles 1, 2, 3 and 4; I start with the circles 1 and 2 arbitrary, and 3 a circle equal to 2: the radical axis, or common chord, of the circles 2 and 3 is thus a line bisecting at right angles the line joining the centres of the circles 2 and 3, say this is the line Ω. We have then four circles, each having its centre in the line Ω and touching the circles 1 and 2: in fact, the locus of the centre of a circle touching the circles 1 and 2 is a pair of conics, each of them having for foci the centres of these circles: the line Ω meets each of these conics in two points, and there are thus on the line Ω four points, each of them the centre of a circle touching the circles 1 and 2. But the equal circles 2 and 3 are symmetrically situate in regard to the line Ω; and it is obvious that the four circles, having their centres on the line Ω, will each of them also touch the circle 3; we have thus the four blue circles, each of them with its centre on the line Ω and touching each of the red circles 1, 2 and 3. And it is moreover clear that, taking the red circle 4 equal to 1 and situate symmetrically therewith in regard to the line Ω, then this circle 4 will touch each of the blue circles: so that we have here the four blue circles, each of them touching the four red circles. As already mentioned, the blue circles have their centre on the line Ω, that is, the line Ω is a common orthotomic of the four blue circles.

4. By inverting in regard to an arbitrary circle we pass to the general case; the line Ω becomes thus a circle Ω, orthotomic to each of the blue circles. Starting *ab initio*, we have here at pleasure the red circles 1, 2, 3: the circle Ω is a circle having for centre a centre of symmetry of the circles 2 and 3, and passing through the points of intersection (real or imaginary) of these two circles; the circles 2 and 3 are thus the inverses (or say the images) each of the other in regard to the circle Ω. We can then find 4 circles each of them orthotomic to Ω, and touching the circles 1 and 2: but a circle orthotomic to Ω is its own inverse or image in regard to Ω; and it will thus touch the circle 3 which is the image of 2 in regard to Ω. We have thus the four blue circles each of them touching the red circles 1, 2 and 3; and then, taking the red circle 4 as the inverse or image of 1 in regard to Ω, this circle 4 will also touch each of the blue circles. Thus starting with the arbitrary red circles 1, 2, 3, we find the four blue circles and the remaining red circle 4, such that each of the blue circles touches each of the red circles. Since in the construction we group together at pleasure the two circles 2, 3 (out of the three circles 1, 2, 3) and use at pleasure either of the two centres of symmetry, it appears that the number of ways in which the figure might have been completed is $= 6$.

5. The blue circles have a common orthotomic circle Ω, that is, the radical axis or common chord of each two of the blue circles passes through one and the same point, the centre of the circle Ω. The figure is symmetrical in regard to the red and blue circles respectively, and thus the red circles have a common orthotomic circle Ω', that is, the radical axis or common chord of each two of the red circles passes through one and the same point, the centre of the circle Ω'.

6. Projecting stereographically on a spherical surface, the four red circles and the four blue circles become circles of the sphere; and then making the general homographic transformation, they become plane sections of a quadric surface; we have thus the theorem that on a given quadric surface it is possible to find four red sections and four blue sections such that each blue section touches each red section; and moreover the capacity of the system is $= 9$; viz. 3 of the red sections may be assumed at pleasure. But (as is well known) the theory of the tangency of plane sections of a quadric surface is far more simple than that of the tangency of circles: the condition in order that two sections may touch each other is simply the condition that the line of intersection of the two planes shall touch the quadric surface. And we construct, as follows, the sections touching each of three given sections: say the given sections are 1, 2, 3; through the sections 1 and 2 we have two quadric cones having for vertices say the points d_{12} and i_{12} (direct and inverse centres of the two sections): similarly, through the sections 1 and 3 we have two quadric cones vertices d_{13} and i_{13} respectively, and through the sections 2 and 3 we have two quadric cones vertices d_{23} and i_{23} respectively; the points d_{12}, i_{12}, d_{13}, i_{13}, d_{23}, i_{23} lie three and three in four intersecting lines or axes, viz. these are $d_{12}d_{13}d_{23}$, $d_{12}i_{13}i_{23}$, $d_{13}i_{12}i_{23}$, $d_{23}i_{12}i_{13}$ respectively. Through any one of these axes, say $d_{12}d_{13}d_{23}$, we may draw to the quadric surface two tangent planes each touching the three cones which have their vertices in the points d_{23}, d_{31}, d_{12} respectively; and the section by either tangent plane is thus a section touching each of the three given sections 1, 2, 3; we have thus the eight tangent sections of these three sections.

7. Taking as three of the red sections the arbitrary sections 1, 2, 3; and grouping together two at pleasure of these sections, say 2 and 3; we may take for the blue sections the two sections through the axis $d_{23}d_{31}d_{12}$, and those through the axis $d_{23}i_{31}i_{12}$; we have thus the four blue sections touching each of the given red sections 1, 2, 3; and this being so, there exists a remaining red section 4 touching each of the blue sections; we have thus the four blue sections touching each of the red sections 1, 2, 3 and 4. This implies that the vertices or points d_{24} and d_{34} lie on the axis $d_{12}d_{13}d_{23}$, and that the vertices or points i_{24} and i_{34} lie on the axis $d_{23}i_{24}i_{13}$; or, what is the same thing, that the four sections 1, 2, 3, 4 have in common an axis $d_{12}d_{13}d_{14}d_{23}$ and also an axis $d_{23}i_{24}i_{13}i_{34}$.

8. If the quadric surface be a flat surface (surface aplatie) or conic, then the red sections become chords of the conic; the axes are lines in the plane of the conic, and thus the tangent planes through an axis each coincide with the plane of the conic, and it would at first sight appear that any theorem as to tangency becomes nugatory. But this is not so; comparing with the last preceding paragraph, we still have the theorem: on a given conic, taking at pleasure any three chords 1, 2, 3, it is possible to find a fourth chord 4, such that the four chords have in common an axis $d_{12}d_{13}d_{14}d_{23}$ and also an axis $d_{23}i_{24}i_{13}i_{34}$. And the analytical theory (although somewhat complex) is extremely interesting. Considering the conic $xz - y^2 = 0$, the coordinates of a point on the conic are given by $x : y : z = 1 : \theta : \theta^2$, or say any point of the conic is determined by its parameter θ; and this being so, considering any three chords 1, 2, 3, I take for the two extremities of 1 the values ϵ, ζ; for those of 2 the values α, β; and for those of 3 the values γ, δ; the remaining chord 4 is to be determined as above, and I take for its two extremities the values E, Z.

9. Starting with the chords 1, 2, 3, we have each of the points d_{23}, &c., as the intersection of two lines, viz. these are

$$\begin{aligned}
&\alpha z - y(\alpha + \delta) + z = 0, \quad \alpha z - y(\alpha + \gamma) + z = 0, \quad \text{for } d_{23}, i_{23},\\
&\alpha z - y(\alpha + \zeta) + z = 0, \quad \alpha z - y(\alpha + \epsilon) + z = 0, \quad \text{for } d_{12}, i_{12},\\
&\gamma z - y(\gamma + \epsilon) + z = 0, \quad \gamma z - y(\gamma + \zeta) + z = 0, \quad \text{for } d_{31}, i_{31},
\end{aligned}$$

and we thence find without difficulty for the axis $d_{23}d_{31}d_{12}$ the equation

$$\begin{aligned}
&a\{\alpha\beta(\gamma\delta-\zeta\epsilon)+\gamma\delta(\zeta\alpha-\epsilon\beta)+\zeta\epsilon(\beta\gamma-\alpha\delta)\}\\
&\quad+y\{(\beta-\alpha)(\gamma\delta-\zeta\epsilon)+(\delta-\gamma)(\zeta\alpha-\epsilon\beta)+(\zeta-\epsilon)(\alpha\beta-\gamma\delta)\}\\
&\qquad\qquad\quad+z\{\qquad -(\gamma\delta-\zeta\epsilon)-(\zeta\alpha-\epsilon\beta)-(\alpha\beta-\gamma\delta)\}=0;
\end{aligned}$$

the equation of the axis $d_{23}i_{12}i_{13}$ is obtained herefrom by the interchange of ϵ and ζ.

10. The points d_{24} and i_{34} will lie upon the first-mentioned axis if only d_{24} lies upon this axis, viz. if we have

$$\begin{aligned}
&\tfrac{1}{2}\{(\beta-\alpha)(\gamma\delta-\zeta\epsilon)+(\delta-\gamma)(\zeta\alpha-\epsilon\beta)+(\zeta-\epsilon)(\alpha\beta-\gamma\delta)\}\,(\epsilon E-\zeta)\\
&+\{\qquad -(\gamma\delta-\zeta\epsilon)-(\zeta\alpha-\epsilon\beta)-(\beta\gamma-\alpha\delta)\}\ \ldots\ldots\ldots\ldots\ldots\ldots
\end{aligned}$$

Reducing this equation, the factor $\alpha-\beta$ divides out, and we finally obtain

$$(\beta-\delta)(\alpha\gamma-\epsilon E)-(\alpha\delta-\beta\gamma)(\zeta-E)+(\alpha-\gamma)(\beta\epsilon-\delta\zeta)=0,$$

say this is

$$A+BE+CZ+DEZ=0,$$

where

$$\begin{aligned}
A&=,\\
B&=-(\gamma-\alpha)\beta\delta\cdot-(\alpha\beta-\gamma\delta)\epsilon+(\beta\delta)\zeta,\\
C&=-(\beta-\delta)\alpha\gamma+(\alpha\beta-\gamma\delta)\epsilon\cdot+(\gamma-\alpha),\\
D&=-(\alpha\delta-\beta\gamma)-(\beta-\delta)\epsilon-(\gamma-\alpha)\zeta;
\end{aligned}$$

viz. this is the condition for the existence of the axis $d_{23}d_{13}d_{12}d_{24}d_{34}$.

We interchange herein ϵ, ζ and also E, Z, and we thus obtain

$$A'+C'E+B'Z+D'EZ=0,$$

where

$$\begin{aligned}
A'&=\quad(\zeta\delta-\alpha)\beta\gamma+(\gamma-\alpha)\zeta\delta\epsilon+(\alpha\delta-\beta\gamma)\zeta,\\
B'&=(\alpha\zeta-\beta\gamma)\epsilon\cdot+(\zeta-\delta),\\
C'&=\cdot-(\alpha\beta-\gamma\delta)\zeta+(\gamma-\alpha),\\
D'&=;
\end{aligned}$$

viz. this is the condition for the existence of the axis $d_{23}i_{13}i_{12}i_{34}i_{24}$.

11. I remark that we have

$$A' = A+a(\epsilon-\zeta),\quad B' = B+b(\epsilon-\zeta),\quad C' = C+c(\epsilon-\zeta),\quad D' = D+d(\epsilon-\zeta),$$

where

$$\begin{aligned} a &= (\beta-\delta)\alpha\gamma - (\gamma-\alpha)\beta\delta = \alpha\beta(\gamma+\delta) - \gamma\delta(\alpha+\beta), \\ b &= c = \gamma\delta - \alpha\beta, \\ d &= \alpha+\beta-\gamma-\delta, \end{aligned}$$

and further that

say this is Π.

12. It thus appears that, for the determination of E, Z, we have

$$\begin{aligned} A + BE + CZ + DEZ = 0, \\ A' + C'E + B'Z + D'EZ = 0. \end{aligned}$$

Eliminating Z, we find

$$\begin{vmatrix} A+BE & C+DE \\ A'+C'E & B'+D'E \end{vmatrix} = 0,$$

that is,

$$(AC' - A'C) + (AD' - A'D + BB' - CC')\,E + (BD' - B'D)\,E^2 = 0;$$

upon reducing the coefficients of this equation it appears that they contain each of them the factor Π, and throwing out this factor, the equation is

$$\epsilon\{\alpha\beta(\gamma-\delta) + \gamma\delta(\beta-\alpha) + (\alpha\delta-\beta\gamma)\} + \; E = 0,$$

this contains obviously the factor $E-\epsilon$, or throwing out this factor, we have for E the simple equation

$$E =$$

In a similar manner it may be shown that the two equations give for Z the like simple equation

$$Z =$$

viz. starting from the chords 1, 2, 3 which depend on the parameters (ϵ,ζ), (α,β), (γ,δ) respectively, these last two equations give the parameters (E,Z) of the chord 4.

948.

REPORT OF A COMMITTEE APPOINTED FOR THE PURPOSE OF CARRYING ON THE TABLES CONNECTED WITH THE PELLIAN EQUATION FROM THE POINT WHERE THE WORK WAS LEFT BY DEGEN IN 1817.

[From the British Association Report, (1893), pp. 73—120.]

We have, on the Pellian Equation, Degen's tables, the title of which is "Canon Pellianus sive Tabula simplicissimam aequationis celebratissimae $y^2 = ax^2 + 1$ solutionem pro singulis numeri dati valoribus ab 1 usque ad 1000 in numeris rationalibus iisdemque integris exhibens." Autore Carolo Ferdinando Degen. Hafniae, apud Gerardum Bonnierum, MDCCCXVII., 8vo. Introductio, pp. v—xxiv. Tabula I. Solutionem aequationis $y^2 - ax^2 - 1 = 0$ exhibens, pp. 3—106. Tabula II. Solutionem aequationis $y^2 - ax^2 + 1 = 0$, quotiescunque valor ipsius a talem admiserit, exhibens, pp. 109—112.

The mode of calculation is explained in the Introduction, and illustrated by the examples of the numbers 209, 173.

As to the first of these, the entry in Table I. is

$$\begin{array}{ccc}
14, & 2,\ 5,\ 3, & (2) \\
1, & 13,\ 5,\ 8, & 11 \\
209 & & \\
 & 3220 & \\
 & & 46551
\end{array}$$

where the first line gives the expression of $\sqrt{209}$ as a continued fraction, viz. we have

$$\sqrt{209} = 14 + \cfrac{1}{2 + \cfrac{1}{5 + \cfrac{1}{3 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{2 + \cfrac{1}{28 + \cdots}}}}}}}}$$

the denominators being 2, 5, 3, (2), 3, 5, 2, then 28, which is the double of the integer part 14, and then again 2, 5, 3, (2), 3, 5, 2, and so on, the parentheses of the (2) being used to indicate that this is the middle term of the period.

The second row gives auxiliary numbers occurring in the calculation of the first row and having a meaning, as will presently appear. Observe that the 11 which comes under the (2) should also be printed in parentheses (11), but this is not done.

The process for the calculation of the x, y is as follows:

209			
14			
1	0	+1	
2	14	1	−13
5	29	2	+5
3	159	11	−8
(2)	506	35	+(11)
3	1171	81	−8
5	4019	278	+5
2	21266	1471	−13
28	46551	3220	+1

viz. writing down as a first column the numbers of the first row, and beginning the second column with 1, 14 (14 the number at the head of the first column), and the third column with 0, 1, we calculate the numbers of the second column, $29 = 2 \cdot 14 + 1$, $159 = 5 \cdot 29 + 14$, $506 = 3 \cdot 159 + 29$, &c., and the numbers of the third column in like manner, $2 = 2 \cdot 1 + 0$, $11 = 5 \cdot 2 + 1$, $35 = 3 \cdot 11 + 2$, &c.; and then writing down as a fourth column the numbers of the second row with the signs $+$, $-$ alternately, we have a series of equations $y^2 - ax^2 = \pm A$, viz.

$$1^2 - 209 \cdot 0^2 = +1,$$
$$14^2 - 209 \cdot 1^2 = -13,$$
$$29^2 - 209 \cdot 2^2 = +5,$$

the last of them being

$$(46551)^2 - 209 \cdot (3220)^2 = +1,$$

this last corresponding as above to the value $+1$, and the numbers 46551 and 3220 being accordingly the y and x given in the fourth and third rows of the table.

As to the second of the foregoing numbers, 173, the only difference is that the period has a double middle term, viz. the entry in the Table I. is

$$\begin{array}{cccc} 13, & 6, & (\ 1, & 1) \\ 1, & 4, & (13, & 13) \\ 190060 & & & \\ 2499849 & & & \end{array}$$

The first row gives the expression of $\sqrt{173}$, viz. that is

$$\sqrt{173} = 13 + \cfrac{1}{6 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{6 + \cfrac{1}{26 + \cdots}}}}}$$

the denominators being 6, 1, 1, 6, then 26 (the double of the integer part 13), and then again 6, 1, 1, 6, and so on. In the second row I remark that Degen prints the parentheses (13, 13) for the double middle term.

The process for the calculation of the x, y is similar to that in the former case, viz. we have

173			
13			
1	0	+1	
6	13	1	−4
1	79	6	+13
1	92	7	−13
6	171	13	+4
26	1118	85	−1

where the second and third columns begin 1, 13 and 0, 1 respectively, and the remaining terms are calculated $79 = 6 \cdot 13 + 1$, $92 = 1 \cdot 79 + 13$, &c., and $6 = 6 \cdot 1 + 0$, $7 = 1 \cdot 6 + 1$, &c.; and then writing down as a fourth column the terms of the second row with the signs $+$, $-$ alternately, we have

$$1^2 - 173 \cdot 0^2 = +1,$$

$$13^2 - 173 \cdot 1^2 = -4,$$

$$79^2 - 173 \cdot 6^2 = +13,$$

the last equation being

$$(1118)^2 - 173 \cdot (85)^2 = -1.$$

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The term for the last equation being always in a case such as the present one, not $+1$, but -1. The final numbers 1118, 85 are consequently entered not in Table I., but in Table II., viz. the entry in this table is

$$\begin{array}{c} 173 \\ 85 \\ 1118 \end{array}$$

and thence we calculate the numbers y, x of Table I., viz. these are

$$2499849 = 2 \cdot (1118)^2 + 1,$$
$$190060 = 2 \cdot 1118 \cdot 85.$$

Generally Table II. gives for each value of a, comprised therein, values of x, y, such that $y^2 = ax^2 - 1$, and then writing $y_1 = 2y^2 + 1$, $x_1 = 2xy$, we have

$$y_1^2 = (2ax^2 - 1)^2 = 4a^2x^4 - 4ax^2 + 1 = a \cdot 4x^2(ax^2 - 1) + 1 = ax_1^2 + 1,$$

so that x_1, y_1 are for the same value of a the values of x, y in Table I.

It is to be remarked that the heading of Table II. is not perfectly accurate, for it purports to give for every value of a, for which a solution exists, a solution of the equation $y^2 = ax^2 - 1$. What it really gives is the solution for each value of a for which the period has a double middle term. But if $a = \alpha^2 + 1$, then obviously we have a solution $y = \alpha$, $x = 1$, and for any such value of a the period has a single middle term, viz. the entry in Table I. is

$$\begin{array}{cc} \alpha^2 + 1 & \\ \alpha, & (2\alpha) \\ 1, & 1 \\ 2\alpha & \\ 2\alpha^2 + 1 & \end{array}$$

and we, in fact, have

$$\begin{array}{r|r|r|r} \alpha^2 + 1 & & & \\ \hline \alpha & & & \\ 1 & 0 & +1 & \\ (2\alpha) & \alpha & 1 & -1 \\ 2\alpha & 2\alpha^2 + 1 & 2\alpha & +1 \end{array}$$

$$
\begin{aligned}
1^2 - (\alpha^2 + 1) \cdot 0^2 &= +1, \\
\alpha^2 - (\alpha^2 + 1) \cdot 1^2 &= -1, \\
(2\alpha^2 + 1)^2 - (\alpha^2 + 1) \cdot (2\alpha)^2 &= +1.
\end{aligned}
$$

The foregoing instances of the calculation of x, y in the case of the numbers 209 and 173 suggest a table which may be regarded as an extended form of Degen's tables; viz. such a table, from $a = 2$ to $a = 99$, is as follows:

SPECIMEN OF EXTENDED FORM OF TABLE IN REGARD TO THE PELLIAN EQUATION.

a	y	x	$y^2 - ax^2$	a	y	x	$y^2 - ax^2$
2	1	0	+1	13	3	1	+1
	(2)	1	1		(1)	1	−4
	3	2	+1		4	1	+3
3	1	0	+1	14	3	1	+1
	(1)	1	−2		(1)	1	−5
	2	1	+1		4	1	+2
5	2	1	+1	15	4	1	+1
	(4)	1	−1	(1)	1	−14	
	9	4	+1	1	4	+1	
6	2	1	+1	17	4	1	+1
	(2)	2	−2		(8)	1	−1
	5	2	+1		33	8	+1
7	2	1	+1	18	4	1	+1
	(1)	1	−3		(4)	1	+2
	3	1	+2		17	4	+1
8	2	1	+1	19	4	1	+1
	(1)	1	−4		(2)	3	−5
	3	1	+1		39	9	+1
10	3	1	+1	20	4	1	+1
	(6)	1	−1		(2)	1	−4
	19	6	+1		9	2	+1
11	3	1	+1	21	4	1	+1
	(3)	1	−2		(1)	2	+3
	10	3	+1		5	1	+4
12	3	1	+1	22	4	1	+1
	(2)	1	−3		(1)	2	−6
	7	2	+1		5	1	+3

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—continued.

a	y	x	$y^2 - ax^2$	a	y	x	$y^2 - ax^2$
23	4	1	+1	27	5	1	+1
	(3)	3	−14		(5)	1	−2
	5	1	+2		26	5	+1
24	4	1	+1	28	5	1	+1
	(1)	4	−8		(3)	2	+1
	5	1	+1		127	24	+1
26	5	1	+1	29	5	1	+1
	(10)	1	+1		(2)	2	−7
	51	10	+1		70	13	−1
30	5	1	+1	31	5	1	+1
	(2)	2	−5		(1)	1	−6
	11	2	+1		6	1	+5
					1520	273	+1
32	5	1	+1	33	5	1	+1
	(1)	1	−7		(2)	1	−8
	6	1	+1		23	4	+1
34	5	1	+1	35	5	1	+1
	(1)	2	−11		(1)	1	−10
	35	6	+1		6	1	+1
37	6	1	+1	38	6	1	+1
	(12)	1	−2		(6)	1	−2
	73	12	+1		37	6	+1
39	6	1	+1	40	6	1	+1
	(4)	2	−3		(3)	1	−4
	25	4	+1		19	3	+1
41	6	1	+1	42	6	1	+1
	(2)	2	+2		(2)	1	−6
	32	5	−1		13	2	+1
	2049	320	+1				
43	6	1	+1	44	6	1	+1
	(3)	2	+1		(1)	1	−8
	3482	531	+1		199	30	+1
45	6	1	+1	46	6	1	+1
	(1)	2	+1		(1)	3	−5
	7	1	+1		5	1	−21
	161	24	−2		24335	3588	+1
	15129	2253	+1				
47	6	1	+1	48	6	1	+1
	(5)	1	−2		(1)	1	−12

a	y	x	$y^2 - ax^2$	a	y	x	$y^2 - ax^2$
	48	7	+1		7	1	+1
50	7 (14) 99	1 1 14	+1 −1 +1	51	7 (7) 50	1 1 7	+1 −2 +1
52	7 (4) 649	1 1 90	+1 +9 +1	53	7 (3) 182	1 1 25	+1 −4 +1
54	7 (2) 485	1 1 66	+1 −5 +1	55	7 (2) 89	1 1 12	+1 −6 +1
56	7 (2) 15	1 1 2	+1 −7 +1	57	7 (1) 8	1 1 1	+1 −8 +1
58	7 (1) 99 2575 19603	1 1 13 338 2574	+1 −9 +3 −2 +1	59	7 (2) 530	1 1 69	+1 −10 +1
60	7 (1) 8 31 231	1 1 1 4 30	+1 −11 +1 −11 +1	61	7 (4) 39 164 29718	1 3 5 21 3805	+1 −5 +4 −5 +1
62	7 (6) 63	1 1 8	+1 −2 +1	63	7 (1) 8	1 1 1	+1 −14 +1
65	8 (16) 129	1 1 16	+1 −1 +1	66	8 (8) 65	1 1 8	+1 −2 +1
67	8 (5) 41 48842	1 1 5 5967	+1 −3 +6 +1	68	8 (4) 33	1 1 4	+1 −4 +1
69	8 (3) 25 7775	1 1 3 936	+1 −5 +4 +1	70	8 (2) 251	1 1 30	+1 −6 +1
71	8 (2) 3480	1 1 413	+1 −7 +1	72	8 (2) 17	1 1 2	+1 −8 +1
73	8 (1) 1068	1 1 125	+1 −9 −1	74	8 (1) 43	1 1 5	+1 −10 +1

a	y	x	$y^2 - ax^2$	a	y	x	$y^2 - ax^2$
	2281249	267000	+1				
75	8	1	+1	76	8	1	+1
	(1)	1	−11		(1)	1	−12
	26	3	+1		57799	6630	+1
77	8	1	+1	78	8	1	+1
	(3)	1	+5		(1)	1	−14
	351	40	+1		53	6	+1
79	8	1	+1	80	8	1	+1
	(1)	1	−15		(1)	1	−16
	80	9	+1		9	1	+1
82	9	1	+1	83	9	1	+1
	(18)	1	+1		(9)	1	−2
	163	18	+1		82	9	+1
84	9	1	+1	85	9	1	+1
	(6)	1	−3		(4)	1	−4
	55	6	+1		378	41	−1
					285769	30996	+1
86	9	1	+1	87	9	1	+1
	(3)	1	−5		(3)	1	−6
	10405	1122	+1		28	3	+1
88	9	1	+1	89	9	1	+1
	(2)	1	−7		(2)	1	−8
	197	21	+1		500	53	+1
90	9	1	+1	91	9	1	+1
	(2)	1	−9		(1)	1	−10
	19	2	+1		1574	165	+1
92	9	1	+1	93	9	1	+1
	(1)	1	−11		(1)	1	−12
	1151	120	+1		12151	1260	+1
94	9	1	+1	95	9	1	+1
	(1)	1	−13		(1)	1	−14
	2143295	221064	+1		39	4	+1
96	9	1	+1	97	9	1	+1
	(1)	1	−15		(1)	1	−16
	49	5	+1		5604	569	+1
98	9	1	+1	99	9	1	+1
	(1)	1	−17		(1)	1	−18
	99	10	+1		10	1	+1

The meaning hardly requires explanation; for each number a, we have a series of pairs of increasing numbers, y, x, satisfying a series of equations $y^2 = ax^2 \pm b$; thus $a = 14$:

y	x	$y^2 - ax^2$
1	0	$+1$
31	9	$-14 \cdot 1 = -5$
4	1	$16 - 14 \cdot 1 = +2$
11	3	$121 - 14 \cdot 9 = -5$
15	4	$225 - 14 \cdot 16 = +1$

The following table, calculated under the superintendence of the Committee, extends from $a = 1001$ to $a = 1500$ (square numbers omitted); it is (with slight typographical variations) nearly but not exactly in the form of Degen's Table I., the chief difference being that for a number a having a double middle term, or of the form $\alpha^2 + 1$ (such number being further distinguished by an asterisk), the x, y entered in the table are the solutions, not of the equation $y^2 = ax^2 + 1$, but of the equation $y^2 = ax^2 - 1$. As remarked above, if we have $y^2 = ax^2 - 1$, then writing $y_1 = 2y^2 + 1$ and $x_1 = 2axy$, we obtain $y_1^2 = ax_1^2 + 1$. Moreover, for each value of a, in the first line, the first term, which is the integer part of $\sqrt{a}$, is separated from the other by a semicolon, and the 1, which is the corresponding first term of the second line, is omitted.

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The calculations were made by C. E. Bickmore, M.A., of New College, Oxford: his values for x and y have been revised as presently mentioned, but it has been assumed that his values for the periods and subsidiary numbers (forming the first and second lines of each division of the table) are accurate; in fact, any error therein would cause the resulting values of x and y to be wildly erroneous; but (except in a single instance which was accounted for) the errors in x and y were in every case in a single figure or two or three figures only.

The values of x and y were in every case examined by substitution in the equation ($y^2 = ax^2 + 1$, or $y^2 = ax^2 - 1$, as the case may be), which should be satisfied by them. These verifications were for the most part made by A. Graham, M.A., of the Observatory, Cambridge. As already mentioned, some errors were detected, and these have been, of course, corrected. The values of x, y given in the table thus satisfy in every case the proper equation $y^2 = ax^2+1$, or $y^2 = ax^2-1$; on the ground above referred to, it is believed that the periods and subsidiary numbers are also accurate.

It may be remarked, in regard to the verification of the equation $y^2 = ax^2 \pm 1$ for large values of a; and y, it is in practice easier and safer to calculate $ax^2 \pm 1$, and then to compare the square root thereof with the given value of y, than to further calculate the value of y^2.

THE TABLE 1001 TO 1500.

a	Continued fraction for $\sqrt{a}$
1001	31; 1, 1, 1, 3, 3, 2, (4) 40, 23, 35, 16, 17, 25, (13)
1002	31; 1, 1, 1, 8, 2, 1, 1, 1, 3, (10) 41, 22, 39, 7, 23, 31, 26, 33, 17, (6)
1003	31; 1, 2, (31) 42, 21, (2)
1004	31; 1, 2, 5, 2, 2, 1, 7, 4, 1, 2, 1, 11, 1, (14) 43, 20, 11, 25, 19, 41, 8, 13, 40, 17, 44, 5, 55, (4)
1005	31; 1, 2, 2, 1, 5, 15, 1, 2, (12) 44, 19, 20, 39, 11, 4, 41, 21, (5)
1006	31; 1, 2, 1, 1, 5, 1, 3, 2, 1, 1, 1, 1, 1, 9, 1, 20, 4, 5, 1, 1, 12, 6, 1, (30) 45, 18, 29, 33, 10, 43, 15, 22, 31, 27, 30, 25, 37, 6, 55, 3, 15, 11, 30, 33, 5, 9, 53, (
1007	31; 1, 2, (1) 46, 17, (38)
1008	31; 1, (2) 47, (16)
1009*	31; 1, (3, 3) 48, (15, 15)
1010*	31; 1, 3, (1, 1) 49, (3, 31)
1011	31; 1, 3, 1, (9) 50, 13, 47, (6)
1012	31; 1, 4, 3, 6, 1, 3, 8, 1, (4) 51, 12, 19, 9, 43, 16, 7, 48, (11)
1013*	31; 1, 4, 1, 4, 15, 1, 2, (2, 2) 52, 11, 44, 13, 4, 43, 19, (23, 23)
1014	31; 1, 5, 2, 1, 1, 1, 1, (20) 53, 10, 23, 30, 29, 25, 38, (3)
1015	31; 1, 6, 10, (2) 54, 9, 6, (29)
1016	31; 1, (6) 55, (8)
1017	31; 1, 6, 1, 1, 5, 2, 1, 1, 1, 2, 1, (12) 56, 7, 8, 49, 9, 13, 41, 16, 11, 31, 32, (9)
1018*	31; 1, 9, 1, 1, 1, 6, 2, (3, 3) 57, 6, 39, 23, 38, 9, 26, (17, 17)
1019	31; 1, 11, 1, 3, 1, 1, 1, 3, 8, 1, 5, 2, (31) 58, 5, 47, 14, 35, 25, 34, 17, 7, 49, 10, 29, (2)
1020	31; 1, (11) 59, (4)

a	Continued fraction for $\sqrt{a}$
1021*	31; 1, 20, 3, 6, 1, 3, 2, 1, 1, 12, 5, 4, 15, 1, 2, 1, 4, 1, 1, 2, 1, 1, 1, (5, 5) 60, 3, 20, 9, 44, 15, 23, 27, 36, 5, 12, 15, 4, 45, 17, 41, 12, 33, 29, 20, 33, 25, 36, (

949.

ON HALPHEN'S CHARACTERISTIC n, IN THE THEORY OF CURVES IN SPACE.

[From Crelle's Journal d. Mathematik, t. cxi. (1893), pp. 347—352.]

If we consider a curve in space without actual singularities, of the order (or degree) d, then this has a number h of apparent double points (adps.), viz. taking as vertex an arbitrary point in space, we have through the curve a cone of the order d, with h nodal lines; and Halphen denotes by n the order of the cone of lowest order which passes through these h lines. For a given value of d, h is at most $= \frac{1}{2}(d-1)(d-2)$, and as shown by Halphen it is at least $= [\frac{1}{4}(d-1)^2]$, if we denote in this manner the integer part of $\frac{1}{4}(d-1)^2$. For given values of d, h, it is easy to see that n must lie within certain limits, viz. if ν be the smallest number such that $\frac{1}{2}\nu(\nu+3)$ equal to or greater than h, then n is at most $=\nu$; and moreover n must have a value such that nd is at least $= 2h$, or say we must have $nd = 2h + \theta$, where θ is $= 0$ or positive. For any given value of d, we thus have a finite number of forms (d, h, n), and we have thus *prima facie* curves in space of the several forms (d, h, n): but it may very well be, and in fact Halphen finds, that when $d = 9$ or upwards, then for certain values of h, n as above, there is not any curve (d, h, n); thus $d = 9$, $h = 17$ the values of n are $n = 4$, $n = 5$, but there is not any curve $d = 9$, $h = 17$ for either of these values of n; or say the curves $(9, 17, 4)$ and $(9, 17, 5)$ are non-existent. And in the Notes and References to the papers 302, 305 in vol. v. of my Collected Mathematical Papers, 4to. Cambridge, 1892, see p. 615, I remarked that, in certain cases for which Halphen finds a curve (d, h, n), such curve does not exist except for special configurations of the h nodal lines not determined by the mere definition of n as the order of the cone of lowest order which passes through the h nodal lines; for instance $d = 9$, $h = 16$, $n = 4$, for which Halphen gives a curve, I find that for the existence of the curve it is not enough that the 16 lines are situate upon a quartic cone, but they must be the 16 lines of intersection of two quartic cones.

In fact, starting from an existing curve, say the complete intersection of two given surfaces of the orders μ, ν respectively, we have, as is known,

$$d = \mu\nu, \quad 2h = \mu\nu(\mu - 1)(\nu - 1);$$

we find also, as will appear, $n = (\mu - 1)(\nu - 1)$: hence also $nd = 2h$, viz. the cone of the order n through the h nodal lines meets the cone of the order d in these lines counting each twice, and in no other lines. I remark also that h is $= \frac{1}{2}n(n + 3)$ if $\mu + \nu = 4$, viz. we have here two nodal lines lying in a plane; but if $\mu + \nu > 4$, then h is greater than $\frac{1}{2}n(n+3)$, viz. in this case the nodal lines are not h lines taken at pleasure, but they are lines subject to the condition of lying in a cone of the order n. But this is not all; the h nodal lines lie not only in this cone of lowest order n, but also in cones of the orders $n + 1$, $n + 2$, $\ldots$, $n + (\mu + \nu - 2)$ respectively: I do not for the moment attempt to determine the number of independent conditions which are hereby imposed upon the h lines.

The last-mentioned theorem constitutes in fact the geometrical interpretation of results contained in Jacobi's paper "De eliminatione variabilis e duabus aequationibus algebraicis," Crelle, t. xv. (1836), pp. 101—124, [Ges. Werke, t. iii. pp. 295—320].

Consider the two equations

$$U = (*\pitchfork x, y, z, w)^{\mu} = 0; \quad V = (*\pitchfork x, y, z, w)^{\nu} = 0;$$

representing surfaces of the orders μ, ν respectively; since the form of the equations is quite arbitrary, we may without loss of generality assume that the point $(x, y, z) = (0, 0, 0)$ is an arbitrary point in space; and this being so, we find the equation of the cone, vertex this point, which passes through the curve of intersection of the two surfaces by the mere elimination of w between the two equations. As the reasoning is exactly the same for a particular case, I write for convenience $\mu = 3$, $\nu = 4$, and consider the two equations

$$A_0w^3 + A_1w^2 + A_2w + A_3 = 0,$$
$$B_0w^4 + B_1w^3 + B_2w^2 + B_3w + B_4 = 0,$$

where the suffixes show the degrees in regard to (x, y, z), viz. A_0, B_0 are mere constants, A_1, B_1 are linear functions $(*\pitchfork x, y, z)^1$, A_2, B_2 quadric functions $(*\pitchfork x, y, z)^2$, and so on. Multiplying the first equation successively by 1, w, w^2, w^3, and the second equation successively by 1, w, w^2, we have 7 equations from which to eliminate 1,

w, w^2, w^3, w^4, w^5, w^6, and the result is

$$\begin{vmatrix} A_0, & A_1, & A_2, & A_3, & & & \\ & A_0, & A_1, & A_2, & A_3, & & \\ & & A_0, & A_1, & A_2, & A_3, & \\ B_0, & B_1, & B_2, & B_3, & B_4, & & \\ & B_0, & B_1, & B_2, & B_3, & B_4, & \\ & & B_0, & B_1, & B_2, & B_3, & B_4 \end{vmatrix} = 0,$$

viz. this is an equation $\Omega = 0$ of the cone of the order $d = 12$, through the curve of intersection of the two surfaces: say this equation is $\Omega = 0$.

But if we only multiply the first equation by 1, w, w^2 successively and the second equation by 1, w successively, then we have 5 equations serving to determine the ratios of w^5, w^4, w^3, w^2, w, 1, viz. we have these quantities proportional to the six determinants which can be formed out of the matrix

$$\begin{matrix} A_0, & A_1, & A_2, & A_3, & & \\ & A_0, & A_1, & A_2, & A_3, & \\ B_0, & B_1, & B_2, & B_3, & B_4, & \\ & B_0, & B_1, & B_2, & B_3, & B_4 \end{matrix}$$

say we have

$$w^5 : w^4 : w^3 : w^2 : w : 1 = L : M : N : P : Q : R,$$

where L, M, N, P, Q, R represent homogeneous functions (∗ ⫛ $x, y, z)^e$, of the degrees 11, 10, 9, 8, 7, 6 respectively. We may if we please write

$$\frac{w^5}{L} = \frac{w^4}{M} = \frac{w^3}{N} = \frac{w^2}{P} = \frac{w}{Q} = \frac{1}{R}$$

or eliminating w, we have the series of equations which may be written

$$\begin{vmatrix} L, & M, & N, & P, & Q \\ M, & N, & P, & Q, & R \end{vmatrix} = 0,$$

viz. we thus denote that the determinants formed with any two columns of this matrix are severally $= 0$. This of course implies that each of the determinants in question is the product of Ω and a factor which is a homogeneous function of the proper degree in

(x, y, z), so that the several equations are all of them satisfied if only $\Omega = 0$. We have for instance $PR - Q^2 = A\Omega$, where A is a quadric function $(*\pitchfork x, y, z)^2$; similarly $NR - PQ = B\Omega$, where B is a cubic function $(*\pitchfork x, y, z)^3$, and the like as regards the other determinants.

If the ratios $x : y : z$ have any given values such that we have for these $\Omega = 0$, then w has a determinate value, that is, on each line of the cone $\Omega = 0$, there is a single point of the curve of intersection of the two surfaces: the only exceptions are when, for the given values of $x : y : z$, the expressions for w assume an indeterminate form, viz. w has then two values, and there are upon the line two points of the curve, or what is the same thing, the line is a nodal line of the cone: the conditions for a nodal line thus are $L = 0$, $M = 0$, $N = 0$, $P = 0$, $Q = 0$, $R = 0$, viz. each of these equations is that of a cone passing through the nodal lines of the cone $\Omega = 0$; the cone of lowest order is $R = 0$, a cone of the order 6 meeting the cone $\Omega = 0$ of the order 12 in 36 lines each twice, which lines are consequently the nodal lines of the cone $\Omega = 0$. The mere condition of the 36 lines lying upon a cone of the order 6 shows that the 36 lines are not arbitrary; and we have moreover, through the 36 lines, cones of the orders 7, 8, 9, 10 and 11 respectively. Obviously the foregoing reasoning is quite general, and for the surfaces of the orders μ, ν we have (as stated above) the cone Ω of the order $\mu\nu$, with $h = \frac{1}{2}\mu\nu(\mu - 1)(\nu - 1)$ nodal lines, the intersections (each counting twice) of this cone with a cone of the order $n = (\mu - 1)(\nu - 1)$; and moreover the h nodal lines lie also in cones of the orders $n + 1$, $n + 2$, $\ldots$, $n + \mu + \nu - 2$ respectively.

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THEORY OF CURVES IN SPACE.

To examine the meaning of the theorem, I form the table

μ, ν	d	h	n	$\ldots$, $n+\mu+\nu-2$
2, 2	4	2	1,	2, 3
2, 3	6	6	2,	3, 4, 5
2, 4	8	12	3,	4, 5, 6, 7
2, 5	10	20	4,	5, 6, 7, 8, 9
3, 3	9	12	4,	5, 6, 7
3, 4	12	18	6,	7, 8, 9, 10, 11
3, 5	15	60	8,	9, 10, 11, 12, 13, 14
4, 4	16	72	9,	10, 11, 12, 13, 14, 15
4, 5	20	120	12,	13, 14, 15, 16, 17, 18, 19

Here μ, $\nu = 2$, 2, there are 2 nodal lines, which are arbitrary, and of course lie on cones of the orders 1, 2, 3 respectively. So μ, $\nu = 2$, 3; there are 6 nodal lines, which are not arbitrary, inasmuch as they lie on a cone of the order 2; but regarding them as arbitrary lines on such a cone, we can through them draw a cone of the order 3 or any higher order, and it is thus no specialisation to say that they lie upon cones of the orders 3, 4, and 5. But going a step further $\mu = 2$, $\nu = 4$: here we have 12 nodal lines which, inasmuch as they lie on a cone of the order 3, are not arbitrary: and they are not arbitrary lines upon this cone, for they lie on a cone of the order 4, and such a cone can be drawn through at most 11 arbitrary lines on a cubic cone. In fact, upon a cone of the order θ, taking at pleasure N lines, the condition that it may be possible through these to draw a cone of the order $\theta + 1$ is $\frac{1}{2}(\theta+1)(\theta+4) = N+3$ at least; for if this number were $= N+2$, then through the N points we have only the improper cone $(x + \beta y + \gamma z)\, U_\theta = 0$, if $U_\theta = 0$ is the cone of the order θ. It thus appears that the 12 nodal lines are not arbitrary lines on a cubic cone, but that they constitute the complete intersection of a cubic cone and a quartic cone. But through such 12 lines we may draw cones of the 5th and higher orders, and it is thus no further condition that the 12 lines lie on cones of the orders 5, 6 and 7 respectively.

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So again $\mu = 3$, $\nu = 3$; we have here 18 nodal lines which, inasmuch as they lie on a cone of the order 4, are not arbitrary: and they are not arbitrary lines on this cone inasmuch as they lie also on a cone of the order 5, and such a cone can be drawn through at most 17 arbitrary lines on the quartic cone: it thus appears that the 18 nodal lines are 18 out of the 20 lines of intersection of a quartic cone and a quintic cone. But there is no further condition, for through such lines we can draw a cone of the order 6 or any higher order and thus the lines lie on cones of the orders 6, 7 and 8 respectively. It appears probable however that, for higher values of μ, ν, it would be necessary to take account not only (as in these examples) of the cones of the orders n and $n+1$, but of those of higher orders $n+2$, &c.; and thus that it is not the true form of the theorem to say that the h nodal lines must be h out of the $n(n+1)$ lines of intersection of two cones of the orders n and $n+1$ respectively.

It appears, by what precedes, that the $h = \frac{1}{2}\mu\nu(\mu-1)(\nu-1)$, lines which are the nodal lines of the cone of arbitrary vertex which passes through the curve of intersection of two surfaces of the orders

μ, ν respectively, form a remarkable special system of lines, which well deserve further study. I remark also that, without having proved the negative, it seems to me clear that given the values of d, h, n it is only in the cases where the h lines form some such special system (and not in the general case where the h nodal lines are any lines whatever on a cone of the order n) that there exists a curve (d, h, n); and thus that the question for further investigation is, for given values of (d, h, n) to determine the conditions necessary for the existence of a curve in space with these characteristics (d, h, n).

950.

ON THE SEXTIC RESOLVENT EQUATIONS OF JACOBI AND KRONECKER.

[From Crelle's Journal d. Mathematik, t. cxm. (1894), pp. 42—49.]

The equations referred to are: the first of them, that given by Jacobi in the paper "Observatiunculae ad theoriam aequationum pertinentes," Crelle, t. xiii. (1835), pp. 340—352, [Ges. Werke, t. iii., pp. 269—284], under the heading "Observatio de aequatione sexti gradus ad quam aequationes sexti gradus revocari possunt," and the second, that of Kronecker in the note "Sur la resolution de l'équation du cinquième degré," Comptes Rendus, t. XLVI. (1858), pp. 1150—1152. Jacobi's equation is closely connected with that obtained by Malfatti in 1771, see Brioschi's paper "Sulla resolvente di Malfatti per l'equazione del quinto grado," Mem. R. Ist. Lomb., t. ix. (1863); but the characteristic property first presents itself in Jacobi's form, and I think the equation is properly described as Jacobi's resolvent equation. The other equation has been always known as Kronecker's resolvent equation; it belongs to the class of equations for the multiplier of an elliptic function considered by Jacobi in the paper "Suite des notices sur les fonctions elliptiques," Crelle, t. iii. (1828), pp. 303—310, see p. 308, [Ges. Werke, t. i., pp. 255—263, see p. 261]: say Kronecker's equation belongs to the class of Jacobi's Multiplier Equations. We have in regard to it the paper by Brioschi, "Sul metodo di Kronecker per la resoluzione delle equazioni di quinto grado," Atti Ist. Lomb., t. i. (1858), pp. 275—282, and see also the "Appendice terza" to his translation of my Elliptic Functions (Milan, 1880): it seems to me however that the theory of Kronecker's equation has not hitherto been exhibited in the clearest form.

I consider the forms

12345	13524
13254	12435
24315	23541
35421	34152
41532	45213
52143	51324

which, if in the first instance the figures are regarded as points, represent the twelve pentagons which can be formed with the points

1, 2, 3, 4, 5; each form in the right-hand column is derived from the corresponding form in the left-hand column by stellation, say we have $13524 = S\,12345$, and so in other cases.

A pentagon is in general reversible, but we sometimes consider it as irreversible (viz. we distinguish between the pentagons 12345 and 15432); when this is so, we write $15432 = R\,12345$, and we have thus twelve new forms, in all twenty-four forms. The symbols R, S are such that $R^2 = 1$, $S^2 = R$, $RS = SR$, $S^4 = 1$. But for a reversible pentagon, there is no occasion to use the symbol R, and we have simply $S^2 = 1$.

In a somewhat different point of view, we may for an irreversible pentagon write

12345 to denote any one of the forms 12345, 23451, 34512, 45123, 51234,

$R\,12345$ to denote any one of the forms 15432, 54321, 43215, 32154, 21543,

and for a reversible pentagon 12345 to denote any one of these same ten forms.

Each pentagon gives thus ten forms, viz. we have in all 120 forms which all are the different arrangements of the five figures. But further regarding the arrangement 12345 as positive, then the forms in the left-hand column are each of them positive, and the ten forms derived from any one of these are each of them positive, that is, the forms in the left-hand column give all the 60 positive arrangements of the five figures: and similarly the forms in the right-hand column give all the 60 negative arrangements of the five figures.

Taking 1, 2, 3, 4, 5 to denote any quantities, or say the five roots x_1, x_2, x_3, x_4, x_5 of a quintic equation, we regard 12345, ..., as denoting functions of these roots: in particular, 12345 may denote a cyclic reversible function, the analogue of the reversible pentagon, or it may denote a cyclic irreversible function, the analogue of the irreversible pentagon.

Jacobi's resolvent equation.

The most simple course is to take 12345 a cyclic reversible function of the roots x; a root of the resolvent equation is then

$12345 - 13524, = (1 - S)\,12345$, and the six roots are

$$\begin{aligned}
z_1 &= (1 - S)\,12345,\\
z_2 &= (1 - S)\,13254,\\
z_3 &= (1 - S)\,24315,\\
z_4 &= (1 - S)\,35421,\\
z_5 &= (1 - S)\,41532,\\
z_6 &= (1 - S)\,52143.
\end{aligned}$$

Here effecting on the roots x any positive substitution whatever, we permute inter se the roots z; but effecting on the roots x any negative substitution whatever, then the roots z are permuted inter se, but at the same time each root z changes its sign: hence the coefficients of the equation having the roots z are not absolutely unaltered, but they are each of them multiplied by a determinate factor (1 or -1) according as the substitution is positive or negative; such an equation is not properly invariantive in regard to the general substitutions of the roots, but only in regard to the positive substitutions: or say it is a *semi-invariant* equation. But observe that the equation having the roots z^2 is invariantive as well in regard to the positive as to the negative substitutions; hence if from the equation in z we eliminate z by means of the equation $y = z^2$, we obtain an invariant equation of the order 6 in y.

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951. Non-Euclidian Geometry.

[From the Transactions of the Cambridge Philosophical Society, vol. xv. (1894), pp. 37—61. Read January 27, 1890.]

...then the values are

$$\lambda = rr_1\sqrt{1-(M-M_1)^2}, \quad \mu = -rr_1\sqrt{1-\ldots}$$

And, this being so, the two systems of values of A, B, C, F, G, H, are

$$\begin{array}{ll}
\lambda(2l+a)+\mu(2l-a), & \lambda(2l+a)-\mu(2l-a),\\
\lambda(S_3+\beta)+\mu(S_3-\beta), & \lambda(S_3+\beta)-\mu(S_3-\beta),\\
\lambda(S_2+\gamma)+\mu(S_2-\gamma), & \\
\lambda(S_1+a)-\mu(S_1-a), & \lambda(S_1+a)+\mu(S_1-a),\\
\lambda(S_3+\beta)-\mu(S_3-\beta), & \lambda(S_3+\beta)+\mu(S_3-\beta),\\
\lambda(S_2+\gamma)+\mu(S_2-\gamma), &
\end{array}$$

viz. the two perpendiculars are reciprocals each of the other.

35. Before going further I notice that, if

$$\frac{a_1+f_1}{a+f} = \frac{b_1+g_1}{b+g} = \frac{c_1+h_1}{c+h}, \quad \text{and} \quad \frac{a_1-f_1}{a-f} = \frac{b_1-g_1}{b-g} = \frac{c_1-h_1}{c-h},$$

then the four equations for (A, B, C, F, G, H) reduce themselves to three equations only: and thus instead of two perpendiculars we have a singly infinite series of perpendiculars (see ante No. 15).

To explain the meaning of the equations, I observe that a line (a, b, c, f, g, h) will be a generating line of the one kind, or say a "generatrix," of the Absolute if

$$a+f=0, \quad b+g=0, \quad c+h=0;$$

and it will be a generating line of the other kind, or say a "directrix," of the Absolute if $a-f=0$, $b-g=0$, $c-h=0$. Or what is the same thing, we have

$$(a, b, c, -a, -b, -c), \text{ where } a^2+b^2+c^2=0 \text{ for a generatrix,}$$

and

$(a, b, c, \quad a, \quad b, \quad c)$, where $a^2+b^2+c^2 = 0$ for a directrix, of the Absolute.

Consider now two directrices (a, b, c, a, b, c) and $(a_1, b_1, c_1, a_1, b_1, c_1)$: if a line (a, b, c, f, g, h) meets each of these, then

$$(a + f)a + (b + g)b + (c + h)c = 0,$$
$$(a + f)a_1 + (b + g)b_1 + (c + h)c_1 = 0,$$

and consequently

$$a + f : b + g : c + h = bc_1 - b_1c : ca_1 - c_1a : ab_1 - a_1b;$$

and similarly if $(a_1, b_1, c_1, f_1, g_1, h_1)$ meets each of the two directrices, then

$$a_1 + f_1 : b_1 + g_1 : c_1 + h_1 = bc_1 - b_1c : ca_1 - c_1a : ab_1 - a_1b,$$

that is, if the lines each of them meet the same two directrices of the Absolute, then

$$\frac{a_1 + f_1}{a + f} = \frac{b_1 + g_1}{b + g} = \frac{c_1 + h_1}{c + h}.$$

Conversely, if these relations are satisfied, then the lines each of them meet two directrices of the Absolute.

In like manner, if the lines each meet two generatrices of the Absolute, then

$$\frac{a_1 - f_1}{a - f} = \frac{b_1 - g_1}{b - g} = \frac{c_1 - h_1}{c - h};$$

and conversely, if these relations are satisfied, then the lines each of them meet the same two generatrices of the Absolute. In the former case, the lines are said to be "right parallels": in the latter case, "left parallels."

A line (a, b, c, f, g, h) meets the Absolute in two points, and through each of these we have a directrix and a generatrix: that is, the line meets two directrices and two generatrices.

Through a given point we may draw, meeting the two directrices, or meeting the two generatrices, a line: that is, through a given point we may draw a line

$$(a_1, b_1, c_1, f_1, g_1, h_1)$$

which is a right parallel, and a line which is a left parallel, to a given line. That is, regarding as given the first line, and also a point of the second line, there are two positions of the second line such that for each of them, the $\perp$s of the pair of lines, instead of being two determinate lines, are a singly infinite series of lines.

36. Reverting to the general case, we have found (A, B, C, F, G, H) the coordinates of either of the lines $\perp$ to the given lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$: supposing that the $\perp$ intersects the first of these lines in the point the coordinates of which are (x, y, z, w), and the second in the point the coordinates of which are (x_1, y_1, z_1, w_1), then we have for each set of coordinates a fourfold expression; the choice of the form is indifferent, and I write

$$x : y : z : w = cB - bC : aC - cA : bA - aB : fA + gB + hC,$$
$$x_1 : y_1 : z_1 : w_1 = c_1B - b_1C : a_1C - c_1A : b_1A - a_1B : f_1A + g_1B + h_1C.$$

We have then, for the distance of these two points,

$$\cos\theta = \dots$$

where $\phi = \delta$ or θ, according to the sign of the radical $\sqrt{}$ contained in the expressions for A, B, C, F, G, H.

I have not succeeded in obtaining in this manner the final formulae for the determination of the distances: these in fact are, by what precedes, given by the equations

$$\sin\delta\sin\theta = M, \quad \cos\delta\cos\theta = M_1.$$

For then, writing ϕ to denote either of the distances δ, θ, at pleasure, we have

$$\sin^2\phi\cos^2\phi = MM_1,$$

that is,

$$\cos^4\phi - \cos^2\phi(M^2 - M_1^2 + 1) + M_1^2 = 0,$$

or

$$\cos^2\phi = \frac{M^2 - M_1^2 + 1 \pm \sqrt{(M^2 - M_1^2 + 1)^2 - 4M_1^2}}{2},$$

which is the expression for the cosine of the distance.

In the case where the two lines intersect $M = 0$, and if δ be the $\perp$ distance which vanishes, then $\delta = 0$, and consequently $\cos\theta = M_1$: the last-mentioned formula, putting therein $M = 0$ and taking the radical to be $= M_1^2 - 1$, gives $\cos^2\phi = M_1^2$, that is, $\phi = \theta$, and $\cos^2\theta = M_1^2$, as it should be.

37. I verify as follows, in the case in question of two intersecting lines,

$$(af_1 + bg_1 + ch_1 + a_1f + b_1g + c_1h = 0),$$

the formula

$$\cos\theta = \frac{xx_1 + yy_1 + zz_1 + ww_1}{\sqrt{x^2 + y^2 + z^2 + w^2}\sqrt{x_1^2 + y_1^2 + z_1^2 + w_1^2}}.$$

We have here

$$\begin{aligned}
A &= S_1 = bc_1 - b_1c + fgh_1 - f_1g_1h,\\
B &= S_2 = ca_1 - c_1a + hfg_1 - h_1f_1g,\\
C &= S_3 = ab_1 - a_1b + ghf_1 - g_1h_1f,\\
F &= \alpha = b_1h - bh_1 - cg_1 + c_1g,\\
G &= \beta = c_1f - cf_1 - ah_1 + a_1h,\\
H &= \gamma = a_1g - ag_1 - bf_1 + b_1f.
\end{aligned}$$

I stop to notice that these formulae may be obtained in a different and somewhat more simple manner: the two lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$ intersect; hence their reciprocals also intersect: the equation of the plane through the two lines and that of the plane through the two reciprocal lines are respectively

$$\begin{aligned}
&(g_1h - gh_1)x + (h_1f - hif)y + (f_1g - fg_1)z + (af_1 + bg_1 + ch_1)w = 0,\\
&(bc_1 - b_1c)x + (ca_1 - c_1a)y + (ab_1 - a_1b)z + (fa_1 + gb_1 + hc_1)w = 0;
\end{aligned}$$

the line (A, B, C, F, G, H) is thus the line of intersection of these two planes, and it is thence easy to obtain the foregoing values.

From the values of A, B, C, F, G, H, we have to find x, y, z, w and x_1, y_1, z_1, w_1 by the formulae given above. We have

$$\begin{aligned}
x = cB - bC &= bc(a_1 - a) + c(fg_1 - f_1g) + c(hf - h_1f)\\
&\quad - ab(b_1) + a_1b^2 - bfg_1 + \ldots\\
&= (bg + ch)f_1 + (h(b^2 + c^2) - b_1hb - c_1hc - bfg_1 - cfh_1)\\
&= -f(af_1 + bg_1 + ch_1) + a_1(b^2 + c^2) - b_1hb - c_1hc;
\end{aligned}$$

or writing here $af_1 + b_1g + c_1h$ in place of $-(af_1 + bg_1 + ch_1)$, this is a linear function of a_1, b_1, c_1; and similarly finding the values of y, z,

w, we have

$$\begin{aligned}
x &= a_1(b^2+c^2+f^2)+b_1(fg-hb)+c_1(hf-ca),\\
y &= a_1(fg-ab)+b_1(c^2+a^2+g^2)+c_1(gh-bc),\\
z &= a_1(hf-ca)+b_1(gh-bc)+c_1(a^2+b^2+h^2),\\
w &= a_1(bh-cg)+b_1(cf-ah)+c_1(ag-bf).
\end{aligned}$$

And in like manner, (I introduce for convenience the sign $-$, as is allowable),

$$\begin{aligned}
-x_1 &= a_1(b^2+c^2+f^2)+b_1(fg_1-a_1h)+c_1(h_1f-c_1a),\\
-y_1 &= a_1(h_1f_1-c_1g_1)+\ldots,\\
-z_1 &= a_1(\ldots)+\ldots+c_1(a^2+b^2+h^2),\\
-w_1 &= a_1(bh_1-c_1g_1)+b_1(c_1f-a_1h)+c_1(ag_1-b_1f).
\end{aligned}$$

38. Write for shortness

$$p=a^2+b^2+c^2,\quad p_1=f^2+g^2+h^2,\quad q=af+bg+ch,$$

$r=f^2+g^2+h^2$, $r_1=a^2+b^2+c^2$, and therefore $q_1=a_1f+b_1g+c_1h$; and also

$$\omega=af_1+bg_1+ch_1,\quad \omega_1=a_1f+b_1g+c_1h.$$

We have

$$\begin{aligned}
x &= cp-cq+h\omega, & x_1 &= -cr+dq+h_1\omega,\\
y &= \ldots, & y_1 &= \ldots,\\
z &= cp-cq+h\omega, & z_1 &= -cr+dq+h_1\omega,\\
w &= -c\ldots, & w_1 &= -c\ldots
\end{aligned}$$

from which we easily obtain

$$\omega^2+x^2+z^2=p(pr-q^2)+(p_1+2p)\omega^2,$$

and by expressing ω^2 in the form of a determinant

$$\omega^2=p(pr-q^2)-p\omega^2,$$

we obtain

$$x^2+y^2+z^2+w^2=(r+r_1)(pr-q^2+\omega^2),$$

and in like manner

$$x_1^2 + y_1^2 + z_1^2 + w_1^2 = (r + r_1)(pr - q^2 + \omega^2).$$

And again

$$xx_1 + yy_1 + zz_1 = q(pr - q^2) + (q_1 + 2q)\omega^2$$

and by expressing ww_1 in the form of a determinant

$$ww_1 = q_1(pr - q^2) - q\omega^2,$$

we find

$$xx_1 + yy_1 + zz_1 + ww_1 = (q + q_1)(pr - q^2 + \omega^2).$$

Hence substituting in

$$\cos\theta = \frac{xx_1 + yy_1 + zz_1 + ww_1}{\sqrt{x^2 + y^2 + z^2 + w^2}\sqrt{x_1^2 + y_1^2 + z_1^2 + w_1^2}}$$

the factor $pr - q^2 + \omega^2$ disappears, and we have

$$\cos\theta = \frac{q + q_1}{r + r_1},$$

the required result.

952. On the Kinematics of a Plane, and in particular on Three-bar Motion: and on a Curve-tracing Mechanism.

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The first part of the present paper, On the Kinematics of a Plane, and on Three-bar Motion, is purely theoretical: the second part contains a brief description of a Curve-tracing Mechanism, which at my suggestion has been constructed by Prof. Ewing in the workshops of the Engineering Laboratory, Cambridge.

Part I.

1. The theory of the motion of a plane, when two given points thereof describe given curves, has been considered by Mr S. Roberts in his paper, "On the motion of a plane under given conditions," Proc. Lond. Math. Soc. t. III. (1871), pp. 286—318, and he has shown that, if for the given curves the order, class, number of nodes, and of cusps, are (m, n, δ, κ) and $(m', n', \delta', \kappa')$ respectively $(n = m^2 - m - 2\delta - 3\kappa$, $n' = m'^2 - m' - 2\delta' - 3\kappa')$, then for the curve described by any fixed point of the plane:

$$\begin{aligned}
\text{order} &= 2mm', \\
\text{class} &= 2(mm' + mn' + nm'), \\
\text{number of nodes} &= mm'(2mm' - m - m') + 2(m\delta' + m'\delta), \\
\text{number of cusps} &= 2(m\kappa' + m'\kappa);
\end{aligned}$$

but he remarks that these formulae require modification when the directrices or either of them pass through the circular points at infinity. And he has considered the case where the two directrices become one and the same curve.

2. It will be convenient to speak of the line joining the two given points as the *link*; the two given points, say B and D, are then the extremities of the link; and I take the length of the link to be $= c$, and the two directrices to be b and d; we have thus the link $c = BD$ moving in such wise that its extremity B describes the curve b of the order m, and its extremity D the curve d of the order m': in Mr Roberts' problem, the locus is that described by a point P rigidly connected with the link, or say by a point P the vertex of the triangle PBD.

3. The points B, D describe of course the directrices b, d respectively: taking on b a point B_1 at pleasure, then if B be at B_1, the corresponding positions of D are the intersections of d by the circle centre B_1 and radius c, viz. there are thus $2m'$ positions of D: and similarly taking on d a point D_1 at pleasure, then if D be at D_1, the corresponding positions of B are the intersections of b by the circle centre D_1 and radius c, viz. there are thus $2m$ positions of B. The motion thus establishes a $(2m, 2m')$ correspondence between the points of the directrices b and d, viz. to a given point on b there correspond $2m'$ points on d, and to a given point on d there correspond $2m$ points on b. Of course, for a given position on either directrix, the corresponding points on the other directrix may be any or all of them imaginary; and thus it may very well be that for either directrix not the whole curve but only a part or detached parts thereof will be actually described in the course of the motion.

In saying that a part is described, we mean described by a continuous motion; say that the point B (the point D remaining always on a part of d) is capable of describing continuously a part of b; it may very well happen that the point B (the point D remaining always on a different part of d) is capable of describing continuously a different part of b, but that it is not possible for B to pass from the one to the other of these parts of b without removing D from the one part and placing it on the other part of d, and thus that we have on b detached parts each of them continuously described by B; and similarly, we may have on d detached parts each of them continuously described by D.

4. But dropping for the moment the question of reality, to a given position of B on b there correspond as was mentioned $2m'$ positions of D on d, or say $2m'$ positions of the link c: in the entire motion of the link it must assume each of these $2m'$ positions, and for each of them the point B comes to assume the position in question on b; the directrix b is thus described $2m'$ times, that is, the locus described by B will be the directrix b repeated $2m'$ times, or say a curve of the order $m \times 2m'$, $= 2mm'$. Similarly, the locus described by D will be the directrix d repeated $2m$ times, or say a curve of the order $m' \times 2m$, $= 2mm'$.

5. In general, if B_1D_1 be any position of the link and if B moves from B_1 along b in a determinate sense, then D will move from D_1 along d in a determinate sense; and if B moves from B_1 along b in the opposite sense, then also D will move from D_1 along d in the opposite sense. Or what is the same thing, we may have B moving in a determinate sense through B_1, and D moving in a determinate sense through D_1; and reversing the sense of B's motion, we reverse also the sense of D's motion. But there are certain critical positions of the link, viz. we have a critical position when the link is a normal at B_1 to the directrix b, or a normal at D_1 to the directrix d.

Say first the link is a normal at B_1 to the directrix b. The infinitesimal element at B_1 may be regarded as a straight line at right angles to the link; hence if for a moment D_1 is regarded as a fixed point, the link may rotate in either direction round D_1, that is, B may move from B_1 along b in either of the two opposite senses, say B_1 is a "two-way point." But if on d we take on opposite sides of D_1 the consecutive points D_1' and D_1'', say $D_1'D_1$ cuts D_1B_1 at an acute angle and $D_1''D_1$ cuts it at an obtuse angle, then D_1' will be nearer to b than was D_1, and thus the circle centre D_1' and radius c will cut b in two real points B_1' and B_1'' near to and on opposite sides of B_1; or as D moves to D_1', B will move from B_1 indifferently to B_1' or B_1''. Contrariwise, D_1'' is further from b than was D_1, and thus the circle centre D_1'' and radius c, will not meet b in any real point near to B_1, and hence D is incapable of moving from D_1 in the sense D_1D_1''. Or what is the same thing, the described portion of d, which includes a point D_1', will terminate at D_1, or say D_1 is a "summit" on the directrix d. We have thus a summit on d, corresponding to the two-way point on b. And of course in like manner, if the link is a normal at D_1 to the directrix d, then D_1 is a two-way point on d, and the corresponding point B_1 is a summit on b.

6. If the link is at the same time a normal at B_1 to b and at D_1 to d, then each of the points B_1, D_1 is a two-way point and also a summit; or more accurately, each of them is a two-way point and also a pair of coincident summits.

But the case requires further investigation. Considering the position B_1D_1 as given, we may take the axis of x coincident with this line, and the origin O in such wise that OB_1, OD_1 are each positive and $OD_1 > OB_1$; say we have $OD_1 = \delta$, $OB_1 = \beta$, and therefore $\delta - \beta = c$. The equation of the curve b in the neighbourhood of B_1 is $y^2 = 2p(x - \beta)$, where p is the radius of curvature at B_1, assumed to be positive when the curve is convex to O, or what is the same thing when the centre of curvature R lies to the right of B_1 $(OR - OB_1 = +)$; and similarly the equation of d in the neighbourhood of D_1 is $y^2 = 2\sigma(x - \delta)$, where σ is the radius of curvature at D_1 assumed to be positive when the curve is convex to O, or what is the same thing when the centre of curvature S lies to the right of D_1

$(OS - OD_1 = +)$.

Consider now (x_1, y_1) the coordinates of a point on b in the neighbourhood of B_1, $y_1^2 = 2p(x_1 - \beta)$, and taking B at this point, let (x_2, y_2) be the coordinates of the corresponding point D on d in the neighbourhood of D_1, $y_2^2 = 2\sigma(x_2 - \delta)$. We have

$$c^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2,$$

and here

$$\begin{aligned} x_1 - x_2 &= \frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma} + (\delta - \beta) = \frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma} + c, \\ (y_1 - y_2)^2 &= y_1^2 - 2y_1y_2 + y_2^2, \end{aligned}$$

whence

$$\left(\frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma}\right)^2 + 2c\left(\frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma}\right) + (y_1 - y_2)^2 = 0.$$

The equation thus becomes

$$\frac{1}{4}\left(\frac{y_1^2}{p} - \frac{y_2^2}{\sigma}\right)^2 + c\left(\frac{y_1^2}{p} - \frac{y_2^2}{\sigma}\right) + (y_1 - y_2)^2 = 0,$$

that is,

$$\frac{y_1^2}{p}\left(\frac{y_1^2}{4p} + c\right) + \frac{y_2^2}{\sigma}\left(\frac{y_2^2}{4\sigma} - c\right) - \frac{y_1^2y_2^2}{2p\sigma} - 2y_1y_2 + y_1^2 + y_2^2 = 0,$$

a quadric equation between y_1 and y_2. Evidently if we had taken D a point on d, coordinates (x_2, y_2), in the neighbourhood of D_1 and had sought for the coordinates (x_1, y_1) of the corresponding point B on b in the neighbourhood of B_1, we should have found the same equation between y_1 and y_2.

7. The equation will have real roots if

$$p\sigma > (p + \beta - \delta)(\sigma + \delta - \beta),$$

viz. p, σ having the same sign, this is

$$p\sigma > (p + \beta - \delta)(\sigma + \delta - \beta) :$$

but p, σ having opposite signs, then

$$p\sigma < (p+\beta-\delta)(\sigma+\delta-\beta).$$

These conditions may be written

$$(OR-OB_1)(OS-OD_1)-(OS-OB_1)(OR-OD_1) > \text{ or } < 0,$$

that is,

$$(OS-OR)(OD_1-OB_1) > \text{ or } < 0.$$

But we have $OD_1-OB_1=+$, and therefore, p, σ having the same sign, the condition of reality is $OS > OR$, i.e. S to the right of R; but p, σ having opposite signs, the condition of reality is $OS < OR$, i.e. S to the left of R. Observe that, S lying to the left of R, we cannot have $p=-$, $\sigma=+$, and that the second alternative thus is $p=+$, $\sigma=-$, then $OS<OR$, or S lies to the left of R.

The condition was investigated as above in order to exhibit more clearly the geometrical signification: but of course the original form, or say the equation

$$\frac{1}{p\sigma}\left[1-\left(1+\frac{\beta-\delta}{p}\right)\left(1+\frac{\delta-\beta}{\sigma}\right)\right] > 0$$

gives at once

$$1-\left(1+\frac{\beta-\delta}{p}\right)\left(1+\frac{\delta-\beta}{\sigma}\right) > 0.$$

8. Writing the quadric equation in the form

$$ay_1^2 + 2hy_1y_2 + by_2^2 = 0,$$

we have

$$a = \frac{1}{p}\left(\frac{y_1^2}{4p} + c\right) + 1, \quad b = \frac{1}{\sigma}\left(\frac{y_2^2}{4\sigma} - c\right) + 1, \quad 2h = -\frac{y_1y_2}{p\sigma} - 2;$$

the two values of $y_1 : y_2$ will have the same sign or opposite signs according as $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ have the same sign or opposite signs, and in the case where these have the same sign, then this is also the sign of each of the two values of $y_1 : y_2$.

Or what is the same thing, if $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ are each of them positive, then the two values of $y_1 : y_2$ are each of them positive; if $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ are each of them negative, then the two values of $y_1 : y_2$ are each of them negative; and if $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ have opposite signs, then the two values of $y_1 : y_2$ have opposite signs. Considering the different cases $p, \sigma = +,+; +,-; -,+; -,-$, we find

$p, \sigma = +,+$, then values of $y_1 : y_2$ are $++$ or $--$, according as DR, DS are $++$ or $--$;

$p, \sigma = -,-$, then values of $y_1 : y_2$ are $++$ or $--$, according as DR, DS are $--$ or $++$;

$p, \sigma = +,-$, then values of $y_1 : y_2$ are $+-$, according as DR, DS are $+-$ or $-+$;

$p, \sigma = -,+$, then values of $y_1 : y_2$ are $+-$, according as DR, DS are $-+$ or $+-$;

and in each case the values of $y_1 : y_2$ are $++$ or $--$, if the two distances referred to have opposite signs: $DR = +$ means that R is to the right of, or beyond, D, and so in other cases.

9. The different cases, two real roots as above, are

	$y_1 : y_2$	
$p, \sigma = ++$	$++$	$R \quad S$ $\delta \quad \beta$
$p, \sigma = --$	$--$	$S \quad R$ $\beta \quad \delta$
$p, \sigma = +-$	$+-$	$\delta \quad R \quad S \quad \beta$
$p, \sigma = -+$	$+-$	$\beta \quad S \quad R \quad \delta$

Obviously the cases $p, \sigma = --$ correspond exactly to the cases $p, \sigma = ++$: the only difference is that the concavities, instead of the convexities, of the two curves are turned towards the point O.

10. If the two roots of the quadratic equation are imaginary, then B_1 is a conjugate or isolated position of the link, and B_1, D_1 are isolated points on the curves b and d respectively.

11. If the roots are real, then the three cases $y_1 : y_2 = ++, --,$ and $+-$, may be delineated as in the annexed figures, viz. taking in each case y_1 as positive, that is, imagining B to move upwards from B_1 through an infinitesimal arc of b, then D moves from D_1 through either of two infinitesimal arcs of d, both upwards, both downwards, or the one upwards and the other downwards, as shown in the figures.

[Figures: $y_1 : y_2 = ++$, $y_1 : y_2 = --$, $y_1 : y_2 = +-$ with D' and D'' indicated]

and where it is to be observed that, reversing the sense of the motion of B from B_1, we reverse also the senses of the motion of D from D_1: moreover that, considering D as moving through an infinitesimal arc of d from D_1, we have the like relations thereto of the two infinitesimal arcs of b described by B from B_1. Thus the points B_1 and D_1 are singular points of like character.

If $y_1 : y_2 = ++$, we may say that B_1 (or D_1) is a *for-forwards point*; if $y_1 : y_2 = --$, then that B_1 (or D_1) is a *back-backwards point*; and if $y_1 : y_2 = +-$, then that B_1 (or D_1) is a *back-forwards point*.

12. The separating case between two imaginary roots and two real roots is that of two equal roots, viz. the condition is

$$p\sigma = (p + \beta - \delta)(\sigma + \delta - \beta),$$

or, as this may be written,

$$\frac{1}{p} + \frac{1}{\sigma} = \frac{1}{\delta - \beta} = \frac{1}{c},$$

a condition independent of the aberrancies of curvature of the curves b and d at these points respectively. If each of the curves is a circle, then the curves are concentric circles, and the link BD moves in suchwise that its direction passes always through the common centre of the two circles—or say so that BD is always a radius of the annulus formed by the two circles—and the foregoing condition is satisfied identically for every position of the link.

13. I consider in particular the case where the directrices b, d are circles. The mechanism is then the "three-bar motion" of Mr Roberts; viz. the link $c = BD$ has its extremities to the arms or radii $AB = b$, and $ED = d$, which rotate about the fixed centres A, E at a distance from each other $= a$. Here a, b, c, d are each of them positive; a, b, d may have any values, but then c is at most $= a + b + d$, and if $a > b + d$ then c is at least $= a - b - d$; but if $a =$ or $< b + d$, then c may be $= 0$, viz. it may have any value from 0 to $a + b + d$. And in either case there will be critical values of c. The cases are very numerous. To make an exhaustive enumeration, we may assume d at most $= b$, and in each of the two cases $d < b$ and $d = b$, considering the centre of the circle d as moving from the right of the centre of the circle b towards this centre, we may in the first instance divide as follows:

$d = b$	$d < b$
$\odot d$ exterior to $\odot b$, touches it externally, cuts it, touches it internally, is concentric and thus coincident with it;	$\odot d$ exterior to $\odot b$, touches it externally, cuts it, touches it internally, lies within it, is concentric with it,

and then, in each of these cases, give to the length c of the link its different admissible values.

14. Considering the case $d < b$, then we have (see Plate I., at p. 516), exterior series, the figures 1, 1–2, 2, 2–3, 3, 3–4, 4, viz.

fig. 1, $c = a - b - d$,
fig. 1–2, intermediate,
fig. 2, $c = a - b + d$,
fig. 2–3, intermediate,
fig. 3, $c = a + b - d$,
fig. 3–4, intermediate,
fig. 4, $c = a + b + d$.

15. In figure 1, the curves described by the extremities B and D respectively are each of them a mere point.

In figure 1–2, we have $a+d>b+c$ and $a+b>d+c$. Hence in the course of the motion the arms b, c come into a right line, giving a position B_1D_1 of the link, where B_1 is a two-way point on b and D_1 a summit on d; or rather, there are two such positions symmetrically situate on opposite sides of the axis Ax. And again, in the course of the motion, the arms d, c come into a right line, giving a position $B_1'D_1'$, where D_1' is a two-way point on d and B_1' a summit on b; or rather, there are two such positions symmetrically situate on opposite sides of the axis Ax. Only an arc of the circle b is described, viz. the arc adjacent to d included between the two summits B_1' on b; and in like manner, only an arc of the circle d is described, viz. the arc adjacent to b included between the two summits B_1' on d. The described portions on b and d respectively are to be regarded each of them as a double line or indefinitely thin bent oval: and it is to be observed that for a given position of B (or D) there are two positions of the link BD, each of these positions being assumed by the link in the course of its motion.

16. In figure 2, the two positions B_1D_1 of the link come to coincide together in a single axial position BD, but we still have the other two positions $B_1'D_1'$ of the link, where B_1' is a summit on b, and D_1' a two-way point on d. As regards BD, this is the configuration p, $\sigma=+-$, R, δ; $y_1:y_2=\pm$, and thus each of the axial points B, D is a back-and-forwards point. Thus only the arc $B_1'B_1$ of the circle b is described by the point B, but the whole circumference of the circle d is described by the point D. If we further examine the motion it will appear that, as B moves from the axial point B say to the upper summit B_1 and returns to B, then D starting from the axial point D may describe (and that in either sense, viz. $y_1=+$, then we have $y_2=\pm$) the entire circumference of d, returning to the axial point D; and similarly, as B moves from the axial point B to the lower summit B_1' and returns to B, then D starting as before from the axial point D may describe (and that in either sense, viz. $y_1=-$, then we have $y_2=+$) the entire circumference of d, returning to the axial point D.

17. In figure 2–3, we have still the two positions B_1D_1 symmetrically situate on opposite sides of the axis, B_1 a two-way point on b, D_1 a summit on d; and also the two positions $B_1'D_1'$ symmetrically situate on opposite sides of the axis, B_1' a summit on b, D_1' a two-way point on d. The described portion of b is the arc $B_1'B_1$, and the described portion of d is the arc $D_1'D_1$, these arcs being thus the nearer portions of the two circles respectively. But we have moreover a new pair of critical positions B_1D_1 situate symmetrically on opposite sides of the axis: the configuration is p, $\sigma = +-$, B, R, S, D; and the values of $y_1 : y_2$ are $+-$; that is, B_1 (or D_1) is a back-forwards point: the described portion of b is still the arc $B_1'B_1$, but that of d is, as before, the entire circumference. It is to be observed (as in fig. 2) that properly it is not the entire arc $B_1'B_1$, but each of the half-arcs BB_1, BB_1', which corresponds to the entire circumference of d.

18. In figure 3, the two positions $B_1'D_1'$ come to coincide in the single axial position BD; we have thus only the two positions B_1D_1 symmetrically situate on opposite sides of the axis; the configuration is p, $\sigma = ++$, B, D, R, S; $y_1 : y_2 = ++$. The described portion of b is the entire circumference, and that of d is, as before, the entire circumference. It is to be observed (as in fig. 2) that properly it is not the entire arc $B_1'B_1$, but each of the half-arcs BB_1, BB_1', which corresponds to the entire circumference of d.

19. The figure 3–4 closely corresponds to fig. 1–2, the only difference being that the arcs B_1B_1 and $D_1'D_1'$, which are the described portions of b and d respectively (instead of being the nearer portions, or those with their convexities facing each other) are the further portions, or those with their concavities facing each other, of the two circles respectively.

Finally, in fig. 4, the described portions of the two circles reduce themselves to the axial points B and D respectively.

20. Still assuming $d < b$, and passing over the case of external contact, we come to that in which the circles intersect each other; but this case has to be subdivided. Since the circles intersect, we have $b + d > a$, consistently herewith we may have:

- b, d each $< a$, A, E each outside the lens common to the two circles,

- $b = a$, $d < a$, A outside, E on boundary of the lens,
- $b > a$, $d < a$, A outside, E inside the lens,
- $b > a$, $d = a$, A on boundary of, E inside the lens,
- b, d each $> a$, A, E, each inside the lens;

and in each case we have to consider the different admissible values of c. I omit the discussion of all these cases.

21. Still assuming $d < b$, and passing over the case of internal contact, we come to that of the circle d included within the circle b: we have here again a subdivision of cases; viz. we may have $d > a$, that is, A inside d; $d = a$, that is, A on the circumference of d; or $d < a$, that is, A outside d. The critical values of c, arranged in order of increasing magnitude in these three cases respectively, are:

$d > a$	$d = a$	$d < a$
$b - 2d$,		
$b - d - a$,	$b - d - a$,	
$b + d - a$,	$b + d - a$,	$b - d - a$,
$b - d + a$,	$b - d + a$,	$b + d - a$,
$b + d + a$,	$b + 2d$,	$b - d + a$,
		$b + d + a$.

I attend only to the first case; we have here (see Plate II., at p. 516), interior series, the figures 1, 1–2, 2, 2–3, 3, 3–4, 4.

22. In figure 1, the curves described by the points B, D are each of them a mere point. In figure 1–2, we have two critical positions $B_1 D_1$ situate symmetrically on opposite sides of the axis, B_1 being a summit on b, and D_1 a two-way point on d, and moreover two critical positions $B_1' D_1'$ situate symmetrically on opposite sides of the axis, B_1' being a two-way point on b, and D_1' a summit on d. The described portion of b is the arc $B_1' B_1$, and the described portion of d is the arc $D_1' D_1$, these two arcs being thus the nearer portions of the two circles respectively.

23. In figure 2, the four critical positions coalesce all of them in the axial position BD; the described portions are thus the entire circumferences of the two circles respectively. This is a remarkable case. The configuration is p, $\sigma = ++$; B, D, R, S; $y_1 : y_2 = ++$. Imagine D to move from the axial point D in a given sense round the circle d, say with uniform velocity, then B moves from the axial point B in the same sense but with either of two velocities round the circle b; one of these velocities is at first small but ultimately increases rapidly, the other is at first large but ultimately decreases rapidly, so that the two revolutions of B from the axial point B round the entire circumference to the axial point B correspond each of them to the revolution of D from the axial point D round the entire circumference to the axial point D. And similarly, if we imagine B to move in a given sense from the axial point B round the circle b, say with uniform velocity, then D moves from the axial point D in the same sense but with either of two velocities round the circle d: one of these velocities is at first small but ultimately increases rapidly, the other is at first large but ultimately decreases rapidly, so that the two revolutions from the axial point D round the entire circumference of d to the axial point D correspond each of them to the revolution of B from the axial point B round the entire circumference of b to the axial point B.

24. In figure 2–3, the four critical positions again exist, viz. we have the two positions B_1D_1 symmetrically situate on opposite sides of the axis, B_1 a two-way point on b, D_1 a summit on d; and also the two positions $B_1'D_1'$ symmetrically situate on opposite sides of the axis, B_1' a summit on b, D_1' a two-way point on d. The described portion of b is the arc $B_1'B_1$, and the described portion of d is the arc $D_1'D_1$, these arcs being thus the nearer portions of the two circles respectively. But we have moreover, as in fig. 2–3 of the exterior series, a new pair of critical positions B_1D_1 situate symmetrically on opposite sides of the axis; the configuration is p, $\sigma = +-$, B, R, S, D; and the values of $y_1 : y_2$ are $+-$; the described portion of b is still the arc $B_1'B_1$, but that of d is, as in fig. 2, the entire circumference.

25. In figure 3, the two positions $B_1'D_1'$ again coalesce in the axial position BD; we have thus only the two positions B_1D_1 symmetrically situate on opposite sides of the axis; the configuration is p, $\sigma = ++$, B, D, R, S; $y_1 : y_2 = ++$. The described portion of b is the entire circumference, and that of d is, as in fig. 2, the entire circumference; but observe that, if we imagine B starting from the axial point B in a given sense round the circle b say with uniform velocity, then D moves from the axial point D round the circle d in the opposite sense, and with either of two velocities.

26. In figure 3–4, we have again the two critical positions B_1D_1 symmetrically situate on opposite sides of the axis, B_1 a summit on b, D_1 a two-way point on d: and also the two critical positions $B_1'D_1'$ symmetrically situate on opposite sides of the axis, B_1' a two-way point on b, D_1' a summit on d. The described portion of b is the arc $B_1'B_1$, and the described portion of d the arc $D_1'D_1'$, these arcs being thus the further portions of the two circles respectively.

Finally, in figure 4, the described portions reduce themselves to the two points B, D respectively.

27. The several forms for $d = b$ can be at once obtained from those for $d < b$; the only difference is that several intermediate forms disappear, and the entire series of divisions is thus not quite so numerous.

Part II.

1. The curve-tracing mechanism was devised with special reference to the curves of three-bar motion, viz. the object proposed was that of tracing the curve described by a point K of the link BD, the extremities whereof B and D describe given circles respectively, or more generally by a point K, the vertex of a triangle KBD, whereof the other vertices B and D describe given circles respectively, and that in suchwise that the points B and D might be free to describe the two entire circumferences respectively: but the principle applies to other motions, and I explain it in a general way as follows.

2. Imagine the cranked link BD, composed of the bars $B\beta$ and $D\delta$, rigidly attached $B\beta$ to the top and $D\delta$ to the bottom of the cylindrical disk K (this same letter K is used to denote the axis of the disk), and where $B\beta$ and $D\delta$ may be either parallel or inclined to each other at any given angle, so that, referring the points B, K, D to a horizontal plane, BKD is either a right line, or else K is the vertex of a triangle the other vertices whereof are B and D. The disk K, with the attached bars $B\beta$ and $D\delta$, moves in a horizontal plane: and if the motion of the point B be regulated in any manner by a mechanism lying wholly below B and supported by the bed of the entire mechanism, and similarly if the motion of the point D be regulated in any manner by a mechanism lying wholly above D and supported by a bridge of sufficient length (resting on the bed of the entire mechanism), then the disk K moves in its own horizontal plane unimpeded by other parts of the mechanism: and if we fit the disk K so as to move smoothly within a circular aperture in the arm of a pentagraph, then the pencil of the pentagraph will trace out on a sheet of paper the curve described by the point K on the axis of the disk, or say by the point K of the beam BKD. Of course for the three-bar motion, all that is required is that the point B shall describe a circle, viz. it must be pivoted on to an arm AB, which is itself pivoted at A to the bed: and that the point D shall describe a circle, viz. it must be pivoted on to an arm DE, which is itself pivoted at E to the bridge.

Special arrangements are required to enable the variation of the several lengths AB, BK, KD, DE and ED, and the mechanism thus unavoidably assumes a form which appears complicated for the object intended to be thereby effected.

3. The form of Pentagraph which I use consists of a parallelogram $ABCD$, pivoted together at the points A, B, C, D, the bars AB and DC being above AD and BC. There is a cradle G, rotating about a fixed centre, and which carries between guides the arm AD, which has a sliding motion, so that the lengths GD and GA may be made to have any given ratio to each other. Above the bar DC and sliding along it, we have the arm KL (where K is the circular aperture which fits on to the disk K of the cranked link): and above AB and sliding along it, we have the arm MP which carries the pencil P: of course, in order that the pentagraph may be in adjustment, the points K, G, P must be in line.

953. On the Nine-Points Circle.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 23—25.]

If from the angles A, B, C of a triangle we draw tangents to a conic H, meeting the opposite sides in the points α, α'; β, β'; γ, γ' respectively, then it is known that these six points lie in a conic. In particular, if the conic H reduce itself to a point-pair OO', then we have the theorem that, if from the angles A, B, C, we draw to the point O lines meeting the opposite sides in the points α, β, γ respectively; and to the point O' lines meeting the opposite sides in the points α', β', γ' respectively, then the six points α, α'; β, β'; γ, γ' lie in a conic.

We may enquire the conditions under which this conic becomes a circle. It may be remarked that one of the points say O' remains arbitrary: for if through the points α', β', γ', we draw a conic (or in particular a circle) meeting the three sides respectively in the remaining points α, β, γ, then (by a converse of the general theorem) the lines $A\alpha$, $B\beta$, $C\gamma$ will meet in a point O.

Using trilinear coordinates (x, y, z) and writing $x : y : z = a : b : c$ for the point O, and $x : y : z = a' : b' : c'$ for the point O', it is at once seen that the equation of the conic through the six points is

$$aa'x^2 + bb'y^2 + cc'z^2 - (bc' + b'c)yz - (ca' + c'a)zx - (ab' + a'b)xy = 0;$$

in fact, writing herein successively $x = 0$, $y = 0$, $z = 0$, we see that the equation is satisfied by $x = 0$, $(by - cz)(b'y - c'z) = 0$; by $y = 0$, $(cz - ax)(c'z - a'x) = 0$; and by $z = 0$, $(ax - by)(a'x - b'y) = 0$ respectively. And it is to be observed that the equation may also be written

$$\begin{aligned}(aa'x + bb'y + cc'z)(x + y + z) - (b + c)(b' + c')yz \\ - (c + a)(c' + a')zx - (a + b)(a' + b')xy = 0.\end{aligned}$$

Suppose now that x, y, z represent areal coordinates, viz. that (x, y, z) are proportional to the perpendicular distances of the point from the sides, each divided by the perpendicular distance of the opposite angle from the same side; or, what is the same thing, coordinates such that the equation of the line infinity is $x+y+z = 0$. Then if A, B, C denote the angles of the triangle, the general equation of a circle is

$$(yz\sin^2 A + zx\sin^2 B + xy\sin^2 C) + (\lambda x + \mu y + \nu z)(x + y + z) = 0,$$

where λ, μ, ν are arbitrary coefficients.

Hence, putting this

$$\begin{aligned} = \Theta\{ & -(b+c)(b'+c')yz - (c+a)(c'+a')zx \\ & - (a+b)(a'+b')xy + (aa'x + bb'y + cc'z)(x+y+z)\}, \end{aligned}$$

we must have

$$\begin{aligned} \Theta(b+c)(b'+c') &= -\sin^2 A, \\ \Theta(c+a)(c'+a') &= -\sin^2 B, \\ \Theta(a+b)(a'+b') &= -\sin^2 C; \end{aligned}$$

and then

$$\Theta aa' = \lambda, \quad \Theta bb' = \mu, \quad \Theta cc' = \nu,$$

which last equations determine the values of λ, μ, ν.

Taking a', b', c' at pleasure, we have

$$\begin{aligned} \frac{1}{\Theta} &= -\frac{\sin^2 A}{(b+c)(b'+c')} = -\frac{\sin^2 B}{(c+a)(c'+a')} = -\frac{\sin^2 C}{(a+b)(a'+b')}, \\ a &= \frac{1}{\Theta}\left(\frac{\sin^2 A}{b'+c'} - \frac{\sin^2 B}{c'+a'} - \frac{\sin^2 C}{a'+b'}\right), \\ b &= \frac{1}{\Theta}\left(-\frac{\sin^2 A}{b'+c'} + \frac{\sin^2 B}{c'+a'} - \frac{\sin^2 C}{a'+b'}\right), \\ c &= \frac{1}{\Theta}\left(-\frac{\sin^2 A}{b'+c'} - \frac{\sin^2 B}{c'+a'} + \frac{\sin^2 C}{a'+b'}\right); \end{aligned}$$

viz. a, b, c having these values, the conic through the six points α, β, γ, α', β', γ' is the circle having for its equation

$$yz\sin^2 A + zx\sin^2 B + xy\sin^2 C + \Theta(aa'x + bb'y + cc'z)(x+y+z) = 0;$$

and we may obviously without loss of generality give to Θ any specific value, say $\Theta = 1$.

If $a' = b' = c'$, $= 1$, then we have

$$\begin{aligned} -4a &= (-\sin^2 A + \sin^2 B + \sin^2 C), \\ -4b &= (\sin^2 A - \sin^2 B + \sin^2 C), \\ -4c &= (\sin^2 A + \sin^2 B - \sin^2 C), \end{aligned}$$

or writing for convenience $\Theta = -\frac{1}{4}$, the values of a, b, c are

$\frac{1}{4}(-\sin^2 A+\sin^2 B+\sin^2 C)$, $\frac{1}{4}(\sin^2 A-\sin^2 B+\sin^2 C)$, $\frac{1}{4}(\sin^2 A+\sin^2 B-\sin^2 C)$

respectively. But we have

$$A + B + C = \pi,$$

and thence

$$\begin{aligned}\sin^2 A + \sin^2 B - \sin^2 C &= \sin^2 A + \sin^2 B - \sin^2(A + B)\\ &= -2\sin A \sin B(\sin A \sin B - \cos A \cos B),\\ &= -2\sin A \sin B \cos(A + B),\\ &= 2\sin A \sin B \cos C,\end{aligned}$$

and we thus have

$$a, b, c = \sin B \sin C \cos A,\ \sin C \sin A \cos B,\ \sin A \sin B \cos C,$$

(or, what is the same thing, $a : b : c = \cot A : \cot B : \cot C$), and the equation of the circle is

$$\begin{aligned}&yz \sin^2 A + zx \sin^2 B + xy \sin^2 C\\ &-\tfrac{1}{4}(x \sin B \sin C \cos A+y \sin C \sin A \cos B+z \sin A \sin B \cos C)(x+y+z) = 0.\end{aligned}$$

We thus have $x : y : z = 1 : 1 : 1$ for the point O', and $x : y : z = \cot A : \cot B : \cot C$ for the point O; viz. O' is the point of intersection of the lines from the angles to the mid-points of the opposite sides respectively; and O is the point of intersection of the perpendiculars from the angles on the opposite sides respectively: and the foregoing equation is consequently that of the *Nine-points Circle.*

954. On the Nine-Points Circle of a Plane Triangle.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 25—27.]

I consider the circle which meets the sides of a triangle ABC in the points F, L; G, M; H, N respectively, where ultimately F, G, H are the feet of the perpendiculars let fall from the angles on the opposite sides, and L, M, N are the mid-points of the sides: but in the first instance, they are taken to be arbitrary points.

Taking the radius of the circle to be unity, the coordinates of the point F may be taken to be $\cos F$, $\sin F$, and these may be expressed rationally in terms of the tangent of the half-angle, $f = \tan \frac{1}{2}F$; and similarly for the other points, viz. we may determine the six points by the parameters f, g, h, l, m, n respectively. The sides of the triangle are the lines joining the points L, F; M, G; N, H respectively: thus the equations of the sides are

$$\begin{array}{rll} \text{for } BC: & x(1-lf)+y(l+f)-(1+lf)=0, & \text{say } U=0, \\ \text{“ } CA: & x(1-mg)+y(m+g)-(1+mg)=0, & \text{“ } V=0, \\ \text{“ } AB: & x(1-nh)+y(n+h)-(1+nh)=0, & \text{“ } W=0. \end{array}$$

We have AF, a line through the intersections of BC and CA; its equation is therefore of the form $BV - CW = 0$, and to determine B, C we have $BV_0 - CW_0 = 0$, if V_0, W_0 are the values of V, W belonging to the point F, the coordinates of which are

$$\frac{1-f^2}{1+f^2}, \quad \frac{2f}{1+f^2};$$

we find

$$V_0 = -2(f-g)(f-m), \quad W_0 = 2(h-f)(f-n);$$

and then $B : C = W_0 : V_0$; we thus find the following equations:

$$\begin{array}{rl} \text{that of } AF \text{ is} & BV - CW = 0, \\ \text{“ } BG \text{ “} & C'W - A'U = 0, \\ \text{“ } CH \text{ “} & A''U - B''V = 0, \end{array}$$

where

$$\begin{aligned} B : C &= -(h-f)(f-n) : (f-g)(f-m), \\ C' : A' &= -(f-g)(g-l) : (g-h)(g-n), \\ A'' : B'' &= -(g-h)(h-m) : (h-f)(h-l). \end{aligned}$$

The condition in order that the three lines may meet in a point is $BC'A'' = CA'B''$, viz. this is

$$(f-n)(g-l)(h-m) + (f-m)(g-n)(h-l) = 0,$$

or, as this may also be written,

$$fgh - gh(m+n) - hf(n+l) - fg(l+m) + mn(g+h) + nl(h+f) + lm(f+g) - 2lmn = 0.$$

Similarly, the equation of

$$\begin{aligned} AL &\text{ is } \mathfrak{B}V - \mathfrak{C}W = 0, \\ BM &\text{ `` } \mathfrak{C}'W - \mathfrak{A}'U = 0, \\ CN &\text{ `` } \mathfrak{A}''U - \mathfrak{B}''V = 0, \end{aligned}$$

where

$$\begin{aligned} \mathfrak{B} : \mathfrak{C} &= -(n-l)(h-l) : (l-m)(g-l), \\ \mathfrak{C}' : \mathfrak{A}' &= -(l-m)(f-m) : (m-n)(h-m), \\ \mathfrak{A}'' : \mathfrak{B}'' &= -(m-n)(g-n) : (n-l)(f-n). \end{aligned}$$

The condition in order that the three lines may meet in a point is $\mathfrak{B}\mathfrak{C}'\mathfrak{A}'' = \mathfrak{C}\mathfrak{A}'\mathfrak{B}''$, viz. this is the same condition as before; that is, if the lines AF, BG, CH meet in a point, then also the lines AL, BM, CN will meet in a point.

In the case of the nine-points circle, we have MN, NL, LM parallel to LF, MG, NH, respectively: the equation of MN is

$$x(1-mn) + y(m+n) - (1+mn) = 0,$$

and this is parallel to LF, if

$$\frac{1-mn}{1-lf} = \frac{m+n}{l+f}, \quad \text{i.e. if } m+n-l-mnf = 0.$$

Hence, for the nine-points circle, we have

$$l+F = M+N, \quad M+G = N+L,$$

or, as these equations may be written,

$$l = \tfrac{1}{2}(M+N+F-G), \quad m = \tfrac{1}{2}(N+L+G-H), \quad n = \tfrac{1}{2}(L+M+H-F);$$

viz. it thus appears that the radii to the points L, M, N respectively, or say the radii L, M, N, bisect the angles made by the radii G and H, H and F, F and G respectively.

It may be added that we have

$$\begin{aligned} m+n-l+lmn &= f\{1-mn+l(m+n)\}, \\ n+l-m+lmn &= g\{1-nl+m(n+l)\}, \\ l+m-n+lmn &= h\{1-lm+n(l+m)\}, \end{aligned}$$

viz. f, g, h are expressible each of them as a rational function of l, m, n.

955. The Numerical Value of $\Pi_i = \Gamma(1+i)$.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 36—38.]

I do not know whether the numerical value of Π_x for an imaginary value of x has ever been calculated; and I wish to calculate it for a simple case $x = i$.

We have

$$\Pi_i = \Gamma(1+i) = i\Gamma(i) = i\int_0^\infty e^{-t}t^{i-1}\,dt;$$

and therefore, writing $t = e^{-\theta}$, $dt = -e^{-\theta}\,d\theta$,

$$\begin{aligned}\Pi_i &= i\int_{-\infty}^0 e^{e^{-\theta}}e^{-\theta(i-1)}(-e^{-\theta}\,d\theta)\\ &= i\int_0^\infty e^{-e^{-\theta}}e^{-i\theta}\,d\theta.\end{aligned}$$

Writing $e^{-e^{-\theta}} = \phi_0 - \phi_1\theta + \phi_2\theta^2 - \&c.$ (where $\phi_0 = e^{-1}$, and the law of the coefficients will be considered presently), we have

$$\Pi_i = i\int_0^\infty (\phi_0 - \phi_1\theta + \phi_2\theta^2 - \&c.)(\cos\theta - i\sin\theta)\,d\theta;$$

and observing that

$$\int_0^\infty \theta^n\cos\theta\,d\theta = n!\cos\tfrac{1}{2}n\pi,\qquad \int_0^\infty \theta^n\sin\theta\,d\theta = n!\sin\tfrac{1}{2}n\pi,$$

we have

$$\Pi_i = i\{(\phi_0 - 1!\phi_2 - 3!\phi_4 + 5!\phi_6 + \&c.) + i(1!\phi_1 - 2!\phi_3 - 4!\phi_5 + 6!\phi_7 + \&c.)\},$$

($\phi_2 = 0$, and in the subsequent terms the imaginary part is taken with a negative sign in order to obtain positive values for ϕ_2, ϕ_4, &c.), $= R(\cos\Theta + i\sin\Theta)$, if R be the modulus and Θ the amplitude.

We have

$$R^2 = (\phi_0 - 1!\phi_2 - 3!\phi_4 + \&c.)^2 + (1!\phi_1 - 2!\phi_3 - 4!\phi_5 + \&c.)^2,$$

which may be calculated directly. The value of Θ admits, however, of a finite expression.

956. On Richelot's Integral of the Differential Equation $\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0$.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 42—47.]

In the Memoir "Einige neue Integralgleichungen des Jacobischen Systems Differentialgleichungen," Crelle, t. xxv. (1843), pp. 97—118, Richelot, working with the more general differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0,$$

where X, Y, Z are quartic functions of x, y, z respectively, obtains the integral of this equation in a form involving the quotients of the two different hyperelliptic integrals belonging to the equation. And he remarks that the result may be applied to the integral of the equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

where X, Y are each of them a quartic function of x, y respectively; in fact, writing herein $z =$ constant, and therefore $dz = 0$, the equation becomes the equation in question. But the result is thereby presented in a form which is not immediately intelligible: and in the memoir "Sur l'intégration de l'équation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

où X, Y sont les mêmes fonctions de x et de y respectivement," Crelle, t. c. (1887), pp. 290—294, I obtained the integral under a more simple form. I propose to reproduce the investigation, with some explanations which were omitted in the original memoir.

1. If in the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

we write $x = \dfrac{a\theta^2 + 2b\theta + c}{a'\theta^2 + 2b'\theta + c'}$, then y will be the same function of ϕ, where ϕ is determined by the equation

$$\frac{d\theta}{\sqrt{\Theta}} + \frac{d\phi}{\sqrt{\Phi}} = 0;$$

and we thus arrive at the standard form belonging to the differential equation in question.

Working with the standard form, if θ, ϕ are coordinates, the differential equation determines a family of curves; taking any curve of the family, we may from any given point of this curve determine a new point thereof by means of the integral equation; and the integral equation is in fact an equation between the parameters θ, ϕ belonging to the two points respectively.

2. I use the accentuate letters a, b, c, f, g, h to denote the coefficients of the sextic function

$$(a, b, c, f, g, h \between \theta, 1)^6,$$

or, what is the same thing, the coefficients of the quartic function $(a, b, c, f, g, h \between x, y)^4$, where $x : y = \theta : 1$.

The differential equation is

$$\frac{d\theta}{\sqrt{\Theta}} + \frac{d\phi}{\sqrt{\Phi}} = 0,$$

where $\Theta = (a, b, c, f, g, h \between \theta, 1)^6$, and Φ is the same function of ϕ.

3. I remark that, starting from the differential equation, and for the moment using u instead of θ, so that the equation is

$$\frac{du}{\sqrt{U}} + \frac{dv}{\sqrt{V}} = 0,$$

then from the first equation we have $v =$ const., and from the second equation we have $u =$ const., and it thus appears that the integral of the differential equation may be written in the form $F(u, v) = 0$, where $F(u, v)$ is a symmetrical function of u, v. But the equation may also be written in the form $v = f(u)$, where f denotes a functional symbol: and from the symmetry we have $u = f(v)$, that is, $f(f(u)) = u$, or f is a periodic function of the second order.

957. Illustrations of Sylow's Theorems on Groups.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 59—62.]

The theorems 1, 2, and 3 in the paper Sylow, "Theoremes sur les groupes de Substitutions," Math. Ann. t. V. (1872), pp. 584—594, apply to groups in general, and not only to groups of substitutions. They are as follows:

THEOREM 1. If n^α be the highest power of the prime number n which divides the order of a group G, this group contains a group g of the order n^α: if, moreover, $n^\alpha\nu$ is the order of the highest group contained in G, the operations whereof are permutable with the group g, then the order of G is of the form $n^\alpha\nu(nk+1)$.

(I write k for Sylow's p, since it is convenient to have p to denote a prime number; and for Sylow's "Substitutions" I write "Operations.")

THEOREM 2. Everything being as in the preceding theorem, the group G contains precisely $nk+1$ distinct groups of the order n^α; and these are obtained by transforming any one of them by the operations of G, each group being given by $n^\alpha\nu$ distinct transformations.

THEOREM 3. If the order of a group is n^α, n being prime, then any operation S whatever of the group may be expressed by the formula

$$S = G_0^{\theta_0} G_1^{\theta_1} G_2^{\theta_2} \dots G_{\alpha-1}^{\theta_{\alpha-1}},$$

where

$$G_\lambda^{n^\lambda} = G_0^{a_0} G_1^{a_1} \dots G_{\lambda-1}^{a_{\lambda-1}},$$

and where

$$G_\mu G_\lambda = G_0^{c_0} G_1^{c_1} \dots G_\lambda^{c_\lambda} \cdot G_\lambda \cdot G_0^{d_0} G_1^{d_1} \dots G_{\mu-1}^{d_{\mu-1}}.$$

But at present I attend only to the theorems 1 and 2.

For instance, consider the group G of the order $N = 6$,

$$1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2 \quad (\alpha^2 = 1, \beta^3 = 1, \alpha\beta^2 = \beta\alpha, \alpha\beta = \beta^2\alpha).$$

Here $n = 2$ or 3: if $n = 2$, we have $N = n^\alpha\nu(nk+1) = 2 \cdot 1 \cdot (2+1)$; if $n = 3$, we have $N = n^\alpha\nu(nk+1) = 3 \cdot 2 \cdot 1$.

First, $n = 2$; we should have a group g of the order 2; one such group is $(1, \alpha)$, and the only group the substitutions whereof are permutable with $(1, \alpha)$ is the group $(1, \alpha)$ itself: for, taking any other operation of the group, for instance β, it is not true that $\beta(1,\alpha) = (1,\alpha)\beta$; in fact, the left-hand is $(\beta,\beta\alpha)$ and the right-hand is $(\beta,\alpha\beta)$ or $(\beta,\beta^2\alpha)$: hence $n^\alpha\nu = 2\nu = 2$, or ν is $= 1$.

Hence also, by theorem 2, there should be 3 groups of the order 2 such as $(1,\alpha)$, viz. these are $(1,\alpha)$, $(1,\alpha\beta)$, $(1,\alpha\beta^2)$, derived from $(1,\alpha)$ as follows:

$$\begin{aligned} 1\cdot(1,\alpha)\cdot 1^{-1} &= (1,\alpha),\\ \alpha\cdot(1,\alpha)\cdot\alpha^{-1} &= (1,\alpha),\\ \beta\cdot(1,\alpha)\cdot\beta^{-1} &= (1,\alpha\beta),\\ \beta^2\cdot(1,\alpha)\cdot\beta^{-2} &= (1,\alpha\beta^2),\\ \alpha\beta\cdot(1,\alpha)\cdot(\alpha\beta)^{-1} &= (1,\alpha\beta^2),\\ \alpha\beta^2\cdot(1,\alpha)\cdot(\alpha\beta^2)^{-1} &= (1,\alpha\beta), \end{aligned}$$

since $\beta^{-1} = \beta^2$, and therefore the second term is $\beta^2\alpha\beta^2 = \alpha\beta^2\cdot\beta^2 = \alpha\beta$,

$\beta^{-2} = \beta$, and therefore the second term is $\beta\alpha\beta = \alpha\beta\cdot\beta = \alpha\beta^2$,

$(\alpha\beta)^{-1} = \alpha\beta$, and therefore the second term is $\alpha\beta\alpha\alpha\beta = \alpha\beta^2\alpha\alpha\beta = \alpha\beta$,

$(\alpha\beta^2)^{-1} = \alpha\beta^2$, and therefore the second term is $\alpha\beta^2\alpha\alpha\beta^2 = \alpha\beta^2\alpha\alpha\beta^2 = \alpha\beta$;

viz. the derivatives are $(1,\alpha)$, $(1,\alpha\beta)$, $(1,\alpha\beta^2)$, each twice.

Secondly, $n = 3$; there should be here a group of the order 3, viz. this is $(1,\beta,\beta^2)$. The group, the substitutions whereof are permutable with $(1,\beta,\beta^2)$, is the entire group $(1,\beta,\beta^2,\alpha,\alpha\beta,\alpha\beta^2)$; in fact, taking any substitution thereof, for instance α, we have $\alpha(1,\beta,\beta^2) = (1,\beta,\beta^2)\alpha$, viz. the left-hand side is $(\alpha,\alpha\beta,\alpha\beta^2)$, and the right-hand side is $(\alpha,\beta\alpha,\beta^2\alpha) = (\alpha,\alpha\beta^2,\alpha\beta)$, which is the left-hand side, the change of order being immaterial; this is the meaning of the expression used, "the operations whereof are permutable with the group g." Hence, we have $n^\alpha\nu = 3\nu = 6$, or $\nu = 2$; and thence also $nk+1 = 3k+1 = 1$, viz. $k = 0$. There is thus only a single group of the order 3, viz. the group $(1,\beta,\beta^2)$.

As another instance, I take the group of the order 12 formed by the positive substitutions of four letters, viz. these are

$$1,\quad ab\cdot cd,\quad ac\cdot bd,\quad ad\cdot bc,$$

$$abc,\quad acb,\quad abd,\quad adb,\quad acd,\quad adc,\quad bcd,\quad bdc.$$

Here, taking $n = 2$, we have $N = n^\alpha \nu(nk+1) = 2^2 \cdot 3 \cdot 1$; there is a group g of the order 4, viz. this is

$$(1, ab \cdot cd, ac \cdot bd, ad \cdot bc),$$

and the greatest group, the substitutions whereof are permutable with this group g, is the entire group of the order 12; thus, considering any substitution thereof, for instance abc, we have

$$abc \left| \begin{matrix} 1 & ab \cdot cd & ac \cdot bd & ad \cdot bc \\ abc & acd & bdc & adb \end{matrix} \right| = \left| \begin{matrix} 1 & ab \cdot cd & ac \cdot bd & ad \cdot bc \\ abc & bdc & adb & acd \end{matrix} \right| abc,$$

viz. the left-hand is (abc, acd, bdc, adb), the right-hand is (abc, bdc, adb, acd); hence $n^\alpha \nu = 4\nu = 12$ or $\nu = 3$; whence also $nk + 1 = 2k + 1 = 1$: and thus the foregoing group g is the only group of the order 4.

Similarly, taking $n = 3$, we have $N = n^\alpha \nu(nk+1) = 3 \cdot 1 \cdot 4$. There is a group g of the order 3, say $(1, abc, acb)$; the greatest group, the substitutions whereof are permutable with g, is the group g itself, viz. we have $n^\alpha \nu = 3\nu = 3$, or $\nu = 1$; and then $n^\alpha(k+1) = 3k+1 = 4$: there are thus 4 groups of the order 3, viz. these are

$$(1, abc, acb), \quad (1, abd, adb), \quad (1, acd, adc), \quad (1, bcd, bdc).$$

Reverting to the before-mentioned group of the order 6, this not only contains each of the groups $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$ of order 2, and the group $(1, \beta, \beta^2)$ of order 3, but it is also the permutable product of any one of the groups $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$ of the order 2 by the group $(1, \beta, \beta^2)$ of the order 3; but it is the permutable product of a group of order 2 by a group of order 3, say it is

$$G = (1, \alpha)(1, \beta, \beta^2) = (1, \beta, \beta^2)(1, \alpha).$$

A group, which is thus a permutable product of two factors, is said to be a *true* product.

958. On the Surface of the Order n which passes through a given Cubic Curve.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 79, 80.]

We may through a given cubic curve draw a surface of any given order n; such surface depends upon a certain number of constants, and the question arises to find the number of the constants. As to this, observe that the number of constants is equal to that of the constants in the equation of the surface of the order n, diminished by the number of the conditions in order that the surface may pass through the given cubic curve. I propose to show that the number of conditions is $= 3n + 1$.

We may through any $3n + 1$ points of the cubic curve draw a surface of the order n; this surface meets the cubic curve in $3n$ points, and therefore in one additional point; or, what is the same thing, the surface contains the cubic curve.

Taking the cubic curve to be the intersection of the quadric surface $y^2 - zx = 0$ and the cubic surface $yw - z^2 = 0$, we may write

$$x : y : z : w = \theta^3 : \theta^2 : \theta : 1,$$

and then any surface of the order n meets the cubic curve in $3n$ points, determined by an equation of the degree $3n$ in θ; hence if the surface pass through $3n + 1$ points of the cubic curve, it passes through the whole curve.

The number of conditions thus is $= 3n + 1$; and the number of constants in the surface of the order n is $= \frac{1}{6}(n + 1)(n + 2)(n + 3)$; whence the number of the constants in the surface is

$$\tfrac{1}{6}(n + 1)(n + 2)(n + 3) - (3n + 1).$$

959. Note on Plucker's Equations.

[From the Messenger of Mathematics, vol. xxIV. (1895), pp. 23, 24.]

It is well known that if

$$A, B, C, D = 2, 3, 6, 8,$$

then the equations

$$\begin{aligned} n &= m^2 - m - A\delta - B\kappa, \\ \iota &= 3m^2 - 6m - C\delta - D\kappa, \\ m &= n^2 - n - A\iota - B\tau, \\ \kappa &= 3n^2 - 6n - C\iota - D\tau \end{aligned}$$

imply

$$\tfrac{1}{2}(m-1)(m-2) - \delta - \kappa = \tfrac{1}{2}(n-1)(n-2) - \iota - \tau,$$

viz. this last equation is the only relation independent of A, B, C, D which exists between the six quantities.

If A, B, C, D are arbitrary, the foregoing equations give

$$\begin{aligned} n - m^2 + m &= -A\delta - B\kappa, \\ 3m^2 - 6m - \iota &= C\delta + D\kappa, \end{aligned}$$

and thence eliminating κ, we have

$$B(3m^2 - 6m - \iota) - D(n - m^2 + m) = (BC - AD)\delta,$$

or say

$$\delta = \frac{B(3m^2 - 6m - \iota) - D(n - m^2 + m)}{BC - AD}.$$

Similarly, the same two equations eliminating δ give

$$\kappa = \frac{C(n - m^2 + m) - A(3m^2 - 6m - \iota)}{BC - AD};$$

and the equations involving ι, τ give in like manner

$$\iota = \frac{B(3n^2 - 6n - \kappa) - D(m - n^2 + n)}{BC - AD},$$

$$\tau = \frac{C(m - n^2 + n) - A(3n^2 - 6n - \kappa)}{BC - AD}.$$

But the relation $\frac{1}{2}(m-1)(m-2) - \delta - \kappa = \frac{1}{2}(n-1)(n-2) - \iota - \tau$ must be satisfied identically; and substituting the foregoing values of δ, κ, ι, τ, the equation becomes identically true.

960. On the Circle of Curvature at any point of an Ellipse.

[From the Messenger of Mathematics, vol. xxIV. (1895), pp. 47, 48.]

Let

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1.$$

The equation of the circle of curvature at the point (x, y) of the ellipse is

$$X^2+Y^2+\frac{2}{a^2b^2}(-b^2Xu^2+a^2Yv^2)+u^2(a^2-2b^2)+v^2(b^2-2a^2)=0,$$

where

$$u^2=\frac{x^2}{a^4}+\frac{y^2}{b^4},\quad u=\frac{x}{a^2},\quad v=\frac{y}{b^2}.$$

Writing $X=x+\xi$, $Y=y+\eta$, and observing that

$$\frac{x\xi}{a^2}+\frac{y\eta}{b^2}=0,$$

the equation becomes

$$\xi^2+\eta^2+\frac{2}{a^2b^2}(-b^2(x+\xi)u^2+a^2(y+\eta)v^2)+\text{terms independent of } \xi,\eta=0,$$

and hence reducing by means of the foregoing relations, we find the equation of the circle of curvature in the form

$$\xi^2+\eta^2-\frac{2(a^2-b^2)}{a^2b^2}\left(\frac{b^2x}{a^2}\xi+\frac{a^2y}{b^2}\eta\right)\frac{1}{u^2}=0.$$

Or, what is the same thing, the circle of curvature at the point (x, y) has for its equation

$$(X-x)^2+(Y-y)^2=\frac{2(a^2-b^2)}{a^2b^2}\left(\frac{b^2x}{a^2}(X-x)+\frac{a^2y}{b^2}(Y-y)\right)\frac{1}{u^2}.$$

The centre is given by

$$X=x-\frac{(a^2-b^2)x}{a^4u^2},\quad Y=y-\frac{(a^2-b^2)y}{b^4u^2},$$

and the radius of curvature is

$$\rho=\frac{(a^2-b^2)^{3/2}}{a^2b^2}\cdot\frac{xy}{u^3}.$$

961. A Trigonometrical Identity.

[From the Messenger of Mathematics, vol. xxIV. (1895), pp. 49—51.]

The following was proposed as a Senate House Problem: Given the equations

$$\begin{aligned} a\cos(\beta+\gamma)+b\cos(\beta-\gamma)+c=0, \\ a\cos(\gamma+\alpha)+b\cos(\gamma-\alpha)+c=0, \\ a\cos(\alpha+\beta)+b\cos(\alpha-\beta)+c=0, \end{aligned}$$

to eliminate α, β, γ, and find the relation between a, b, c.

Writing the equations in the forms

$$\begin{aligned} (a+b)\cos\beta\cos\gamma+(-a+b)\sin\beta\sin\gamma+c=0, \\ (a+b)\cos\gamma\cos\alpha+(-a+b)\sin\gamma\sin\alpha+c=0, \\ (a+b)\cos\alpha\cos\beta+(-a+b)\sin\alpha\sin\beta+c=0, \end{aligned}$$

and writing for shortness

$$p=a+b, \quad q=-a+b, \quad r=c,$$

the equations become

$$\begin{aligned} p\cos\beta\cos\gamma+q\sin\beta\sin\gamma+r=0, \\ p\cos\gamma\cos\alpha+q\sin\gamma\sin\alpha+r=0, \\ p\cos\alpha\cos\beta+q\sin\alpha\sin\beta+r=0. \end{aligned}$$

We hence find

$$\begin{aligned} \cos\alpha:\sin\alpha:1 &= (q\sin\beta\sin\gamma+r):-(p\cos\beta\cos\gamma+r):(p\cos\beta\sin\gamma-q\sin\beta\cos\gamma \\ \cos\beta:\sin\beta:1 &= (q\sin\gamma\sin\alpha+r):-(p\cos\gamma\cos\alpha+r):(p\cos\gamma\sin\alpha-q\sin\gamma\cos\alpha \\ \cos\gamma:\sin\gamma:1 &= (q\sin\alpha\sin\beta+r):-(p\cos\alpha\cos\beta+r):(p\cos\alpha\sin\beta-q\sin\alpha\cos \end{aligned}$$

Writing $x=\tan\frac{1}{2}\alpha$, $y=\tan\frac{1}{2}\beta$, $z=\tan\frac{1}{2}\gamma$, we have

$$\cos\alpha=\frac{1-x^2}{1+x^2}, \quad \sin\alpha=\frac{2x}{1+x^2}, \quad \cos\beta=\frac{1-y^2}{1+y^2}, \quad \sin\beta=\frac{2y}{1+y^2}, \quad \cos\gamma=\frac{1-}{1+}$$

Substituting these values, the first equation becomes

$$p(1-x^2)(1-y^2)(1-z^2)+4qxyz+r(1+x^2)(1+y^2)(1+z^2)=0,$$

and similarly the other two equations may be obtained by cyclical interchanges of x, y, z.

The elimination of x, y, z from these three equations leads to the required relation between a, b, c.

We have a, b, $2c$ equal to the values given by the determinant; and then, substituting in the first equation, we have

$$x(1+2y^2z^2)(x^2-yz)+x(y^2+z^2)(-1+x^2yz)+(1-x^2)(yz^2+yz^2)=0,$$

which is

$$(x-y)(x-z)[x+y+z-(yz+zx+xy)xyz]=0,$$

viz. our relation between x, y, z is

$$x+y+z-(yz+zx+xy)xyz=0,$$

or, what is the same thing,

$$\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=xyz(\frac{1}{yz}+\frac{1}{zx}+\frac{1}{xy}).$$

The elimination hence gives

$$a^2-b^2+2bc=0,$$

which is the required relation.

The equation to be proved may also be written

$$\begin{vmatrix} \cos(\beta+\gamma) & \cos(\beta-\gamma) & 1 \\ \cos(\gamma+\alpha) & \cos(\gamma-\alpha) & 1 \\ \cos(\alpha+\beta) & \cos(\alpha-\beta) & 1 \end{vmatrix}=0,$$

which is at once seen to be true.

962. Coordinates versus Quaternions.

[From the Proceedings of the Royal Society of Edinburgh, vol. xx. (1895), pp. 271—275. Read July 2, 1894.]

It is contended that Quaternions (as a method) are more comprehensive and less laborious than Cartesian Coordinates. It is not easy to meet this contention by a formal reply; but it is very easy to show that, as a matter of fact, in the solution of geometrical problems, the method of Coordinates is more powerful and more convenient than that of Quaternions.

The question was raised in the Society in connexion with a theorem, "Given the edges of a tetrahedron, find the radius of the circumscribed sphere." I gave a solution by coordinates; and Professor Tait, while admitting that the coordinates were competent to deal with the problem, gave as his view, that they were inappropriate to it. I do not understand the appropriateness or inappropriateness of a method; the question is, which method gives the more easily the result required.

The quaternion solution depends on the identity

$$V\alpha\beta\gamma = S\alpha\beta\gamma S\alpha\beta - S\alpha\gamma S\beta + S\beta\gamma S\alpha,$$

and the condition, in order that the vectors α, β, γ may be coplanar, is $S\alpha\beta\gamma = 0$, and Professor Tait contrasts this with the prolixity of the corresponding coordinate formula.

But the vector formula $V\alpha\beta\gamma = \ldots$ is not the formula which should be used in the solution of the problem. The proper formula is

$$\alpha^2\beta^2\gamma^2 = (S\beta\gamma)^2\alpha^2 + (S\gamma\alpha)^2\beta^2 + (S\alpha\beta)^2\gamma^2 - 2S\beta\gamma S\gamma\alpha S\alpha\beta,$$

which, when $S\alpha\beta\gamma = 0$, reduces to

$$0 = (S\beta\gamma)^2\alpha^2 + (S\gamma\alpha)^2\beta^2 + (S\alpha\beta)^2\gamma^2 - 2S\beta\gamma S\gamma\alpha S\alpha\beta.$$

This is the formula which should be used; and when translated into coordinates, it gives at once the required result.

I would further remark that the quaternion formula, even when it is the most appropriate, requires to be translated into coordinates in order to be understood. As an illustration, I compare a quaternion formula to a pocket-map—a capital thing to put in one's pocket, but which for use must be unfolded: the formula, to be understood, must be translated into coordinates.

I remark that the imaginary of ordinary algebra—for distinction call this θ—has no relation whatever to the quaternion symbols i, j, k; in fact, in the general quaternion $w+ix+jy+kz$, the coefficients w, x, y, z are, and must be, ordinary real or imaginary magnitudes.

And for the comparison of the two solutions, we have

$$\alpha = i(x_2 - x_1) + j(y_2 - y_1), \quad \beta = i(x_4 - x_3) + j(y_4 - y_3).$$

But this example of a plane theorem is a trivial one, given only for the sake of completeness.

963. Note on Dr Muir's Paper, "A Problem of Sylvester's in Elimination."

[From the Proceedings of the Royal Society of Edinburgh, vol. xx. (1895), pp. 306—308.]

The problem is, given the equations

$$\begin{aligned} U &= ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0, \\ V &= a'x^2 + b'y^2 + c'z^2 + 2f'yz + 2g'zx + 2h'xy = 0, \\ W &= a''x^2 + b''y^2 + c''z^2 + 2f''yz + 2g''zx + 2h''xy = 0, \end{aligned}$$

to eliminate (x, y, z), and obtain the condition for the coexistence of the three equations.

Dr Muir obtains the resultant in the form

$$\begin{vmatrix} A & H & G & A' & H' & G' \\ H & B & F & H' & B' & F' \\ G & F & C & G' & F' & C' \\ A' & H' & G' & A'' & H'' & G'' \\ H' & B' & F' & H'' & B'' & F'' \\ G' & F' & C' & G'' & F'' & C'' \end{vmatrix} = 0,$$

where A, B, C, F, G, H are the inverse coefficients of U, and similarly for V and W.

This is a perfectly correct result; but it is to be observed that the resultant is of the degree 12 in the coefficients of each of the functions U, V, W, and is therefore of the degree 36 in all. But the resultant of three quadric surfaces is known to be of the degree 24 only; and the foregoing form thus contains the resultant as a factor, multiplied by an extraneous factor of the degree 12.

The explanation is as follows: the condition that the three quadrics may have a common point is not that the determinant may vanish, but that the determinant and its first minors may all of them vanish; or, what is the same thing, that the matrix

$$\begin{pmatrix} A & H & G & A' & H' & G' \\ H & B & F & H' & B' & F' \\ G & F & C & G' & F' & C' \\ A' & H' & G' & A'' & H'' & G'' \\ H' & B' & F' & H'' & B'' & F'' \\ G' & F' & C' & G'' & F'' & C'' \end{pmatrix}$$

may be of rank 3.

But Dr Muir's result may be transformed as follows: writing

$$L = -ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy,$$
$$M = ax^2 - by^2 + cz^2 + 2fyz + 2gzx + 2hxy,$$
$$N = ax^2 + by^2 - cz^2 + 2fyz + 2gzx + 2hxy,$$

the equations $U = 0$, $V = 0$, $W = 0$ may be replaced by $L = 0$, $M = 0$, $N = 0$; and we have identically

$$L + fP = x(Ax + Hy + Gz),$$
$$M + gQ = y(Hx + By + Fz),$$
$$N + hR = z(Gx + Fy + Cz),$$

where P, Q, R are certain quadric functions. But L, M, N are sums of mere numerical multiples of U, V, W, viz. we have

$$2L = -aU + bV + cW,$$
$$2M = aU - bV + cW,$$
$$2N = aU + bV - cW,$$

so that the original equations $U = 0$, $V = 0$, $W = 0$ are equivalent to and may be replaced by $L = 0$, $M = 0$, $N = 0$.

Muir shows that we have identically

$$L - fP = x(Ax + Hy + Gz),$$
$$M - gQ = y(Hx + By + Fz),$$
$$N - hR = z(Gx + Fy + Cz),$$

and hence the condition for a common point is obtained in the required form.

0.5 954. On the Nine-Points Circle of a Plane Triangle (concluded)

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 26–38.]

where

$$\begin{aligned} B' : C' &= -(h-f)(f-n) : (f-g)(f-m), \\ C' : A' &= -(f-g)(g-l) : (g-h)(g-n), \\ A'' : B'' &= -(g-h)(h-m) : (h-f)(h-l). \end{aligned}$$

The condition in order that the three lines may meet in a point is $B'C'A'' = C'A'B''$, viz. this is

$$(f-n)(g-l)(h-m) + (f-m)(g-n)(h-l) = 0,$$

or, as this may also be written,

$$fgh - gh(m+n) - hf(n+l) - fg(l+m) + mn(g+h) + nl(h+f) + lm(f+g) - 2lmn = 0.$$

Similarly, the equation of

$$\begin{array}{rcl} AL & \text{is} & \mathfrak{B}'F - \mathfrak{C}'W = 0, \\ BM & ,, & \mathfrak{C}'W - \mathfrak{A}'U = 0, \\ CN & ,, & \mathfrak{A}'U - \mathfrak{B}''V = 0, \end{array}$$

where

$$\begin{aligned} \mathfrak{B}' : \mathfrak{C}' &= -(n-l)(h-l) : (l-m)(g-l), \\ \mathfrak{C}' : \mathfrak{A}' &= -(l-m)(f-m) : (m-n)(h-m), \\ \mathfrak{A}'' : \mathfrak{B}'' &= -(m-n)(g-n) : (n-l)(f-n). \end{aligned}$$

The condition in order that the three lines may meet in a point is $\mathfrak{B}\mathfrak{B}'\mathfrak{A}'' = \mathfrak{C}\mathfrak{A}'\mathfrak{B}''$, viz. this is the same condition as before; that is, if the lines AF, BG, CH meet in a point, then also the lines AL, BM, CN will meet in a point.

In the case of the nine-points circle, we have MN, NL, LM parallel to LF, MG, NH, respectively: the equation of MN is

$$x(l-mn) + y(m+n) - (l+mn) = 0,$$

and this is parallel to LF, if ....

Hence, for the nine-points circle, we have

$$L + F = M + N, \quad M + G = N + L,$$

or, as these equations may be written, ...

viz. it thus appears that the radii to the points L, M, N respectively, or say the radii L, M, N, bisect the angles made by the radii G and H, H and F, F and G respectively.

It may be added that we have

$$\begin{aligned}
m + n - l + lmn &= f\{1 - mn + l(m + n)\},\\
n + l - m + lmn &= g\{1 - nl + m(n + l)\},\\
l + m - n + lmn &= h\{1 - lm + n(l + m)\},
\end{aligned}$$

viz. f, g, h are expressible each of them as a rational function of l, m, n.

C. XIII. 66

0.6 955. The Numerical Value of $\Pi i = \Gamma(1+i)$

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 36–38.]

I do not know whether the numerical value of Πx for an imaginary value of x has ever been calculated; and I wish to calculate it for a simple case $x = i$.

We have

$$\frac{1+z}{z}\Pi z = \frac{1+z}{z} \cdot \frac{z}{1} \cdot \frac{z+1}{2} \cdot \frac{z+2}{3} \cdots ,$$

here hl denotes the hyperbolic logarithm. Hence, in particular, when $z = i$, we have

$$\begin{aligned} &\frac{1+i}{i}\Pi i = \frac{1+i}{i} \cdot \frac{i}{1} \cdot \frac{i+1}{2} \cdot \frac{i+2}{3} \cdots \\ &= \sqrt{(1+i)} \cdot \frac{\cos\theta_1 + i\sin\theta_1}{\cos\phi_1 - i\sin\phi_1} \cdot \frac{\cos\theta_2 + i\sin\theta_2}{\cos\phi_2 - i\sin\phi_2} \cdot \frac{\cos\theta_3 + i\sin\theta_3}{\cos\phi_3 - i\sin\phi_3} \cdots \end{aligned}$$

($\phi_1 = 0$, and in the subsequent terms the imaginary part is taken with a negative sign in order to obtain positive values for ϕ_2, ϕ_3, &c.), $= H(\cos\Theta + i\sin\Theta)$, if H be the modulus and Θ the sum.

We have

$$H^2 = \frac{1}{i\Pi i \cdot \Pi(-i)} = \frac{\sin \pi i}{\pi i} = \frac{e^{\pi} - e^{-\pi}}{2\pi},$$

which may be calculated directly. The value of H admits, however, of a finite expression, viz. we have

$$H^2 = \frac{1}{i\Pi i \cdot \Pi(-i)} = \frac{\sin \pi i}{\pi i} = \frac{e^{\pi} - e^{-\pi}}{2\pi},$$

the approximate numerical value is $H = 1.9173$, viz. we have

$$e^{\pi} - e^{-\pi} = 23.141 - 0.043 = 23.098 : \quad \log = 1.3635744, \quad -\log 2\pi = 1.201819,$$

$$\log H^2 = 0.5653935, \quad \log H = 0.2826967, \quad \text{or } H = 1.9173.$$

$$\tan\theta_1 = 1, \quad \tan\theta_2 = \tfrac{1}{2}, \quad \tan\theta_3 = \tfrac{1}{3}, \quad \text{\&c.},$$

whence

$$\theta_n = \arctan\frac{1}{n}.$$

We have also

$$\tan\phi_n = \frac{1}{n}\tan\theta_n,$$

where M is the modulus for the Briggian logarithms,

$$M = 0.4342944 \quad \log = 1.6377843,$$

$$\pi = 3.1415926 \quad „ \quad = 0.4971499,$$

$$180 = 2.2552755,$$

$$\frac{180^\circ}{M\pi}\log\frac{1}{n} = \arcsin\frac{1}{n}, \quad \&\text{c.},$$

whence

$$\frac{180^\circ}{M\pi} = 212.03383,$$

$$\frac{180^\circ\cdot\log 2}{M\pi} = 131^\circ.9284.$$

We hence calculate the succession of values of θ and ϕ as follows:

n	θ_n	tan	arc	
1	45°			
2	$26^\circ\,34'$	$\log\frac{1}{2} = 0.3010300$	$= 39^\circ\,43'$	$-13^\circ\,9'$
3	$18^\circ\,26'$	0.1760913	$23^\circ\,14'$	$4^\circ\,48'$
4	$14^\circ\,2'$	0.1249387	$16^\circ\,29'$	$2^\circ\,27'$
5	$11^\circ\,19'$	0.0969100	$12^\circ\,47'$	$1^\circ\,28'$
6	$9^\circ\,28'$	0.0791813	$10^\circ\,26'$	$0^\circ\,58'$
7	$8^\circ\,8'$	0.0669467	$8^\circ\,50'$	$0^\circ\,42'$
8	$7^\circ\,8'$	0.0579920	$7^\circ\,39'$	$0^\circ\,31'$
9	$6^\circ\,20'$	0.0511525	$6^\circ\,44'$	$0^\circ\,24'$
10	$5^\circ\,43'$	0.0457575	$6^\circ\,2'$	$0^\circ\,19'$

The sum of all the negative arcs $\theta_2 - \phi_2$, $\theta_3 - \phi_3$, ... as far as calculated, that is, up to $\theta_{10} - \phi_{10}$ is $= 24^\circ\,46'$, or, writing x for the sum of the remaining arcs $\theta_n - \phi_n$ to infinity, we have

$$\Pi i = 1.9173(\cos\Theta + i\sin\Theta),$$

where

$$\Theta = 45^\circ - 24^\circ\,46' - x = 20^\circ\,14' - x.$$

It would not be difficult to calculate a limit to the value of x.

0.7 956. On Richelot's Integral of the Differential Equation $\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0$

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 42–47.]

In the Memoir "Einige neue Integralgleichungen des Jacobischen Systems Differentialgleichungen," Crelle, t. xxv. (1843), pp. 97—118, Richelot, working with the more general problem of a system of $n-1$ differential equations between n variables, obtains a result which in the particular case $n = 2$ (that is, for the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0, \quad X = a + bx + cx^2 + dx^3 + ex^4,$$

and Y the same function of y), is in effect as follows: an integral is

$$\frac{\sqrt{\Omega_1} + \sqrt{\Omega_2} + \sqrt{\Omega_3} + \sqrt{\Omega_4}}{\sqrt{D}} = \text{const.},$$

where D, θ are arbitrary constants, and Ω denotes the quartic function

$$\Omega = a + b\theta + c\theta^2 + d\theta^3 + e\theta^4;$$

viz. this is theorem 3, p. 107 (t. c.), taking therein $n = 2$, and writing θ, D for Richelot's α and const.

The peculiarity is that the integral contains apparently two arbitrary constants, and it is very interesting to show how these really reduce themselves to a single arbitrary constant.

Observe that, on the right-hand side, there are terms in θ^4, θ^3 whereas no such terms present themselves on the left-hand side. But by changing the constant D, we can get rid of these terms, and so bring each side to contain only terms in θ^2, θ, 1; viz. writing

$$D = -2e\theta^2 - d\theta - c + C,$$

where C is a new arbitrary constant, the equation becomes

$$\begin{aligned}\theta^2 &\left[e(x-y)^2\right] \\ &+ \theta\left[-2exy(x+y) - dxy - (C-c)(x+y) + b\right] \\ &\qquad + \left[ex^2y^2 + (C-c)xy + a\right],\end{aligned}$$

which still contains the two arbitrary constants θ, C.

But this gives the three equations

$$\begin{aligned}
\frac{\sqrt{X}+\sqrt{Y}}{x-y} &= C,\\
\frac{\sqrt{X}-\sqrt{Y}}{x-y} &= \theta\cdot e(x+y)+\frac{d\theta+(C-c)}{x-y},\\
\frac{y\sqrt{X}-x\sqrt{Y}}{x-y} &= \theta\cdot exy+\text{etc.}
\end{aligned}$$

The first of these is Lagrange's integral containing the arbitrary constant C; and it is necessary that the three equations shall be one and the same equation: viz. the second and third equations must be each of them a mere transformation of the first equation.

It is easy to verify that this is so. Starting from the first equation, we require, first the value of

$$\frac{\sqrt{X}-\sqrt{Y}}{x-y}$$

for a moment. We form a rational combination, or combination without any term in $\sqrt{XY}$; this is

$$\frac{(\sqrt{X}-\sqrt{Y})^2}{(x-y)^2}=\frac{X+Y-2\sqrt{XY}}{(x-y)^2},$$

where the left-hand side is

$$\frac{(x-y)(X-Y)}{(x-y)^2}=\frac{X-Y}{x-y},$$

which is

$$=e(x^3+x^2y+xy^2+y^3)+d(x^2+xy+y^2)+c(x+y)+b,$$

and we thence have for

$$\frac{\sqrt{X}-\sqrt{Y}}{x-y}$$

the value given by the second equation.

Secondly, starting again from the first equation, and proceeding in like manner to find the value of

$$\frac{y\sqrt{X}-x\sqrt{Y}}{x-y}$$

for a moment, we form a rational combination

$$\frac{(y\sqrt{X} - x\sqrt{Y})^2}{(x-y)^2},$$

where the left-hand side is

$$\frac{(x-y)(-yX+xY)}{(x-y)^2} = \frac{-yX+xY}{x-y},$$

which is

$$= -exy(x^2+xy+y^2) - dxy(x+y) - cxy + a;$$

and we thence have for

$$\frac{y\sqrt{X} - x\sqrt{Y}}{x-y}$$

the value given by the third equation.

In conclusion, I give what is in effect the process by which Richelot obtained his integral. The integral is $v = D$, where

$$v = \frac{\sigma + \tau}{1 - x \cdot x - y \quad \theta - y \cdot y - x},$$

if, for shortness,

$$\sigma = \frac{\sqrt{X}}{\theta - x}, \quad \tau = \frac{\sqrt{Y}}{\theta - y},$$

and it is required thence to show that

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

or, what is the same thing, to show that v satisfies the partial differential equation

$$\sqrt{X}\frac{\partial v}{\partial x} + \sqrt{Y}\frac{\partial v}{\partial y} = 0.$$

We have

$$\frac{\partial v}{\partial x} = \frac{\partial \sigma}{\partial x} \cdot \frac{1}{\tau} + \sigma \cdot \frac{-1}{\tau^2}\frac{\partial \tau}{\partial x},$$

and thence, attending to the value of Ω,

$$\frac{\partial v}{\partial x} = \frac{\sqrt{X}}{\Omega}\frac{\partial v}{\partial \theta},$$

or say

$$\sqrt{X}\frac{\partial v}{\partial x} = \frac{X}{\Omega}\frac{\partial v}{\partial \theta},$$

and it is consequently to be shown that the function on the right-hand side is symmetrical in $(x, \sqrt{X})$ and $(y, \sqrt{Y})$, viz. that we have

$$\frac{X}{\Omega}\frac{\partial v}{\partial \theta} + \frac{Y}{\Omega}\frac{\partial v}{\partial \theta} = 0,$$

which is the equation to be proved.

0.8 957. Illustrations of Sylow's Theorems on Groups

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 47–48.]

The theorems of Sylow relate to a group G of the order N, and a prime number n which divides N: writing $N = n^\alpha \nu(nk+1)$, where n does not divide ν, then

Theorem 1. There exists a group g of the order n^α, contained in the group G.

Theorem 2. Writing $n^\alpha \nu$ for the order of the greatest group, the substitutions whereof are permutable with the group g, then k has a determinate value, and the number of groups of the order n^α contained in G is $= nk+1$.

And where

But at present I attend only to the theorems 1 and 2.

For instance, consider the group G of the order $n = 6$,

$$1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2 \quad (\alpha^2 = 1, \beta^3 = 1, \alpha\beta^2 = \beta\alpha, \alpha\beta = \beta^2\alpha).$$

Here $n_1 = 2$ or 3: if $n = 2$, we have $N = n^\alpha \nu(nk+1) = 2.1(2+1)$; if $n = 3$, we have $N = n^\alpha \nu(nk+1) = 3.2.1$.

First, $n_1 = 2$; we should have a group g of the order 2; one such group is $(1, \alpha)$, and the only group the substitutions whereof are permutable with $(1, \alpha)$ is the group $(1, \alpha)$ itself: for, taking any other operation of the group, for instance β, it is not true that $\beta(1, \alpha) = (1, \alpha)\beta$; in fact, the left-hand is $(\beta, \beta\alpha)$ and the right-hand is $(\beta, \alpha\beta)$ or $(\beta, \beta^2\alpha)$: hence $n^\alpha \nu = 2\nu = 2$, or $\nu = 1$.

Hence also, by theorem 2, there should be 3 groups of the order 2 such as $(1, \alpha)$, viz. these are $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$, derived from $(1, \alpha)$ as follows:

$$\begin{array}{rl}
1 & (1, \alpha)1^{-1} = (1, \alpha), \\
\alpha & (1, \alpha)\alpha^{-1} = (1, \alpha), \\
\beta & (1, \alpha)\beta^{-1} = (1, \alpha\beta), \\
\beta^2 & (1, \alpha)\beta^{-2} = (1, \alpha\beta^2), \\
\alpha\beta & (1, \alpha)(\alpha\beta)^{-1} = (1, \alpha\beta^2), \\
\alpha\beta^2 & (1, \alpha)(\alpha\beta^2)^{-2} = (1, \alpha\beta),
\end{array}$$

since $\beta^{-1}=\beta^2$, and therefore the second term is $\beta\alpha\beta^2=\alpha\beta^2\cdot\beta^2=\alpha\beta$,

$\beta^{-2}=\beta$, „ $\beta\alpha\beta=\alpha\beta\cdot\beta=\alpha\beta^2$,
$(\alpha\beta)^{-1}=\alpha\beta$, „ $\alpha\beta\alpha\alpha\beta=\alpha^2\beta\cdot\beta=\alpha\beta^2$,
$(\alpha\beta^2)^{-1}=\alpha\beta^2$, , $\alpha\beta^2\alpha\alpha\beta^2=\alpha^2\beta^2\cdot\beta^2=\alpha\beta$;

viz. the derivatives are $(1,\alpha)$, $(1,\alpha\beta)$, $(1,\alpha\beta^2)$, each twice.

Secondly, $n=3$; there should be here a group of the order 3, viz. this is $(1,\beta,\beta^2)$. The group, the substitutions whereof are permutable with $(1,\beta,\beta^2)$, is the entire group $(1,\beta,\beta^2,\alpha,\alpha\beta,\alpha\beta^2)$; in fact, taking any substitution hereof, for instance α, we have $\alpha(1,\beta,\beta^2)=(1,\beta,\beta^2)\alpha$, viz. the left-hand side is $(\alpha,\alpha\beta,\alpha\beta^2)$, and the right-hand side is $(\alpha,\beta\alpha,\beta^2\alpha)=(\alpha,\alpha\beta^2,\alpha\beta)$, which is the left-hand side, the change of order being immaterial; this is the meaning of the expression used, "the operations whereof are permutable with the group g." Hence, we have $n^\alpha\nu=3\nu=6$, or $\nu=2$; and thence also $nk+1=3k+1=1$, viz. $k=0$. There is thus only a single group of the order 3, viz. the group $(1,\beta,\beta^2)$.

As another instance, I take the group of the order 12 formed by the positive substitutions of four letters, viz. these are

$$\begin{array}{ccc} 1, & ab\cdot cd, & abc, \\ & ac\cdot bd, & acb, \\ & ad\cdot bc, & abd, \\ & & adb, \\ & & acd, \\ & & adc, \\ & & bcd, \\ & & bdc. \end{array}$$

Here, taking $n=2$, we have $N=n^\alpha\nu(nk+1)=2^2\cdot3\cdot1$; there is a group g of the order 4, viz. this is

$$(1,ab\cdot cd,ac\cdot bd,ad\cdot bc),$$

and the greatest group, the substitutions whereof are permutable with this group g, is the entire group of the order 12; thus, considering any substitution thereof, for instance abc, we have

$$abc\begin{pmatrix}1 & ab\cdot cd\\ ac\cdot bd & ad\cdot bc\end{pmatrix}=\begin{pmatrix}abc & acd\\ bdc & adb\end{pmatrix},$$

viz. the left-hand is $\begin{pmatrix} abc \\ acd \end{pmatrix}$, the right-hand is $\begin{pmatrix} abc & bdc \\ adb & acd \end{pmatrix}$; hence $n^\alpha \nu = 4\nu = 12$ or $\nu = 3$; whence also $nk + 1 = 2k + 1 = 1$: and thus the foregoing group g is the only group of the order 4.

Similarly, taking $n = 3$, we have $N = n^\alpha \nu(nk + 1) = 3 \cdot 1 \cdot 4$. There is a group g of the order 3, say $(1, abc, acb)$; the greatest group, the substitutions whereof are permutable with g, is the group g itself, viz. we have $n^\alpha \nu = 3\nu = 3$, or $\nu = 1$; and then $nk + 1 = 3k + 1 = 4$: there are thus 4 groups of the order 3, viz. these are

$$(1, abc, acb), \quad (1, abd, adb), \quad (1, acd, adc), \quad (1, bcd, bdc).$$

Reverting to the before-mentioned group of the order 6, this not only contains each of the groups $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$ of order 2, and the group $(1, \beta, \beta^2)$ of order 3; but it is the permutable product of a group of order 2 by a group of order 3, say it is

$$G = (1, \alpha)(1, \beta, \beta^2) = (1, \beta, \beta^2)(1, \alpha).$$

A group, which is thus a permutable product of two factors, is said to be a *true product*; and when it cannot be thus expressed as a permutable product of two factors, it is *prime* or *simple*. A group, the order of which is equal to a prime number p (the cyclical group of the order p) is simple; but the order may be a composite number and yet the group be simple—it was remarked by Galois, Liouville, t. XI. (1865), p. 409, that the order of the lowest simple group of composite order is 60, $= 2^2 \cdot 3 \cdot 5$, and it has been recently shown, Hölder, "Die einfachen Gruppen im ersten und zweiten Hundert der Ordnungszahlen," Math. Ann. t. XL. (1892), pp. 55—88, that the only other composite order of a simple group in the first 200 numbers is 168. Moreover, in the paper Cole, "Simple groups from order 201 to order 500," Amer. Math. Jour. t. XIV. (1892), pp. 378—388, it is shown that within these limits the only numbers which can give a simple group or groups are 360 and 432. I take the opportunity of referring to two other important papers, Young, "On the determination of groups whose order is a power of a prime," Amer. Math. Jour. t. XV. (1893), pp. 124—178, and Cole and Glover, "On groups whose orders are products of three prime factors," ib. pp. 191—220.

0.9 958. On the Surface of the Order n which Passes through a Given Cubic Curve

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 79, 80.]

It is natural to assume that, taking A, B, C to denote the general functions $(x, y, z, w)^{n-2}$ of the order $n-2$, the general surface of the order n which passes through the curve

$$x : y : z : w = 1 : \theta : \theta^2 : \theta^3$$

(or, what is the same thing, the curve $x : y : z : w = 1 : \theta : \theta^2 : \theta^3$), has for its equation

$$A \cdot P + B \cdot Q + C \cdot R = 0;$$

but the formal proof is not immediate. Writing the equation in the form

$$U = \Sigma a x^\alpha y^\beta z^\gamma w^\delta = 0, \quad (\alpha + \beta + \gamma + \delta = n),$$

then U must vanish on writing therein $x : y : z : w = 1 : \theta : \theta^2 : \theta^3$: a term a becomes $= a\theta^p$, where $p = \beta + 2\gamma + 3\delta$ is the weight of the term reckoning the weights of x, y, z, w as 0, 1, 2, 3 respectively; and hence the condition is that, for each given weight p, the sum Σa of the coefficients of the several terms of this weight shall be $= 0$. Using any such equation to determine one of the coefficients thereof in terms of the others, the function U is reduced to a sum of duads $a(x^\alpha y^\beta z^\gamma w^\delta - x^{\alpha'} y^{\beta'} z^{\gamma'} w^{\delta'})$, where in each duad the two terms are of the same degree and of the same weight, and where a is an arbitrary coefficient; it ought therefore to be true that each such duad $x^\alpha y^\beta z^\gamma w^\delta - x^{\alpha'} y^{\beta'} z^{\gamma'} w^{\delta'}$ has the property in question—or writing P, Q, $R = yw - z^2$, $zy - xw$, $xz - y^2$, say that each such duad is of the form $AP + BQ + CR$.

Suppose for a moment that α' is greater than α, but that β', γ', δ' are each less than β, γ, δ respectively: the duad is $x^{\alpha'} y^{\beta'} z^{\gamma'} w^{\delta'}(x^\lambda - y^\mu z^\nu w^\rho)$, where λ, μ, ν, ρ are each positive, and hence $x^\lambda - y^\mu z^\nu w^\rho$ is a duad having the property in question, or changing the notation say $x^\alpha - y^\beta z^\gamma w^\delta$ has the property in question; and in like manner, by considering the several cases that may happen, we have to show that each of the duads has the property in question; it being of course understood that, in each of these duads, the two terms have the same degree and the same weight.

The first form cannot exist; for we must have therein $\alpha = \beta + \gamma + \delta$ and $0 = \beta + 2\gamma + 3\delta$, which is inconsistent with α, β, γ, δ each of them positive. For the second form $\beta = \alpha + \gamma + \delta$, $\beta = 2\gamma + 3\delta$: this is $\alpha = \gamma + 2\delta$ or the duad is $y^{\gamma+2\delta} - x^{\gamma+\delta}w^{\delta}$, $= (y^2)^{\gamma}y^{\delta} - (xz)^{\gamma}(xw)^{\delta}$. Writing $y^2 = xz - R$, we have terms containing the factor R, and a residual term $(xz)^{\gamma}\{y^{\delta} - (xw)^{\delta}\}$, and writing herewith $y^{\delta} - (xw)^{\delta} = (y - xw)(\quad)$, we have the factor Q, and a residual term which ultimately depends on $y^2 - xw$, that is, on Q. The process is a little different according as δ is even or odd, but in either case the required form is obtained; and the like for the other cases.

67—2

0.10 959. A Trigonometrical Identity

[From *the Messenger of Mathematics*, vol. XXIII. (1894), p. 80.]

Writing for shortness $S = \alpha + \beta + \gamma + \delta$, the identity is

$$\begin{aligned}
&\sin(\alpha+\beta)\sin(\alpha+\gamma)\sin(\alpha+\delta)\\
&\qquad + \sin(\beta+\alpha)\sin(\beta+\gamma)\sin(\beta+\delta)\\
&\qquad + \sin(\gamma+\alpha)\sin(\gamma+\beta)\sin(\gamma+\delta)\\
&\qquad + \sin(\delta+\alpha)\sin(\delta+\beta)\sin(\delta+\gamma)\\
&= \sin(\alpha+\beta+\gamma+\delta)\sin(\alpha+\beta)\sin(\gamma+\delta)\\
&\qquad + \sin(\alpha+\gamma)\sin(\beta+\delta)\sin(\alpha+\delta)\sin(\beta+\gamma),
\end{aligned}$$

where observe that the left-hand side is

$$\sin S \Sigma \sin(\alpha+\beta)\sin(\alpha+\gamma)\sin(\alpha+\delta)/\sin S.$$

But the more convenient form is

$$\begin{aligned}
&\sin(\alpha+\beta)\sin(\alpha+\gamma)\sin(\alpha+\delta)\\
&\qquad = \sin(\alpha+\beta)\sin(\alpha+\gamma)\sin(\alpha+\delta)\\
&\qquad\qquad - \sin(\beta+\gamma)\sin(\beta+\delta)\sin(\beta+\alpha)\\
&\qquad - \sin(\gamma+\alpha)\sin(\gamma+\beta)\sin(\gamma+\delta)\\
&\qquad\qquad - \sin(\delta+\alpha)\sin(\delta+\beta)\sin(\delta+\gamma),
\end{aligned}$$

viz. this is

$$\begin{aligned}
&\sin(\alpha+\beta)\sin(\alpha+\gamma)\sin(\alpha+\delta)\\
&\quad = \tfrac{1}{2}\sin(2\alpha+\beta+\gamma+\delta)[\cos(\beta-\gamma)-\cos(\beta+\gamma+2\delta)]\\
&\quad - \tfrac{1}{2}\sin(2\beta+\gamma+\delta+\alpha)[\cos(\gamma-\delta)-\cos(\gamma+\delta+2\alpha)]\\
&\quad - \tfrac{1}{2}\sin(2\gamma+\delta+\alpha+\beta)[\cos(\delta-\alpha)-\cos(\delta+\alpha+2\beta)]\\
&\qquad - \tfrac{1}{2}\sin(2\delta+\alpha+\beta+\gamma)[\cos(\alpha-\beta)-\cos(\alpha+\beta+2\gamma)],
\end{aligned}$$

where the right-hand side should be $= \sin(\alpha+\beta)\sin(\alpha+\gamma)\sin(\alpha+\delta)$.

0.11 960. Note on a Formula in Elliptic Functions

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 80, 81.]

The formula

$$\frac{\operatorname{sn}(u+v)}{\operatorname{sn} u \operatorname{sn} v} = \frac{\operatorname{sn}' u \operatorname{sn}' v}{\operatorname{sn}(u+v)} + \text{etc.}$$

might, I think, have found a place in the elementary treatment of elliptic functions; it presents itself in the theory of the addition of arguments, but it does not appear to be obtainable by any easy transformation from the original formula for $\operatorname{sn}(u+v)$.

Starting from the original formula,

$$\operatorname{sn}(u+v) = \frac{\operatorname{sn}^2 u - \operatorname{sn}^2 v}{\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v - \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u},$$

and representing for a moment the denominator by Δ, we have

$$\frac{\operatorname{sn}(u+v)}{\operatorname{sn} u \operatorname{sn} v} = \frac{\operatorname{sn}^2 u - \operatorname{sn}^2 v}{\Delta \cdot \operatorname{sn} u \operatorname{sn} v},$$

and

$$\frac{\Delta}{\operatorname{sn}(u+v)} = \frac{\Delta^2}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}.$$

The value of Δ^2 is

$$\Delta^2 = (\operatorname{sn} u \operatorname{cn} v \operatorname{dn} v - \operatorname{sn} v \operatorname{cn} u \operatorname{dn} u)^2,$$

and reducing by means of the relations $\operatorname{cn}^2 u = 1 - \operatorname{sn}^2 u$, $\operatorname{dn}^2 u = 1 - k^2 \operatorname{sn}^2 u$, we find

$$\begin{aligned}\Delta^2 = \operatorname{sn}^2 u(1-\operatorname{sn}^2 v)(1-k^2\operatorname{sn}^2 v) + \operatorname{sn}^2 v(1-\operatorname{sn}^2 u)(1-k^2\operatorname{sn}^2 u)\\ - 2\operatorname{sn} u \operatorname{sn} v \operatorname{cn} u \operatorname{cn} v \operatorname{dn} u \operatorname{dn} v.\end{aligned}$$

0.12 961. A Trigonometrical Identity

[From *the Messenger of Mathematics*, vol. XXIII. (1894), p. 81.]

If $A+B+C+D=360^\circ$, then

$$\cos A+\cos B+\cos C+\cos D \\ = -4\cos\tfrac{1}{2}(A+B)\cos\tfrac{1}{2}(A+C)\cos\tfrac{1}{2}(A+D).$$

In fact, the left-hand side is

$$\begin{aligned} &= \cos A+\cos B+2\cos\tfrac{1}{2}(C+D)\cos\tfrac{1}{2}(C-D) \\ &= \cos A+\cos B+2\cos\tfrac{1}{2}(C+D)\cos\tfrac{1}{2}(C-D), \end{aligned}$$

and the right-hand side is

$$\begin{aligned} &= -4\cos\tfrac{1}{2}(A+B)\cos\tfrac{1}{2}(A+C)\cos\tfrac{1}{2}(A+D) \\ &= -2\cos\tfrac{1}{2}(A+B)[\cos\tfrac{1}{2}(2A+C+D)+\cos\tfrac{1}{2}(C-D)] \\ &= -2\cos\tfrac{1}{2}(A+B)[\cos(180^\circ-\tfrac{1}{2}B)+\cos\tfrac{1}{2}(C-D)] \\ &= -2\cos\tfrac{1}{2}(A+B)[-\cos\tfrac{1}{2}B+\cos\tfrac{1}{2}(C-D)] \\ &= 2\cos\tfrac{1}{2}(A+B)\cos\tfrac{1}{2}B-2\cos\tfrac{1}{2}(A+B)\cos\tfrac{1}{2}(C-D), \end{aligned}$$

where $2\cos\frac{1}{2}(A+B)\cos\frac{1}{2}B=\cos A+\cos B$; and we have also

$$\cos\tfrac{1}{2}(C+D)=\cos(180^\circ-\tfrac{1}{2}(A+B))=-\cos\tfrac{1}{2}(A+B),$$

so that the other terms also agree.

0.13 962. Coordinates versus Quaternions

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 82—84.]

I wish to call attention to a theorem in coordinates which corresponds precisely to the quaternion formula

$$\nabla \cdot QR = \nabla Q \cdot \nabla R + SQ \cdot R - SQ \cdot SR,$$

and which is, in fact, the same theorem expressed in coordinate language.

In (x, y, z) coordinates, if we have two lines through the origin

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \text{and} \quad \frac{x}{a'} = \frac{y}{b'} = \frac{z}{c'},$$

then the cosine of the angle between them is

$$\frac{aa' + bb' + cc'}{\sqrt{a^2 + b^2 + c^2}\sqrt{a'^2 + b'^2 + c'^2}},$$

and the sine is given by

$$\frac{\sqrt{(bc' - b'c)^2 + (ca' - c'a)^2 + (ab' - a'b)^2}}{\sqrt{a^2 + b^2 + c^2}\sqrt{a'^2 + b'^2 + c'^2}}.$$

The coordinate theorem is as follows: taking (x, y, z) as current coordinates, the equation of the plane through the origin and the two lines is

$$\begin{vmatrix} x & y & z \\ a & b & c \\ a' & b' & c' \end{vmatrix} = 0.$$

The corresponding quaternion formula is obtained by writing

$$Q = ai + bj + ck, \quad R = a'i + b'j + c'k,$$

and the theorem then expresses that the scalar and vector parts of the product QR are connected with the angle between the lines represented by Q and R.

0.14 963. Note on Dr Muir's Paper on Elimination

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 84—87.]

Dr Muir's paper "On the Transformation of the Determinantal Equation of a Matrix into an Equation in Differences or Differential Coefficients" contains some very interesting results which I should like to notice.

The fundamental theorem is as follows: If M be a matrix of the order n,

$$M = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix},$$

and if we form the determinant $|M - \lambda I| = 0$, then this determinantal equation can be transformed into an equation involving the differences or differential coefficients of the elements.

For instance, in the case $n = 2$, we have

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0,$$

that is,

$$(a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0,$$

or

$$\lambda^2 - \lambda(a_{11} + a_{22}) + (a_{11}a_{22} - a_{12}a_{21}) = 0.$$

In the case $n = 3$, writing the matrix as

$$\begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix},$$

the equation is

$$\begin{vmatrix} a-\lambda & h & g \\ h & b-\lambda & f \\ g & f & c-\lambda \end{vmatrix} = 0,$$

or, expanded,

$$\lambda^3 - \lambda^2(a+b+c) + \lambda(bc+ca+ab-f^2-g^2-h^2) - abc + af^2 + bg^2 + ch^2 - 2fgh = 0.$$

Dr Muir shows that these equations can be expressed in terms of the differences of the elements, and that the transformation is connected with the theory of continuants.

0.15 964. On the Nine-Points Circle of a Spherical Triangle

[From *the Messenger of Mathematics*, vol. XXIII. (1894), pp. 87—94.]

The nine-points circle of a spherical triangle is a circle which passes through nine points connected with the triangle. The investigation of its properties presents some interesting features.

Taking A, B, C for the angles and a, b, c for the sides of the spherical triangle, we have the fundamental relations

$$\cos a = \cos b \cos c + \sin b \sin c \cos A, \quad \&c.$$

The nine-points circle is defined as follows: Let D, E, F be the mid-points of the sides BC, CA, AB respectively; let L, M, N be the feet of the perpendiculars from A, B, C on the opposite sides; and let P, Q, R be the mid-points of the arcs joining the vertices to the orthocentre. Then the circle through any three of these nine points passes through all nine.

The equation of the nine-points circle may be obtained as follows. If (x, y, z) are the trilinear coordinates of a point on the sphere, referred to the triangle ABC as fundamental triangle, then the equation of a circle is

$$la^2 + mb^2 + nc^2 = 0,$$

where l, m, n are constants; and for the nine-points circle we have particular values of these constants.

Secondly, for the points L', M', N', we have

$$\frac{\sin BL'}{\sin CL'} = \frac{r}{q} = \frac{\cos BA'}{\cos CA'},$$

and similarly

$$\frac{\sin CM'}{\sin AM'} = \frac{p}{r} = \frac{\cos CB'}{\cos AB'},$$
$$\frac{\sin AN'}{\sin BN'} = \frac{q}{p} = \frac{\cos AC'}{\cos BC'};$$

viz. the sides $B'C'$, $C'A'$, $A'B'$ are by the points L', M', N' divided each into two parts such that for any side the sines of the two parts are proportional to the cosines of the other two sides. We have

$$\sin BL' \cdot \sin CM' \cdot \sin AN' = \sin CL' \cdot \sin AM' \cdot \sin BN',$$

viz. the arcs AL', BM', CN' meet in a point K' which may be called the *cos-centre* of the triangle $A'B'C'$ (where observe that, for a, b, c indefinitely small, i.e. for a plane triangle, the points L', M', N' are the mid-points of the sides, and the centre K' is the c. G. or median point of the triangle).

But further, we have

$$\sin BL = \frac{L}{q^2 + r^2 + 2pqr},$$

that is,

$$\sin BL = \frac{L}{q^2 + r^2 + 2pqr}, \quad \sin CL = \frac{L}{q^2 + r^2 + 2pqr},$$

and thence

$$\sin BL \cdot \sin CM \cdot \sin AN = \sin CL \cdot \sin AM \cdot \sin BN,$$

viz. the arcs AL, BM, CN meet in a point, which is obviously not the cos-centre of the triangle ABC.

We have thus the construction of the nine-points circle as a six-points circle, by means of the points F, G, H, L, M, N; and by way of recapitulation we may say that the nine-points circle meets the sides BC, CA, AB in the points F, L; G, M; H, N respectively, where the points F, G, H depend on the ortho-centre of the semi-triangle, and the points L, M, N depend on the cos-centre of the semi-triangle.

The triangle ABC has an inscribed circle and three escribed circles, and we have (as is known) the theorem that the nine-points circle touches each of these four circles.

The circles BC, CA, AB and the nine-points circle form a tetrad of circles, and the inscribed circle and the three escribed circles a tetrad of circles, or say the eight circles form a *bitetrad*, such that each circle of the one tetrad touches each circle of the other tetrad.

0.16 965. On the Sixty Icosahedral Substitutions

[From *the Quarterly Journal of Pure and Applied Mathematics*, vol. XXVII. (1895), pp. 236—242.]

The Sixty Icosahedral Substitutions were obtained in an elegant form by Gordan in the paper "Ueber endliche Gruppen linearer Transformationen einer Veränderlichen," Math. Ann. t. XII. (1877), pp. 23—46, see p. 45, where the group is exhibited in the canonical form

$$x_1 : y_1 = \frac{\epsilon^\nu(\epsilon^2+\epsilon^4)\eta+\epsilon^\rho}{-\epsilon^{-\rho}\eta-(\epsilon+\epsilon^3)}, \quad x_2 : y_2 = \frac{\epsilon^\nu\eta+\epsilon^\rho}{\epsilon^{-\rho}(\epsilon^2+\epsilon^4)\eta+\epsilon^\nu},$$

(ϵ an imaginary fifth root of unity), agreeing with the developed form given in my paper, "On the Schwarzian Derivative and the Polyhedral Functions," Camb. Phil. Trans., t. XIII. Part 1 (1881), [745], pp. 5—68, see pp. 55—56, [this Collection, vol. XI, pp. 204—205].

But a slight change of form is desirable. I write $\begin{pmatrix} a, b \\ c, d \end{pmatrix}$ to denote the unimodular matrix

$$\frac{1}{\sqrt{(ad-bc)}}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \frac{a}{\sqrt{(ad-bc)}} & \frac{b}{\sqrt{(ad-bc)}} \\ \frac{c}{\sqrt{(ad-bc)}} & \frac{d}{\sqrt{(ad-bc)}} \end{pmatrix},$$

so that the value is independent of the absolute magnitudes of a, b, c, d, but depends only on their ratios; and observe that $m\begin{pmatrix} a, b \\ c, d \end{pmatrix}$ denotes $\begin{pmatrix} ma, b \\ c, d \end{pmatrix}$, the factor m applying to the top-line only. As before, ϵ denotes an imaginary fifth root of unity, and I write $A = \epsilon+\epsilon^4$, $B = \epsilon^2+\epsilon^3$, $A' = \epsilon-\epsilon^4$, $B' = \epsilon^2-\epsilon^3$, but, in fact, I scarcely need more than the first of these symbols.

The substitutions are

$$(x_1, y_1) = \begin{pmatrix} a, b \\ c, d \end{pmatrix} (x, y),$$

that is, $x_1 = ax + by$, $y_1 = cx + dy$; and we then have sixty matrices $\begin{pmatrix} a, b \\ c, d \end{pmatrix}$, forming the group in question.

Giving to ρ the values 0, 1, 2, 3, 4, I write

$$K_\rho = \begin{pmatrix} \epsilon^{2\rho}, & 0 \\ 0, & \epsilon^{-\rho} \end{pmatrix}, \quad L_\rho = \begin{pmatrix} \epsilon^{\rho}, & 0 \\ 0, & -\epsilon^{-\rho} \end{pmatrix},$$

$$M_\rho = \begin{pmatrix} 0, & \epsilon^{\rho} \\ \epsilon^{-\rho}, & A\epsilon^{\rho} \end{pmatrix}, \quad N_\rho = \begin{pmatrix} 0, & \epsilon^{\rho} \\ -\epsilon^{-\rho}, & -A\epsilon^{\rho} \end{pmatrix},$$

so that K_0 is the matrix unity $\begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}$; then the matrices are

$$\begin{array}{ll} K_0, & K_1, K_2, K_3, K_4, \\ & \epsilon M_0, \epsilon^2 M_1, \epsilon^3 M_2, \epsilon^4 M_3, M_4, \\ & \epsilon N_0, \epsilon^2 N_1, \epsilon^3 N_2, \epsilon^4 N_3, N_4, \\ & \epsilon^2 N_0, \epsilon^3 N_1, \epsilon^4 N_2, N_3, \epsilon N_4, \\ & \epsilon^3 N_0, \epsilon^4 N_1, N_2, \epsilon N_3, \epsilon^2 N_4, \\ & \epsilon^4 N_0, N_1, \epsilon N_2, \epsilon^2 N_3, \epsilon^3 N_4; \end{array}$$

viz. $5 + 25 + 25 + 5 = 60$ matrices in all, including the matrix unity.

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It is to be observed that the system must remain unaltered if we change ϵ into ϵ^2, ϵ^3, or ϵ^4: this is at once obvious for the change ϵ into ϵ^4, but less immediately so for the other two changes. Suppose ϵ changed into ϵ^2, then M_ρ becomes

$$\begin{pmatrix} 0, & \epsilon^{2\rho} \\ \epsilon^{-2\rho}, & A\epsilon^{2\rho} \end{pmatrix},$$

or changing ρ into $-\frac{1}{2}\rho$, which has the same values, this is $=$ $\epsilon^{2\rho}\begin{pmatrix} 0, & 1 \\ \epsilon^{-\rho}, & A\epsilon^{\rho} \end{pmatrix}$, viz. M_ρ is equal to a power of ϵ multiplied by N_ρ; and similarly N_ρ becomes equal to a power of ϵ multiplied by M_ρ. And the like as regards the change ϵ into ϵ^3.

I find that the sixty matrices may be coordinated with the sixty positive substitutions of the letters *abcde* as follows:

No.	Matrix	Substitution
1	$K_0 = \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}$	1
2	$L_0 = \begin{pmatrix} 1, & 0 \\ 0, & -1 \end{pmatrix}$	$ab \cdot cd$
3	$L_1 = \begin{pmatrix} \epsilon, & 0 \\ 0, & -\epsilon^4 \end{pmatrix}$	$ab \cdot ce$
4	$L_2 = \begin{pmatrix} \epsilon^2, & 0 \\ 0, & -\epsilon^3 \end{pmatrix}$	$ac \cdot be$
5	$L_3 = \begin{pmatrix} \epsilon^3, & 0 \\ 0, & -\epsilon^2 \end{pmatrix}$	$ad \cdot bc$
6	$L_4 = \begin{pmatrix} \epsilon^4, & 0 \\ 0, & -\epsilon \end{pmatrix}$	$ae \cdot cd$
7	$\epsilon K_1 = \begin{pmatrix} \epsilon^4, & 0 \\ 0, & \epsilon^2 \end{pmatrix}$	$ac \cdot bd$
8	$\epsilon K_2 = \begin{pmatrix} \epsilon^3, & 0 \\ 0, & \epsilon \end{pmatrix}$	$ae \cdot bd$
9	$\epsilon K_3 = \begin{pmatrix} \epsilon^2, & 0 \\ 0, & 1 \end{pmatrix}$	$ae \cdot bc$
10	$\epsilon K_4 = \begin{pmatrix} \epsilon, & 0 \\ 0, & \epsilon^4 \end{pmatrix}$	$be \cdot de$
11	$\epsilon M_0 = \begin{pmatrix} 0, & \epsilon \\ 1, & A\epsilon \end{pmatrix}$	$ac \cdot de$
12	$\epsilon M_1 = \begin{pmatrix} 0, & \epsilon^2 \\ \epsilon^4, & A\epsilon^2 \end{pmatrix}$	$ab \cdot cd$
13	$\epsilon^2 M_0 = \begin{pmatrix} 0, & \epsilon^2 \\ \epsilon, & A\epsilon^2 \end{pmatrix}$	$ad \cdot be$
14	$\epsilon^2 M_1 = \begin{pmatrix} 0, & \epsilon^3 \\ \epsilon^2, & A\epsilon^3 \end{pmatrix}$	$ab \cdot ce$
15	$\epsilon^3 M_0 = \begin{pmatrix} 0, & \epsilon^3 \\ \epsilon^2, & A\epsilon^3 \end{pmatrix}$	$bc \cdot cd$
16	$\epsilon^3 M_1 = \begin{pmatrix} 0, & \epsilon^4 \\ \epsilon^3, & A\epsilon^4 \end{pmatrix}$	$ad \cdot ce$

No.	Matrix	Substitution
17	$\epsilon^4 M_0 = \begin{pmatrix} 0, & \epsilon^4 \\ \epsilon^3, & A\epsilon^4 \end{pmatrix}$	$abcde$
18	$\epsilon^4 M_1 = \begin{pmatrix} 0, & 1 \\ \epsilon^4, & A \end{pmatrix}$	adc
19	$\epsilon M_2 = \begin{pmatrix} 0, & \epsilon^3 \\ \epsilon, & A\epsilon^3 \end{pmatrix}$	adb
20	$\epsilon^2 M_2 = \begin{pmatrix} 0, & \epsilon^4 \\ \epsilon^2, & A\epsilon^4 \end{pmatrix}$	acb
21	$\epsilon^3 M_2 = \begin{pmatrix} 0, & 1 \\ \epsilon^3, & A \end{pmatrix}$	bce
22	$\epsilon^4 M_2 = \begin{pmatrix} 0, & \epsilon \\ \epsilon^4, & A\epsilon \end{pmatrix}$	abc
23	$\epsilon M_3 = \begin{pmatrix} 0, & \epsilon^4 \\ \epsilon, & A\epsilon^4 \end{pmatrix}$	bce
24	$\epsilon^2 M_3 = \begin{pmatrix} 0, & 1 \\ \epsilon^2, & A \end{pmatrix}$	cde
25	$\epsilon^3 M_3 = \begin{pmatrix} 0, & \epsilon \\ \epsilon^3, & A\epsilon \end{pmatrix}$	acd
26	$\epsilon^4 M_3 = \begin{pmatrix} 0, & \epsilon^2 \\ \epsilon^4, & A\epsilon^2 \end{pmatrix}$	abd
27	$\epsilon M_4 = \begin{pmatrix} 0, & 1 \\ \epsilon, & A \end{pmatrix}$	aec
28	$\epsilon^2 M_4 = \begin{pmatrix} 0, & \epsilon \\ \epsilon^2, & A\epsilon \end{pmatrix}$	bcd
29	$\epsilon^3 M_4 = \begin{pmatrix} 0, & \epsilon^2 \\ \epsilon^3, & A\epsilon^2 \end{pmatrix}$	abc
30	$\epsilon^4 M_4 = \begin{pmatrix} 0, & \epsilon^3 \\ \epsilon^4, & A\epsilon^3 \end{pmatrix}$	aed

No.	Matrix	Substitution
31	$\epsilon N_0 = \begin{pmatrix} 0, & \epsilon \\ -1, & -A\epsilon \end{pmatrix}$	bcd
32	$\epsilon N_1 = \begin{pmatrix} 0, & \epsilon^2 \\ -\epsilon^4, & -A\epsilon^2 \end{pmatrix}$	bde
33	$\epsilon^2 N_0 = \begin{pmatrix} 0, & \epsilon^2 \\ -\epsilon, & -A\epsilon^2 \end{pmatrix}$	acb
34	$\epsilon^2 N_1 = \begin{pmatrix} 0, & \epsilon^3 \\ -\epsilon^2, & -A\epsilon^3 \end{pmatrix}$	ade
35	$\epsilon^3 N_0 = \begin{pmatrix} 0, & \epsilon^3 \\ -\epsilon^2, & -A\epsilon^3 \end{pmatrix}$	ace
36	$\epsilon^3 N_1 = \begin{pmatrix} 0, & \epsilon^4 \\ -\epsilon^3, & -A\epsilon^4 \end{pmatrix}$	ace
37	$\epsilon^4 N_0 = \begin{pmatrix} 0, & \epsilon^4 \\ -\epsilon^3, & -A\epsilon^4 \end{pmatrix}$	aed
38	$\epsilon^4 N_1 = \begin{pmatrix} 0, & 1 \\ -\epsilon^4, & -A \end{pmatrix}$	$adceb$
39	$\epsilon N_2 = \begin{pmatrix} 0, & \epsilon^3 \\ -\epsilon, & -A\epsilon^3 \end{pmatrix}$	$abecd$
40	$\epsilon^2 N_2 = \begin{pmatrix} 0, & \epsilon^4 \\ -\epsilon^2, & -A\epsilon^4 \end{pmatrix}$	$acbde$
41	$\epsilon^3 N_2 = \begin{pmatrix} 0, & 1 \\ -\epsilon^3, & -A \end{pmatrix}$	$adecb$
42	$\epsilon^4 N_2 = \begin{pmatrix} 0, & \epsilon \\ -\epsilon^4, & -A\epsilon \end{pmatrix}$	$acdeb$
43	$\epsilon N_3 = \begin{pmatrix} 0, & \epsilon^4 \\ -\epsilon, & -A\epsilon^4 \end{pmatrix}$	$acebd$
44	$\epsilon^2 N_3 = \begin{pmatrix} 0, & 1 \\ -\epsilon^2, & -A \end{pmatrix}$	$abdce$
45	$\epsilon^3 N_3 = \begin{pmatrix} 0, & \epsilon \\ -\epsilon^3, & -A\epsilon \end{pmatrix}$	$adcbe$

No.	Matrix	Substitution
46	$\epsilon^4 N_3 = \begin{pmatrix} 0, & \epsilon^2 \\ -\epsilon^4, & -A\epsilon^2 \end{pmatrix}$	$aebcd$
47	$\epsilon N_4 = \begin{pmatrix} 0, & 1 \\ -\epsilon, & -A \end{pmatrix}$	$abced$
48	$\epsilon^2 N_4 = \begin{pmatrix} 0, & \epsilon \\ -\epsilon^2, & -A\epsilon \end{pmatrix}$	$abedc$
49	$\epsilon^3 N_4 = \begin{pmatrix} 0, & \epsilon^2 \\ -\epsilon^3, & -A\epsilon^2 \end{pmatrix}$	$adbec$
50	$\epsilon^4 N_4 = \begin{pmatrix} 0, & \epsilon^3 \\ -\epsilon^4, & -A\epsilon^3 \end{pmatrix}$	$aecdb$
51	$\epsilon^2 N_1 = \begin{pmatrix} 0, & \epsilon^3 \\ -\epsilon^2, & -A\epsilon^3 \end{pmatrix}$	$acbed$
52	$\epsilon^3 N_1 = \begin{pmatrix} 0, & \epsilon^4 \\ -\epsilon^3, & -A\epsilon^4 \end{pmatrix}$	$aedbc$
53	$\epsilon^4 N_1 = \begin{pmatrix} 0, & 1 \\ -\epsilon^4, & -A \end{pmatrix}$	$abcde$
54	$\epsilon N_2 = \begin{pmatrix} 0, & \epsilon^3 \\ -\epsilon, & -A\epsilon^3 \end{pmatrix}$	$aecbd$
55	$\epsilon^2 N_2 = \begin{pmatrix} 0, & \epsilon^4 \\ -\epsilon^2, & -A\epsilon^4 \end{pmatrix}$	$acdbe$
56	$\epsilon^3 N_2 = \begin{pmatrix} 0, & 1 \\ -\epsilon^3, & -A \end{pmatrix}$	$abdec$
57	$\epsilon^4 N_2 = \begin{pmatrix} 0, & \epsilon \\ -\epsilon^4, & -A\epsilon \end{pmatrix}$	$adbce$
58	$\epsilon N_3 = \begin{pmatrix} 0, & \epsilon^4 \\ -\epsilon, & -A\epsilon^4 \end{pmatrix}$	$aebdc$
59	$\epsilon^2 N_3 = \begin{pmatrix} 0, & 1 \\ -\epsilon^2, & -A \end{pmatrix}$	$adebc$
60	$\epsilon^3 N_3 = \begin{pmatrix} 0, & \epsilon \\ -\epsilon^3, & -A\epsilon \end{pmatrix}$	$aedcb$

As an example of the composition of the forms, take 56, 57: we have

$$56 = adbce \cdot abdec,$$
$$57 = aebdc \cdot abdec,$$

whence

$$56 \cdot 57 = cde = 24,$$

and we have

$$56 \cdot 57 = \begin{pmatrix} -\epsilon^2, & A\epsilon \\ A\epsilon^3, & \epsilon^4 \end{pmatrix} \begin{pmatrix} \epsilon A, & \epsilon^4 \\ -\epsilon^3, & -A \end{pmatrix},$$

viz. taking out the factor $A(\epsilon^2 - \epsilon^4)$, we ought to have

$$\frac{1}{A(\epsilon^2-\epsilon^4)} \begin{pmatrix} -A, & 1 + A\epsilon^3 \\ A(1-\epsilon^2), & A(\epsilon^2-\epsilon^4) \end{pmatrix} = \frac{1}{A(\epsilon^2-\epsilon^4)} \begin{pmatrix} -A\epsilon^4, & A(\epsilon^2-\epsilon^4) \\ A(\epsilon^2-\epsilon^4), & \epsilon^4 + A\epsilon \end{pmatrix},$$

a relation which will be identically true if only

$$\begin{aligned} -A &= -A\epsilon^4, \\ 1 + A\epsilon^3 &= A(\epsilon^2 - \epsilon^4), \\ A(1-\epsilon^2) &= A(\epsilon^2 - \epsilon^4). \end{aligned}$$

These are

$$\begin{aligned} 1 &= A^2(1 - \epsilon^2 - \epsilon^3), \\ \epsilon^4 &= A^2(-\epsilon - \epsilon^2 + \epsilon^4 + 1), \end{aligned}$$

which are satisfied in virtue of $A = \epsilon + \epsilon^4$, $A^2 = \epsilon^2 + \epsilon^3 + 2 = 1 + A'$, where $A' = \epsilon^2 + \epsilon^3 = -1$.

The invariants of the icosahedral group are connected with the form

$$f = y_1 y_2 (y_1^{10} + 11 y_1^5 y_2^5 - y_2^{10}),$$

which remains unaltered by the substitutions. Observe that we have $AB = -1$, so that, omitting common factors, the substitutions are

$$\begin{array}{lll} y_1, \quad y_2 & \text{multiplied by} & Ay_1 + y_2, \quad y_1 - Ay_2, \\ x_1, \quad x_2 & \text{multiplied by} & -(x_1 - Ax_2), \quad Ax_1 + x_2, \end{array}$$

where it is to be noticed that (x_1, x_2) are neither cogredient nor contragredient with (y_1, y_2); but they are what may be termed *sociogredient*, viz. the substitution for (y_1, y_2) determines uniquely that for (x_1, x_2).

The verification of the invariance of f might be effected rather more simply by means of the last-mentioned forms, but it is interesting to use the original forms

$$\begin{array}{lll} y_1, \quad y_2 & \text{multiplied by} & Ay_1 + y_2, \quad y_1 - Ay_2, \\ x_1, \quad x_2 & \text{multiplied by} & Bx_1 + x_2, \quad x_1 - Bx_2. \end{array}$$

Making the substitution, the whole coefficient of y_1^3 is found to be

$$\begin{aligned} &-A^3B^3 - 3A^4B + 8AB^2 + 2B^3 \\ &\qquad + (A^3 - 2A^3B^2 + 3A^2B^4 + 8AB - B^3 + \cdots)x^2 \\ &\qquad\qquad + (-A^3B - A^2B^3 + A \cdot B^4)x^2, \end{aligned}$$

where, reducing by $AB = -1$, the several coefficients are

$$\begin{aligned} A + A^4 - B^2 - B &= 0, \\ B &= 0, \\ &= 0, \end{aligned}$$

in virtue of the relations $A^2 + A = 1$, $B^2 + B = 1$. There are similar reductions for the coefficients of $y_1^2 y_2$, $y_1 y_2^2$ and y_2^3; and the whole thus becomes $= 5(A - B)$, multiplied by

$$\begin{vmatrix} y_1^3 & y_1^2 y_2 & y_1 y_2^2 & y_2^3 \\ -1 & A & -B & 1 \end{vmatrix};$$

but there is a denominator $-A'^3B'^3$, and we have $A'B' = B - A$, or this denominator is $(A-B)^3$, $= (A-B)(A-B)^2$, where

$$(A-B)^2 = A^2 - 2AB + B^2 = 1 + 2 + 1 = 4 - 5 = 5,$$

viz. the denominator is $= 5(A-B)$. Thus the new value of f is equal to its original value, which is the theorem in question.

0.17 966. Note on a Memoir in Smith's Collected Papers

[From *the Bulletin of the American Mathematical Society*, Ser. 2, vol. I. (1895), pp. 94—96.]

Among the most noticeable papers in the Collected Mathematical Papers of H. J. S. Smith we have the hitherto unpublished "Memoir on the Theta and Omega Functions," XLIII. (vol. II. pp. 415—623), written in connexion with Dr Glaisher's Tables of the Theta-Functions and originally intended as an Introduction thereto. It appears that in 1873 or 1874 Dr Glaisher asked him, as a member of the British Association committee for the calculation of the Tables, whether he would contribute an Introduction. His reply was that he did not see his way to writing anything appropriate to the tables themselves, but that he "could say something with respect to the constants at the head of the pages." These constants were K, K', E, J, J', &c., the numerical values whereof were given for every minute of the modular angle. The memoir grew in extent, and it was finally decided that it should follow these yet unpublished tables with the before-mentioned title, "Memoir on the Theta and Omega Functions," but fortunately it has at length appeared in the Collected Papers as above.

In explanation of the title and scope of the memoir, it will be remembered that the Theta-Functions are functions of two arguments, x and q; so that giving to x the value zero or any numerical value, or any value depending on that of q, we obtain a series of functions containing the single argument q, or writing as usual $q = e^{i\pi\omega}$, say the single argument ω; and in the memoir, the attention is directed chiefly but not exclusively to these functions of a single argument which are termed Omega Functions. The functions chiefly considered under this designation are Hermite's functions $\phi\omega$, $\psi\omega$, $\chi\omega$, which represent the values of $\sqrt{k}$, $\sqrt{k'}$, and $1/\sqrt{kk'}$ considered as functions of $q = e^{i\pi\omega}$. To fix the ideas, it may be mentioned that the actual values (in one of their very numerous forms) are

$$\phi\omega = 2q^{1/4}\frac{1+q^{2.2}+q^{3.3}+q^{4.4}+\cdots}{1+2q^{1.1}+2q^{2.2}+2q^{3.3}+\cdots},$$
$$\psi\omega = \frac{1-2q^{1.1}+2q^{2.2}-2q^{3.3}+\cdots}{1+2q^{1.1}+2q^{2.2}+2q^{3.3}+\cdots},$$
$$\chi\omega = 2q^{1/4}\frac{1+q^{1.2}+q^{2.3}+q^{3.4}+\cdots}{1-2q^{1.1}+2q^{2.2}-2q^{3.3}+\cdots},$$

each of them a one-valued function of ω: any rational and integral function of $(\phi\omega, \psi\omega, \chi\omega)$ is termed a *Modular Function.* It is right to add that the definitions extend only to those values of q for which the series are convergent, or (what is the same thing) to ω regarded as a point must be situate within a certain region of the upper infinite half-plane $y = +\infty$.

The requisite formulae for the Theta-Functions are obtained from Jacobi's fundamental formula for the multiplication of four Theta-Functions; and the Elliptic Functions are introduced by means of their definitions in terms of the Theta-Functions: and the whole theory of Elliptic Functions is thus brought into connexion with the Theta and Omega Functions. The theory of Transformation depends in a great measure on the arithmetical and geometrical theory of binary matrices, of which the constituents are integral numbers; this theory plays an extensive part throughout the memoir.

An abstract of the contents of the memoir is as follows:

Arts.	
1—14.	Definitions and Elementary Properties of the Theta, Omega, and Elliptic Functions.
15—23.	Arithmetical Theory of Binary Matrices.
24—34.	The Transformation of the Theta and Omega Functions.
35—45.	Geometrical Representation of Binary Quadratic Forms.
46—51.	Geometrical Representation of the Modular Functions $3\chi\omega$ and $\Psi\omega$ $(=\phi\chi\cdot\psi^8\omega)$.
52—58.	The Modular Equation.
59—62.	The Equation of the Multiplier.
63—73.	The Modular Curves.
74—82.	Theory of the Modular Functions $\phi\chi\omega$ and $\psi^3\omega$.
83—88.	Theory of the Modular Function $T\omega=(1-\chi^{16}\omega)^3-\chi^{48}\omega$.
89—90.	The Differential Equation of the Modular Equations and Curves (this last section somewhat incomplete).

A good deal of the same ground is gone over in Weber's *Elliptische Functionen und Algebraische Zahlen* (8vo. Brunswick, 1891), a work which exhibits in a very compendious form the higher parts of the theory of Elliptic Functions, and which well deserves to be carefully studied.

0.18 967. An Elementary Treatise on Elliptic Functions

[From *the Messenger of Mathematics*, vol. XXIV. (1895), pp. 1—2.]

The first edition was published in 1876; the second edition was in the press at the time of Professor Cayley's death in 1895, and it was published in the course of that year.

END OF VOL. XIII.

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Note on Pages 591–594

Pages 591–594 of the original volume are blank or contain library circulation notices and are omitted from this transcription.